



# 第五届数学建模国际赛 小美赛0奖论文集

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主办方：数学中国 全球建模能力认证中心



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### **Problem A(MCM): Predictions of Ozone Depletion**

Ozone depletion includes several phenomena observed since the late 1970s: a steady decline in the total amount of ozone in Earth's stratosphere (the ozone layer), and a much larger springtime decrease in stratospheric ozone — as known as ozone hole — around Earth's polar regions. In addition to these well-known stratospheric phenomena, there are also springtime polar tropospheric ozone depletion events.

It is believed that the primary cause of ozone depletion is the presence of chlorine-containing source gases, including CFCs and related halocarbons, nitrous oxide, etc. In 1985, 20 nations signed the Vienna Convention for the Protection of the Ozone Layer. In 1987, representatives from 43 nations signed the Montreal Protocol. At Montreal, the participants agreed to freeze production of CFCs at 1986 levels and to reduce production by 50 percent by 1999.

Since the adoption and strengthening of the Montreal Protocol has led to reductions in the emissions of CFCs, atmospheric concentrations of the most-significant compounds have been declining. Please develop a mathematical model to predict the ozone levels over the next 50 years.



### **Problem B(MCM): The Direct Metro Lines**

In current metro network of a megapolis, it usually takes a long time to transport passengers between two far apart stations. We may suggest that set a direct shuttle train between two long-distance stations to save time of the longhaul passengers.

**Part I.** Please develop a mathematical framework to optimize the choice of the end stations, and evaluate the expected result of the suggest.

**Part II.** Explore any advantages you may achieve in terms of economic and social benefits if you adjust the interval between trains or build a special direct line.

**Part III.** Write a two-page letter to the Mayor of your city summarizing your analysis. Include your train schedule as the second page of your letter.

**Problem C(ICM): The War on Superbugs**

These days there's a lot of talk about our antibiotic resistance crisis. Bacteria that have evolved to withstand antibiotics kill 700,000 people each year, and ever more powerful strains are spreading around the world. Researchers are worried that we will enter a post-antibiotic age, in which we are infected by bacteria that can defeat every drug medicine has to offer. Next week, the United Nations will convene a high-level meeting to coordinate the global fight against these invisible enemies.

Although mostly forgotten today, Batten and other like-minded scientists were warning of a coming antibiotic crisis over 60 years ago. They were right, but they were ignored. Their failure offers some important lessons for today's new crusaders.

Just a year after the World War II's end, though, the discoverer of penicillin warned that it might become useless. Accepting his Nobel Prize in 1945, Alexander Fleming mapped out how the failure could happen. "There is the danger that the ignorant man may easily underdose himself," Fleming said, "and by exposing his microbes to non-lethal quantities of the drug make them resistant."

It's not that other scientists denied the reality of evolution. It just didn't seem to matter very much. Scientists at the time were discovering even more antibiotics. If bacteria became resistant to penicillin, they'd always be able to switch to a different weapon.

In a struggle that lasted over a decade, reformers pushed the Food and Drug Administration to regulate antibiotics more strictly. And they succeeded, getting a number of the offending drugs off the market, with more rigorous rules for allowing new ones onto it.

Meanwhile, the true threat of antibiotic resistance was growing even clearer. In the early 1960s, scientists discovered that once a resistance gene evolved in one strain of bacteria, the microbes could donate it to other strains. Microbes could load these donated genes together on a single piece of DNA, accelerating the spread of resistance even further.

In 1966, an article in the magazine *Look* described this new vision of antibiotic resistance with near-apocalyptic rhetoric. "Are Germs Winning the War against People?" ran the headline. The article referred to these germs with a new nickname: superbug.

Yet organizing a fight against antibiotic resistance proved much harder than against ineffective or dangerous drugs. The World Health Organization organized meetings about antibiotic resistance as early as the 1950s, but they fizzled out. The experts who came to the meetings got bogged down in arguments over how to measure resistance and what level to consider a threat to public

health.

It was also harder to figure out what to do to fight antibiotic resistance. To get bad drugs off the market, reformers had just one target: the FDA. To get good drugs used less often, reformers had to reach many different targets at once: doctors, hospital administrators, patients, governments, pharmaceutical companies — even to farmers who began feeding antibiotics to livestock to get them to grow bigger.

Making matters worse, the campaign to regulate antibiotics had left doctors angry. They were saying, "Who's the FDA to be taking these drugs away? I've used these drugs for 30 years and my patients seem to be getting better."

The UN has asked your team, the ICM-FDA to help develop a better understanding of the factors involved with antibiotic resistance crisis.

Task 1: Develop a model that provides a interest chain for use of antibiotics and the sensitivity of parties. You may need to consider the dynamic nature of the factors that affect both supply and demand in your modeling process.

Task 2: Predict the development trend towards the crisis under perfect competitive markets.

Task 3: Design a portable management system for the use of antibiotics is the most urgent affair for superbug. ICM-FDA has been asked to attend a policy strategy meeting where your team is asked to write a report on your model and propose a set of policies.

Task 4: Develop a reward policy for formers to encourage them to raise cattle without antibiotic drugs.

**\*Your ICM submission should consist of a 1 page Summary Sheet and your solution cannot exceed 20 pages for a maximum of 21 pages.**

# Predictions of Ozone Depletion

## Abstract

*This paper deals with the prediction of ozone depletion. In the process of formation and change of ozone hole, the process of atmospheric power transmission plays the most important role. The aerodynamic effect makes the ozone concentration in the local area decrease obviously, the chlorofluorocarbons gas compound causes the ozone content in the atmosphere to decrease, Circulation regulates the ozone concentration between the high and low latitudes towards equilibrium.*

*The study in this paper can be divided into two parts. First, annual mean ozone data of global atmosphere was used to analyses total atmospheric ozone features of the spatial and temporal distribution. Singular Spectrum Analysis (SSA) was applied to derive the long-term variability of total column ozone at the global scale during the time period of 1979-2015. Based on the divided long-term variability, the total column ozone variability in the past decades has been divided into three different phases, with ozone depletion during 1979-1994 and transition during 1995-2005 as well as ozone recovery since 2006.*

*The second part studies the prediction of total atmospheric ozone. Assuming a constant response of total column ozone to the implementation of Montreal Protocol in the coming decades, forecasted results from DHR models indicate that the ozone layer in the Northern Hemisphere will recover to its 1980's level by about 2030 while approximately by 2040 in the Southern Hemisphere. Furthermore, the model was established based on the combination of wavelet decomposition and advanced Least Square Support Vector Machine(LSSVM) Regression. This can be approached in three steps: The observations were decomposed into several different frequencies signal subsets; the independent prediction models of decomposed signals with Takens delay embedding theorem and Least—Squares Support Vector Machine(LSSVM) were set up; independent predicted results were integrated as the final Prediction with wavelet reconstruction.*

**Key words:** *Atmospheric Motion Equation, Wavelet Decomposition, Least Squares Support Vector Machine, Time Series Prediction, SSA, DHR.*

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# **I. Introduction**

## **1.1 Problem Description**

*Ozone depletion includes several phenomena observed since the late 1970s: a steady decline in the total amount of ozone in Earth's stratosphere (the ozone layer), and a much larger springtime decrease in stratospheric ozone — as known as ozone hole — around Earth's polar regions. In addition to these well-known stratospheric phenomena, there are also springtime polar tropospheric ozone depletion events.*

*It is believed that the primary cause of ozone depletion is the presence of chlorine-containing source gases, including CFCs and related halocarbons, nitrous oxide, etc.*

*In 1985, 20 nations signed the Vienna Convention for the Protection of the Ozone Layer. In 1987, representatives from 43 nations signed the Montreal Protocol. At Montreal, the participants agreed to freeze production of CFCs at 1986 levels and to reduce production by 50 percent by 1999.*

*Since the adoption and strengthening of the Montreal Protocol has led to reductions in the emissions of CFCs, atmospheric concentrations of the most-significant compounds have been declining.*

## **1.2 Terminology and Definitions**

### **What is stratospheric ozone ?**

*Atmospheric ozone that is relatively concentrated in the lower stratosphere in a layer between 9 and 18 miles (15 and 30 kilometers) above the earth's surface, and plays a critical role for the biosphere by absorbing the damaging ultraviolet radiation with wavelengths 320 nanometers and lower. Also, known as ozone layer*

### **What is The Dobson Unit (DU)?**

*The Dobson Unit (DU) is the unit of measure for total ozone. If you were to take all the ozone in a column of air stretching from the surface of the earth to space, and bring all that ozone to standard temperature (0 °Celsius) and pressure (1013.25 millibars, or one atmosphere, or “atm”), the column would be about 0.3 centimeters thick. Thus, the total ozone would be 0.3 atm-cm. To make the units easier to work with, the “Dobson Unit” is defined to be 0.001 atm-cm. Our 0.3 atm-cm would be 300 DU.*

### **What is the polar vortex?**

*The polar vortex is a region in the stratosphere that is contained by the strong jet stream that circles the pole. The vortex spins up in the autumn as the winds increase and then breaks up when the winds die down in the spring. Inside the vortex is a coherent air mass*

*that is colder and that has a chemical composition that is substantially different from the air outside of it in the mid latitudes.*

### **1.3 Stratospheric “good” ozone and Tropospheric “bad” ozone**

*Ozone in the stratosphere absorbs most of the ultraviolet radiation from the Sun. Without ozone, the Sun’s intense UV radiation would sterilize the Earth’s surface. Ozone screens all of the most energetic, UV-c, radiation, and most of the UV-b radiation. Ozone only screens about half of the UV-a radiation. Excessive UV-b and UV-a radiation can cause sunburn and can lead to skin cancer and eye damage.*

*Although ozone high up in the stratosphere provides a shield to protect life on Earth, direct contact with ozone is harmful to both plants and animals (including humans). Ground-level, “bad,” ozone forms when nitrogen oxide gases from vehicle and industrial emissions react with volatile organic compounds (carbon-containing chemicals that evaporate easily into the air, such as paint thinners).*

*In the troposphere near the Earth’s surface, the natural concentration of ozone is about 10 parts per billion (0.000001 percent). According to the Environmental Protection Agency, exposure to ozone levels of greater than 70 parts per billion for 8 hours or longer is unhealthy<sup>[1]</sup>. Such concentrations occur in or near cities during periods where the atmosphere is warm and stable. The harmful effects can include throat and lung irritation or aggravation of asthma or emphysema.*

### **1.4 Our work**

*We study the effects of solar radiation, atmospheric circulation and chlorine-containing source gases on ozone depletion. In order to study the influence of atmospheric circulation on the distribution of ozone, we establish the equation of atmospheric motion. We also analyze mechanism of ozone and chlorine-containing source gases.*

## **II. Problem Analysis**

### **2.1 The Description of the Problem**

*We need to analyze factors which impact the ozone levels and make a systemic analysis of its mechanism of action. Finally, we should develop a mathematical model to predict the ozone levels over the next 50 years.*

### **2.2 Which factors impact ozone concentration?**

*The ozone levels not only relate to emission of harmful gases, but to the nature factors. Now, after look up literatures about ozone depletion, it is found that the main factors*

*impacting the ozone levels are: solar radiation, atmospheric circulation and chlorine-containing source gases, including CFCs and related halocarbons, nitrous oxide, etc.*

*According to the study, the ozone concentration responds primarily to solar radiation in the atmosphere, because the formation of ozone is closely related to solar radiation. In the stratosphere, sunlight breaks apart an oxygen molecule ( $O_2$ ), producing two oxygen (O) atoms, which then attack an oxygen molecule next to it to make ozone. Ozone radiation is broken down into an oxygen molecule and an oxygen atom or attack an active oxygen atom to make two oxygen molecules by intense ultraviolet.*

*The poleward transport of ozone due to general circulation of atmosphere and effect of stratospheric winds. During the polar winter, there is no solar radiation and the poleward transport of local ozone absolutely due to the winds from the equator to the polar. Overall, the distribution of global ozone become balanced because of its function. There are strong seasonal vortexes, namely the polar vortex in the Polar Regions. The polar vortex of the Antarctic is stronger than that of the North Pole because of its ice field. A strong cold air mass forms a closed air system around the Antarctic, which appears as a polar vortex at the beginning of winter on the Antarctic. Around November, when the temperature rises, the polar vortex will collapse. The polar vortex blocked the transport process of ozone from the equator to the polar. A large amount of ozone is broken up by mechanism of positive feedback between the polar vortex and stratospheric cloud of the Polar Regions.*

*In the stratosphere, Freon release a chlorine atom under the effect of strong ultraviolet radiation, which then constantly consume and disintegrate ozone.*

## **2.3 How to predict the ozone levels over the next 50 years?**

*This can be approached in three steps: The observations were decomposed into several different frequencies signal subsets; the independent prediction models of decomposed signals with Takens delay embedding theorem and Least—Squares Support Vector Machine(LSSVM) were set up; independent predicted results were integrated as the final Prediction with wavelet reconstruction.*

# **III. Models**

## **3.1 Atmospheric dynamic process and ozone hole**

*The ozone changes are the combination of chemical destruction and natural variations. The decline in ozone is bound to reduce the absorption of solar radiation in the stratosphere, increase the ultraviolet radiation reaching the ground, lower the*



stratospheric temperature in the polar and high latitudes and it cause warming of the troposphere. Besides it also affects the global atmospheric circulation<sup>[1]</sup>.

Therefore, despite the worldwide reduction in the amount of atmospheric ozone is associated with chemical substances to destroy the ozone process of Freon group participation represented. But, the atmospheric dynamics also play a significant role in the change of regional atmospheric ozone<sup>[3]</sup>. The process of power transmission plays a major role and the chemical process partially compensates for ozone depletion caused by the transport process in the formation and variation of the ozone trough.

### 3.1.1 The temporal and Spatial Distributions of Atmospheric Ozone

The atmosphere is driven by an exogenous source and it is dissipated by a gravitational field. It is a non-uniform rotating<sup>[7]</sup> fluid system. Its motion is controlled by a complex set of equations of momentum, mass and energy nonlinearity. So, the motion of the atmosphere is non-linear in nature.

It is generally believed that the area in which ozone concentrations drop below the historical threshold of 220 Dobson Units, it is considered to be the emergence of the ozone hole. According to this standard, the total spring ozone over the Antarctic although the beginning of the late 1970s markedly reduced, but in October 1982 was the first time less than 220 Dobson unit. In the last five years, the ozone levels in the ozone hole above the Antarctic are below 120 Dobson units.

In this paper, the TOMS satellite data provided by NASA is used.

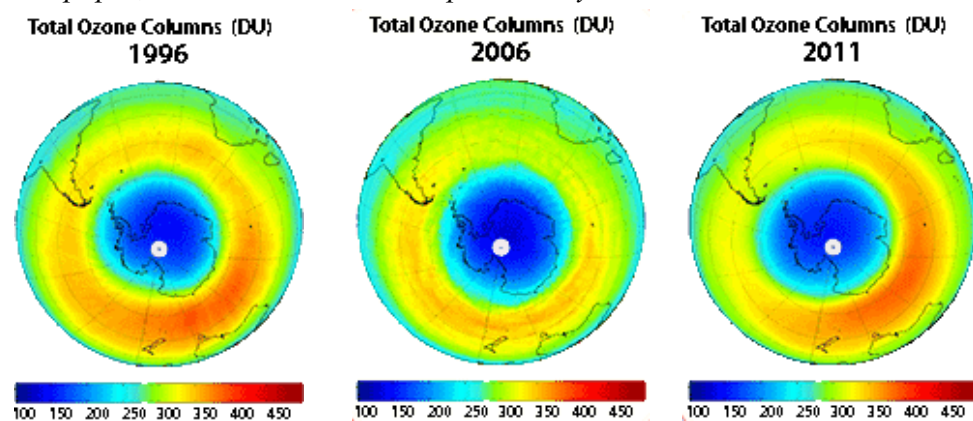


Figure1. The changes in the Antarctic ozone layer

The TOMS data were used to analyze the seasonal variation of total atmospheric ozone in the polar regions<sup>[1]</sup>. The results show that the decrease of ozone in the polar regions is the fastest in spring, and the decrease of atmospheric ozone will result in the difference of the temperature difference between high and low layer.

The strong absorption of ozone by solar ultraviolet radiation is the main heat source of

stratospheric atmospheric movement. The presence of stratospheric ozone plays a decisive role in the stratospheric temperature, air pressure and wind.

In the stratosphere, it is modulated by the ozone to change the conductivity of the middle atmosphere. The flare activity may first affect the ozone change, and then affect the temperature change. Through the interaction between the upper and lower layers, it will affect the global circulation of the atmosphere <sup>[6]</sup>. When the total amount of ozone is reduced, the thermal budgets of the northern and southern hemispheres will inevitably be changed, which will affect the thermal conditions of the atmosphere and the ground in the northern and southern hemispheres, thus affecting the global atmospheric circulation and climate change.

### 3.1.2 The dynamic equations of the atmosphere

#### 1.The original atmospheric dynamics equation

The dynamic equations of the atmosphere <sup>[4]</sup> can be written in this form:

$$\frac{dx}{dt} = X(x, y, \varepsilon) \quad (1)$$

$x$  : It represents a characteristic quantity of the atmospheric state.

$y$  : It represents the role of the outside world in the atmosphere

$\varepsilon$  : The small parameter variable of the relative change rate of the atmosphere.

The initial value of  $y(t_0)$  for the outside world is implied by the assumption that it is invariant in the forecast time interval  $\Delta t$  and a linear regression equation can be established to predict the time-averaged atmospheric characteristics  $\hat{x}(t_0 + \Delta t)$ :

$$\hat{x}(t_0 + \Delta t) = Ay(t_0) \quad (2)$$

Introducing a slow change time  $\tau = \varepsilon \cdot t$ , then the (1) can be transformed into:

$$\varepsilon \frac{dx}{d\tau} = X(x, y, \varepsilon) \quad (3)$$

Assume that the atmosphere adapts to the external action at time  $\varepsilon^{-1}$  which is not long period free oscillations. Then, the external role has not changed. So, with respect to the climate of the atmospheric state linearization of this (3) formula, get the regression equation such as this (2).

It is assumed that the atmospheric flow equation is a poor two nonlinear ordinary differential equations.

$$\varepsilon \frac{\partial x}{\partial \tau} = \sum_j \sum_k K_{jk}^i \delta_{jk} + \sum_j K_j^i \mu_j + y_i \quad (4)$$

$$\varepsilon \frac{\partial \delta_{ij}}{\partial \tau} = \sum_k \sum_l (K_{kl}^i \tau_{ikl} + K_{kl}^j \tau_{jkl}) + \sum_k (K_k^i \delta_{ik} + K_k^j \delta_{jk}) + y_i \mu_j + y_j \mu_i \quad (5)$$

## 2.The Atmospheric Dynamics Equations

The atmosphere is an infinite dimensional dynamical system, it's evolution can be expressed by a set of partial differential equations. Atmospheric dynamics studies and numerical weather prediction are organized around this set of equations and their various simplified forms.

*A Special Operator Equation in Hilbert Space:*

Using spherical coordinate system  $(\lambda, \theta, r)$ . Then introduce the vector function  $\varphi$ .

The complete equations of wet atmospheric motion can be transformed into the following equivalent operator equations.

$$\begin{cases} \frac{\partial \varphi}{\partial t} + [N(\varphi) + L(\varphi)]\varphi = \xi(\varphi), \\ |\varphi|_{t=0} = \varphi_0. \end{cases} \quad (6)$$

In the Hilbert space, the  $N(\varphi), L(\varphi)$  has the following properties.

The  $N(\varphi)$  is anti-companion operator. The  $L(\varphi)$  is self-adjoint operator.

The  $N(\varphi)$  is antisymmetric. The  $L(\varphi)$  is self-symmetric.

The  $L(\varphi)$  is a positive operator.

Atmospheric circulation wheel test and observation facts all show that the atmospheric circulation can have multiple equilibrium states under the same external forcing, and the equilibrium problem, that is, the boundary value problem of the stationary atmospheric motion equations<sup>[5]</sup>. So, the equation (6) can be transformed into:

$$N(\varphi)\varphi + L(\varphi)\varphi = \xi(\varphi) \quad (7)$$

The change of total ozone amount is controlled by atmospheric horizontal and vertical movement and solar radiation. And the main factor of the change of the ozone field is the direction of the air flow in the horizontal movement. It forms high ridge trough on isobaric surface. When the high-altitude trough transit area, the total amount of ozone increased, the area of high-altitude ridge transit, the total amount of ozone decreased. From the statistical results, the best negative correlation between the total ozone and 300hPa isobaric surface<sup>[6]</sup>.

The primitive equations of Navier-Stokes "shallow atmosphere" in the spherical Z coordinate system.

$$\begin{aligned}
\frac{du}{dt} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + f_v - \tilde{f}_w + (uv \cdot \operatorname{tg} \varphi - uw) / a_0 + F_u \\
\frac{dv}{dt} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} - f_u + (u^2 \operatorname{tg} \varphi - vw) / a_0 + F_v \\
\frac{dw}{dt} &= -\frac{1}{\rho} \frac{\partial p}{\partial z} - g + \tilde{f}_u + (u^2 + v^2) / a_0 + F_w
\end{aligned} \tag{8}$$

$$\begin{aligned}
\frac{d \ln p}{dt} &= \frac{-1}{1-\kappa} \nabla \cdot \vec{V} \\
\text{Possion: } d \ln T &= \kappa \cdot d \ln p \\
\frac{dq}{dt} &= 0
\end{aligned}$$

$p$  : Pressure;

$T$  : Temperature;

$q$  : Specific humidity;

$t$  : Time;

$u, v$  : horizontal wind speed;

$w$  : Vertical velocity;

$a_0$  : Earth radius;

$\lambda, \varphi$  : Latitude and longitude;

$r(a_0 + z)$  : This piece of distance to the center of the earth.

Then,

$$\begin{cases}
\delta x = a_0 \cdot \cos \varphi \delta \lambda \\
\delta y = a_0 \cdot \delta \varphi \\
\delta r = \delta z \\
f = 2\Omega \sin \varphi \\
\tilde{f} = 2\Omega \cos \varphi
\end{cases} \tag{9}$$

### 3.2 Atmospheric chemistry and the ozone hole

The ozone depletion process in the stratosphere is very complex. It is one of the key research topics in atmospheric dynamics, atmospheric chemistry and geophysics. And the chemical mechanism of ozone depletion involves hundreds of chemical reactions. The main reasons for the direct depletion of ozone is NO-NO<sub>2</sub>, HOx, Cl-ClO and Br-BrO and other catalytic cycle. While CFCs and AOX are important chemical substances that destroy ozone<sup>[9]</sup>.

### 3.2.1 The ozone depletion chemistry in the stratosphere.

Human production of chlorine-containing chemicals, such as CFCs, has added an additional factor that destroys ozone. CFCs are molecules made up of chlorine, fluorine and carbon. Because they are extremely stable molecules, CFCs do not react with other chemicals in the lower atmosphere, but exposure to ultraviolet radiation in the stratosphere breaks them apart, releasing chlorine atoms.

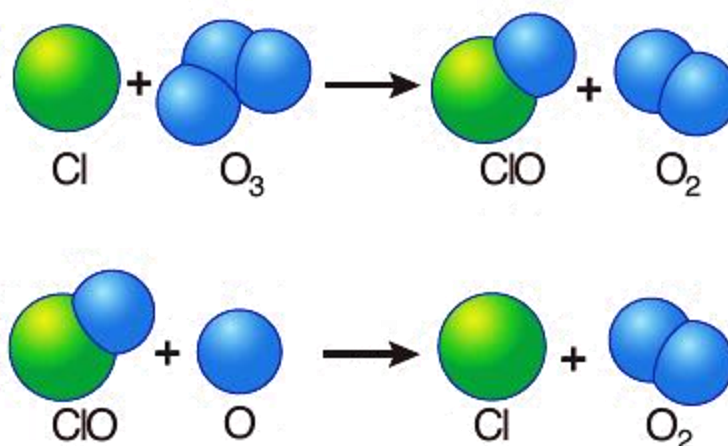


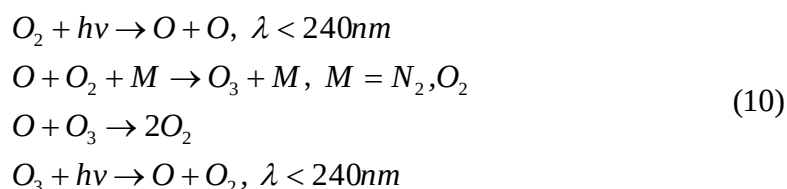
Figure2. Cl-CIO catalytic cycle

Free chlorine (Cl) atoms then react with ozone molecules, taking one oxygen atom to form chlorine monoxide (ClO) and leaving an oxygen molecule (O<sub>2</sub>). If each chlorine atom released from a CFC molecule destroyed only one ozone molecule, CFCs would pose very little threat to the ozone layer. However, when a chlorine monoxide molecule encounters a free atom of oxygen, the oxygen atom breaks up the chlorine monoxide, stealing the oxygen atom and releasing the chlorine atom back into the stratosphere to destroy another ozone molecule. These two reactions happen over and over again, so that a single atom of chlorine, acting as a catalyst, destroys many molecules (about 100,000) of ozone.

Fortunately, chlorine atoms do not remain in the stratosphere forever. Free chlorine atoms react with gases, such as methane (CH<sub>4</sub>), and get bound up into hydrogen chloride (HCl) molecules. These molecules eventually end up back in the troposphere where they are washed away by rain. Therefore, if humans stop putting CFCs and other ozone-destroying chemicals into the stratosphere, stratospheric ozone will eventually return to its earlier, higher values.

### 3.2.2 Catalytic cycle for the destruction of ozone.

*The mechanism of the generation and consumption of ozone in the stratosphere. The ground state oxygen atom and the oxygen molecule are formed by the oxygen molecule to generate the ozone molecule. The new generation of ozone molecules are photolyzed by the sun's ultraviolet light to produce oxygen atoms and oxygen molecules, or ozone molecules and oxygen atom reaction to generate oxygen molecules<sup>[9]</sup>.*



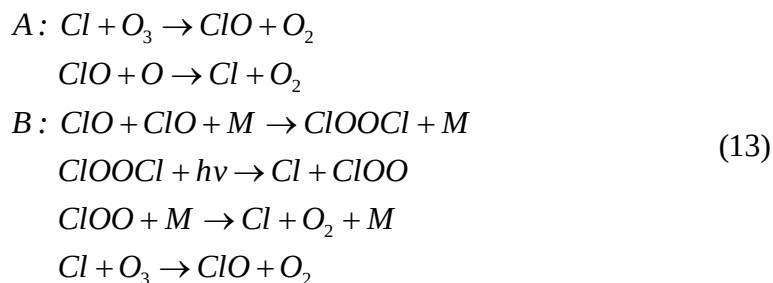
#### 1. The NO-NO<sub>2</sub> catalytic cycle



#### 2. The HOx catalytic cycle



#### 3. The Cl-CIO catalytic cycle



#### 4. The Br-BrO catalytic cycle



*In 1987, the Montreal Protocol on the Limitation of the Use of CFCs was signed under the Global Environmental Protection Treaty. From the global stratospheric ozone depletion perspective. The chemicals most damaging to ozone are chlorofluorocarbons (CFCs) and then is olfactory methane. Its ozone-depleting capacity is estimated to account for 10 per cent of global ozone depletion.*

Table1. The global pollutant emissions

| Year | CFC-11 | CH <sub>3</sub> CCl <sub>3</sub> | CFC-113 | ODGI(new) Mid-Lat |
|------|--------|----------------------------------|---------|-------------------|
| 1992 | 379    | 271                              | 69      | 81.8              |
| 1993 | 382    | 266                              | 71      | 88.4              |
| 1994 | 382    | 250                              | 72      | 93.9              |
| 1995 | 381    | 225                              | 73      | 97.6              |
| 1996 | 380    | 200                              | 73      | 99.6              |
| 1997 | 378    | 172                              | 73      | 100               |
| 1998 | 376    | 145                              | 73      | 99.2              |
| 1999 | 374    | 121                              | 72      | 98                |
| 2000 | 372    | 101                              | 72      | 96.6              |

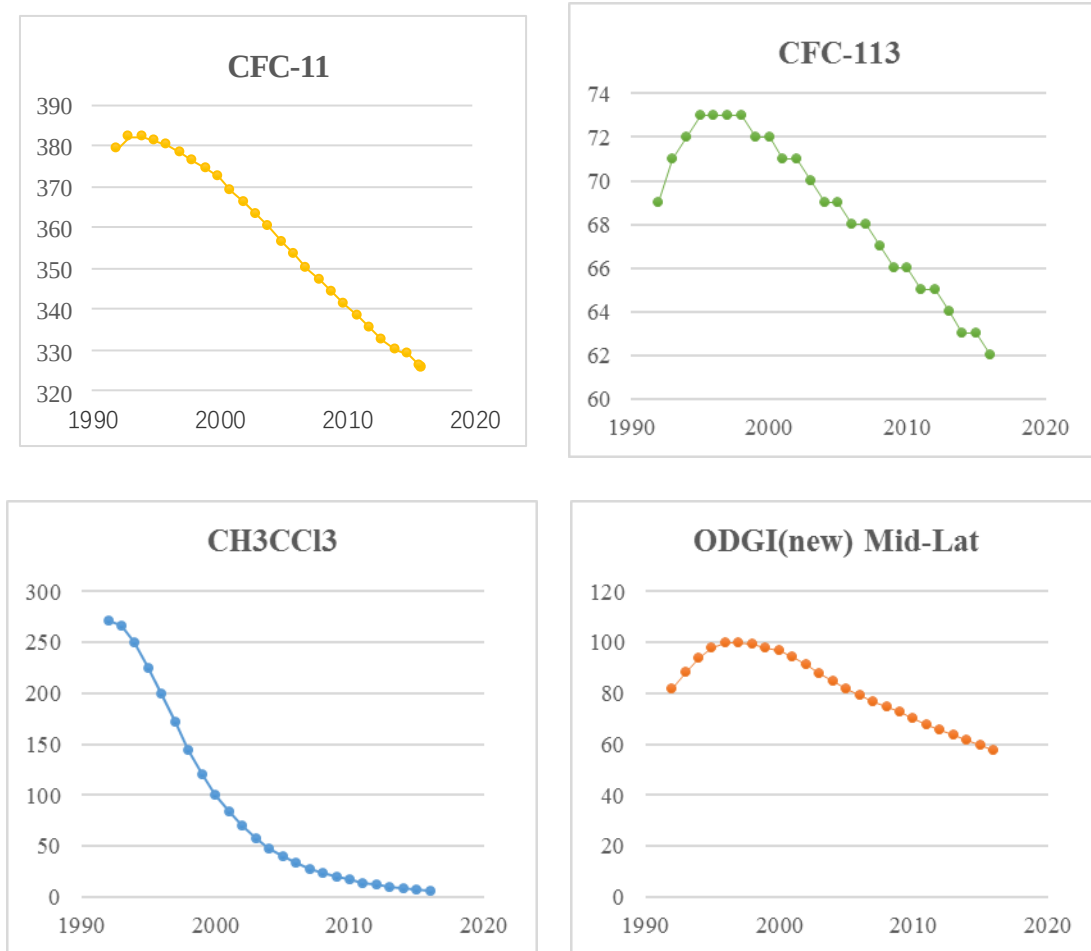


Figure3. The curve of global pollutant emissions

*After the Montreal agreement, the concentration changes of pollutants are shown in the following figure.*

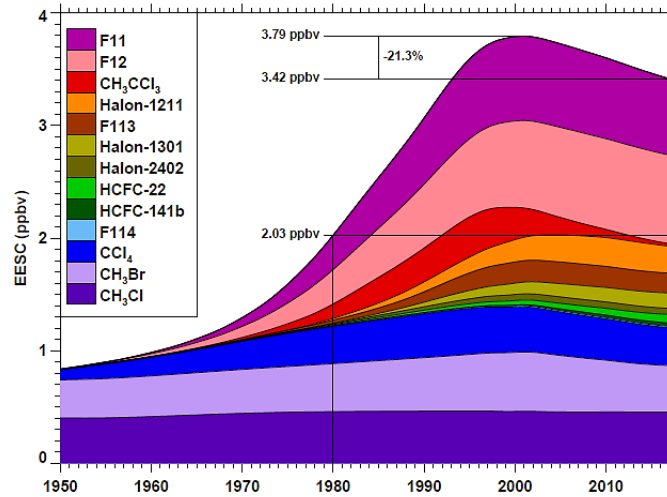


Figure4. The change of global pollutants

### 3.3 The ozone changes in the area

#### 3.3.1 The polar variation characteristics of total ozone

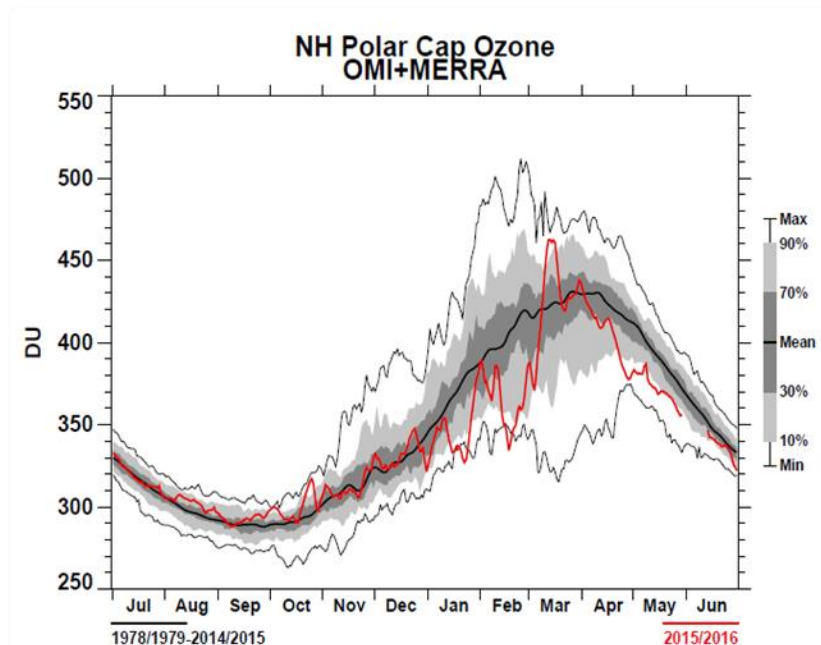


Figure5. 2015/2016 Arctic OMI and MERRA Ozone

*The lines and shading follow a progression from bottom to top as the minimum (thin black line), the 10th percentile (light gray shading), 30th percentile (dark gray shading), mean (thick black line), 70th percentile (dark gray shading), 90th percentile (light gray shading), and the maximum (thin black line) over the climatological period.*

*From the trend of seasonal evolution, the four seasons showed a downward trend. And it is the same as the North Pole. The fastest season for ozone decline occurs in spring, then in*



winter and summer, and the lowest in the autumn. And after the fall of 1985 ozone was almost unchanged. In general, the Antarctic and the Arctic have the same characteristics of the seasonal trend of ozone evolution in the Arctic, the total amount of ozone is greater than the Antarctic. And the Arctic ozone difference in the four seasons is very big, it reached 130 DU. And the difference of the total ozone amount in the four seasons is relatively small, only about 40 DU.

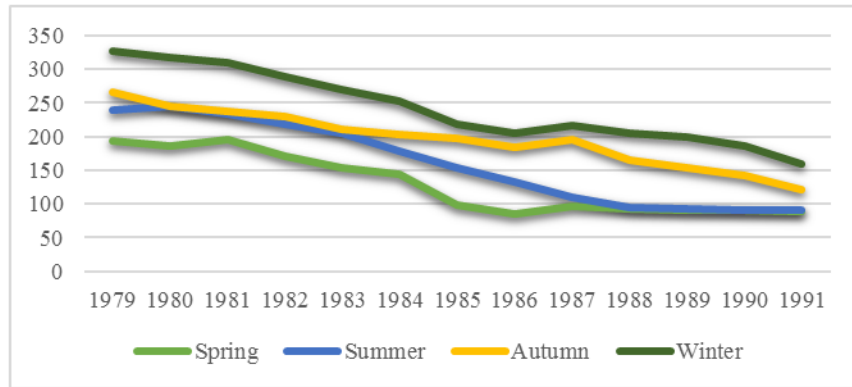


Figure6. The seasonal trends in the ozone content of the Antarctic atmosphere

In summary, the total ozone in the Antarctic and Arctic seasons are declining. One of the fastest decline in spring, then is in winter and summer, the fall of ozone is very slow. Especially after 1985, the Antarctic ozone almost unchanged in autumn. The difference of the total ozone in the Arctic is very big, and the difference of total ozone in the Antarctic is small.

It is a seasonal change in the ozone content over the Antarctic<sup>[8]</sup>. The spring of the year will fall by about 70%. With the new method, it will more accurately estimate the changes of man-made chemicals the Antarctic chlorine or bromine with time. And their effect on the ozone loss. And then calculate the size of the ozone hole.

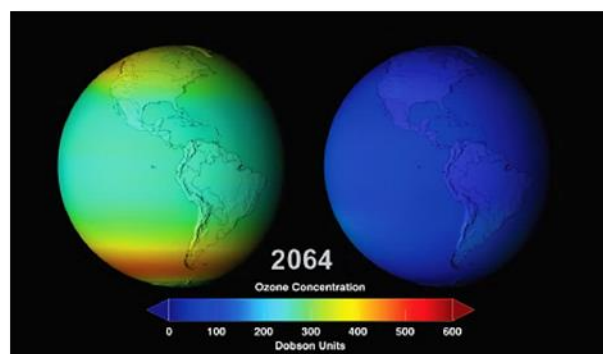


Figure7. predictions of the state of the ozone layer in 2064

These maps show computer model predictions of the state of the ozone layer in 2064 with (above left) and without (above right) the effects of international agreements to curb

ozone-destroying chemicals in the 1980s and 90s. (NASA images by the GSFC Scientific Visualization Studio.)

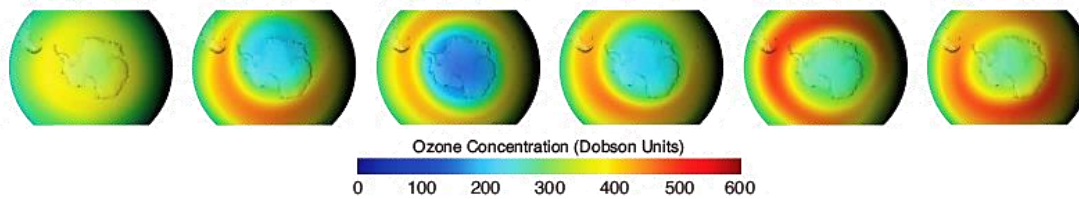


Figure8. The Antarctic ozone hole

The Antarctic ozone hole (blue areas), which first appeared in the early 1980s and peaked in the 2000s, is expected to shrink markedly by 2064. International agreements successfully mitigated the threat posed by CFCs and other ozone-destroying chemicals. (NASA images by the GSFC Scientific Visualization Studio.)

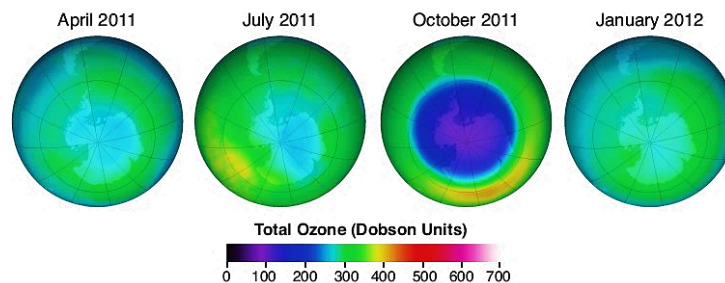


Figure9. Southern Hemisphere ozone concentration for four months

The ozone destruction continues within the polar vortex until the ozone levels approaches zero at the altitudes where reactions on the thin clouds have released chlorine atoms. Once ozone has reached such a low level, the chlorine atoms react with methane, filling the vortex with HCl. The low ozone persists until the vortex weakens and breaks apart. Then the ozone levels in the polar stratosphere begin to return to pre-September levels due to the increase in solar UV and the mixing of polar and nonpolar air.

### 3.3.2 Maximal Ozone Hole Area and Minimum Ozone

Table2. Ozone Hole Area and Minimum Ozone

| Ozone Hole Area<br>(million km <sup>2</sup> )<br>Maximum Daily |        |       | Minimum Ozone<br>(DU)<br>Minimum Daily |       |
|----------------------------------------------------------------|--------|-------|----------------------------------------|-------|
| Year                                                           | Date   | Value | Date                                   | Value |
| 1979                                                           | 17-Sep | 1.1   | 17-Sep                                 | 194   |
| 1980                                                           | 21-Sep | 3.3   | 16-Oct                                 | 192   |
| 1981                                                           | 10-Oct | 3.1   | 10-Oct                                 | 195   |
| 1982                                                           | 2-Oct  | 10.8  | 2-Nov                                  | 170   |
| 1983                                                           | 17-Oct | 12.2  | 6-Oct                                  | 154   |
| 1984                                                           | 24-Sep | 14.7  | 3-Oct                                  | 144   |
| 1985                                                           | 3-Oct  | 18.8  | 24-Oct                                 | 124   |

|             |        |      |        |     |
|-------------|--------|------|--------|-----|
| <b>1986</b> | 6-Oct  | 14.4 | 6-Oct  | 140 |
| <b>1987</b> | 29-Sep | 22.4 | 5-Oct  | 109 |
| <b>1988</b> | 20-Sep | 13.8 | 30-Sep | 162 |
| <b>1989</b> | 3-Oct  | 21.7 | 7-Oct  | 108 |
| <b>1990</b> | 19-Sep | 21.1 | 5-Oct  | 111 |
| <b>1991</b> | 4-Oct  | 22.6 | 6-Oct  | 94  |

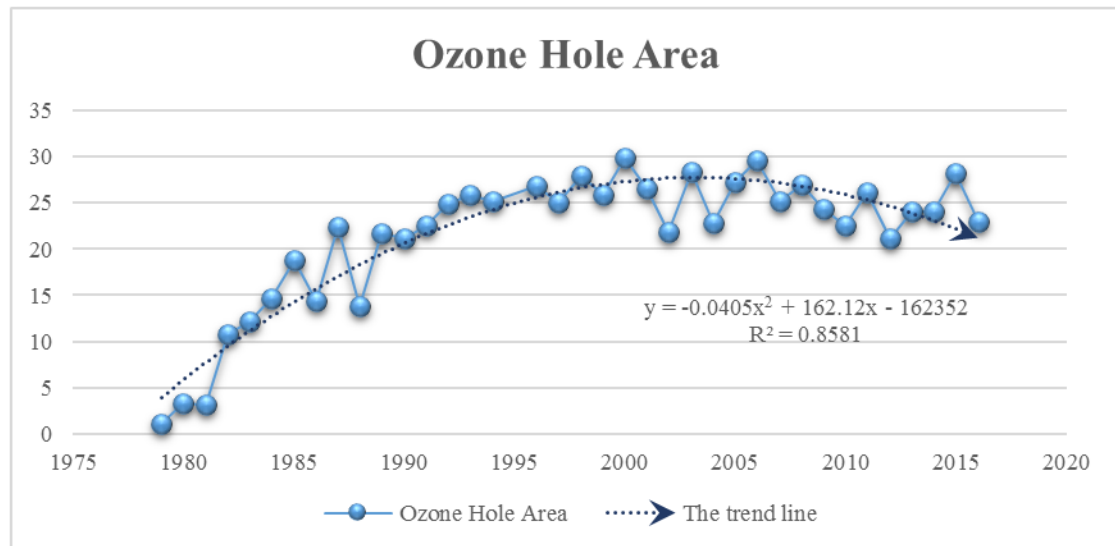


Figure10. Maximum Daily Ozone Hole Area

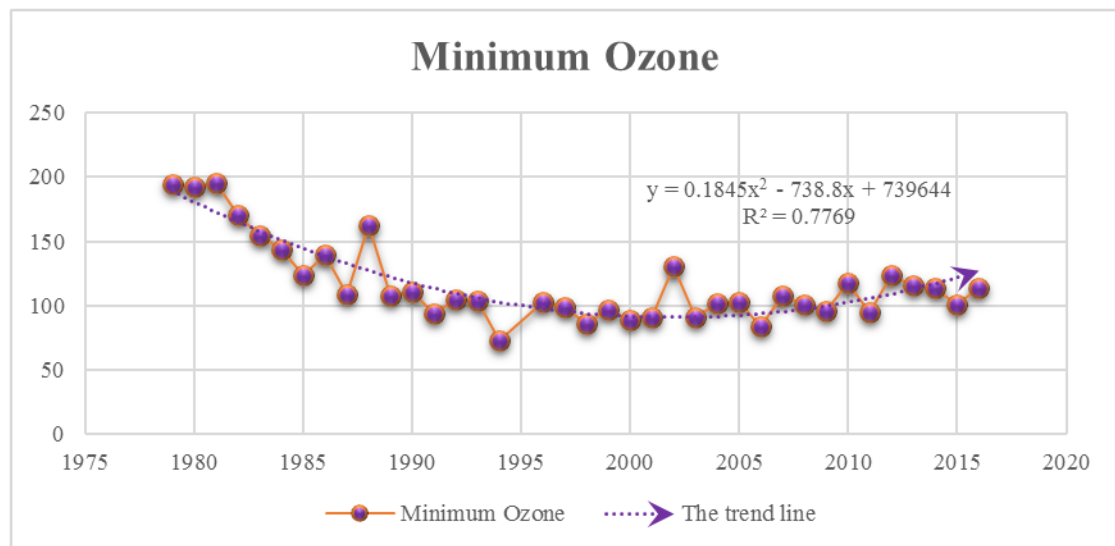


Figure11. Minimum Ozone(DU)

According to Figure 6 and Figure 7, we can know that during the period from 1979 to 2016, the annual maximum area of the ozone hole showed the trend of growth first, then buffer, and then decreased with the time. At the same time, the very low concentration of ozone also increased with time, it showed the first decrease, and then after the recovery of the trend. Through the 1979 to 2016 period of ozone hole area of the average value and the

concentration value changes. We will find that it has the same law. On this basis, we carry out qualitative analysis and quantitative prediction. As shown in the following figure.

Table3. Average of Ozone Hole Area and Minimum Ozone

| Year | Ozone Hole Area<br>(million km <sup>2</sup> ) | Minimum Ozone<br>(DU)        |
|------|-----------------------------------------------|------------------------------|
|      | 7-Sep<br>through 13 October                   | 21-Sep<br>through 16 October |
| 1979 | 0.1                                           | 225                          |
| 1980 | 1.4                                           | 203                          |
| 1981 | 0.6                                           | 209.5                        |
| 1982 | 4.8                                           | 185                          |
| 1983 | 7.9                                           | 172.9                        |
| 1984 | 10.1                                          | 163.6                        |
| 1985 | 14.2                                          | 146.5                        |
| 1986 | 11.3                                          | 157.8                        |
| 1987 | 19.3                                          | 123                          |
| 1988 | 10                                            | 171                          |
| 1989 | 18.7                                          | 127                          |
| 1990 | 19.2                                          | 124.2                        |
| 1991 | 18.8                                          | 119                          |

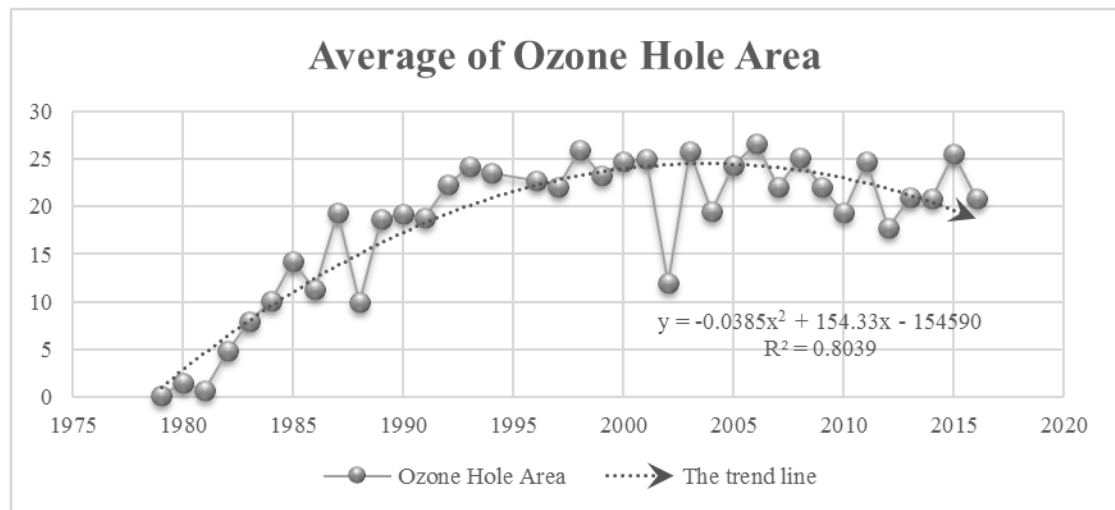


Figure12. Average of Ozone Hole Area

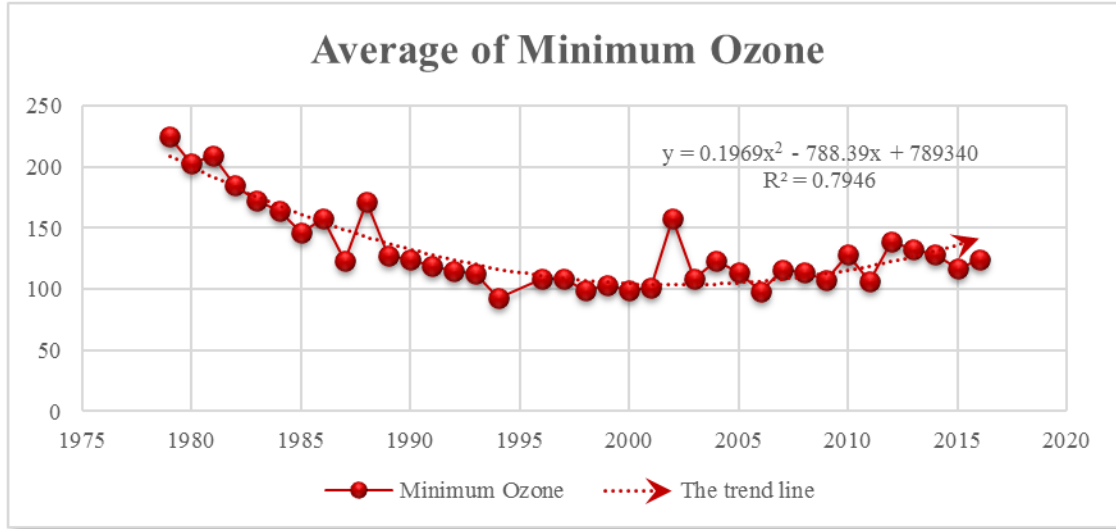


Figure13. Average of Minimum Ozone

### 3.4 Singular Spectrum Analysis

*In order to analyze the long-term variation of global atmospheric ozone in 1979-2015. In this research, we select a time series analysis tool with adaptive features which is the Singular Spectrum Analysis. It is used to extract the total ozone amount of each grid for ten years. On the basis of the ten-year trend of SSA extraction and the rate of the change of annual total ozone. In this study, we found that the changes in global atmospheric ozone over the last 36 years can be divided into three stages. The significant decline in the total ozone level was between 1979 and 1994, with the transition between 1995 and 2005 and the ozone recovery phase between 2006 and 2015<sup>[12]</sup>.*

*The implementation of this SSA algorithm is mainly based on the following two steps which is the time series decomposition and reconstruction.*

#### 1. The trajectory matrix $X$ is constructed

$$X = [X(1), \dots, X(k)] = \begin{pmatrix} x(1) & \dots & x(k) \\ \vdots & \ddots & \vdots \\ x(L) & \dots & x(N) \end{pmatrix} \quad (15)$$

And

$$X(i) = [x(i), \dots, x(i+L-1)]^T, i = 1, \dots, k \quad (16)$$

**2. Singular value decomposition of vector matrix. And then extract the different components according to the singular value size**

$$X = \sum_i \sqrt{\lambda(i)} U(i) V(i)^T \quad (17)$$

$\lambda(i) = \|X(i)\|_F^2 = \|Q(i)\|^2$  : The singular value of  $XX^T$ .

$U(i)$ : The left eigenvector after  $X$  decomposition.

$V(i)$ : The right eigenvector after  $X$  decomposition.

### 3.Reconstruction of the original time series $X(t)$

The time series reconstruction in SSA begins with the clustering of singular values and the characteristic vector after the singular value decomposition. The vectors and singular values are divided into different sub classes according to their characteristics.

$$\begin{aligned} X &= X(I(1)) + \dots + X(I(m)) \\ X(I) &= \sum_{i \in I} X(i) = X(1) + \dots + X(L) \end{aligned} \quad (18)$$

Finally, we convert each subclass  $X(I(m))$  into the trajectory matrix  $Y$  of  $L \times K$ . And then through the diagonal average  $Y$ . A new time series  $\{F(k), k = 1, 2, \dots, m\}$  of length  $N$  can be obtained. The relationship between any reconstructed time series  $F(k)$  and the original time series  $x(t)$  is shown below.

$$x = F(1) + \dots + F(m) \quad (19)$$

The original time series  $x(t)$  can eventually be decomposed into sub-classes of different features. Then each subclass represents the major variation of the time series at different scales<sup>[12]</sup>.

### 3.5 Dynamic Harmonic Regression

In order to predict the spatial and temporal variation characteristics of global ozone in the coming decades. We use the method of Dynamic Harmonic Regression. This method is similar to the Auto Regression Integrated Moving Average model, which is ARIMA model. It is based on the frequency domain characteristics of the original time series to simulate the future changes<sup>[12]</sup>. It's specific principle is that we decompose a time series  $X(t)$  into a number of components which cannot be directly observed Unobserved Components. As follows.

$$X(t) = T(t) + S(t) + e(t) = \sum_{k=0}^N [a_{k,t} \cos(w_k t) + b_{k,t} \sin(w_k t) + e(t)] \quad (20)$$

$T(t)$ : The long-term trend of the original time series  $X(t)$ .

$S(t)$ : The periodic component of the original time series  $X(t)$ .

$e(t)$ : Residual and Gauss noise.

Algorithm Principle of DHR Model<sup>[12]</sup>. So, it can be used for signal feature extraction, time series smoothing, missing data interpolation and prediction of unknown data.

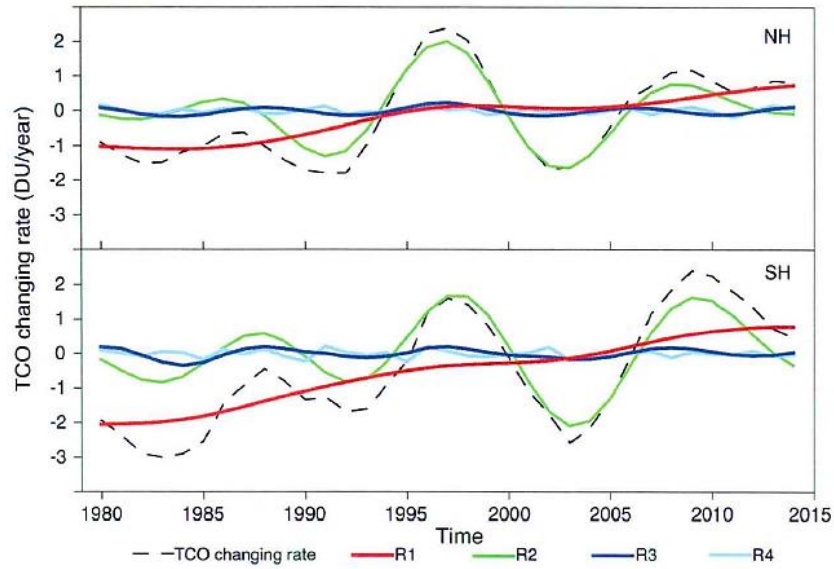


Figure14. SSA Decomposition Results of Annual Ozone Total Variation Rate in the Northern and Southern Hemispheres

*According to the prediction of DHR, the annual average ozone growth in the southern hemisphere is faster than that in the northern hemisphere, and the fluctuation range is small. Feedback to the total ozone on the southern hemisphere, the ozone layer will generally be in 2029-2032 years to restore to the level before 1980. In the northern hemisphere, the atmospheric ozone layer will be in 2022-2030 years to restore to the level before 1980.*

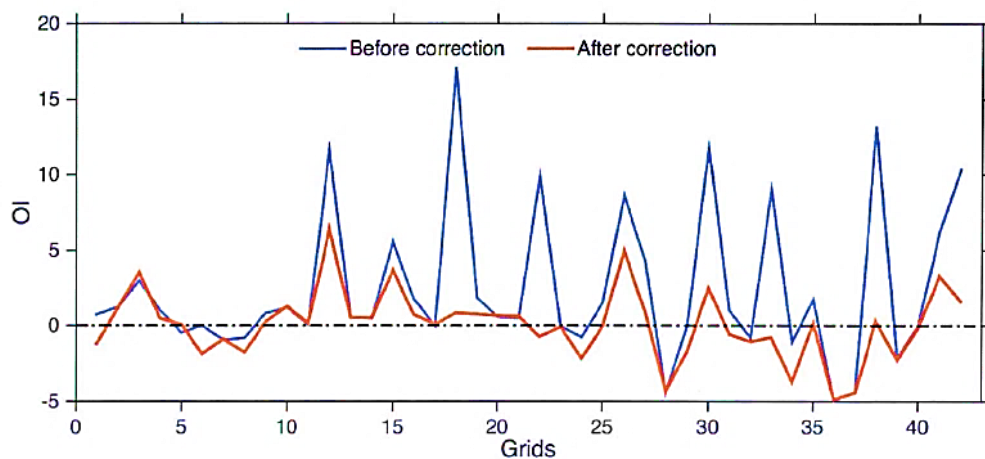


Figure15. The difference of the total ozone was significant before and after the prediction.

*During the period from 1995 to 2005, the global total atmospheric ozone has experienced a first rise and then decline in the trend of change. Since 2006, the global total atmospheric ozone began to rise star trend<sup>[12]</sup>.*



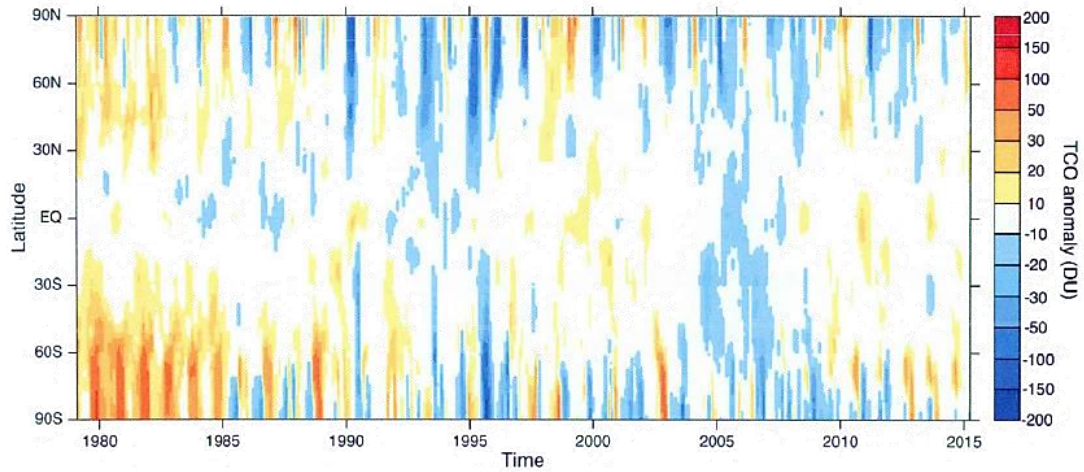


Figure16. Temporal and Spatial Variation of Global Monthly Ozone Total from 1979 to 2015

*Based on the Dynamic Harmonic Regression model, the prediction of total ozone change in the future is based on the premise that the total amount of atmospheric ozone will not change to the Montreal Protocol. The ozone layer in the northern hemisphere will return to its 1980 level by around 2030. The ozone layer in the southern hemisphere will return to its 1980 level around 2040. The changes of total ozone in the Antarctic and the wind field in the southern hemisphere were investigated. We can find that the influence of the Antarctic ozone hole on the climate of the Southern Hemisphere has been changed to the atmospheric circulation pattern of the Southern Hemisphere. Which led to the general atmospheric circulation pattern gradually moving to high latitudes in the southern hemisphere, thus changing the regional climate.*

### 3.6 WA-LSSVM Time Series Prediction

#### 3.6.1 Phase-Space Reconstruction theory

*The evolution of any component in the system is determined by the other components interacting with it, so the information of these related components is implied in the development of any component. We can reconstruct an equivalent phase space by simply observing a component, that is in some fixed time delay points, and treating it as a new dimension. As a word, a model of the system can be reconstructed by a component in the system.*

*For a given total monthly observation time series  $(x_1, x_2, \dots, x_N)$  of ozone, the time interval is  $\Delta t$ . If the embedding dimension is  $m$  and the reconstruction time delay is  $\tau$ , the reconstructed  $Nm$   $m$ -dimensional vectors are:*



### 3.6.2 Least Squares Support Vector Machine

*LSSVM is to constrain the inequality in the standard SVM algorithm into equality constraints and changes the empirical risk from the first power of the deviation to the second power, so that the quadratic programming problem is transformed into a linear system of equations. It improves the speed and accuracy of solving the problem.*

*SVM approximation method is called support vector regression (SVR). For a given training data  $(x_i, y_i)$ ,  $i = 1, \dots, l$ ,  $x_i \in R^n$ ,  $y_i \in R$ , the regression problem is to find a mapping of the input space to the output space ( $R^n \rightarrow R, y = f(x)$ ).*

*Using the least squares support vector machine (LSSVM) method based on statistical theory is used to predict ozone content time series. Firstly, the phase space of the chaotic dynamical system is reconstructed by using the time series of ozone concentration data <sup>[11]</sup>. Then, the ozone content short-term prediction model is established by using the least square support vector machine <sup>[13]</sup>.*

*For non-linear regression problems, we have a training sample ( $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_l, y_l)\}$ ). The nonlinear regression function is:*

$$f(x) = w^T \cdot \varphi(x) + b \quad (21)$$

*$w, x \in R^n$ ,  $b \in R$ ;  $T$  is the transpose of matrix;  $\varphi$  is a non-linear mapping of the data set to a high-dimensional feature space.*

*According to the structural risk minimization principle, considering the complexity of the function and the fitting error, the original space optimization problem and its constraint condition*

*The constraints are:*

$$\begin{aligned} \min_{w, b, \xi} J(w, \xi) &= \frac{1}{2} w^T w + \frac{1}{2} \gamma \sum_{i=1}^N \xi_i^2 \\ y_i &= w^T \varphi(x_i) + b + \xi_i, i = 1, 2, \dots, l \end{aligned} \quad (22)$$

*$w$  is the weight vector;  $\xi_i$  is the regression error;  $b$  is a constant deviation;  $\gamma$  is the regularization parameter.*

*Introduce Lagrange function:*

$$L(w, b, \xi_i, \alpha_i, \beta_i) = \frac{1}{2} w^T w + \frac{\gamma}{2} \sum_{i=1}^N \xi_i^2 - \sum_{i=1}^N [\alpha_i (w^T \varphi(x_i) + b + \xi_i - y_i)] \quad (23)$$

*$\alpha_i, \beta_i \in R$ —Lagrange multiplier.*

According to the least squares method, for the  $w$ ,  $b$ ,  $\xi_i$ ,  $\alpha_i$ ,  $\beta_i$  respectively, for the partial derivative of  $L$  is 0. At the same time, according to KKT complementary conditions, the following relationship can be obtained:

$$\begin{cases} w = \sum_{j=1}^l (\alpha_j + \beta_j) \phi(x_j) - a \sum_{j=1}^l \beta_j \phi(-x_j), \\ \sum_{j=1}^l \alpha_j = 0, \\ \gamma \xi_i = \alpha_i, \\ y_i = w^T \phi(x_i) + b + \xi_i, i = 1, 2, \dots, l. \end{cases} \quad (24)$$

After eliminating  $\xi_i$  and  $w$ , the following system of linear equations can be obtained:

$$\begin{bmatrix} 0 & I^T \\ I & \Omega + \frac{1}{\gamma} I \end{bmatrix} \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} 0 \\ y \end{bmatrix} \quad (25)$$

$$y = [y_1, \dots, y_l]^T, I = [1, \dots, 1]^T, \alpha = [\alpha_1, \dots, \alpha_l]^T, \psi_{i,j} = k(x_i, x_j) = \phi(x_i)^T \cdot \phi(x_j), \quad i, j = 1, 2, \dots, l.$$

$k(x_i, x_j)$  is a kernel function that satisfies the Mercer condition. Finally, we obtain the following regression LSSVM model:

$$y(x) = \sum_{i=1}^l \alpha_i k(x_j, x) + b \quad (26)$$

### 3.6.3 Wavelet Decomposition

For most signals. The low frequency portion gives the signal characteristics. The high frequency part, noise and disturbance have mutual relations among the three. So, the low-frequency signal space for detailed decomposition, so that the low-frequency part of the resolution level is getting higher and higher, the basic characteristics of the signal will be retained. The final goal of wavelet decomposition is to construct an orthogonal wavelet basis which is close to the original signal in frequency. Different frequency resolution of the orthogonal wavelet basis is equivalent to the bandwidth of the band-pass filter.

In this paper, it uses three-layer wavelet decomposition method. The following figure shows an exploded view. And the  $J$  is the decomposition level. For the general signal  $f(t)$ , after decomposition, it becomes  $f(t) = s + a_3 + d_3 + d_2 + d_1$ . Among them, the  $a_3$  said third low frequency signals  $d_1$ ,  $d_2$ ,  $d_3$ , which respectively, said the wavelet decomposition of the first, second, third layer of high frequency signals<sup>[13]</sup>.

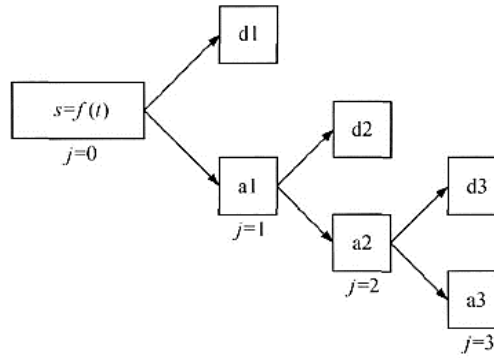


Figure17. Schematic diagram of three-scale wavelet decomposition

### 3.6.4 The Prediction Model of WT - LSSVM

The model structure of wavelet decomposition and least square support vector machine is shown in Figure 2. It is divided into the following three stages.

(1) The three-dimensional wavelet decomposition of the monthly total ozone observation sequence was carried out. The signal sequences of each frequency band are obtained by extracting three different scale components.

(2) The least squares support vector machine is used to model and predict the wavelet sequences on each layer. For the high frequency signal and the low frequency signal, they all use this Gaussian radial basis kernel function to construct the LSSVM model.

(3) The predicted signal is added and reconstructed. We can get the average monthly ozone content of the final forecast.

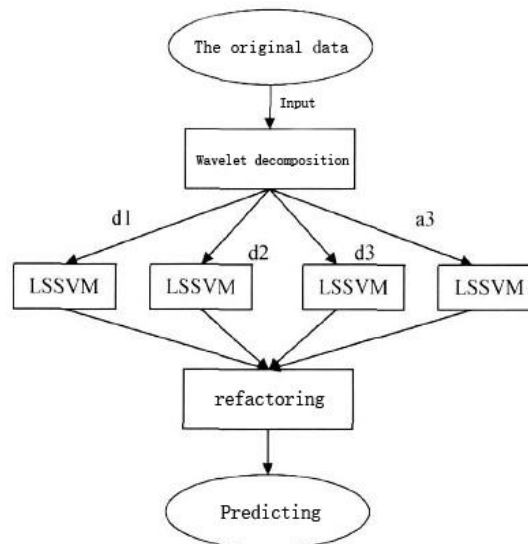


Figure18. The model structure of WT-LSSVM

## **IV. Conclusions**

### **4.1 Effects of Atmospheric Hydrodynamics on Ozone Concentration**

- 1) *Atmospheric circulation and the role of stratospheric wind, so that the ozone to the poles transported in the polar winter during the night without solar radiation, the local ozone entirely by the wind from the equator to the polar transport. On the whole, the role of the global ozone level distribution tends to balance.*
- 2) *Note that the addition of winter ozone to the poles depends on the transport of the wind from the equator to the poles, and if there is a strong airflow around the poles that blocks the ozone transport from the equator to the poles, there will not be much change.*
- 3) *Polar regions do have a strong seasonal vortex, that is polar vortex. The ice sheet surface makes the Antarctic polar vortex stronger than the Arctic. About 5,6 months each year, at the beginning of the Antarctic winter, the formation of a strong cold air mass around the South Pole closed wind, a polar vortex. About November, when the temperature rises, the polar vortex will collapse.*
- 4) *Polar vortex not only blocks the ozone transport process from the equator to the polar region, but also with the positive feedback mechanism of polar stratospheric clouds, which makes ozone decompose.*

### **4.2 Freon leads to a general reduction of ozone in the atmosphere**

- 1) *The average concentration of CFCs in the northern hemisphere is higher than that in the southern hemisphere. CFCs discharge in the atmosphere after the rapid spread of the north and south hemispheres of the atmosphere, to cross the equator completely mixed, takes about 2 years. In the northern hemisphere, the average annual increase rate of CFCs in the atmosphere is 4-5%, while the average concentration of CFCs in the southern hemisphere is about 8-10% lower than that in the northern hemisphere. Therefore, the CFCs in the southern hemisphere are about 2 years behind the northern hemisphere. The rate is increasing.*
- 2) *Between 45 and 65 degrees' north latitude, in 1992 to 1993 at the turn of the winter and spring, ozone content is the lowest over the years. In 1994, the ozone layer above the northern hemisphere was thinner than ever before, with ozone levels averaging 10-15 per cent lower in Europe and North America, and by 35 per cent over Siberia.*

### 4.3 Global long - term trend of atmospheric ozone

*Based on the long-term trend and the rate of change of global annual ozone, we found that the atmospheric ozone layer has experienced three different stages, namely, the period of total ozone decrease from 1979 to 1994, and the transition period from 1995 to 2005 And the post-2006 recovery phase of the ozone layer. At the same time, the Montreal Protocol, implemented in 1986, effectively controlled the rate of decline in global ozone.*

### 4.4 Forecast of global atmospheric ozone levels over the next 50 years

*Under consideration of only the Montreal Protocol as an anthropogenic factor, the northern hemisphere ozone layer will return to its 1980 level by about 2030, and the ozone layer in the southern hemisphere will return to the 1980 level by around 2040.*

*The global average annual atmospheric ozone levels for the next 50 years are forecast as follows:*

Table4. Annual average atmospheric ozone level prediction results

| Year | Ozone Hole Area<br>(million km <sup>2</sup> ) | Minimum Ozone<br>(DU) |
|------|-----------------------------------------------|-----------------------|
| 2017 | 24.5                                          | 113.6                 |
| 2018 | 20.9                                          | 106.5                 |
| 2019 | 24.1                                          | 100.3                 |
| 2020 | 18.7                                          | 116.2                 |
| 2021 | 23.8                                          | 99.3                  |
| 2022 | 20.7                                          | 116.2                 |
| 2023 | 23.4                                          | 92.8                  |
| 2024 | 16.9                                          | 87.5                  |
| 2025 | 23.1                                          | 90.2                  |
| 2026 | 11.9                                          | 79.5                  |
| 2027 | 22.7                                          | 93.6                  |
| 2028 | 19.5                                          | 90.7                  |
| 2029 | 19.1                                          | 77.5                  |
| 2030 | 19.2                                          | 74.1                  |

*For figure19 and figure 20, it can be seen from 2017 to 2066, the global maximum atmospheric ozone hole into the decline in the area, and we have expected that the ozone hole area size of 2060 is basic close to 1980's. For the global minimum concentration of ozone also entered the recovery period as the same time, and the minimum concentration level has reached the level of the last century 90's after 50 years, and it is indicating that the ozone concentration has been significantly restored.*

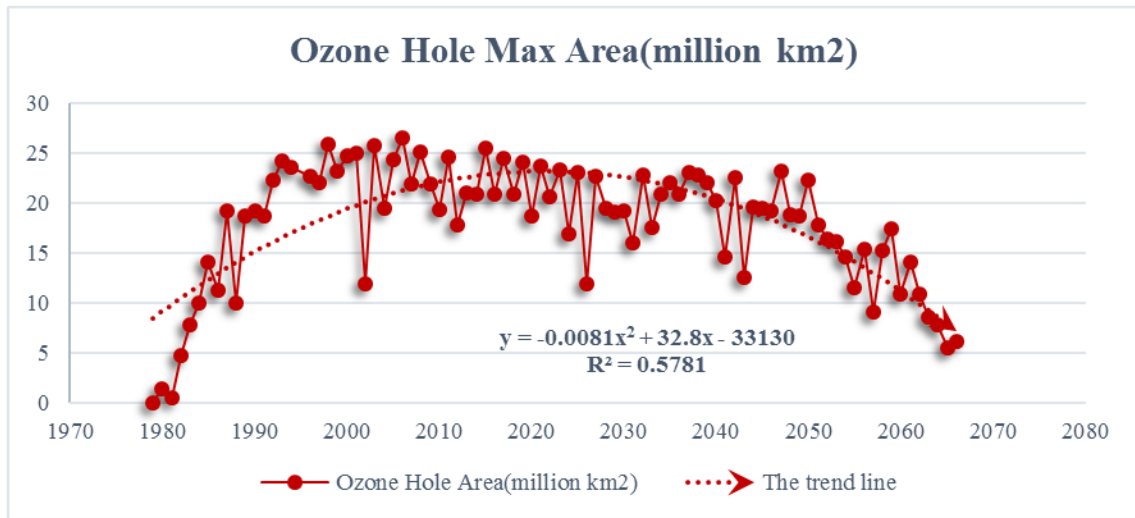


Figure19. Annual mean ozone hole maximum

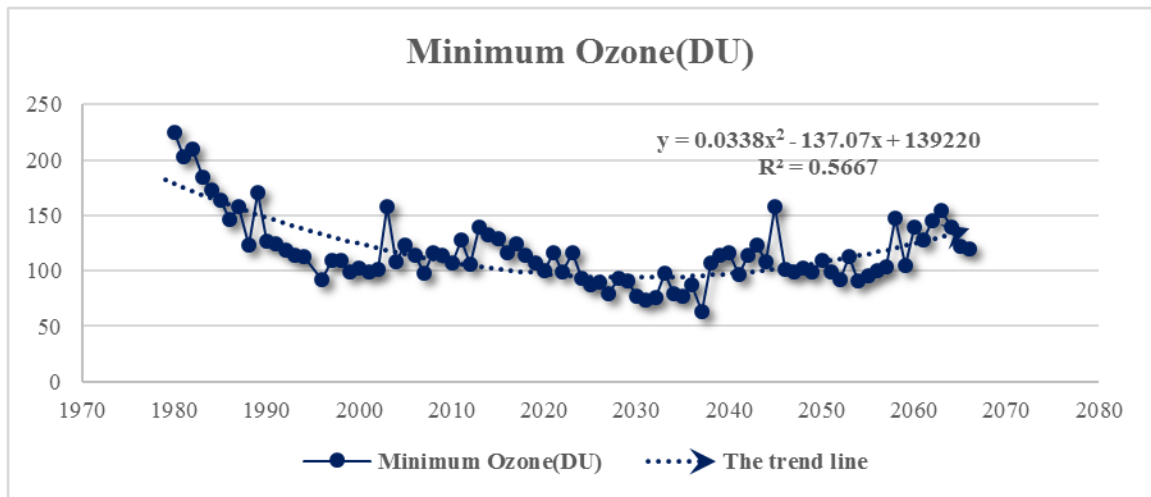


Figure20. Annual average ozone concentration minimum

## V. Future Work

### 5.1 Model Advantages

- 1) *Wavelet analysis is a kind of available time-frequency analysis method, which has upstanding localize peculiarity in both time field and frequency field. We use the method to decompose the time series of ozone level into more simple high frequency and low frequency signals, so as to isolate the nonlinear effects between different scales. It can evidently decrease error of prediction.*
- 2) *Applying phase space reconstruction method to divide time series into segments, we have mapped original series into multidimensional data space. Then developing a model along the phase trajectory in this space, the physical significance of the model*

become more clear. In addition, it could obviously improve the precision of forecasting by the method.

- 3) Ozone level on the past time is made as input, ozone level on the future time is be used as output, and the given the nonlinear mapping relation of network realization is completed by the least squared support vector machine. This method could obviously improve the precision of forecasting due to its good performance in small sample and nonlinear problem.
- 4) We using the method of direct prediction in the process, there is no need to rely on the prediction results of the front to forecast, which can avoid the error accumulation and improve the precision of forecasting.
- 5) In this prediction mode, what we need is only the simple time series of ozone content. It is more convenient and applicable, because there is no need of other meteorological data.

## 5.2 Model limitations

- 1) Data missing affects directly affect the precision of forecasting, because the quality requirements of ozone content data are relatively high. In this paper, using interpolation method to deal with the missing data. However, it is necessary to make a deep research on the interpolation method of missing data when we cannot reconstruct the missing data reasonably because of too much data missing.
- 2) There are many methods for determining the two parameters of embedded dimension and time delay when we reconstruct the phase space of zone content time series. There is no conclusive effective method, which needs more study.
- 3) The selection of kernel function is a key factor that directly affects the implementation and effect of the algorithm. Up to now, the research in this respect is not enough and how to choose kernel function is also a lack of appropriate theoretical basis. In this paper, we adopt radial basis function for its better permeance. It is necessary to further study prediction performance when we choose different kernel function.
- 4) This model is based on one dimensional time series of ozone content, without considering the role of other factors. In order to further improve the model, we can consider adding variables to investigate the impact of different factors on the results of prediction in the future.

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## VII. Appendix

**Table 1: The harmful gas**

| Year | CFC-12 | CFC-11 | CH <sub>3</sub> Cl | CH <sub>3</sub> Br | CCl <sub>4</sub> | CH <sub>3</sub> CCl <sub>3</sub> | halons | CFC-113 | HCFCs | WMO Minor |
|------|--------|--------|--------------------|--------------------|------------------|----------------------------------|--------|---------|-------|-----------|
| 1992 | 230    | 379    | 242                | 326                | 242              | 271                              | 140    | 69      | 13    | 39        |
| 1993 | 234    | 382    | 242                | 326                | 240              | 266                              | 150    | 71      | 14    | 41        |
| 1994 | 237    | 382    | 242                | 326                | 238              | 250                              | 161    | 72      | 16    | 42        |
| 1995 | 239    | 381    | 244                | 326                | 236              | 225                              | 171    | 73      | 18    | 44        |
| 1996 | 242    | 380    | 241                | 327                | 234              | 200                              | 177    | 73      | 20    | 45        |
| 1997 | 245    | 378    | 238                | 326                | 232              | 172                              | 184    | 73      | 22    | 46        |
| 1998 | 246    | 376    | 244                | 333                | 230              | 145                              | 190    | 73      | 24    | 46        |
| 1999 | 248    | 374    | 248                | 333                | 228              | 121                              | 196    | 72      | 26    | 46        |
| 2000 | 249    | 372    | 245                | 320                | 225              | 101                              | 201    | 72      | 28    | 45        |
| 2001 | 249    | 369    | 240                | 305                | 222              | 84                               | 204    | 71      | 30    | 45        |
| 2002 | 250    | 366    | 238                | 296                | 220              | 70                               | 207    | 71      | 32    | 44        |
| 2003 | 250    | 363    | 239                | 294                | 217              | 58                               | 208    | 70      | 33    | 44        |
| 2004 | 250    | 360    | 238                | 288                | 215              | 48                               | 211    | 69      | 35    | 44        |
| 2005 | 249    | 356    | 238                | 283                | 213              | 40                               | 214    | 69      | 36    | 43        |
| 2006 | 249    | 353    | 238                | 277                | 210              | 34                               | 214    | 68      | 37    | 43        |
| 2007 | 248    | 350    | 239                | 273                | 207              | 28                               | 214    | 68      | 39    | 42        |
| 2008 | 247    | 347    | 240                | 269                | 205              | 24                               | 212    | 67      | 40    | 41        |
| 2009 | 246    | 344    | 240                | 259                | 201              | 20                               | 210    | 66      | 42    | 41        |
| 2010 | 244    | 341    | 238                | 254                | 199              | 17                               | 208    | 66      | 43    | 40        |
| 2011 | 243    | 338    | 237                | 255                | 196              | 14                               | 206    | 65      | 45    | 40        |
| 2012 | 242    | 335    | 237                | 253                | 193              | 12                               | 203    | 65      | 46    | 39        |
| 2013 | 241    | 332    | 239                | 249                | 191              | 10                               | 201    | 64      | 48    | 39        |
| 2014 | 239    | 330    | 239                | 241                | 189              | 8                                | 197    | 63      | 49    | 38        |
| 2015 | 238    | 329    | 242                | 240                | 186              | 7                                | 193    | 63      | 50    | 38        |
| 2016 | 236    | 326    | 246                | 243                | 184              | 6                                | 190    | 62      | 51    | 37        |

**Table 2: Ozone Hole Area and Minimum Ozone**

| Year | Ozone Hole Area            |       | Minimum Ozone |       |
|------|----------------------------|-------|---------------|-------|
|      | (million km <sup>2</sup> ) |       | (DU)          |       |
|      | Maximum Daily              |       | Minimum Daily |       |
|      | Date                       | Value | Date          | Value |
| 1979 | 17-Sep                     | 1.1   | 17-Sep        | 194   |
| 1980 | 21-Sep                     | 3.3   | 16-Oct        | 192   |
| 1981 | 10-Oct                     | 3.1   | 10-Oct        | 195   |
| 1982 | 2-Oct                      | 10.8  | 2-Nov         | 170   |
| 1983 | 17-Oct                     | 12.2  | 6-Oct         | 154   |
| 1984 | 24-Sep                     | 14.7  | 3-Oct         | 144   |
| 1985 | 3-Oct                      | 18.8  | 24-Oct        | 124   |
| 1986 | 6-Oct                      | 14.4  | 6-Oct         | 140   |
| 1987 | 29-Sep                     | 22.4  | 5-Oct         | 109   |

|             |        |      |        |     |
|-------------|--------|------|--------|-----|
| <b>1988</b> | 20-Sep | 13.8 | 30-Sep | 162 |
| <b>1989</b> | 3-Oct  | 21.7 | 7-Oct  | 108 |
| <b>1990</b> | 19-Sep | 21.1 | 5-Oct  | 111 |
| <b>1991</b> | 4-Oct  | 22.6 | 6-Oct  | 94  |
| <b>1992</b> | 27-Sep | 24.9 | 11-Oct | 105 |
| <b>1993</b> | 19-Sep | 25.8 | 25-Sep | 104 |
| <b>1994</b> | 30-Sep | 25.2 | 30-Sep | 73  |
| <b>1996</b> | 7-Sep  | 26.9 | 5-Oct  | 103 |
| <b>1997</b> | 27-Sep | 25.1 | 24-Sep | 99  |
| <b>1998</b> | 19-Sep | 27.9 | 6-Oct  | 86  |
| <b>1999</b> | 15-Sep | 25.8 | 29-Sep | 97  |
| <b>2000</b> | 9-Sep  | 29.9 | 29-Sep | 89  |
| <b>2001</b> | 17-Sep | 26.5 | 22-Sep | 91  |
| <b>2002</b> | 19-Sep | 21.9 | 20-Sep | 131 |
| <b>2003</b> | 24-Sep | 28.4 | 26-Sep | 91  |
| <b>2004</b> | 22-Sep | 22.8 | 4-Oct  | 102 |
| <b>2005</b> | 11-Sep | 27.3 | 30-Sep | 103 |
| <b>2006</b> | 24-Sep | 29.6 | 8-Oct  | 84  |
| <b>2007</b> | 13-Sep | 25.2 | 24-Sep | 108 |
| <b>2008</b> | 12-Sep | 27   | 4-Oct  | 101 |
| <b>2009</b> | 17-Sep | 24.4 | 25-Sep | 96  |
| <b>2010</b> | 25-Sep | 22.6 | 1-Oct  | 118 |
| <b>2011</b> | 12-Sep | 26.1 | 8-Oct  | 95  |
| <b>2012</b> | 22-Sep | 21.1 | 1-Oct  | 124 |
| <b>2013</b> | 16-Sep | 24   | 29-Sep | 116 |
| <b>2014</b> | 11-Sep | 24.1 | 30-Sep | 114 |
| <b>2015</b> | 2-Oct  | 28.2 | 4-Oct  | 101 |
| <b>2016</b> | 28-Sep | 23   | 1-Oct  | 114 |

**Table3: Annual mean Ozone Hole Area and Minimum Ozone**

| <b>Year</b> | <b>Ozone Hole Area</b>      | <b>Minimum Ozone</b>         |
|-------------|-----------------------------|------------------------------|
|             | (million km <sup>2</sup> )  | (DU)                         |
|             | 7-Sep<br>through 13 October | 21-Sep<br>through 16 October |
| <b>1979</b> | 0.1                         | 225                          |
| <b>1980</b> | 1.4                         | 203                          |
| <b>1981</b> | 0.6                         | 209.5                        |
| <b>1982</b> | 4.8                         | 185                          |
| <b>1983</b> | 7.9                         | 172.9                        |
| <b>1984</b> | 10.1                        | 163.6                        |
| <b>1985</b> | 14.2                        | 146.5                        |
| <b>1986</b> | 11.3                        | 157.8                        |
| <b>1987</b> | 19.3                        | 123                          |
| <b>1988</b> | 10                          | 171                          |

|             |      |       |
|-------------|------|-------|
| <b>1989</b> | 18.7 | 127   |
| <b>1990</b> | 19.2 | 124.2 |
| <b>1991</b> | 18.8 | 119   |
| <b>1992</b> | 22.3 | 114.3 |
| <b>1993</b> | 24.2 | 112.6 |
| <b>1994</b> | 23.6 | 92.3  |
| <b>1996</b> | 22.7 | 108.8 |
| <b>1997</b> | 22.1 | 108.8 |
| <b>1998</b> | 25.9 | 98.8  |
| <b>1999</b> | 23.2 | 102.9 |
| <b>2000</b> | 24.8 | 98.7  |
| <b>2001</b> | 25   | 100.9 |
| <b>2002</b> | 12   | 157.4 |
| <b>2003</b> | 25.8 | 108.7 |
| <b>2004</b> | 19.5 | 123.5 |
| <b>2005</b> | 24.4 | 113.8 |
| <b>2006</b> | 26.6 | 98.2  |
| <b>2007</b> | 22   | 116.2 |
| <b>2008</b> | 25.2 | 114.1 |
| <b>2009</b> | 22   | 107.5 |
| <b>2010</b> | 19.4 | 128   |
| <b>2011</b> | 24.7 | 106.2 |
| <b>2012</b> | 17.8 | 139.1 |
| <b>2013</b> | 21   | 132.5 |
| <b>2014</b> | 20.9 | 128.6 |
| <b>2015</b> | 25.6 | 116.5 |
| <b>2016</b> | 20.9 | 124.3 |

**Table4: The prediction of Ozone Hole Area and Minimum Ozone**

| <b>Year</b> | <b>Ozone Hole Area (million km2)</b> | <b>Minimum Ozone(DU)</b> |
|-------------|--------------------------------------|--------------------------|
| <b>2017</b> | 24.5                                 | 113.6                    |
| <b>2018</b> | 20.9                                 | 106.5                    |
| <b>2019</b> | 24.1                                 | 100.3                    |
| <b>2020</b> | 18.7                                 | 116.2                    |
| <b>2021</b> | 23.8                                 | 99.3                     |
| <b>2022</b> | 20.7                                 | 116.2                    |
| <b>2023</b> | 23.4                                 | 92.8                     |
| <b>2024</b> | 16.9                                 | 87.5                     |
| <b>2025</b> | 23.1                                 | 90.2                     |
| <b>2026</b> | 11.9                                 | 79.5                     |
| <b>2027</b> | 22.7                                 | 93.6                     |
| <b>2028</b> | 19.5                                 | 90.7                     |
| <b>2029</b> | 19.1                                 | 77.5                     |
| <b>2030</b> | 19.2                                 | 74.1                     |

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|             |      |       |
|-------------|------|-------|
| <b>2031</b> | 16   | 76.2  |
| <b>2032</b> | 22.9 | 98.2  |
| <b>2033</b> | 17.6 | 79.64 |
| <b>2034</b> | 20.9 | 77.36 |
| <b>2035</b> | 22.1 | 87.6  |
| <b>2036</b> | 20.9 | 62.8  |
| <b>2037</b> | 23.1 | 107.5 |
| <b>2038</b> | 22.9 | 114.1 |
| <b>2039</b> | 22.1 | 116.2 |
| <b>2040</b> | 20.3 | 96.2  |
| <b>2041</b> | 14.6 | 113.8 |
| <b>2042</b> | 22.6 | 123.5 |
| <b>2043</b> | 12.6 | 108.7 |
| <b>2044</b> | 19.7 | 157.4 |
| <b>2045</b> | 19.5 | 100.9 |
| <b>2046</b> | 19.3 | 98.7  |
| <b>2047</b> | 23.2 | 102.9 |
| <b>2048</b> | 18.9 | 98.8  |
| <b>2049</b> | 18.8 | 108.8 |
| <b>2050</b> | 22.4 | 98.9  |
| <b>2051</b> | 17.9 | 92.3  |
| <b>2052</b> | 16.4 | 112.6 |
| <b>2053</b> | 16.2 | 90.5  |
| <b>2054</b> | 14.6 | 95.2  |
| <b>2055</b> | 11.6 | 100.4 |
| <b>2056</b> | 15.4 | 103.2 |
| <b>2057</b> | 9.1  | 147.2 |
| <b>2058</b> | 15.3 | 104.7 |
| <b>2059</b> | 17.5 | 139.5 |
| <b>2060</b> | 10.9 | 128.2 |
| <b>2061</b> | 14.1 | 145.3 |
| <b>2062</b> | 10.9 | 154.6 |
| <b>2063</b> | 8.6  | 138.9 |
| <b>2064</b> | 7.9  | 122.3 |
| <b>2065</b> | 5.5  | 119.2 |
| <b>2066</b> | 6.2  | 154.8 |

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# Global Ozone Status and the Prediction of Future Trends

## Abstract

Ozone, as a trace gas in the atmosphere, has a strong absorption of the high-energy ultraviolet radiation emitted by the sun, making it an important barrier to protecting the health of human beings and other living beings on Earth. The main factors affecting the ozone concentration of solar ultraviolet radiation intensity and seasonal changes, human emissions of chlorofluorocarbon gas compounds, the impact of atmospheric circulation. It is believed that the main cause of ozone depletion is the presence of chlorine source gases, including chlorofluorocarbons and related halocarbons, nitrous oxide, and the like. Since the adoption and strengthening of the Montreal Protocol to reduce emissions of CFCs, the concentration of the most important compounds in the atmosphere has been declining. With the smooth implementation of the agreements such as Montreal, the atmospheric ozone layer also began to recover in the global, the recovery will lead to a new round of regional climate change. This will be an important guide for accurately predicting future global and regional climate change and developing appropriate climate change response measures. The main work of this paper is as follows:

(1) In order to analyze the long-term trend of global atmospheric ozone in the period 1979-2015, this paper chooses a time-series analysis tool with self-adaptive characteristics: Singular Spectrum Analysis, it is used to extract the ten-year trend of total ozone in each grid.

(2) The past ozone observations and model simulations were established based on the ten-year trend of SSA extraction and quantification of the annual change rate of total ozone.

(3) Future total ozone was predicted based on non-linear regression analysis, BP neural network and optical climate model (CCMs). Under the premise that the total amount of ozone in the atmosphere will remain unchanged for the Montreal Protocol, the ozone layer in the northern hemisphere will return to its 1980 level by about 2030; the ozone layer in the southern hemisphere will return to its 1980 level around 2040.

(4) The effect of ozone hole on the regional climate change in the Antarctic region was simulated in the study by using the observed increase in precipitation in the southern hemisphere which is based on the Maximum Covariance Analysis (MCA) and the Extreme Learning Machine (ELM).

The ozone hole in the polar regions is the main factor leading to the increase of rainfall in the middle and high latitudes of the southern hemisphere, at the same time, the main cause of ozone depletion is the presence of chlorine source gas. Due to the people's control, the ozone level will rise to a certain level from the trough.

**Key words:** Solar radiation ; Atmospheric motion ; CFCs ; SSA ; Linear regression+BP- neural network

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# **I.Introduction**

The solar ultraviolet radiation intensity and seasonal variation are the main factors influencing the variation of ozone concentration in the stratosphere. Anthropogenic chlorofluorocarbon gaseous compounds result in an overall reduction in atmospheric ozone levels. Atmospheric circulation regulates the ozone concentration between the high and low latitudes tends to balance. The dynamic effect of air to the local area of the ozone concentration was significantly reduced.

## **1.1 Ozone Hole and Climate Change**

Ground-based observation records show that the trend of ozone depletion in the atmosphere began in the late 1950s, and the British Antarctic expedition team in the Antarctic circle observations found the ozone hole for the first time in 1985, causing the scientific community of great concern. With the arrival of the southern hemisphere spring (September - November) and the solar ultraviolet radiation enhancement and temperature rise, in the sunlight, the ice dissolves and decomposes to produce a large number of free chlorine free radicals, and then the ozone molecules combined with a variety of chemical Reaction, resulting in stratospheric ozone concentration drastically reduced to form a large area of ozone thin area, resulted in the so-called South Pole ozone hole phenomenon. With the continuous reduction of ozone in the ozone layer and the continuous expansion of the ozone hole, a large number of high-energy solar ultraviolet radiation will be able to penetrate the atmosphere to reach the ground, thus the Earth's ecosystems and human and other biological organisms have a very adverse health effects. Among them, the ozone hole on the Earth's ecological system, especially regional climate change, the most scientific concern. The reason is that abnormal weather conditions caused by climate change, such as drought, floods, heavy rain, etc., will have the most direct and significant impact in people's lives and property safety and daily production and life.

## **1.2 The Ozone Content of the Factors**

### **1.2.1 The Impact of Solar Radiation on the Size of the Ozone Concentration**

Solar radiation is the most important factor to affect the concentration of ozone in the atmosphere, because the formation of ozone is closely related to solar radiation. In the stratosphere of the atmosphere, the solar ultraviolet radiation is more, the oxygen molecules break down into oxygen atoms, oxygen atoms and adjacent oxygen molecules react to ozone, ozone decomposes into oxygen molecules and an oxygen atom or with the active oxygen atoms to form two oxygen molecules by the strong ultraviolet radiation. The above-mentioned generation and decomposition process maintains a delicate and fragile balance.

To the upper atmosphere, the solar ultraviolet radiation is stronger, the material exists in the atomic state; to the lower atmosphere, solar ultraviolet radiation is weak,

oxygen molecules can not be broken down into oxygen atoms. So the concentration of ozone in the upper atmosphere and the troposphere is extremely low.

In the stratospheric atmosphere, the intensity of solar UV radiation determines the amount of ozone. Sunspot activity peak years, UV radiation intensity is large, ozone increased, some people think that more than 3%.

With the increase of the latitude of the earth, the solar ultraviolet radiation intensity will decrease, and the equator obtains the maximum intensity and the minimum of the two poles. Considering only solar ultraviolet radiation, the concentration of ozone in the atmosphere should be weakened with the increase of the latitude of the earth. However, the reality is that the concentration of ozone in the atmosphere is far greater than that of the equator.

### **1.2.2 Effect of Atmospheric Dynamics on Ozone Concentration**

1. By the atmospheric circulation and the role of stratospheric wind, the ozone transported to the poles, during the night, there isn't solar radiation in the polar winter, the local ozone completely by the wind from the equator to the polar transport. On the whole, the role of the global ozone level distribution tends to balance.

Note that the addition of winter ozone to the poles depends on the transport of the wind from the equator to the poles, and if there is a strong airflow around the poles that blocks the ozone transport from the equator to the poles, there will be no major changes.

2. Polar regions do have a strong seasonal vortex, that is polar vortex. The ice sheet surface makes the Antarctic polar vortex stronger than the Arctic. About 5,6 months each year, at the beginning of the Antarctic winter, a strong cold air mass forms a closed wind system around the South Pole, and a polar vortex appears. About November, when the temperature rises, the polar vortex will collapse.

3. Polar vortex not only blocks the ozone transport process from the equator to the polar region, but also with the positive feedback mechanism of polar stratospheric clouds, which makes ozone decompose.

4. The airflow increased strongly in the equatorial and the troposphere top was higher. The rise of the lower atmosphere diluted the ozone in the top troposphere, resulting in a lower concentration of ozone over the equator.

### **1.2.3 Freon leads to an Overall Decrease in Atmospheric Ozone**

Freon was developed in the 1930s. Belonging to the chlorofluorocarbons (CFCs), Freon is its trade name. It is not easy to burn, non-corrosive, non-toxic, stable and inexpensive. As an industrial chemical, Freon is widely used in various refrigerants for refrigerating and air conditioning, cleaning solvents for electronics and optical components, cosmetics and other sprays, as well as foaming agents for foamed PU, PS and PE. From the early 1930s to the 1950s and 1960s, humans produced a total of 15 million tons of CFCs.

Prior to the control of Freon, the world discharged to the atmosphere of Freon has reached 20 million tons. Due to their average lifetime in the atmosphere for



centuries most of the emissions remain in the atmosphere, and most of them remain in the troposphere, a small fraction moving into the stratosphere as the atmosphere moves.

Freon into the stratosphere in the strong ultraviolet radiation, the release of a chlorine atom:  $\text{CCl}_3\text{F} + h\nu \rightarrow \text{CCl}_2\text{F} + \text{Cl}$ . Chlorine atoms continue to consume and catalyze the decomposition of ozone. Since chlorine atoms can also produce substances such as methane ( $\text{CH}_4$ ) and nitrogen dioxide ( $\text{NO}_2$ ) and other chlorine storage materials, so purely from a chemical point of view, Freon damage to ozone is limited.

Due to the northern hemisphere of the major industrial countries in the world, the average concentrations of CFCs in the northern hemisphere are higher than those in the southern hemisphere. CFCs discharge in the atmosphere after the rapid spread of the north and south hemispheres of the atmosphere, to cross the equator completely mixed, takes about 2 years. In the northern hemisphere, the average annual increase rate of CFCs in the atmosphere is 4-5%, while the average concentration of CFCs in the southern hemisphere is about 8-10% lower than that in the northern hemisphere. Therefore, the CFCs in the southern hemisphere are about 2 years behind the northern hemisphere, at the same rate to increase. A large range of ozone concentrations continue to reduce, the culprit is human emissions of freon.

## II. The Description of Problem

Ozone depletion includes several phenomena observed since the late 1970s: a steady decline in the total amount of ozone in Earth's stratosphere (the ozone layer), and a much larger springtime decrease in stratospheric ozone — as known as ozone hole — around Earth's polar regions. In addition to these well-known stratospheric phenomena, there are also springtime polar tropospheric ozone depletion events.

It is believed that the primary cause of ozone depletion is the presence of chlorine-containing source gases, including CFCs and related halocarbons, nitrous oxide, etc. In 1985, 20 nations signed the Vienna Convention for the Protection of the Ozone Layer. In 1987, representatives from 43 nations signed the Montreal Protocol. At Montreal, the participants agreed to freeze production of CFCs at 1986 levels and to reduce production by 50 percent by 1999.

Since the adoption and strengthening of the Montreal Protocol has led to reductions in the emissions of CFCs, atmospheric concentrations of the most-significant compounds have been declining. Please develop a mathematical model to predict the ozone levels over the next 50 years.

## III. Assumptions and Symbols

### 3.1 Assumptions

1. Suppose that the solar ultraviolet radiation intensity decreases with increasing

latitude of the earth, and will not be artificially changed.

2. Assume that the effect of atmospheric dynamics on ozone concentration does not appear to be abrupt.

3. Assume that the human factors affect the ozone content of the material just freon.

### 3.2 Symbols

| 4.symbol | 5.Description             |
|----------|---------------------------|
| 6.x      | 7.Input data              |
| 8.x1     | 9.Input data transpose    |
| 10.net   | 11.Neural network         |
| 12.R     | 13.Return value of row    |
| 14.Q     | 15.Return value of column |
| 16.w     | 17.Weight coefficient     |
| 18.b     | 19.Bias coefficient       |
| 20.an    | 21.Trained data.          |
| 22.t     | 23.Time                   |

## IV.Models

### 4.1 Description of Models

#### 4.1.1 Extract the Total Ozone Amount of Each Grid Model

In order to analyze the long-term trend of global atmospheric ozone in 1979-2015, in this study, we selected a time-series analysis tool with adaptive features - Singular Spectrum Analysis, which can be used to extract the annual trend of total ozone in each grid. In this study, we found that the changes in global atmospheric ozone can be divided into three stages over the past 36 years, that is the total amount of ozone decreased significantly stage (1979-1994), transition phase (1995-2005) and ozone recovery phase (2006-2015).

#### 4.1.2 Models for Predicting Changes in Future Ozone levels

Based on nonlinear regression analysis+BP neural network and the optical climate model tool (CCMs) are used to predict the future ozone change: under the premise that the total amount of ozone in the atmosphere will remain unchanged for the Montreal Protocol, the ozone layer will return to its 1980 level by about 2030 in the northern hemisphere; the ozone layer will return to its 1980 level around 2040 in the southern hemisphere.

## 4.2 The Principle of the Model

### 4.2.1 Time-series Analysis Based on Principal Component Analysis (Singular Spectrum Analysis, SSA)

SSA is a time series analysis method based on principal component analysis (PCA), first proposed by Broomhead & King (1986). The aim of this paper is to decompose any time series into many components with different characteristics without prior knowledge; these components have the same length of time as the original time series, and each component has a different physical meaning, such as the frequency of the different components can be divided into slow fluctuations in the long-term trends, periodic fluctuations and high frequency noise (Hassani, 2007). Based on SSA, Marques et al. (2006) analyzed and predicted long-term trend of regional precipitation; Macias et al. (2014), by SSA decomposition of global annual mean temperature data, it was found that the rate of global warming was a significant decline in the late 1980s, and once stopped warming. The discovery of global climate change research put forward a new argument, providing a new simulation scenario for accurate prediction of future global climate change.

SSA algorithm is mainly based on the following two steps: time series decomposition and reconstruction. Taking the time series  $\{x(t) : t = 1, \dots, N\}$  containing  $N$  samples as an example, first construct a  $L \times K$  vector trajectory matrix based on  $x(t)$ , where  $L$  is the size of the time window, that is the scale of the time series to be extracted, such as month, year, or decade and so on;  $K = N - L + 1$  represents another segment of the time series. The constructed trajectory matrix  $X$  can be expressed as:

$$X = [X(1), \dots, X(k)] = \begin{pmatrix} x(1) & \cdots & x(k) \\ \vdots & \ddots & \vdots \\ x(L) & \cdots & x(N) \end{pmatrix}$$

Among them

$$X(i) = [x(i), \dots, x(i + L - 1)]^T, i = 1, \dots, k$$

As is shown in the above formula, the rows and columns of the constructed trajectory matrix  $X$  are all sub-segments of the original time series  $x(t)$ , that is, converting  $x(t)$  into two-dimensional vector spaces. After constructing the trajectory matrix  $X$ , the singular value decomposition of the vector matrix needs to be performed (Singular Value Decomposition, SVD), so according to the singular value size to extract different components:

$$X = \sum_i \sqrt{\lambda(i)} U(i) V(i)^T$$

Among them,  $\lambda(i) = \|X(i)\|_F^2 = \|Q(i)\|^2$  is the singular value of  $XX^T$ .  $U(i)$  denotes the  $X$  decomposed left eigenvector, and  $V(i)$  denotes the  $X$  decomposed right eigenvector.

After completing the singular value decomposition of the trajectory matrix, the original time series  $x(t)$  needs to be reconstructed according to the singular value, each time series reconstructed is similar to the principal component which is orthogonal to each other in principal component analysis, that is, each time series has its unique physical meaning. The time series reconstruction in SSA begins with the clustering of singular values and eigenvectors after singular value decomposition, that is, the decomposed vector and the singular value are divided into different subclasses according to their characteristics. For example, the original time series  $x(t)$  is decomposed in SSA according to the  $L$  time scale, it can be divided into  $m$  different sub-categories, the mathematical process can be described as:

$$X = X(I(1)) + \dots + X(I(m))$$

$$X(I) = \sum_{i \in I} X(i) = X(1) + \dots + X(L)$$

That is the track matrix  $X$  is divided into a  $m$ -type sub-matrix assembly. Finally, each subclass  $X(I(m))$  is converted to  $L \times K$  Trajectory matrix  $Y$ , By diagonally averaging  $Y$  to get a new time series of length  $\{F(K), k = 1, 2, \dots, m\}$ . The relationship between any reconstructed time series  $F(k)$  and the original time series  $x(t)$ :

$$x = F(1) + \dots + F(m)$$

That is, the original time series  $x(t)$  can finally be decomposed into  $m$  sub-classes of different features. Each subclass represents the major variation of the time series at different scales.

#### 4.2.2 Based on Non-linear Regression Analysis +BP Neural Network and the Use of Optical Climate Model Tools

BP Neural network<sup>[4]</sup> has the ability to approximate the non-linear function, it is based on Kolmogorov mapping network theory of existence, it is also the most widely used math mapping in the neural network information processing arithmetic. Given an input vector  $X$ , the network should produce an output vector  $Y = \psi(X)$ , the basic characteristic of the network is to extract and identify the

necessary parameters from the complex high-dimensional data. *Kolmogorov* theory that, as long as the processing unit is an arbitrary continuous increase function of the input variable or the sum of several variables, Then an input vector  $X$  can be mapped to an arbitrary output function  $Y = \psi(X)$ .

1989 *Robert Hecht-Nielsen* proved that for any continuous function in the closed interval can be a hidden layer *BP* network to approach, thus a three-layer *BP* network can perform any  $N$ -dimensional to  $M$ -dimensional mapping.

Figure 1 shows the  $j$ th basic *BP* neuron<sup>[4]</sup>(node), it only imitates the biological neurons have the three most basic and most important function: weighting、summing and transfer. Where  $x_1, x_2, \dots, x_j, \dots, x_n$  represents the input from neuron 1, 2, ...,  $j$ , ...,  $n$ , respectively ;  $w_{j1}, w_{j2}, \dots, w_{ji}, \dots, w_{jn}$  represents the connection between neurons 1, 2, ...,  $j$ , ...,  $n$  and  $j$ th neurons intensity, respectively, that is weight;  $b_j$  is the threshold;  $f(\cdot)$  is the transfer function;  $y_j$  is the output of the  $j$ th neuron.

The net input value  $S_j$  of the  $j$ th neuron:

$$S_j = \sum_{i=1}^n w_{ji} * x_i + b_j = W_j * X + b_j$$

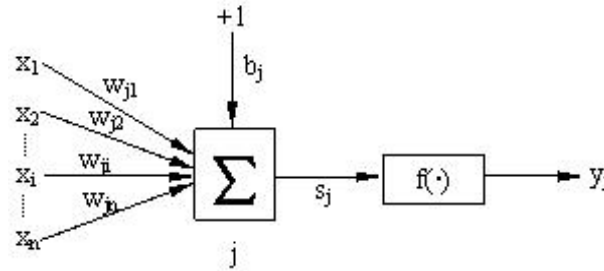


Figure 1 Neurons

If  $x_0 = 1$ ,  $x_{j0} = b_j$ , let  $x$  and  $w_j$  include  $x_0 = w_{j0}$  and  $w_{j0}$  and then

$$X = [x_1, x_2, \dots, x_i, \dots, x_n]^T, \quad W = [w_{j1}, w_{j2}, \dots, w_{ji}, \dots, w_{jn}]$$

Then the net input  $S_j$  of the node  $j$  can be expressed as:

$$S_j = \sum_{i=0}^n w_{ji} * x_i = W_j * X$$

After the net input  $S_j$  passes the transfer function  $f(\cdot)$ , we can get the output of the  $j$ th neurons  $y_j$ :

$$y_j = f(S_j) = f\left(\sum_{i=0}^n w_{ji} * x_i\right) = F(W_j * X)$$

Formula  $f(\cdot)$  is a monotonically increasing function, and it must be bounded function, because the cell-transmitted signal can not be infinitely increased, there must be a maximum.

### 1) BP network

BP algorithm<sup>[1,4]</sup> consists of the forward computation (forward propagation) of the data stream and the backward propagation of the error signal. Positive propagation, the propagation direction is the input layer  $\rightarrow$  hidden layer  $\rightarrow$  output layer, the state of each layer of neurons only affects the next layer of neurons. If the desired output is not available at the output layer, the steering error signal is propagated backwards. Through the alternation of these two processes, a gradient descent strategy is implemented in the weight vector space, and a set of weight vectors are searched dynamically to minimize the network error function, and the information extraction and memory process is completed.

### 2) Forward transmission

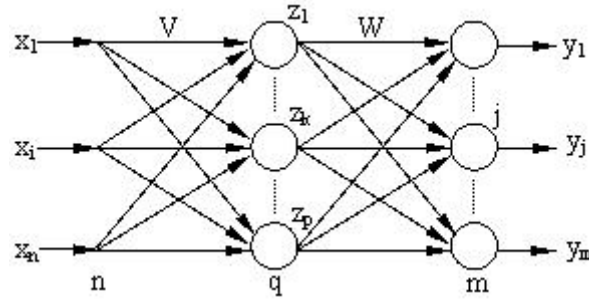


Figure 2 Topological structure of three-layer neural network

Suppose there are  $n$  nodes in the input layer of BP network<sup>[4]</sup>,  $q$  nodes in hidden layer,  $m$  nodes in output layer, and the weight between input layer and hidden layer is  $w_{jk}$ , as shown in Figure 2. The transfer function of the hidden layer is  $f_1(\cdot)$ , The transfer function of the output layer is  $f_2(\cdot)$ , The output of the hidden layer node is (writes the threshold value in the summing term):

$$z_k = f_1\left(\sum_{j=0}^n v_{jk} * x_i\right) \quad k = 1, 2, \dots, q$$

The output of the output layer node:

$$y_j = f_2\left(\sum_{k=0}^n w_{jk} * z_k\right) \quad j = 1, 2, \dots, m$$

At this point, the *BP* network completes the approximate mapping of  $n$ -dimensional space vector to  $m$ -dimensional space.

### 3) Back propagation

#### 1. Model for predicting ozone content

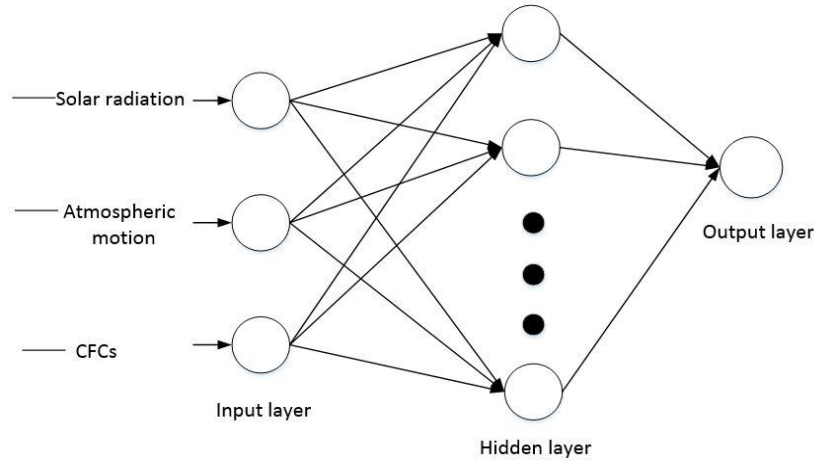


Figure 3 Model for predicting ozone content

2. Enter  $p$  learning samples for  $x^1, x^2, \dots, x^j, \dots, x^p$ . The  $p$ th sample is input to the network and the output  $y_j^p$   $j = 1, 2, \dots, m$  is obtained. Using the square error function, then we can get the error of the  $p$ th samples  $E_p$ :

$$E_p = \frac{1}{2} \sum_{j=1}^m \left( t_j^p - y_j^p \right)^2$$

Where  $t_j^p$  is the desired output.

For  $p$  samples, the global error is:

$$E = \frac{1}{2} \sum_{i=1}^p \sum_{j=1}^m \left( t_j^p - y_j^p \right)^2 = \sum_{i=1}^p E_p$$

#### 3. Output layer weight changes

Using the cumulative error *BP* algorithm to adjust  $w_{jk}$ , so that the global error  $E$  becomes small, let  $\eta$  be the learning efficiency, then

$$\Delta w_{jk} = -\eta \frac{\partial E}{\partial w_{jk}} = -\eta \frac{\partial}{\partial w_{jk}} \left( \sum_{i=1}^p E_p \right) = \sum_{i=1}^p \left( -\eta \frac{\partial E_p}{\partial w_{jk}} \right)$$

The error signal is defined as:

$$\delta_{yj} = -\frac{\partial E_p}{\partial S_j} = -\frac{\partial E_p}{\partial y_j} * \frac{\partial y_j}{\partial S_j}$$

One of the first:

$$\frac{\partial E_p}{\partial y_j} = \frac{\partial}{\partial y_j} \left[ \frac{1}{2} \sum_{j=1}^m (t_j^p - y_j^p)^2 \right] = -\sum_{j=1}^m (t_j^p - y_j^p)$$

Second:

$$\frac{\partial y_j}{\partial S_j} = f'_2(S_j)$$

is the partial derivative of the transfer function of the output layer. Then:

$$\delta_{yj} = \sum_{j=1}^m (t_j^p - y_j^p) f'_2(S_j)$$

From the chain theorem:

$$\frac{\partial E_p}{\partial w_{jk}} = \frac{\partial E_p}{\partial S_j} * \frac{\partial S_j}{\partial w_{jk}} = -\delta_{yj} * z_k = -\sum_{j=1}^m (t_j^p - y_j^p) f'_2(S_j) * z_k$$

The weight adjustment formula of Hidden layer neurons:

$$\Delta w_{jk} = \sum_{i=1}^p \sum_{j=1}^m \eta (t_j^p - y_j^p) f'_2(S_j) z_k$$

#### 4. Hidden layer weight changes

$$\Delta v_{jk} = -\eta \frac{\partial E}{\partial v_{jk}} = -\eta \frac{\partial}{\partial v_{jk}} \left( \sum_{i=1}^p E_p \right) = \sum_{i=1}^p \left( -\eta \frac{\partial E_p}{\partial v_{jk}} \right)$$

The error signal is defined as:

$$\delta_{zk} = -\frac{\partial E_p}{\partial S_k} = -\frac{\partial E_p}{\partial z_k} * \frac{\partial z_k}{\partial S_k}$$

One of the first:

$$\frac{\partial E_p}{\partial z_k} = \frac{\partial}{\partial z_k} \left[ \frac{1}{2} \sum_{j=1}^m (t_j^p - y_j^p)^2 \right] = -\sum_{j=1}^m (t_j^p - y_j^p) \frac{\partial y_j}{\partial z_k}$$

According to chain theorem are:

$$\frac{\partial y_j}{\partial z_k} = \frac{\partial y_j}{\partial S_j} * \frac{\partial S_j}{\partial z_k} = f'_2(S_j) w_{jk}$$

Second:



$$\frac{\partial z_k}{\partial S_k} = f'_1(S_k)$$

is the partial derivative of the hidden layer transfer function. Then:

$$\delta z_k = \sum_{j=1}^m (t_j^p - y_j^p) f'_2(S_j) w_{jk} f'_1(S_k)$$

From the chain theorem:

$$\frac{\partial E_p}{\partial v_{jk}} = \frac{\partial E_p}{\partial S_k} * \frac{\partial S_k}{\partial v_{jk}} = -\delta z_k * x_i = -\sum_{j=1}^m (t_j^p - y_j^p) f'_2(S_j) w_{jk} f'_1(S_k) * x_i$$

The weight adjustment formula of hidden layer neurons:

$$\Delta v_{jk} = \sum_{i=1}^p \sum_{j=1}^m \eta (t_j^p - y_j^p) f'_2(S_j) w_{jk} f'_1(S_k) * x_i$$

## V. Past ozone in Observations and Model Simulations

### 5.1 Data Sources

In order to analyze and forecast the trends of total atmospheric ozone at the global scale, this study uses the European Center for Mesoscale Weather Forecast (European Centre for Medium-Range Weather Forecasts, ECMWF) Based on ERA-Interim 1979-2015 data on the total atmospheric ozone reanalysis data. However, ERA-Interim Ozone Total Reanalysis data were used in this study, primarily because ERA-Interim Ozone Total Reanalysis data can be used to conduct continuous polar ozone trends analysis. ERA-Interim reanalysis data provide complete global coverage, the total amount of ozone over the polar region is still estimated even during the polar night. Which can not be achieved by a single satellite observation inversion (No ozone total inversion during the night) In addition, the data assimilation method used in the reanalysis data can effectively remove the deviation of total atmospheric ozone from the multi-source satellite remote sensing, thus ensuring its continuity and consistency on the time scale.

ERA-Interim reanalysis data is a new generation of reanalysis data set created by the European Center for Mesoscale Weather Forecasting, and it provides data assimilation results for various parameters of the atmosphere, land and ocean surface from 1979 to the current global scale. In order to improve the accuracy of the total ozone reanalysis data, the European Center for Mesoscale Weather Forecasting has assimilated the observational data of various satellite platforms and ground-based ozone in the ERA-Interim project.

Table 1 Merged total ozone column data

| Data Set                      | Instruments                                                         | Record Length                                    | Reference                                                     |
|-------------------------------|---------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------|
| Ground-Based                  | Dobson, Brewer, SAOZ                                                | 01/1964 to 12/2013                               | Fioletov et al., 2002, 2008; Hendrick et al., 2011            |
| SBUV MOD V8                   | BUV Nimbus-4, SBUV/TOMS Nimbus-7, SBUV/2 NOAA 9 to 19, OMI, TOMS EP | 11/1978 to 12/2012                               | Stolarski and Frith, 2006                                     |
| SBUV V8.6 NASA (MOD V8.6)     | BUV Nimbus-4, SBUV Nimbus-7, SBUV/2 NOAA 9 to 19                    | BUV: 01/1970 to 05/1976 SBUV: 11/1978 to 12/2013 | Chiou et al., 2014; Labow et al., 2013; McPeters et al., 2013 |
| SBUV V8.6 NOAA                | SBUV Nimbus-7, SBUV/2 NOAA 9 to 19                                  | 11/1978 to 12/2013                               |                                                               |
| GSG - Bremen                  | GOME, SCIAMACHY, GOME2                                              | 07/1995 to 12/2013                               | Weber et al., 2013                                            |
| GTO - ESA/DLR                 | GOME, SCIAMACHY, GOME2                                              | 07/1995 to 12/2013                               | Chiou et al., 2014; Lerot et al., 2014                        |
| Multiple Satellite Reanalysis | assimilated TOMS, SBUV, GOME, SCIAMACHY, OMI                        | 11/1978 to 12/2013                               | van der A et al., 2010                                        |

The HARMonized data set of OZone profiles (HARMOZ: Sofieva et al., 2013) produced by the European Space Agency Ozone Climate Change Initiative (ESA O3-CCI). HARMOZ consists of quality-screened European individual limb sounder data sets that are provided in common ozone units and altitude grid. HARMOZ consists of ozone data from the Optical Spectrograph and Infrared Imaging System (OSIRIS), the Sub-Millimetre Radiometer (SMR), both on the Odin satellite since 2001, and from the Global Ozone Monitoring by Occultation of Stars (GOMOS), the Michelson Interferometer for Passive Atmospheric Sounding (MIPAS), and the Scanning Imaging Absorption Spectrometer for Atmospheric Chartography (SCIAMACHY), all on the Envisat platform from 2002 to 2012.

## 5.2 Changes in Total Column Ozone

Table 2 Main ozone profile data

| Data Set                        | Instruments Most Relevant for Trend                                  | Record Length                                                   | References                                                              |
|---------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| SBUV V8.6<br>NASA<br>(MOD V8.6) | BUV Nimbus-4, SBUV<br>Nimbus-7, SBUV/2<br>NOAA 9 to 19               | BUV: 01/1970 to<br>05/1976<br>SBUV: 11/1978 to<br>07/2013       | Kramarova et al.,<br>2013a; McPeters et al.,<br>2013                    |
| SBUV V8.6<br>NOAA               | SBUV Nimbus-7,<br>SBUV/2 NOAA 9 to 19                                | 11/1978 to 12/2012                                              |                                                                         |
| GOZCARDS                        | SAGE I, SAGE II,<br>HALOE, MLS-Aura                                  | 02/1979 to 12/2013<br>Gap: 11/1981 to<br>10/1984                | R. Wang et al., 2013                                                    |
| SAGE II,<br>V7.0                | SAGE II                                                              | 10/1984 to 08/2005                                              | Damadeo et al., 2013                                                    |
| HARMOZ                          | ODIN: OSIRIS, SMR<br>ACE-FTS; Envisat:<br>GOMOS, MIPAS,<br>SCIAMACHY | 10/2001 to 12/2012<br>08/2002 to 12/2011                        | Sofieva et al., 2013                                                    |
| NDACC<br>ground-based           | lidars, microwave<br>radiometers, FTIRs<br>about 5 stations          | late 1980s/early 1990s<br>to 2012/2013,<br>depending on station | Steinbrecht et al.,<br>2009; Vigoroux et al.,<br>2008                   |
| Umkehr                          | about 5 stations                                                     | 1956 (Arosa) 1984<br>(Lauder) to 2012                           | Petropavlovskikh et<br>al., 2005 ; Tourpali et                          |
| Ozonesondes                     | about 50 stations                                                    | Late 1960s / mid-<br>1990s to 2012/2013<br>depending on station | Smit et al., 2007;<br>Deshler et al., 2008;<br>Thompson et al.,<br>2012 |

The gray range of CCMVal-2 simulations in Figure4 gives the multi-model mean anomaly (from the 1998 to 2008 baseline) and 2 standard deviations of individual annual mean anomalies. The multi-model mean (and standard deviation) comes from 15 CCMs. Two models (MRI and CNRM-CMM) with known deficiencies were omitted (see Oman et al., 2010b; Michou, 2011). This approach to presenting the CCMVal-2 results differs from the last Assessment, where the complex Time Series Adaptive Method (Scinocca et al., 2010; WMO, 2011) was used to define the range of simulated ozone values. Because the TSAM method cannot readily be applied to observed time series, the simpler approach of computing means and standard deviation over sliding five-year windows and multiple models defines the range of simulated interannual, and, to a lesser

degree, intermodel variability.

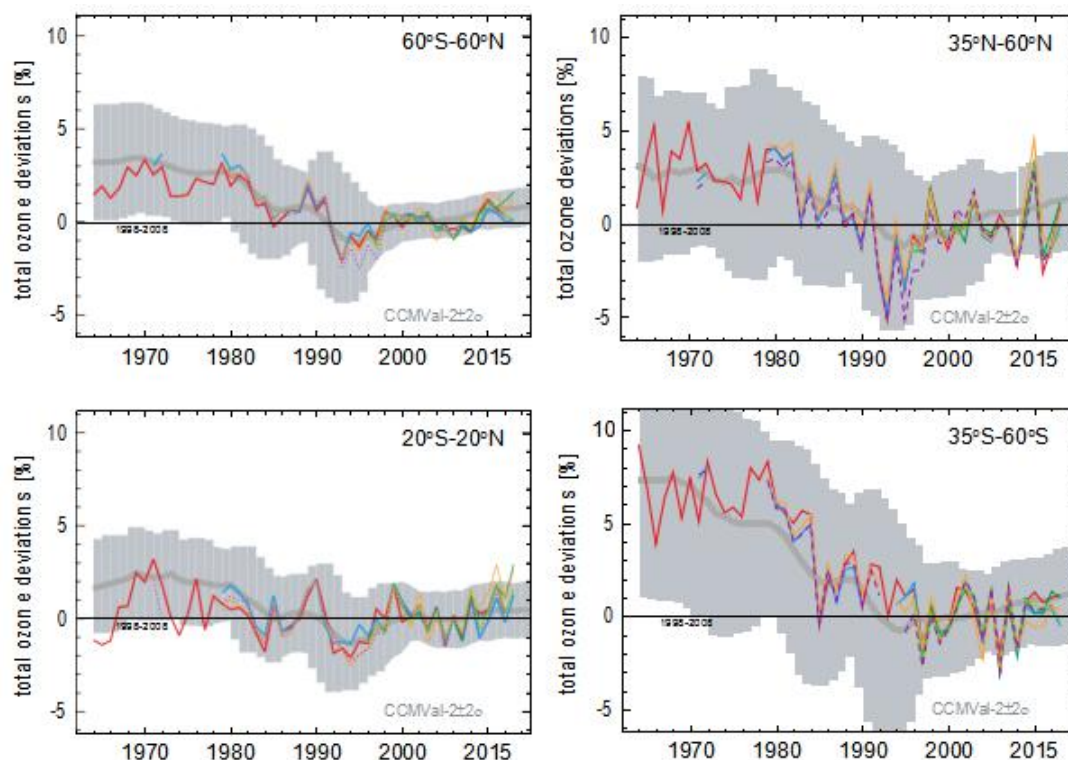


Figure 4 Total column ozone annual mean anomalies for different data sets

Anomalies are with respect to the 1998 to 2008 mean of each individual data set. Top left: 60°S–60°N (global), top right: 35°N–60°N (Northern Hemisphere), bottom left: 20°S–20°N (tropics), bottom right: 35°S–60°S (Southern Hemisphere). Colored lines give observed results for the data sets from Table 2. Gray line and gray range give multi-model mean and  $\pm 2$  standard deviation range of annual mean anomalies simulated by CCMVal-2 models. The model simulations account for changing ODSs, GHGs, and a QBO. Up to about 2000, all simulations also account for the 11-year solar cycle, volcanic aerosol, observed sea surface temperatures, and sea ice coverage. After about 2000, sea surface temperature and sea ice are prescribed from other model simulations, and no solar cycle is included for most models.

### 5.3 Time Series Analysis of Principal Component Analysis (SSA)

SSA is a time series analysis method based on principal component analysis (PCA), which was first proposed by Broomhead & King (1986). It aims to decompose any arbitrary time series into many components with different characteristics without prior knowledge, and these components have the same length of time as the original time series, and each component has a different physical meaning, such as the frequency can be divided into different types of long-term trend of slow fluctuations, cyclical fluctuations and high frequency Sub-noise. The findings provide a new perspective for global climate change research, providing a new simulation scenario for accurately predicting future global climate change.

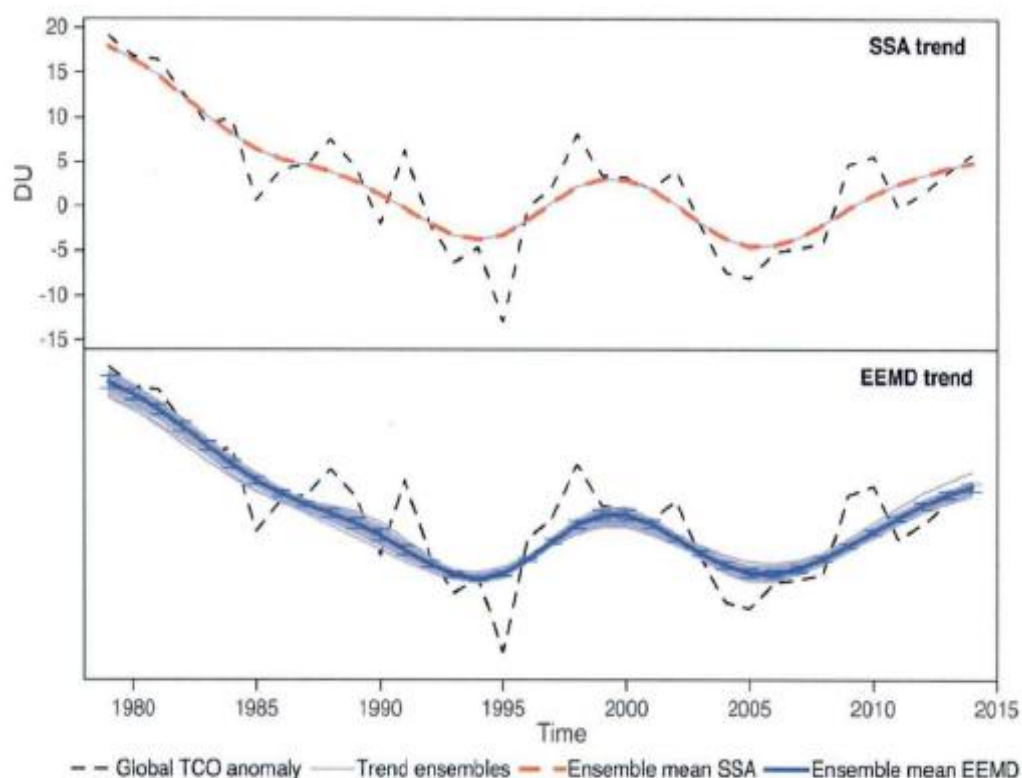


Figure 5 Comparison of long-term trend stability of SSA and EEMD extraction

The above-mentioned comparative analysis confirms that SSA is more stable than EEMD in time series. In addition, compared with EEMD the computing speed of SSA is faster, about 10 times that of EEMD. Based on this conclusion, SSA will be selected as the time series analysis method to extract the long-term trend of global ozone in 1979-2015.

## 5.4 Trends in Ozone Profiles

### 5.4.1 Ozone Trends up to 1999

Figure 6 shows resulting linear ozone trends for the declining period up to 1997, for the GOZCARDS data set (in pressure coordinates), which is largely based on SAGE I and SAGE II (V6.20) over this trend period from 1978 to 1997, for the SBUV-NASA data set (trend period 1970 to 1997, in pressure coordinates), and for the SAGE II V7.0 data set (trend period 1984 to 1997, in altitude coordinates). The corresponding pattern from the CCMVal-2 model simulations (trend period 1970 to 1997) is shown as well. Note that small differences between trends can arise from the use of pressure or altitude in different observing systems.

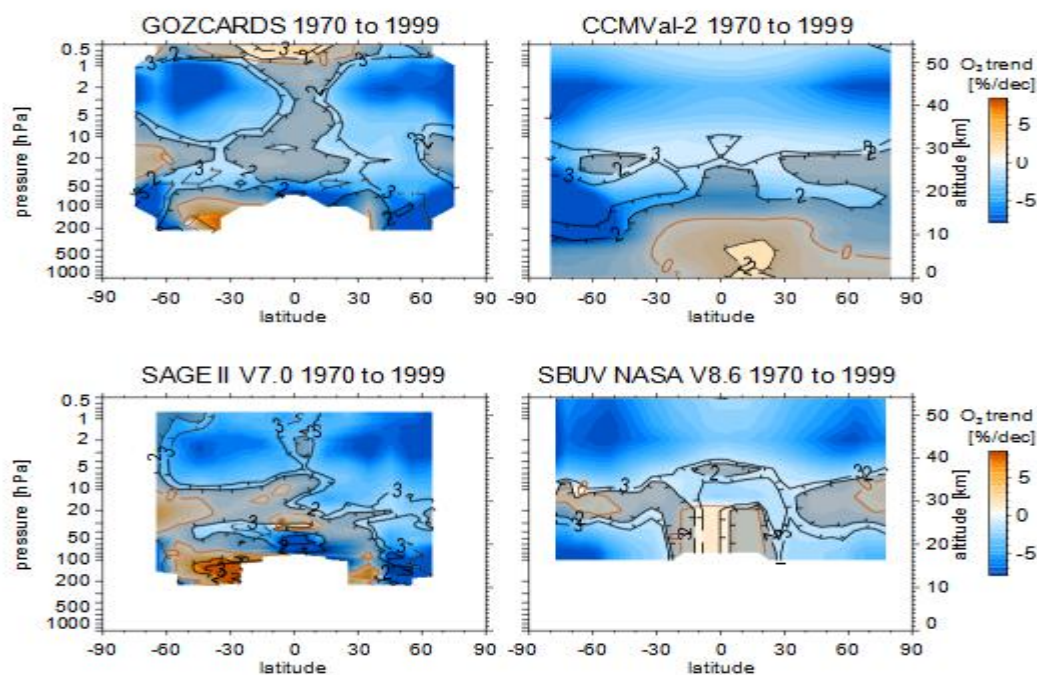


Figure 6 Ozone trends (in % per decade) before 1999 as a function of latitude and pressure

#### 5.4.2 Ozone Trends since 2000

Confirming these studies, Figure 7 gives ozone trends since 2000 for three independent observational data sets and for the CCMVal-2 simulations.

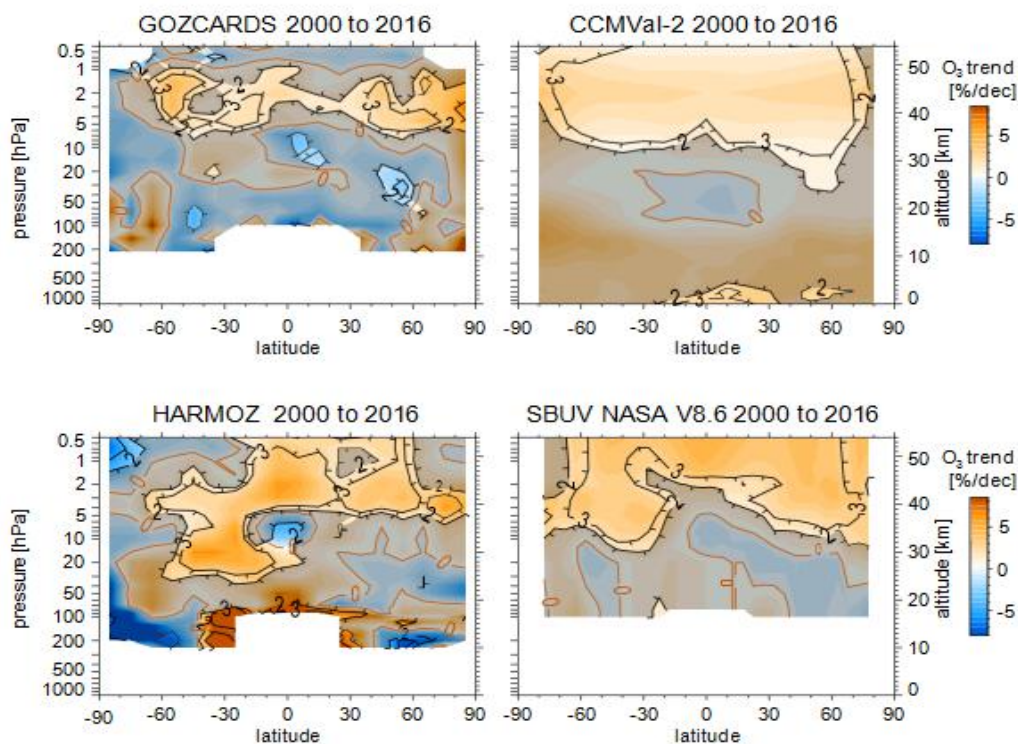


Figure 7 ozone trends (in % per decade) since 2000



In the mid- and lower stratosphere, ozone trends since 2000 are generally not significant. In the SH middle stratosphere (around 30 km/15 hPa), HARMOZ and SBUV show significant ozone increases, which are also evident, but not significant, in GOZCARDS.

## 5.5 Total Ozone Trends

Here we use MLR on the basis of annual means for total ozone and monthly means for profile data. ODS values reached their maximum around 1997 and there are different approaches to express ODS changes in the regression.

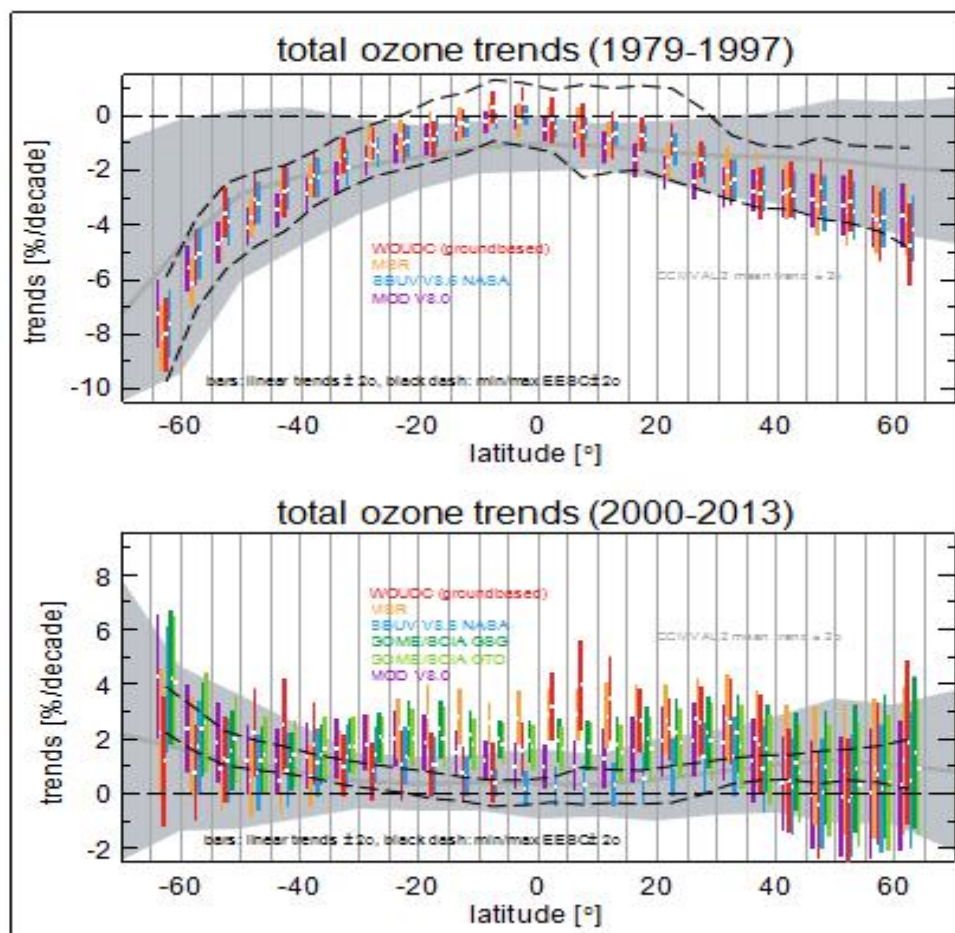


Figure8 Ozone trends for different merged total ozone data sets

Top: Trends from 1979 to 1997. Bottom: Trends from 2000 to 2013. Observed trends are derived using multiple linear regression (MLR) to account for solar cycle. The white dots give the linear trend in % per decade, the colored vertical bars indicate the  $\pm 2\sigma$  uncertainty range from the regression. Gray areas give average trend and  $\pm 2$  standard deviation range of individual model trends from CCMVal-2 simulations. The black dashed lines give the range of linear trends arising from MLR fits to the observations that include Equivalent Effective Stratospheric Chlorine (EESC) as a proxy for ODS changes. The fitted EESC coefficients usually give a negative ozone trend while EESC increases from 1979

to 1997, and a positive ozone trend while EESC declines from 2000 to 2013.

## VI. Updates on Natural Ozone Variations

### 6.1 Solar Radiation

Recent studies have confirmed a 2–4% variation of stratospheric ozone (3% in total ozone) in phase with the 11-year solar cycle. However, the exact shape of the solar response profile depends on the type of data and/or analysis, the length of data records, and the time periods under investigation. There is evidence that SORCE SSI strongly overestimates UV solar cycle variability.

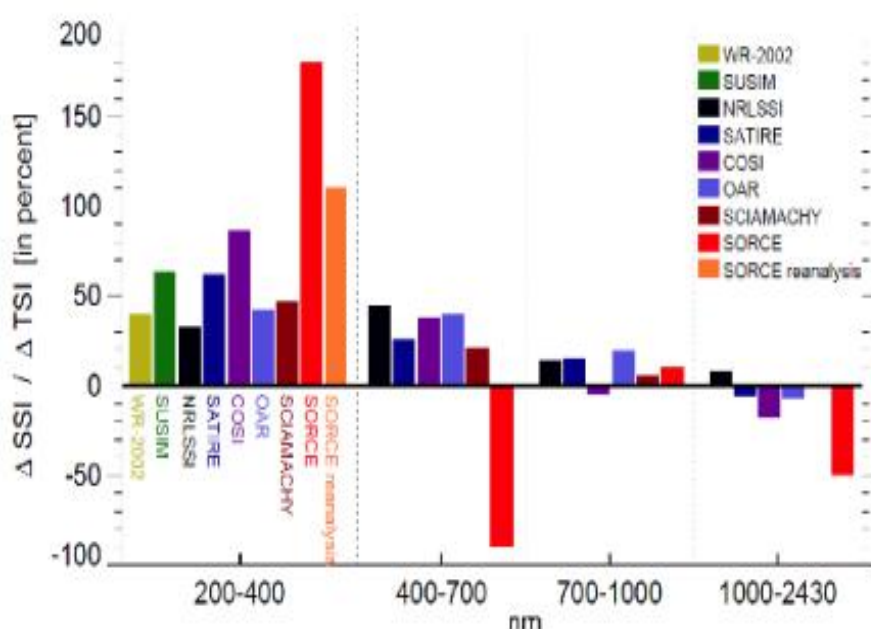


Figure 9 Spectral solar irradiance (SSI) changes over a solar cycle and in various spectral regions

SSI changes are normalized to total solar irradiance changes (TSI, solar constant), and are given for several data sets. WR-2002 (Woods and Rottman, 2002), SORCE, Solar Ultraviolet Spectral Irradiance Monitor (SUSIM), and SCIAMACHY are derived from observations. SATIRE, COSI, and OAR are solar models. NRLSSI is a solar reconstruction from Lear.

### 6.2 Effects of Atmospheric Motion on the Stratosphere

Previous Assessments have described how stratospheric sulfate aerosols impact ozone concentrations through both direct impacts on heterogeneous chemical processes and indirect impacts on temperature and transport.



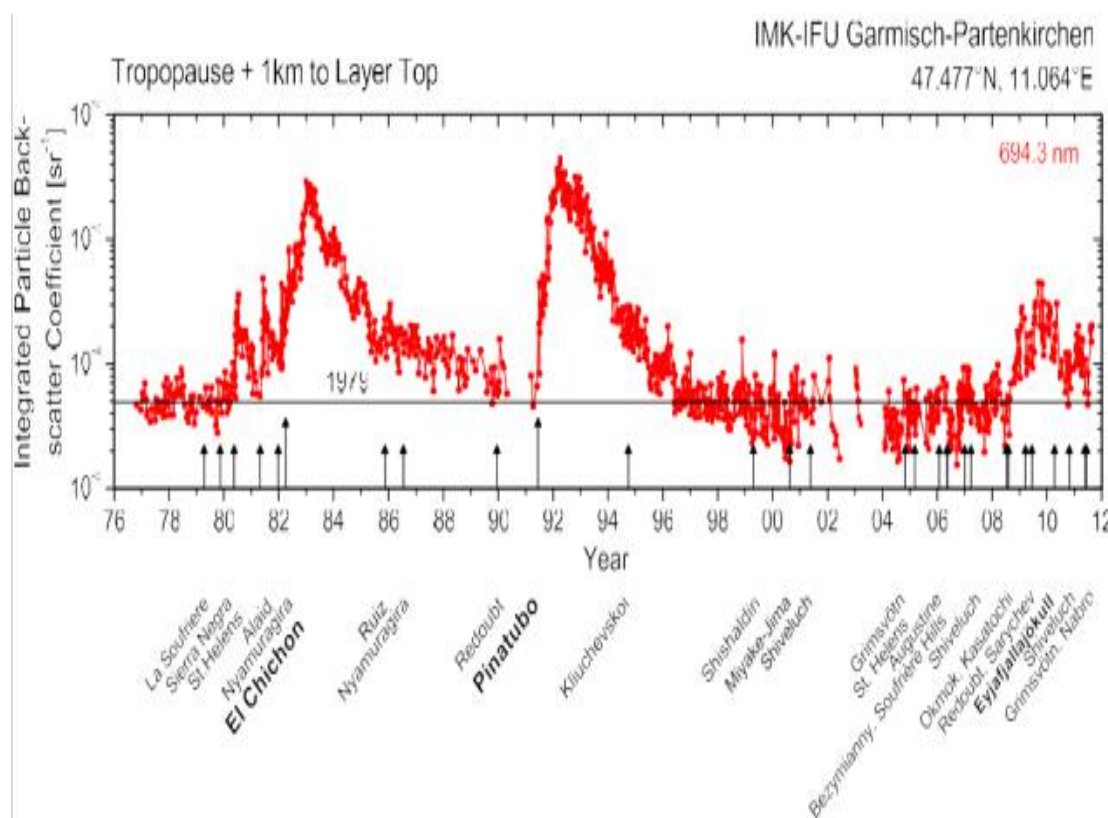


Figure 10 Aerosol backscatter coefficient measured since 1976 by lidar at Garmisch-Partenkirchen ( $47.5^{\circ} N, 11^{\circ} E$ ) as a measure of stratospheric aerosol loading.

## VII.2016-2066 Year's Global Atmospheric Ozone

### Total Rorecast

#### 7.1 Effects of Future Stratospheric Temperature and Circulation

##### Changes

Future increases in greenhouse gases (GHGs), especially carbon dioxide (CO<sub>2</sub>) are important for global ozone because they result in stratospheric cooling, slowing some chemical ozone destruction rates in the middle and upper stratosphere. According to model simulations, GHG- forced climate change leads to increased upwelling in the tropics and to changes in planetary waves that drive an increase in the strength of the BDC.

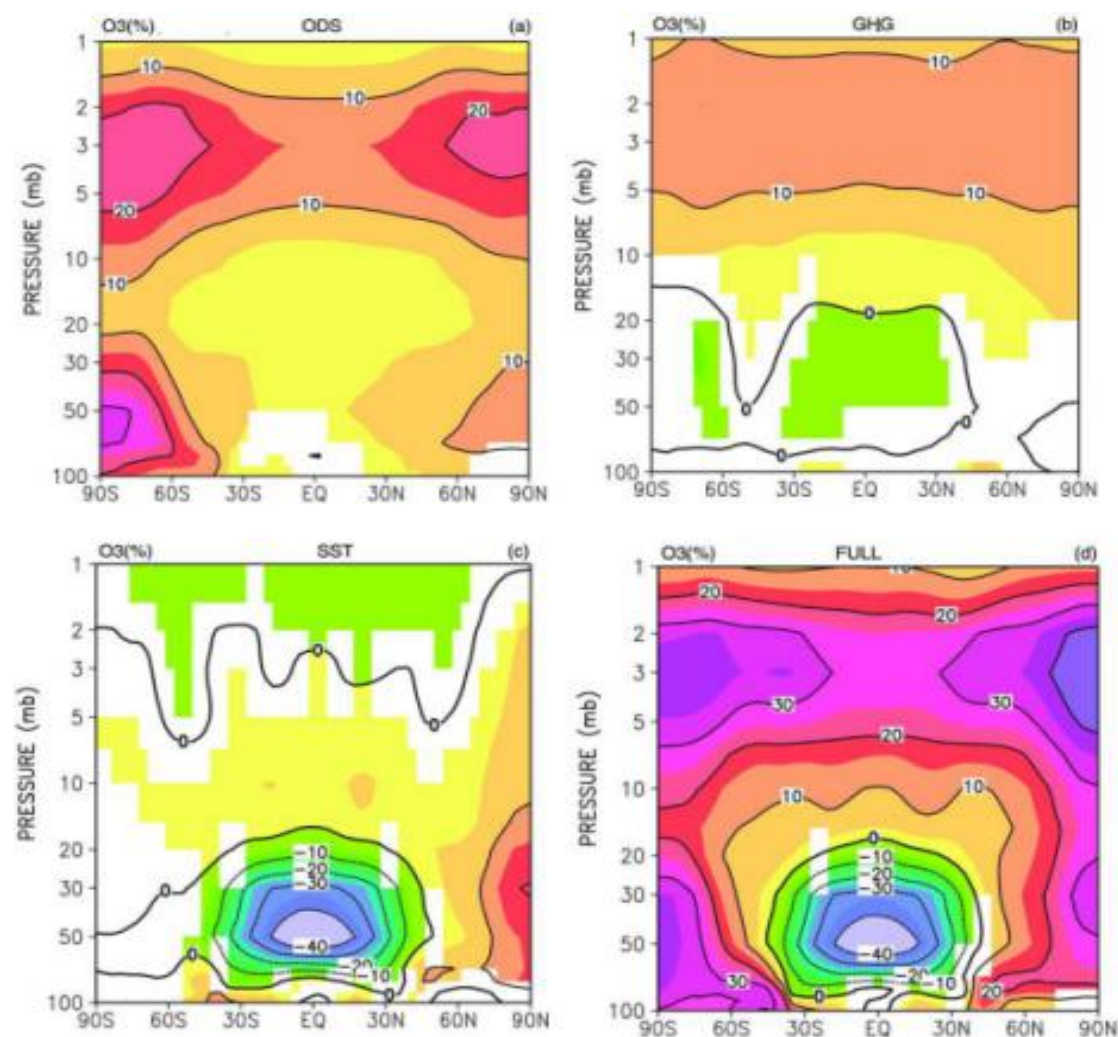


Figure 11 annual and ensemble mean percentage changes of the ozone mixing ratio between different timeslice integrations of the SOCOL CCM for 2100 and for 2000 conditions

(a) Ozone changes from 2000 to 2100 when only ODSs are changed.

(b) Ozone changes from 2000 to 2100 due to GHG changes only.

(c) Ozone changes due to changes in sea surface temperature and sea ice, driven by tropospheric warming.

(d) Ozone changes when ODS, GHG, sea surface temperature, and sea ice are all changed together. White regions indicate nonsignificant changes at the 5% confidence level. Adapted from Zubov et al.

## 7.2 Expected Return to 1960 or 1980 Levels and Ozone Recovery

When building a mathematical model to predict ozone levels over the next 50 years, the input data for the following table (Some data are listed).

Table3 Some data as input data

| Time      | Data  | Minimum | 10%   | 30%   | Mean  | 70%   | 90%   | Maximum |
|-----------|-------|---------|-------|-------|-------|-------|-------|---------|
| 1979/1/1  | 253   | 221     | 235   | 246   | 250.5 | 254.7 | 267.8 | 280     |
| 1979/1/2  | 259   | 227     | 235.1 | 247.6 | 254   | 257.4 | 264   | 278     |
| 1979/1/3  | 274   | 220     | 237.4 | 244.2 | 251   | 258   | 265   | 277     |
| 1979/1/4  | -9999 | 224     | 237.1 | 243   | 249   | 256.7 | 261.9 | 270     |
| 1979/1/5  | 272   | 222     | 230.4 | 243.6 | 251   | 255   | 263.8 | 272     |
| 1979/1/6  | 269   | 223     | 230.4 | 244.6 | 247   | 256.4 | 267.2 | 274     |
| 1979/1/7  | 269   | 221     | 233   | 245.6 | 250   | 258   | 262.8 | 278     |
| 1979/1/8  | -9999 | 227     | 232.2 | 243   | 248.5 | 255   | 264.7 | 277     |
| 1979/1/9  | 260   | 228     | 234.4 | 242.2 | 250   | 256.8 | 262.8 | 267     |
| 1979/1/10 | 265   | 224     | 235.2 | 243.9 | 250.5 | 259.7 | 265.9 | 270     |

The past two Ozone Assessments (WMO, 2007; 2011) have used chemistry-climate models (CCMs) as the primary tool for future ozone projections. WMO (2011) based their future projections on the simulations of 17 models, all of which participated in CCMVal-2 (Eyring et al., 2010). This ensemble remains the most comprehensive set of coordinated simulations of past and future stratospheric ozone changes. It is therefore the basis of future projections in the current Assessment. These CCMs have been extensively evaluated by CCMVal and the results are documented in the SPARC CCMVal Report (2010). Recent new simulations by four CCMVal-2 models using the revised estimates of ODS lifetimes (SPARC, 2013; also Chapter 1) are shown in Figure 11. These simulations indicate that the expected future evolution of near-global total ozone columns changes very little when using the revised ODS lifetimes. Changes in expected recovery dates to 1980 levels, for example, are typically less than a few years and are usually not significant given the large interannual variability and the intermodel differences. The estimates of return dates in different regions (Table 3) are not updated from WMO (2011). At this point there is not enough new evidence

to motivate a revision of the estimates of ozone recovery presented in the previous Assessment.

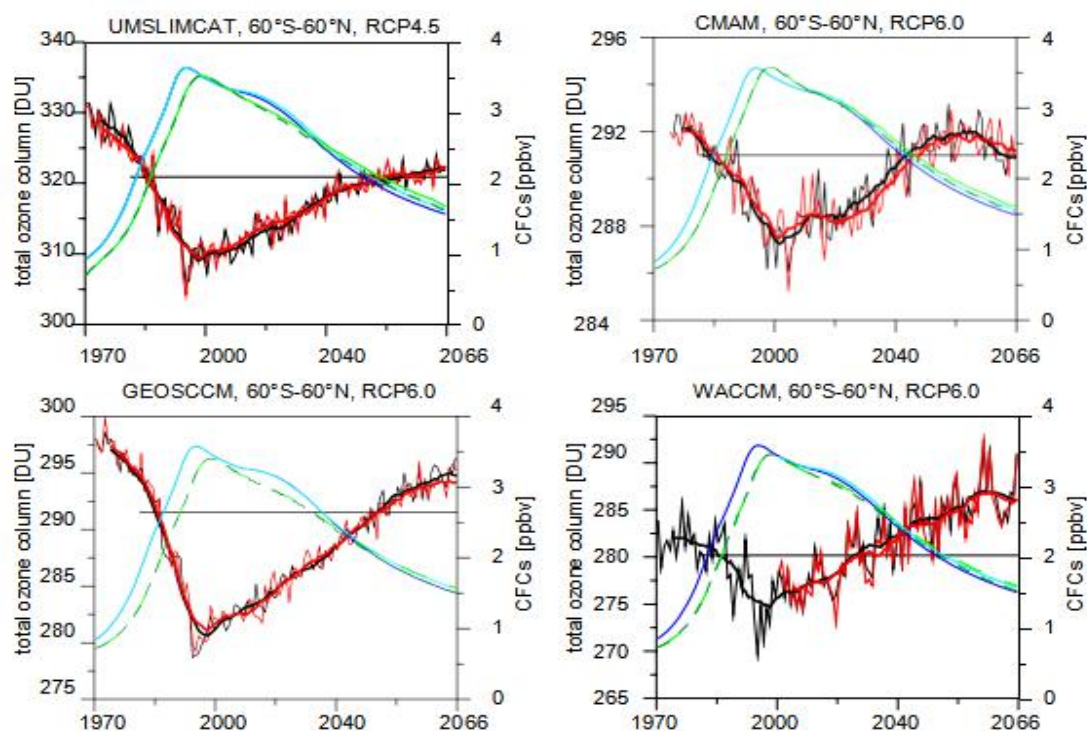


Figure12 Global atmospheric ozone total forecast

Douglass et al. (2012) showed that, in agreement with expectation based on photochemical theory of upper stratospheric ozone, models with cold biases in upper stratospheric temperatures have higher ozone levels and stronger ozone sensitivity to chlorine changes. Thus, intermodel spread is not always an appropriate measure of uncertainty in future upper stratospheric ozone projections, but is also indicative of systematic errors that depend on model biases in simulated temperature, ozone, and reactive nitrogen climatologies.

Table 4 Expected return years when ozone columns will return to 1960 or 1980 values.

| Region                           | Date of | Mean Estimate | Lower Bound | Upper Bound |
|----------------------------------|---------|---------------|-------------|-------------|
| Global annual mean               | 1960    | 2053          | 2046        | 2064        |
|                                  | 1980    | 2032          | 2027        | 2038        |
| Tropics annual mean              | 1960    | ---           | ---         | ---         |
|                                  | 1980    | 2042          | 2028        | ---         |
| Northern midlatitude annual mean | 1960    | 2029          | 2024        | 2036        |
|                                  | 1980    | 2021          | 2017        | 2026        |
| Southern midlatitude annual mean | 1960    | 2055          | 2049        | 2064        |
|                                  | 1980    | 2035          | 2030        | 2040        |

The global ozone trend was analyzed and predicted in this study which is based

on the global monthly ERA-Interim ozone reanalysis data from 1970 to 2015. In this study, we analyzed the temporal and spatial variation of atmospheric ozone over the past decades and found that the change of total atmospheric ozone was mainly influenced by human factors, such as the resumption of ozone levels resulting from the implementation of the Montreal Protocol. Based on this result and assuming that the future global total atmospheric ozone response to the agreement remains unchanged, that the total amount of ozone under this agreement will continue to maintain growth and is not affected by other additional factors, this study predicts the temporal and spatial characteristics of the future total ozone at the global scale under this ideal condition, and then we get the conclusion that the ozone level tends to rise.

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## IX.Appendix

### 1. ssa 程序

%主程序

Main.m

```
x=Load('data.txt');
```

```
x1=x';
```

```
ssa(x1,10000);
```

%子程序

```
function [y,r,vr]=ssa(x1,L)
```

```
% x1 Original time series (column vector form)
```

```
% L Window length
```

```
% y Reconstructed time series 重构时间序列
```

```
% r Residual time series r=x1-y 剩余时间序列
```

```
% vr Relative value of the norm of the approximated trajectory matrix with respect
```

```
%to the original trajectory matrix 相对于原始轨迹矩阵的近似轨道矩阵范数的相对值
```

```
% The program output is the Singular Spectrum of x1 (must be a column vector),
```

```
% using a window length L. You must choose the components be used to reconstruct
```

```
%the series in the form [i1,i2:ik,...,iL], based on the Singular Spectrum appearance.
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
% Step1 : Build trayectory matrix
```

```
N=length(x1);
```

```
if L>N/2;L=N-L;end
```

```
K=N-L+1;
```

```
X=zeros(L,K); for i=1:K
```

```
    X(1:L,i)=x1(i:L+i-1);
```

```
end
```

```
% Step 2: SVD
```

```
S=X*X';
```

```
[U,autoval]=eig(S); [d,i]=sort(-diag(autoval));
```

```
d=-d; U=U(:,i);sev=sum(d);
```

```
plot((d./sev)*100),hold on,plot((d./sev)*100,'rx');
```

```
title('Singular Spectrum');xlabel('Eigenvalue
```

```
Number');ylabel('Eigenvalue (% Norm of trajectory matrix retained)')
```

```
V=(X')*U;
```

```
rc=U*V';
```

```

% Step 3: Grouping
I=input('Choose the agrupation of components to reconstruct the series
in the form I=[i1,i2:ik,...,iL] ');
Vt=V';
rca=U(:,I)*Vt(I,:);
% Step 4: Reconstruction
y=zeros(N,1);
Lp=min(L,K);
Kp=max(L,K);
for k=0:Lp-2
    for m=1:k+1;
        y(k+1)=y(k+1)+(1/(k+1))*rca(m,k-m+2);
    end
end
for k=Lp-1:Kp-1
    for m=1:Lp;
        y(k+1)=y(k+1)+(1/(Lp))*rca(m,k-m+2);
    end
end
for k=Kp:N
    for m=k-Kp+2:N-Kp+1
        y(k+1)=y(k+1)+(1/(N-k))*rca(m,k-m+2);
    end
end
figure;subplot(2,1,1);hold on;xlabel('Data point');ylabel('Original and
reconstructed series')
plot(x1,'b');grid on;plot(y,'r')
% plot(t,y)
r=x1-y;
subplot(2,1,2);plot(r,'g');xlabel('Data point');ylabel('Residual
series');grid on
vr=(sum(d(I))/sev)*100;

```

## 2. 非线性回归+神经网络

```

clear;
%导入原始数据测量数据
set(0,'defaultfigurecolor','w');
x=load('data.txt');
p=[x(:,3) x(:,4) x(:,5)]';
t=x(:,2)';
% 对原始数据进行规范化处理
[pn,meanp,stdp,tn,meant,stdt]=prestd(p,t);
[ptrans,transMat]=prepca(pn,0.001);
[R,Q]=size(ptrans);
% 将原始数据分成几部分作为不同用途

```

```

iitst=2:4:Q;
iival=4:4:Q;
iitr=[1:4:Q 3:4:Q];
vv.T=tn(:,iival);vv.P=ptrans(:,iival);vt.P=ptrans(:,iitst);
vt.T=tn(:,iitst);ptr=ptrans(:,iitr);ttr=tn(:,iitr);
[pn,minp,maxp,tn,mint,maxt]=premnmx(p,t);
%建立网络
net=newff(minmax(ptr),[10,1],{'tansig','purelin'},'trainlm');
%训练网络
[R,Q]=size(pn);[S2,Q]=size(tn);S1=10;
w1=rands(S1,R);
b1=rands(S1,R);
w2=rands(S2,S1);
b2=rands(S2,S1);
net.trainParam.show=1000; %每间隔 1000 步显示一次训练结果
net.trainParam.Lr=0.05; %学习速率为 0.05
net.trainParam.epochs=50000; %允许最大训练步数 50000 步
net.trainParam.goal=0.65*10^(-3); %均方误差
[net,tr]=train(net,pn,tn);
%开始训练, 其中 pn,tn 分别为输入输出样本
view(net);
%利用原始数据对 BP 网络仿真
an=sim(net,pn); %用训练好的模型进行仿真
a=postmnmx(an,mint,maxt); % 把仿真得到的数据还原为原始的数量级;
a=a'
%绘制训练过程中各误差的变化曲线
plot(tr.epoch,tr.perf,'r',tr.epoch,tr.vperf,'g',tr.epoch,tr.tperf,'-b');
legend('训练','确证','测试',-1);ylabel('平方误差');xlabel('时间');
%结果分析, 并绘制曲线
an=sim(net,ptrans);a=poststd(an,meant,stdt);
sst=sum((t-mean(t)).^2);sse=sum((t-a).^2);
ssr=sum((a-mean(t)).^2);
hh=(1-sse/sst)
hh2=ssr/sst
ppp=[a;t];sssssssssss=sse/109
% 寻找优化源程序
for i=1:10000

    if hh>0.85
        break;
    end
end
end

```



|                     |                     |                     |
|---------------------|---------------------|---------------------|
| For office use only | Team Control Number | For office use only |
| T1 _____            | <b>2389</b>         | F1 _____            |
| T2 _____            |                     | F2 _____            |
| T3 _____            | Problem Chosen      | F3 _____            |
| T4 _____            | <b>A</b>            | F4 _____            |

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**2016**  
**Mathematical Contest in Modeling (MCM/ICM) Summary Sheet**

## Ozone Forecasting Model Based On Grey Prediction

### Summary

Despite of the small amount in the atmosphere, ozone is one of the most critical atmospheric component as it protects human beings and any other life forms on the Earth from the high frequency ultraviolet radiation of the Sun. Due to uncontrolled anthropogenic emissions of the substances, ozone amount in the atmosphere has been greatly depleted, in particularly over the Antarctica. However, due to the implementation of the Montreal Protocol, the total amount of ozone depleting substances in the atmosphere has been prominent reduced, and in turn the ozone layer starts to recovery. Within this context, investigations of the long—term variability of global total column ozone in the past decades to project the future ozone recovery would be much helpful in better prediction of the climate in the future.

In this thesis, we use GM model to predict the ozone levels over the next 50 years, results and conclusions can be generally summarized as follows:

- According to the result of every station in ESRL Global Monitoring Division, we find that the change trend of Ozone is falling based on the data before 1987. On the contrary, the change trend is rising based on the data after 1987 except for very few stations such as the Antarctica and the equator.
- According to the result of the global ozone level, we find the same discipline, and that is to say the total Ozone is rising gradually from 1987. And from the simulated picture of GM model, we could determine the Ozone content is between 295DU and 300DU in 2066.
- The change trend of global ozone levels possesses regional difference. The Montreal Protocol has a positive effect on the Northern Hemisphere..
- To develop a novel family of grey forecasting methods instead of time series and regressive methods. All of these may be referred to as an approach to deal with grey process.
- The Montreal Protocol has a huge impact on the change trend of global ozone levels, and the ozone level is recovering over the 50 years.

All the results above show that our model has a positive effect on the economy, geography and environment. And we can easily analyze the climate change in the future.

**Keywords:** Ozone Level    Grey System    GM Model    Residual Tests    Posterior Check

# Ozone Forecasting Model Based On Grey Prediction

— { L<sup>A</sup>T<sub>E</sub>X Studio } —

November 28, 2016

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## 1 Introduction

With the boom of the Internet and the rise of e-commerce, online shopping has become the primary choice for many consumers. However, the speed of delivery always brings a big trouble for Electric business enterprise account for the complaint of delivery speed. Therefore, building warehouses has become a priority for improving. In this case, how to meet the largest consumer's demand with minimal warehouse quantity and the optimal location is the question we will discuss in this article. In order to solve this problem and minimize the tax liability to the most customers as far as possible, many researches were down for finding out optimal warehouses' locations based on Genetic algorithm or Super network optimization, and this methods could also cope with the location of warehouse with minimum cost and proper location. However, all of solutions above could just deal with the issue which only consider the cost and the location while not thinking about the other constraint conditions. So we develop a mathematical model to settle the location of warehouse for best location and minimum cost. Besides, based on the first model we analysis the constraint condition that requires the minimal taxes for consumers and maximal clothing sales for shop. What is more, we talk about how to get the result which is the optimal one in the above two conditions based on the consumption data of all states.

## 2 Symbol and definitions

|       |                                    |
|-------|------------------------------------|
| 40S   | Forty degrees south                |
| 30S   | Thirty degrees south               |
| 20N   | Twenty degrees north               |
| 40N   | Forty degrees north                |
| 50N   | Fifty degrees north                |
| 60N   | Sixty degrees north                |
| 70N   | Seven degrees north                |
| $x^n$ | Developing coefficient in GM model |
| $a$   | Developing coefficient in GM model |
| $b$   | Grey input in GM model             |

$x^0(k)$  A grey derivative which maximizes the information density for a given series to be modelled

$b_i$  An  $i$  th influence coefficient

$GM(1,N)$  A grey differential equation having  $N$  variables

$Pval$  The probability of residual in Posterior Check

$Cval$  Variance ratio in Posterior Check

### 3 General Assumptions

For the sake of convenience of the following discussions, we firstly assume that :

- The future trend of Ozone level would be stable based on the historical data, and the data from the station is accurate and not exists error.
- There will be no major natural disasters happen in the future to influence the Ozone level.
- Considering that Ozone level is stable during the year, we regard the arithmetic mean, the maximum and the minimal of the whole year Ozone level as the desired value of this year respectively.
- Through literature review, we consider Ozone levels have latitudinal distributions. We divide from  $90^\circ$  south latitude to  $90^\circ$  north latitude into 19 parts. Each part represents 10 degrees of latitude.
- At each part of every 10 degrees of latitude, there are many Dobson sites of ESRL Global Monitor Division. We select one site of them to represent Ozone level of this part randomly.

### 4 Analyze of the Problem

The requirement of this problem is to predict the trend changes of Ozone over the 50 years. However, we find there are two kind of data resources of Ozone. And one is from ground observation station, another is satellite remote sensing data. Obviously, the data from satellite is more accurate. Given the availability and access of data, we decide to use the data from ground observation station Dobson. (The Dobson Ozone Spectrophotometer has been used to study total ozone since its development in the 1920's. The observations of total ozone, the total amount of ozone in a column from the surface to the edge of the atmosphere, by this instrument is one of the longest geophysical measurements series (at Arosa, Switzerland) in existence.)) And the location of Dobson is shown in figure 1.



Figure 1: location of Dobson

Through checking the data, we find there are dozens of stations which could be chosen, and different stations will be influenced by different factors, such as latitude, longitude and the geographical position. We need to choose the most important factor as our main consideration of station selection.

Besides, we also need to consider the difference of time because representatives from 43 nations signed the Montreal Protocol for the protection of the Ozone Layer in 1987. Hence, there must be a big effect on the total Ozone. And we may compare the time before 1987 and the time after 1987.

Then, in order to calculate easily and efficiently, we may consider getting a yearly symbol data of every station with the statistical techniques.

After that, we may develop a proper model to predict the trends of Ozone over the 50 years and modify it according to the results of statistical method.

## 5 Mathematical model

### 5.1 Grey System theory

Grey System theory was initiated in 1982 by professor Deng. As far as information is concerned, the system which lacks information, such as structure message, operation mechanism and behavior document, are referred to as Grey Systems. For example, the human body, agriculture, economy, etc. are Grey Systems.

Usually, on the grounds of existing grey relations, grey elements, grey numbers one can identify which Grey System is, where “grey” means poor, incorporate, uncertain, etc. The goal of Grey System and its applications is to bridge the gap existing between social science and natural science. Thus, one can say that the Grey System theory is inter-disciplinary, cutting across a variety of specialized fields, and it is evident that Grey System theory stands the test of time since 1982.

The aims of Grey System theory are to provide theory, techniques, notions and ideas for resolving or analyzing latent and intricate systems, for example :

- To establish a non—function model instead of regressive analyzing.
- To define and constitute Grey Process replacing the stochastic process and to find the real time techniques instead of statistical model to deal with the grey process, in order to obtain an approach to modelling with few data, avoiding searching for data in quantities.
- To turn the disorderly raw data into a more regular series by grey generating techniques for the benefit of modelling instead of modelling with original data.
- To build a differential model—so called grey model(GM)—by using the least 4 data to replace difference modelling in vast quantities of data
- To develop a novel family of grey forecasting methods instead of time series and regressive methods—all of these may be referred to as an approach to deal with grey process.
- To develop innovational techniques and concepts for decision making—so called grey decision making..
- To develop novel control techniques. e.g., the grey forecasting control replacing classical control which is referred to as afterward control, also relational control, generating control and programming control
- To study the mechanism theory, including grey sequence theory and grey structure theory;
- To study feeling and emotion functions and fields with whitening functions;
- Based on grey relations, grey elements and grey numbers to be used to study grey mathematics instead of classical mathematics study.

The essential contents and topics of Grey System theory encompass the following areas: grey relational space, grey generating space, grey forecasting, grey decision making, grey control, grey mathematics, and grey theory.

## 5.2 GM Model

On the subject of differential modeling, T.C Hxia follows that, for biological phenomena, economy and biological medicine, one had better build a differential model, but this has been impossible in the past. In Grey System theory a dynamic model with a group of differential equations is developed, which is called grey differential model. To do this Deng inferred :

- A stochastic process whose amplitudes vary with time is referred to as a grey process.
- The grey modelling is based on the generating series rather than on the raw one.

- The grey derivative, parallel shooting and grey differential equation are defined and proposed in order to build a GM.
- To build a GM model, only a few data(as few as 4) are needed to distinguish it. non-function model instead of regressive analyzing

$$x^{(1)} = AGOx^{(0)}x^{(0)}(k) + az^{(1)}(k) = b, k = 1, 2, \dots, n, ..$$

Is a grey differential model, called GM(1,1) as it includes only one variable, where

$$z^{(1)}(k) + 0.5x^{(1)}(k) + 0.5x^{(1)}(k-1), k = 1, 2, \dots, n, ..$$

a, b, are the coefficients; in Grey System theory terms, a is said to be a developing coefficient and b the grey input. According to the least square method, we have

$$\vec{a} = \begin{pmatrix} a \\ b \end{pmatrix} = (B^T B)^{-1} B^T y_N, B = \begin{bmatrix} -z^{(1)}(2) & 1 \\ \cdots & \cdots \\ -z^{(n)} & 1 \end{bmatrix}, y_N = \begin{bmatrix} x^{(0)}(2) \\ \cdots \\ x^{(0)}(n) \end{bmatrix}$$

Here B is called a data matrix. By regarding the following equation

$$\frac{dx^{(1)}}{dt} + a \otimes (x^{(0)}) = bx^{(0)}(k) + az^{(1)}(k) = b$$

Means the parallel morphism or shooting from  $x(1)$  to the components of  $dx(1)/dt$ , then the response equation for GM(1,1) are as follows

$$x^{(1)}(k+1) = (x^{(0)}(1) - \frac{b}{a})e^{(-ab)} + \frac{b}{a}x^{(0)}(k+1) = x^{(1)}(k+1) - x^{(1)}(k)$$

A grey differential equation having N variables is called GM(1,N), whose expression can be written as follows:

$$x_1^{(0)}(k) + az_1^{(1)}(k) = \sum_{i=2}^N b_i x_i^{(1)}(K), \quad k = 1, 2, \dots$$

Then we have

$$\vec{a} = [a \quad b_2 \quad \cdots \quad b_n]$$

Model GM(1,1) plays an important role in grey forecasting, grey programming and grey control. Model GM(1,N) has laid an important foundation for regional economic programming and grey multivariable control.

### 5.3 The Foundation of Model

#### 5.3.1 The model for every station

Based on the Grey Forecasting Model, we select observation data from nine ground stations. For starting date of some data is different and they are accurate to the day, we take the annual average, maximum value and minima value as the symbol of this year respectively. 30S is shown as follow.



| Year                | 1984     | 1985     | 1986     | 1987     | 1988     | 1989     | 1990     | 1991     | 1992     | 1993     |
|---------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| The Arithmetic Mean | 313.4258 | 279.8362 | 285.4177 | 287.0063 | 293.9057 | 297.4321 | 287.4406 | 291.6145 | 292.6204 | 283.0737 |
| The MAX value       | 389      | 335      | 351      | 365      | 374      | 367      | 373      | 356      | 351      | 375      |
| The MIN value       | 271      | 242      | 242      | 247      | 250      | 247      | 245      | 245      | 245      | 232      |

Figure 2: 30s

| 1994     | 1995     | 1996     | 1997     | 1998    | 1999     | 2000     | 2001     | 2002     | 2003     | 2004     |
|----------|----------|----------|----------|---------|----------|----------|----------|----------|----------|----------|
| 284.8895 | 285.7515 | 288.4643 | 273.0677 | 286.234 | 280.5424 | 286.4481 | 285.8641 | 283.8854 | 287.7029 | 273.4763 |
| 354      | 354      | 346      | 337      | 363     | 350      | 387      | 356      | 359      | 378      | 351      |
| 242      | 245      | 245      | 226      | 238     | 234      | 221      | 239      | 241      | 233      | 227      |

Figure 3: 30s

| 1994     | 1995     | 1996     | 1997     | 1998    | 1999     | 2000     | 2001     | 2002     | 2003     | 2004     |
|----------|----------|----------|----------|---------|----------|----------|----------|----------|----------|----------|
| 284.8895 | 285.7515 | 288.4643 | 273.0677 | 286.234 | 280.5424 | 286.4481 | 285.8641 | 283.8854 | 287.7029 | 273.4763 |
| 354      | 354      | 346      | 337      | 363     | 350      | 387      | 356      | 359      | 378      | 351      |
| 242      | 245      | 245      | 226      | 238     | 234      | 221      | 239      | 241      | 233      | 227      |

Figure 4: 30s

Because the mean value of the whole year can reflect the overall situation better, we use the Grey Forecasting Model to predict the trend of nine stations we have chosen. Given the effect of the Montreal Protocol, we set the 1987 as the node of Ozone trend, and we predict the Ozone change trend of nine station based on the data before 1987 and after 1987 respectively. And the result is as follow :

### 5.3.2 The Model for the Global

To get a global Ozone trend better, we calculate the mean value of nine station for every year and use the same way to predict the trend of Ozone over 50 years based on the mean value we get before 1987 and after 1987.

## 5.4 Result and Analysis

### 5.4.1 The Result for every station

The predictive results of every station over the 50 years is show as follow, and the detailed data is on the appendix.

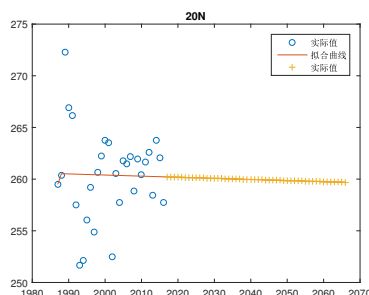


Figure 5: 20N after 1987

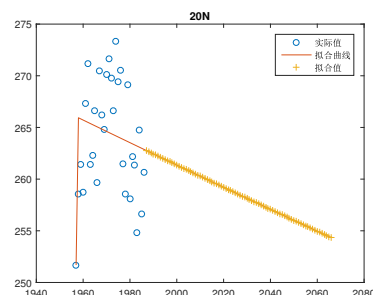


Figure 6: 20N before 20N

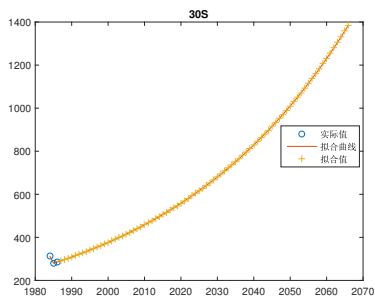


Figure 7: 30S after 1987

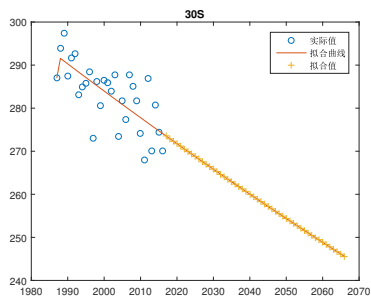


Figure 8: 30S before 30S

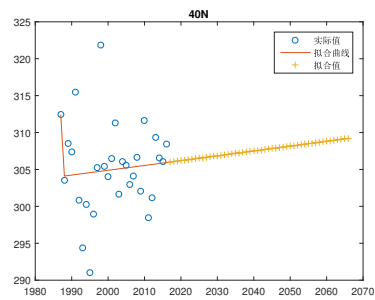


Figure 9: 40N after 1987

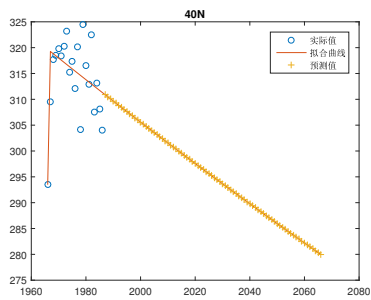


Figure 10: 40N before 1987

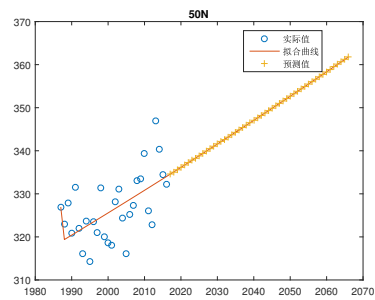


Figure 11: 50N after 1987

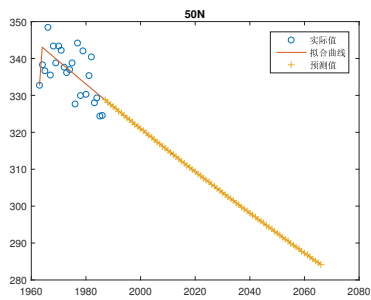


Figure 12: 50N before 1987

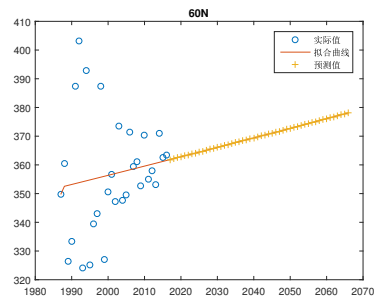


Figure 13: 60N after 1987

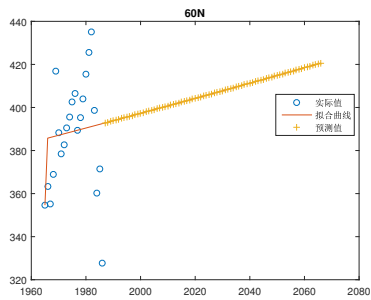


Figure 14: 60N before 1987

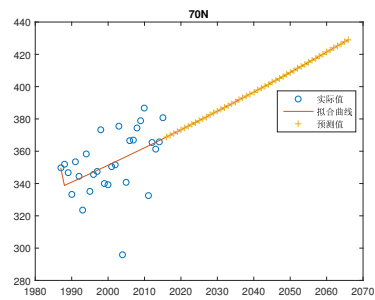


Figure 15: 70N after 1987

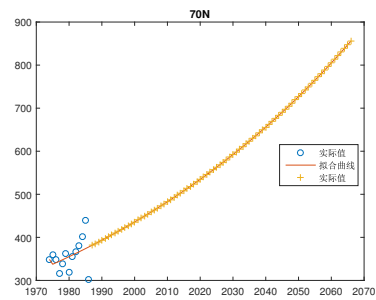


Figure 16: 70N before 1987

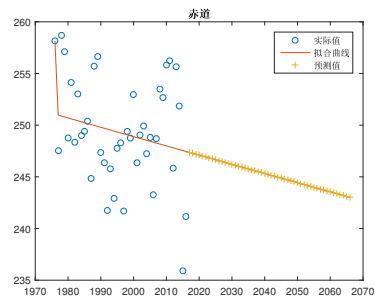


Figure 17: the equator after 1987

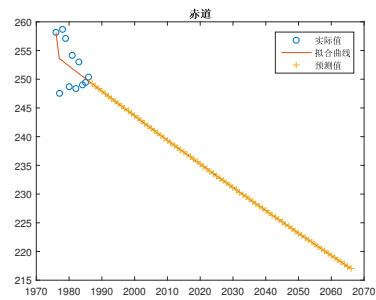


Figure 18: the equator before 1987

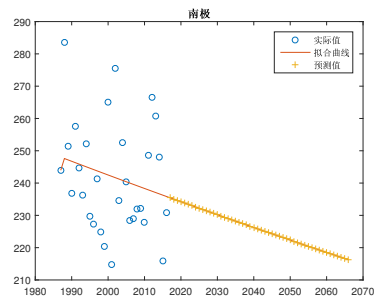


Figure 19: the Antarctic after 1987

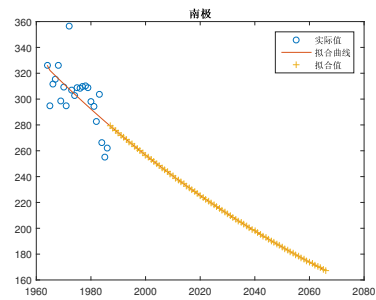


Figure 20: the Antarctic before 1987

### 5.4.2 The result for the global

The predictive results of our GM(1,1) model for the global over the 50 years is shown as follow :

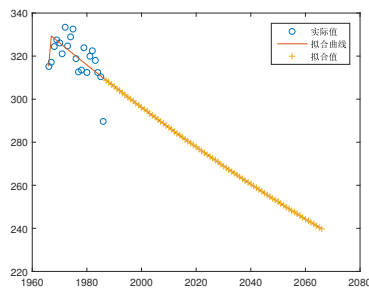


Figure 21: the global after 1987

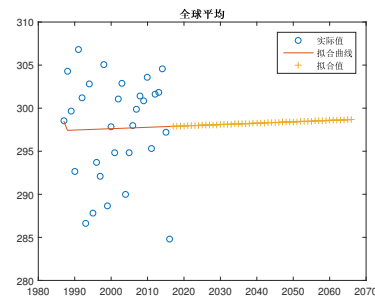


Figure 22: the global before 1987

### 5.4.3 Analysis

Based on the simulated pictures of the GM model, we conclude analysis results as follow:

- The trend of ozone levels of every sites of ESRL can be divided into two parts: before 1987, the trend of ozone levels is on the decline remarkably, while after 1987, the trend of ozone levels has a increasing tendency slowly except for very few stations such as the Antarctica and the equator.
- According to the simulated pictures of global of GM model, we find the same discipline, and that is to say the total Ozone is rising gradually from 1987. And from the picture, we could determine the Ozone content is between 295DU and 300DU in 2066.
- The change trend of global ozone levels possesses regional difference. From the simulated pictures based on the data after 1987, we can figure out that the ozone levels of Northern Hemisphere have a remarkably increasing tendency. So we can know that the Montreal Protocol has a positive effect on the Northern Hemisphere.
- By comparing the change trend of ozone levels based on the data before 1987 and after 1987 respectively, we can conclude that the Montreal Protocol has a huge impact on the change trend of Ozone, and the Ozone level is recovering over the 50 years.

## 6 The Verification of Our Models

The verification of GM can be divided into three parts : Residual Tests, Correlation Degree Test, Posterior Check. Residual Tests is based on points, this method indicates the correlation between prediction values and actual values. Correlation Degree Test examines the similarity between established model and actual function. Posterior Check is the test of statistical properties of residual analysis. The most frequently used methods are Residual Tests and Posterior Check.

## 6.1 Posterior Check

Posterior Check is the test of statistical properties of residual analysis. We get the rank index of forecast accuracy based on the previous statement, the rank table and test result are as follow :

| PVal <sub>o</sub>  | CVal <sub>o</sub>  | forecast accuracy <sub>o</sub> |
|--------------------|--------------------|--------------------------------|
| >0.95 <sub>o</sub> | <0.35 <sub>o</sub> | Excellent <sub>o</sub>         |
| >0.70 <sub>o</sub> | <0.65 <sub>o</sub> | Qualified <sub>o</sub>         |
| <0.70 <sub>o</sub> | >0.65 <sub>o</sub> | Unqualified <sub>o</sub>       |

Figure 23: the rank table

|                   |            |           |           |
|-------------------|------------|-----------|-----------|
| Dobson site       | South Pole | S40       | S30       |
| Pval              | 0.7358     | 0.9995    | 0.7879    |
| Cval              | 0.641      | 0.6333    | 0.5959    |
| forecast accuracy | Qualified  | Excellent | Qualified |
| Dobson site       | Equator    | N20       | N40       |
| Pval              | 0.9873946  | 0.9064    | 0.9179943 |
| Cval              | 0.4878     | 0.5333    | 0.62745   |
| forecast accuracy | Excellent  | Qualified | Qualified |
| Dobson site       | N50        | N60       | N70       |
| Pval              | 1.242326   | 0.9237    | 0.9954    |
| Cval              | 0.5        | 0.5385    | 0.5952    |
| forecast accuracy | Excellent  | Qualified | Excellent |

Figure 24: the result table

According to the results, our model is qualified.

## 6.2 Residual Tests

Residual Tests is the test of residual between model value and actual value point by point. Through accessing to relevant information, we know that if the model's relative error under 5%, this model is qualified.

The result is as follow:

|              |             |             |             |
|--------------|-------------|-------------|-------------|
| Dobson site  | 50N         | 60N         | 70N         |
| Max relative | 0.018994    | 0.054664    | 0.050183    |
| Rank         | Qualified   | Unqualified | Unqualified |
| Dobson site  | 40N         | 20N         | Equator     |
| Max relative | 0.01706     | 0.015703    | 0.015907    |
| Rank         | Qualified   | Qualified   | Qualified   |
| Dobson site  | South Pole  | 30S         | 40S         |
| Max relative | 0.059426    | 0.018593    | 0.018322    |
| Rank         | Unqualified | Qualified   | Qualified   |

Figure 25: the result of test

We can figure out that Dobson site: 50N, 40N, 20N, Equator, 30S, 40S are qualified. And relative errors of others (60N, 70N, Antarctic) are very close to 5%. All in all, our model is qualified.

## 7 Strengths and Weaknesses of the Model

### 7.1 Strengths

**Simplicity:** The mathematical model we construct is simple enough to be understood with a small amount of mathematical skill. And, the calculation of it can be solved by simple software such as C language program and simulated by MATLAB.

**Operability:** Our model only needs to collect the data from different resources and put it with a single calculation. Besides, combine the matlab and GM(1,1), we could easily simplify the algorithm.

**Accuracy:** In order to get the accurate results, we divide stations into two kinds. One is the polar region and another is the other districts so that we could analyze the difference and deviation. **Comparability:** Through analyzing the different trend before 1987 and after 1987, we could easily get the effect.

**Developed from data in real life:** The observed data of each station that we used are from Dobson site's Ozone level's database of ESRL Global Monitoring Division.

### 7.2 Weakness

**Assumptions:** we just choose the data from some of ground stations and the selection of station is random and not representative. Furthermore, we select 10 degrees of latitude as a symbol of all stations around, which may lead to a deviation. And ESRL Global

Monitoring Division do not have Dobson sites in the whole latitude from 70 degrees south to 70 degrees north.

**Input:** through consulting the literature, we find the data from the satellite is more accurate than that of ground station, we still choose the ground data. And because of the data missing for some year in some station, we supply the data by ourselves based on the trend of the past years, which may have a serious effect to our predict. In addition, we make the mean value of a year of Ozone as yearly data, and this statistical approach is not typical.

**Output:** due to the mean value of input data for every year, we could only get the value over 50 years. In addition, the result is only the yearly data, therefore, the trend of line chart is not very precise.

**Model:** Grey Forecast Model could only predict the trends in a short or middle term, but our purpose is to foresee the results of Ozone over 50 years. Besides, this model weakens the random changes and strengthens the trend of the forecasted object, which lead to the ignorance of effect of the Montreal Protocol.

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# Appendices

## Appendix A First appendix

Here are programmes of Grey Forecasting Model we used in our model as follow.

### Input matlab source:

---

```
function gml(x)
sizexd2=size(x,2);
k=0;
for y1=x
    k=k+1;
    if k>1
        x1(k)=x1(k-1)+x(k);
        z1(k-1)=-0.5*(x1(k)+x1(k-1));
        yn1(k-1)=x(k);
    else
        x1(k)=x(k);
    end
end
size1=size(z1,2);
z2=z1';
z3=ones(1,size1)';
YN=yn1';
B=[z2,z3];
au0=inv(B'*B)*B'*YN;
au=au0';
afor=au(1);
ufor=au(2);
ua=au(2)./au(1);
constant1=x(1)-ua;
afor1=-afor;
x1t1='x1(t+1)';
estr='exp';
tstr='t';
leftbra='(';
rightbra=')';
strcat(x1t1,'=',num2str(constant1),estr,leftbra,num2str(afor1),tstr,rightbra,'+',leftbra,num2str(1),rightbra);

k2=0;
for y2=x1
    k2=k2+1;
    if k2>k
        else
            zel(k2)=exp(-(k2-1)*afor);
        end
    end
end

sizezel=size(zel,2);
```



```

z4=ones(1,sizezel)';
G=[zel',z4];
X1=x1';
au20=inv(G'*G)*G'*X1;
au2=au20';
Aval=au2(1);
Bval=au2(2);
strcat(xlt1,'=',num2str(Aval),estr,leftbra,num2str(afor1),tstr,rightbra,'+',leftbra,num2str(Bval),tstr,rightbra);

nfinal = sized2-1 + 1;
for k3=1:nfinal
    x3fcast(k3)=constant1*exp(afor1*k3)+ua;
end

for k31=nfinal:-1:0
    if k31>1
        x31fcast(k31+1)=x3fcast(k31)-x3fcast(k31-1);
    else
        if k31>0
            x31fcast(k31+1)=x3fcast(k31)-x(1);
        else
            x31fcast(k31+1)=x(1);
        end
    end
end
x31fcast

for k4=1:nfinal
    x4fcast(k4) = Aval*exp(afor1*k4)+Bval;
end

%x4fcast

for k41=nfinal:-1:0
    if k41>1
        x41fcast(k41+1) = x4fcast(k41)-x4fcast(k41-1);
    else
        if k41>0
            x41fcast(k41+1) = x4fcast(k41)-x(1);
        else
            x41fcast(k41+1) = x(1);
        end
    end
end

c=x41fcast;
x41fcast,x

k5=0;
for y5=x
    k5=k5+1;
    if k5>sized2
    else
        errl(k5)=x(k5)-x41fcast(k5);
    end
end

xavg=mean(x);

errlavg=mean(errl);

```

---

```

k5=0;
slttotal=0;
for y5=x;
    k5=k5+1;
    if k5>sizexd2
    else
        slttotal=slttotal+(x(k5)-xavg)^2;
    end
end
slsuqare=slttotal./sizexd2;
slsqr=sqrt(slsuqare);

k5=0;
s2total=0;
for y5=x
    k5=k5+1;
    if k5>sizexd2
    else
        s2total=s2total+(errl(k5)-errlavg)^2;
    end
end

s2suqare = s2total ./ sizexd2;

Cval=sqrt(s2suqare./slsuqare);
Cval

k5=0;
pnum=0;
for y5=x
    k5=k5+1;
    if abs(errl(k5)-errlavg)<0.6745*slsqr
        pnum=pnum+1;
    else
    end
end
pval=pnum./sizexd2;
pval

```

---

## Appendix B Second appendix

Here are programmes of testing availability of Grey Forecasting Model we used in our model as follow.

### Input matlab source:

---

```

clc;clear
syms a b;
c=[a,b]';
A=[348.8095238 359.0989583 348.0584416 316.5058824 338.2857143 361.8043478 318.8275862 355.019
];
B=cumsum(A);
n=length(A);
for i=1:(n-1)
    C(i)=(B(i)+B(i+1))./2;
end

```

```
D=A;
D(1)=[];
D=D';
E=[-C;ones(1,n-1)];
c=inv(E*E')*E*D;
c=c';
a=c(1);
b=c(2);
F=[];
F(1)=A(1);
for i=2:(n+50)
F(i)=(A(1)-b/a)/exp(a*(i-1))+b/a;
end
G=[];
G(1)=A(1);
for i=2:(n+50)
G(i)=F(i)-F(i-1);
end
t1=1974:2015;
t2=1974:2065;
t3=2016:2065;
G
plot(t1,A,'o',t2,G)
% x0=A;
x1=G(1,43:92);
hold on
plot(t3,x1,'+')
% title('Äïjñ');
% legend('ÊïjËÖť','Ôd'šäÖť');
```

---

## Appendix C Third appendix

Here are pictures of data we have caculated in our model as follow.

| Year | Total_Ozone | Year | Total_Ozone | Year | Total_Ozone | Year | Total_Ozone |
|------|-------------|------|-------------|------|-------------|------|-------------|
| 1976 | 258.1513    | 2004 | 248.532567  | 2032 | 246.0290346 | 2060 | 243.5507209 |
| 1977 | 250.9708114 | 2005 | 248.442718  | 2033 | 245.9400907 | 2061 | 243.462673  |
| 1978 | 250.880081  | 2006 | 248.3529015 | 2034 | 245.8511789 | 2062 | 243.3746568 |
| 1979 | 250.7893833 | 2007 | 248.2631174 | 2035 | 245.7622992 | 2063 | 243.2866725 |
| 1980 | 250.6987184 | 2008 | 248.1733658 | 2036 | 245.6734517 | 2064 | 243.19872   |
| 1981 | 250.6080863 | 2009 | 248.0836467 | 2037 | 245.5846364 | 2065 | 243.1107992 |
| 1982 | 250.5174869 | 2010 | 247.9939599 | 2038 | 245.4958531 | 2066 | 243.0229103 |
| 1983 | 250.4269203 | 2011 | 247.9043056 | 2039 | 245.4071019 |      |             |
| 1984 | 250.3363865 | 2012 | 247.8146838 | 2040 | 245.3183828 |      |             |
| 1985 | 250.2458854 | 2013 | 247.7250943 | 2041 | 245.2296958 |      |             |
| 1986 | 250.155417  | 2014 | 247.6355372 | 2042 | 245.1410408 |      |             |
| 1987 | 250.0649813 | 2015 | 247.5460125 | 2043 | 245.0524179 |      |             |
| 1988 | 249.9745783 | 2016 | 247.4565201 | 2044 | 244.963827  |      |             |
| 1989 | 249.8842079 | 2017 | 247.3670601 | 2045 | 244.8752682 |      |             |
| 1990 | 249.7938703 | 2018 | 247.2776325 | 2046 | 244.7867414 |      |             |
| 1991 | 249.7035653 | 2019 | 247.1882371 | 2047 | 244.6982465 |      |             |
| 1992 | 249.613293  | 2020 | 247.0988741 | 2048 | 244.6097837 |      |             |
| 1993 | 249.5230532 | 2021 | 247.0095434 | 2049 | 244.5213529 |      |             |
| 1994 | 249.4328462 | 2022 | 246.920245  | 2050 | 244.432954  |      |             |
| 1995 | 249.3426717 | 2023 | 246.8309789 | 2051 | 244.3445871 |      |             |
| 1996 | 249.2525298 | 2024 | 246.7417451 | 2052 | 244.2562521 |      |             |
| 1997 | 249.1624205 | 2025 | 246.6525435 | 2053 | 244.167949  |      |             |
| 1998 | 249.0723438 | 2026 | 246.5633741 | 2054 | 244.0796779 |      |             |
| 1999 | 248.9822997 | 2027 | 246.474237  | 2055 | 243.9914387 |      |             |
| 2000 | 248.8922881 | 2028 | 246.3851321 | 2056 | 243.9032314 |      |             |
| 2001 | 248.802309  | 2029 | 246.2960594 | 2057 | 243.815056  |      |             |
| 2002 | 248.7123625 | 2030 | 246.207019  | 2058 | 243.7269124 |      |             |
| 2003 | 248.6224485 | 2031 | 246.1180107 | 2059 | 243.6388008 |      |             |

Figure 26: predicted data of GM Forecast Model(The Equator)

| Year | Total_Ozone | Year | Total_Ozone | Year | Total_Ozone | Year | Total_Ozone |
|------|-------------|------|-------------|------|-------------|------|-------------|
| 1964 | 326.2083333 | 1993 | 260.4597779 | 2022 | 210.7799739 | 2051 | 170.5760397 |
| 1965 | 319.5086568 | 1994 | 258.5659363 | 2023 | 209.2473616 | 2052 | 169.3357561 |
| 1966 | 317.1854621 | 1995 | 256.6858652 | 2024 | 207.7258931 | 2053 | 168.1044908 |
| 1967 | 314.8791597 | 1996 | 254.8194643 | 2025 | 206.2154874 | 2054 | 166.8821781 |
| 1968 | 312.5896268 | 1997 | 252.9666343 | 2026 | 204.7160641 | 2055 | 165.6687531 |
| 1969 | 310.3167414 | 1998 | 251.1272765 | 2027 | 203.2275433 | 2056 | 164.4641511 |
| 1970 | 308.0603824 | 1999 | 249.3012929 | 2028 | 201.7498458 | 2057 | 163.268308  |
| 1971 | 305.8204298 | 2000 | 247.4885863 | 2029 | 200.2828928 | 2058 | 162.08116   |
| 1972 | 303.5967643 | 2001 | 245.6890602 | 2030 | 198.8266063 | 2059 | 160.9026439 |
| 1973 | 301.3892673 | 2002 | 243.9026187 | 2031 | 197.3809086 | 2060 | 159.732697  |
| 1974 | 299.1978214 | 2003 | 242.1291667 | 2032 | 195.9457229 | 2061 | 158.5712569 |
| 1975 | 297.0223098 | 2004 | 240.3686098 | 2033 | 194.5209725 | 2062 | 157.4182619 |
| 1976 | 294.8626166 | 2005 | 238.6208541 | 2034 | 193.1065818 | 2063 | 156.2736504 |
| 1977 | 292.718627  | 2006 | 236.8858065 | 2035 | 191.7024753 | 2064 | 155.1373616 |
| 1978 | 290.5902265 | 2007 | 235.1633748 | 2036 | 190.3085782 | 2065 | 154.0093349 |
| 1979 | 288.477302  | 2008 | 233.4534671 | 2037 | 188.9248164 | 2066 | 152.8895102 |
| 1980 | 286.3797409 | 2009 | 231.7559925 | 2038 | 187.5511161 |      |             |
| 1981 | 284.2974315 | 2010 | 230.0708604 | 2039 | 186.1874042 |      |             |
| 1982 | 282.2302628 | 2011 | 228.3979811 | 2040 | 184.8336081 |      |             |
| 1983 | 280.1781248 | 2012 | 226.7372656 | 2041 | 183.4896556 |      |             |
| 1984 | 278.1409082 | 2013 | 225.0886254 | 2042 | 182.1554752 |      |             |
| 1985 | 276.1185046 | 2014 | 223.4519728 | 2043 | 180.8309958 |      |             |
| 1986 | 274.1108061 | 2015 | 221.8272204 | 2044 | 179.5161468 |      |             |
| 1987 | 272.1177059 | 2016 | 220.2142819 | 2045 | 178.2108584 |      |             |
| 1988 | 270.1390978 | 2017 | 218.6130713 | 2046 | 176.9150609 |      |             |
| 1989 | 268.1748765 | 2018 | 217.0235033 | 2047 | 175.6286853 |      |             |
| 1990 | 266.2249373 | 2019 | 215.4454933 | 2048 | 174.3516632 |      |             |
| 1991 | 264.2891764 | 2020 | 213.8789573 | 2049 | 173.0839265 |      |             |
| 1992 | 262.3674907 | 2021 | 212.3238118 | 2050 | 171.8254076 |      |             |

Figure 27: predicted data of GM Forecast Model(Antarctic)

| Year | Total_Ozone |  | Year | Total_Ozone |  | Year | Total_Ozone |
|------|-------------|--|------|-------------|--|------|-------------|
| 1984 | 313.4258065 |  | 2013 | 276.9336691 |  | 2042 | 263.6609844 |
| 1985 | 290.3822973 |  | 2014 | 276.465056  |  | 2043 | 263.2148307 |
| 1986 | 289.8909272 |  | 2015 | 275.997236  |  | 2044 | 262.7694319 |
| 1987 | 289.4003885 |  | 2016 | 275.5302075 |  | 2045 | 262.3247869 |
| 1988 | 288.9106799 |  | 2017 | 275.0639693 |  | 2046 | 261.8808943 |
| 1989 | 288.4218    |  | 2018 | 274.5985201 |  | 2047 | 261.4377528 |
| 1990 | 287.9337473 |  | 2019 | 274.1338585 |  | 2048 | 260.9953611 |
| 1991 | 287.4465205 |  | 2020 | 273.6699831 |  | 2049 | 260.5537181 |
| 1992 | 286.9601182 |  | 2021 | 273.2068927 |  | 2050 | 260.1128223 |
| 1993 | 286.4745389 |  | 2022 | 272.7445859 |  | 2051 | 259.6726727 |
| 1994 | 285.9897813 |  | 2023 | 272.2830615 |  | 2052 | 259.2332678 |
| 1995 | 285.5058439 |  | 2024 | 271.8223179 |  | 2053 | 258.7946065 |
| 1996 | 285.0227255 |  | 2025 | 271.3623541 |  | 2054 | 258.3566874 |
| 1997 | 284.5404245 |  | 2026 | 270.9031685 |  | 2055 | 257.9195094 |
| 1998 | 284.0589397 |  | 2027 | 270.44476   |  | 2056 | 257.4830712 |
| 1999 | 283.5782697 |  | 2028 | 269.9871271 |  | 2057 | 257.0473714 |
| 2000 | 283.098413  |  | 2029 | 269.5302687 |  | 2058 | 256.612409  |
| 2001 | 282.6193683 |  | 2030 | 269.0741833 |  | 2059 | 256.1781825 |
| 2002 | 282.1411342 |  | 2031 | 268.6188697 |  | 2060 | 255.7446909 |
| 2003 | 281.6637093 |  | 2032 | 268.1643265 |  | 2061 | 255.3119327 |
| 2004 | 281.1870923 |  | 2033 | 267.7105525 |  | 2062 | 254.8799069 |
| 2005 | 280.7112819 |  | 2034 | 267.2575463 |  | 2063 | 254.4486121 |
| 2006 | 280.2362766 |  | 2035 | 266.8053067 |  | 2064 | 254.0180471 |
| 2007 | 279.762075  |  | 2036 | 266.3538324 |  | 2065 | 253.5882107 |
| 2008 | 279.2886759 |  | 2037 | 265.903122  |  | 2066 | 253.1591017 |
| 2009 | 278.8160778 |  | 2038 | 265.4531743 |  |      |             |
| 2010 | 278.3442795 |  | 2039 | 265.003988  |  |      |             |
| 2011 | 277.8732794 |  | 2040 | 264.5555617 |  |      |             |
| 2012 | 277.4030764 |  | 2041 | 264.1078943 |  |      |             |

Figure 28: predicted data of GM Forecast Model(30 Degrees South)

| Year                | 1964    | 1965    | 1966    | 1967    | 1968    | 1969    | 1970    | 1971    | 1972    | 1973    | 1974    | 1975    |
|---------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| The Arithmetic Mean | 326.208 | 294.922 | 311.582 | 315.603 | 326.355 | 298.711 | 309.371 | 294.605 | 356.409 | 306.789 | 302.981 | 309.049 |
| The MAX value       | 439     | 360     | 392     | 424     | 467     | 372     | 387     | 361     | 382     | 506     | 430     | 386     |
| The MIN value       | 240     | 227     | 242     | 237     | 225     | 196     | 238     | 249     | 313     | 236     | 181     | 223     |
| Year                | 1976    | 1977    | 1978    | 1979    | 1980    | 1981    | 1982    | 1983    | 1984    | 1985    | 1986    | 1987    |
| The Arithmetic Mean | 308.277 | 309.953 | 310.418 | 308.827 | 298.028 | 294.524 | 282.617 | 303.814 | 266.503 | 254.953 | 262.036 | 243.966 |
| The MAX value       | 396     | 433     | 406     | 409     | 393     | 385     | 362     | 429     | 340     | 340     | 396     | 332     |
| The MIN value       | 241     | 217     | 223     | 246     | 200     | 207     | 206     | 250     | 194     | 148     | 114     | 113     |
| Year                | 1988    | 1989    | 1990    | 1991    | 1992    | 1993    | 1994    | 1995    | 1996    | 1997    | 1998    | 1999    |
| The Arithmetic Mean | 283.517 | 251.325 | 236.9   | 257.519 | 244.684 | 236.188 | 252.106 | 229.722 | 227.285 | 241.261 | 224.786 | 220.386 |
| The MAX value       | 389     | 315     | 375     | 387     | 302     | 338     | 363     | 315     | 334     | 321     | 300     | 297     |
| The MIN value       | 195     | 139     | 137     | 129     | 138     | 99      | 109     | 85      | 112     | 126     | 105     | 102     |
| Year                | 2000    | 2001    | 2002    | 2003    | 2004    | 2005    | 2006    | 2007    | 2008    | 2009    | 2010    | 2011    |
| The Arithmetic Mean | 265.097 | 214.756 | 275.531 | 234.586 | 252.612 | 240.444 | 228.471 | 229.023 | 231.891 | 232.192 | 227.779 | 248.629 |
| The MAX value       | 336     | 280     | 383     | 315     | 341     | 346     | 320     | 299     | 309     | 318     | 331     | 282     |
| The MIN value       | 128     | 97      | 147     | 101     | 138     | 118     | 91      | 117     | 94      | 97      | 106     | 212     |
| Year                | 2012    | 2013    | 2014    | 2015    | 2016    |         |         |         |         |         |         |         |
| The Arithmetic Mean | 266.617 | 260.772 | 248.036 | 215.901 | 230.857 |         |         |         |         |         |         |         |
| The MAX value       | 349     | 333     | 329     | 302     | 321     |         |         |         |         |         |         |         |
| The MIN value       | 117     | 133     | 116     | 91      | 103     |         |         |         |         |         |         |         |

Figure 29: actual data of GM Model(Antarctic)

|                         |         |         |         |         |         |         |         |         |         |         |         |         |
|-------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Year                    | 1964    | 1965    | 1966    | 1967    | 1968    | 1969    | 1970    | 1971    | 1972    | 1973    | 1974    | 1975    |
| Primary fitting of Mean | 319.509 | 317.185 | 314.879 | 312.59  | 310.317 | 308.06  | 305.82  | 303.597 | 301.389 | 299.198 | 297.022 | 294.863 |
| Primary fitting of Max  | 419.483 | 416.809 | 414.153 | 411.514 | 408.891 | 406.286 | 403.697 | 401.124 | 398.568 | 396.028 | 393.504 | 390.996 |
| Primary fitting of Min  | 259.048 | 253.826 | 248.709 | 243.695 | 238.782 | 233.968 | 229.251 | 224.629 | 220.101 | 215.663 | 211.316 | 207.055 |
| Year                    | 1976    | 1977    | 1978    | 1979    | 1980    | 1981    | 1982    | 1983    | 1984    | 1985    | 1986    | 1987    |
| Primary fitting of Mean | 292.719 | 290.59  | 288.477 | 286.38  | 284.297 | 282.23  | 280.178 | 278.141 | 276.119 | 274.111 | 272.118 | 270.139 |
| Primary fitting of Max  | 388.505 | 386.029 | 383.569 | 381.124 | 378.696 | 376.282 | 373.884 | 371.502 | 369.134 | 366.782 | 364.444 | 362.122 |
| Primary fitting of Min  | 202.881 | 198.791 | 194.783 | 190.857 | 187.009 | 183.239 | 179.545 | 175.925 | 172.378 | 168.903 | 165.498 | 162.162 |
| Year                    | 1988    | 1989    | 1990    | 1991    | 1992    | 1993    | 1994    | 1995    | 1996    | 1997    | 1998    | 1999    |
| Primary fitting of Mean | 268.175 | 266.225 | 264.289 | 262.367 | 260.46  | 258.566 | 256.686 | 254.819 | 252.967 | 251.127 | 249.301 | 247.489 |
| Primary fitting of Max  | 359.814 | 357.521 | 355.243 | 352.979 | 350.73  | 348.495 | 346.274 | 344.067 | 341.874 | 339.696 | 337.531 | 335.38  |
| Primary fitting of Min  | 158.892 | 155.689 | 152.55  | 149.475 | 146.462 | 143.509 | 140.616 | 137.781 | 135.003 | 132.282 | 129.615 | 127.002 |
| Year                    | 2000    | 2001    | 2002    | 2003    | 2004    | 2005    | 2006    | 2007    | 2008    | 2009    | 2010    | 2011    |
| Primary fitting of Mean | 245.689 | 243.903 | 242.129 | 240.369 | 238.621 | 236.886 | 235.163 | 233.453 | 231.756 | 230.071 | 228.398 | 226.737 |
| Primary fitting of Max  | 333.243 | 331.119 | 329.009 | 326.912 | 324.829 | 322.759 | 320.702 | 318.658 | 316.628 | 314.61  | 312.605 | 310.613 |
| Primary fitting of Min  | 124.441 | 121.933 | 119.474 | 117.066 | 114.706 | 112.393 | 110.127 | 107.907 | 105.732 | 103.6   | 101.512 | 99.4651 |
| Year                    | 2012    | 2013    | 2014    | 2015    | 2016    |         |         |         |         |         |         |         |
| Primary fitting of Mean | 225.089 | 223.452 | 221.827 | 220.214 | 218.613 |         |         |         |         |         |         |         |
| Primary fitting of Max  | 308.633 | 306.667 | 304.712 | 302.77  | 300.841 |         |         |         |         |         |         |         |
| Primary fitting of Min  | 97.4599 | 95.4951 | 93.5699 | 91.6835 | 89.8352 |         |         |         |         |         |         |         |

Figure 30: fitting data of GM Model(Antarctic)

|                     |            |            |             |           |             |             |            |            |            |             |           |           |
|---------------------|------------|------------|-------------|-----------|-------------|-------------|------------|------------|------------|-------------|-----------|-----------|
| Year                | 1976       | 1977       | 1978        | 1979      | 1980        | 1981        | 1982       | 1983       | 1984       | 1985        | 1986      | 1987      |
| The Arithmetic Mean | 258.151316 | 247.554717 | 258.6690141 | 257.09901 | 248.7297297 | 254.1320132 | 248.360927 | 253.036789 | 248.989865 | 249.4102564 | 250.37247 | 244.8664  |
| The MAX value       | 301        | 280        | 293         | 292       | 281         | 293         | 278        | 292        | 295        | 287         | 283       | 303       |
| The MIN value       | 238        | 227        | 234         | 236       | 215         | 233         | 230        | 237        | 225        | 219         | 224       | 208       |
| Year                | 1988       | 1989       | 1990        | 1991      | 1992        | 1993        | 1994       | 1995       | 1996       | 1997        | 1998      | 1999      |
| The Arithmetic Mean | 255.714286 | 256.634259 | 247.3306947 | 246.35385 | 241.751073  | 245.7477477 | 242.9163   | 247.743478 | 248.311966 | 241.68125   | 249.37433 | 248.78261 |
| The MAX value       | 278        | 289        | 275         | 283       | 272         | 278         | 266        | 275        | 282        | 270         | 286       | 276       |
| The MIN value       | 231        | 214        | 225         | 231       | 217         | 221         | 230        | 230        | 228        | 219         | 222       | 232       |
| Year                | 2000       | 2001       | 2002        | 2003      | 2004        | 2005        | 2006       | 2007       | 2008       | 2009        | 2010      | 2011      |
| The Arithmetic Mean | 252.946108 | 246.356757 | 249.0223214 | 249.93749 | 247.2222222 | 248.839779  | 243.238906 | 248.674286 | 253.466981 | 252.6576577 | 255.82353 | 256.24138 |
| The MAX value       | 294        | 306        | 273         | 287       | 303         | 272         | 274        | 278        | 287        | 288         | 270       | 291       |
| The MIN value       | 232        | 228        | 222         | 227       | 228         | 230         | 213        | 226        | 230        | 233         | 241       | 233       |
| Year                | 2012       | 2013       | 2014        | 2015      | 2016        |             |            |            |            |             |           |           |
| The Arithmetic Mean | 245.835443 | 255.622378 | 251.8320611 | 235.91667 | 241.1851852 |             |            |            |            |             |           |           |
| The MAX value       | 265        | 340        | 339         | 246       | 252         |             |            |            |            |             |           |           |
| The MIN value       | 231        | 208        | 219         | 226       | 225         |             |            |            |            |             |           |           |

Figure 31: actual data of GM Forecast Model(The Equator)

|                         |          |          |          |          |          |          |          |          |          |          |          |          |
|-------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Year                    | 1976     | 1977     | 1978     | 1979     | 1980     | 1981     | 1982     | 1983     | 1984     | 1985     | 1986     | 1987     |
| Primary fitting of Mean | 250.9708 | 250.8801 | 250.7894 | 250.6987 | 250.6081 | 250.5175 | 250.4269 | 250.3364 | 250.2459 | 250.1554 | 250.0650 | 249.9746 |
| Primary fitting of Max  | 285.4100 | 285.3786 | 285.3473 | 285.3160 | 285.2847 | 285.2534 | 285.2221 | 285.1908 | 285.1595 | 285.1282 | 285.0969 | 285.0656 |
| Primary fitting of Min  | 226.3744 | 226.3552 | 226.3359 | 226.3167 | 226.2975 | 226.2783 | 226.2591 | 226.2399 | 226.2207 | 226.2015 | 226.1823 | 226.1631 |
| Year                    | 1988     | 1989     | 1990     | 1991     | 1992     | 1993     | 1994     | 1995     | 1996     | 1997     | 1998     | 1999     |
| Primary fitting of Mean | 249.8842 | 249.7939 | 249.7036 | 249.6133 | 249.5230 | 249.4328 | 249.3427 | 249.2525 | 249.1624 | 249.0723 | 248.9823 | 248.8923 |
| Primary fitting of Max  | 285.0343 | 285.0030 | 284.9717 | 284.9405 | 284.9092 | 284.8779 | 284.8467 | 284.8154 | 284.7841 | 284.7529 | 284.7216 | 284.6904 |
| Primary fitting of Min  | 226.1439 | 226.1247 | 226.1055 | 226.0863 | 226.0671 | 226.0479 | 226.0287 | 226.0095 | 225.9903 | 225.9711 | 225.9519 | 225.9327 |
| Year                    | 2000     | 2001     | 2002     | 2003     | 2004     | 2005     | 2006     | 2007     | 2008     | 2009     | 2010     | 2011     |
| Primary fitting of Mean | 248.8023 | 248.7124 | 248.6224 | 248.5326 | 248.4427 | 248.3529 | 248.2631 | 248.1734 | 248.0836 | 247.9939 | 247.9043 | 247.8147 |
| Primary fitting of Max  | 284.6591 | 284.6279 | 284.5967 | 284.5654 | 284.5342 | 284.5030 | 284.4717 | 284.4405 | 284.4093 | 284.3781 | 284.3469 | 284.3157 |
| Primary fitting of Min  | 225.9136 | 225.8944 | 225.8752 | 225.8560 | 225.8368 | 225.8177 | 225.7985 | 225.7793 | 225.7602 | 225.7410 | 225.7218 | 225.7027 |
| Year                    | 2012     | 2013     | 2014     | 2015     | 2016     |          |          |          |          |          |          |          |
| Primary fitting of Mean | 247.7251 | 247.6355 | 247.5460 | 247.4565 | 247.3670 |          |          |          |          |          |          |          |
| Primary fitting of Max  | 284.2845 | 284.2533 | 284.2221 | 284.1909 | 284.1597 |          |          |          |          |          |          |          |
| Primary fitting of Min  | 225.6835 | 225.6643 | 225.6452 | 225.6260 | 225.6069 |          |          |          |          |          |          |          |

Figure 32: fitting data of GM Model(The Equator)

# TITLE

## Abstract:

Based on the analytic hierarchy process (AHP), the paper classifies the qualitative and quantitative methods. For the qualitative analysis, the origin of ozone depleting substances and the origin of ozone depletion substances in the air to study the way of transmission. For the quantitative analysis, the method of large data prediction is used to study. Mainly used factor analysis, principal component analysis, Gaussian misty rain model, BP neural network, time series and other methods to predict the global future 50 years of ozone content.

From the aspect of origin, the material of ozone depletion was searched, and the principal component analysis of factor analysis was used to solve the problem. The extraction rates of the substances were respectively. The main factors influencing the ozone loss are.

From the aspect of communication, taking Gaussian misty rain model as the main influencing factor, MATLAB is used to solve the problem. When the gas reaches the Antarctic or the Arctic, the ice cloud theory and the role of atmospheric circulation, the gas will remain in the Antarctic or Arctic.

From the aspect of prediction, the ozone content of 1978 to 2016 is searched (see attached table), the broken line chart is drawn by MATLAB, the date data is analyzed and the time series model is used to forecast the 38 years data, and SPSS22.0 is used to obtain the prediction result. See table), the average of 38 years of data, the use of neural network model, using MATLAB to get the results (see table).

The influence factors、 modes of propagation and forecasting results、 ozone depletion were analyzed. With the reduction of the emission factors, the transmission mode slowed down and the ozone content increased, the consumption of ozone decreased year by year.

**Key words:** Fuzzy analytic hierarchy process, Principal Component Analysis  
Gaussian Smoke Rain Model, BP Neural Network, Time Series

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## 1 The problem repeat

Ozone layer refers to the relatively high concentrations of ozone in the stratosphere, the atmosphere of the part, its main effect is absorbing shortwave ultraviolet light. Atmospheric chemical substances (such as chlorofluorocarbons, relevant halon, nitrous oxide) in the destruction of stratospheric ozone, the thinning ozone layer, appear even phenomenon that ozone depletion in the ozone layer. Ozone in the atmosphere each reduced by 1%. Ultraviolet (u v) radiation to the ground is increased by 2%, skin cancer is increased by 3%, also by diseases such as cataracts, the immune system defects and arrested development. By extension, if all the ozone layer is destroyed, the sun's ultraviolet will kill all terrestrial life, humans have been "extinction", the earth would be no life of unproductive. It is clear that the hole in the ozone layer has a serious threat to the survival of mankind. Ozone depletion including observed since the 1970 s of the steady decline in the earth the amount of ozone in the stratosphere (the ozone layer) and a larger spring stratospheric ozone reduce wait for a phenomenon. Since 1985, a series of conventions signed from all countries should hand in hand governance system specification ozone depletion of the serious problems.

## 2 Problem analysis

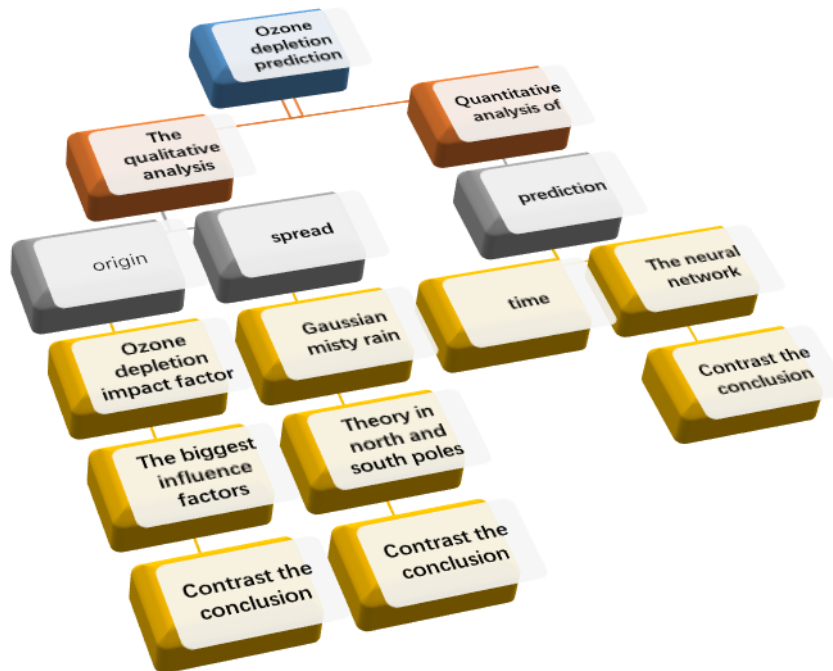


Figure 1 (The problem analysis diagram)

### 3 model assumptions

- 1, the concentration of pollutants in the y, z axis of the distribution is Gaussian distribution (normal distribution);
- 2, the source of the pollution source is continuous and uniform, the initial concentration of the cloud within the time, the temperature was evenly distributed;
- 3, the diffusion process does not take into account the temperature changes within the cloud, ignoring the heat transfer, thermal convection and thermal radiation;
- 4, the leakage of gas is the ideal gas, to comply with the ideal gas equation of state;
- 5, in the horizontal direction, the atmospheric diffusion coefficient was isotropic;
- 6, take the x-axis as the average wind speed direction, the whole process of diffusion of wind speed, the direction of the same does not change with the location, time changes;
- 7, the surface of the gas from the leakage effect of total reflection, absorption or adsorption does not occur;
- 8, the whole process, the leakage of gas does not occur settlement, decomposition, does not occur any chemical reaction.

### 4 symbols that

| symbol    | instructions                                     |
|-----------|--------------------------------------------------|
| $r$       | The correlation coefficient                      |
| $f$       | Of undetermined coefficients                     |
| $x_i$     | Standard deviation or the mean square error      |
| $\bar{x}$ | Average of the independent variables             |
| $x_j$     | A sign of the dependent variable sequence values |

## 5 the establishment of the model and solving

### 5.1 model

#### 5.1.1 Factor Analysis

Factor analysis of the basic purpose is to use a few factors to describe the link between the many indicators or factors, is more closely related several variables in the same class, each kind of variable becomes a factor (referred to as factor, because it is not to be observed, that is not specific variables), with fewer factors reflect most of the original data information. Factor analysis is to find the common factor model analysis method, it is on the basis of the principal components to build several meaning more clear common factor, in the framework of their decomposition of the original variables, to examine the relationship and difference between the original variables.

By adopting the technology of this kind of research, we can easily find out the main factors affecting consumer purchase, consumption and satisfaction is what, as well as their influence (weight) by adopting the technology of this kind of research, we can also do preliminary analysis for market segmentation.

#### 5.1.2 Principal Component Analysis

Principal Component Analysis(PCA) is a kind of statistical method. Through orthogonal transformation converts a set of possible correlation between variables to a set of linear irrelevant variables, the transformed this set of variables called principal components.

In the practical project, in order to analyze the problem, often put forward a lot of related variables (or factors), because each variable in different extent reflect the subject of some information.

Principal component analysis is first and foremost by k. Pearson to the introduction of a random variable, then h. Hotelling the method to random vector. The size of the information is usually measured by the sum of squared residuals or variance.

##### ① Covariance matrix calculation

Calculate the covariance matrix sample data:  $\Sigma = (s_{ij})_{p \times p}$ , Among them,

$$s_{ij} = \frac{1}{n-1} \sum_{k=1}^n \left( x_{ki} - \bar{x}_i \right) \left( x_{kj} - \bar{x}_j \right) \quad i, j = 1, 2, 3 \cdots p \quad (1)$$

##### ② Calculate $\Sigma$ eigenvalues of $\lambda_i$ and The corresponding feature vector

orthogonalization unit  $a_i$ .

Σfirst m larger eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \lambda_m > 0$ , m before is a principal component corresponds to the variance,  $\lambda_i$  characteristic vector  $a_i$  unit is the main component of the original variable coefficient of  $f_i$ , the original variables for the ith a principal component of  $f_i$ :

$$f_i = a_i' x \quad (2)$$

The variance contribution rates of the principal component is used to reflect the size of the amount of information for  $a_i$ :

$$a_i = \frac{\lambda_i}{\sum_{i=1}^m \lambda_i} \quad (3)$$

### ③ Select principal component

Ultimately want to choose a few principal components, that is  $f_1, f_2, \dots, f_m$ , determination of m in the  $f_m$  is determine by the cumulative variance contribute rate.

$$G(m) = \frac{\sum_{i=1}^m \lambda_i}{\sum_{k=1}^p \lambda_k} \quad (4)$$

When the cumulative contribution rate is more than 85%, is thought to reflect the original variable enough information, the corresponding sampling before m, m is a main component

### ④ Calculate principal component loads

Principal component load is principal component  $f_i$  and correlation degree between the original variables  $x_j$ , The original variable  $x_j$  ( $j=1, 2, \dots, p$ ) in the load on the principal component  $f_j$  ( $j=1, 2, \dots, p$ ),  $1_{ij}$  ( $i=1, 2, \dots, m; j=1, 2, \dots, p$ )

$$I(Z_i, X_j) = \sqrt{\lambda_i} a_{ij} \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, p) \quad (5)$$

⑤ Calculate principal component scores

Calculation sample in m on the principal component score:

$$F_i = a_{1i}x_1 + a_{2i}x_2 + \dots + a_{pi}x_p \quad i = 1, 2, \dots, m \quad (6)$$

Practical application, the index of often have different dimension, so before the main composition calculation should be to eliminate the influence of the dimension. There are so many ways to eliminate the dimension of data, the commonly used method is to standardize the original data, to do data transformation as follows:

$$x_{ij}^2 = \frac{x_{ij} - \bar{x}_j}{s_j} \quad i = 1, 2, \dots, m; j = 1, 2, \dots, p \quad (7)$$

Among them,  $\bar{x}_i = \frac{1}{n} \sum_{i=1}^n x_{ij}, \quad s_j^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$

## 5.2 Solution of the model 1

Based on factor analysis of principal component analysis (pca) and correlation between factors respectively in the table below

Table 1 Principal component analysis (pca)

|             |                  | Sulfur dioxide | Nitrogen dioxide | freon | Carbon monoxide |
|-------------|------------------|----------------|------------------|-------|-----------------|
| related     | Sulfur dioxide   | 1.000          | 0.805            | 0.678 | 0.657           |
|             | Nitrogen dioxide | 0.805          | 1.000            | 0.727 | 0.625           |
|             | freon            | 0.678          | 0.727            | 1.000 | 0.586           |
|             | Carbon monoxide  | 0.657          | 0.625            | 0.586 | 1.000           |
| significant | Sulfur dioxide   |                | 0.000            | 0.000 | 0.000           |
|             | Nitrogen dioxide | 0.000          |                  | 0.000 | 0.000           |
|             | freon            | 0.000          | 0.000            |       | 0.000           |
|             | Carbon monoxide  | 0.000          | 0.000            | 0.000 |                 |

## 6 model establishment and solving of the two

### 6.1 model

#### 6.1.1 Establishment of the model

1, the coordinate system

The coordinate system of the Gaussian model is shown in Figure 2.1. The origin is the emission point (if the source is an elevated source, the origin is the projection of the discharge point on the ground), the axis is the wind speed direction, the axis is perpendicular to the axis in the horizontal plane, Of the left side, the axis perpendicular to the horizontal plane, up for the positive. In this coordinate system, the centerline of the smoke stream or the centerline of the smoke stream overlaps the axis in the plane.

By the normal distribution assumption can export under the wind direction at any point function of leakage gas concentration are as follows:

$$X(x, y, z) = A(x) e^{-ay^2} e^{-bz^2} \quad (8)$$

By probability statistics theory can write expressions of variance are as follows:

$$\left\{ \begin{array}{l} \sigma_y^2 = \frac{\int_0^\infty y^2 X dy}{\int_0^\infty X dy} \\ \sigma_z^2 = \frac{\int_0^\infty z^2 X dz}{\int_0^\infty X dz} \end{array} \right. \quad (9)$$

By the assumption can write source strong integral formula:

$$Q = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u X dy dz \quad (10)$$

Type:, as leaking gas in standard deviation, y, z direction distribution unit for m; X (X, y, z) for any leaked gas concentration, the unit is kg/m; U for average wind speed, the unit is m/s; Q for the source (that is, the leakage velocity), the unit is kg/s;

Will (1) into (2), integral available:

$$\begin{cases} a = \frac{1}{2\sigma_y^2} \\ b = \frac{1}{2\sigma_z^2} \end{cases} \quad (11)$$

Will (1) and (4) into (3), integral available:

$$A(x) = \frac{Q}{2\pi u \sigma_y \sigma_z} \quad (12)$$

Then (4) and (5) into (1), available:

$$X(x, y, z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left[-\left(\frac{y^2}{2\sigma_y^2} + \frac{z^2}{2\sigma_z^2}\right)\right] \quad (13)$$

Unbounded space in order to carry on the continuous point source diffusion formula of gaussian model in practice, however, because of the ground, the spread of the plume is bounded. According to the hypothesis can put the ground as a mirror, total reflection effect to the leakage of gas, and processing, as the source method principle as shown in figure 2.2. Can take any point p at the concentration as a contribution to the sum of two parts: there was no part of the ground caused by the leakage concentration; Is partly a result of the ground reflection effect increase the concentration of the leakage. It is equivalent to the leakage concentration does not exist when the ground by the actual source of (0, 0, H) and the (0, 0, -H) as source at point P space caused by the leakage of the sum of concentration.

Numerical calculation of FREON in the atmosphere for the migration and diffusion of basically can be divided into two steps. First step according to the theory of atmospheric dynamics for stroke care about area calculation, its theoretical basis is the atmospheric motion equation, continuity equation and state equation, the equation of thermodynamics and the equation of the water vapor constitute the basic equations. In the field of atmospheric science research, several practical pattern of atmospheric circulation. The second step to known wind field FREON migration and diffusion of computation, can use similar to deal with the method of atmospheric pollution, assuming that FREON does not affect the air flow velocity and temperature, solving the continuity equation of FREON, FREON diffusion simulation diagram as shown in figure 2.

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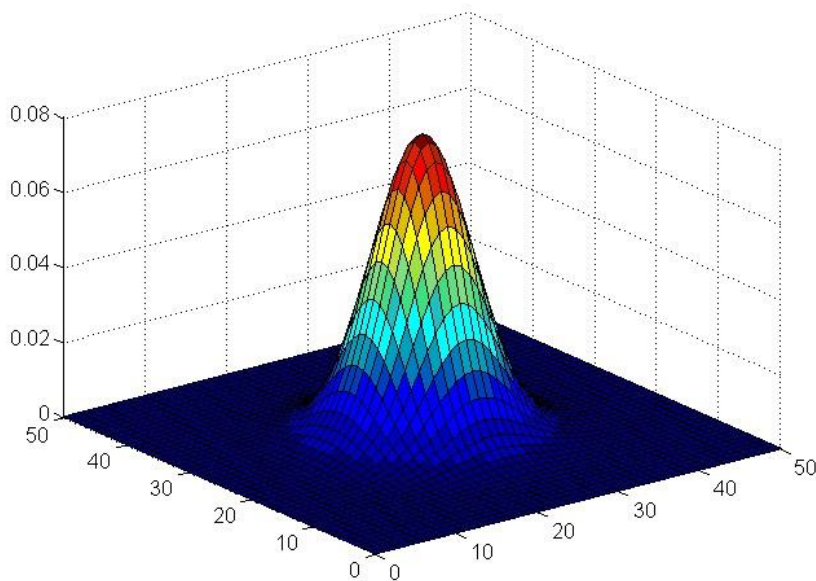


Figure 2 (FREON diffusion simulation diagram)

When the wind speed of KMS/continuous point source using gaussian diffusion model analysis of monitoring stations around the change of FREON concentrations. The discharge point source is a boundary point, the actual height of H discharge point source. To discharge projection point as the origin of coordinates of a point source in the ground, in the direction of the wind direction, for the x axis, vertical direction of z axis, and x axis vertical plane of the three-dimensional coordinate system is established with the y axis, due to the diffusion process of concentration change in y and z axis distribution conforms to the gaussian



distribution, so under the direction of any point Cxyz (,,) the concentration of the function is:

$$C(xyz)=Axe(x)^{-ay^2}e^{-bz^2} \quad (14)$$

Among them, the actual source of contributions as follows:

$$X_1(x, y, z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{1}{2} \frac{y^2}{\sigma_y^2}\right) \exp\left(-\frac{1}{2} \frac{(z-H)^2}{\sigma_z^2}\right) \quad (15)$$

As a source of contributions as follows:

$$X_2(x, y, z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{1}{2} \frac{y^2}{\sigma_y^2}\right) \exp\left(-\frac{1}{2} \frac{(z+H)^2}{\sigma_z^2}\right) \quad (16)$$

Is it the actual concentration as follows:

$$X(x, y, z) = X_1(x, y, z) + X_2(x, y, z) \quad (17)$$

By the above formula can be elevated continuous point source diffusion of gaussian plume model formula is:

$$X(x, y, z, t, H) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{1}{2} \frac{y^2}{\sigma_y^2}\right) \times \\ \left[ \exp\left(-\frac{1}{2} \frac{(z-H)^2}{\sigma_z^2}\right) + \exp\left(-\frac{1}{2} \frac{(z+H)^2}{\sigma_z^2}\right) \right] \quad (18)$$

Among them: X (X, y, z) as the downwind X m, transverse y m, above ground, z m the diffusion of gas concentration, the unit is kg/m; Q for the source (i.e., the source release rate), the unit is kg/s; U for average wind speed, the unit is m/s; For horizontal diffusion parameters, the unit is m; For vertical diffusion parameters, the unit is m; T as leaking after it was time, unit for s; H as leaking source effective height, the unit is m; Y is the horizontal distance, the unit is m; Z is the vertical distance, the unit is m.

Such as: type, z = 0, can be ground gas concentration calculation formula:

$$X(x, y, 0, H) = \frac{Q}{\pi u \sigma_y \sigma_z} \exp\left[-\frac{1}{2} \left(\frac{y}{\sigma_y}\right)^2\right] \exp\left(\frac{H^2}{2\sigma_z^2}\right) \quad (19)$$

Make = 0, y = z axis can be ground gas concentration calculation formula:

$$X = \frac{Q}{\pi u \sigma_y \sigma_z} \exp\left(-\frac{H^2}{2\sigma_z^2}\right) \quad (20)$$

Among them, the  $X(X, y, 0)$  as the downwind  $X$  m, the ground of the transverse  $y$  m diffusion gas concentration, the unit is kg/m. If the  $y = 0$ , then can get the concentration distribution of the wind center line.

Leakage source effective height of lifting height calculation (smoke) :

Leakage source effective height refers to the leakage of the above type gas cloud basically become a horizontal gas cloud center height from the ground. In fact, the leakage source effective height equals leakage source geometric height increased leakage cloud lifting height.

There are many factors that affect the lift height of the smoke cloud, including the initial velocity and direction of the leakage gas, the initial temperature, the diameter of the leakage port, the rate of change of the environmental wind speed and the wind age, the ambient temperature and the atmospheric stability.

Effective source height:

$$H = H_s + \Delta H \quad (21)$$

Among them, for the leakage source high geometric frame, lifting height of smoke.

Experiments show that the leakage source lifting height can be approximate calculation with the following formula:

$$\Delta H = 2.4V_s d / V \quad (22)$$

Among them, is a gas cloud release rate, the unit is m/s; Leakage is outlet diameter, the unit is m;  $V$  wind speed for the environment, the unit is m/s;

is in the late 1970 s and early 80 s, Wilson is based on the experimental formula of the pipeline rupture experiments. Experiments of gas injection direction perpendicular to wind speed and gas injection no obstacle on the path. The experimental results show that when the gas injection direction is vertical, the ratio of the predicted value and actual value within the wash gangle 2.

After lifting height, calculate the leakage smoke leakage source lifting height and leakage source actual geometry highly together, get the leakage source effectively.

The selection of diffusion coefficient:

Diffusion coefficient, and the atmospheric turbulence structure, with the size of the height from the ground, the ground roughness, leak duration, sampling interval, wind speed and leave the leakage source distance and other factors. Atmospheric turbulence structure and wind speed are considered in the atmospheric stability.

Atmospheric stability consists of more than 10 meters high wind speed, solar radiation during the day or night is decided by the parameters such as cloud cover.

According to the classification of Pasquill method, with the increase of stability of meteorological condition, atmospheric stability can be divided into A, B, C, D, E, F six classes. Where A, B, C three kinds of means meteorological condition is not stable, E, F two kinds said weather conditions stable, class D said neutral meteorological conditions, that is to say, the stability of meteorological condition between stable and unstable. A, B, C three types of stability of A class represents meteorological conditions extremely unstable, class B said moderate unstable weather condition, class C said weak unstable weather conditions. E and F, the stability of the two types of class E said weakly stable weather conditions, F class represents moderate stable weather conditions. The atmospheric stability classification methods, tables, see table vii.

Table 2 (Pasquill determination of atmospheric stability

| The ground wind speed (m/s) | Sunshine during the day |        |      | Nighttime conditions                                     |                            |
|-----------------------------|-------------------------|--------|------|----------------------------------------------------------|----------------------------|
|                             | strong                  | medium | weak | It's cloudy and thin clouds, or a low cloud cover is 4/8 | Cloud cover the sky is 3/8 |
| < 2                         | A                       | A-B    | B    |                                                          |                            |
| 2 ~ 3                       | A-B                     | B      | C    | E                                                        | F                          |
| 3 ~ 4                       | B                       | B-C    | C    | D                                                        | E                          |
| 4 ~ 6                       | C                       | C-D    | D    | D                                                        | D                          |
| > 6                         | C                       | D      | D    | D                                                        | D                          |

Table 3 The intensity of sunlight

| Clouds in the sky                                              | Sun Angle > 60° | Sun Angle < 60° and > 35° | Sun Angle > 15° and < 35° |
|----------------------------------------------------------------|-----------------|---------------------------|---------------------------|
| It's cloudy and thin clouds, or a low cloud cover is 4/8       | Sun Angle       | medium                    | weak                      |
| Cloud cover the sky for 5/8-7/8, cloud height is 2134-4877 - m | medium          | weak                      | weak                      |
| Cloud cover the sky for 5/8-7/8, clouds height < 2134 m        | weak            | weak                      | weak                      |

The cloud cover in Tables 7 and 8 refers to the coverage of the local sky layer. For example, a cloud of 3/8 means that 3/8 of the sky is covered by clouds. Sunshine angle refers to the local sun rays and the horizon between the degree of folder. For example, the sunshine angle when the sun is shining vertically is 90 °

## 6.2 Model establishment: Optimize Gaussian plume model

### Determination of wind speed

The wind speed changes with altitude in the monotonically rising type mostly, and the wind speed changes with height in accordance with the exponential law:

$$U = u_s \left( \frac{Z_m}{10} \right)^p \quad (23)$$

Where:  $u_s$  is the mean wind speed from the ground,  $U$  is the average wind speed at 10 m from the ground, and  $p$  is the wind speed index.

Table 3 Wind speed high index P value table

| Pasquale stability | Strong unstable | unstable | Weak unstable | neutral | The weak stability and stability |
|--------------------|-----------------|----------|---------------|---------|----------------------------------|
| P value            | 0.1             | 0.15     | 0.2           | 0.25    | 0.3                              |

The distribution of pollutants in the space is Gaussian (normal distribution) diffusion model, the so-called normal distribution function in the form of the following relationship:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-u)^2}{2\sigma^2}\right] \quad (-\infty < x < +\infty) \quad (11) \quad (24)$$

When there is wind, the mean wind direction (ie direction) of the turbulent diffusion size is less than the average wind speed () the average delivery rate, so the direction of turbulent diffusion can be ignored. The model uses the ground concentration diffusion model of the continuous point source; but when the wind speed is small (), the diffusion mode of the continuous point source under the breeze condition is adopted.

(1) small winds and winds (average wind speed)

Diffusion of the direction can not be ignored, the current small wind and static wind conditions under the application of the atmosphere diffusion model is more integral diffusion model, simplified integral smoke group model, 360 ° uniform distribution model. Here, the model selects the integral chimney diffusion model with the following expression:

$$C(x, y, z, H_m) = \int_0^{+\infty} \frac{2Q}{(2\pi)^{\frac{3}{2}}} dt \exp\left[-\frac{(x-Ut)^2}{2\sigma_x^2}\right] \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \exp\left(-\frac{H_m^2}{2\sigma_z^2}\right) \quad (25)$$

In the formula, the integral can be replaced by the integral of a finite interval. According to experience, if the diffusion time is small, the integration limit can be reached very accurately.

Where the diffusion parameters

$$\sigma_x = \sigma_y = \gamma_1 t, \quad \sigma_z = \gamma_2 t \quad (26)$$

$$Q = C_h V_h \quad (27)$$

According to the above formula, the average wind speed is m / s, the horizontal direction is evenly distributed on the concentric circle with the pollution source as the center, and then using the continuity condition integral, we can get the wind condition under the static wind condition. The model of continuous point source contaminant surface concentration is:

$$C(x, y, z, H_m) = \frac{2Q}{(2\pi)^{\frac{3}{2}} RU \sigma_z} \exp\left(-\frac{H_m^2}{2\sigma_z^2}\right) \quad (28)$$

Where: is the straight-line distance of the point source;

(2) wind, and the average wind speed

To the chimney's ground position as the origin, down to the ground any point, less than 24 hours sampling time concentration:

$$C(x, y, z, H_m) = \left(\frac{Q}{2\pi U \sigma_y \sigma_z}\right) \exp\left(-\frac{y^2}{2\sigma_y^2}\right) A \quad (29)$$

Where: is the concentration of pollutant at any point in space, is the amount of emission per unit time, is the average wind speed of the discharge port, x is the distance from the source to any point upwards, m is the distance between the point and the passage through the chimney. The average vertical axis of the wind in the horizontal plane of the vertical distance; for the height from the ground to any point; is perpendicular to the average wind direction of the horizontal diffusion coefficient; for the vertical diffusion coefficient; for the effective height of the chimney. The formula is as follows:

$$A = \sum_{n=-k}^{+k} \left\{ \exp\left[-\frac{(2nh - H_m)^2}{2\sigma_y^2}\right] + \exp\left[-\frac{(2nh + H_m)^2}{2\sigma_z^2}\right] \right\} \quad (30)$$

Where: is the thickness of the mixed layer, and is the number of reflections.

Where the diffusion parameters, according to the following formula:

$$\sigma_y = \gamma_1 x^{\alpha_1}, \quad \sigma_z = \gamma_2 x^{\alpha_2} \quad (31)$$

Among them, for the horizontal, the vertical diffusion parameter regression coefficient; for the horizontal, vertical diffusion parameter index;

For the horizontal distance from the chimney downwind. According to the following formula:

$$H_m = H + \Delta H \quad (32)$$

Flue gas lift height using the formula:

(1) Neutral and unstable atmospheric conditions when there is wind:

① When the flue gas heat release rate, and the flue gas temperature and ambient temperature difference, the flue gas lift height calculation formula:

$$\Delta H = n_0 Q_h^{n_1} H^{n_2} U^{-1} \quad (33)$$

Which is the height of flue gas, the thermal condition of flue gas and the coefficient of surface condition, the index of heat release rate of flue gas, and the index of flue gas height of exhaust pipe.

$$\Delta H = \Delta H_1 + (\Delta H_2 - \Delta H_1)(Q_h - 1700)/400 \quad (34)$$

$$\Delta H_1 = 2(1.5V_s D + 0.01Q_h)/U - 0.048(Q_h - 1700)/U \quad (35)$$

Where: for the exit of the chimney flue gas emissions rate; for the chimney outlet diameter,; calculated by ①.

③  $Q_h \leq 1700 \text{ KJ / s}$  and  $\Delta T < 35 \text{ K}$  then:

$$\Delta H = 2(1.5V_s D + 0.01Q_h)/U \quad (36)$$

(1) the wind, the stable atmospheric conditions, the flue gas lift height formula:

$$\Delta H = Q_h^{1/3} (dT_a/dZ + 0.0098)^{-1/3} U^{-1/3} \quad (37)$$

Where: for the chimney height above the geometric height of the atmospheric temperature gradient,  $K/m$

(2) static wind and small wind, using the following formula:

$$\Delta H = 5.50 Q_h^{1/4} (dT_a/dZ + 0.0098)^{-3/8} \quad (38)$$

Where the value should be less than. At that time, take, when, according to ① calculation.

### 6.3 Solving the model

In this paper, we can regard the point source model as an elevated point source, and point the source point on the ground as the coordinate origin, and the effective source is located at the point on the axis. If the contaminant is assumed to be completely absorbed after reaching the ground and there is no accumulation of the reflection concentration, then the pollutant concentration is calculated as:

$$C(x, y, z, H) = \frac{Q}{\pi U \sigma_y \sigma_z} \exp\left[-\frac{1}{2}\left(\frac{y^2}{\sigma_y^2} + \frac{(z-H)^2}{\sigma_z^2}\right)\right] \quad (39)$$

To calculate the ground concentration formula for an elevated point source,

$$C(x, y, 0, H) = \frac{Q}{\pi U \sigma_y \sigma_z} \exp\left[-\frac{1}{2}\left(\frac{y^2}{\sigma_y^2} + \frac{H^2}{\sigma_z^2}\right)\right] \quad (40)$$

According to the above formula, if further order, you can get along the x-axis concentration distribution formula:

$$C(x, 0, 0, H) = \frac{Q}{\pi U \sigma_y \sigma_z} \exp\left(-\frac{H^2}{2\sigma_z^2}\right) \quad (41)$$

Equations (37) and (38) are often used to estimate air pollution, and the estimated values are consistent with the environmental monitoring data in the vicinity of the isolated elevated point source.

#### Selection of Diffusion Coefficient

According to national standards (GB / T13201-1991) to develop local air pollutant discharge standards of technical methods, the level of atmospheric stability is divided into six A-F, A is very unstable, F and stable. The diffusion coefficient is related to the atmospheric turbulence structure, height and height, surface roughness, discharge duration, sampling interval, wind speed, distance from source, etc. According to Pasquill's classification method, with the stability of meteorological conditions Increase, the atmospheric stability can be divided into six AF, A is extremely unstable, F and stable. The determination of the diffusion coefficient (R) of rheumatism using Briggs gives a set of diffusion parameters. According to the meteorological data of Hebei Province, we choose the diffusion coefficient with atmospheric stability D.

Table 5 Diffusion coefficient calculation method

| The atmospheric stability | $\sigma_y$                | $\sigma_z$                |
|---------------------------|---------------------------|---------------------------|
| A                         | $0.22x/(1+0.0001x)^{0.5}$ | $0.2x$                    |
| B                         | $0.16x/(1+0.0001x)^{0.5}$ | $0.12x$                   |
| C                         | $0.11x/(1+0.0001x)^{0.5}$ | $0.08x/(1+0.0002x)^{0.5}$ |
| D                         | $0.08x/(1+0.0001x)^{0.5}$ | $0.06x/(1+0.0015x)^{0.5}$ |
| E                         | $0.06x/(1+0.0001x)^{0.5}$ | $0.03x/(1+0.0003x)$       |
| F                         | $0.04x/(1+0.0001x)^{0.5}$ | $0.016x/(1+0.0003x)$      |

8:00 am and 9:00 pm are not emissions time, assuming the wind speed unchanged, in the direction of a normal distribution, at 12:00 in the discharge of pollutants in time, you can use the established model to solve. Using Matlab software, you can obtain the air pollution concentration distribution and air quality grade at 8:00 am, 12:00 pm and 9:00 pm respectively.

Table 6 Period of air pollution concentration distribution and air quality grade table

| time  | distance                       | 1     | 2     | 3     | 4     | 5     |
|-------|--------------------------------|-------|-------|-------|-------|-------|
| 8     | Nitrogen concentration (ug/m3) | 0.34  | 0.433 | 0.127 | 0.11  | 0.043 |
| 12:00 | Nitrogen concentration (ug/m3) | 1.539 | 0.875 | 0.236 | 0.13  | 0.09  |
| 21:00 | Nitrogen concentration(ug/m3)  | 0.25  | 0.41  | 0.38  | 0.151 | 0.06  |

In order to get more direct nitrogen dioxide emissions, the Gaussian smoke plume diffusion model with Matlab software to make the 12 noon nitrogen oxide concentration distribution map:



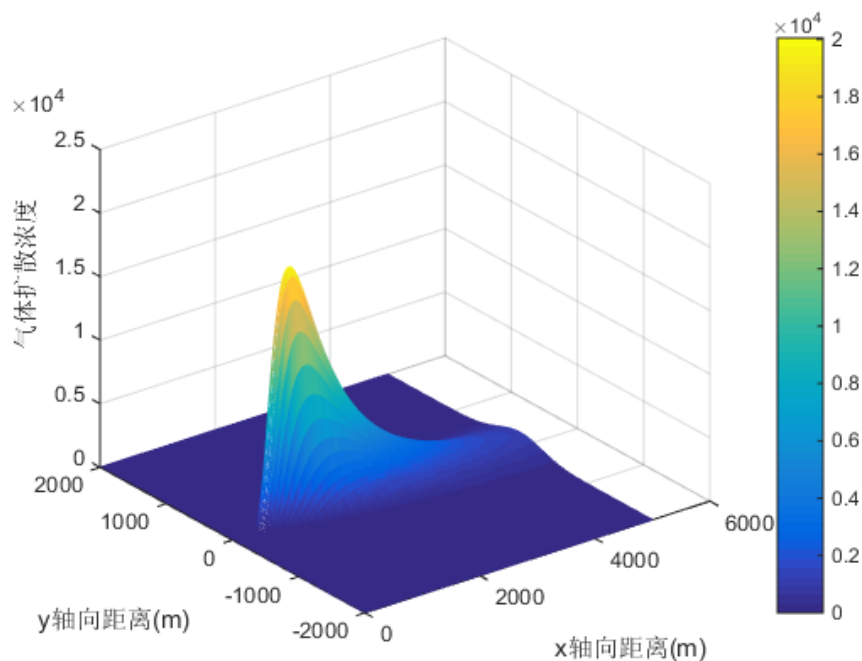


Figure 3 based on gauss smoke plume simulation

Assuming the wind direction is north, wind speed is 40km / h, attenuation coefficient is 0.00001, effective height is 50m, FREON initial concentration for each site FREON concentration of the maximum value of Annex 2 and Annex 3 FREON data into the Gaussian plume model, the use of MATLAB simulation, the following FREON diffusion simulation as shown in Figure 6-5, And the spread of data, such as Table XII, Table XIII (high-voltage switch factory, for example):

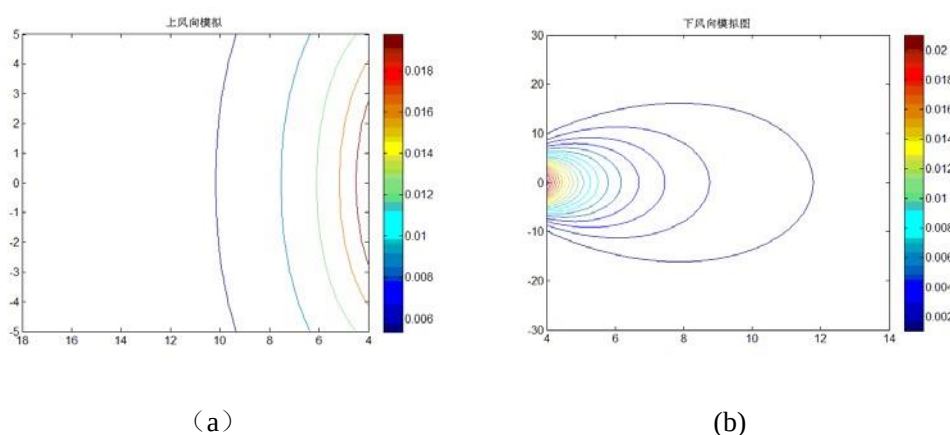


Fig.4 Simulation of FREON diffusion, where Fig. (A) and (b) show the prediction

## **7 The establishment of the model of three**

### **7.1 The establishment of the model**

Use methods such as observation, investigation and statistics, sampling time series dynamic data obtained by observation system.

According to the dynamic data for relevant figure, correlation analysis, autocorrelation function. Correlation diagram can show the change trend and cycle, and can be found in hops and inflection point. Hops is inconsistent with other data observed value. If the hops is the correct observation, should be taken into account when modeling, if it is abnormal phenomenon, should adjust the jumping point to the expectations. Inflection point is time series from rising suddenly into a downward trend. If there is a turning point, when modeling must use different models to segmenting fitting of this time series, such as using threshold regression model.

Identify appropriate stochastic model, carries on the curve fitting, which is general stochastic model to fit time series observation data. Available for short or short time series, and seasonal trend model fitting model with error. For stationary time series, can use generic model ARIMA (autoregressive moving average) model and its special cases of autoregressive models, moving average model or combination - ARIMA model for fitting. When the observed value more than 50 are generally using ARIMA model. For non-stationary time series is the first time sequence of the observed difference operation, into a stationary time series, and then use the appropriate model to fitting the difference sequence.

Time series is a special kind of random process, take of nonnegative integer, can represent each moment, can be regarded as time series (time series), therefore, when a random process can be viewed as time series, we can use the existing model of time series modeling analysis of the characteristics of the stochastic process.

ARIMA model is also called the autoregressive moving average model, for peace when the time series itself is not stable, if its incremental, namely a difference, steady near zero, can be regarded as stationary series. In practical problems encountered by the majority of nonstationary sequence can be by one or more difference after become stationary time series, can build a model:

This shows that any number of nonstationary sequence as long as through proper order difference arithmetic to realize difference after steady, can be the difference sequences after fitting of the ARIMA model.

Model refers to the highest order autocorrelation for after order difference, moving average of the highest order to model, usually independent of an unknown factor: it contains. It can be predicted with minimum mean square error principle implementation:

7.2 solution of the model

According to the time series model, get the following table 7 and figure 5

Table 7 Model fitness

| Goodness of fit statistics | The average | The minimum value | The maximum | The percentile |       |       |       |       |       |       |
|----------------------------|-------------|-------------------|-------------|----------------|-------|-------|-------|-------|-------|-------|
|                            |             |                   |             | 5              | 10    | 25    | 50    | 75    | 90    | 95    |
| Smooth R square            | 0.068       | 0.068             | 0.068       | 0.068          | 0.068 | 0.068 | 0.068 | 0.068 | 0.068 | 0.068 |
| R square                   | 0.959       | 0.959             | 0.959       | 0.959          | 0.959 | 0.959 | 0.959 | 0.959 | 0.959 | 0.959 |

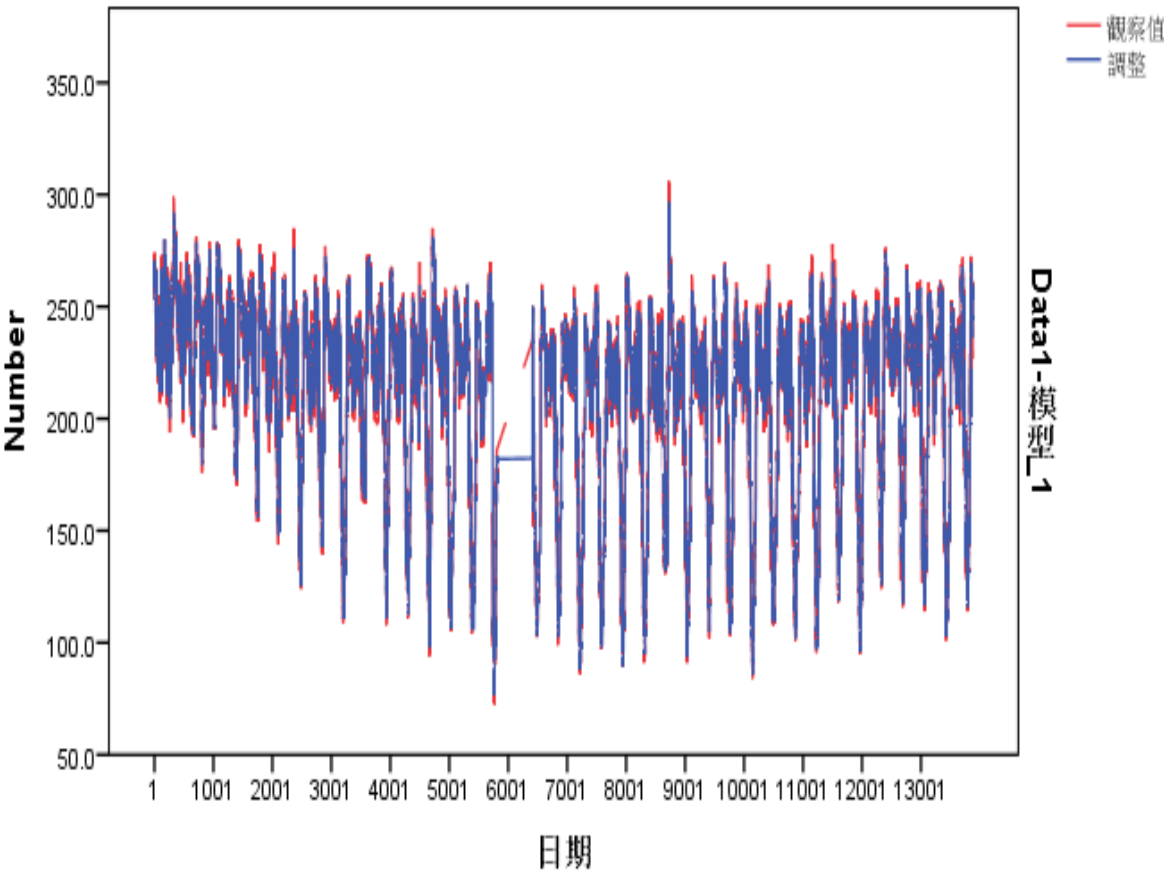


Figure 5 in the southern hemisphere

According to the time series model, get the following table 8 and figure 6

Table 8 Model fitness

| Goodness of fit statistics | The average | The minimum value | The maximum | The percentile |       |       |       |       |       |       |
|----------------------------|-------------|-------------------|-------------|----------------|-------|-------|-------|-------|-------|-------|
|                            |             |                   |             | 5              | 10    | 25    | 50    | 75    | 90    | 95    |
| Smooth R square            | 0.203       | 0.203             | 0.203       | 0.203          | 0.203 | 0.203 | 0.203 | 0.203 | 0.203 | 0.203 |
| R square                   | 0.996       | 0.996             | 0.996       | 0.996          | 0.996 | 0.996 | 0.996 | 0.996 | 0.996 | 0.996 |

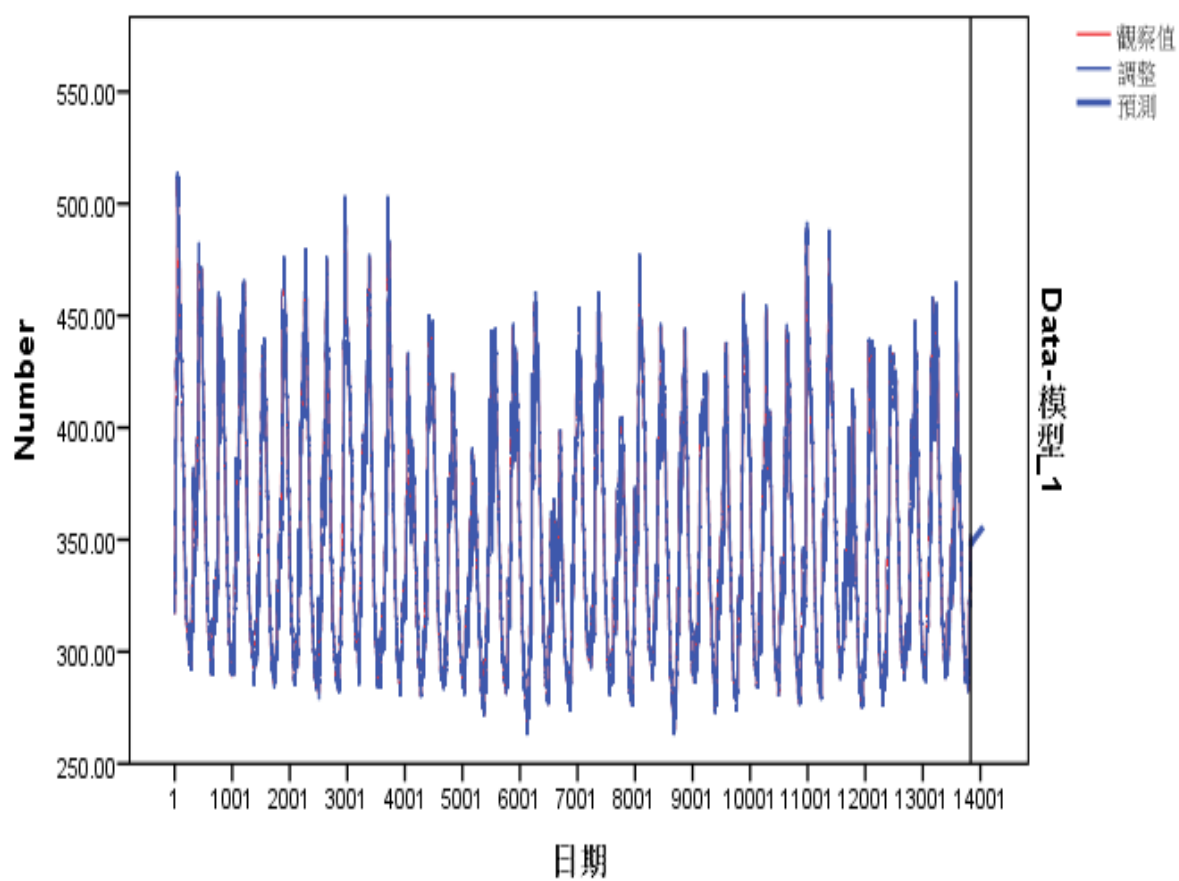


Figure 6 the predictive value of the northern hemisphere

## **8 Evaluation and promotion of the model**

### **8.1 The advantages of the model**

- 1, a model using principal component analysis model, better able to analyze one of the most influence on ozone depletion substances;
- 2, model 2 by gauss misty rain, can calculate the expand the calculate the rate of gas in the air;
- 3, model using the neural network model, three good forecasts the 50 years of data, the data with time sequence comparison;
- 4, model by using the time series model, four good forecasts the 50 years of data every day, to the average prediction and neural network prediction model.
- 5, this article adopts the way of 4 model are connected, from the origin of the ozone, transmission, predict good combination, can better predict the nearly fifty years of ozone, and make contrast, mutual support, analysis the prediction error.

### **8.2 Disadvantages of the model**

- 1, model a data volume is not enough, may have a certain impact on the impact of factors.
- 2, In the second model, due to the larger model, the atmospheric motion equation may be error.

### **8.3 Model promotion**

This article, we use a variety of methods to solve each problem, which includes the principal component analysis, single factor variance analysis, multiple linear fitting, partial correlation analysis method and model. Principal component analysis (pca) is to multiple variables through linear transformation to choose less more important variable of a multivariate statistical analysis method, it can also be used to sFiguredy the change of tea scent, thus found that the main compounds, affect their scent changes of aroma compounds on the aroma contribution degree. Factor analysis refers to the sFiguredy of common factors extracted from the variable group of statistical techniques. Factor analysis can find hidden in many variable factors of representative. The same variable in the naFigureere of a factor, can reduce the number of variables, but also test the hypothesis that relationships between variables. Factor analysis can also be used to the loneliness of old people, etc. Multiple linear regression model is an extension of the unary linear regression model, its basic principle is similar to a yuan linear regression model, multivariate linear regression model can be used to sFiguredy the net income of rural residents of simulation and prediction, can also be used to analyze the material content of underground conditions. Partial correlation refers to the two variables associated with one or more other variables of common parts are put forward after the relevance.

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## Appendix

### Appendix a (factor analysis of principal component analysis (pca) code)

```
GET DATA /TYPE=XLSX
  /FILE='C:\Users\Administrator\Desktop\2015.xlsx'
  /SHEET=name 'Sheet5'
  /CELLRANGE=full
  /READNAMES=on
  /ASSUMEDSTRWIDTH=32767.
EXECUTE.
DATASET NAME Data set 1 WINDOW=FRONT.
FACTOR
  /VARIABLES Data1 Minimum V4 V5 Mean V7 V8 Maximum
  /MISSING LISTWISE
  /ANALYSIS Data1 Minimum V4 V5 Mean V7 V8 Maximum
  /PRINT INITIAL CORRELATION SIG EXTRACTION ROTATION FSCORE
  /PLOT EIGEN ROTATION
  /CRITERIA MINEIGEN(1) ITERATE(25)
  /EXTRACTION PC
  /CRITERIA ITERATE(25)
  /ROTATION VARIMAX
  /SAVE REG(ALL)
  /METHOD=CORRELATION.
```

### Appendix 2 (misty rain gaussian model code)

```
clearall;
[x,y]=meshgrid(0:20:5000,-2000:20:2000);
Q=406.92;%NOx
sigy=0.08*x.*(1+0.0001*x).^(-0.5);
sigz=0.06*x.*(1+0.0015*x).^(0.5);
H=[130.5362776132.7538462119.9039121120.3678051120.4165713120.85393
26126.8520578132.2781406];
u=1./[2.1370786522.0472440942.7065015482.6753993612.6721712542.643564
3562.3051282052.065873016];
f=[0.0335365850.0426829270.0792682930.3841463410.2408536590.08231707
30.0670731710.070121951];
```

```

s=size(u);
fori=1:s(2)
Qpi=Q./(pi*u(i)*sigy.*sigz+eps);
ex1=exp(-0.5*(y./(sigy+eps)).^2);
ex2=exp(-0.5*(H(i)./(sigz+eps)).^2);
X1(:,i)=Qpi.*ex1.*ex2*f(i);
end
X=X1;
forj=2:s(2)
X(:,j)=X(:,:(j-1))+X(:,j);%
end
X2=1000000*X(:,s(2));
mesh(x,y,X2);

```

### Appendix 3 (BP) neural network code

```

clc
clear
%p=load('shuru.txt');
%p=p';
p=[1979 2016];
%t=load('shuchu.txt');
%t=t';
t=[124.175 117.0666667 108.3333333 102.8083333 99.225 98.6 100.35 100.725
99.41666667 101.1666667 103.9 101.8166667 101.4666667 104.7666667 105.9
99.29166667 103.32255];
% plot(p,t)

[pn,minp,maxp,tn,mint,maxt]=premnmx(p,t);
dx=[-1,1];

net=newff(dx,[5,1],{'tansig','tansig','purelin'},'traingdx');
net.trainParam.show=1000;
net.trainParam.Lr=0.05;
0.05
net.trainParam.epochs=3000;

```



```

net.trainParam.goal=1e-5;

net=train(net,pn,tn);
an=sim(net,pn);
a=postmnmx(an,mint,maxt);
x=1994:2010;
newk=a(1,:);
figure;
plot(x,newk,'r-o',x,t,'b--+');
pnew=[2017:2062];
pnewn=tramnmx(pnew,minp,maxp);

```

## Appendix 4 code (time series)

```

GET DATA /TYPE=XLSX
  /FILE='C:\Users\Administrator\Desktop\.xls x'
  /SHEET=name '南半球'
  /CELLRANGE=full
  /READNAMES=on
  /ASSUMEDSTRWIDTH=32767.
EXECUTE.
DATASET NAME WINDOW=FRONT.
PREDICT THRU END.
TSMODEL
  /MODELSUMMARY PRINT=[MODELFIT RESIDACF]
PLOT=[ SRSQUARE RSQUARE]
  /MODELSTATISTICS DISPLAY=YES MODELFIT=[ SRSQUARE
RSQUARE]
  /SERIESPLOT OBSERVED FORECAST FIT
  /OUTPUTFILTER DISPLAY=ALLMODELS
  /SAVE PREDICTED
  /AUXILIARY CILEVEL=95 MAXACFLAGS=24
  /MISSING USERMISSING=EXCLUDE
  /MODEL DEPENDENT=Data1
  PREFIX=

```

/ARIMA AR=[0] DIFF=0 MA=[0]  
 TRANSFORM=NONE CONSTANT=YES  
 /AUTOOUTLIER DETECT=ON TYPE=[ADDITIVE LEVELSHIFT].

## Appendix 5(Some predict ozone levels)

| month / year | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2014 | 2015 |
|--------------|-------|-------|-------|-------|-------|-------|-------|------|------|
| 1/1          | 226.2 | 238.1 | 247.7 | 246.7 | 244.3 | 244.1 | 236   | 242  | 247  |
| 1/2          | 226.8 | 231.7 | 245.6 | 243   | 238.6 | 244   | 235.2 | 243  | 245  |
| 1/3          | 232.5 | 234.3 | 245.9 | 244.6 | 246   | 234.6 | 233.5 | 249  | 247  |
| 1/4          | 233.7 | 238.8 | 249.9 | 243.3 | 251.5 | 234.1 | 232.3 | 252  | 253  |
| 1/5          | 235.5 | 238.2 | 253.9 | 244.6 | 248.8 | 237.2 | 236   | 249  | 246  |
| 1/6          | 238.2 | 238.8 | 250.8 | 242.6 | 250.5 | 234.7 | 238.4 | 249  | 250  |
| 1/7          | 240.4 | 240.5 | 243.1 | 242.1 | 254   | 239.6 | 237.3 | 251  | 256  |
| 1/8          | 236.2 | 240.1 | 237.5 | 235.8 | 256.4 | 244.6 | 232.4 | 249  | 251  |
| 1/9          | 239.2 | 243.2 | 238.7 | 239.1 | 241.2 | 248.1 | 230.5 | 245  | 256  |
| 1/10         | 243.7 | 243   | 245.2 | 244.5 | 241.8 | 246.4 | 236.4 | 249  | 255  |
| 1/11         | 247.1 | 250.9 | 238   | 240.2 | 245.1 | 242.2 | 236.1 | 244  | 251  |
| 1/12         | 245.4 | 254.9 | 248.2 | 239.3 | 235.6 | 242   | 236   | 251  | 253  |
| 1/13         | 238.8 | 256.5 | 244.1 | 245.3 | 234.3 | 237.3 | 236.8 | 253  | 253  |
| 1/14         | 231.9 | 258.5 | 229.1 | 249.8 | 237.2 | 232.4 | 236.2 | 250  | 257  |
| 1/15         | 236.7 | 251   | 225.9 | 246.8 | 233.1 | 238.4 | 237.6 | 250  | 255  |
| 1/16         | 238.5 | 253.4 | 233   | 244.6 | 230.7 | 235.7 | 237.9 | 248  | 246  |
| 1/17         | 238.1 | 248.4 | 243.2 | 238.6 | 234.9 | 236.7 | 231.7 | 240  | 253  |
| 1/18         | 234.1 | 254.4 | 236.8 | 241.3 | 238.1 | 230.7 | 228   | 242  | 252  |
| 1/19         | 227   | 253.3 | 234.6 | 242.6 | 238.8 | 232.5 | 228   | 240  | 250  |
| 1/20         | 225.4 | 245.2 | 231   | 241.4 | 234.3 | 238.4 | 239   | 235  | 246  |
| 1/21         | 218   | 241.1 | 226.3 | 240.3 | 227.8 | 236.5 | 231.2 | 229  | 244  |
| 1/22         | 221.1 | 246.5 | 219   | 243.2 | 235   | 242.4 | 223.2 | 233  | 249  |
| 1/23         | 225.7 | 250   | 218.2 | 245.4 | 243.6 | 241.3 | 219.1 | 242  | 246  |
| 1/24         | 228.3 | 246.1 | 221.2 | 241.2 | 239.2 | 235.6 | 214.3 | 243  | 245  |
| 1/25         | 225.7 | 252.3 | 222.6 | 233.2 | 235.1 | 238.3 | 210.1 | 240  | 243  |
| 1/26         | 222.8 | 253.6 | 215.9 | 233   | 233.5 | 237.3 | 210.8 | 240  | 237  |
| 1/27         | 229.2 | 253.9 | 216   | 234.6 | 240.9 | 233.9 | 213.3 | 238  | 247  |
| 1/28         | 233.8 | 250.8 | 228.5 | 234.1 | 248   | 237.1 | 213.1 | 235  | 246  |
| 1/29         | 235.5 | 250.2 | 228.9 | 238.7 | 251.1 | 233.9 | 216.2 | 246  | 246  |
| 1/30         | 236.7 | 246.9 | 229   | 230.3 | 247.9 | 234   | 218.4 | 223  | 237  |
| 1/31         | 234.6 | 237.6 | 225.9 | 235.6 | 240.9 | 239.5 | 222   | 224  | 238  |
| 2/1          | 231   | 231.6 | 205.6 | 235.1 | 247.3 | 235.2 | 218.9 | 230  | 238  |
| 2/2          | 232.6 | 228   | 191   | 230.3 | 251   | 211.5 | 214.3 | 232  | 236  |
| 2/3          | 234.5 | 226.4 | 194.9 | 237.1 | 248.6 | 209.5 | 208.6 | 221  | 232  |
| 2/4          | 231   | 226.9 | 201.3 | 237.1 | 247.4 | 206.8 | 202.6 | 231  | 238  |

|      |       |       |       |       |       |       |       |     |     |
|------|-------|-------|-------|-------|-------|-------|-------|-----|-----|
| 2/5  | 232.6 | 232.5 | 185.4 | 238.6 | 244.7 | 213.2 | 203.7 | 225 | 239 |
| 2/6  | 232.9 | 220.3 | 181.9 | 238.1 | 240.2 | 217.8 | 211   | 232 | 238 |
| 2/7  | 233.8 | 216.9 | 177.3 | 234.1 | 240.8 | 219.5 | 216.5 | 228 | 237 |
| 2/8  | 233.2 | 218.6 | 193.5 | 227.7 | 237.8 | 216.8 | 217.7 | 239 | 235 |
| 2/9  | 236.2 | 222   | 203.3 | 223.2 | 226.2 | 202.8 | 210.9 | 237 | 234 |
| 2/10 | 250.9 | 259.7 | 203.1 | 227   | 221.3 | 206.9 | 210.2 | 240 | 234 |
| 2/11 | 222   | 244.2 | 205.4 | 234.8 | 222.6 | 214.8 | 206.9 | 235 | 231 |
| 2/12 | 211.8 | 240.1 | 212.9 | 227.9 | 231.6 | 227.5 | 206.2 | 244 | 239 |
| 2/13 | 218.2 | 243.2 | 208.3 | 232.7 | 231.9 | 234.2 | 200.6 | 243 | 240 |
| 2/14 | 225.1 | 246.2 | 220.6 | 228.2 | 233.5 | 234   | 204.8 | 243 | 235 |
| 2/15 | 229.7 | 250   | 207.6 | 234.3 | 218.2 | 230.9 | 214.4 | 245 | 234 |
| 2/16 | 237.8 | 248.4 | 224.4 | 230.1 | 204.7 | 227.1 | 214.9 | 241 | 232 |
| 2/17 | 238   | 247.3 | 210.7 | 223.8 | 208.1 | 220.7 | 214.2 | 223 | 235 |
| 2/18 | 239.6 | 248.6 | 210.2 | 209   | 225.3 | 228.8 | 221.9 | 235 | 233 |
| 2/19 | 223.4 | 245.8 | 207.7 | 220   | 236   | 227.4 | 226.7 | 237 | 235 |
| 2/20 | 222.3 | 238.1 | 215   | 228.6 | 232.1 | 226.3 | 223   | 225 | 232 |
| 2/21 | 226   | 231   | 213.4 | 239.1 | 226.5 | 219.8 | 218.3 | 223 | 234 |
| 2/22 | 234.6 | 227.1 | 218.6 | 235.9 | 230   | 221.5 | 227.5 | 240 | 237 |
| 2/23 | 240.4 | 234.1 | 219.7 | 236.8 | 233.1 | 229   | 233.4 | 226 | 235 |
| 2/24 | 222.8 | 237.9 | 221.5 | 233.8 | 229.1 | 220.4 | 226.8 | 230 | 228 |
| 2/25 | 215.9 | 237.2 | 222.7 | 232.4 | 225.9 | 228.7 | 224.6 | 233 | 225 |
| 2/26 | 208.9 | 231.6 | 226.1 | 226.6 | 224.4 | 229.7 | 217.9 | 231 | 226 |
| 2/27 | 216.8 | 241.3 | 219.7 | 223.8 | 226.4 | 222.1 | 218   | 239 | 227 |
| 2/28 | 223.2 | 242.6 | 222.3 | 232.6 | 229.2 | 225.2 | 214.9 | 234 | 228 |
| 3/1  | 226.2 | 242.1 | 236.2 | 216.4 | 229.1 | 234.4 | 227.5 | 227 | 225 |
| 3/2  | 231.5 | 238.9 | 237.6 | 227.9 | 229.8 | 234.1 | 223.2 | 243 | 216 |
| 3/3  | 215.4 | 231.9 | 244.2 | 238.2 | 226.8 | 230.1 | 218.3 | 231 | 218 |
| 3/4  | 226.9 | 219.4 | 233.1 | 242   | 217.5 | 226.9 | 216.5 | 220 | 225 |
| 3/5  | 230.9 | 225.4 | 232.2 | 234.9 | 208.5 | 230.9 | 207.5 | 217 | 229 |
| 3/6  | 236.5 | 232.9 | 232   | 226.4 | 205   | 235.7 | 216.5 | 218 | 229 |
| 3/7  | 242.4 | 236.9 | 233.6 | 226.9 | 215.2 | 237.5 | 223.2 | 215 | 226 |
| 3/8  | 244.4 | 242.5 | 230.8 | 232.5 | 223.7 | 234   | 220.7 | 218 | 224 |
| 3/9  | 233.1 | 245.2 | 213.7 | 219.5 | 216.1 | 230.1 | 216.2 | 225 | 220 |
| 3/10 | 226.7 | 237.2 | 217.9 | 227.7 | 220.7 | 229.2 | 220   | 229 | 224 |
| 3/11 | 236.4 | 226.1 | 209.3 | 225.6 | 229.6 | 229   | 222.3 | 231 | 225 |
| 3/12 | 228.2 | 237.8 | 214.6 | 229   | 238.5 | 229.8 | 228.3 | 227 | 218 |
| 3/13 | 225.7 | 236.4 | 207.1 | 228.2 | 230.3 | 230.7 | 228.1 | 225 | 218 |
| 3/14 | 221.2 | 236.1 | 215.6 | 225.7 | 236.3 | 225.5 | 221   | 226 | 217 |
| 3/15 | 215.6 | 237.6 | 223.8 | 223.6 | 230.6 | 226.7 | 228   | 228 | 211 |
| 3/16 | 219.8 | 230.1 | 230.2 | 224.7 | 231.7 | 224.6 | 221   | 223 | 210 |
| 3/17 | 215.3 | 224.5 | 223.8 | 228.9 | 225.7 | 221   | 212.4 | 221 | 209 |
| 3/18 | 216.6 | 219.4 | 224   | 229.8 | 238.5 | 229.6 | 208.2 | 218 | 200 |
| 319  | 211.4 | 211.2 | 225.6 | 238.6 | 239.7 | 229.9 | 204.1 | 217 | 188 |

|      |       |       |       |       |       |       |       |     |     |
|------|-------|-------|-------|-------|-------|-------|-------|-----|-----|
| 3/20 | 211.1 | 200.9 | 232.2 | 235   | 232.9 | 238.6 | 210.3 | 216 | 204 |
| 3/21 | 225.9 | 226.9 | 229.7 | 224.8 | 236.1 | 238.1 | 218.7 | 219 | 219 |
| 3/22 | 236.2 | 214.4 | 231.5 | 230.4 | 238.4 | 233.3 | 217.4 | 226 | 204 |
| 3/23 | 241.5 | 215.7 | 235   | 237.9 | 243.6 | 220.5 | 224.1 | 229 | 202 |
| 3/24 | 238   | 225.3 | 234.2 | 243.5 | 240.8 | 214.6 | 233.4 | 226 | 203 |
| 3/25 | 246.6 | 226.6 | 249   | 237.6 | 228.4 | 191.4 | 240.2 | 218 | 207 |
| 3/26 | 245.3 | 231.6 | 252.1 | 232.4 | 228.1 | 195.8 | 241.6 | 214 | 218 |
| 3/27 | 242.7 | 233.5 | 238.7 | 229.7 | 228   | 210.9 | 235.6 | 209 | 216 |
| 3/28 | 240.6 | 239.4 | 249.1 | 219   | 237.4 | 216.5 | 239.8 | 221 | 222 |
| 3/29 | 235.4 | 244.6 | 245.9 | 225.3 | 249.6 | 224.7 | 240.8 | 228 | 224 |
| 3/30 | 264.9 | 244.1 | 248.3 | 216.4 | 246   | 230.4 | 246.4 | 223 | 219 |
| 3/31 | 242.2 | 250.3 | 244.9 | 221.6 | 246   | 229.3 | 235.9 | 204 | 223 |
| 4/1  | 242   | 247.7 | 245.8 | 225   | 246   | 233   | 239.9 | 203 | 231 |
| 4/2  | 243.6 | 248.7 | 231   | 219.5 | 245.2 | 236.1 | 237.6 | 217 | 223 |
| 4/3  | 246.3 | 252.9 | 231   | 221.5 | 248.2 | 236.8 | 233.2 | 236 | 225 |
| 4/4  | 244.5 | 231.8 | 237.3 | 225   | 246.5 | 233   | 234.6 | 227 | 216 |
| 4/5  | 251.9 | 230.4 | 240.2 | 233.6 | 246.9 | 215   | 238.1 | 227 | 225 |
| 4/6  | 249.6 | 241.8 | 244   | 222.2 | 238.3 | 217.3 | 233.3 | 224 | 228 |
| 4/7  | 257   | 249   | 240.1 | 228.3 | 235.7 | 225.7 | 234.6 | 218 | 224 |
| 4/8  | 255.4 | 255.3 | 230.6 | 246.9 | 233.6 | 239.3 | 236.5 | 220 | 217 |
| 4/9  | 249.6 | 234.7 | 227.8 | 242.3 | 230   | 235.1 | 225.1 | 218 | 220 |
| 4/10 | 247.6 | 232.6 | 240.5 | 246.8 | 237.8 | 228   | 221.1 | 223 | 219 |
| 4/11 | 241.6 | 241.5 | 241.7 | 233.6 | 230.9 | 238.2 | 221   | 230 | 229 |
| 4/12 | 235.6 | 256.8 | 237.2 | 233.9 | 245.1 | 242   | 224.9 | 234 | 235 |
| 4/13 | 242.2 | 260.1 | 237.8 | 237.9 | 250.5 | 227.9 | 234.4 | 224 | 236 |
| 4/14 | 247.5 | 242.8 | 238.7 | 241.9 | 247   | 211.5 | 241.9 | 223 | 238 |
| 4/15 | 240.8 | 237.5 | 249.1 | 236.5 | 246.2 | 210.3 | 239.6 | 221 | 235 |
| 4/16 | 239.4 | 239.5 | 250.6 | 229.8 | 242.9 | 203.8 | 238.4 | 221 | 234 |
| 4/17 | 237.5 | 249.3 | 247   | 228.4 | 246.9 | 199.2 | 238.9 | 228 | 236 |
| 4/18 | 236.3 | 253   | 251.7 | 233.6 | 256.4 | 212.4 | 225.6 | 216 | 231 |
| 4/19 | 233.7 | 245.9 | 251.9 | 240.2 | 249.8 | 223.8 | 226.7 | 227 | 227 |
| 4/20 | 230   | 244.4 | 244.1 | 233   | 241.3 | 223.2 | 238.7 | 232 | 226 |
| 4/21 | 223.7 | 242.5 | 251.9 | 232.2 | 235.6 | 223.8 | 236.6 | 224 | 226 |
| 4/22 | 227.9 | 255.4 | 252.8 | 231.3 | 244.5 | 221.6 | 225.9 | 221 | 233 |
| 4/23 | 227.9 | 243.3 | 246.7 | 237.3 | 244.9 | 229.8 | 236.2 | 204 | 230 |
| 4/24 | 233.5 | 232.1 | 235.2 | 248.1 | 259.1 | 224.5 | 228.2 | 201 | 227 |
| 4/25 | 240.9 | 239.1 | 229.5 | 255.9 | 252   | 217   | 227.3 | 214 | 218 |
| 4/26 | 247.3 | 246.9 | 231.5 | 248.1 | 237.1 | 221.7 | 231   | 226 | 231 |
| 4/27 | 247.3 | 248.5 | 232.7 | 243.3 | 247.2 | 229.8 | 232.6 | 227 | 235 |
| 4/28 | 242.4 | 231.6 | 239.2 | 242.3 | 249.4 | 237   | 233.7 | 227 | 229 |
| 4/29 | 244.4 | 222.5 | 251.6 | 250.7 | 249.9 | 240.1 | 219.8 | 232 | 226 |
| 4/30 | 237.8 | 220.5 | 253.5 | 247.8 | 245.3 | 243.2 | 221.5 | 232 | 227 |
| 5/1  | 237.8 | 233.5 | 251.5 | 242.5 | 253.7 | 243.8 | 214.1 | 223 | 227 |

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|------|-------|-------|-------|-------|-------|-------|-------|-----|-----|
| 5/2  | 253.6 | 236.2 | 243.3 | 241.3 | 241.4 | 247.1 | 215.6 | 214 | 225 |
| 5/3  | 258.6 | 240.8 | 250.1 | 243.4 | 234   | 254.1 | 235.5 | 221 | 222 |
| 5/4  | 257.4 | 251.9 | 243.8 | 243.9 | 234   | 243   | 239   | 224 | 221 |
| 5/5  | 257.4 | 254.3 | 247.1 | 245.5 | 244.2 | 233.6 | 235.9 | 224 | 219 |
| 5/6  | 257.4 | 256.4 | 255.6 | 243.5 | 255   | 240.2 | 240.7 | 236 | 222 |
| 5/7  | 257.4 | 251.4 | 247.3 | 247.8 | 242.5 | 244   | 251.9 | 220 | 227 |
| 5/8  | 257.4 | 250.3 | 239.2 | 235.4 | 236.6 | 236.9 | 254.3 | 217 | 229 |
| 5/9  | 257.4 | 239.1 | 243   | 220.2 | 238.5 | 240.1 | 278.4 | 225 | 220 |
| 5/10 | 257.4 | 241.4 | 236.7 | 212.2 | 246.7 | 227.5 | 257.7 | 233 | 223 |
| 5/11 | 275.1 | 241.1 | 238.5 | 226.9 | 249.3 | 243.6 | 241.5 | 233 | 228 |
| 5/12 | 264   | 236.3 | 238.9 | 238   | 233.4 | 241.6 | 223.1 | 239 | 229 |
| 5/13 | 264   | 240.8 | 250.7 | 240.3 | 232.3 | 252.1 | 231.6 | 231 | 234 |
| 5/14 | 270.3 | 257.4 | 260.4 | 251.8 | 243   | 264.6 | 235.1 | 237 | 236 |
| 5/15 | 266.9 | 253.2 | 256.2 | 248.8 | 251.6 | 250   | 240.5 | 235 | 231 |
| 5/16 | 256   | 245.2 | 246.6 | 238   | 252.7 | 247.6 | 247.2 | 234 | 221 |
| 5/17 | 252.9 | 238   | 225   | 243.5 | 251.4 | 261.3 | 246.3 | 241 | 234 |
| 5/18 | 253.8 | 245.1 | 235.2 | 243.1 | 262.8 | 259.5 | 244.5 | 229 | 239 |
| 5/19 | 253.9 | 255.2 | 242.9 | 253.2 | 248.1 | 255.2 | 234.7 | 235 | 240 |
| 5/20 | 242.2 | 248.8 | 245.3 | 253.8 | 239.4 | 251.9 | 224.7 | 249 | 241 |
| 5/21 | 232.6 | 244.2 | 242.7 | 261.8 | 230.5 | 245.7 | 207.7 | 245 | 232 |
| 5/22 | 233.7 | 246.4 | 245.3 | 246.3 | 249.7 | 243.6 | 214.2 | 238 | 242 |
| 5/23 | 246.5 | 236.7 | 240.4 | 223.3 | 260.1 | 240   | 225.8 | 226 | 237 |
| 5/24 | 248.5 | 233   | 236.9 | 233.3 | 263.2 | 244.7 | 236.2 | 230 | 253 |
| 5/25 | 251.2 | 239.3 | 240.1 | 240.9 | 264.6 | 238.7 | 234.5 | 241 | 242 |
| 5/26 | 251.1 | 250   | 233   | 240.2 | 251.6 | 227.9 | 241.2 | 239 | 240 |
| 5/27 | 251.8 | 255.5 | 237.7 | 255.7 | 255.8 | 232.7 | 233.2 | 247 | 240 |
| 5/28 | 242.5 | 270   | 238.7 | 253.6 | 247.3 | 227.4 | 239.3 | 245 | 235 |
| 5/29 | 246.8 | 270.8 | 235.8 | 261   | 251.8 | 239.6 | 232.8 | 238 | 228 |
| 5/30 | 232.1 | 273.3 | 243.8 | 257.1 | 257.4 | 246.2 | 220.4 | 239 | 219 |
| 5/31 | 241.4 | 269.9 | 236.1 | 244.5 | 250.8 | 245.3 | 231.9 | 241 | 225 |
| 6/1  | 238   | 267.6 | 238.4 | 238.6 | 243.1 | 257.6 | 236.7 | 239 | 221 |
| 6/2  | 239.6 | 264   | 246.7 | 246.8 | 228.9 | 258.7 | 242.4 | 242 | 224 |
| 6/3  | 249.3 | 252.2 | 256.3 | 252.4 | 221.1 | 256.6 | 246.8 | 250 | 237 |
| 6/4  | 255.3 | 256.8 | 251.4 | 244.3 | 225.7 | 256.9 | 242.3 | 244 | 233 |
| 6/5  | 249.6 | 254.6 | 254.2 | 246.4 | 233   | 246   | 232.6 | 254 | 222 |
| 6/6  | 253.8 | 258   | 265   | 242.2 | 229.1 | 249.9 | 229   | 261 | 234 |
| 6/7  | 258.7 | 254.1 | 256.4 | 244.4 | 214.9 | 260.2 | 225.9 | 222 | 225 |
| 6/8  | 248.7 | 260.3 | 257.7 | 241.7 | 219.7 | 264   | 225.2 | 236 | 237 |
| 6/9  | 236.4 | 257.7 | 261.8 | 235.7 | 228.6 | 242.8 | 239.9 | 248 | 243 |
| 6/10 | 238.4 | 242.3 | 263.5 | 234.4 | 236.7 | 238.3 | 250.2 | 254 | 259 |
| 6/11 | 248.3 | 254.6 | 268.6 | 226.2 | 244.8 | 230.2 | 244.5 | 239 | 242 |
| 6/12 | 248.8 | 258.8 | 276.8 | 237.8 | 244.2 | 231.6 | 245.7 | 247 | 256 |
| 6/13 | 252.1 | 254.3 | 271.5 | 246.6 | 241.7 | 231.9 | 238.9 | 249 | 245 |

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|------|-------|-------|-------|-------|-------|-------|-------|-----|-----|
| 6/14 | 237.9 | 254.8 | 248.4 | 240.6 | 230.2 | 217.1 | 224.1 | 246 | 244 |
| 6/15 | 252.9 | 251.8 | 251.2 | 230.7 | 236.3 | 261.7 | 221.7 | 250 | 226 |
| 6/16 | 262.4 | 251.2 | 259.7 | 244.3 | 250.2 | 244.7 | 213.3 | 236 | 259 |
| 6/17 | 266.8 | 247.9 | 257.6 | 236.2 | 252.4 | 236.3 | 227.2 | 234 | 253 |
| 6/18 | 254.4 | 255   | 261   | 232.1 | 253.7 | 261.9 | 240.4 | 221 | 253 |
| 6/19 | 255.7 | 257.4 | 266.5 | 227.3 | 257.1 | 270.6 | 232.2 | 222 | 253 |
| 6/20 | 241.8 | 256.3 | 260.6 | 236.5 | 253.1 | 253.7 | 230.5 | 235 | 252 |
| 6/21 | 245.9 | 252.1 | 260.1 | 251   | 251.5 | 251.6 | 226.2 | 245 | 254 |
| 6/22 | 252.2 | 256   | 263.2 | 249.4 | 248.7 | 244.1 | 230.7 | 257 | 251 |
| 6/23 | 248.1 | 254.4 | 262.3 | 238.1 | 257.6 | 229.9 | 238.8 | 241 | 259 |
| 6/24 | 245.7 | 252.5 | 268.3 | 228.6 | 261   | 227.6 | 249.2 | 248 | 251 |
| 6/25 | 238.9 | 243.5 | 260.2 | 231.3 | 247.7 | 238.1 | 242   | 243 | 262 |
| 6/26 | 251.5 | 246.2 | 259.3 | 235   | 250.3 | 245.9 | 231.8 | 230 | 250 |
| 6/27 | 244.1 | 247.6 | 255.1 | 237.3 | 261.8 | 244.4 | 236.7 | 218 | 246 |
| 6/28 | 242.4 | 249.5 | 255   | 226.1 | 253.3 | 240.9 | 233   | 230 | 239 |
| 6/29 | 229.5 | 251.5 | 255.8 | 215   | 253.1 | 240.2 | 221.2 | 230 | 237 |
| 6/30 | 211.1 | 248.7 | 266.2 | 239.3 | 248.3 | 225.1 | 221.8 | 229 | 239 |
| 7/1  | 224.4 | 248.2 | 263.7 | 243.8 | 245.7 | 216.4 | 216.5 | 228 | 225 |
| 7/2  | 228.8 | 252.7 | 261.6 | 240   | 253.8 | 222.4 | 227.1 | 237 | 222 |
| 7/3  | 211.7 | 265.5 | 246.2 | 253.3 | 247.7 | 221.3 | 234.1 | 240 | 237 |
| 7/4  | 233.1 | 261.2 | 237.4 | 253.9 | 244   | 210.1 | 233.2 | 241 | 242 |
| 7/5  | 240.1 | 255.5 | 237.1 | 254.8 | 251.8 | 224.9 | 233.1 | 231 | 252 |
| 7/6  | 251   | 248.1 | 233.9 | 251   | 242.6 | 228.1 | 228.3 | 229 | 239 |
| 7/7  | 253.4 | 248.8 | 246.5 | 227.5 | 246.8 | 220.2 | 233.6 | 218 | 241 |
| 7/8  | 246.8 | 234.8 | 246.9 | 243.6 | 229.7 | 215.3 | 242.5 | 226 | 237 |
| 7/9  | 235.2 | 230.3 | 247.8 | 244.7 | 226.8 | 223.7 | 239.8 | 239 | 238 |
| 7/10 | 248.4 | 245.7 | 242.5 | 247.3 | 239.5 | 228.6 | 234.5 | 236 | 243 |
| 7/11 | 252   | 248.3 | 229.5 | 251   | 239.1 | 221.1 | 231.7 | 205 | 246 |
| 7/12 | 241.8 | 261.4 | 226.8 | 237.7 | 234.3 | 220.2 | 221   | 201 | 255 |
| 7/13 | 260.8 | 251.7 | 246.6 | 239.5 | 202.7 | 216.9 | 202.2 | 243 | 251 |
| 7/14 | 254.7 | 241.7 | 242.2 | 233.6 | 214.7 | 216.2 | 195   | 240 | 245 |
| 7/15 | 257.3 | 234.1 | 239.7 | 233.1 | 212.6 | 206.6 | 209.9 | 248 | 255 |
| 7/16 | 247.6 | 234.8 | 243.9 | 215.8 | 222.3 | 212.4 | 211.5 | 234 | 239 |
| 7/17 | 243.2 | 238.9 | 240   | 212.8 | 216.6 | 202.7 | 222.1 | 239 | 241 |
| 7/18 | 240.7 | 239   | 252.6 | 213   | 221.6 | 214.7 | 225.9 | 240 | 228 |
| 7/19 | 238.6 | 234.3 | 252.9 | 218.5 | 232.1 | 223.6 | 217.4 | 250 | 254 |
| 7/20 | 244.4 | 236.4 | 254.5 | 215   | 236.7 | 212.1 | 217.1 | 242 | 260 |
| 7/21 | 239.4 | 240   | 252.5 | 227.5 | 222   | 212   | 220.9 | 236 | 252 |
| 7/22 | 225   | 229.8 | 226.2 | 233.4 | 219.7 | 192.4 | 221   | 253 | 257 |
| 7/23 | 209.3 | 218.2 | 215.9 | 227.6 | 223.1 | 192.9 | 220.2 | 262 | 235 |
| 7/24 | 214.6 | 212.6 | 219.9 | 234.2 | 222.2 | 207.1 | 210.6 | 242 | 239 |
| 7/25 | 215.7 | 218.4 | 238   | 234.8 | 218.1 | 214.1 | 207   | 242 | 236 |
| 7/26 | 229.3 | 218.9 | 242.7 | 229.5 | 221.2 | 215.6 | 207.8 | 230 | 253 |

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| 7/27 | 228.3 | 226.8 | 236.7 | 216.6 | 226.5 | 212.8 | 194.6 | 237 | 249 |
| 7/28 | 228.8 | 223.8 | 237.7 | 209.8 | 224.5 | 198   | 190.2 | 242 | 254 |
| 7/29 | 235.2 | 229.5 | 234.8 | 215.5 | 210.8 | 206.6 | 183.8 | 245 | 250 |
| 7/30 | 221.7 | 228.3 | 238.9 | 213.5 | 211.7 | 203.8 | 185.5 | 245 | 259 |
| 7/31 | 230.6 | 232   | 234.3 | 209.2 | 212.7 | 192.2 | 189   | 243 | 248 |
| 8/1  | 235.6 | 220.2 | 230.9 | 209   | 208.2 | 177.9 | 194.5 | 239 | 242 |
| 8/2  | 234.3 | 222.4 | 231.8 | 205.9 | 193.9 | 183.5 | 191   | 236 | 258 |
| 8/3  | 225.4 | 205.6 | 224.9 | 204.4 | 195.6 | 192.5 | 185.5 | 214 | 239 |
| 8/4  | 218.8 | 214.5 | 217.1 | 210.4 | 192   | 187.4 | 178.8 | 226 | 242 |
| 8/5  | 231.5 | 215.7 | 222.5 | 210.9 | 197.5 | 194.1 | 171.1 | 218 | 225 |
| 8/6  | 231.9 | 212   | 218.2 | 208.6 | 186.9 | 190.1 | 183.6 | 206 | 229 |
| 8/7  | 213.2 | 211.2 | 222   | 199.5 | 187   | 187.7 | 177.6 | 203 | 240 |
| 8/8  | 209.1 | 207.1 | 217.3 | 195.2 | 183.9 | 187.9 | 172.4 | 175 | 238 |
| 8/9  | 197.3 | 212.5 | 222.6 | 191.1 | 188.7 | 188   | 170.5 | 229 | 234 |
| 8/10 | 203.3 | 212.1 | 226.8 | 188.7 | 191.3 | 188   | 172.5 | 214 | 230 |
| 8/11 | 213.3 | 207.3 | 231.7 | 188.1 | 196.6 | 189.6 | 171.3 | 220 | 229 |
| 8/12 | 215.4 | 199.2 | 226.4 | 189.6 | 189.8 | 186.8 | 169.5 | 236 | 233 |
| 8/13 | 215.9 | 202.2 | 225.3 | 202.5 | 182.1 | 187   | 173.8 | 215 | 226 |
| 8/14 | 212.8 | 198.9 | 220.4 | 199.7 | 189.9 | 184.6 | 171.6 | 198 | 225 |
| 8/15 | 213   | 197.4 | 220.1 | 189.7 | 190.8 | 184.1 | 161.7 | 190 | 227 |
| 8/16 | 217.7 | 210.4 | 216.9 | 183.7 | 188.6 | 180.9 | 162.7 | 203 | 221 |
| 8/17 | 223.4 | 210.1 | 216.2 | 194.1 | 185.8 | 181   | 163.7 | 211 | 229 |
| 8/18 | 220.7 | 207.7 | 216.8 | 179.1 | 187.5 | 174.7 | 163.9 | 179 | 213 |
| 8/19 | 217   | 205.6 | 213   | 183.7 | 184.8 | 172.6 | 156.9 | 179 | 204 |
| 8/20 | 216.2 | 201.2 | 213.8 | 188.6 | 185.7 | 170.6 | 152.3 | 173 | 202 |
| 8/21 | 227   | 196.3 | 210   | 188.1 | 183.6 | 169.3 | 152.1 | 161 | 191 |
| 8/22 | 233.3 | 210.2 | 216.3 | 182.5 | 178.4 | 169.9 | 143.4 | 163 | 210 |
| 8/23 | 235.4 | 201.4 | 216.8 | 181.3 | 181.2 | 165.3 | 141.5 | 127 | 176 |
| 8/24 | 239   | 201.1 | 209.9 | 183.4 | 178.7 | 160.4 | 134.1 | 134 | 164 |
| 8/25 | 239   | 203.4 | 207.6 | 181.5 | 176.6 | 162.4 | 136.4 | 160 | 167 |
| 8/26 | 232.7 | 202.3 | 208.7 | 182.7 | 174.6 | 148   | 144.7 | 130 | 173 |
| 8/27 | 231.4 | 206   | 208.2 | 184.5 | 164.7 | 147.2 | 149.6 | 152 | 202 |
| 8/28 | 231.4 | 206   | 208.8 | 184.9 | 156.3 | 152.5 | 142.9 | 171 | 190 |
| 8/29 | 224   | 203.6 | 209   | 183.4 | 156.9 | 157.6 | 140.6 | 170 | 216 |
| 8/30 | 220.9 | 204.7 | 205.9 | 181.5 | 162.5 | 154   | 138.6 | 176 | 169 |
| 8/31 | 221   | 202.6 | 202.8 | 180.3 | 162.1 | 157.1 | 142   | 192 | 167 |
| 9/1  | 216.3 | 212.3 | 197.5 | 185.6 | 165.2 | 156.2 | 145.9 | 171 | 161 |
| 9/2  | 216.8 | 213.6 | 195.5 | 188.3 | 165.8 | 152.9 | 146.8 | 166 | 165 |
| 9/3  | 222.5 | 209.2 | 195.9 | 188.8 | 160.5 | 149.8 | 146.2 | 169 | 183 |
| 9/4  | 229.9 | 198.9 | 196.8 | 189   | 157.7 | 156.2 | 138.2 | 150 | 184 |
| 9/5  | 226.9 | 194.3 | 195.4 | 193.7 | 163.4 | 161.5 | 139.6 | 162 | 175 |
| 9/6  | 230.9 | 195.6 | 204.5 | 186.9 | 162.3 | 161.9 | 134.4 | 170 | 163 |
| 9/7  | 236.5 | 194.4 | 215.9 | 176   | 158.1 | 167.5 | 138   | 153 | 162 |

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| 9/8   | 235.3 | 192.5 | 223.8 | 175.2 | 159.6 | 161.6 | 133.3 | 145 | 156 |
| 9/9   | 226.4 | 202.3 | 216.1 | 180.5 | 159.1 | 161.9 | 133.8 | 157 | 157 |
| 9/10  | 228.4 | 205.2 | 213.7 | 184.8 | 155.9 | 164.3 | 134   | 159 | 159 |
| 9/11  | 233.6 | 206.6 | 212.4 | 189.7 | 161.5 | 168   | 135.6 | 143 | 159 |
| 9/12  | 241.8 | 200.6 | 212.1 | 200.9 | 168.9 | 166.4 | 134.3 | 148 | 166 |
| 9/13  | 247.4 | 204.1 | 214.4 | 206.5 | 164.3 | 162.2 | 132.5 | 148 | 155 |
| 9/14  | 260.4 | 207.9 | 219.6 | 210.8 | 164.1 | 162.8 | 130.5 | 152 | 146 |
| 9/15  | 262.4 | 208   | 216   | 214.1 | 164.8 | 159.8 | 130.1 | 153 | 140 |
| 9/16  | 257.4 | 201.7 | 212.9 | 206.2 | 165.7 | 160   | 125.3 | 147 | 144 |
| 9/17  | 257.9 | 200.4 | 211.4 | 192.7 | 170.7 | 166.3 | 126.6 | 148 | 154 |
| 9/18  | 259.5 | 194.6 | 209.5 | 189   | 174.8 | 168.4 | 128.5 | 146 | 145 |
| 9/19  | 259.9 | 193.3 | 207.5 | 186.7 | 177.3 | 172.8 | 130.5 | 139 | 143 |
| 9/20  | 258.4 | 200.9 | 207.1 | 183   | 178.6 | 172.2 | 131.7 | 134 | 143 |
| 9/21  | 255   | 201.8 | 213.3 | 185.4 | 177.4 | 176   | 139.8 | 135 | 141 |
| 9/22  | 244   | 198   | 211.5 | 186.6 | 176.3 | 173.6 | 149.4 | 133 | 140 |
| 9/23  | 242.4 | 195.7 | 214.2 | 186.9 | 177.6 | 184.9 | 157.7 | 126 | 139 |
| 9/24  | 254.6 | 192.8 | 217.2 | 176   | 172.4 | 192   | 161.9 | 133 | 136 |
| 9/25  | 258.1 | 196.9 | 222.5 | 171.3 | 182.3 | 196.7 | 166.7 | 129 | 130 |
| 9/26  | 258   | 208.7 | 227.6 | 174.2 | 196.2 | 194.6 | 169.3 | 128 | 127 |
| 9/27  | 254.9 | 211.3 | 227.1 | 196   | 207   | 196.5 | 170.6 | 130 | 123 |
| 9/28  | 250.3 | 211.1 | 227.8 | 209.3 | 219.6 | 193.8 | 172.5 | 126 | 122 |
| 9/29  | 250.8 | 219.6 | 228.7 | 207.5 | 232.5 | 212.8 | 172.1 | 118 | 120 |
| 9/30  | 251.8 | 222.3 | 228.2 | 203.2 | 248.6 | 206.7 | 171.2 | 114 | 124 |
| 10/1  | 253.5 | 216.6 | 227.3 | 192.1 | 250.5 | 211.6 | 175   | 116 | 120 |
| 10/2  | 259.4 | 213   | 224.7 | 189.7 | 244.6 | 215.1 | 172.6 | 121 | 113 |
| 10/3  | 260.7 | 224   | 220.2 | 190.7 | 248.1 | 223.6 | 176.1 | 118 | 107 |
| 10/4  | 260.1 | 230.3 | 215.3 | 194.9 | 244.1 | 215.3 | 183.1 | 124 | 101 |
| 10/5  | 259.2 | 234   | 222.1 | 195.8 | 253.4 | 199.4 | 197.9 | 126 | 106 |
| 10/6  | 267.7 | 229.3 | 237.7 | 214   | 253.9 | 202.2 | 197.2 | 128 | 109 |
| 10/7  | 261.7 | 233.8 | 246.6 | 219.5 | 256.3 | 212.2 | 193.9 | 129 | 108 |
| 10/8  | 254.9 | 237.1 | 251.6 | 222.2 | 257.6 | 216.8 | 196.3 | 128 | 103 |
| 10/9  | 257.3 | 245.6 | 257.4 | 225.2 | 249.3 | 217.7 | 192.1 | 129 | 103 |
| 10/10 | 268.1 | 252.2 | 257.1 | 225.8 | 248.3 | 216.4 | 183.4 | 133 | 106 |
| 10/11 | 279.8 | 254.4 | 261.7 | 224.4 | 248.8 | 217.6 | 177.6 | 132 | 103 |
| 10/12 | 294.9 | 251.7 | 265.1 | 224.1 | 249.8 | 219.5 | 181.1 | 134 | 106 |
| 10/13 | 293.4 | 257.4 | 263.4 | 234.2 | 249.2 | 211.3 | 177.1 | 137 | 112 |
| 10/14 | 289.9 | 256.3 | 261.5 | 239.5 | 252.2 | 208.7 | 174.7 | 138 | 113 |
| 10/15 | 286.1 | 244.3 | 260.3 | 242.3 | 255.2 | 216.8 | 174.9 | 139 | 105 |
| 10/16 | 279.7 | 249.6 | 262.4 | 247.5 | 259   | 220.9 | 178.9 | 139 | 111 |
| 10/17 | 272.1 | 256.2 | 259.7 | 252.6 | 253.5 | 217.1 | 185.3 | 147 | 116 |
| 10/18 | 275.9 | 255.3 | 272.5 | 253.7 | 256.2 | 213.9 | 193.7 | 146 | 113 |
| 10/19 | 271.3 | 254.3 | 267.4 | 251.6 | 260   | 218.7 | 194.7 | 136 | 111 |
| 10/20 | 261.6 | 254.8 | 260.8 | 258.2 | 263.9 | 207.9 | 197.3 | 129 | 113 |



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| 10/21 | 273.7 | 252.6 | 272.7 | 271.4 | 275   | 222.1 | 205.7 | 133 | 111 |
| 10/22 | 285.7 | 250.6 | 275.3 | 274.2 | 275   | 224.4 | 211.4 | 136 | 115 |
| 10/23 | 285.9 | 253.3 | 277.4 | 278.8 | 271.1 | 224.1 | 208.7 | 132 | 109 |
| 10/24 | 281.3 | 253.8 | 275.5 | 277.4 | 264.7 | 220.9 | 208.2 | 133 | 110 |
| 10/25 | 281.8 | 264.2 | 275.1 | 255.9 | 270.4 | 217.1 | 204.9 | 137 | 111 |
| 10/26 | 278.8 | 266.4 | 271.1 | 260.7 | 269.3 | 209.2 | 209.7 | 138 | 114 |
| 10/27 | 284.5 | 269.2 | 271   | 266.4 | 268.3 | 218.4 | 213.9 | 140 | 119 |
| 10/28 | 283.3 | 269   | 263.9 | 266.9 | 264.1 | 223.6 | 221   | 140 | 127 |
| 10/29 | 284.6 | 270.6 | 275.8 | 264.6 | 256.2 | 231   | 232   | 143 | 124 |
| 10/30 | 286.5 | 270.1 | 275.2 | 261   | 249   | 241.2 | 246.9 | 136 | 120 |
| 10/31 | 285.3 | 266.1 | 275.8 | 257.1 | 242.7 | 246.5 | 242.3 | 139 | 121 |
| 11/1  | 281.9 | 266   | 274.4 | 254.7 | 242.9 | 239.8 | 231.1 | 148 | 124 |
| 11/2  | 286.7 | 269.9 | 267.8 | 267.5 | 245.3 | 240.7 | 245.9 | 147 | 126 |
| 11/3  | 276.7 | 278.6 | 271.1 | 268.7 | 256.8 | 238.6 | 249.1 | 149 | 130 |
| 11/4  | 272.2 | 277.3 | 259.3 | 266.6 | 258.5 | 242   | 250.6 | 150 | 132 |
| 11/5  | 273.6 | 273.2 | 253.6 | 271.6 | 266   | 244.4 | 254   | 159 | 136 |
| 11/6  | 274.7 | 266.8 | 252.3 | 273.5 | 269.9 | 239.4 | 249.3 | 158 | 133 |
| 11/7  | 276.5 | 254.4 | 255.2 | 261.3 | 268.4 | 249.3 | 246.7 | 159 | 126 |
| 11/8  | 267.5 | 257.2 | 260.5 | 257.9 | 267.3 | 257.7 | 250.1 | 160 | 131 |
| 11/9  | 273.4 | 255.5 | 271.9 | 254.8 | 265.5 | 243   | 238.3 | 164 | 153 |
| 11/10 | 271.5 | 259.8 | 266.5 | 262   | 265.1 | 253.2 | 241.2 | 168 | 154 |
| 11/11 | 275   | 265.4 | 270   | 267.5 | 267.4 | 253   | 248.1 | 167 | 149 |
| 11/12 | 275.8 | 262.7 | 269.2 | 270.2 | 267.1 | 253   | 247.2 | 170 | 146 |
| 11/13 | 266.5 | 268.4 | 268.3 | 271.6 | 263.9 | 249.9 | 252.5 | 179 | 149 |
| 11/14 | 269.3 | 265   | 266.5 | 275.1 | 253.8 | 257.8 | 254.5 | 181 | 149 |
| 11/15 | 280.8 | 266.6 | 273.2 | 271.9 | 249.2 | 259.5 | 256.5 | 187 | 150 |
| 11/16 | 282.5 | 271.6 | 269.1 | 268.1 | 257.7 | 262.3 | 256.1 | 186 | 143 |
| 11/17 | 271.1 | 270.3 | 273.7 | 269.6 | 259.5 | 255   | 251.3 | 184 | 144 |
| 11/18 | 271.8 | 269.3 | 277.1 | 267.6 | 263   | 247.9 | 253.4 | 184 | 149 |
| 11/19 | 279.8 | 264.4 | 273.1 | 271.8 | 266.1 | 259.8 | 255.4 | 182 | 152 |
| 11/20 | 276.8 | 266.4 | 266   | 270.4 | 266.8 | 259.9 | 255.9 | 184 | 153 |
| 11/21 | 275.4 | 262.2 | 260.5 | 266.9 | 271.7 | 256.9 | 256   | 186 | 153 |
| 11/22 | 273.5 | 261.3 | 266.4 | 269.3 | 268.8 | 256.2 | 252.9 | 187 | 146 |
| 11/23 | 278.6 | 259.5 | 268.4 | 269.9 | 266.6 | 253.7 | 244.3 | 190 | 145 |
| 11/24 | 278.1 | 259.9 | 269.7 | 268.4 | 266.1 | 248.4 | 245.6 | 196 | 150 |
| 11/25 | 277.2 | 263.1 | 260.5 | 264.2 | 258.2 | 248.9 | 246.7 | 195 | 156 |
| 11/26 | 266.1 | 267.7 | 261.7 | 264.8 | 254.9 | 250.5 | 253.2 | 192 | 158 |
| 11/27 | 260.5 | 269.5 | 267.4 | 261.8 | 245.6 | 247.8 | 250.7 | 197 | 160 |
| 11/28 | 263.3 | 271.5 | 269.4 | 254.9 | 241.2 | 251.9 | 255.6 | 193 | 161 |
| 11/29 | 261.5 | 268.7 | 273   | 261.3 | 245.7 | 254.3 | 255.9 | 206 | 158 |
| 11/30 | 272.1 | 268.9 | 276.9 | 262.6 | 255.4 | 259.6 | 251.3 | 215 | 159 |
| 12/1  | 261.8 | 267.4 | 277   | 259   | 257.4 | 257.6 | 247.9 | 215 | 162 |
| 12/2  | 264.3 | 264   | 267.6 | 262.9 | 258.7 | 257.9 | 250.3 | 217 | 162 |

|       |       |       |       |       |       |       |       |     |     |
|-------|-------|-------|-------|-------|-------|-------|-------|-----|-----|
| 12/3  | 269.6 | 268.7 | 264.8 | 264.6 | 255.8 | 259.5 | 246.9 | 206 | 157 |
| 12/4  | 266   | 263.4 | 262.6 | 261.8 | 256.7 | 258.3 | 248.6 | 212 | 156 |
| 12/5  | 254.2 | 270.9 | 264.5 | 246.3 | 246   | 251.8 | 246.6 | 222 | 159 |
| 12/6  | 251.7 | 266.3 | 266.5 | 241.4 | 246.8 | 250.4 | 246.1 | 225 | 155 |
| 12/7  | 258.2 | 264.5 | 266.1 | 247.4 | 246.2 | 254.8 | 246.8 | 229 | 151 |
| 12/8  | 260.4 | 259.4 | 255.8 | 251.8 | 226.4 | 255.7 | 246.2 | 224 | 152 |
| 12/9  | 256.9 | 252   | 246.6 | 255.1 | 229.2 | 257.5 | 244.5 | 214 | 161 |
| 12/10 | 248.4 | 260.6 | 251.6 | 256.6 | 230.6 | 262.6 | 241.7 | 216 | 170 |
| 12/11 | 245.7 | 253.9 | 253.5 | 250.6 | 234.1 | 263.7 | 243.5 | 215 | 171 |
| 12/12 | 252.2 | 250.8 | 256.2 | 250.9 | 237.9 | 259.2 | 236.8 | 231 | 180 |
| 12/13 | 250.5 | 247   | 260.8 | 259.6 | 256   | 259   | 235.4 | 238 | 185 |
| 12/14 | 254   | 249.4 | 259.4 | 256   | 255.2 | 255.1 | 245.3 | 251 | 189 |
| 12/15 | 246.9 | 252.2 | 252.8 | 252.9 | 256.6 | 253.5 | 245.8 | 250 | 204 |
| 12/16 | 249.3 | 253.6 | 249   | 242   | 258.5 | 253.1 | 242.8 | 261 | 200 |
| 12/17 | 253.8 | 256.3 | 246.7 | 249.8 | 257.3 | 248.3 | 239   | 251 | 202 |
| 12/18 | 258.7 | 252.9 | 241.4 | 247.6 | 252.4 | 241.8 | 239   | 253 | 205 |
| 12/19 | 257.4 | 247.5 | 237.2 | 253.4 | 248.9 | 231.8 | 231.9 | 253 | 203 |
| 12/20 | 255.5 | 247.9 | 234.7 | 256.2 | 252.9 | 229.6 | 231.2 | 245 | 211 |
| 12/21 | 252.8 | 256.6 | 239.6 | 256.8 | 255.3 | 240.1 | 234.2 | 256 | 223 |
| 12/22 | 246.7 | 245.9 | 234.4 | 249.9 | 257.4 | 239.2 | 234.8 | 243 | 234 |
| 12/23 | 241.4 | 244.4 | 231   | 253.9 | 254   | 242.2 | 236.5 | 240 | 238 |
| 12/24 | 251.3 | 251.1 | 234.1 | 252.4 | 245.4 | 246.7 | 229.1 | 250 | 235 |
| 12/25 | 248.7 | 259.7 | 234   | 252.9 | 237.2 | 246.9 | 233.7 | 257 | 236 |
| 12/26 | 246.6 | 255.2 | 232.4 | 251.4 | 237.1 | 246.2 | 232.4 | 254 | 239 |
| 12/27 | 255.5 | 248.8 | 229.7 | 244   | 241.7 | 249.2 | 234.4 | 254 | 250 |
| 12/28 | 260.6 | 254.4 | 233.1 | 237.7 | 241.2 | 249.8 | 231   | 251 | 243 |
| 12/29 | 258.6 | 254.1 | 234.6 | 247.4 | 239.5 | 245.3 | 227.9 | 251 | 249 |
| 12/30 | 249.5 | 248.5 | 236.5 | 247.9 | 244.6 | 241.1 | 229.5 | 248 | 250 |
| 12/31 | 242   | 246.5 | 238.5 | 249.5 | 241   | 239.5 | 233.8 | 253 | 252 |

# **Problem A: Predictions of Ozone Depletion**

## **Abstract**

It is believed that the primary cause of ozone depletion is the presence of chlorine-containing source gases under the context of Montreal protocol. Accomplishing the prediction of ozone level in our model, several works have been done: data collection for the prediction, definition of the ozone level, how to interpret the factors related to ozone level, and reliability and feasibility of the prediction methodology.

We examine several indexes that reflect ozone level: Total ozone concentration, ozone hole area, equivalent effective stratospheric chlorine, and ozone depleting gas index. The index of ozone itself is bound to massive detected data, which fails to provide concise results yet causes instability of the prediction. By comparison and compromise, we analyzed to decide EESC and ODGI as indexes of ozone level.

In basic model, by applying ARIMA algorithm, we test and get the parameters for prediction of EESC and ODGI at mid latitude and the Antarctic: for EESC-A,  $p=1, d=2, q=0$ ; EESC-ML:  $p=1, d=1, q=1$ ; ODGI-A:  $p=2, d=1, q=2$ ; ODGI-ML:  $p=1, d=1, q=1$ . The results of ARIMA prediction coincide with the Montreal Protocol. EESC at the Antarctic is declining in 50 years, and it will drop below 2,500 ppt after 2050, which indicates EESC-A level can be as less as EESC at mid latitude. ODGI at the Antarctic has a massive drop during 2020-2030, and steady decline after 2040. The ozone hole will be recovered gradually in the following half centuries.

In improved model, not only the indexes of ozone depletion substances but also ozone hole area have been examined. By applying RBF model, we get the changing trend about EESC and ODGI value, which coincide with basic model. However the results of improved model draw more elaborative conclusion with detailed values: We predict the Antarctic ozone hole area can recover to  $5.2 \times 10^6 \text{ km}^2$  in 2066 and will disappear after 2075. The Antarctica EESC values will return to 1980's level reaching 2075. The Antarctic EESC will decline to 1982's level and the mid latitude EESC values will return to 1980's level when reaching 2049. Mid latitude Ozone hole will disappear after 2049.

**Key words: Ozone Level Prediction; EESC; ODGI; ARIMA; RBF**

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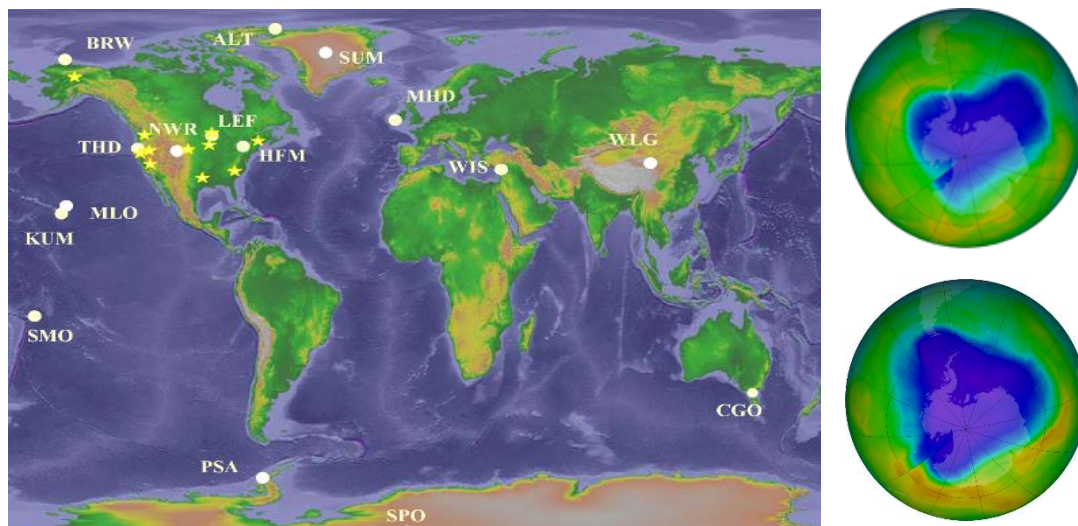
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## I. Introduction

The international concern following the discovery of Antarctic stratospheric ozone depletion has prompted unprecedented international action by governments to control the production, sales and usage of a range of ozone-depleting chemicals. These international treaty obligations include the Montreal Protocol and its London and Copenhagen Amendments. In order to indicate the origin of ozone depletion, the following background is worth mentioning.

### 1.1 Ozone depletion phenomena

Ozone depletion includes several phenomena observed since the late 1970s: Vernazza et al<sup>[1]</sup> analyzed the ozone observation data of 1957-1975 on the Northern Hemisphere, and it turned out that the minimum amount of ozone during September to October has increased, and in the seasonal variation of the maximum amount (from February to April) didn't have such obvious change. Stolarski et al<sup>[2]</sup> dealt with TOMS data from 1978 to 1990, to calculate average annual change of ozone of different latitude worldwide. Near the equator the value is approximate to zero, which increases gradually to the poles. Ozone decreases mainly in the winter and spring season. Randcl et al<sup>[3]</sup> analyzed the global ozone changes from 1991 to 1994, and found that in the four years, the biggest change area is the mid-latitude of northern hemisphere during winter and spring season every year especially during the period of 1992 to 1993. Xu et al<sup>[4]</sup> analyzed total polar atmospheric ozone variation characteristics, and the results showed that the polar zone of total atmospheric ozone have been falling during the past ten years, but exists inflection phenomenon<sup>[5][6]</sup>. However, they have different geographic distribution characteristics, existing periodic oscillation phenomena of the arctic and Antarctic area.



**Figure 1.1** Locations where ozone-depleting chemicals are regularly made by NOAA/ESRL, and it combines NASA ozone watch on the left. (The Fig is from <http://www.esrl.noaa.gov/gmd/odgi/>)

## 1.2 Montreal Protocol

The Montreal Protocol on Substances that Deplete the Ozone Layer (a protocol to the Vienna Convention for the Protection of the Ozone Layer) is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances that are responsible for ozone depletion<sup>[7]</sup>. It was agreed on 26 August 1987, and entered into force on 26 August 1989, followed by a first meeting in Helsinki, May 1989. Since then, it has undergone eight revisions, in 1990 (London), 1991 (Nairobi), 1992 (Copenhagen), 1993 (Bangkok), 1995 (Vienna), 1997 (Montreal), 1998 (Australia), 1999 (Beijing) and 2016 (Kigali, adopted, but not in force). As a result of the international agreement, the ozone hole in Antarctica is slowly recovering. Climate projections indicate that the ozone layer will return to 1980 levels between 2050 and 2070.

The two ozone treaties have been ratified by 197 parties, which includes 196 states and the European Union, making them the first universally ratified treaties in United Nations history.

These truly universal treaties have also been remarkable in the expedience of the policy making process at the global scale, where bare 14 years lapsed between a basic scientific research discovery (1973) and the international agreement signed (1985 & 1987) <sup>[8]</sup>. When comparing this very success story with attempts to establish an international policy on the Earth's climate or atomic energy, the entire process from a problem formulation to a global acceptance supported by a legal framework took less than a quarter of a single human generation life span.

## 1.3 Prime concern on chlorofluorocarbons and chlorocarbons

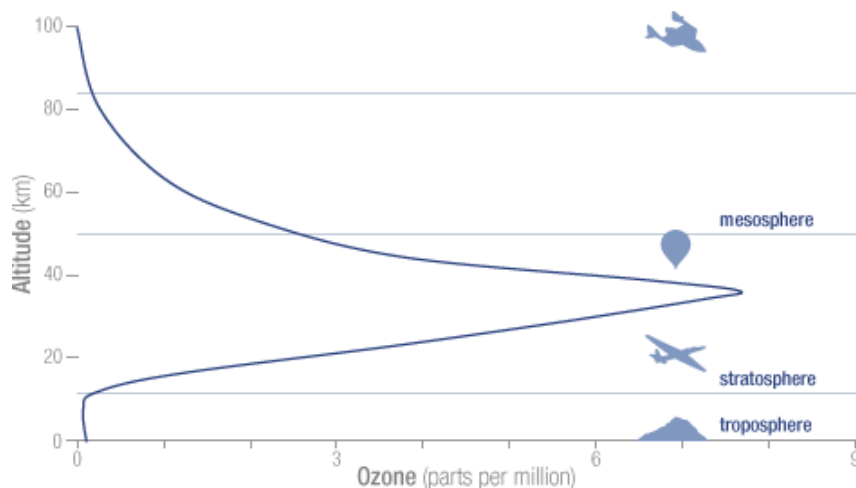
The principal man-made halocarbons that Montreal Protocol brought up considering ozone depleting substances include:

- CFC-11, (or  $\text{CCl}_3\text{F}$  or trichlorofluoromethane),
- CFC-12, (or  $\text{CCl}_2\text{F}_2$  or dichlorofluoromethane),
- CFC-113, (or  $\text{CCl}_2\text{FCClF}_2$  or 1,1,2-trichloro-1,2,2-trifluoromethane),
- $\text{CCl}_4$ , (or carbon tetrachloride),
- $\text{CH}_3\text{CCl}_3$ , (or methyl chloroform or 1,1,1-trichloroethane).

In order to do the prediction of ozone level in the next 50 years, it is believed that previous chlorofluorocarbons and chlorocarbons data are required. Because the ozone depletion phenomena mainly happened in Antarctic and Arctic zone, we collected the data in polar zones and mid latitude area for comparison.

## 1.4 Ozone facts with quantified indexes

Ninety percent of the ozone in the atmosphere sits in the stratosphere, the layer of atmosphere between about 10 and 50 kilometers altitude. The natural level of ozone in the stratosphere is a result of a balance between sunlight that creates ozone and chemical reactions that destroy it.

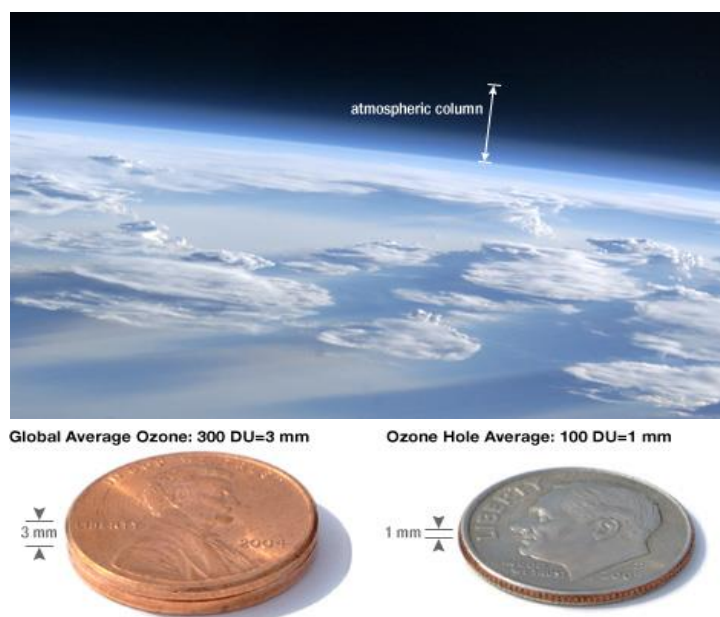


**Fig 1.2 The concentration of ozone varies with altitude. Peak concentrations, an average of 8 molecules of ozone per million molecules in the atmosphere, occur between an altitude of 30 and 35 kilometers.(The Figure is from <https://www.nasa.gov/ozone/>)**

It is very important for us to quantify the hazard of ODS, therefore the following indexes are of great help for the prediction of ozone level.

1. Total ozone concentration (DU)

Over the Earth's surface, the ozone layer's average thickness is about 300 Dobson Units or a layer that is 3 millimeters thick. The Dobson Unit is the most common unit for measuring ozone concentration. One Dobson Unit is the number of molecules of ozone that would be required to create a layer of pure ozone 0.01 millimeters thick at a temperature of 0 degrees Celsius and a pressure of 1 atmosphere (the air pressure at the surface of the Earth). Expressed another way, a column of air with an ozone concentration of 1 Dobson Unit would contain about  $2.69 \times 10^{16}$  ozone molecules for every square centimeter of area at the base of the column<sup>[9]</sup>.



**Fig 1.3 the ozone in the atmosphere would form a layer about the height of two pennies stacked together. Levels in the ozone hole are much less—only the height of a single dime. (The picture is from [http://ozonewatch.gsfc.nasa.gov/facts/dobson\\_SH.html](http://ozonewatch.gsfc.nasa.gov/facts/dobson_SH.html))**

## 2. Ozone hole area (km<sup>2</sup>)

Each year for the past few decades during the Southern Hemisphere spring, chemical reactions involving chlorine and bromine cause ozone in the southern polar region to be destroyed rapidly and severely. This depleted region is known as the “ozone hole”.

## 3. EESC (ppt or ppbv)

Equivalent effective stratospheric chlorine (EESC) is a convenient parameter we use to estimate the amount of stratospheric ozone depletion due to these compounds. The higher the values of EESC the more chlorine is available for the destruction of ozone. The units of EESC are in parts per billion by volume (ppbv, volume of equivalent chlorine to the volume of air).

## 4. ODGI

The Ozone Depleting Gas Index (ODGI) is divided into ODGI-A and ODGI-ML. The ODGI-A is defined by the observed decline in halogen abundance from its peak in Antarctica (ODGI = 100). Similar to ODGI-A, the ODGI-ML is defined as 100 at the peak in EESC-ML, and zero at the 1980 benchmark EESC-ML level, corresponding to when substantial ozone-layer recovery might be expected in the mid-latitude stratosphere if all other factors were to remain constant.

The above factors are closely reflecting the ozone level.

## 1.5 ARIMA and RBF methodology

An autoregressive integrated moving average model (also called ARIMA model) is effective in time series analysis, and it can also be used for predicting the future points of the time series data with the known series, using time as a random variable<sup>[10]</sup>. The autocorrelation of such random variables indicates extension of predicted objects' trend.

On the other hand, Freedman (2006) proposed that the physical relationship between databases of inputs and outputs can be approximated by a mapping function which is a linear combination of Gaussian RBFs. The coefficients of the mapping function are calibrated by using the measured dataset in the laboratory. The mapping function can be considered as a multi-dimensional interpolation between input and output databases<sup>[11]</sup>. For a sample not included in the database, we can estimate the required output by the mapping function when only input measurements are known. Figure 1 shows the schematic diagram of two dimensional RBF interpolation. For multi-dimension RBF interpolation, the principle is same. The RBF interpolation function is similar to a soft membrane that passes through all known data points of various samples<sup>[12]</sup>. The shape of the membrane can be roughly determined from these known experimental data points to create an interpolation function expression. As long as the experimental data is sufficient and is uniformly distributed in a wide range, the constituted interpolation function almost represents the unknown theoretical expression.



## II . The Description of the Problem

### 2.1 How do we understand the whole course of ozone level prediction?

The problem A gave us general idea and background of the ozone depletion issue, under the context of Montreal protocol.

It is believed that the primary cause of ozone depletion is the presence of chlorine-containing source gases, including CFCs and related halocarbons, nitrous oxide, etc. Since the adoption and strengthening of the Montreal Protocol has led to reductions in the emissions of CFCs, atmospheric concentrations of the most-significant compounds have been declining.

In our perspective, the problem require the participants to consider the following aspects for accomplishing the ozone level prediction task.

- Data collection for the prediction
- Definition of the ozone level
- How to interpret the factors related to ozone level
- Reliability and feasibility of the prediction methodology

### 2.2 How do we define the ozone level?

- 1) **From the perspective of ozone depletion phenomena:** it is believed the level of ozone depletion substances in the atmosphere could reflect ozone level; there're several chlorofluorocarbons and chlorocarbons listed in introduction part, and it is crucial to give out a comprehensive index to reflect the effects of ODSs.
- 2) **From the perspective of the ozone concentration:** The Dobson Unit (DU) is the most common unit for measuring ozone concentration. However, using total ozone column is bound to massive detected data, which can't provide concise results but causes instability of the prediction.

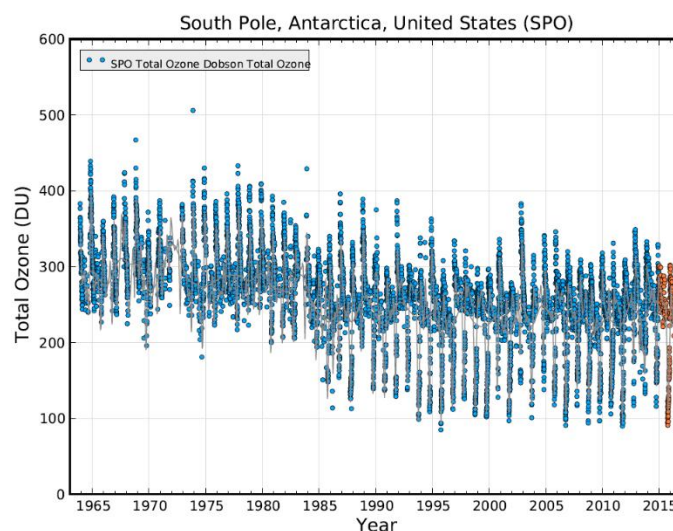


Fig 2.1 Massive data of Total column ozone(the data is from <http://www.esrl.noaa.gov/>)

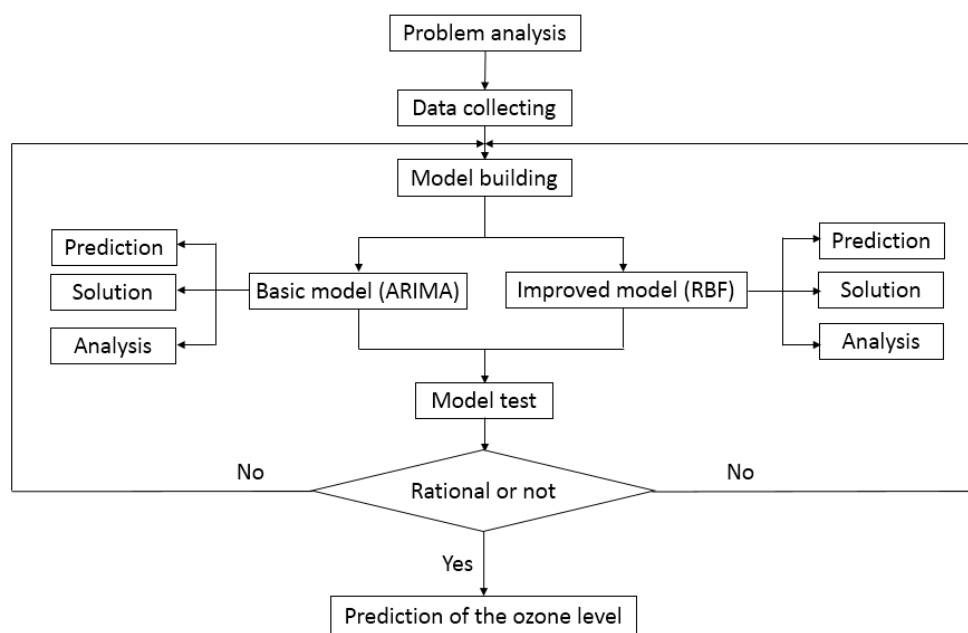
**3) Compromise:** Consequently, we use integrated indexes, ODGI as well as EESC, from the perspective of ozone depletion substances to analyze the ozone level and give out the concise and relatively accurate prediction.

Finally, the predicted values of the two indicators are used to determine which historical time the ozone level will reach in the next 50 years.

## 2.3 The differences in locations of ozone level

According to NASA Goddard Space Flight Center, owing to wind and temperature factors, ozone hole occurs frequently in the South pole, which will be the focus in our prediction model with the comparison to the level of mid latitude.

## 2.4 The essay construction



**Fig 2.2 the structure of the essay**

### III. Assumption and Symbol Definition

#### *Assumption*

- The data we collected for the model are valid.
- Considering the sequence of time series is a random series.
- Assume full adherence to controls on production and consumption of ODSs in the fully revised and amended Montreal Protocol.
- Effects of external environment such as wind speed and temperature are invalid to the level of ozone.

#### *Symbol Definition*

- EESC-A: Equivalent effective stratospheric chlorine at the Antarctic
- EESC-ML: Equivalent effective stratospheric chlorine at the mid latitude
- ODGI-A: Ozone Depleting Gas Index at the Antarctic
- ODGI-ML: Ozone Depleting Gas Index at the mid latitude.
- $p$ : autoregression
- $d$ : difference
- $q$ : Moving average number
- $F(x_i)$ : The mapping function
- $c_i$ : The real coefficients (weight),
- Obs: observed data
- $\gamma$ : regularization parameter.(real and positive)
- $\varepsilon$ : The error of the prediction
- $I$ : identity matrix
- $\varphi$ : The nonlinear function

## IV. Models

### 4.1 Basic Model-ARIMA

An autoregressive integrated moving average model (also called ARIMA model) is effective in time series analysis, and it can also be used for predicting the future points of the time series data with the known series, using time as a random variable. The autocorrelation of such random variables indicates extension of predicted objects' trend. Once the autocorrelation is described by mathematical model, future points can be predicted from the past and present values. The research performed analysis on relative errors based on data issued by national environmental monitoring website and concluded the error range was in 5%-15%, so that the prediction model and data were reliable.

#### 4.1.1 Predicted indicators

##### (1) Equivalent effective stratospheric chlorine (EESC)

In order to calculate and summarize the sum of the effects of different ODSs on stratospheric ozone, the concept of EESC is introduced, See equation (1). Due to the complexity of ozone column concentration variation, EESC is not a perfect quantitative measure of ozone change, but it is a commonly used indicator in the current scientific research. It expresses the ability of ODS to destroy ozone with the equivalent capacity of CFC-11 for the total destructive capacity of the ozone layer.

EESC, as a function of time  $t$ , is defined as

$$EESC(t) = C_0 \sum_{Cl} n_m f_m \rho_m + C_0 \alpha \sum_{Br} n_m f_m \rho_m \quad (1)$$

where  $n$  is the number of chlorine or bromine atoms of a particular source gas  $m$ ,  $f$  represents the efficiency of stratospheric halogen release of the source gas, and  $\rho_m$  is the source gas mixing ratio in the stratosphere (Daniel et al., 1995). Summations are over the chlorine- and bromine-containing halocarbons. The leading factor,  $C_0$ , can be an arbitrary value, or it can be the fractional release value of chlorofluorocarbon-11 (CFC-11) so that the EESC quantity accurately represents the amount of inorganic chlorine ( $Cl_y$ ) and bromine ( $Br_y$ ) in some region of the stratosphere. In the rest of this manuscript, we use  $C_0=1.0$ . Equivalent effective chlorine (ECl) represents the same quantity as EESC but with no consideration of the transport time to the stratosphere. In the classic EESC,  $\rho_m$  is calculated assuming a simple time lag,  $\Gamma$ , from the surface observations

$$\rho_m = \rho_{m,entry}(t - \Gamma) \quad (2)$$

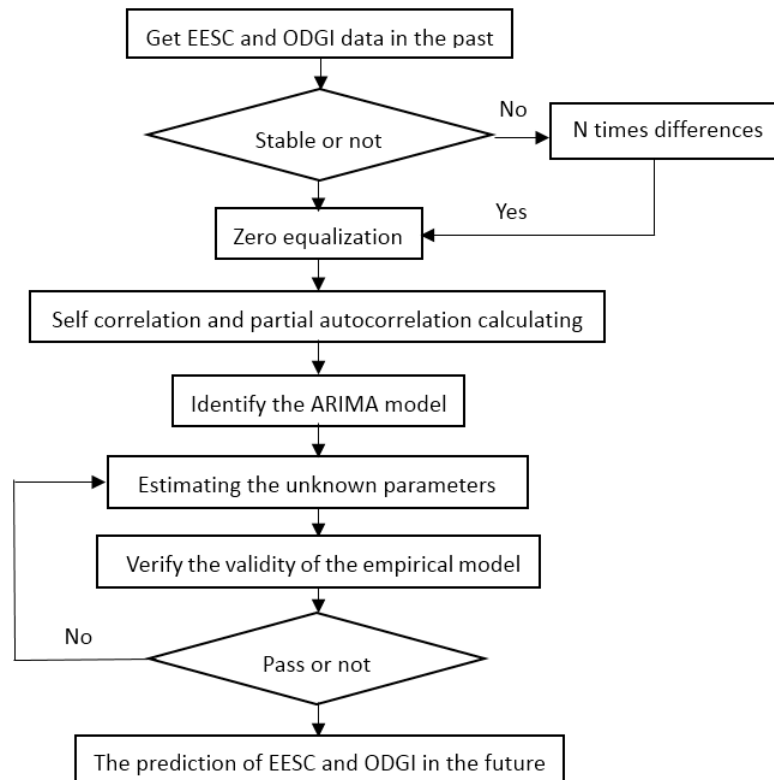
Where  $\rho_{m,entry}(t)$  is the surface observation at time  $t$ . Many previous studies estimated this classic EESC assuming  $\Gamma=3y$  to obtain a value appropriate for relating to globally averaged ozone loss.

The relative effectiveness of bromine compared to chlorine for ozone depletion ( $\alpha$  in Eq. 1, we adopt a value of 60 for  $\alpha$ ), arises from the residence of inorganic bromine in more active compounds for ozone destruction. This relative effectiveness is usually

presented for global ozone depletion although it is a function of altitude, latitude, and background chlorine and bromine amount.

#### 4.1.2 The Foundation of ARIMA Model

- **The algorithm flow chart:**



**Fig 4.1.1 the process of EESC and ODGI prediction based on ARIMA**

- **The utility function module**

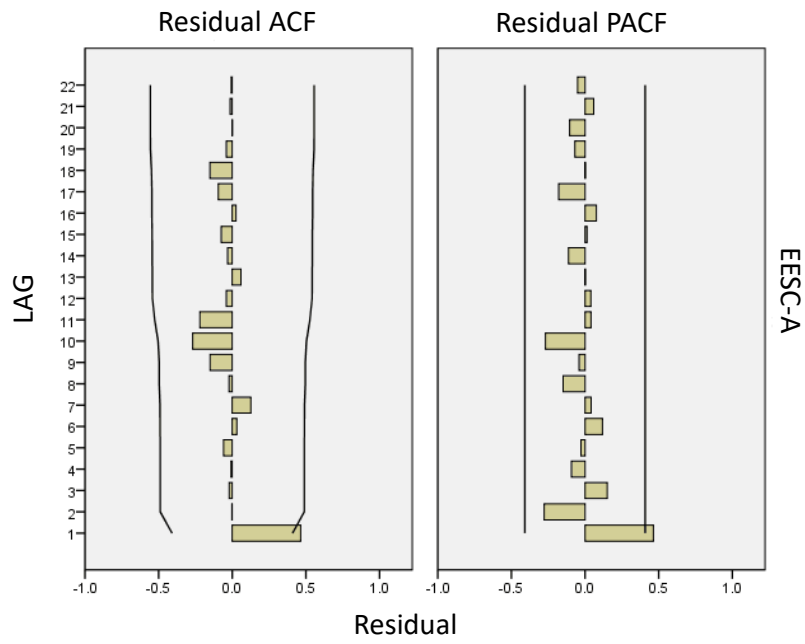
According to ARIMA, we can apply SPSS to calculate the EESC and ODGI as follows.

From NOAA ESRL website, we can get the data of the contributions of ozone depleting chemicals and groups of chemicals to ozone-depleting halogen in the Antarctic atmosphere as well as in the mid-latitude stratosphere:

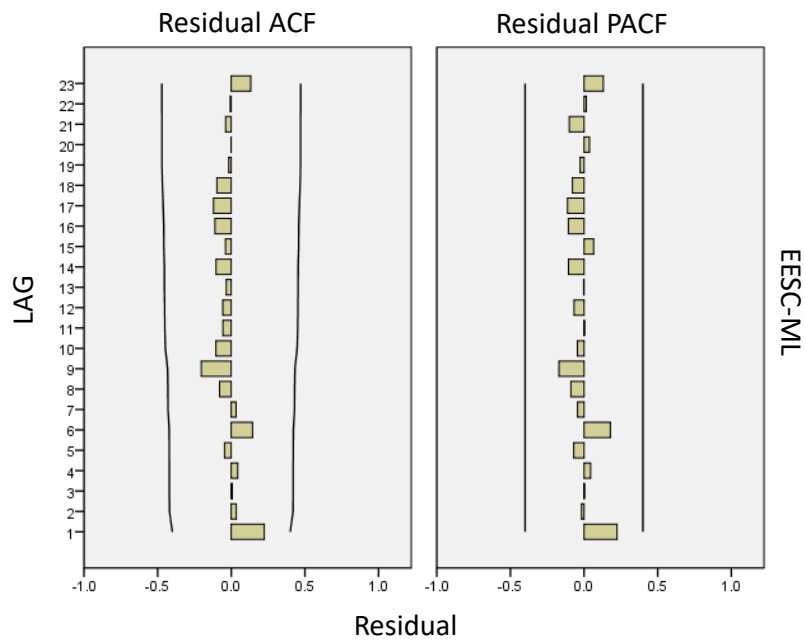
Create an ARIMA case, we set the data of EESC and ODGI as dependent variable, year as independent variable. Pre-testing is very important to ARIMA method, so we examine the ACF and PACF to justify the index.

The results are:

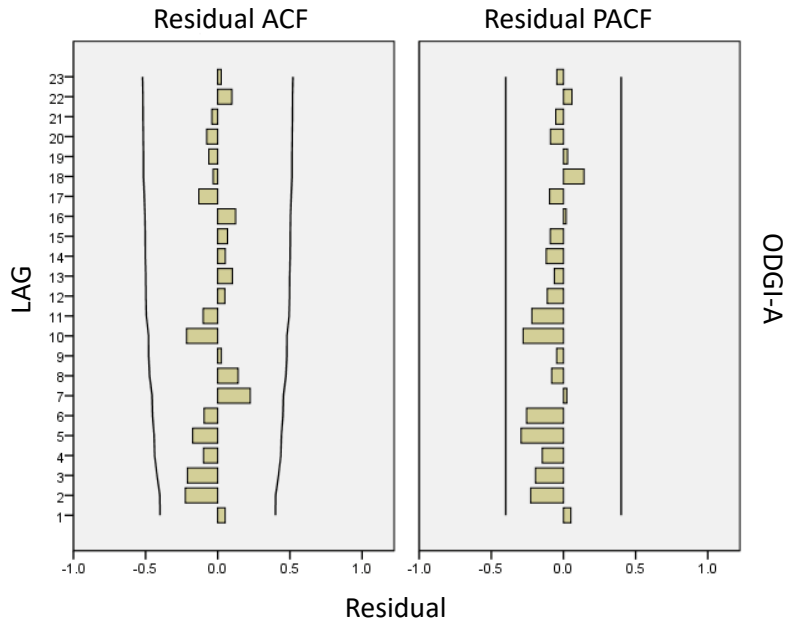
- p=1, d=2, q=0 are reasonable for EESC-A;
- p=1, d=1, q=1 are reasonable for EESC-ML;
- p=2, d=1, q=2 are reasonable for ODGI-A;
- p=1, d=1, q=1 are reasonable for ODGI-ML;



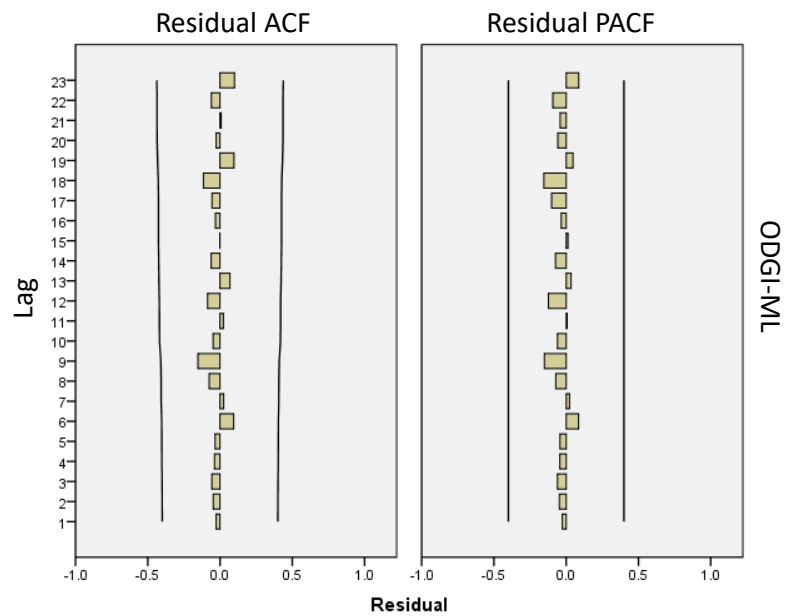
**Fig 4.1.2 residual of EESC-A Model**



**Fig 4.1.3 residual of EESC-ML Model**



**Fig 4.1.4 residual of ODGI-A Model**



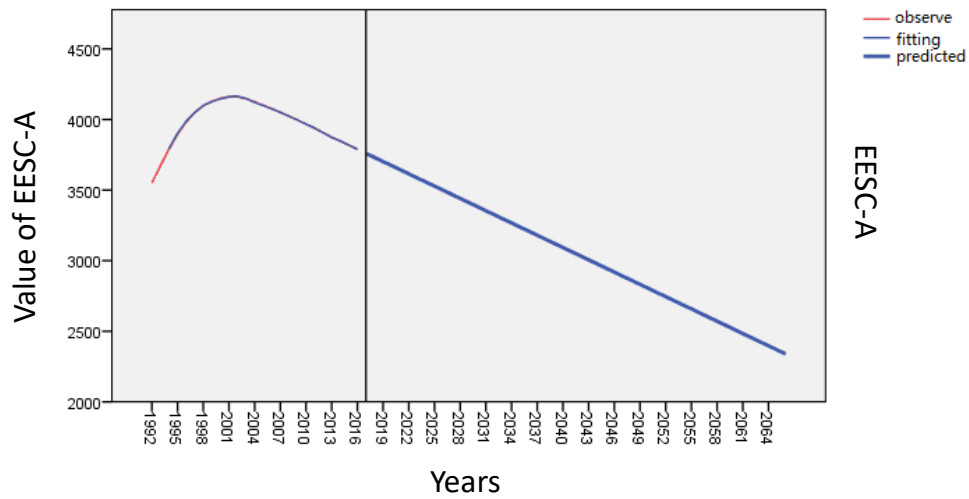
**Fig 4.1.5 residual of ODGI-ML Model**

The ARIMA model required sequence is a stationary series. We conduct stability analysis of the data by autocorrelation and partial autocorrelation analysis as is shown above. Referring to residual ACF and PACF figures (Fig 3.1-3.4), we can find the curves are smooth, so the ARIMA (p,d,q) are reasonable under the circumstances..

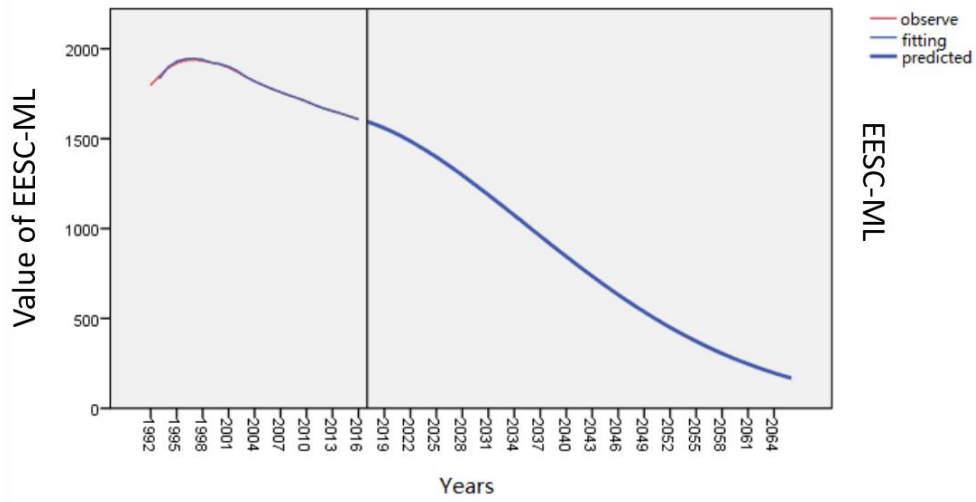
### 4.1.3 Solutions and Results

1) The solution of the programming:

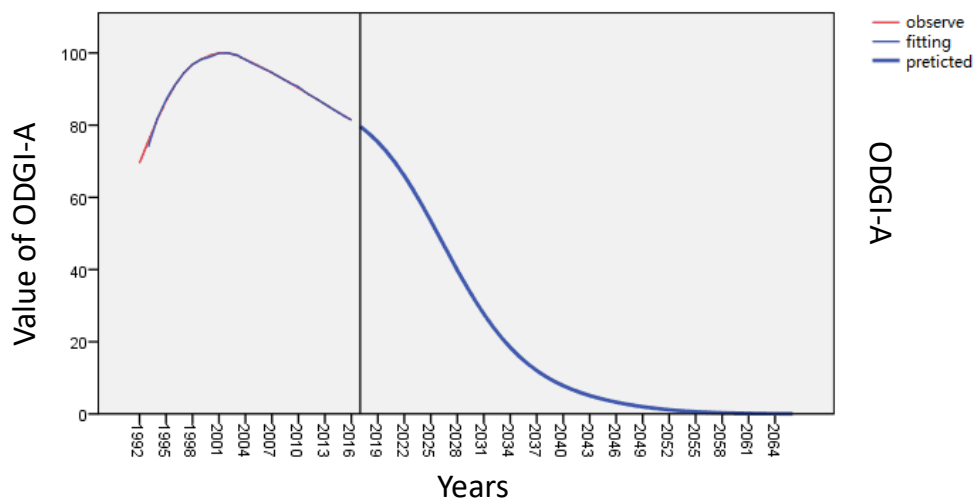
We get the diagram of prediction of EESC as well as ODGI in 50 years.



**Fig 4.1.6 the prediction of EESC-A in 50 years**

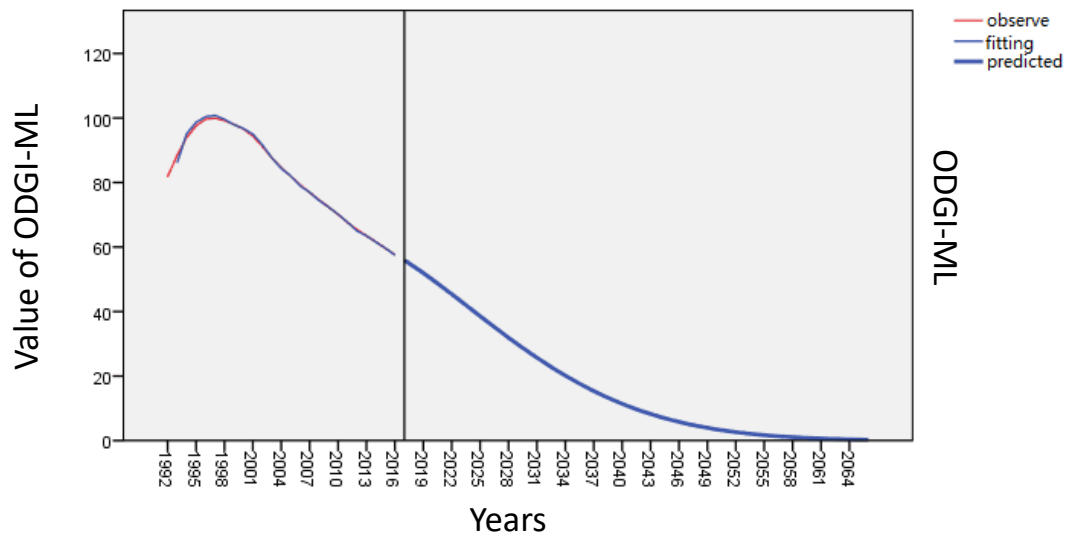


**Fig 4.1.7 the prediction of EESC-ML in 50 years**



**Fig 4.1.8 the prediction of ODGI-A in 50 years**





**Fig 4.1.9 the prediction of ODGI-ML in 50 years**

## 2) Results:

From SPSS, we can get a list of table related to Model's reliability, and we can analyze these data to find out the feasibility of the model.

**Table 4.1.1 ODGI-A Model Fitting**

| Fitting Statistics | Mean Value | Minimum Value | Maximum Value | Percentage |       |       |       |       |       |       |
|--------------------|------------|---------------|---------------|------------|-------|-------|-------|-------|-------|-------|
|                    |            |               |               | 5          | 10    | 25    | 50    | 75    | 90    | 95    |
| Stable R's Square  | .977       | .977          | .977          | .977       | .977  | .977  | .977  | .977  | .977  | .977  |
| R's Square         | .997       | .997          | .997          | .997       | .997  | .997  | .997  | .997  | .997  | .997  |
| RMSE               | .418       | .418          | .418          | .418       | .418  | .418  | .418  | .418  | .418  | .418  |
| MAPE               | .213       | .213          | .213          | .213       | .213  | .213  | .213  | .213  | .213  | .213  |
| MaxAPE             | 2.085      | 2.085         | 2.085         | 2.085      | 2.085 | 2.085 | 2.085 | 2.085 | 2.085 | 2.085 |
| MAE                | .181       | .181          | .181          | .181       | .181  | .181  | .181  | .181  | .181  | .181  |
| MaxAE              | 1.576      | 1.576         | 1.576         | 1.576      | 1.576 | 1.576 | 1.576 | 1.576 | 1.576 | 1.576 |
| Normal State BIC   | -.952      | -.952         | -.952         | -.952      | -.952 | -.952 | -.952 | -.952 | -.952 | -.952 |

**Table 4.1.2 Model Statistics**

| Model  | Predicted Number | R's Square | Ljung-Box Q(18) |    |      | Number of Outliers |
|--------|------------------|------------|-----------------|----|------|--------------------|
|        |                  |            | Statistics      | DF | Sig. |                    |
| ODGI-A | 1                | .997       | 5.569           | 14 | .976 | 0                  |

**Table 4.1.3 EESC-A Model Fitting**

| Fitting Statistics | Mean Value | Minimum Value | Maximum Value | Percentage |       |       |       |       |       |       |
|--------------------|------------|---------------|---------------|------------|-------|-------|-------|-------|-------|-------|
|                    |            |               |               | 5          | 10    | 25    | 50    | 75    | 90    | 95    |
| Stable R's Square  | .765       | .765          | .765          | .765       | .765  | .765  | .765  | .765  | .765  | .765  |
| R's Square         | .999       | .999          | .999          | .999       | .999  | .999  | .999  | .999  | .999  | .999  |
| RMSE               | 3.503      | 3.503         | 3.503         | 3.503      | 3.503 | 3.503 | 3.503 | 3.503 | 3.503 | 3.503 |
| MAPE               | .061       | .061          | .061          | .061       | .061  | .061  | .061  | .061  | .061  | .061  |
| MaxAPE             | .240       | .240          | .240          | .240       | .240  | .240  | .240  | .240  | .240  | .240  |
| MAE                | 2.454      | 2.454         | 2.454         | 2.454      | 2.454 | 2.454 | 2.454 | 2.454 | 2.454 | 2.454 |
| MaxAE              | 9.346      | 9.346         | 9.346         | 9.346      | 9.346 | 9.346 | 9.346 | 9.346 | 9.346 | 9.346 |
| Normal State BIC   | 2.644      | 2.644         | 2.644         | 2.644      | 2.644 | 2.644 | 2.644 | 2.644 | 2.644 | 2.644 |

**Table 4.1.4 Model Statistics**

| Model  | Predicted Number | Ljung-Box Q(18) |    |      | Number of Outliers |
|--------|------------------|-----------------|----|------|--------------------|
|        |                  | Statistics      | DF | Sig. |                    |
| EESC-A | 0                | 17.117          | 17 | .446 | 0                  |

**Table 4.1.5 EESC-ML Model Fitting**

| Fitting Statistics | Mean Value | Minimum Value | Maximum Value | Percentage |        |        |        |        |        |        |
|--------------------|------------|---------------|---------------|------------|--------|--------|--------|--------|--------|--------|
|                    |            |               |               | 5          | 10     | 25     | 50     | 75     | 90     | 95     |
| Stable R's Square  | .941       | .941          | .941          | .941       | .941   | .941   | .941   | .941   | .941   | .941   |
| R's Square         | .998       | .998          | .998          | .998       | .998   | .998   | .998   | .998   | .998   | .998   |
| RMSE               | 5.605      | 5.605         | 5.605         | 5.605      | 5.605  | 5.605  | 5.605  | 5.605  | 5.605  | 5.605  |
| MAPE               | .195       | .195          | .195          | .195       | .195   | .195   | .195   | .195   | .195   | .195   |
| MaxAPE             | .842       | .842          | .842          | .842       | .842   | .842   | .842   | .842   | .842   | .842   |
| MAE                | 3.603      | 3.603         | 3.603         | 3.603      | 3.603  | 3.603  | 3.603  | 3.603  | 3.603  | 3.603  |
| MaxAE              | 15.567     | 15.567        | 15.567        | 15.567     | 15.567 | 15.567 | 15.567 | 15.567 | 15.567 | 15.567 |
| Normal State BIC   | 3.977      | 3.977         | 3.977         | 3.977      | 3.977  | 3.977  | 3.977  | 3.977  | 3.977  | 3.977  |

**Table 4.1.6 Model Statistics**

| Model   | Predicted Number | Ljung-Box Q(18) |    |      | Number of Outliers |
|---------|------------------|-----------------|----|------|--------------------|
|         |                  | Statistics      | DF | Sig. |                    |
| EESC-ML | 1                | 9.140           | 16 | .908 | 0                  |

**Table 4.1.7 ODGI-ML Model Fitting**

| Fitting Statistics | Mean Value | Minimum Value | Maximum Value | Percentage |       |       |       |       |       |       |
|--------------------|------------|---------------|---------------|------------|-------|-------|-------|-------|-------|-------|
|                    |            |               |               | 5          | 10    | 25    | 50    | 75    | 90    | 95    |
| Stable R's Square  | .943       | .943          | .943          | .943       | .943  | .943  | .943  | .943  | .943  | .943  |
| R's Square         | .998       | .998          | .998          | .998       | .998  | .998  | .998  | .998  | .998  | .998  |
| RMSE               | .733       | .733          | .733          | .733       | .733  | .733  | .733  | .733  | .733  | .733  |
| MAPE               | .475       | .475          | .475          | .475       | .475  | .475  | .475  | .475  | .475  | .475  |
| MaxAPE             | 2.633      | 2.633         | 2.633         | 2.633      | 2.633 | 2.633 | 2.633 | 2.633 | 2.633 | 2.633 |
| MAE                | .418       | .418          | .418          | .418       | .418  | .418  | .418  | .418  | .418  | .418  |
| MaxAE              | 2.328      | 2.328         | 2.328         | 2.328      | 2.328 | 2.328 | 2.328 | 2.328 | 2.328 | 2.328 |
| Normal State BIC   | -.090      | -.090         | -.090         | -.090      | -.090 | -.090 | -.090 | -.090 | -.090 | -.090 |

**Table 4.1.8 Model Statistics**

| Model   | Predicted Number | R's Square | Ljung-Box Q(18) |    |      | Number of Outliers |
|---------|------------------|------------|-----------------|----|------|--------------------|
|         |                  |            | Statistics      | DF | Sig. |                    |
| ODGI-ML | 1                | .998       | 4.546           | 16 | .998 | 0                  |

#### 4.1.4 Analysis of the Results

##### (1) *R-squared*

From the table 4.1.1-4.1.8, all the R-squared are around 0.997-0.998, indicating the fittings are of good quality. Stable R-squared reveal the prediction's reliability, and we can tell the prediction is statistically reasonable and reliable.

##### (2) *Trend*

- EESC at the Antarctic is declining in 50 years, and it will drop below 2,500 ppt after 2050, which indicates EESC-A level can be as less as EESC at mid latitude. It helps to predict the ozone hole will be closed around 2050, and it coincides with the Montreal Protocol.
- EESC at the mid latitude also declines yet in a milder way. It converges at a relatively low level in 50 years, which verify the validity of Montreal Protocol.
- ODGI at the Antarctic has a massive drop during 2020-2030, and steady decline after 2040. It drops to 0 eventually according to our prediction, which is consistent to the result of EESC-A as well as EESC-ML.
- ODGI at the mid latitude has the similar trend to ODGI-A, but in a steadier way. It also converges to zero in 50 years.

##### (3) *Comparison*

The EESC and ODGI prediction coincide with each other, casting a positive view for the future ozone level. Because of less ozone depletion substances in the atmosphere.

#### 4.1.5 Strength and Weakness

##### (1) *Strength*

The model has proved that the ozone level will be ascending in the next 50 years, which coincides with Montreal Protocol. Moreover, we have drawn some useful conclusions about EESC at the Antarctic is declining in 50 years, and it will drop below 2,500 ppt after 2050, which indicates EESC-A level can be as less as EESC at mid latitude. ODGI at the Antarctic has a massive drop during 2020-2030, and steady decline after 2040. The model is fit for problem related to time series forecasting, with the advantage of handy, statistically reliable, and interpretively concise results.

##### (2) *Weakness*

This model just applies to the situation where we can find some historical data. As we have stated, the model relies much on statistics. And the ODGI converges to zero might need further modification. That's just what we should do in the improved model.

### 4.2 Improved Model-RBF

Freedman proposed that the physical relationship between databases of inputs and outputs can be approximated by a mapping function which is a linear combination of Gaussian RBFs. The coefficients of the mapping function are calibrated by using the measured dataset in the laboratory. The mapping function can be considered as a multi-dimensional interpolation between input and output databases. For a sample not included in the database, we can estimate the required output by the mapping function when only input measurements are known. Figure 1 shows the schematic diagram of two dimensional RBF interpolation. For multi-dimension RBF interpolation, the principle is same. The RBF interpolation function is similar to a soft membrane that passes through all known data points of various samples. The shape of the membrane can be roughly determined from these known experimental data points to create an interpolation function expression. As long as the experimental data is sufficient and is uniformly distributed in a wide range, the constituted interpolation function almost represents the unknown theoretical expression<sup>[13][14]</sup>.

#### 4.2.1 Extra Predicted indicator

##### 1. Ozone-hole area

Predictive indicators (EESC & ODGI) above are still valid. Here is an extra evaluation standard-Ozone hole area.

The area of the ozone hole is determined from a map of total column ozone. It is calculated from the area on the Earth that is enclosed by a line with a constant value of 220 Dobson Units. The value of 220 Dobson Units is chosen since total ozone values of less than 220 Dobson Units were not found in the historic observations over Antarctica prior to 1979. Also, from direct measurements over Antarctica, a column ozone level of less than 220 Dobson Units is a result of the ozone loss from chlorine and bromine compounds.

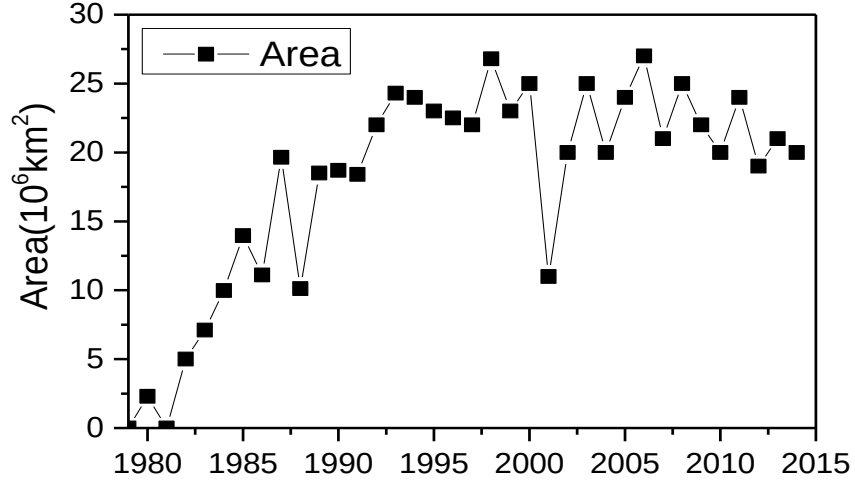


Fig.4.2.1 Mean area of Antarctic ozone hole(7 September-13 October)

#### 4.2.2 The Foundation of RBF Model

An RBF is a real-valued function with a value depending on only the distance from the origin to measuring point, i.e.,  $\varphi(x) \equiv \varphi(\|x\|)$  or on the distance from the measuring point to a different point  $x_0$  (known as a center); thus  $\varphi(x, x_0) \equiv \varphi(\|x - x_0\|)$ . The norm is usually Euclidean distance.

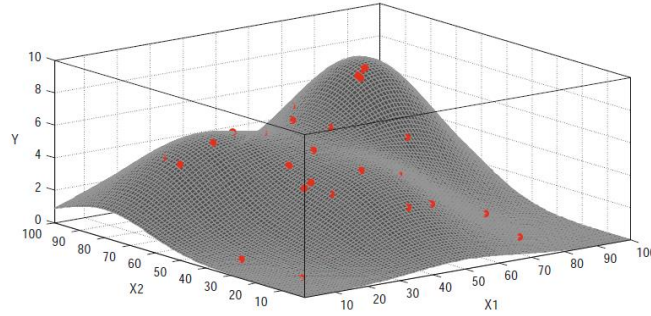


Fig.4.2.2 Schematic diagram of 2D radial basis function (RBF).

The calibration dataset contains  $N$  measurements with  $N$  input parameters,  $x_1, x_2, \dots, x_N$ , and  $N$  output physical properties,  $y_1, y_2, \dots, y_N$ . The interpolation problem is to construct a mapping function  $F(x_i)$  that satisfies the following equation:

$$F(x_i) = y_i, \quad i=1,2,\dots,N \quad (3)$$

RBF interpolation chooses a norm-approximating function or mapping function ,

$$F(x) = \sum_{i=1}^N c_i \varphi(\|x - x_i\|) \quad (4)$$

where the approximating function  $F(x)$  is the weighted sum of  $N$  radial basis functions, each associated with a different center  $x_i$  and weighted by an appropriate coefficient  $c_i$ . The weights  $c_i$  can be estimated by using the matrix methods of linear least squares because the approximating function is the linear weighted average of each norm.

The nonlinear function,  $\varphi$ , is RBF. The RBF parameter depends on the Euclidean norm in  $n$ -dimensional space: i.e.,

$$\|x - x_i\| = \sqrt{\sum_{m=1}^n (x_m - x_{i,m})^2} \quad (5)$$

The real coefficients (weight),  $c_i$ , are determined by substituting equations (4) and (5) into equation (3). Therefore, the coefficients are a linear combination of the given function values,

$$c_i = \sum_{j=1}^N \Phi_{i,j}^{-1} y_j \quad (6)$$

where  $\Phi_{i,j}^{-1} = \varphi\|x_i - x_j\|$  is an  $N \times N$  interpolation matrix. If the condition number of the interpolation matrix is large or the given function values are noisy, the coefficients determined from equation (6) can be unstable. Therefore, equation (6) should be replaced by the regularized solution,

$$c_i = \sum_{j=1}^N (\Phi + \gamma I)_{i,j}^{-1} y_j \quad (7)$$

where  $I$  is an  $N \times N$  identity matrix and  $\gamma$  is a real and positive regularization parameter. One of the most commonly used RBFs is the normalized Gaussian RBF (Freedman, 2006), as

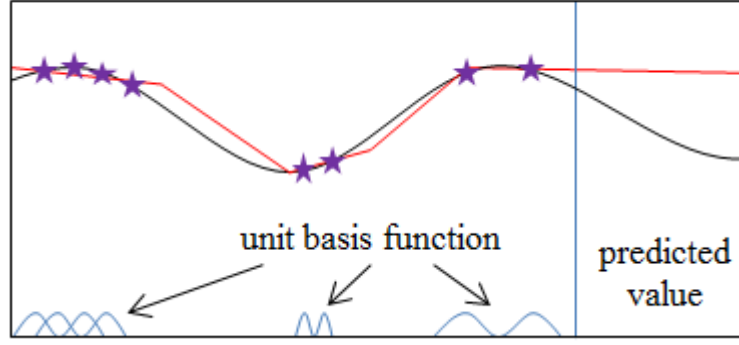
$$\varphi\|x - x_i\| = \frac{\exp\left(-\frac{\|x - x_i\|^2}{2s_i^2}\right)}{\sum_{j=1}^N \exp\left(-\frac{\|x - x_j\|^2}{2s_j^2}\right)} \quad (8)$$

The RBF value lies between 0 and 1 after the normalization scale. Substituting the normalized Gaussian RBF in equation (4), the mapping function is given as

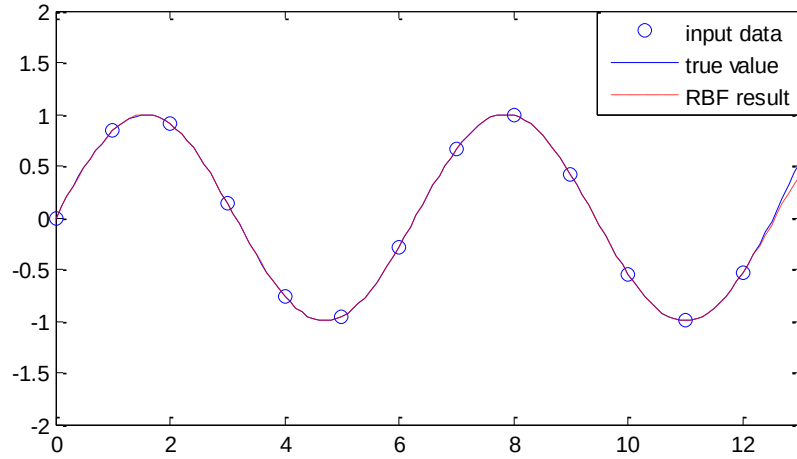
$$F(x) = \frac{\sum_{i=1}^N c_i \exp\left(-\frac{\|x - x_i\|^2}{2s_i^2}\right)}{\sum_{i=1}^N \exp\left(-\frac{\|x - x_i\|^2}{2s_i^2}\right)} \quad (9)$$

Hence, the mapping function is a linear combination of Gaussian functions whose centers are located at the input database.

The Gaussian RBFs widths,  $s_i$ , determine the topographical behavior of the interpolating function. For optimal interpolation, the widths of the Gaussian RBFs are chosen such that they are proportional to the Euclidian nearest neighbor distances in the input space. This choice ensures that the input space filled by basis functions has an overlap with the nearest neighbors to a certain extent and negligible overlap with more distant neighbors.



**Fig.4.2.3 Schematic diagram of the radial basis function (RBF) interpolation approach implementation.**



**Fig.4.2.4 Radial basis function (RBF) approach application example.**

Figure 4.2.3 schematically shows the manner in which the RBF method works. The input parameters are the unit basis functions located in the space-domain axis, whereas the measured physical properties are the data points in the curves. The black curve is the result by the RBF method, where the unit basis functions are multiplied by coefficients  $c_i$ . If there are three distant clusters of data points, five red lines are produced after other method calculation, and each is determined respectively by its nearest neighbor's data values and differ significantly from the true functional curve.

Figure 4.2.4 accurately shows that a good prediction result is obtained by fitting the 13 points using RBF approach. Therefore, RBF method is more reliable than other methods. The RBF approach can be used to predict the desired true value when more calibration points are available.

To test the accuracy of RBF prediction, we use the infinite norm of the difference between the predicted value and the true value:

$$\varepsilon = \|y_i^p - y_i^{true}\|_{\infty}, i = 1, 2, \dots, n \quad (10)$$

The error of the prediction example results in Figure 4.2.4 is  $4.551914 \times 10^{-15}$ , this value is much smaller than  $10 \times 10^{-6}$ . If the error is within the range of  $10 \times 10^{-6}$  we believe that the result of this method of prediction is accurate.

Hence, we can infer that the RBF model prediction is acceptable.

### 4.2.3 Solutions and Results

#### 1. ODSs concentrations prediction

Past concentrations are derived from NOAA measurements of both chlorine- and bromine-containing chemicals; assume full adherence to controls on production and consumption of ODSs in the fully revised and amended Montreal Protocol.

Measured tropospheric changes are indicated with special symbol points, while predicted stratospheric changes are indicated as dashed curves. Estimates are provided for different regions: the mid-latitude stratosphere (lower part) and the Antarctic stratosphere (upper part). Observed ODSs concentrations and predicted ODSs concentrations based on the RBF model is as follows:

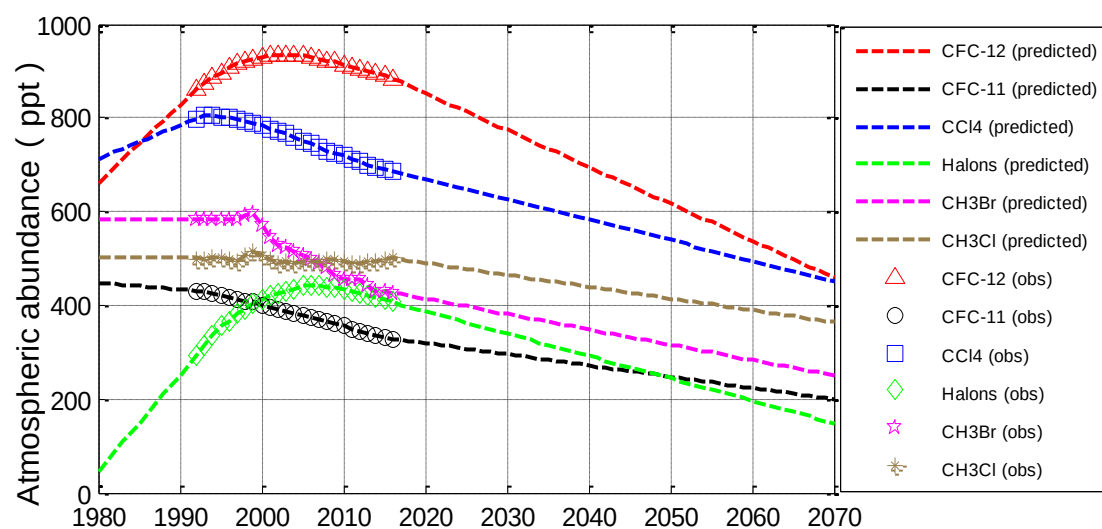


Fig.4.2.5 Observed data and predictive future changes of ODS in Antarctica.

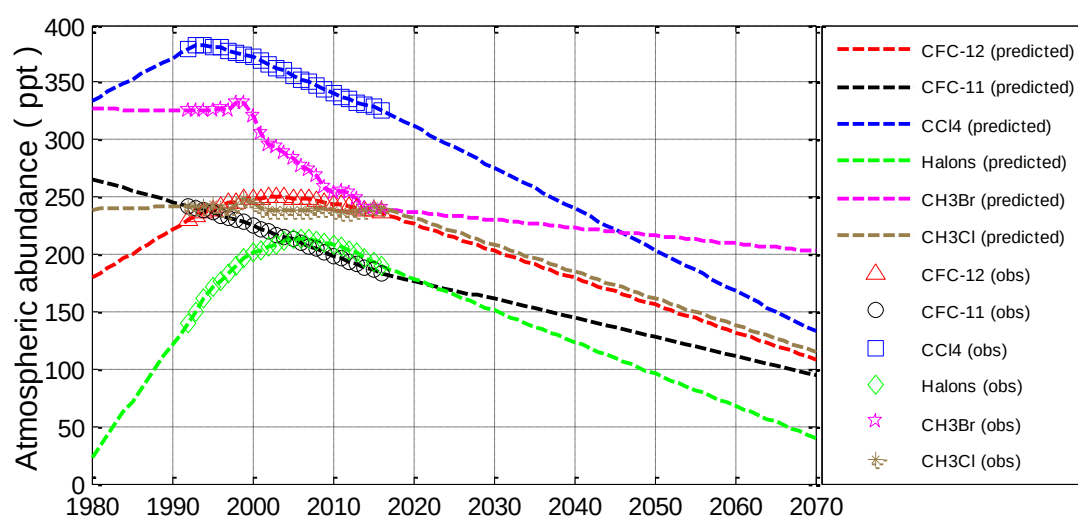


Fig.4.2.6 Observed data and predictive future changes of ODS in Mid-latitude.



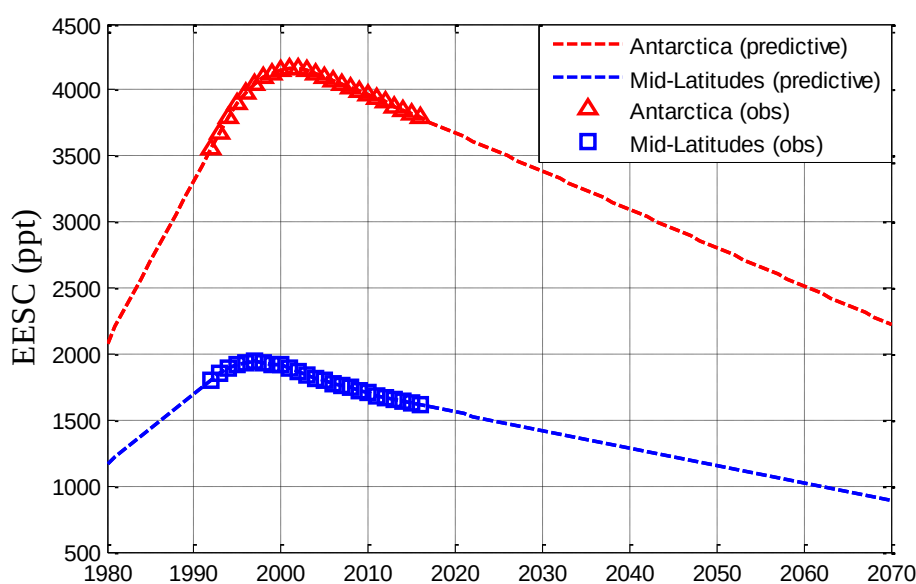
The error of the prediction ODSs results in Antarctica and Mid-latitude are as follow:

**Table 4.2.1 Prediction error of ODSs concentrations**

| ODSs                      | CFC-12 | CFC-11 | CCl4 | Halons | CH3Br | CH3Cl |
|---------------------------|--------|--------|------|--------|-------|-------|
| Error(Antarctica)/ e-12   | 2.72   | 7.27   | 5.45 | 9.09   | 13.18 | 31.8  |
| Error(Mid-latitude)/ e-12 | 1.82   | 2.73   | 1.82 | 1.82   | 11.8  | 7.28  |

## 2. EESC prediction results

The EESC values calculated from the observed ODSs data are indicated with special symbol points, while predicted EESC changes are indicated as dashed curves. Estimates are provided for different regions.



**Fig.4.2.7 Observed data and predictive future changes of EESC.**

## 4.2.4 Analysis of the Results

It can be seen from the data in the table that all the prediction errors are less than  $10e-6$ , so all the data predicted from the RBF model are reliable.

From the two sets of different regions results can draw the following conclusions:

- The concentrations of all the major man-made halocarbons are now steadily declining.
- Global sources of CFC-11, CFC-12, CH3Br, CH3Cl and CCl4 have remained highly significant during 1996~2006.
- Halocarbon emissions have declined in response to international treaty obligations, though they were still significant throughout 1996 and a phase-out in atmospheric release has yet to be achieved.
- New scenarios for future ODS regulation in addition to the baseline scenario have been formulated because parties to the Montreal Protocol have acknowledged interest in increasing its dual benefit to ozone and climate protection.
- The mid-latitude EESC values are steadily declining each year over the next 50 years and will return to 1980's level reaching 2049. On the other hand, The

Antarctica EESC values are also steadily declining each year over the next 50 years and will return to 1980's level reaching 2075.

- Indirectly reflecting the level of ozone in order to return to 1980 levels, the Antarctic region needs to 2075 and the mid-latitude region just needs to 2049
- The mid-latitude EESC will reach 939.9981 (ppt) in 2066. At the same time, Mid-latitude Ozone hole will disappear after 2049.
- The Antarctic EESC will reach 2339.346 (ppt) in 2066 approaching 1982's level. Therefore, compared to the ozone hole area in 1982 we predict the ozone hole area can recover to  $5.2 \times 10^6 \text{ km}^2$  in Antarctic. At the same time, Antarctic Ozone hole will disappear after 2075.

**Table.4.2.2 Observed and predictive EESC in Antarctica**

| Significance        | Years | EESC(ppt) | Approach    |
|---------------------|-------|-----------|-------------|
| Ozone hole appeared | 1980  | 2080.794  | Prediction  |
| Peak                | 2001  | 4162      | Observation |
| Now                 | 2016  | 3789      | Observation |
| The first decade    | 2026  | 3499.069  | Prediction  |
| The Second Decade   | 2036  | 3209.139  | Prediction  |
| The Third Decade    | 2046  | 2919.208  | Prediction  |
| The Fourth Decade   | 2056  | 2629.277  | Prediction  |
| The Fifth Decade    | 2066  | 2339.346  | Prediction  |
| 1980' level         | 2075  | 2078.409  | Prediction  |

**Table.4.2.3 Observed and predictive EESC in Mid-latitude**

| Significance        | Years | EESC(ppt) | Approach    |
|---------------------|-------|-----------|-------------|
| Ozone hole appeared | 1980  | 1169.613  | Prediction  |
| Peak                | 1997  | 1940      | Observation |
| Now                 | 2016  | 1610      | Observation |
| The first decade    | 2026  | 1476      | Prediction  |
| The Second Decade   | 2036  | 1341.999  | Prediction  |
| The Third Decade    | 2046  | 1207.999  | Prediction  |
| 1980's level        | 2049  | 1167.799  | Prediction  |
| The Fourth Decade   | 2056  | 1073.999  | Prediction  |
| The Fifth Decade    | 2066  | 939.9981  | Prediction  |

#### 4.2.5 Strength and Weakness

##### (1) Strength

The improved model can fit the observed data perfectly and can make a good prediction with super small errors. The RBF model results seem to declare that this model is more reasonable than the ARIMA model and much more effective than the existing design.

##### (2) Weakness

Due to limit of database, we just obtain the concentrations of ODSs from the year 1992 to 2016, if there are more data, the forecast results will be more accurate.

## **V. Conclusions**

### **Conclusions of the problem**

- New scenarios for future ODS regulation in addition to the baseline scenario have been formulated because parties to the Montreal Protocol have acknowledged interest in increasing its dual benefit to ozone and climate protection.
- EESC-A at the Antarctic is declining in 50 years, and it will drop below 2,500 ppt after 2050, which indicates EESC-A level can be as less as EESC at mid latitude.
- ODGI at the Antarctic has a massive drop during 2020-2030, and steady decline after 2040.
- The ozone hole will be recovered gradually in the following half centuries
- The mid latitude EESC values are steadily declining each year over the next 50 years and will return to 1980's level reaching 2049, and the Antarctica EESC values will return to 1980's level reaching 2075.
- Indirectly reflecting the level of ozone in order to return to 1980 levels, the Antarctic region needs to 2075 and the mid latitude region just needs to 2049.
- The mid latitude EESC will reach 939.9981 (ppt) in 2066. At the same time, mid latitude Ozone hole will disappear after 2049. The Antarctic EESC will reach 2339.346 (ppt) in 2066 approaching 1982's level.
- Compared to the ozone hole area in 1982 we predict the Antarctic ozone hole area can recover to  $5.2 \times 10^6$  km<sup>2</sup> in 2066 and will disappear after 2075.

### **Methods used in our models**

- ARIMA algorithm
- RBF algorithm
- Error analysis

### **Applications of our models**

- The basic model is fit for problem related to time series forecasting, with the advantage of handy, statistically reliable, and interpretively concise results.
- The improved model has good predicted results with minimal error, the predicted result using RBF is more accurate than the basic model with a perfect theoretical basis.

## **VI. Future Work**

### **6.1 General overview of our model**

In this modeling, these methods have a scientific and reliable prediction of the ozone concentration in the next fifty years, which are satisfactory.

However, in this modeling process, there are still some problems we properly didn't handled well, which may bring a small part of the impact to the results of the forecast. For further research in the future, we will improve our model mainly from the following aspects.

### **6.2 Limitations and the way to improved it**

- First of all, the data collection, induction, and other preliminary work need to be further optimized. Due to the limitation of time, we have not been able to search overall information, which can reflect the changes in the concentration of ozone and confirms the reliability of our model.
- In terms of modeling, the ozone concentration is predicted in these respects. In addition, it can be predicted by regression analysis, wavelet transform theory, phase space reconstruction and so on, which may get a better result.
- Some physical factors such as temperature, wind speed, pressure and so on can be further considered in our model, and the accuracy of the prediction of local area and the overall anti interference ability of the model can be enhanced.

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## VIII. Appendix

**Table 8.1 contributions of ozone depleting chemicals and groups of chemicals-Antarctic**

| Year | CFC-12 | CFC-11 | CH <sub>3</sub> Cl | CH <sub>3</sub> Br | CCl <sub>4</sub> | CH <sub>3</sub> CCl <sub>3</sub> | halons | CFC-113 | HCFCs | EESC<br>SUM<br>(ppt) | ODGI(new)-A |
|------|--------|--------|--------------------|--------------------|------------------|----------------------------------|--------|---------|-------|----------------------|-------------|
| 1992 | 861    | 798    | 500                | 583                | 431              | 400                              | 292    | 215     | 41    | 3550                 | 69.5        |
| 1993 | 876    | 804    | 500                | 583                | 429              | 393                              | 313    | 219     | 44    | 3672                 | 75.6        |
| 1994 | 885    | 805    | 501                | 583                | 425              | 370                              | 336    | 224     | 48    | 3790                 | 81.5        |
| 1995 | 895    | 803    | 504                | 583                | 422              | 333                              | 357    | 226     | 53    | 3895                 | 86.7        |
| 1996 | 907    | 800    | 499                | 585                | 418              | 296                              | 367    | 227     | 59    | 3981                 | 91          |
| 1997 | 915    | 797    | 492                | 582                | 414              | 254                              | 381    | 226     | 65    | 4049                 | 94.4        |
| 1998 | 920    | 792    | 505                | 594                | 410              | 214                              | 391    | 225     | 70    | 4097                 | 96.8        |
| 1999 | 926    | 788    | 513                | 596                | 407              | 179                              | 404    | 224     | 76    | 4129                 | 98.3        |
| 2000 | 930    | 783    | 506                | 572                | 401              | 149                              | 414    | 222     | 81    | 4150                 | 99.4        |
| 2001 | 932    | 777    | 496                | 545                | 397              | 124                              | 419    | 221     | 87    | 4162                 | 100         |
| 2002 | 934    | 772    | 491                | 529                | 392              | 103                              | 424    | 220     | 92    | 4161                 | 100         |
| 2003 | 934    | 765    | 494                | 525                | 388              | 86                               | 428    | 218     | 96    | 4147                 | 99.3        |
| 2004 | 933    | 758    | 491                | 515                | 384              | 72                               | 434    | 216     | 100   | 4126                 | 98.2        |
| 2005 | 932    | 751    | 492                | 505                | 380              | 60                               | 441    | 214     | 103   | 4102                 | 97          |
| 2006 | 931    | 744    | 492                | 496                | 375              | 50                               | 442    | 212     | 107   | 4077                 | 95.8        |
| 2007 | 927    | 737    | 495                | 488                | 370              | 42                               | 442    | 210     | 111   | 4051                 | 94.5        |
| 2008 | 923    | 730    | 497                | 481                | 366              | 35                               | 440    | 208     | 116   | 4024                 | 93.1        |
| 2009 | 919    | 724    | 496                | 463                | 360              | 29                               | 437    | 206     | 121   | 3996                 | 91.8        |
| 2010 | 914    | 719    | 493                | 454                | 355              | 25                               | 435    | 204     | 125   | 3968                 | 90.3        |
| 2011 | 909    | 712    | 490                | 456                | 350              | 21                               | 431    | 202     | 129   | 3938                 | 88.8        |
| 2012 | 905    | 706    | 491                | 453                | 345              | 17                               | 427    | 200     | 134   | 3906                 | 87.3        |
| 2013 | 900    | 700    | 495                | 444                | 341              | 14                               | 423    | 198     | 138   | 3876                 | 85.8        |
| 2014 | 895    | 695    | 493                | 430                | 337              | 12                               | 417    | 196     | 141   | 3847                 | 84.3        |
| 2015 | 891    | 692    | 500                | 429                | 332              | 10                               | 411    | 195     | 145   | 3818                 | 82.9        |
| 2016 | 884    | 688    | 508                | 435                | 329              | 8                                | 406    | 194     | 147   | 3789                 | 81.4        |

**Table 8.2 contributions of ozone depleting chemicals and groups of chemicals- mid-latitude**

| Year | CFC-12 | CFC-11 | CH <sub>3</sub> Cl | CH <sub>3</sub> Br | CCl <sub>4</sub> | CH <sub>3</sub> CCl <sub>3</sub> | halons | CFC-113 | HCFCs | SUM*<br>(ppt) | EESC<br>SUM<br>(ppt) | ODGI(new)<br>Mid-Lat |
|------|--------|--------|--------------------|--------------------|------------------|----------------------------------|--------|---------|-------|---------------|----------------------|----------------------|
| 1992 | 230    | 379    | 242                | 326                | 242              | 271                              | 140    | 69      | 13    | 1951          | 1798                 | 81.8                 |
| 1993 | 234    | 382    | 242                | 326                | 240              | 266                              | 150    | 71      | 14    | 1966          | 1849                 | 88.4                 |
| 1994 | 237    | 382    | 242                | 326                | 238              | 250                              | 161    | 72      | 16    | 1967          | 1892                 | 93.9                 |
| 1995 | 239    | 381    | 244                | 326                | 236              | 225                              | 171    | 73      | 18    | 1957          | 1921                 | 97.6                 |
| 1996 | 242    | 380    | 241                | 327                | 234              | 200                              | 177    | 73      | 20    | 1939          | 1936                 | 99.6                 |
| 1997 | 245    | 378    | 238                | 326                | 232              | 172                              | 184    | 73      | 22    | 1915          | 1940                 | 100                  |

|      |     |     |     |     |     |     |     |    |    |      |      |      |
|------|-----|-----|-----|-----|-----|-----|-----|----|----|------|------|------|
| 1998 | 246 | 376 | 244 | 333 | 230 | 145 | 190 | 73 | 24 | 1906 | 1933 | 99.2 |
| 1999 | 248 | 374 | 248 | 333 | 228 | 121 | 196 | 72 | 26 | 1892 | 1924 | 98   |
| 2000 | 249 | 372 | 245 | 320 | 225 | 101 | 201 | 72 | 28 | 1856 | 1913 | 96.6 |
| 2001 | 249 | 369 | 240 | 305 | 222 | 84  | 204 | 71 | 30 | 1819 | 1896 | 94.4 |
| 2002 | 250 | 366 | 238 | 296 | 220 | 70  | 207 | 71 | 32 | 1792 | 1871 | 91.2 |
| 2003 | 250 | 363 | 239 | 294 | 217 | 58  | 208 | 70 | 33 | 1777 | 1844 | 87.7 |
| 2004 | 250 | 360 | 238 | 288 | 215 | 48  | 211 | 69 | 35 | 1757 | 1820 | 84.7 |
| 2005 | 249 | 356 | 238 | 283 | 213 | 40  | 214 | 69 | 36 | 1741 | 1798 | 81.9 |
| 2006 | 249 | 353 | 238 | 277 | 210 | 34  | 214 | 68 | 37 | 1723 | 1778 | 79.3 |
| 2007 | 248 | 350 | 239 | 273 | 207 | 28  | 214 | 68 | 39 | 1708 | 1759 | 76.9 |
| 2008 | 247 | 347 | 240 | 269 | 205 | 24  | 212 | 67 | 40 | 1692 | 1742 | 74.6 |
| 2009 | 246 | 344 | 240 | 259 | 201 | 20  | 210 | 66 | 42 | 1669 | 1725 | 72.4 |
| 2010 | 244 | 341 | 238 | 254 | 199 | 17  | 208 | 66 | 43 | 1651 | 1706 | 70.1 |
| 2011 | 243 | 338 | 237 | 255 | 196 | 14  | 206 | 65 | 45 | 1639 | 1687 | 67.6 |
| 2012 | 242 | 335 | 237 | 253 | 193 | 12  | 203 | 65 | 46 | 1626 | 1670 | 65.4 |
| 2013 | 241 | 332 | 239 | 249 | 191 | 10  | 201 | 64 | 48 | 1612 | 1655 | 63.5 |
| 2014 | 239 | 330 | 239 | 241 | 189 | 8   | 197 | 63 | 49 | 1592 | 1640 | 61.6 |
| 2015 | 238 | 329 | 242 | 240 | 186 | 7   | 193 | 63 | 50 | 1585 | 1624 | 59.6 |
| 2016 | 236 | 326 | 246 | 243 | 184 | 6   | 190 | 62 | 51 | 1582 | 1610 | 57.7 |

#### RBF model MATLAB codes:

```

1.rbftest1.m:
clear all
y1=[3550    3672    3790    3895    3981    4049    4097    4129    4150    4162
4161    4147    4126    4102    4077    4051    4024    3996    3968    3938    3906
3876    3847    3818    3789];
y2=[1798    1849    1892    1921    1936    1940    1933    1924    1913    1896
1871    1844    1820    1798    1778    1759    1742    1725    1706    1687
1670    1655    1640    1624    1610];
x = 1992:1:2016;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Antarctic
% y1=[861    876 885 895 907 915 920 926 930 932 934 934 933 932 931 927 923
919 914 909 905 900 895 891 884];
% y2=[798    804 805 803 800 797 792 788 783 777 772 765 758 751 744 737 730
724 719 712 706 700 695 692 688];
% y6=[500    500 501 504 499 492 505 513 506 496 491 494 491 492 492 495 497
496 493 490 491 495 493 500 500.1];
% y5=[583    583 583 583 585 582 594 596 572 545 529 525 515 505 496 488 481
463 454 456 453 444 430 429 427];
% y3=[431    429 425 422 418 414 410 407 401 397 392 388 384 380 375 370 366
360 355 350 345 341 337 332 329];
% y4=[292    313 336 357 367 381 391 404 414 419 424 428 434 441 442 442 440
437 435 431 427 423 417 411 406];
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% mid altitude

```

```

% y1=[230 234 237 239 242 245 246 248 249 249 250 250 250 249 249 248 247
246 244 243 242 241 239 238 236];
% y2=[379 382 382 381 380 378 376 374 372 369 366 363 360 356 353 350 347
344 341 338 335 332 330 329 326];
% y6=[242 242 242 244 241 238 244 248 245 240 238 239 238 238 238 239 240
240 238 237 237 239 239 242 241];
% y5=[326 326 326 326 327 326 333 333 320 305 296 294 288 283 277 273 269
259 254 255 253 249 241 240 239.7];
% y3=[242 240 238 236 234 232 230 228 225 222 220 217 215 213 210 207 205
201 199 196 193 191 189 186 184];
% y4=[140 150 161 171 177 184 190 196 201 204 207 208 211 214 214 214 212
210 208 206 203 201 197 193 190];
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
xi =1980:1:2090;

op3=rbfcreate(x, y2,'RBFFunction', 'cubic'); rbfcheck(op3);
op4=rbfcreate(x, y1,'RBFFunction', 'cubic'); rbfcheck(op4);
% % op5=rbfcreate(x, y3,'RBFFunction', 'cubic'); rbfcheck(op5);
% % op6=rbfcreate(x, y4,'RBFFunction', 'cubic'); rbfcheck(op6);
% % op7=rbfcreate(x, y5,'RBFFunction', 'cubic'); rbfcheck(op7);
% % op8=rbfcreate(x, y6,'RBFFunction', 'cubic'); rbfcheck(op8);

fi3 = rbfinterp(xi, op3);
fi4 = rbfinterp(xi, op4);
% fi5 = rbfinterp(xi, op5);
% fi6 = rbfinterp(xi, op6);
% fi7 = rbfinterp(xi, op7);
% fi8 = rbfinterp(xi, op8);
plot(xi, fi4,'--r');
% hold on
% plot(xi, fi5,'--k');
hold on
plot(xi, fi3,'--b');
% hold on
% plot(xi, fi6,'--g');
% hold on
% plot(xi, fi7,'--m');
% hold on
% plot(xi, fi8,'--','Color',[0.6 0.5 0.3]);

set(findall(gcf,'type','line'),'linewidth',2)
grid on
plot(x, y1,'^r');
hold on

```



```

% plot(x, y3, 'ok');
% hold on
plot(x, y2, 'Squareb');
hold on
% plot(x, y4, 'Diamondg');
% hold on
% plot(x, y5, 'Pentagramm');
% hold on
% plot(x, y6, '*', 'Color', [0.6 0.5 0.3]);
% legend('CFC-12 (predicted)', 'CFC-11 (predicted)', 'CCl4
(predicted)', 'Halons (predicted)', 'CH3Br (predicted)', 'CH3Cl
(predicted)', 'CFC-12 (obs)', 'CFC-11 (obs)', 'CCl4 (obs)', 'Halons
(obs)', 'CH3Br (obs)', 'CH3Cl (obs)');
legend('Antartica (predictive)', 'Mid-Lantitudes (predictive)', 'Antartica
(obs)', 'Mid-Lantitudes (obs)')

```

# The establishment of direct metro lines

## Abstract:

Nowadays, passengers usually spend a long time on travelling especially between two far apart stations in present metro network of large cities. Thus, it seems to be a good choice to set a direct metro line to save time.

The direct metro line is a kind of multi-index problem, which means the result cannot be obtained by accurate calculation with plenty of data directly. This paper aims to analyze that this model about the direct metro line is built to make a comprehensive evaluation of whether build a special direct line or not.

As for Task 1, we describe metrics in six aspects: *the traffic interchange of line, the size of the surrounding schools, peripheral business scale, the employment post, tourism value, surrounding community scale* based on surroundings nearby subway stations after discussion. We build this model based on Analytic Hierarchy Process (AHP). AHP is a decision-making method for multi-index problems which is particularly suitable in situation where results cannot be obtained by accurate calculation directly. Firstly, we make a weight matrix and construct a function to work out the weight of these 6 indexes. Secondly, we attempt to figure out the radius  $r$  of influence including direct and indirect influence to optimize the station choices with setting a standard. Finally, we obtain some suitable stations as shown in the text below.

As for Task 2, we need to compare adjusting the interval between trains to building a special direct line in consideration of their expenses and benefits to society and economy. To building a special direct line, we decide to construct a function about expense and find the cost of building per kilometer. Likewise, we easily think to do linear programming in order to find the cost of adjusting schedules besides profit. On this basis, we can determine whether a station suits for adjusting or building. Finally we decide to choose station  $B_{1-1}$  and  $B_{1-10}$  as representative for analyzing the problem because others can be similarly obtained according to this rule.

As for Task 3, we attempt to write a letter with a modified time- table to the mayor by evaluating our model and discussing its strengths and weaknesses based on the previous basis.

**Key words:** Direct metro line, Analytic Hierarchy Process, Cost model, Linear programming, Schedule

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# **I. Introduction**

## **1.1 Problem Background**

In past 30 years, China's urban rail transit gradually entered a steady, orderly and rapid development stage, especially in the past 10 years. By the end of year 2015, China will have 158 lines in 38 cities, operating a total of 4190 kilometers. With the rapid improvement of traffic in various cities, it becomes more and more convenient for people to go outside. However, it usually takes a long time to transport passengers between two distant stations while there are only a few special direct lines.

We face mainly three problems:

- Articulate and set our own metrics and build a mathematical model.
- Set up an evaluation system for the model with plenty data.
- Analyze the influences of the models especially economic and social benefits.

## **1.2 Previous Research**

Some newspapers, magazines and websites discuss about subway whether the schedule should be advanced or not without scientific and rational theories. And there are also some papers talking about track construction but little about building special direct lines.

## **1.3 Our Work**

In our paper, we try to establish an integrated model for this problem.

For Part 1, we build a AHP model and set up selection and evaluation system.

For Part 2, we build a linear programming and a cost model.

For Part 3, we summarize our study and write a letter to mayor with new schedules.

## **II. The Description of the Problem**

### **2.1 Restatement of the problem.**

In current metro network of metropolis, it usually takes a long time to transport passengers between two far apart stations. We may suggest that set a direct shuttle train between two long-distance stations to save time of the busy passengers.

Part I. Please develop a mathematical framework to optimize the choice of the end stations, and evaluate the expected result of the suggest.

Part II. Explore any advantages you may achieve in terms of economic and social benefits if you adjust the interval between trains or build a special direct line.

Part III. Write a two-page letter to the Mayor of your city summarizing your analysis. Include your train schedule as the second page of your letter

### **2.2 The problem analysis.**

#### **2.2.1 The analysis to question 1.**

First of all, we choose a first-tier city as our main object of study. For the end station of choice, we must first identify a few indicators as a measure of the index in comparison to the muscle too much, too little can't make an accurate judgment, at the same time, these indicators should have representative and scientific, comprehensive, we consider the Analytic Hierarchy Process (AHP) is adopted to establish the model of initial site selection; And then according to this model and we collected data to evaluate the points on the subway, it is concluded that the optimal starting point. Finally predict the choice of the starting point of the results, and evaluate.

#### **2.2.2 The analysis to question 2.**

Firstly, we need to discuss two projects we made as follows:

Project 1: We can build a new subway line as a direct line.

Project2: We can adjust the schedule based on previous subway's situation.

To project1, we need to consider the cost of building a new direct line and build a model which can calculate it. Then we refer to much data about how subway building costs and substituted to the cost model. Finally, we obtain how much it costs per kilometer if we try to build a direct line.

To project2, we can add more direct subways and adjust the time based on previous situations.

In the end, we try to figure out what benefits two projects do to society and economy.

### III. Models

#### QUESTION 1

##### 3.1 Symbols and Meanings.

| Symbols    | Meanings                                                                                        |
|------------|-------------------------------------------------------------------------------------------------|
| $a_{ij}$   | The relative importance of the factor i to the factor j in the Analytic Hierarchy Process (AHP) |
| CI         | Consistency index                                                                               |
| RI         | Average random consistency index                                                                |
| CR         | Consistency ratio                                                                               |
| $W_i$      | The ranking weight of the total target of the target layer                                      |
| $W_j$      | The weight of the target layer on the target layer                                              |
| $W_{ij}$   | Single layer ranking weights of the scheme layer to the criterion layer                         |
| A          | Alternate station A                                                                             |
| B          | Alternate station B                                                                             |
| C          | Alternate station C                                                                             |
| $E_i$      | Index                                                                                           |
| $\omega$   | Weight vector                                                                                   |
| $B_i$      | Combined weight vector                                                                          |
| k          | From 1 to 6                                                                                     |
| $K(x_i)$   | Weighted value of indexes                                                                       |
| F          | Index weight function                                                                           |
| $D_i$      | Relative to the weight of the maximum of the index statistics                                   |
| A1-1~A1-22 | Stations of line A1                                                                             |
| B1-1~B1-20 | Stations of line B1                                                                             |
| C1-1~C1-26 | Stations of line C1                                                                             |
| D1-1~D1-8  | Stations of line D1                                                                             |

|       |                                                      |
|-------|------------------------------------------------------|
| $S_m$ | Radiant area of a center station (m<br>from 1 to 38) |
|-------|------------------------------------------------------|

TABLE 1

## 3.2 Assumptions

- (1) Difference of each community is small.
- (2) Difference of every kind of school is small.
- (3) Difference of every mall is small.

## 3.3 Basic models

### 3.3.1 Analytic Hierarchy Process model

Analytical Hierarchy Process(AHP), a decision-making methodology developed by Saaty is an attempt in this direction. AHP can be used in any situation where the presence of multiple influencing factors and decision criteria make it difficult to understand the interactions among them intuitively. In such cases AHP offers a structured approach to reduce the complexity and help us making a decision objectively. This process can be divided into 4 parts: firstly, modelling hierarchical structure model; building judgment matrix; single hierarchical arrangement and consistency test; comprehensive hierarchical arrangement and consistency test. The detailed steps are:

#### Step1.modelling hierarchical structure model

Firstly, we should make the question methodize and hierarchic by building a structural model which has clear hierarchical relation if we want to use analytic hierarchy process to solve the problem.

Hierarchical structure usually has target layer, criterion layer and schematic layer. Criterion layer is to analyze the intended target and ideal results of the problem and always has one factor in this hierarchical. Target layer contains intermediate link of the goal and it always has many hierarchical. Schematic layer contains the elective measures and decisions to achieve the goal.

#### Step2.building judgment matrix

In fact, we are usually faced with the difficulties of comparing every factor how they do to the problem and draw a wrong conclusion when we are put in a double squeeze if there are a lot of influencing factors. But if we decide to compare by one by one, the things may seem to be easier. The details are: using  $a_{ij}$  ( $i, j=1,2,\dots, n$ ) to show the relative importance of factor  $i$  relative to factor  $j$ .  $a_{ij}$  can be chosen from one to nine and their reciprocal numbers.

#### Step3..Level simple sequence and consistency test

Level simple sequence is the process that we figure out every factor's relative sequential weight to their target hierarchy by comparing judgement matrix in their domains.

The way to solve the problem includes eigenvalue method, summation method and square root method.

Making judgement matrix may reduce the effect of other factors, but it is likely to contain a certain extent nonuniformity if we compare the result from its comprehensive, and elements of a matrix satisfy the condition if want to the results by comparing uniformly:

$$a_{ij} \times a_{jk} = a_{ik}, \forall i, j, k = 1, 2, 3, 4 \dots, n \quad (3.1)$$

The step of testing the uniformity of judgement matrix:

(1) Calculate the CI which is the index of uniformity

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (3.2)$$

(2) Find the corresponding average random consistency index RI. When  $n=1, 2, \dots, 9$ , the corresponding values of RI are shown in TABLE 2:

| n  | 1 | 2 | 3    | 4   | 5    | 6    | 7    | 8    | 9    |
|----|---|---|------|-----|------|------|------|------|------|
| RI | 0 | 0 | 0.58 | 0.9 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 |

TABLE 2

(3) Calculate the consistency ratio CR

$$CR = \frac{CI}{RI} \quad (3.3)$$

When  $CR < 0.1$ , the consistency of the judgment matrix can be accepted, or to make appropriate adjustments to the discriminant matrix.

#### Step4: Hierarchical total ordering and consistency checking.

Hierarchical order is a set of elements on the corresponding element of the corresponding elements of the weight vector, we will eventually get the total level of the total sort, the weight of the total ranking order to synthesize the weight of a single ranking.

Let the weight of the target layer is  $W_j$ , let program layer to the standard layer of the single sort weight is  $W_{ij}$ , the ranking weight of the total target of the target layer is  $W_i$ . And

$$W_i = \sum_{j=1}^m W_{ij} \times W_j, i = 1, 2, \dots, n. \quad (3.4)$$

Consistency checking is also needed in order to make the level of the hierarchy, and the elements that are related to the target layer are tested by means of consistency.



And the test index is  $CI(j), (j=1, 2, \dots, m)$ , the corresponding average random consistency index is  $RI(j)$ , and the total order of the criteria layer is the random consistency ratio is :

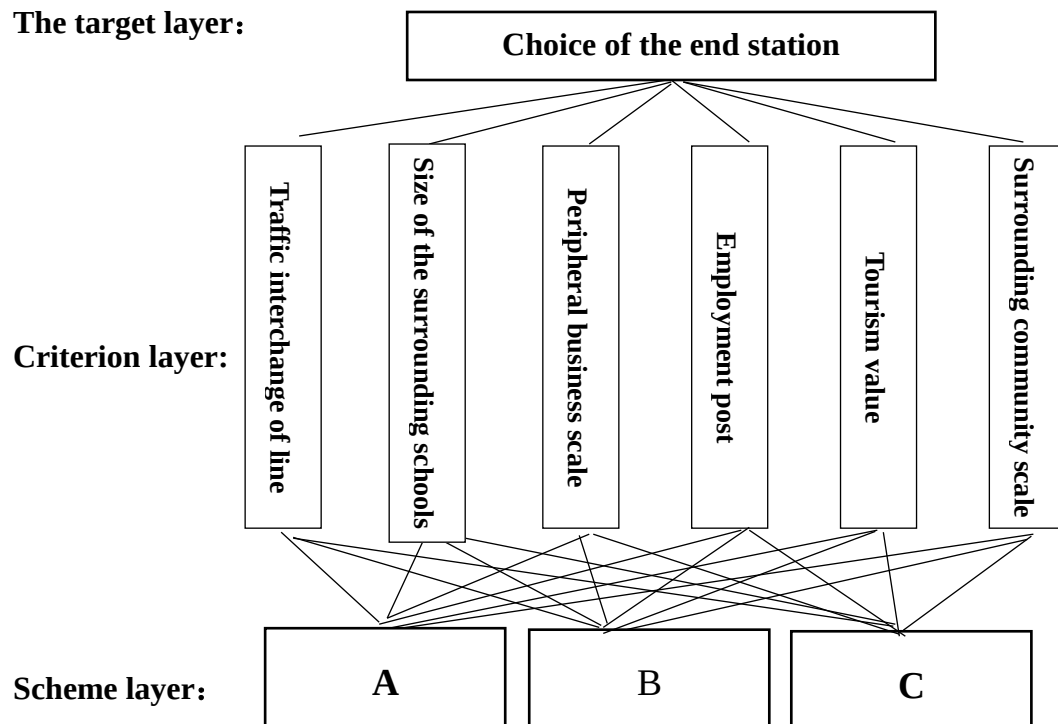
$$CR = \frac{\sum_{j=1}^m CI(j) a_j}{\sum_{j=1}^m RI(j) a_j} \quad (3.5)$$

When  $CR < 0.1$ , it is considered that the level of the total ranking has a satisfactory consistency and accepting the result.

### Step5: Solving model

Reference to the subway network planning and selection of subway line main station location, we consider whether the subway station transport interchange, the six indexes of subway stations near the school scale, business scale, employment, tourism value, the surrounding area scale, basically can satisfy the comprehensive index requirements. Thus, we establish the hierarchical structure as follows:

The target layer:



We let:

$E_1$ : Traffic interchange of line

$E_2$ : Size of the surrounding schools

$E_3$ : Peripheral business scale

$E_4$ : Employment post

$E_5$ : Tourism value

$E_6$ : Surrounding community scale

$a_{ij}$ : importance of  $E_i$  : importance of  $E_j$

Through access to information, the use of various levels indicators, comparing the degree of influences on the targets of the higher corresponding values in accordance with the rules, it is concluded that the comparison discriminant matrix as shown below:

$$A = \begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 2 \\ 1/3 & 1 & 1 & 1/2 & 1/2 & 1/2 \\ 1/3 & 1 & 1 & 1/2 & 1/2 & 1/2 \\ 1/2 & 2 & 2 & 1 & 1 & 1 \\ 1/2 & 2 & 2 & 1 & 1 & 1 \\ 1/2 & 2 & 2 & 1 & 1 & 1 \end{pmatrix}$$

Calculating the maximum eigenvalue by using MATLAB,  $\lambda_{max} = 6.0138$ ,

Corresponding normalized feature vectors is:

$$\omega = (0.3106, 0.0894, 0.0894, 0.1702, 0.1702, 0.1702)^T.$$

Consistency index CI by formula (3.2):  $CI = 0.00276$  (This time  $n=6$ ), Check list was random consistency index  $RI=1.24$ , consistency ratio index  $CR$  through the formula (3.3):  $CR=0.00223 < 0.1$ , Through consistency test, so it can be a weight vector.

We use the same method to construct the pairwise comparison matrix for each of the criteria of the scheme layer. Let them are  $B_k$  ( $k=1,2,3,4,5,6$ ):

$$B_1 = \begin{pmatrix} 1 & 1/2 & 1/4 \\ 2 & 1 & 1/2 \\ 4 & 2 & 1 \end{pmatrix} \quad B_2 = \begin{pmatrix} 1 & 1/2 & 1/3 \\ 2 & 1 & 1/2 \\ 3 & 2 & 1 \end{pmatrix} \quad B_3 = \begin{pmatrix} 1 & 1/2 & 1/3 \\ 2 & 1 & 1/2 \\ 3 & 2 & 1 \end{pmatrix}$$

$$B_4 = \begin{pmatrix} 1 & 1/2 & 1/2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix} \quad B_5 = \begin{pmatrix} 1 & 1/2 & 1/4 \\ 2 & 1 & 1/2 \\ 4 & 2 & 1 \end{pmatrix} \quad B_6 = \begin{pmatrix} 1 & 1/2 & 1/2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}$$

Getting the Table3 by using MATLAB:

| k            | 1                          | 2                          | 3                          | 4                          | 5                          | 6                          |
|--------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| $\omega_k^T$ | 0.1429<br>0.2857<br>0.5714 | 0.1634<br>0.2970<br>0.5396 | 0.1634<br>0.2970<br>0.5396 | 0.2000<br>0.4000<br>0.4000 | 0.1429<br>0.2857<br>0.5714 | 0.2000<br>0.4000<br>0.4000 |
| $\lambda_k$  | 3.0000                     | 3.0092                     | 3.0092                     | 3.0000                     | 3.0000                     | 3.0000                     |
| $CI_k$       | 0.0000                     | 0.0046                     | 0.0046                     | 0.0000                     | 0.0000                     | 0.0000                     |

**TABLE 3**

It is not difficult to see that all of the values of  $CI_k$  can be tested by consistency.

According to the above calculation, 6 indexes and their weights of selection of the end stations as shown in TABLE 4.

| sequence number | index                           | weighted value |
|-----------------|---------------------------------|----------------|
| 1               | Traffic interchange             | 0.3106         |
| 2               | Size of the surrounding schools | 0.0894         |
| 3               | Peripheral business scale       | 0.0894         |
| 4               | Employment post                 | 0.1702         |
| 5               | Tourism value                   | 0.1702         |
| 6               | Surrounding community scale     | 0.1702         |

**TABLE 4**

### 3.3.2 Establishment of function model

Let  $K(x_i)$  (i=1, 2, 3, 4, 5, 6): weighted value of each index;

F: sum of the weighted value of each index of the station.

$D_i$ : Relative to the weight of the maximum of the index statistics

And expression for establishment of function model is:

$$F = 0.3106 * D_1 + 0.0894 * D_2 + 0.0894 * D_3 + 0.1702 * D_4 + 0.1702 * D_5 + 0.1702 * D_6 \quad (3.6)$$

Above the model shows that higher of the value of the function, higher of the degree of optional.

When  $D_i = K(x_i)$ , this time,  $F = F_0 = 0.1993612$ , and we think this is a critical value of F.

We obtain:

If  $F > 0.1993612$ , we select this station as the end station. Or  $F = 0$  we select this station as the end station(it has large potential for development.)

Else if  $0 < F < 0.1993612$ , we don't select it.

### 3.3.3 Weight function value of each station

We select Tianjin subway lines to analyze. And we consider line 9 is a light rail so that we have no statistics on it. We label the subway route map of subway's route and calculate the distance between the two stations.



Figure 1

We collect the following data through “Tianjin Rail Transit net”, and making the following TABLES by excel(in Appendix TABEL 1~TABEL 8).

Now we are going to quantize ‘YES ’ and ‘ NO ’ from the tables and let ‘ YES ’ be 1 while ‘ NO ’ be 0. We can also normalize all the other indexes because we consider that the maximum number (10) from all the data from the tables should be treated as radix when quantizing.

Then through the statistic and collected data, we can get a conclusion that a primary school usually has 2000 students, and a middle school usually has 3000 students, and a college has 5000 students or so while a university has 30000 students or so. Now if we let the population of one school divide all schools’ population and we can obtain:

1. the primary school’s contribution to this index:

$$2000/(2000+3000+5000+30000)=0.05;$$

2. the middle school’s contribution to this index:

$$3000/(2000+3000+5000+30000)=0.075;$$

3. the college's contribution to this index:

$$5000/(2000+3000+5000+30000)=0.125;$$

4. the university's contribution to this index:

$$30000/(2000+3000+5000+30000)=0.75;$$

According to formula (3.6) and quantizing process we can obtain all the F of every route and station by excel tables. We get F as **TABLE 5~TABLE8** by using excel.

| Station | weighted value (wv)of school | wv of community | wv of Workplace | wv of mall | wv of Tourist spots | wv of interchange | F        |
|---------|------------------------------|-----------------|-----------------|------------|---------------------|-------------------|----------|
| A1-1    | 0.0011175                    | 0.01702         | 0.01702         | 0          | 0                   | 0                 | 0.035158 |
| A1-2    | 0.0145275                    | 0.06808         | 0.03404         | 0          | 0                   | 0                 | 0.116648 |
| A1-3    | 0.0011175                    | 0               | 0               | 0          | 0                   | 0                 | 0.001118 |
| A1-4    | 0.000447                     | 0.11914         | 0.03404         | 0.01788    | 0.01702             | 0                 | 0.188527 |
| A1-5    | 0.000447                     | 0.03404         | 0               | 0.00894    | 0                   | 0                 | 0.043427 |
| A1-6    | 0.000894                     | 0.05106         | 0.0851          | 0.03576    | 0                   | 0                 | 0.172814 |
| A1-7    | 0.001788                     | 0.11914         | 0.06808         | 0.0447     | 0                   | 0                 | 0.233708 |
| A1-8    | 0.0011175                    | 0.13616         | 0.03404         | 0.01788    | 0.05106             | 0                 | 0.240258 |
| A1-9    | 0.008493                     | 0               | 0.03404         | 0.06258    | 0.05106             | 0                 | 0.156173 |
| A1-10   | 0.0033525                    | 0.01702         | 0.01702         | 0.08046    | 0.01702             | 0.3106            | 0.445473 |
| A1-11   | 0.0029055                    | 0.11914         | 0.05106         | 0.03576    | 0.01702             | 0                 | 0.225886 |
| A1-12   | 0.000447                     | 0.03404         | 0.05106         | 0.03576    | 0.03404             | 0                 | 0.155347 |
| A1-13   | 0.002682                     | 0.05106         | 0.03404         | 0.02682    | 0                   | 0.3106            | 0.425202 |
| A1-14   | 0                            | 0               | 0               | 0.00894    | 0                   | 0                 | 0.00894  |
| A1-15   | 0.0037995                    | 0.06808         | 0.01702         | 0.02682    | 0.03404             | 0                 | 0.14976  |
| A1-16   | 0.000447                     | 0               | 0               | 0          | 0.03404             | 0                 | 0.034487 |
| A1-17   | 0.008493                     | 0.0851          | 0               | 0.02682    | 0                   | 0                 | 0.120413 |
| A1-18   | 0.005364                     | 0.06808         | 0.10212         | 0          | 0                   | 0                 | 0.175564 |
| A1-19   | 0.00894                      | 0.10212         | 0.01702         | 0          | 0                   | 0                 | 0.12808  |
| A1-20   | 0.000447                     | 0.1702          | 0               | 0.00894    | 0                   | 0                 | 0.179587 |
| A1-21   | 0.0015645                    | 0.06808         | 0               | 0.0447     | 0.01702             | 0                 | 0.131365 |
| A1-22   | 0.0011175                    | 0.10212         | 0               | 0          | 0                   | 0                 | 0.103238 |

**TABLE 5**

| station | weighted value (wv)of school | wv of community | wv of Workplace | wv of mall | wv of Tourist spots | wv of interchange | F        |
|---------|------------------------------|-----------------|-----------------|------------|---------------------|-------------------|----------|
| B1-1    | 0                            | 0               | 0               | 0          | 0                   | 0                 | 0        |
| B1-2    | 0                            | 0.01702         | 0.03404         | 0          | 0                   | 0                 | 0.05106  |
| B1-3    | 0                            | 0.01702         | 0               | 0          | 0.01702             | 0                 | 0.03404  |
| B1-4    | 0                            | 0.0851          | 0.01702         | 0          | 0                   | 0                 | 0.10212  |
| B1-5    | 0.0011175                    | 0.0851          | 0.01702         | 0.00894    | 0                   | 0                 | 0.112178 |
| B1-6    | 0.0011175                    | 0.03404         | 0               | 0          | 0                   | 0                 | 0.035158 |
| B1-7    | 0.0006705                    | 0.10212         | 0.03404         | 0          | 0                   | 0                 | 0.136831 |
| B1-8    | 0                            | 0.0851          | 0.01702         | 0.00894    | 0                   | 0                 | 0.11106  |
| B1-9    | 0.0006705                    | 0.05106         | 0               | 0.00894    | 0                   | 0                 | 0.060671 |
| B1-10   | 0.0011175                    | 0.0851          | 0.0851          | 0.01788    | 0.01702             | 0.3106            | 0.516818 |
| B1-11   | 0                            | 0.05106         | 0.03404         | 0          | 0.03404             | 0                 | 0.11914  |
| B1-12   | 0                            | 0               | 0               | 0          | 0.01702             | 0                 | 0.01702  |
| B1-13   | 0                            | 0.03404         | 0.01702         | 0.02682    | 0.03404             | 0.3106            | 0.42252  |
| B1-14   | 0                            | 0.03404         | 0               | 0.02682    | 0                   | 0                 | 0.06086  |
| B1-15   | 0.0006705                    | 0.05106         | 0.01702         | 0.00894    | 0                   | 0.3106            | 0.388291 |
| B1-16   | 0                            | 0.03404         | 0.03404         | 0.01788    | 0.01702             | 0                 | 0.10298  |
| B1-17   | 0.003576                     | 0.06808         | 0               | 0          | 0                   | 0                 | 0.071656 |
| B1-18   | 0.0006705                    | 0.05106         | 0.01702         | 0          | 0                   | 0                 | 0.068751 |
| B1-19   | 0                            | 0.0851          | 0               | 0          | 0                   | 0                 | 0.0851   |
| B1-20   | 0                            | 0               | 0.03404         | 0.00894    | 0.01702             | 0                 | 0.06     |

**TABLE 6**

| station | weighted value (wv) of school | wv of community | wv of Workplace | wv of mall | wv of Tourist spots | wv of interchange | F        |
|---------|-------------------------------|-----------------|-----------------|------------|---------------------|-------------------|----------|
| C1-1    | 0                             | 0.01702         | 0.01702         | 0          | 0                   | 0                 | 0.03404  |
| C1-2    | 0                             | 0               | 0.01702         | 0          | 0                   | 0                 | 0.01702  |
| C1-3    | 0                             | 0               | 0.05106         | 0          | 0                   | 0                 | 0.05106  |
| C1-4    | 0                             | 0.06808         | 0.01702         | 0          | 0                   | 0                 | 0.0851   |
| C1-5    | 0                             | 0.06808         | 0               | 0          | 0                   | 0                 | 0.06808  |
| C1-6    | 0.0006705                     | 0.03404         | 0               | 0          | 0                   | 0                 | 0.034711 |
| C1-7    | 0.001788                      | 0.03404         | 0               | 0          | 0.01702             | 0                 | 0.052848 |
| C1-8    | 0.001788                      | 0.01702         | 0               | 0          | 0.01702             | 0                 | 0.035828 |
| C1-9    | 0.0033525                     | 0.01702         | 0               | 0.01788    | 0.01702             | 0                 | 0.055273 |
| C1-10   | 0.0020115                     | 0.03404         | 0               | 0.00894    | 0                   | 0                 | 0.044992 |
| C1-11   | 0.0011175                     | 0               | 0.0851          | 0.01788    | 0                   | 0.3106            | 0.414698 |
| C1-12   | 0                             | 0               | 0               | 0          | 0.01702             | 0                 | 0.01702  |
| C1-13   | 0                             | 0               | 0               | 0.03576    | 0.05106             | 0                 | 0.08682  |
| C1-14   | 0.001341                      | 0.01702         | 0.01702         | 0.08046    | 0.01702             | 0.3106            | 0.443461 |
| C1-15   | 0.000447                      | 0.0851          | 0               | 0          | 0.01702             | 0                 | 0.102567 |
| C1-16   | 0.021903                      | 0.05106         | 0.01702         | 0.00894    | 0                   | 0                 | 0.098923 |
| C1-17   | 0                             | 0.06808         | 0               | 0.00894    | 0.06808             | 0                 | 0.1451   |
| C1-18   | 0.006705                      | 0.01702         | 0               | 0          | 0.03404             | 0                 | 0.057765 |
| C1-19   | 0.0073755                     | 0.0851          | 0               | 0.00894    | 0                   | 0.3106            | 0.412016 |
| C1-20   | 0.006705                      | 0.06808         | 0               | 0.00894    | 0.01702             | 0                 | 0.100745 |
| C1-21   | 0                             | 0               | 0.03404         | 0.03576    | 0                   | 0                 | 0.0698   |
| C1-22   | 0.02682                       | 0               | 0               | 0          | 0                   | 0                 | 0.02682  |
| C1-23   | 0                             | 0.01702         | 0.01702         | 0          | 0                   | 0                 | 0.03404  |
| C1-24   | 0                             | 0               | 0.1702          | 0          | 0                   | 0                 | 0.1702   |
| C1-25   | 0                             | 0.05106         | 0               | 0          | 0                   | 0                 | 0.05106  |
| C1-26   | 0                             | 0               | 0               | 0          | 0                   | 0                 | 0        |

**TABLE 7**

| station | weighted value(wv) of school | wv of community | wv of Workplace | wv of Mall | wv of Tourist spots | wv of interchange | F        |
|---------|------------------------------|-----------------|-----------------|------------|---------------------|-------------------|----------|
| D1-1    | 0.000447                     | 0.0851          | 0.06808         | 0.01788    | 0                   | 0.3106            | 0.482107 |
| D1-2    | 0                            | 0.06808         | 0.06808         | 0          | 0                   | 0                 | 0.13616  |
| D1-3    | 0                            | 0.06808         | 0               | 0.00894    | 0                   | 0                 | 0.07702  |
| D1-4    | 0                            | 0.03404         | 0.03404         | 0          | 0                   | 0                 | 0.06808  |
| D1-5    | 0                            | 0.1702          | 0               | 0          | 0                   | 0                 | 0.1702   |
| D1-6    | 0                            | 0.10212         | 0               | 0.00894    | 0                   | 0.3106            | 0.42166  |
| D1-7    | 0                            | 0.10212         | 0               | 0          | 0                   | 0                 | 0.10212  |
| D1-8    | 0                            | 0.06808         | 0.01702         | 0          | 0.03404             | 0                 | 0.11914  |

**TABLE 8**

We find that

A1-7,A1-8,A1-10,A1-11,A1-13,B1-1,B1-10,B1-15,C1-11,C1-14,C1-19 ,C1-26,D1-1, D1-6 can be selected as the end station.

### 3.3.4 Final selection of the end station

#### ■ The kinds of the radius of subway station' influence

##### ① The radius of the direct influence

The radius of subway station' direct influence defines how long dwellers can go away if they come to subway station because walking is simpler and the most original way than other way to go to other places.

##### ② The radius of the indirect influence

The radius of subway station' indirect influence defines passengers can walk and then get to the subway station by other transportation. For example, we can take 'walking + public bicycles', 'walking + public bicycles + walking', 'walking + bus', 'walking + one bus + another bus', 'walking + bus + walking'.

#### ■ The way to get the radius of the subway station's influence

#### The radius $r_1$ of the direct influence

1. First, we need to consider the speed of walking. It is usually considered as 5 kilometers per hour from international standard. The step size should be 0,694m if we take 2 steps per second as the standard to calculate the walking speed. We can find a

benchmark which defines that step size/height is 0.407 for male and is 0.397 for female from the internet. And it is suit to a man who is 1.71 meters high and for a woman who is 1.75 meters high. In fact, Asian frame is slightly smaller than Europe and American's which leads to our country's average height is shorter than international. We also can find the accurate data that 1.67 meters high for Chinese male whose step size can be 0.68 meters long and 1.56 meters high for Chinese female whose step size can be 0.62 meters long.

We may find that it is faster than 2 steps per hour for almost people if we consider 5 kilometers per hour as the standard. After investigation, we can get a conclusion that much people may feel tired in this situation. In consideration of this research take subway passengers as subjects and we also need to consider women and children. In this study we take 0.625 meters long for each step and 2 steps per second as the standard from much observing and comparing and we can get a conclusion that our usual walking speed can reach 4.5 kilometers per hour and take it as the standard for our study.

2. Find how long people may feel comfortable without tired when they are walking and how long may they can reach in this situation

From the reference and data from the internet we can find that people may walk 15 minutes and feel comfortable without tired in urban which is based on the fact our big city has wonderful public transit which is improving gradually. Hence, we can take 15 minutes as the standard because of people walking to destination without tired and it seem to be a far place if people need to walk more than 15 minutes and usually people may choose another efficient way to go there.

From the distance formula

$$x = vt + x_0 \quad (3.7)$$

We can take  $x_0$  as 0 in this model and calculate that we usually can walk 1125 meters away in 15 minutes. And we can take it as 1200 meters for calculate easier from individual differences.

We elicit that a conclusion that 1200 meters is the best one for our study in this model as the best radius of the subway station's direct influence

### **The radius $r_2$ of the indirect influence**

As we all know, other public transit bus, public bicycle, taxi and so forth in urban may also increase the radius. Just because of these public transit, we can know that the radius of the subway station's indirect influence may also be much affected. But these are based on local the public transportation, construction of traffic network and also includes the tradition and life styles of local people. Because of lack of enough survey data, it is difficult to get a specific standard of quantification.

### **The whole radius $r$ of every kind of influence**

According to general solution, we usually focus on the subway station, and take one number as the radius to delimit the circular range. So we can easily get the

relation  $r=r_1+r_2$  and also the relation schema as follows:

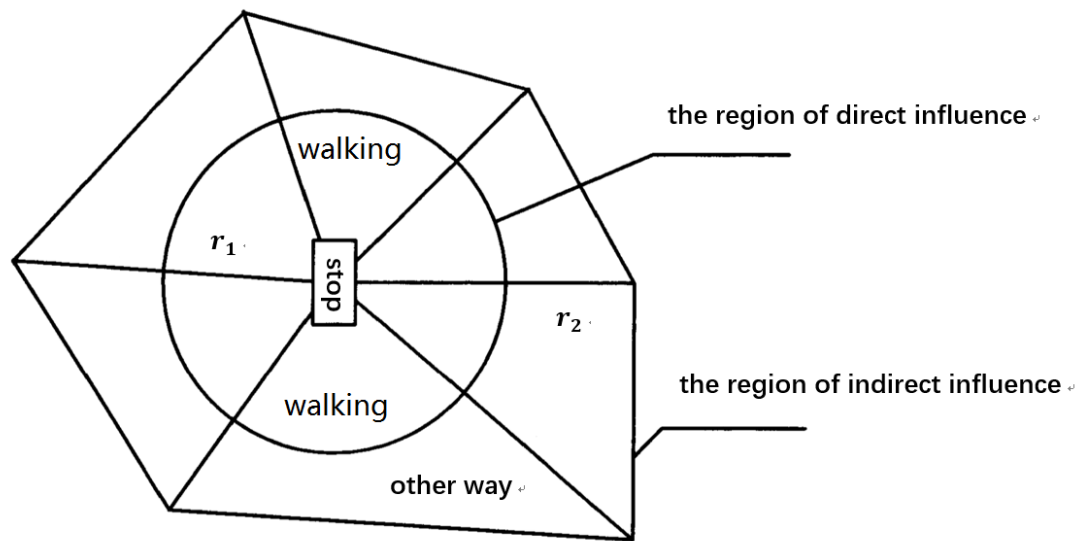


Figure 2

Conclusion:

Because every place may have different public-transit situation, it may be too complicated and irregularly if we consider the indirect influence too much. It also will be so untypical that it can't be popularized to other cities. From the relation  $r=r_1+r_2 \geq r_1$ , we can easily know that it is at least 1200 meters. And we can know that two subway station has no difference if the distance between two stations of subway less than 1200 meters. Whence, we can focus on one subway station, and take 1200 meters as the radius radiate to outward and form a circle. We can consider that the value of this subway station belongs to the center station if there are other subway stations in this area expect this center station.

#### The rules of choosing the center subway station:

We can take one subway station as the center one by one on the subway route, and its range of radiation which is a circle which radius is 1200 meters contains other stations 's value belonged to center station. To one designated subway station, we can make it belong to a station which has fewer distance for this designated station if this designated station satisfy the condition for two adjacent subway stations at the same time. Also, we take one subway station as the center station which is farther away from a group of other center stations if we are facing even numbers subway stations forming an area.

- According to this rule, finally we choose these points as the center subway stations as follows:

**The black points mean that it is the center subway station.**

**Each circle means the radiant area of the center station.**



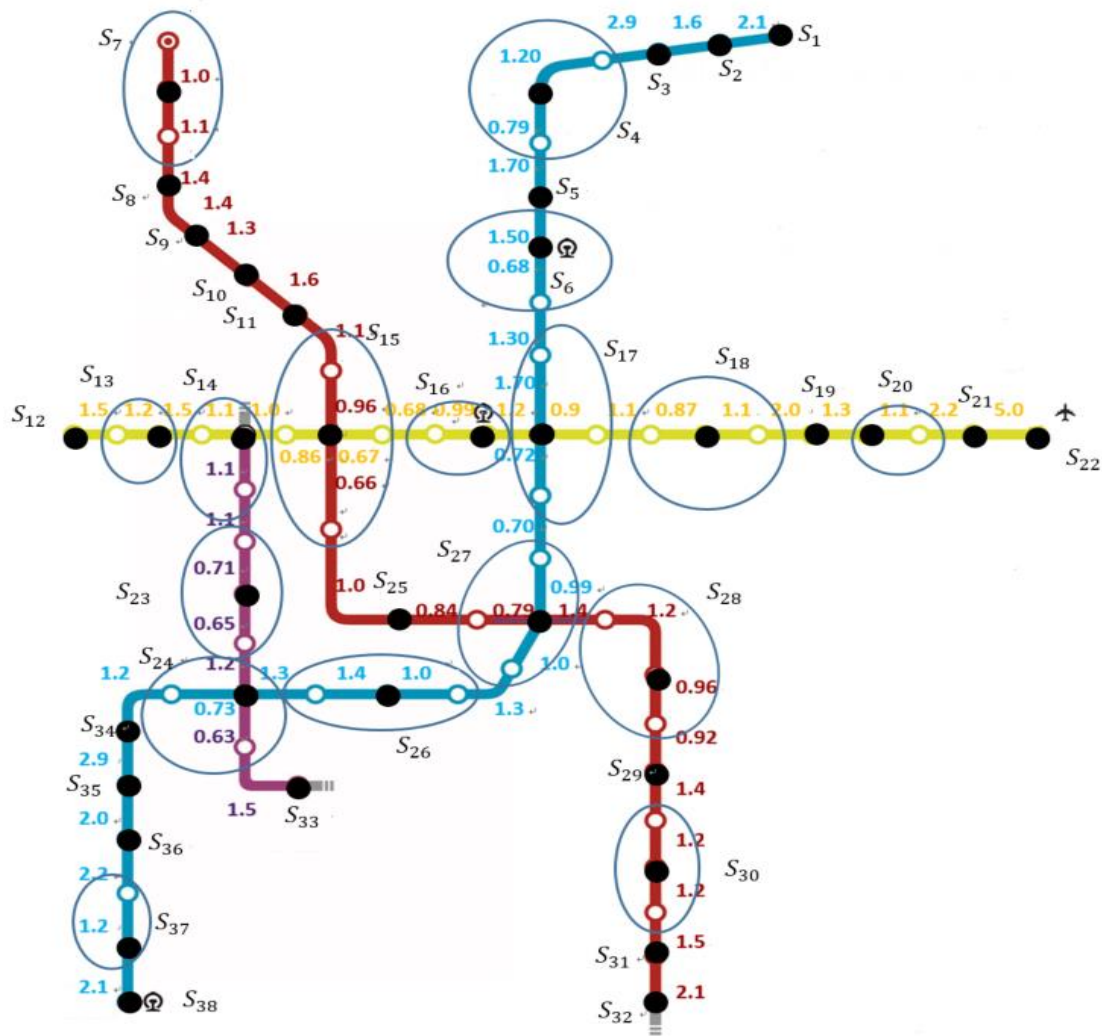


Figure 3

We find  $S_{17}, S_{15}, S_{28}, S_{24}, S_{14}, S_{22}, S_{38}, S_{27}$  can be selected, and let the center station as the station. Selecting A1-8, A1-10, B1-1, B1-10, B1-15, C1-19, C1-26 as the end station.

**Working out the distance between points, and working out the average value of these. (data in the appendix)**

Then the average of these distances is 10.186km.

We think distance of two points at least is 10.186km to establish a direct station.

### 3.4 Evaluation of expected results

First, we can find that almost every alternative point are close by intersections of subway stations. With the data from investigation, we can get a conclusion that there are plenty of residential areas, busiest streets, many schools and jobs nearby. If we plan to build non-stop subway nearby, it will not only do good to local economy and many talented people can be transported to the other side of non-stop subway's station, but also can bring much economic benefits and social benefits with easing the traffic

pressure. On the one hand, it will be more convenient for people to go out. On the other hand, it also do good to transport construction if we plan to build non-stop subway nearby airport, port and railway stations. From here we see that building non-stop subway, and it does good to local development, and can also relieve the pressure of subway.

But it is obvious to know it may increase degree of difficulty of dispatching other subways and decrease the safety which can't be ignored if we decide to build a non-stop subway on the previous routes. It may need spend more manpower and more materials, wasting too much resource and cost too much money if we decide to build a special direct line.

In general, we may confront with problems but can overcome if we plan to build these non-stop stations and special direct lines. Although it seem to be it will cost too much money, it can bring inestimable value.

Personal speaking, we think that the advantages outweigh the disadvantages if we decide to build a special direct line.

## QUESTION 2

### 3.5 Symbols and Meanings

| Symbols    | Meanings                      |
|------------|-------------------------------|
| $C_1$      | Carriage purchase cost        |
| $C_2$      | Material purchase cost        |
| $C_3$      | Rental cost of shield machine |
| $C_4$      | Salary and welfare            |
| $C_5$      | Management expenses           |
| $C_6$      | Financial expenses            |
| $C_7$      | Other cost                    |
| $C_T$      | Total cost                    |
| $\alpha_1$ | Cost each one's average       |

|            |                                                                       |
|------------|-----------------------------------------------------------------------|
| $\alpha_2$ | Cost of track building per kilometer                                  |
| $\alpha_3$ | Cost of cement and other materials per kilometer                      |
| $\alpha_4$ | Rent of per shield machine                                            |
| $\alpha_5$ | Cost of salary and welfare                                            |
| $\alpha_6$ | Proportion of management expenses divided workers' wages and benefits |
| $\alpha_7$ | Loans rate for building subway                                        |
| $\alpha_8$ | Proportion of other cost comparing to direct expenses                 |
| $P_0$      | Loan remaining at the end of year after subway constructions          |
| M          | maximum cross-section traffic a day, unit: person                     |
| I          | driving time, unit: minute                                            |
| W          | the real carrying capacity of trains a day, unit: person              |
| Q          | the general train size, unit: people                                  |
| F          | fare per kilometer, unit: Yuan                                        |
| t          | tax rate, unit:%                                                      |
| P          | profit rate, unit: %                                                  |
| C          | cost per kilometer, unit: Yuan / person                               |
| m          | number of trains opened a day                                         |
| L          | the minimum design criteria for driving time, unit: minute            |

**TABLE 9**

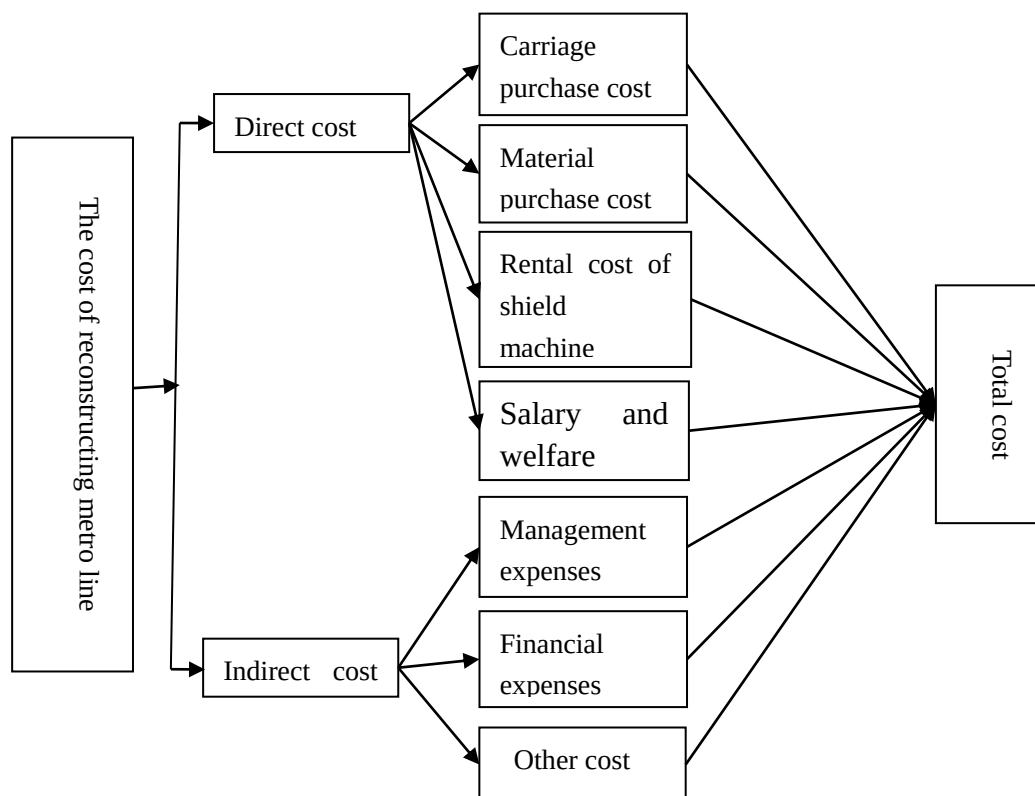
### 3.6 Assumptions

- (1) According to daily runtime, we can divide it into each hour as a period and obtain some periods while each period is without scheduling strategy change. And passengers arrive and leave evenly.
- (2) Train marshalling is 6 and the train run evenly without accidents during its running.
- (3) The patience of waiting trains is equal to the departure interval which depends on the traffic volume. And the minimum allowable starting interval is not less than 4 minutes while the maximum patience time is no more than 10 minutes
- (4) We define the peak is that the peak hourly traffic volume is more 20,000 times and the maximum cross-sectional traffic volume is more than 10,000 times and less than 20,000 times. And the maximum waiting time is no mora than 7 minutes in the peak and 10 minutes in the normal period.

### Project 1

### 3.7 Cost model of building a new subway line as a direct line

- For building a new direct line, we naturally divide it into 2 parts for direct expenses and indirect expenses. As **Figure 4**:



**Figure 4**

#### ■ Direct expenses

1. Carriage purchase cost

We should purchase some new carriages after building direct lines. The cost formula is as following:

$$C_1 = \alpha_1 N_1 \quad (3.8)$$

## 2. Material purchase cost

We need to consider the material purchase cost including track costs, cement and other materials costs if we try to build special direct lines. This kind of cost formula is as following:

$$C_2 = \alpha_2 + \alpha_3 \quad (3.9)$$

## 3. Rental cost of shield machine

Rental cost of shield machine includes the shield machines and the salary of drivers. This kind of cost formula is as following:

$$C_3 = \alpha_4 N_2 \quad (3.10)$$

## 4. Salary and welfare

It includes salary and welfare. This kind of cost formula is as following:

$$C_4 = \alpha_5 N_3 \quad (3.11)$$

## ■ Indirect expenses

### 5. Management expenses

Management expenses includes the expenses of the management of company personnel's daily life and work. This kind of cost formula is as following:

$$C_5 = \alpha_6 C_0 \quad (3.12)$$

### 6. Financial expenses

Financial expenses is that subway company spend money on interest expense, exchange net loss, auxiliary expenses to financial institution and other expenses on raising money. This kind of cost formula is as following:

$$C_6 = \alpha_7 P_0 \quad (3.13)$$

### 7. Other cost

It means that other expenses expect what we mentioned above in direct expenses. They may include the fee of the temporary facility, material consumption, low value and easy damaged objects, transportation, drawing designing, monitoring, water and electricity, heating, labor protection and office business and so forth. Other cost should be determined by the present of direct expenses. The formula is as following:

$$C_7 = \alpha_8 (C_1 + C_2 + C_3 + C_4) \quad (3.14)$$

**The final cost is shown as following:**

$$C_T = a*(C_0 + C_2 + C_3 + C_4 + C_5 + C_7 + C_8) + C_6 \quad (3.15)$$

## ■ Calculation of cost

From the data on building fees which we worked out before, we can obtain an

approximate fee of every carriage purchasing, the cost of material purchasing per kilometer, the rent  $\alpha_4$  per shield machine, the salary and welfare for one worker, management fees per kilometer, financial cost per kilometer and other expenses. We refer to some information about the cost of building one subway line locally. We can obtain 10.186 kilometer is the perfect distance for building a special direct line considering the discussion about question 1.

The final cost is shown as following:

$$C_T = 10.186 * (C_0 + C_2 + C_3 + C_4 + C_5 + C_7 + C_8) + C_6 = 4.3642 \text{ billion.}$$

This program show that we need spend 0.4285 billion on per kilometer building. We also can find that it should spend about 0.4 million on per kilometer building in this city previously by referring the data from the internet. **This shows that the model is reasonable.**

Comparing to normal subway lines which have many stations, a special direct line only has 2 stations. As we all know, the most expensive expenses on building new subway lines is the cost of building subway stations nowadays. We decide to build a special direct line which can save the money on building stations. A new station cost about 0.2 billion. It is shown that it is logical in a way if we plan to build a special direct line.

## Project 2 (As Figure 5)

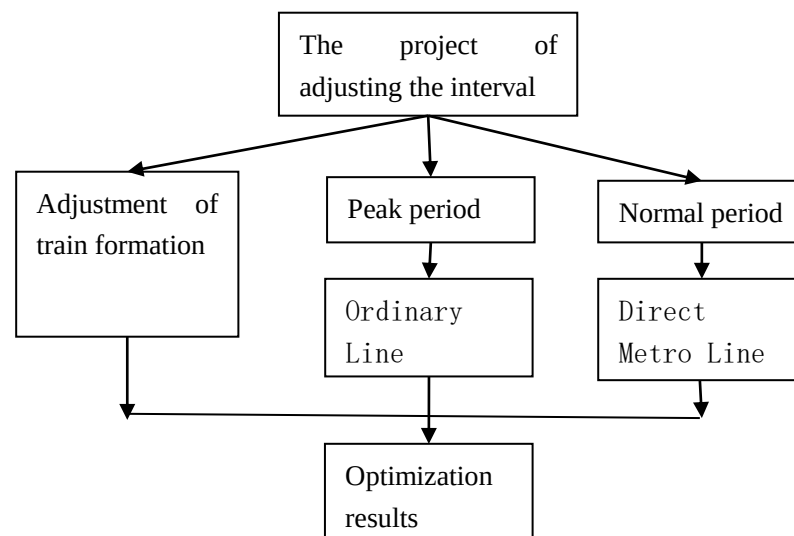


Figure 5

## 3.8 Model of Linear programming and Calculation of time interval

### 3.8.1 Model of Linear programming

By referring to relevant information, we can know that inbound and the outbound traffic will not affect the departure intervals but the cross-section traffic which reflects the subway in a section of the actual carrying capacity.

Then we can make a function:

$$\min(Z_1) = (W - Q)^2 \quad (3.16)$$

From the point of view of economic and the social benefits, the revenue may be good or bad according to the average profit margins and cost.

Then we can make another function:

$$\max(Z_2) = \frac{WF}{CQ} \quad (3.17)$$

According to the standard of fares constitute, per kilometer fare is equal to kilometers operating and costs plus profits and taxes, whence

$$F = C \times (t + p + L) \quad (3.18)$$

The constraints of the model are:

1. The subway are allowed to exceed the staff in the peak to improve vehicle utilization, but the real passenger capacity cannot be exceed the train capacity of 1.1 times in order to relief the congestion, namely  $W \leq 1.1Q$ .
2. The driving time should not be allowed less than the design standard of the system, nor more than the waiting time, Hence,  $L \leq I \leq 8$ .

Now, we obtain:

$$\min Z_1 = \left( \frac{MI}{60} - Q \right)^2 \quad (3.19)$$

$$\max Z_2 = \frac{MI(t+P+L)}{60Q} \quad (3.20)$$

$$\text{s. t. } \begin{cases} MI \leq 65Q \\ I_{\min} \leq I \leq I_{\max} \end{cases} \quad (3.21)$$

Firstly, we should figure out the best  $\min Z_1$  and  $\max Z_2$  under the constraints.

We can obtain the best  $z_1^*$  of objective function  $\min Z_1$  and the best  $z_2$  of objective function  $\max Z_2$ , and the best  $z_2^*$  of objective function  $\min Z_2$  and the best  $z_1$  of objective function  $\max Z_1$ . We also can obtain  $d_i$  which means the upper and lower limits of each objective function and the expansive index.

$$d_i = |z_i^* - z_i|, \quad i = 1, 2 \quad (3.22)$$

Secondly, the parameter  $\lambda$  is introduced to minimize the objective function which is equivalent to find the maximum of the parameter  $\lambda$ .

$$f_i + \lambda d_i \leq z_i \quad (3.23)$$

Finally, introduce the constraint C of multi-objective function which is equivalent to solving the single objective programming problem as following:

$$\max Z_3 = \lambda \quad (3.24)$$

$$\begin{cases} f_1 + \lambda d_1 \leq z_1 \\ f_2 + \lambda d_2 \leq z_2 \\ C \\ \lambda \geq 0 \end{cases} \quad (3.25)$$

The  $f_1$  and the  $f_2$  are the original objective functions. We can reduce  $d_1$  along if we think  $Z_1$  is more important while we can reduce  $d_2$  along if we think  $Z_2$  is more important.

### 3.8.2 Calculation of time interval

Case analysis:

It is known that the Tianjin Metro Line 2 adopts 6 trains as the grouping scheme for Class B. The maximum cross-section traffic volume is 17,000 times and the train capacity is 1470 with a minimum travel interval of 6 minutes. We assume that the firm's profit rate is 20 percent while tax rate is 5 percent. We take the passengers which in Tianjin Station from 6:30 to 7:30 as an example and substitute the data into (3.19),(3.20),(3.21). Through the Lingo: we can obtain the objective function of the optimal solution under the constraints is

$$I=5.188235, z_1^* = 0, z_2 = 1.25 \quad (3.26)$$

By the same way, we can obtain the optimal solution of the objective function is:

$$I=5.620588, z_2^* = 1.354, z_1 = 14965.4 \quad (3.27)$$

According to the multi- objective coordination optimization algorithm, we should calculate the telescopic index if we want to make the two objective function get the

best compromise value, and substitute  $d_1 = 14965.4$ ,  $d_2 = 1.25$  into (3.25)

$$\begin{cases} \left( \frac{17000I}{60} - 1470 \right)^2 + \lambda d_1 \leq 14965.4 \\ \frac{1.25 \times 17000I}{60 \times 1470} - \lambda d_2 \leq 1.25 \\ 17000I \leq 65 \times 1470 \\ 4 \leq I \leq 10 \\ \lambda \geq 0 \end{cases} \quad (3.28)$$

Through the Lingo, we can obtain the optimal solution is  $I=5.188236$ .

Therefore, starting interval should be 5 minutes 11 seconds in Tianjin subway station during 6:30-7:30.

By the same way, we can easily work out the intervals of the peak, normal time and other periods (as **TABEL 10**).



Taking into account the need to spend more money on the peak period of the subway traffic, we decided to open the traffic interval of five minutes or more time to open direct line.(Select subway lines: Binhai airport→Tianjin Railway Station)

| Binhai Airport→Tianjin Station |                                                 |                                         |
|--------------------------------|-------------------------------------------------|-----------------------------------------|
| Time                           | Trains interval not open the Direct Metro Lines | Trains operation mode should be adopted |
| 6:00-6:30                      | 7min8s                                          | Ordinary operation                      |
| 6:30-7:30                      | 5min11s                                         |                                         |
| 7:30-8:30                      | 5min3s                                          |                                         |
| 8:30-9:30                      | 5min23s                                         |                                         |
| 9:30-10:30                     | 6min35s                                         | Direct Metro Line                       |
| 10:30-11:30                    | 5min53s                                         | Ordinary operation                      |
| 11:30-12:30                    | 5min40s                                         |                                         |
| 12:30-14:30                    | 5min36s                                         |                                         |
| 14:30-15:30                    | 6min42s                                         | Direct Metro Line                       |
| 15:30-16:30                    | 6min37s                                         | Ordinary operation                      |
| 16:30-17:30                    | 4min55s                                         |                                         |
| 17:30-18:30                    | 5min6s                                          |                                         |
| 18:30-19:30                    | 5min43s                                         |                                         |
| 19:30-20:30                    | 5min54s                                         | Direct Metro Line                       |
| 20:30-21:30                    | 6min21s                                         |                                         |
| 21:30-22:00                    | 6min15s                                         |                                         |

**TABEL 10**

### 3.9 Benefits to society and economy

#### 1. Saving time

Not only it saves much time on subway stopping at stations because building a special direct line just has end stations with reducing the number of stations, but also increase the efficiency of working and traffic and bring a lot benefits to society and economy.

#### 2. Relieving the pressure

It can link residential areas with center areas and other places which have plenty people, increase the coverage of subway, saving the money about the management of traffic jams, and reapportion passengers each line with increasing traffic capacity.

#### 3. Creating jobs and developing commerce nearby

It can create more jobs and promote social development, strengthen the communication between 2 areas and boost the economic development.

## QUESTION 3

### ● The letter

Dear the Mayor:

Nowadays, the transportation between two distant subway stations in many cities' metro transport networks usually takes up a long time for passengers. Our group has studied this issue and has obtained the following results:

Firstly, our idea is to develop an evaluation system with setting some indexes, then we can find some suitable stations as a direct station and this process is our initial screening. Then we figure out the maximum distance that people are willing to walk away. We obtain the radiation range of a station. According to this number the average distance between these two stations, we determine how long is suitable to be used as the direct station. Eventually we obtain some final stations which are suitable.

It seems to be more suitable for the establishment of direct lines in those places which have large population, commercial developed areas, a lot of schools, railway stations, the airport and so forth. If these conditions are met, zones can be divided into some parts. And further screening of suitable stations for building direct stations is possible. Moreover, we calculate the distance between the two direct sites at least 10 kilometers away. After that we can determine whether the two points can be built as direct stations or not. After these steps, we can know which stations should be built.

Then, we have two measures to set up direct lines, one is to rebuild a direct line, and the other is on the basis of previous subway lines to adjust train travel intervals for direct lines. For the first project, we can obtain that the cost of construction is about 420 million per kilometer, as well as the human input. But, for the development between the two points will play a significant effect after construction. For the second method, not only because maintenance costs, and wages will increase the expenses, but also this project need more trains and these led to cost increasing when compared to the first method is small. The implementation of this method will increase the difficulty in scheduling the vehicle and the security of subway will reduce.

For adjusting schedules to establish a direct line, taking the operation of direct lines into account will make the previous operation greatly affected, we believe that only a few specific periods need direct line operation. Of course, these periods to stagger the peak of the previous lines. First, we need to make the all first subways run normally on previous lines. And then after morning peak, the last bus before this time point run directly with non- stop at the end station of this line while there are only the normal lines are still running at the same time. And then for today's first direct train operation, the normal operation of the line will return to normal after 15 minutes. Afternoon's is similar to the morning's but before the peak and it is about eight or nine o'clock to run the third direct line operation in the evening. Now we can design a schedule as following:

**Metro operation mode and time table**

| Metro operation                   | Station | First vehicle hour |       | Final vehicle hour |       |
|-----------------------------------|---------|--------------------|-------|--------------------|-------|
|                                   |         | 1→10               | 10→1  | 1→10               | 10→1  |
| 6:00–9:30<br>Ordinary operation   | 1       | 6:00               | 6:20  | 9:00               | 9:24  |
|                                   | 2       | 6:01               | 6:18  | 9:02               | 9:22  |
|                                   | 3       | 6:03               | 6:16  | 9:05               | 9:19  |
|                                   | 4       | 6:05               | 6:14  | 9:07               | 9:16  |
|                                   | 5       | 6:07               | 6:12  | 9:10               | 9:13  |
|                                   | 6       | 6:10               | 6:09  | 9:13               | 9:10  |
|                                   | 7       | 6:13               | 6:07  | 9:16               | 9:07  |
|                                   | 8       | 6:15               | 6:05  | 9:19               | 9:05  |
|                                   | 9       | 6:17               | 6:02  | 9:22               | 9:02  |
|                                   | 10      | 6:19               | 6:00  | 9:24               | 9:00  |
| 9:30–10:30<br>Direct Metro Line   | 1       | 9:30               | 9:40  | 10:15              | 10:26 |
|                                   | 10      | 9:41               | 9:30  | 10:25              | 10:15 |
| 10:30–14:30<br>Ordinary operation | 1       | 10:30              | 10:51 | 14:00              | 14:22 |
|                                   | 2       | 10:31              | 10:49 | 14:02              | 14:19 |
|                                   | 3       | 10:33              | 10:47 | 14:05              | 14:16 |
|                                   | 4       | 10:35              | 10:44 | 14:07              | 14:14 |
|                                   | 5       | 10:37              | 10:42 | 14:10              | 14:11 |
|                                   | 6       | 10:40              | 10:39 | 14:14              | 14:09 |
|                                   | 7       | 10:43              | 10:37 | 14:16              | 14:06 |
|                                   | 8       | 10:45              | 10:35 | 14:19              | 14:04 |
|                                   | 9       | 10:47              | 10:32 | 14:22              | 14:02 |
|                                   | 10      | 10:49              | 10:30 | 14:24              | 14:00 |
| 14:30–16:30<br>Direct Metro Line  | 1       | 14:30              | 14:40 | 16:15              | 16:26 |
|                                   | 10      | 14:41              | 14:30 | 16:25              | 16:15 |
| 16:30–20:30<br>Ordinary operation | 1       | 16:30              | 16:51 | 20:00              | 20:22 |
|                                   | 2       | 16:31              | 16:49 | 20:02              | 20:19 |
|                                   | 3       | 16:33              | 16:47 | 20:05              | 20:16 |
|                                   | 4       | 16:35              | 16:44 | 20:07              | 20:20 |
|                                   | 5       | 16:37              | 16:42 | 20:10              | 20:11 |
|                                   | 6       | 16:40              | 16:39 | 20:20              | 20:09 |
|                                   | 7       | 16:43              | 16:37 | 20:16              | 20:06 |
|                                   | 8       | 16:45              | 16:35 | 20:19              | 20:04 |
|                                   | 9       | 16:47              | 16:32 | 20:22              | 20:02 |
|                                   | 10      | 16:49              | 16:30 | 20:24              | 20:00 |
| 20:30–22:00<br>Direct Metro Line  | 1       | 20:30              | 20:40 | 22:00              | 22:11 |
|                                   | 10      | 20:41              | 20:30 | 22:10              | 22:00 |

## IV. Conclusions

In our paper, we propose a suitable model and develop a complete evaluation system with some indexes for the direct metro line by the AHP and linear programming and so forth. And we may decide to build a direct line nearby bustling places and run the direct lines three times a day during the peak.

Now we are going to analyze the strengths and weaknesses of our models.

### **Strengths:**

- Reliability Rationality: When we articulate our own metrics for assessment, we try our best to include all the important elements of setting direct shuttles to make the ranking more accurate.
- Reliability: The mathematical model we built is based on large amounts of data from many official websites and referring to many references.
- Operability: Plans of different shapes - rectangular to circular and other shapes in between have been taken into account, hence, the scope of application is extensive. Weights are everywhere in the model, but some weight assignments might not be the best scheme.

### **Weaknesses:**

- Weights are everywhere in the model, but some weight assignments might not be the best scheme.

## V. Future Work

In our evaluation model, the metrics for assessment are relatively objective. In fact, some weight assignments might not be the best scheme, we can do some experiments to improve our model and get better method. If it is possible, we could also expand evaluation index system and introduce other assessment technique. And there may be other wonderful models for us to study in order to best assessing the direct metro lines.

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## VII Appendix

### 1. Codes of MATLAB (Calculation of Weight Matrix and Check of Consistency of AHP)

```
A=[1 3 3 2 2 2
    1/3 1 1 1/2 1/2 1/2
    1/3 1 1 1/2 1/2 1/2
    1/2 2 2 1 1 1
    1/2 2 2 1 1 1
    1/2 2 2 1 1 1];
[V,D]=eig(A);a=diag(D);b=max(a);b
w1=V(:,1)/sum(V(:,1))
CI1=(b-6)/5;CI1
RI1=CI1/1.24
B1=[1 1/2 1/4;2 1 1/2;4 2 1];
[V,D]=eig(B1);a=diag(D);b=max(a);b
w21=V(:,1)/sum(V(:,1))
CI21=(b-3)/2;CI21
RT21=CI21/0.58
B2=[1 1/2 1/3;2 1 1/2;3 2 1];
[V,D]=eig(B2);a=diag(D);b=max(a);b
w22=V(:,1)/sum(V(:,1))
CI22=(b-3)/2;CI22
RT22=CI22/0.58
B3=[1 1/2 1/3;2 1 1/2;3 2 1];
[V,D]=eig(B3);a=diag(D);b=max(a);b
w23=V(:,1)/sum(V(:,1))
CI23=(b-3)/2;CI23
RT23=CI23/0.58
B4=[1 1/2 1/2;2 1 1;2 1 1];
[V,D]=eig(B4);a=diag(D);b=max(a);b
w24=V(:,1)/sum(V(:,1))
CI24=(b-3)/2;CI24
RT24=CI24/0.58
B5=[1 1/2 1/4;2 1 1/2;4 2 1];
[V,D]=eig(B5);a=diag(D);b=max(a);b
w25=V(:,1)/sum(V(:,1))
CI25=(b-3)/2;CI25
RT25=CI25/0.58
B6=[1 1/2 1/2;2 1 1;2 1 1];
[V,D]=eig(B6);a=diag(D);b=max(a);b
```

```

w26=V(:,1)/sum(V(:,1))
CI26=(b-3)/2;CI26
RT26=CI26/0.58
W=[w21,w22,w23,w24,w25,w26]*w1
CI=[CI21,CI22,CI23,CI24,CI25,CI26];
CR=CI*w1/sum(0.58*w1)

```

## 2.Codes of lingo (Calculation of Optimum Traveling Distance)

```

model:
min=(17000*I/60-1470)^2;
17000*I<65*1470;
I>4;
I<10;
end
model:
max=1.25*17000*I/(60*1470);
17000*I<65*1470;
I>4;
I<10;
end
model:
max=a;
(17000*I/60-1470)^2+a*14965.4<14965.4;
1.25*17000*I/(60*1470)-a*1.25<1.25;
17000*I<65*1470;
I>4;
I<10;
a>0;
end

```

### 3.Data of Lines by excel

**Line A1**

| Station | Number of primary schools | Number of high schools | Number of universities | Number of Colleges |
|---------|---------------------------|------------------------|------------------------|--------------------|
| A1-1    | 0                         | 0                      | 0                      | 1                  |
| A1-2    | 0                         | 0                      | 2                      | 1                  |
| A1-3    | 0                         | 0                      | 0                      | 1                  |
| A1-4    | 1                         | 0                      | 0                      | 0                  |
| A1-5    | 1                         | 0                      | 0                      | 0                  |
| A1-6    | 2                         | 0                      | 0                      | 0                  |
| A1-7    | 1                         | 2                      | 0                      | 0                  |
| A1-8    | 1                         | 1                      | 0                      | 0                  |
| A1-9    | 1                         | 2                      | 1                      | 0                  |
| A1-10   | 3                         | 3                      | 0                      | 0                  |
| A1-11   | 5                         | 1                      | 0                      | 0                  |
| A1-12   | 1                         | 0                      | 0                      | 0                  |
| A1-13   | 0                         | 4                      | 0                      | 0                  |
| A1-14   | 0                         | 0                      | 0                      | 0                  |
| A1-15   | 1                         | 5                      | 0                      | 0                  |
| A1-16   | 1                         | 0                      | 0                      | 0                  |
| A1-17   | 1                         | 2                      | 1                      | 0                  |
| A1-18   | 2                         | 0                      | 0                      | 4                  |
| A1-19   | 0                         | 0                      | 1                      | 2                  |
| A1-20   | 1                         | 0                      | 0                      | 0                  |
| A1-21   | 1                         | 0                      | 0                      | 1                  |
| A1-22   | 1                         | 1                      | 0                      | 0                  |

**Surrounding schools of each station of line A1**

**TABLE 1**

| Station | Surrounding communities | Workplace | Mall | Tourist spots | Whether traffic interchange of line |
|---------|-------------------------|-----------|------|---------------|-------------------------------------|
| A1-1    | 1                       | 1         | 0    | 0             | NO                                  |
| A1-2    | 4                       | 2         | 0    | 0             | NO                                  |
| A1-3    | 0                       | 0         | 0    | 0             | NO                                  |
| A1-4    | 7                       | 2         | 2    | 1             | NO                                  |
| A1-5    | 2                       | 0         | 1    | 0             | NO                                  |
| A1-6    | 3                       | 5         | 4    | 0             | NO                                  |
| A1-7    | 7                       | 4         | 5    | 0             | NO                                  |
| A1-8    | 8                       | 2         | 2    | 3             | NO                                  |
| A1-9    | 0                       | 2         | 7    | 3             | NO                                  |
| A1-10   | 1                       | 1         | 9    | 1             | YES                                 |
| A1-11   | 7                       | 3         | 4    | 1             | NO                                  |
| A1-12   | 2                       | 3         | 4    | 2             | NO                                  |
| A1-13   | 3                       | 2         | 3    | 0             | YES                                 |
| A1-14   | 0                       | 0         | 1    | 0             | NO                                  |
| A1-15   | 4                       | 1         | 3    | 2             | NO                                  |
| A1-16   | 0                       | 0         | 0    | 2             | NO                                  |
| A1-17   | 5                       | 0         | 3    | 0             | NO                                  |
| A1-18   | 4                       | 6         | 0    | 0             | NO                                  |
| A1-19   | 6                       | 1         | 0    | 0             | NO                                  |
| A1-20   | 10                      | 0         | 1    | 0             | NO                                  |
| A1-21   | 4                       | 0         | 5    | 1             | NO                                  |
| A1-22   | 6                       | 0         | 0    | 0             | NO                                  |

**Other indexes of each station of line A1**

**TABLE 2**



### Line B1

| station | Number of primary schools | Number of high schools | Number of universities | Number of Colleges |
|---------|---------------------------|------------------------|------------------------|--------------------|
| B1-1    | 0                         | 0                      | 0                      | 0                  |
| B1-2    | 0                         | 0                      | 0                      | 0                  |
| B1-3    | 0                         | 0                      | 0                      | 0                  |
| B1-4    | 0                         | 0                      | 0                      | 0                  |
| B1-5    | 1                         | 1                      | 0                      | 0                  |
| B1-6    | 0                         | 0                      | 0                      | 1                  |
| B1-7    | 0                         | 1                      | 0                      | 0                  |
| B1-8    | 0                         | 0                      | 0                      | 0                  |
| B1-9    | 0                         | 1                      | 0                      | 0                  |
| B1-10   | 1                         | 1                      | 0                      | 0                  |
| B1-11   | 0                         | 0                      | 0                      | 0                  |
| B1-12   | 0                         | 0                      | 0                      | 0                  |
| B1-13   | 0                         | 0                      | 0                      | 0                  |
| B1-14   | 0                         | 0                      | 0                      | 0                  |
| B1-15   | 0                         | 1                      | 0                      | 0                  |
| B1-16   | 0                         | 0                      | 0                      | 0                  |
| B1-17   | 0                         | 2                      | 0                      | 2                  |
| B1-18   | 0                         | 1                      | 0                      | 0                  |
| B1-19   | 0                         | 0                      | 0                      | 0                  |
| B1-20   | 0                         | 0                      | 0                      | 0                  |

### Surrounding schools of each station of line B1

TABLE 3

| station | Surrounding communities | Mall | Tourist spots | Whether traffic interchange of line |
|---------|-------------------------|------|---------------|-------------------------------------|
| B1-1    | 0                       | 0    | 0             | NO                                  |
| B1-2    | 1                       | 0    | 0             | NO                                  |
| B1-3    | 1                       | 0    | 1             | NO                                  |
| B1-4    | 5                       | 0    | 0             | NO                                  |
| B1-5    | 5                       | 1    | 0             | NO                                  |
| B1-6    | 2                       | 0    | 0             | NO                                  |
| B1-7    | 6                       | 0    | 0             | NO                                  |
| B1-8    | 5                       | 1    | 0             | NO                                  |
| B1-9    | 3                       | 1    | 0             | NO                                  |
| B1-10   | 5                       | 2    | 1             | YES                                 |
| B1-11   | 3                       | 0    | 2             | NO                                  |
| B1-12   | 0                       | 0    | 1             | NO                                  |
| B1-13   | 2                       | 3    | 2             | YES                                 |
| B1-14   | 2                       | 3    | 0             | NO                                  |
| B1-15   | 3                       | 1    | 0             | YES                                 |
| B1-16   | 2                       | 2    | 1             | NO                                  |
| B1-17   | 4                       | 0    | 0             | NO                                  |
| B1-18   | 3                       | 0    | 0             | NO                                  |
| B1-19   | 5                       | 0    | 0             | NO                                  |
| B1-20   | 0                       | 1    | 1             | NO                                  |

### Other indexes of each station of line B1

TABLE 4

### Line C1

| station | Number of primary schools | Number of high schools | Number of universities | Number of Colleges |
|---------|---------------------------|------------------------|------------------------|--------------------|
| C1-1    | 0                         | 0                      | 0                      | 0                  |
| C1-2    | 0                         | 0                      | 0                      | 0                  |
| C1-3    | 0                         | 0                      | 0                      | 0                  |
| C1-4    | 0                         | 0                      | 0                      | 0                  |
| C1-5    | 0                         | 0                      | 0                      | 0                  |
| C1-6    | 0                         | 1                      | 0                      | 0                  |
| C1-7    | 0                         | 1                      | 0                      | 1                  |
| C1-8    | 1                         | 2                      | 0                      | 0                  |
| C1-9    | 0                         | 5                      | 0                      | 0                  |
| C1-10   | 2                         | 0                      | 0                      | 1                  |
| C1-11   | 1                         | 1                      | 0                      | 0                  |
| C1-12   | 0                         | 0                      | 0                      | 0                  |
| C1-13   | 0                         | 0                      | 0                      | 0                  |
| C1-14   | 3                         | 0                      | 0                      | 0                  |
| C1-15   | 1                         | 0                      | 0                      | 0                  |
| C1-16   | 1                         | 2                      | 3                      | 0                  |
| C1-17   | 0                         | 0                      | 0                      | 0                  |
| C1-18   | 0                         | 0                      | 1                      | 0                  |
| C1-19   | 0                         | 1                      | 1                      | 0                  |
| C1-20   | 0                         | 0                      | 1                      | 0                  |
| C1-21   | 0                         | 0                      | 0                      | 0                  |
| C1-22   | 0                         | 0                      | 4                      | 0                  |
| C1-23   | 0                         | 0                      | 0                      | 0                  |
| C1-24   | 0                         | 0                      | 0                      | 0                  |
| C1-25   | 0                         | 0                      | 0                      | 0                  |
| C1-26   | 0                         | 0                      | 0                      | 0                  |

### Surrounding schools of each station of line C1

TABLE 5

| station | Surrounding communities | Workplace | Mall | Tourist spots | Whether traffic interchange of line |
|---------|-------------------------|-----------|------|---------------|-------------------------------------|
| C1-1    | 1                       | 1         | 0    | 0             | NO                                  |
| C1-2    | 0                       | 1         | 0    | 0             | NO                                  |
| C1-3    | 0                       | 3         | 0    | 0             | NO                                  |
| C1-4    | 4                       | 1         | 0    | 0             | NO                                  |
| C1-5    | 4                       | 0         | 0    | 0             | NO                                  |
| C1-6    | 2                       | 0         | 0    | 0             | NO                                  |
| C1-7    | 2                       | 0         | 0    | 1             | NO                                  |
| C1-8    | 1                       | 0         | 0    | 1             | NO                                  |
| C1-9    | 1                       | 0         | 2    | 1             | NO                                  |
| C1-10   | 2                       | 0         | 1    | 0             | NO                                  |
| C1-11   | 0                       | 5         | 2    | 0             | YES                                 |
| C1-12   | 0                       | 0         | 0    | 1             | NO                                  |
| C1-13   | 0                       | 0         | 4    | 3             | NO                                  |
| C1-14   | 1                       | 1         | 9    | 1             | YES                                 |
| C1-15   | 5                       | 0         | 0    | 1             | NO                                  |
| C1-16   | 3                       | 1         | 1    | 0             | NO                                  |
| C1-17   | 4                       | 0         | 1    | 4             | NO                                  |
| C1-18   | 1                       | 0         | 0    | 2             | NO                                  |
| C1-19   | 5                       | 0         | 1    | 0             | YES                                 |
| C1-20   | 4                       | 0         | 1    | 1             | NO                                  |
| C1-21   | 0                       | 2         | 4    | 0             | NO                                  |
| C1-22   | 0                       | 0         | 0    | 0             | NO                                  |
| C1-23   | 1                       | 1         | 0    | 0             | NO                                  |
| C1-24   | 0                       | 10        | 0    | 0             | NO                                  |
| C1-25   | 3                       | 0         | 0    | 0             | NO                                  |
| C1-26   | 0                       | 0         | 0    | 0             | NO                                  |

### Other indexes of each station of line C1

**TABLE 6**

| station | Number of primary schools | Number of high schools | Number of universities | Number of Colleges |
|---------|---------------------------|------------------------|------------------------|--------------------|
| D1-1    | 1                         | 0                      | 0                      | 0                  |
| D1-2    | 0                         | 0                      | 0                      | 0                  |
| D1-3    | 0                         | 0                      | 0                      | 0                  |
| D1-4    | 0                         | 0                      | 0                      | 0                  |
| D1-5    | 0                         | 0                      | 0                      | 0                  |
| D1-6    | 0                         | 0                      | 0                      | 0                  |
| D1-7    | 0                         | 0                      | 0                      | 0                  |
| D1-8    | 0                         | 0                      | 0                      | 0                  |

### Surrounding schools of each station of line D1

**TABLE 7**

| station | Surrounding communities | Workplace | Mall | Tourist spots | Whether traffic interchange of line |
|---------|-------------------------|-----------|------|---------------|-------------------------------------|
| D1-1    | 5                       | 4         | 2    | 0             | YES                                 |
| D1-2    | 4                       | 4         | 0    | 0             | NO                                  |
| D1-3    | 4                       | 0         | 1    | 0             | NO                                  |
| D1-4    | 2                       | 2         | 0    | 0             | NO                                  |
| D1-5    | 10                      | 0         | 0    | 0             | NO                                  |
| D1-6    | 6                       | 0         | 1    | 0             | YES                                 |
| D1-7    | 6                       | 0         | 0    | 0             | NO                                  |
| D1-8    | 4                       | 1         | 0    | 2             | NO                                  |

### Other indexes of each station of line D1

**TABLE 8**

## 4. Data of distance

According to GAODE map, we know:

Linear distance of between A1-8 and A1-10 is 2.53 km.

Linear distance of between A1-8 and B1-1 is 12.26 km.

Linear distance of between A1-8 and B1-10 is 3.69km.

Linear distance of between A1-8 and B1-15 is 6.3km.

Linear distance of between A1-8 and C1-19 is 6.6km.

Linear distance of between A1-8 and C1-26 is 14.95 km.

Linear distance of between A1-10 and B1-1 is 14.06km.

Linear distance of between A1-10 and B1-10 is 2.34km.

Linear distance of between A1-10 and B1-15 is 3.70km.

Linear distance of between A1-10 and C1-19 is 5.18km.

Linear distance of between A1-10 and C1-26 is 13.74km.

Linear distance of between B1-1 and B1-10 is 12.93km.

Linear distance of between B1-1 and B1-15 is 17.13km.

Linear distance of between B1-1 and C1-19 is 18.19km.

Linear distance of between B1-1 and C1-26 is 27.19km.

Linear distance of between B1-10 and B1-15 is 4.28km.

Linear distance of between B1-10 and C1-19 is 7.36km.

Linear distance of between B1-10 and C1-26 is 15.68km.

Linear distance of between B1-15 and C1-19 is 5.13km.

Linear distance of between B1-15 and C1-26 is 12.32km.

Linear distance of between C1-19 and C1-26 is 8.35km.

# The Optimal Selection of The Direct Metro

## ——Creating Maximum Value for The Whole Society

### Abstract

Urban metro transit network has always been the core of a city's infrastructure system. Our paper aims to find out the direct metro line that shorten the maximum traveling time for the whole society. And the general society can redistribute the saved time, bringing considerable social utility and economic utility.

We choose the Shanghai metro transit network as the object of the study. Our team addresses the problem by formulating the traveling time model, passenger flow volume model, and interval time of direct metro model. Utilizing the optimal saved time model, we manage to acquire the traveling time spent between any two stations. Based on this, we calculate the time that is saved when traveling by a direct line. We obtain the optimal route: choosing the **MinHang Developed Zone** and **North Jiangyang Rd** as the initial station and the terminal station. This route can shorten **57 minutes**. Combined with the passenger flow volume model, we make out the direct line from the Perspective of maximum social utility. This route can maximize the general saved time of the society. Under this circumstance, we choose **Lianhua Rd.** and **Pudong Avenue** as the terminal stations. We can save **88,468 minutes** at most. In the end, we use the interval time of direct metro model to figure out the train interval time and make out the train timetable. The interval time of direct metro is **12 minutes** during the fastigium, while the interval time is **30 minutes** during the flat peak period.

Aiming at question one, we divide the traveling time into two parts: the subway running time and the transfer time. According to the Shanghai Metro Official Website, we can acquire the running time (including the stopover time) and transfer time between adjoining stations. Grounded on the **Dijkstra Algorithm**, we can make out that the optimal direct metro line travels between **Minhang Develop Zone** and **North Jiangyang Rd**. The corresponding saved time is up to **57 minutes**. To evaluate the accuracy of the model, we compare the result in Gaode map and our model. According to the result of the test, the deviation is **4.4%**, which verifies our model's accuracy is high.

Aiming at question two, we analyze the passenger flow volume of every station. In this way, we calculate the passenger flow volume between any two stations. Based on the question, we multiply the passenger flow volume and the saved time to get the general saved time of the whole society. The optimal route is between **Lianhua Rd.** and **Pudong Avenue**. The corresponding general saved time of the society is **88468 minutes**.

Aiming at question three, we build the interval time of the direct metro model. Compared with the timetable of subway, we consider the shunt effect, economic cost and waiting choices as the determining factors. On the condition of taking the direct metro, we obtain the result that the interval time of direct metro is **12 minutes** during the fastigium, while the interval time is **30 minutes** during the flat peak period.

In the letter to mayor, we present the feasibility, personal benefit and social utility in our suggestion. The feasibility factor includes average cost, income and the influence on the urban transit system.

**Key word: Dijkstra Algorithm; The shortest Time Algorithm; Passenger Flow Volume Model; The Interval Time of The Direct Metro Model**

## Context

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## I. Introduction

In order to indicate the origin of the direct metro lines problems, the following background is worth mentioning.

### 1.1 Why should we set direct metro lines?

There is no denying that the current metro network in a metropolis like Shanghai is highly-developed. However, the transportation demand in the downtown area is so large that a mass of residents still find it inconvenient to travel from one place to another. Since a non-stop train doesn't exist, they have to transfer several subway lines to get to their destination. To meet the public's needs, we should set a direct metro line between two far apart stations. In this way, passengers may save more time and travel in a more convenient way.

### 1.2 The existing solutions

To solve the problem, the government has introduced a few policies. A direct metro line has been set between Jinshan and Shanghai South Railway Station. It is the first suburban train in China under the railway networks. It has helped improved the city's traffic and driven the tourism industry in the suburbs of Shanghai. Therefore, setting more direct lines may not only meet the citizens' travel demands but also do great benefits to the urban integration of the city.

### 1.3 The object of study

Shanghai, a metropolis, has a mature metro network. It has many stations and considerable passenger flow volume. Correspondingly, the problem of time wasted in the travel is acute and required immediate solutions. What's more, huge amounts of data is propitious to improve our model accuracy.

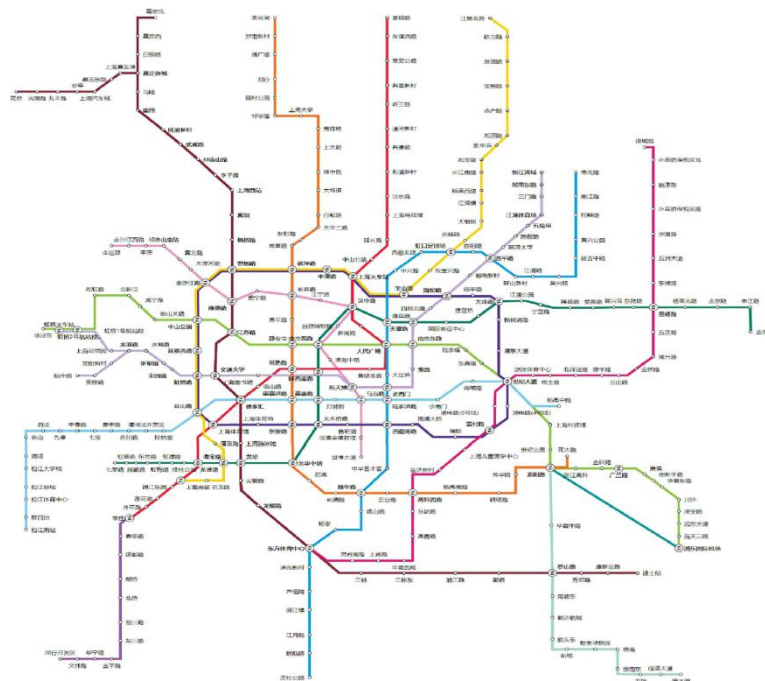


figure 1 Shanghai Metro Map

## **II. The Description of the Problem**

### **2.1 Restatement of the problem**

In order to fulfill the passengers' long-distance travel requirements, the goal of our team seems to be very clear. Transferring between metro lines makes it very inconvenient for passengers who have to travel long-distance frequently. As for office workers who live in the suburbs but work in downtown, they have to get up early to catch the morning bus. The same applies to travel back from work. As a result, we should set more direct metro lines and forge a high-end level of rail transit. The content of our paper is aiming at (developing a model to figure out the best choice of the end stations, evaluate the expected results and find out the intervals that benefits most.)

Question I. Develop a mathematical framework to choose the most proper end stations to save the longhaul passengers' time at the greatest extent. And give a evaluation of the expected result if building this route.

Question II. Talk about the strengths we may achieve in terms of economic and social benefits, when we build a brand-new direct route and remake the intervals between departure time of the trains.

Question III. Work out an ideal train schedule that may meet the passengers' needs.

Question IV. Write a two-page letter to the Mayor of our city to state our suggestion and the advantages we have, in which our train schedule is included on the second page.

## 2.2 The problem analysis



Generally speaking, it is an optimization problem on the current metro network of Shanghai. The goal of setting a direct metro line is to save the maximum time for the mass covering long distances.

Now the problem into the following specific analysis, subject to the requirements can be divided into three subsections:

1. Evaluate the possible time length to travel from one station to another.(considering various influence factors like waiting time and transfer time)



2. Estimate the distribution model of the daily passengers flow volume.
3. Set a time table for the direct metro line which maximum the economic and social benefits.

### **2.2.1 How do we approximate the optimal choice of the end stations?**

In our humble opinion, the optimal choice of the end stations depends on the total time that is saved due to the set of the direct metro lines. As for the first question, we simulate the time it needs to travel from one station to another and work out the time that is saved. We take the time interval between each two stations in one metro line into account and structure a matrix. What's more, we can estimate the passenger flow rate. By multiplying the estimated saved time and the simulated passenger flow rate, we can draw a cursory conclusion of the routine that saves the maximum time.

### **2.2.2 How to maximum both economic and social benefits?**

The goal of our model is to help the metro lines realize the shunt effect during the peak time. Meanwhile, economic benefits should also be considered. Suppose that the direct metro line only carries one passenger to his destination, it saves time but it is absolutely not economical. As a result, we should develop a metro line that balance the benefits between economic and social benefits.

### **2.2.3 How to determine the interval time of the direct metro?**

Aimed at different passenger flow volume at different time, we divide the operating time into two parts: the fastigium and the flat peak period. During the fastigium, our primary intention is to shunt the large passenger flow volume. So during the interval time, all the passengers getting into the station are supposed to take the direct metro. First, we determine the volume of the direct metro which can satisfy the number and the need of passengers. Then, we figure out the optimal interval time so that every passengers can save the time.

During the flat peak period, we take the passenger flow volume and economy into consideration. So we set a longer interval time which can carry as many passengers as possible and also reduce the cost.

### III. Models

#### 3.1 Model I——The Shortest Time Algorithm

##### 3.1.1 The analysis of Model I

The time to travel between any two station mainly consists of the passing time of the metros (including the stopping time) and the transfer time of the passengers.

Obviously, every metro will have a short stay at each platform. We should take the duration of stay into account to get a more accurate figure.

We choose the metro lines in Shanghai as the study object. Since there are

approximately 363 subway stations in Shanghai, we need nearly  $P_{363}^2$  groups of data

to simulate the time to travel between two stations. This will make it difficult for us to search for the statistic in need and do the operation. Hence we choose the Dijkstra algorithm to cut down the amount of the needed data.

tra Arithmetic to deal with the shortest path algorithm, whicIn this model , we mainly consider the traveling time between two stations and the transfer time of the passengers to simulate the time to travel from one station to another.

##### 3.1.2 Terms, Definitions and Symbols

The signs and definitions are mostly generated from *queuing theory*.

| Sign              | Definition                                                                                                                                                                       |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$          | The station serial number of passengers to get on the bus station.                                                                                                               |
| $\beta$           | The station serial number of passengers to get off the bus station.                                                                                                              |
| $i$               | The station number of passengers to get on the bus station.                                                                                                                      |
| $j$               | The station number of passengers to get off the bus station.                                                                                                                     |
| $a_{\alpha\beta}$ | The time of passing adjoining stations(with every stop time included) or the time of different lines passing the same station.It should follow the relation:<br>$\alpha < \beta$ |
| $s_{\alpha\beta}$ | The time of passengers to transfer in the same station from one line to another.                                                                                                 |
| $c_{\alpha\beta}$ | The time of passengers to travel from station $\alpha$ to station $\beta$ .                                                                                                      |

|                   |                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------|
| $t_{\alpha\beta}$ | when $\alpha$ and $\beta$ are adjoining stations.It should follow the relation:<br>$\alpha < \beta$                            |
| $d_{\alpha\beta}$ | The time of passing adjoining stations(with every stop time included) or the time of different lines passing the same station. |
| $t_{save}$        | saving time when taking the direct metro line                                                                                  |
| $t_{all}$         | Before the optimization,the shortest time used between two stations                                                            |
| $t_{real}$        | After the optimization,the shortest time used between two stations                                                             |
| $y$               | the shortest path between two stations                                                                                         |
| $length(y)$       | The amount of passing stations                                                                                                 |
| $t_g$             | The actual time measured by Gaode Map                                                                                          |
| $t_d$             | The time measured by the Dijkstra Algorithm                                                                                    |

### 3.1.3 The Model I Assumptions

- **We only take the time cost into consideration.**

Due to the question mentioned,we just need to save time of long passengers,so we choose the time span as their only determining factor. Ignore the travel cost,the crowded degree of stream of people,the real distance and so on.

- **The stop time is assumed to be the same.In practice,there is a slight difference.**

To simplify the model,we do not take the slight difference into consideration.Instead,we choose 1minute as the unified stop-time.

- **We do not take the time of train to switch between two rails into the consideration.**

Our special train is “ideal” in the existing metro system.We suppose the train can run smoothly along the metro line with no delay of the transfer nodes.

### 3.1.4 The Establishment of Model I

According to the official website of Shanghai Metro, we can have a clear idea of the interval time between every two adjoining stations in a specific metro line. ( as can been seen in the attachment )

To begin with, we define  $\alpha$  as the symbol of the station that the passenger get onto the bus.(and  $\beta$  as the symbol of the station that the passenger get off the bus)

We divide  $a_{\alpha\beta}$  into two parts: the time of passing adjoining stations (with every stop time included) and the time of different lines passing the same station.

We define the distance between two different lines passing the same station as zero. From the attachment, we can get the data of the adjoining stations in one metro line:

$$a_{\alpha\beta} = \begin{cases} t_{\alpha\beta} & \text{when } \alpha \text{ and } \beta \text{ are adjoining stations} \\ 0 & \text{when people are able to transfer at station } \alpha \text{ and } \beta \end{cases} \quad (1.1)$$

Then, we define  $s_{\alpha\beta}$  as the time of passengers to transfer in the same station from one line to another.

We have to simulate bilateral travel, including from A to B and from B to A. Therefore, we let:

$$d_{\alpha\beta} = a_{\alpha\beta} + a_{\alpha\beta}^T \quad (1.2)$$

To avoid choosing the wrong  $a_{ij}$  (only occurs during the origin station and terminal station), we select the shortest time from  $\alpha$  to  $\beta$

By the way, we save the shortest time in the “ $t_{all}$ ” array.

We define the numbers of the stations that stops as  $number_{\alpha\beta}$ .

Obviously, the number of the passed stations equals the length of the stations minus one, which means:

$$number_{\alpha\beta} = length(y) - 1 \quad (1.3)$$

( $y$  represents the amount of the stations)

Since the total saved time consists of transfer time and stopping time, we suppose that the metro stop for one minute on average. And we define the numbers of the stations that stops as  $number_{\alpha\beta}$ .

By using the Dijkstra Arithmetic, we can work out the minimize time between any two stations.

We define “ $t_{save}$ ” as the time that we may save while traveling between station  $\alpha$  and  $\beta$ .

Then we compare the elements in  $t_{save}$  and work out the shortest time algorithm.

### 3.1.5 The Model I optimization

In the former assumption, we have put forward that we treat the same station of different lines as two separate stations. Therefore, it may bring some problems when calculating the shortest travel time between two stations. Since it is quite common for two metro lines to cross, then we may have symbol  $\alpha$  and  $\beta$  for the same station A. When we mean to travel from station A to B, the computer may doubt whether we refers to  $\alpha$  or  $\beta$ . As a result, we should work out a plan to avoid adding the undesired transfer time.

Following discussion only exists in this condition that station A is chosen as the initial station or the terminal station.

For example, suppose that we are going to travel from Shanghai South Railway Station to Nanjing East Road. Having a look at the metro map, we may find that there are both metro line 1 and line 3 in Shanghai South Railway Station, and both line 2 and line 10 in Nanjing East Road. As what has been mentioned above, since they are considered to be four separate stations, we may find four routes between them. Either route indicates a different traveling time, which may confuse the passengers. As a result, we should optimize the shortest time algorithm as following:

We should make special discussions in terms of the starting station and the terminal station. If such problem occurs (passenger find out four different answer to the time to travel between two selected places), we ought to remove the transfer time from the total result. We know  $i, j$  represents the serial number of the same station in different lines.

$$t_{real} = t_{all} - S_{ij} \quad (1.4)$$

Finally, we compare the value among all the element in  $t_{save}$  and find out the optimal choice of the end station.

### 3.1.6 The Result of Model I

We put our sorted data into the MATLAB and obtain the result by running our model.

Here is the top 10 of the time saved most. In our model, we can save 57 minutes at most.

|                        |    |    |    |    |    |
|------------------------|----|----|----|----|----|
| Order number           | 1  | 2  | 3  | 4  | 5  |
| The time saved(minute) | 57 | 56 | 56 | 56 | 55 |
| Order number           | 6  | 7  | 8  | 9  | 10 |
| The time saved(minute) | 55 | 55 | 55 | 55 | 54 |

This table is the top 10 of the time saved and their route.

| The Sequence of the Time Saved | The Time Saved | The Route                                       | The transfer time |
|--------------------------------|----------------|-------------------------------------------------|-------------------|
| 1                              | 57             | Minhang Development Zone To North Jiangyang Rd. | 4                 |
| 2                              | 56             | Gangcheng Rd. To Aerospace Museum               | 3                 |
| 3                              | 56             | Wenjing Rd. To North Jiangyang Rd.              | 4                 |
| 4                              | 56             | Minhang Development Zone To Tieli Rd.           | 4                 |
| 5                              | 55             | North Waigaoqiao Free Trade Zone To Meilanhua   | 3                 |
| 6                              | 55             | Gangcheng Rd. To Luonan Rd.                     | 3                 |
| 7                              | 55             | Huaning Rd. To North Jiangyang Rd.              | 4                 |
| 8                              | 55             | Wenjing Rd. To Tieli Rd.                        | 4                 |
| 9                              | 55             | Minhang Development Zone To Youyi Rd.           | 4                 |

|    |    |                        |   |
|----|----|------------------------|---|
| 10 | 54 | Hangjin Rd. To Meilanh | 3 |
|----|----|------------------------|---|

By analyzing the result, we can find the longer the distance are, the more time can we save by our model. What's more, in the top ten routes of the time saved most, six of them need 4 times transfer. The other four routes also need 3 times transfer. So our model can effectively resolve the multiple transferring. This result accord with our expectation.

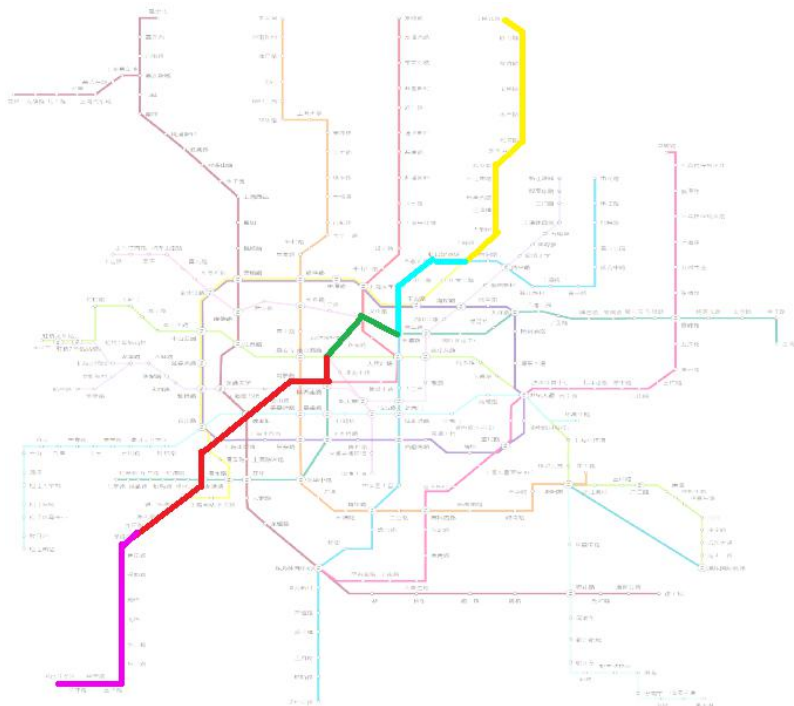


figure 2 The metro line between Minhang Develop Zone and North Jiangyang Rd.

### 3.1.7 The advantages and disadvantages of Model I

#### Advantages:

- (1) The shortest time algorithm can achieve the goal of working out the time that can be saved between any two stations in an effective, fast and automatic manner.
- (2) What's more, there is no denying that the station is a huge amount of information. Therefore, before we process the data, we should work out a way to make data amount smaller and unify the format of the data. This model uses the Dijkstra Arithmetic to decrease the amount of the needed data, and does a perfect job.
- (3) Although many influence factors exists, the result of the model is quite accurate within the limit of error.
- (4) The Improved Model aims to make up for the neglect of adding up the time to travel between the same station in different metro lines. That may increase the result

of the traveling time . The result seems to declare that this model is more reasonable than the Basic Model.

#### Disadvantages:

(1) Since the transfer time plays an important part in the result, the accuracy of the transferring between two metro line remains to be improved.

(2) The result of model one only reflects the maximum time that can be saved personally. Instead, we should take the whole society into consideration. The need of the public should also be considered.

### 3.1.8 The Test of Model I

To test our model accuracy, we choose following stations as the object of study.

| Initial Station  | Shanghai Railway Station | Changshu Rd.     | North Jiangyang Rd.           | Yuntai Rd.             | Huaning Rd.     | Zhenping Rd.      |
|------------------|--------------------------|------------------|-------------------------------|------------------------|-----------------|-------------------|
| Terminal Station | Huangxing Park           | West Nanjing Rd. | Wujiaochang                   | Oriental Sports Center | Beiyangjing Rd. | South Huangpi Rd. |
| $t_g$            | 27                       | 10               | 47                            | 11                     | 74              | 18                |
| $t_d$            | 29                       | 11               | 49                            | 12                     | 77              | 17                |
| Initial Station  | Songyuan Rd.             | Dishui Lake      | Shanghai West Railway Station | Wulian Rd              | Yangshupu Rd.   | Chifeng Rd.       |
| Terminal Station | South Changjiang Rd.     | Baoshan Rd.      | Donglu Rd.                    | Laoximen               | Dapuqiao        | Guilin Rd         |
| $t_g$            | 40                       | 101              | 45                            | 32                     | 20              | 37                |
| $t_d$            | 42                       | 106              | 47                            | 36                     | 21              | 37                |



By using the fitting model, analyze the interaction between  $t_2$  and  $t_1$ . We portray the equation and find it can be approximately seen as a straight line. This result means we have a pretty accuracy. Here is the graph and the equation

$$t_2 = 1.0448t_1 + 0.1087 \quad (1.5)$$

This result means the deviation is 4.4%, which verifies our model accuracy is high.

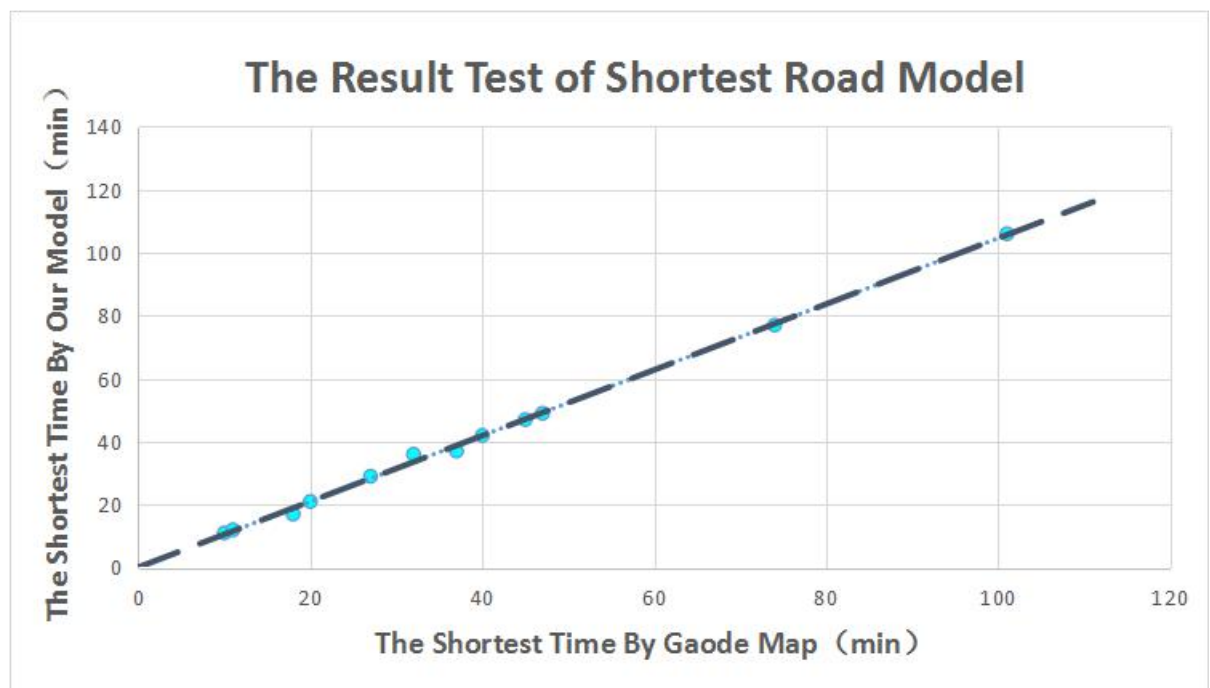


figure 3 The Result Test of Shortest Road Model

## 3.2 Model II——Passenger Flow Volume Model

### 3.2.1 The analysis of Model II

From the perspective of social maximum utility, we choose the product of passenger flow and the time saved in each route as the evaluating indicator. This can perfectly represent the social total time economization.

The time saved index can be obtained directly by the model I. So we establish a new model to get the passenger flow between any two stations. We use two index: the number of passenger into the station and the possibility to get off at partial stations. The two index help us simulate the passenger flow.

While simulating the passenger flow, we made some simplifications. Stations that locate in remote areas without huge demands to go out is sure to be neglected. Therefore, we analyzed the chart of the passenger flow rate rankings and drew a conclusion that a tremendous number of people flow into the downtown in the morning peak and head back in the evening. Under that circumstance, we may choose one end station inside the circle of metro line 4 (the inner ring of Shanghai) and the other outside the circle (where people living far away have great demands to get out). Therefore, we can simplify the procedure of inputting the data.

Combined with the data via model I, we establish the social total time economization model. Using this model, we choose the maximum result as the route. On the basis of this, we acquire the timetable of the special train.

### 3.2.2 Extra Symbols

| Symbol            | Definition                                               |
|-------------------|----------------------------------------------------------|
| $P_{\beta}$       | the possibility of choosing Station B as the end station |
| $Q_{A_{\alpha}}$  | Passenger flow into station A                            |
| $Q_{B_{\beta}}$   | outbound passenger flow of station B                     |
| $Q_{\alpha\beta}$ | passenger flow between station A and B                   |
| $T_{save}$        | social all time saved                                    |

### 3.2.3 The assumption of model II

- **We only talk about the morning peak and evening peak when choosing the maximum social total time saved.**

The problem is to remit phenomenon: people hurry to their destinations. This problem is acute when it comes to the morning peak and the evening peak. Also, our model can help the subway realize the shunt effect during the peak time.

- **To simplify the model, we suppose all the people choose stations outside the inner ring as the initial station and the inside ones as the terminal station in the morning peak.**

Shanghai is a fast-growing city. Considering the price of residential area and combining the collecting the data of every station passenger flow (see at appendix), we find most people live outside the inner ring. What's more, due to the office building and business circle gathers around the downtown, people choose the stations inside the inner ring as the destination. So, we suppose they don't go off the metro until they reach the inner ring borderline-line 4. In reality, there are still a flow of people who has other purposes, however, compared to the morning peak, this flow is negligible.

- **The possibility of the choice of destination has a positive direct ratio to the number of outbound passengers in every main station.**

Since the statistic data is not available, we choose the number of outbound passengers in every station as the measurable standard.

- **Partial stations are not taken into the whole model.**

we cancel out the number of passenger flow which is negligible and it must not be the overall optimal solution.

### 3.2.4 The establishment of Model II

Based on the establishment of model I, we supplement the model by adding matrix A and B.

We analyze the chart of the passenger flow rate rankings and drew a conclusion that a tremendous number of people flow into the downtown in the morning peak and head back in the evening. Under that circumstance, we may choose one terminal station inside the circle of metro line 4 (the inner ring of Shanghai) and the other outside the circle (where people living far away have great demands to get out to workplace).

Collect the average data in a month of every station. It includes people flowing into this station in the morning peak and evening peak and the outbound passenger number of the station (see at appendix). Data of partial station are negligible. They must not be the optimal solution, with the restrict of passenger flow. After preprocess data, we get a matrix A. It shows the passenger flow into stations, which are outside the inner ring. In the meantime, by adding up the outbound passenger number, we get the total number of people whose destinations are inside the inner ring. Since we assume that the possibility of the choice of destination has a positive direct ratio to the number of outbound passengers in every main station, we make a division operation and get the possibility to be chosen as the terminal of every main station. Make sure that every transfer station has the same passenger flow without the influence of different station numbers. In our algorithm, we accurately adjust every station numerical value in the matrix B.

Here is the possibility equation of station  $i$  chosen to be the terminal station:

$$P_j = \frac{Q_{B_j}}{\sum_{j=1}^{33} Q_{B_j}} \quad (2.1)$$

$j$ :the number of main station.  $Q_{A_i}$ :the passenger flow into station  $i$ .  $Q_{B_j}$ :the outbound passenger flow of station  $j$ .

Then,the passenger flow of directed specified distance can be calculated.We stored up the numerical value in the matrix C.It equals matrix A multiplies matrix B.

$$Q_{ij} = Q_{A_i} \times P_j \quad (2.2)$$

$Q_{ij}$ :the passenger flow between station  $i$  and station  $j$ (with the direction from station  $i$  to station  $j$ ).

So,the social all time saved equals matrix C multiplies the time saved per person.

$$T_{save} = t_{save} \cdot Q_{ij} \quad (2.3)$$

Now,we should sort the numerical value in the matrix D to get the maximum value and its route.

### 3.2.5 The Result of Model II

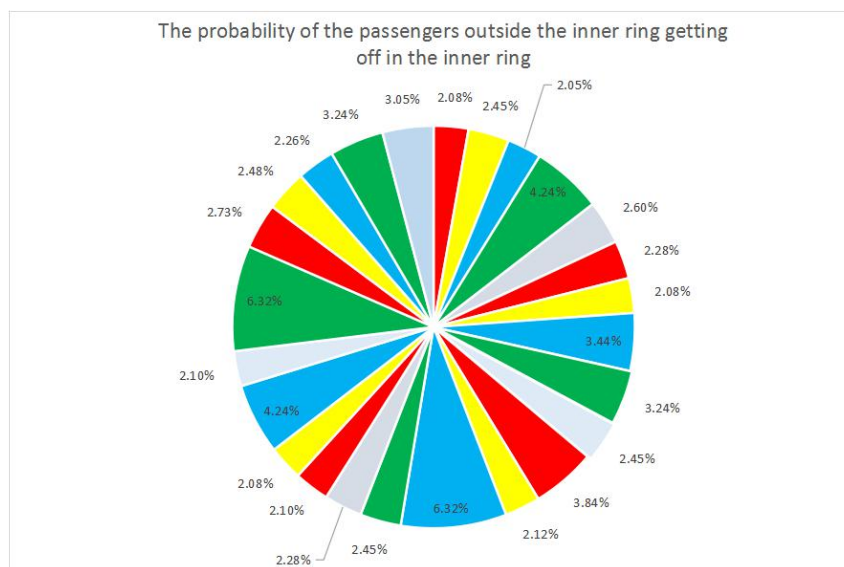


figure 4 The probability of getting off in the inner ring

Input the data outcome after preprocessing into the matrix A B C. We acquire the result by running our model in MATLAB.

Here is the top 10 of the social total time saved and their routes.

| Sequence            | 1             | 2              | 3                              | 4               | 5             |
|---------------------|---------------|----------------|--------------------------------|-----------------|---------------|
| Initial Station     | Lianhua Rd    | Jiuting        | Shanghai South Railway Station | Longyang Rd.    | Tonghe Xincun |
| Terminal Station    | Pudong Avenue | Pudong Avenue  | Pudong Avenue                  | Pudong Avenue   | Pudong Avenue |
| $T_{save}$ (minute) | 88468         | 85623          | 80042                          | 64363           | 63953         |
| Sequence            | 6             | 7              | 8                              | 9               | 10            |
| Initial Station     | Penpu Xincun  | Zhongshan Park | Songhong Rd.                   | Beiyangjing Rd. | Gonfu Xincun  |
| Terminal Station    | Pudong Avenue | Pudong Avenue  | Pudong Avenue                  | Pudong Avenue   | Pudong Avenue |
| $T_{save}$ (minute) | 61075         | 58380          | 55258                          | 53999           | 53218         |

From this table, we can see the optimal route is between Lianhua Road and Pudong Avenue. We can save 88468 minutes for the whole society.

This conclusion also verifies our assumption. Namely, the effective solution is to choose one station outside the inner ring, and the other is inside the inner ring (including its boundary).

From the perspective of society, passengers around the initial and terminal station can choose this special train to shorten their travel time. Effectively decrease the passenger flow in the adjoining station. Help the subway system realize the shunt effect to the utmost degree. Based on this, we enhance the efficiency of the dynamic model of urban network traffic flow. The time totally saved can be utilized in more meaningful spheres and create more social treasure.

From the perspective of economic, thanks to our model containing the passenger flow index, empty operations do not exist. Resource in our model will not be idle. This can also bring a considerable government receipt.

This graph depicts the optimal route in our model.

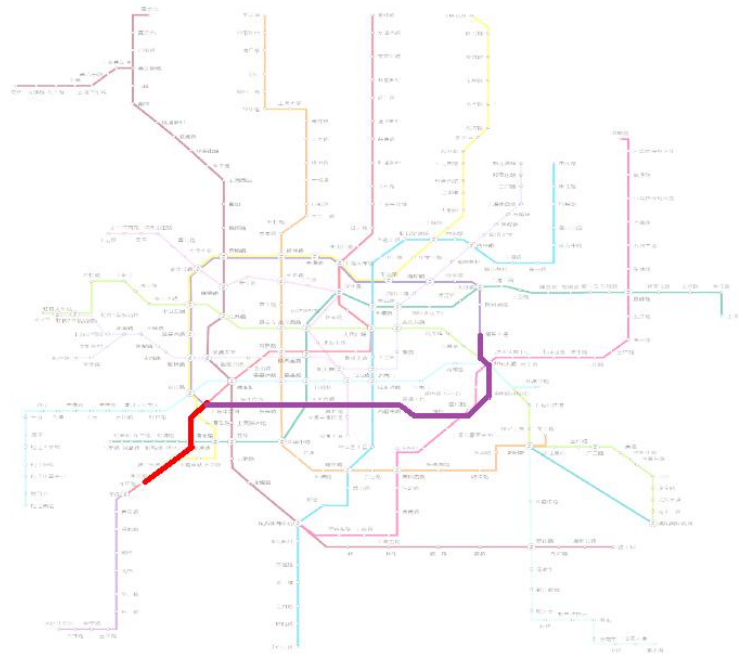


figure 5 The metro line between Lianhua Rd. and Pudong Avenue

### 3.2.6 The advantages and disadvantages of Model II

#### Advantages:

(1) We choose one terminal station inside the circle of metro line 4 (the inner ring of Shanghai) and the other outside the circle. In this way, we simply the question and avoid considering the unnecessary conditions.

(2) We multiply  $Q_{A_i}$  and  $P_j$  to figure out the passenger flow from Station A to Station B. In this way, we get the relatively accurate data in theory.

#### Disadvantages:

(1) There is a chance that passengers getting on outside the inner ring don't head to the inner ring.

(2) According to the database, stations along metro line 4 have big passenger flows of both entering the station and getting out of the station. It leads to the decreasing of the accuracy of our calculation.

### 3.3 Model III—The interval of the direct metro line

#### 3.3.1 The analysis of Model III

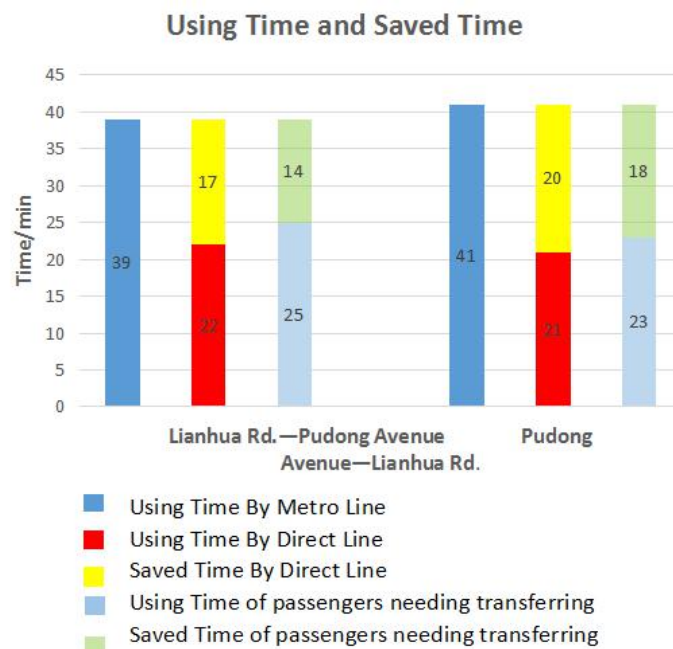
In order to make the timetable of the direct metro, we develop model three to figure out the best interval time of the direct metro.

1. According to the database, we set the former data of the direct metro line:

|                                   | the seating capacity/people | The maximum number of passengers that can be held/people |
|-----------------------------------|-----------------------------|----------------------------------------------------------|
| each carriage of the metro        | 300                         | 420                                                      |
| each carriage of the direct metro | 150                         | 300                                                      |
| carriages of the direct metro     | 300                         | 600                                                      |

According to the database, the interval time of metro line one during the fastigium is  $T_1 = 3 \text{ min}$ , while that of the flat peak period is  $T_2 = 6 \text{ min}$ . Considering the best shunt effect (as for the metro line) and comfort level, we decide the direct metro line to consist of two carriages. Then we may have  $n = 300, n_{\max} = 600$

2. From model one we can know that



(We assume that transferers spend 3 mins on transferring the metro line.)

figure 6 The chart of using time and saved time

|                                                                                                            | Lianhua Rd.—Pudong Avenue | Pudong Avenue—Lianhua Rd. |
|------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|
| Using Time By Metro Line/min                                                                               | 39                        | 41                        |
| Using Time By Direct Line/min                                                                              | 22                        | 20                        |
| Saving Time By Direct Line $t_{save1}$ /min                                                                | 17                        | 21                        |
| Saving Time of passengers going to transfer from the direct metro line to the underground $t_{save2}$ /min | 14                        | 18                        |

We let

$$t_{save} = \min\{t_{save1}, t_{save2}\} = 14 \text{ min} \quad (3.1)$$

3. According to the passenger flow volume we get from model two, and the chart of The Passenger Flow of Each Period of Time Each Day

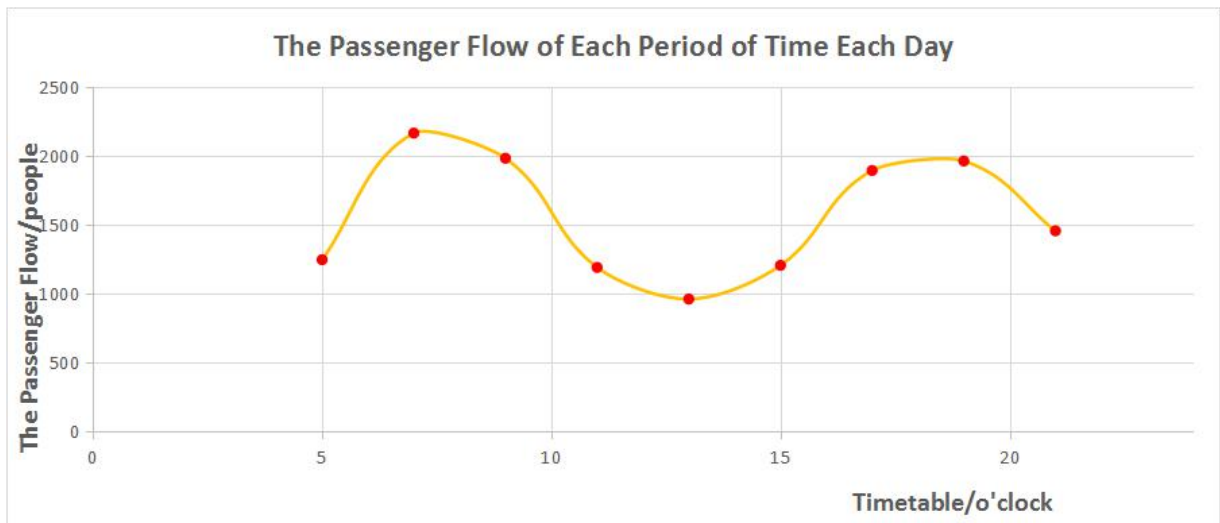


figure 7 The Passenger Flow of Each Period of Time Each Day



| Passenger Flow<br>(people/<br>Period minute) | Lianhua Rd.—Pudong Avenue ( $V_1$ ) | Around Lianhua Rd.—Around Pudong Avenue ( $V_2$ ) | Passengers need to take the direct metro lines<br>( $V = V_1 + V_2$ ) |
|----------------------------------------------|-------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------|
| Fastigium                                    | 10                                  | 23                                                | 33                                                                    |
| Flat peak period                             | 5                                   | 13                                                | <del>                    </del>                                       |

*(The same applies to traveling from Pudong Avenue to Lianhua Road in the evening peak)*

- The time of setting our first direct metro line will keep pace with the original metro line. We set the interval of our direct metro line to be on the multiple of  $k$  as the interval of the metro, name it

$$T = kT_l \quad (3.2)$$

It makes sure that the frequency of the metros between the two direct metro lines is the same.

#### 4.① Fastigium

To bypass the passenger flow to the greatest extent, we let all the passengers from Lianhua Rd. to Pudong Avenue take the direct metro .

#### ② Flat peak period

The passenger flow during the flat peak period is not large. It is no need to let all the passengers take the direct metro line. So passengers will decide the vehicle on the basis of the timetable of the metro and direct metro. Between the time of the sum of waiting for the metro and saving time by direct metro ,and of waiting for the direct metro,passengers will choose the shorter one.

### 3.3.2 Additional Symbols

| Sign        | Definition                                                                                |
|-------------|-------------------------------------------------------------------------------------------|
| $T$         | the interval of the direct metro line                                                     |
| $T_1$       | the interval time of metro line one during the fastigium                                  |
| $T_2$       | the interval time of metro line one during the flat peak period                           |
| $n$         | the seating capacity of the direct metro                                                  |
| $n_{max}$   | the maximum number of passengers that can be held in the direct metro                     |
| $t_{save1}$ | Saving Time By Direct Line                                                                |
| $t_{save2}$ | Saving Time of passengers going to transfer from the direct metro line to the underground |
| $V_1$       | the passenger flow from Lianhua Rd. to Pudong Avenue                                      |
| $V_2$       | the passenger flow from around Lianhua Rd. to around Pudong Avenue                        |
| $V$         | The flow of passengers need to take the direct metro lines                                |
| $k$         | the multiple of the interval of the direct metro as that of the metro                     |
| $T_{wm}$    | passengers' time of waiting for the metro                                                 |

|          |                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------|
| $T_{ws}$ | passengers' time of waiting for the direct metro                                                           |
| $t_l$    | the longest time of passengers from Lianhua Rd. to Pudong Avenue waiting for the direct line               |
| $t_2$    | the longest time of passengers from around Lianhua Rd. to around Pudong Avenue waiting for the direct line |

### 3.3.3 The assumption of model III

- We assume that the passenger flow volume per minute is the same. We know the total volume of the passenger flow during a specific period. We know the length of the specific period, too. So we can figure out the average volume of the passenger flow per minute. In order to simplify the problem, the passenger flow volume per minute is assumed to be unchanged.
- We assume the interval of our direct metro line to be on the multiple of  $k$  as the interval of the metro. If the interval of the direct metro and that of the metro aren't in proportion, there are uncertain number of times of the metro between the two direct metro. In that way, it is difficult for us to make out the timetable.

### 3.3.4 The Foundation of Model III

#### 3.3.4.1 Fastigium

- ① To make passengers choose direct metro instead of metro, the  $kT_l$  should follow the relation of :

$$kT_l < t_{save} \quad (3.3)$$

- ② To make the direct metro carry passengers as many as possible, we should follow the relation of:

$$n \leq V \times (kT_l) \leq n_{max} \quad (k \in N^*) \quad (3.4)$$

#### 3.3.4.2 Flat peak period

When  $T_{wm} + t_{save} < T_{ws}$  , passengers choose to take the metro. (3.5)

When  $T_{wm} + t_{save} \geq T_{ws}$  , passengers choose to take the direct metro. (3.6)

So it follows the relation :

$$n \leq V_1 t_1 + V_2 t_2 \leq n_{\max} \quad (3.7)$$

$t_1$  —the longest time of passengers from Lianhua Rd. to Pudong Avenue waiting for the direct line

$t_2$  —the longest time of passengers from around Lianhua Rd. to around Pudong Avenue waiting for the direct line

### 3.3.5 The solution and Result of Model III

#### 3.3.5.1. Fastigium

• Take the travel from Lianhua Rd. to Pudong Avenue for example:

① To make passengers choose direct metro instead of metro, the  $kT_l$  should follow the relation of :

$$kT_l < t_{\text{save}} \quad (3.8)$$

According to the *Shanghai Metro Official Website*, we can know that the interval of the metro is  $T_l = 3\text{min}$

From Model One, we get  $t_{\text{save}1}$  and  $t_{\text{save}2}$ , from  $t_{\text{save}} = \min\{t_{\text{save}1}, t_{\text{save}2}\}$ , we can know

$$t_{\text{save}} = 14\text{min}$$

$$\text{So we can figure out } k < \frac{t_{\text{save}}}{T_l} = \frac{14}{3}$$

② To make the direct metro carry passengers as many as possible, we should follow the relation of:

$$n \leq V \times (kT_l) \leq n_{\max} \quad (k \in N^*) \quad (3.9)$$

According to the passenger flow volume we get from model two, we can know

$$V = (V_1 + V_2) = 33$$

According to the database of the direct metro line, we can know

$$n = 300, n_{\max} = 600$$

$$\text{So we can figure out } \frac{100}{3} \leq k \leq \frac{200}{3}$$

To summarize ①②, we can figure out  $k = 4$ .

• When it comes to the travel from Pudong Avenue to Lianhua Rd. ,in a similar way we can figure out  $k = 4$  or  $5$ .

To make more people save time and make the social benefits larger ,we choose

$$k = 4 \text{ and } T = kT_l = 12$$

### 3.3.5.2.Flat peak period

It follows the relation :  $n \leq V_1 t_1 + V_2 t_2 \leq n_{\max}$

$t_1$  —the longest time of passengers from Lianhua Rd. to Pudong Avenue waiting for the direct line

$t_2$  —the longest time of passengers from around Lianhua Rd. to around Pudong Avenue waiting for the direct line

- Take the travel from Lianhua Rd. to Pudong Avenue for example:  
(the same as the travel from Pudong Avenue to Lianhua Rd.)

First, we make  represent the direct metro,  represent the metro.

Then,when  $k = 1$  ,from  $T_{wm} + t_{save} < T_{ws}$

We can know that there is no answer to the inequality,  
So we will figure out :  $t_1 = 6, t_2 = 6$ .



figure 8 interval of direct metros when  $k=1$

Then in a similiar way:

①when  $k = 2$  , we can figure out  $t_1 = 12, t_2 = 12$  .

Then  $V_1 t_1 + V_2 t_2 = 5 \times 12 + 13 \times 12 = 216 < 300$  .



figure 9 interval of direct metros when  $k=2$

② when  $k = 3$ , we can figure out  $t_1 = 18, t_2 = 18$ .

Then  $V_1 t_1 + V_2 t_2 = 5 \times 18 + 13 \times 18 = 324 > 300$



figure 10 interval of direct metros when  $k=3$

③ when  $k = 4$ , we can figure out  $t_1 = 24, t_2 = 21$ .

Then  $V_1 t_1 + V_2 t_2 = 5 \times 24 + 13 \times 21 = 393 > 300$



figure 11 interval of direct metros when  $k=4$

④ When  $k \geq 5$ , we can figure out  $t_1 = 27, t_2 = 21$ .

Then  $V_1 t_1 + V_2 t_2 = 5 \times 27 + 13 \times 21 = 408 > 300$ .



figure 12 interval of direct metros when  $k \geq 5$

To sum up, in order to let more passengers to take the direct metro and save money at the same time, we choose

$$k = 5 \quad \text{and} \quad T = kT_2 = 30 \text{ min}$$

So we can get the conclusion:

|                                       | Fastigium | Flat peak period |
|---------------------------------------|-----------|------------------|
| Interval time of the direct metro/min | 12        | 30               |

### 3.3.6 The advantages and disadvantages of Model III

#### Advantages:

(1) During the fastigium, our model bypasses the passenger flow to the greatest extent. Meanwhile, we make the most use of the room in the direct metro.

During the flat peak period, our model reduces the prime cost of running. Meanwhile, we share the burden of the metro to some extent.

(2) We set the interval of our direct metro line to be on the multiple of  $k$  as the interval of the metro. On one hand, it is convenient to consider the bypass flow in the station. On the other hand, it is convenient to set the schedule.

(3) Our model makes the passengers save time and feel comfortable at the same time.

**Disadvantages:**

(1) We take the average value of the passenger flow during a specific period. But the passenger flows have some small changes between different times.

#### **IV.The Letter to the Mayor**

Dear mayor,

We are writing this letter to make a brief introduction of our suggestions on the improvement of the current metro network.

Owing to the strict governance and regulations of the Party, the metro network in Shanghai is so highly-developed that people can almost reach every corner of the city though the metro lines. However, as for the passengers who have to cover long distances, the traveling time would be too tedious, for they have to transfer several metro lines and wait for the intervals between stations and stations.

In our humble opinion , a direct metro line should be set to save the time of the longhual passengers. Therefore, we build several models and discover that building a direct metro line between Lianhua Road and Pudong Avenue saves the maximum general time for the public, considering the amount of people with the traveling need. Here is the summary of our analysis:

From the perspective of the government, both economic and social benefits should be taken into consideration, which means we can bring convenience to the society without resource consumption at the same time.

Although there are many ways of carrying a flock of people from one place to another, they doesn't produce the best results.

Setting a direct metro line with the same routine as the current ones means we don't have to pay excessive construction fees.Instead, we can reach the destination by switching the rails or connecting metro line 1 and line 4 at Shanghai Railway station, which may only cost little.

By the way, the construction of a completely new route will inevitably affect the current transit system and the daily life of the citizen alongside. But putting the direct metro line into practice will avoid such problem.

Our model optimize the interval time between two direct metro lines. The interval time ensures that the passengers quantity reach the minimum value that allows the train to profit and the shunting action to be apparent, maintaining the balance between the economic and social benefits. The specific time table of the direct metro line is enclose with the letter.

From what has been mentioned above, we can draw a conclusion that it is actually a wise decision to set a direct line between Lianhua Road and Pudong Avenue, which may save the maximum amount of time for the general public. As the saying goes, 'Time is money' . By saving the maximum time can we create more social value and improve the social efficiency.



We feel deeply grateful for your sparing time to finish reading our letter. And we hope that our suggestion will be adopted.

Yours,

Sincerely

**The Timetable of the Direct Metro**

| Metro Number | Lianhua Rd. → Pudong Avenue |              | Pudong Avenue → Lianhua Rd. |              |
|--------------|-----------------------------|--------------|-----------------------------|--------------|
|              | Departure time              | Arrival time | Departure time              | Arrival time |
| 1            | 7:30                        | 7:52         | 7:30                        | 7:50         |
| 2            | 7:42                        | 8:04         | 8:00                        | 8:20         |
| 3            | 7:54                        | 8:16         | 8:30                        | 8:50         |
| 4            | 8:06                        | 8:28         | 9:00                        | 9:20         |
| 5            | 8:18                        | 8:40         | 9:30                        | 9:50         |
| 6            | 8:30                        | 8:52         | 10:00                       | 10:20        |
| 7            | 8:42                        | 9:04         | 10:30                       | 10:50        |
| 8            | 8:54                        | 9:16         | 11:00                       | 11:20        |
| 9            | 9:06                        | 9:28         | 11:30                       | 11:50        |
| 10           | 9:18                        | 9:40         | 12:00                       | 12:20        |
| 11           | 9:30                        | 9:52         | 12:30                       | 12:50        |
| 12           | 10:00                       | 10:22        | 13:00                       | 13:20        |
| 13           | 10:30                       | 10:52        | 13:30                       | 13:50        |
| 14           | 11:00                       | 11:22        | 14:00                       | 14:20        |
| 15           | 11:30                       | 11:52        | 14:30                       | 14:50        |
| 16           | 12:00                       | 12:22        | 15:00                       | 15:20        |
| 17           | 12:30                       | 12:52        | 15:30                       | 15:50        |
| 18           | 13:00                       | 13:22        | 16:00                       | 16:20        |
| 19           | 13:30                       | 13:52        | 16:30                       | 16:50        |
| 20           | 14:00                       | 14:22        | 17:00                       | 17:20        |
| 21           | 14:30                       | 14:52        | 17:12                       | 17:32        |
| 22           | 15:00                       | 15:22        | 17:24                       | 17:44        |
| 23           | 15:30                       | 15:52        | 17:36                       | 17:56        |
| 24           | 16:00                       | 16:22        | 17:48                       | 18:08        |
| 25           | 16:30                       | 16:52        | 18:00                       | 18:20        |
| 26           | 17:00                       | 17:22        | 18:12                       | 18:32        |
| 27           | 17:30                       | 17:52        | 18:24                       | 18:44        |
| 28           | 18:00                       | 18:22        | 18:36                       | 18:56        |
| 29           | 18:30                       | 18:52        | 18:48                       | 19:08        |
| 30           | 19:00                       | 19:22        | 19:00                       | 19:20        |
| 31           | 19:30                       | 19:52        | 19:30                       | 19:50        |

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## VI. Appendix

Shanghai metro line 1:

| Number | Station                        | Time  | Using time between adjacent stations (min) |
|--------|--------------------------------|-------|--------------------------------------------|
| 1      | Xinzhuang                      | 22:32 |                                            |
| 2      | Waihuan Rd.                    | 22:34 | 2                                          |
| 3      | Lianhua Rd.                    | 22:36 | 2                                          |
| 4      | Jinjiang Park                  | 22:39 | 3                                          |
| 5      | Shanghai South Railway Station | 22:42 | 3                                          |
| 6      | Caobao Rd.                     | 22:45 | 3                                          |
| 7      | Shanghai indoor Stadium        | 22:48 | 3                                          |
| 8      | Xujiahui                       | 22:50 | 2                                          |
| 9      | Hengshan Rd.                   | 22:52 | 2                                          |
| 10     | Changshu Rd.                   | 22:54 | 2                                          |
| 11     | South Shanxi Rd.               | 22:56 | 2                                          |
| 12     | South Huangpi Rd.              | 22:58 | 2                                          |
| 13     | People' s Square               | 23:01 | 3                                          |
| 14     | Xinzha Rd.                     | 23:03 | 2                                          |
| 15     | Hanzhong Rd.                   | 23:05 | 2                                          |
| 16     | Shanghai Railwat Station       | 23:08 | 3                                          |
| 17     | North Zhongshan Rd.            | 23:10 | 2                                          |
| 18     | Yanchang Rd.                   | 23:13 | 3                                          |
| 19     | Shanghai Circus World          | 23:15 | 2                                          |
| 20     | Wenshui Rd.                    | 23:17 | 2                                          |
| 21     | Penpu Xincun                   | 23:20 | 3                                          |
| 22     | Gongkong Rd.                   | 23:23 | 3                                          |
| 23     | Tonghe Xincun                  | 23:25 | 2                                          |
| 24     | Hulan Rd.                      | 23:27 | 2                                          |
| 25     | Gonfu Xincun                   | 23:30 | 3                                          |
| 26     | Bao'an Highway                 | 23:33 | 3                                          |
| 27     | West Youyi Rd.                 | 23:35 | 2                                          |
| 28     | Fujin Rd.                      | 23:37 | 2                                          |

Shanghai metro line 2:

| Number | Station                              | Time  | Using time between adjacent stations (min) |
|--------|--------------------------------------|-------|--------------------------------------------|
| 1      | Pudong International Airport         | 22:00 |                                            |
| 2      | Haitian Three Way                    | 22:03 | 3                                          |
| 3      | The Far East Avenue                  | 22:10 | 7                                          |
| 4      | Volley Rd.                           | 22:15 | 5                                          |
| 5      | Chuansha                             | 22:18 | 3                                          |
| 6      | East Huaxia Rd.                      | 22:23 | 5                                          |
| 7      | Mid Chuangxin Rd.                    | 22:26 | 3                                          |
| 8      | Tang Towm                            | 22:29 | 3                                          |
| 9      | Guanglan Rd.                         | 22:45 | 16                                         |
| 10     | Jinke Rd.                            | 22:48 | 3                                          |
| 11     | Zhangjiang Hi-Tech Park              | 22:50 | 2                                          |
| 12     | Longyang Rd.                         | 22:54 | 4                                          |
| 13     | Century Park                         | 22:56 | 2                                          |
| 14     | Shanghai Science & Technology Museum | 22:58 | 2                                          |
| 15     | Century Avenvue                      | 23:01 | 3                                          |
| 16     | Dongchang Rd.                        | 23:03 | 2                                          |
| 17     | Lujiazui                             | 23:06 | 3                                          |
| 18     | East Nanjing Rd.                     | 23:08 | 2                                          |
| 19     | People Square                        | 23:11 | 3                                          |
| 20     | West Nanjing Rd.                     | 23:13 | 2                                          |
| 21     | Jing'an Temple                       | 23:16 | 3                                          |
| 22     | Jiangsu Rd.                          | 23:18 | 2                                          |
| 23     | Zhongshan Park                       | 23:21 | 3                                          |
| 24     | Loushanguan Rd.                      | 23:24 | 3                                          |
| 25     | Weining Rd.                          | 23:26 | 2                                          |
| 26     | Beixinjing                           | 23:29 | 3                                          |
| 27     | Songhong Rd.                         | 23:31 | 2                                          |
| 28     | Hongqiao Terminal 2                  | 23:38 | 7                                          |
| 29     | Hongqiao Railway Station             | 23:40 | 2                                          |
| 30     | East Xujing                          | 23:43 | 3                                          |

Shanghai metro line 3:

| Number | Station                        | Time  | Using time between adjacent stations (min) |
|--------|--------------------------------|-------|--------------------------------------------|
| 1      | Shanghai South Railway Station | 05:25 |                                            |
| 2      | Shilong Rd.                    | 05:27 | 2                                          |
| 3      | Longcao Rd.                    | 05:30 | 3                                          |
| 4      | Caoxi Rd.                      | 05:32 | 2                                          |
| 5      | Yishan Rd.                     | 05:34 | 4                                          |
| 6      | Hongqiao Rd.                   | 05:36 | 2                                          |
| 7      | West Yan'an Rd.                | 05:39 | 3                                          |
| 8      | Zhongshan Park                 | 05:41 | 2                                          |
| 9      | Jinshajiang Rd.                | 05:43 | 2                                          |
| 10     | Caoyang Rd.                    | 05:45 | 2                                          |
| 11     | Zhenping Rd.                   | 05:48 | 3                                          |
| 12     | Zhongtan Rd.                   | 05:50 | 2                                          |
| 13     | Shanghai Railway Station       | 05:53 | 3                                          |
| 14     | Baoshan Rd.                    | 05:56 | 3                                          |
| 15     | Dongbaoxing Rd.                | 05:58 | 2                                          |
| 16     | Hongkou Football Stadium       | 06:01 | 3                                          |
| 17     | Chifeng Rd.                    | 06:03 | 2                                          |
| 18     | Dabaishu                       | 06:05 | 2                                          |
| 19     | Jiangwan Town                  | 06:08 | 3                                          |
| 20     | West Yingao Rd.                | 06:10 | 2                                          |
| 21     | South Changjiang Rd.           | 06:11 | 1                                          |
| 22     | Songfa Rd.                     | 06:13 | 2                                          |
| 23     | Zhuanghuabang                  | 06:15 | 2                                          |
| 24     | Songbin Rd.                    | 06:18 | 3                                          |
| 25     | Shuichan Rd.                   | 06:20 | 2                                          |
| 26     | Baoyang Rd.                    | 06:25 | 5                                          |
| 27     | Youyi Rd.                      | 06:27 | 2                                          |
| 28     | Tieli Rd.                      | 06:29 | 2                                          |
| 29     | North Jiangyang Rd.            | 06:31 | 2                                          |

Shanghai metro line 4:

| Number | Station                  | Time  | Using time between adjacent stations (min) |
|--------|--------------------------|-------|--------------------------------------------|
| 1      | Yishan Road              | 05:30 |                                            |
| 2      | Shanghai Indoor Stadium  | 05:32 | 2                                          |
| 3      | Shanghai Stadium         | 05:34 | 2                                          |
| 4      | Dong'an Rd.              | 05:36 | 2                                          |
| 5      | Damuqiao Rd.             | 05:37 | 1                                          |
| 6      | Luban Rd.                | 05:39 | 2                                          |
| 7      | South Xizang Rd.         | 05:42 | 3                                          |
| 8      | Nanpu Bridge             | 05:44 | 2                                          |
| 9      | Tangqiao                 | 05:47 | 3                                          |
| 10     | Lancun Rd.               | 05:49 | 2                                          |
| 11     | Pudian Rd.               | 05:51 | 2                                          |
| 12     | Century Avenue           | 05:53 | 2                                          |
| 13     | Pudong Avenue            | 05:56 | 3                                          |
| 14     | Yangshupu Rd.            | 05:58 | 2                                          |
| 15     | Dalian Rd.               | 06:00 | 2                                          |
| 16     | Linping Rd.              | 06:02 | 2                                          |
| 17     | Hailun Rd.               | 06:04 | 2                                          |
| 18     | Baoshan Rd.              | 06:07 | 3                                          |
| 19     | Shanghai Railway Station | 06:10 | 3                                          |
| 20     | Zhongtan Rd.             | 06:13 | 3                                          |
| 21     | Zhenping Rd.             | 06:15 | 2                                          |
| 22     | Caoyang Rd.              | 06:18 | 3                                          |
| 23     | Jinshajiang Rd.          | 06:19 | 1                                          |
| 24     | Zhongshan Park           | 06:22 | 3                                          |
| 25     | West Yan'an Rd.          | 06:24 | 2                                          |
| 26     | Hongqiao Rd.             | 06:26 | 2                                          |

Shanghai metro line 5:

| Number | Station                  | Time  | Using time between adjacent stations (min) |
|--------|--------------------------|-------|--------------------------------------------|
| 1      | Xinzhuang                | 06:00 |                                            |
| 2      | Chunshen Rd.             | 06:02 | 2                                          |
| 3      | Yindu Rd.                | 06:04 | 4                                          |
| 4      | Zhuanqiao                | 06:08 | 3                                          |
| 5      | Beiqiao                  | 06:11 | 3                                          |
| 6      | Jianchuan Rd.            | 06:14 | 2                                          |
| 7      | Dongchuan Rd.            | 06:16 | 3                                          |
| 8      | Jinping Rd.              | 06:19 | 2                                          |
| 9      | Huaning Rd.              | 06:21 | 3                                          |
| 10     | Wenjing Rd.              | 06:24 | 3                                          |
| 11     | Minhang Development Zone | 06:26 | 2                                          |

Shanghai metro line 6:

| Number | Station                            | Time  | Using time between adjacent stations (min) |
|--------|------------------------------------|-------|--------------------------------------------|
| 1      | Gangcheng Rd.                      | 05:30 |                                            |
| 2      | North Waigaoqiao Free Trade Zone   | 05:32 | 3                                          |
| 3      | Hangjin Rd.                        | 05:35 | 3                                          |
| 4      | South Waigaoqiao Free Trade Zone   | 05:38 | 3                                          |
| 5      | Zhouhai Rd.                        | 05:41 | 2                                          |
| 6      | Wuzhou Avenue                      | 05:43 | 2                                          |
| 7      | Dongjing Rd.                       | 05:45 | 2                                          |
| 8      | Jufeng Rd.                         | 05:47 | 2                                          |
| 9      | Wulian Rd.                         | 05:49 | 3                                          |
| 10     | Boxing Rd.                         | 05:52 | 2                                          |
| 11     | Jinqiao Rd.                        | 05:54 | 2                                          |
| 12     | Yunshan Rd.                        | 05:56 | 3                                          |
| 13     | Deping Rd.                         | 05:59 | 2                                          |
| 14     | Beiyangjing Rd.                    | 06:01 | 2                                          |
| 15     | Minsheng Rd.                       | 06:03 | 3                                          |
| 16     | Yuanshen Stadium                   | 06:06 | 2                                          |
| 17     | Century Avenue                     | 06:08 | 3                                          |
| 18     | Pudian Rd.                         | 06:11 | 2                                          |
| 19     | Lancun Rd.                         | 06:13 | 3                                          |
| 20     | Shanghai Children's Medical Center | 06:16 | 2                                          |
| 21     | Linyi Xincun                       | 06:18 | 3                                          |
| 22     | West Gaoke Rd.                     | 06:21 | 2                                          |
| 23     | Dongming Rd.                       | 06:23 | 3                                          |
| 24     | Gaoqing Rd.                        | 06:26 | 2                                          |

|    |                        |       |   |
|----|------------------------|-------|---|
| 25 | West Huaxia Rd.        | 06:28 | 2 |
| 26 | Shangnan Rd.           | 06:30 | 2 |
| 27 | South Lingyan Rd.      | 06:32 | 2 |
| 28 | Oriental Sports Center | 06:34 | 2 |

Shanghai metro line 7:

| Number | Station             | Time  | Using time between adjacent stations (min) |
|--------|---------------------|-------|--------------------------------------------|
| 1      | Huamu Rd.           | 05:30 |                                            |
| 2      | Longyang Rd.        | 05:33 | 3                                          |
| 3      | Fanghua Rd.         | 05:35 | 2                                          |
| 4      | Jinxiu Rd.          | 05:38 | 3                                          |
| 5      | South Yanggao Rd.   | 05:40 | 2                                          |
| 6      | West Gaoke Rd.      | 05:43 | 3                                          |
| 7      | Yuntai Rd.          | 05:45 | 2                                          |
| 8      | Yaohua Rd.          | 05:47 | 2                                          |
| 9      | Changqing Rd.       | 05:49 | 2                                          |
| 10     | Houtan              | 05:51 | 2                                          |
| 11     | Mid Longhua Rd.     | 05:54 | 3                                          |
| 12     | Dong'an Rd.         | 05:56 | 2                                          |
| 13     | Zhaojiabang Rd.     | 05:58 | 2                                          |
| 14     | Changshu Rd.        | 06:01 | 3                                          |
| 15     | Jing'an Temple      | 06:03 | 2                                          |
| 16     | Changping Rd.       | 06:06 | 3                                          |
| 17     | Changshou Rd.       | 06:08 | 2                                          |
| 18     | Zhenping Rd.        | 06:10 | 2                                          |
| 19     | Lan'gao Rd.         | 06:13 | 3                                          |
| 20     | Xincun Rd.          | 06:15 | 2                                          |
| 21     | No.3 Dahua Rd.      | 06:17 | 2                                          |
| 22     | Xingzhi Rd.         | 06:19 | 2                                          |
| 23     | Dachang Town        | 06:21 | 2                                          |
| 24     | Changzhong Rd.      | 06:24 | 3                                          |
| 25     | Shangda Rd.         | 06:26 | 2                                          |
| 26     | Nanchen Rd.         | 06:28 | 2                                          |
| 27     | Shanghai University | 06:31 | 3                                          |
| 28     | Gucun Park          | 06:37 | 6                                          |
| 29     | Liuhang             | 06:40 | 3                                          |
| 30     | Panguang Rd.        | 06:42 | 2                                          |
| 31     | Luonan Rd.          | 06:46 | 4                                          |
| 32     | Meilanhu            | 06:48 | 2                                          |



Shanghai metro line 8:

| Number | Station                  | Time  | Using time between adjacent stations (min) |
|--------|--------------------------|-------|--------------------------------------------|
| 1      | AerospaceMuseum          | 07:30 |                                            |
| 2      | Lianhang Rd.             | 07:32 | 2                                          |
| 3      | Jiangyue Rd.             | 07:34 | 2                                          |
| 4      | Pujiang Towm             | 07:36 | 4                                          |
| 5      | Luheng Rd.               | 07:40 | 3                                          |
| 6      | Lingzhao Xincun          | 07:43 | 3                                          |
| 7      | Oriental Sports Center   | 07:46 | 3                                          |
| 8      | Yangsi                   | 07:49 | 2                                          |
| 9      | Chengshan Rd.            | 07:51 | 2                                          |
| 10     | Yaohua Rd.               | 07:53 | 3                                          |
| 11     | Zhoujiadu                | 07:56 | 2                                          |
| 12     | South Xizang Rd.         | 07:58 | 2                                          |
| 13     | Lujiabang Rd.            | 08:00 | 2                                          |
| 14     | Laoximen                 | 08:02 | 2                                          |
| 15     | Dashijie                 | 08:04 | 2                                          |
| 16     | People's Square          | 08:06 | 2                                          |
| 17     | Qupu Rd.                 | 08:08 | 2                                          |
| 18     | Zhongxing Rd.            | 08:10 | 3                                          |
| 19     | Northe Xizang Rd.        | 08:13 | 2                                          |
| 20     | Hongkou Football Stadium | 08:15 | 3                                          |
| 21     | Quyang Rd.               | 08:18 | 2                                          |
| 22     | Siping Rd.               | 08:20 | 2                                          |
| 23     | Anshan Xincun            | 08:22 | 2                                          |
| 24     | Jiangpu Rd.              | 08:24 | 2                                          |
| 25     | Huangxing Rd.            | 08:26 | 2                                          |
| 26     | Middle Yanji Rd.         | 08:28 | 2                                          |
| 27     | Huangxing Park           | 08:30 | 4                                          |
| 28     | Xiangyin Rd.             | 08:32 | 2                                          |
| 29     | Nenjiang Rd.             | 08:34 | 2                                          |
| 30     | Shiguang Rd.             | 08:35 | 1                                          |

Shanghai metro line 9:

| Number | Station             | Time  | Using time between adjacent stations (min) |
|--------|---------------------|-------|--------------------------------------------|
| 1      | Middle Yanggao Road | 05:30 |                                            |
| 2      | Century Avenue      | 05:34 | 4                                          |
| 3      | Shangcheng Rd.      | 05:36 | 2                                          |
| 4      | Xiaonanmen          | 05:40 | 4                                          |
| 5      | Lujiabang Rd.       | 05:43 | 3                                          |
| 6      | Madang Rd.          | 05:45 | 2                                          |

|    |                            |       |   |
|----|----------------------------|-------|---|
| 7  | Dapuqiao                   | 05:47 | 2 |
| 8  | Jiangshan Rd.              | 05:49 | 2 |
| 9  | Zhaojiabang Rd.            | 05:51 | 2 |
| 10 | Xujiahui                   | 05:54 | 3 |
| 11 | Yishan Rd.                 | 05:57 | 3 |
| 12 | Guilin Rd.                 | 06:00 | 3 |
| 13 | Caohejing Development Zone | 06:03 | 3 |
| 14 | Hechuan Rd.                | 06:05 | 2 |
| 15 | Xingzhong Rd.              | 06:08 | 3 |
| 16 | Qibao                      | 06:11 | 3 |
| 17 | Zhongchun Rd.              | 06:14 | 3 |
| 18 | Jiuting                    | 06:18 | 4 |
| 19 | Sijing                     | 06:24 | 6 |
| 20 | Sheshan                    | 06:29 | 5 |
| 21 | Dongjing                   | 06:32 | 3 |
| 22 | Songjiang University Town  | 06:36 | 4 |
| 23 | Songjiang Xincheng         | 06:40 | 4 |
| 24 | Songjiang Sports Center    | 06:43 | 3 |
| 25 | Drunken Bai Pond           | 06:46 | 3 |
| 26 | South Songjiang Station    | 06:50 | 4 |

Shanghai metro line 10:

| Number | Station             | Time  | Using time between adjacent stations (min) |
|--------|---------------------|-------|--------------------------------------------|
| 1      | New Jiangwan Town   | 05:35 |                                            |
| 2      | East Yin'gao Rd.    | 05:36 | 1                                          |
| 3      | Sanmen Rd.          | 05:38 | 2                                          |
| 4      | Jiangwan Stadium    | 05:41 | 3                                          |
| 5      | Wujiaochang         | 05:43 | 2                                          |
| 6      | Guoquan Rd.         | 05:45 | 2                                          |
| 7      | Tongji University   | 05:47 | 2                                          |
| 8      | Siping Rd.          | 05:49 | 2                                          |
| 9      | Youdian Xincun      | 05:51 | 2                                          |
| 10     | Hailun Rd.          | 05:53 | 2                                          |
| 11     | North Sichuan Rd.   | 05:56 | 3                                          |
| 12     | Tiantong Rd.        | 05:58 | 2                                          |
| 13     | East Nanjing Rd.    | 06:00 | 2                                          |
| 14     | Yuyuan Garden       | 06:03 | 3                                          |
| 15     | Laixinmen           | 06:05 | 2                                          |
| 16     | Xintiandi           | 06:08 | 3                                          |
| 17     | Souh Shanxi Rd.     | 06:11 | 3                                          |
| 18     | Shanghai Library    | 06:13 | 2                                          |
| 19     | Jiaotong University | 06:16 | 3                                          |

|    |                          |       |   |
|----|--------------------------|-------|---|
| 20 | Hongqiao Rd.             | 06:18 | 2 |
| 21 | Songyuan Rd.             | 06:20 | 2 |
| 22 | Yili Rd.                 | 06:23 | 3 |
| 23 | Shuicheng Rd.            | 06:25 | 2 |
| 24 | Longxi Rd.               | 06:27 | 2 |
| 25 | Shanghai Zoo             | 06:29 | 2 |
| 26 | Hongqiao Airport 1       | 06:32 | 3 |
| 27 | Hongqiao Airport2        | 06:36 | 4 |
| 28 | Hongqiao Railway Station | 06:37 | 1 |

Shanghai metro line 11:

| Number | Station                       | Time  | Using time between adjacent stations (min) |
|--------|-------------------------------|-------|--------------------------------------------|
| 1      | Disneyland                    | 22:30 |                                            |
| 2      | Kangxin Highwat               | 22:34 | 4                                          |
| 3      | Xiuyan Rd.                    | 22:37 | 3                                          |
| 4      | Luoshan Rd.                   | 22:40 | 3                                          |
| 5      | Yuqiao                        | 22:43 | 3                                          |
| 6      | Pusan Rd.                     | 22:47 | 4                                          |
| 7      | East Sanlin                   | 22:50 | 3                                          |
| 8      | Sanlin                        | 22:52 | 2                                          |
| 9      | Oriental Sports Center        | 22:57 | 5                                          |
| 10     | Longyao Rd.                   | 23:00 | 3                                          |
| 11     | Yunjin Rd.                    | 23:02 | 2                                          |
| 12     | Longhua                       | 23:04 | 2                                          |
| 13     | Shanghai Swimming Stadium     | 23:06 | 2                                          |
| 14     | Xujiahui                      | 23:10 | 4                                          |
| 15     | Jiaotong University           | 23:12 | 2                                          |
| 16     | Jiangsu Rd.                   | 23:16 | 4                                          |
| 17     | Longde Rd.                    | 23:18 | 2                                          |
| 18     | Caoyang Rd.                   | 23:21 | 3                                          |
| 19     | Fengqiao Rd.                  | 23:22 | 1                                          |
| 20     | Zhenru                        | 23:25 | 3                                          |
| 21     | Shanghai West Railway Station | 23:27 | 2                                          |
| 22     | Liziyuan                      | 23:30 | 3                                          |
| 23     | Qilianshan Rd.                | 23:32 | 2                                          |
| 24     | Wuwei Rd.                     | 23:34 | 2                                          |
| 25     | Taopu Xincun                  | 23:37 | 3                                          |
| 26     | Nanxiang                      | 23:41 | 4                                          |
| 27     | Malu                          | 23:46 | 5                                          |
| 28     | Jiading TXincheng             | 23:50 | 4                                          |
| 29     | Baiyin Rd.                    | 23:52 | 2                                          |
| 30     | Jiading West                  | 23:56 | 4                                          |
| 31     | Jiading North                 | 23:59 | 3                                          |

Shanghai metro line 12:

| Number | Station                                  | Time  | Using time between adjacent stations (min) |
|--------|------------------------------------------|-------|--------------------------------------------|
| 1      | Jinhai Rd.                               | 05:30 |                                            |
| 2      | Shenjiang Rd.                            | 05:34 | 4                                          |
| 3      | Jinjing Rd.                              | 05:36 | 2                                          |
| 4      | North Yanggao Rd.                        | 05:38 | 3                                          |
| 5      | Jufeng Rd.                               | 05:41 | 2                                          |
| 6      | Donglu Rd.                               | 05:43 | 3                                          |
| 7      | Fuxing Island                            | 05:46 | 2                                          |
| 8      | Aiguo Rd.                                | 05:48 | 2                                          |
| 9      | Longchang Rd.                            | 05:50 | 3                                          |
| 10     | Ningguo Rd.                              | 05:53 | 2                                          |
| 11     | Jiangpu Park                             | 05:55 | 2                                          |
| 12     | Dalian Rd.                               | 05:57 | 2                                          |
| 13     | Tilan Qiao                               | 05:59 | 2                                          |
| 14     | International Passenger Transport Center | 06:01 | 3                                          |
| 15     | Tiantong Rd.                             | 06:04 | 3                                          |
| 16     | Qufu Rd.                                 | 06:07 | 3                                          |
| 17     | Hanzhong Rd.                             | 06:10 | 3                                          |
| 18     | West Nanjing Rd.                         | 06:13 | 3                                          |
| 19     | South Shanxi Rd.                         | 06:16 | 2                                          |
| 20     | Jiashan Rd.                              | 06:18 | 3                                          |
| 21     | Damuqiao Rd.                             | 06:21 | 2                                          |
| 22     | Mid Longhua Rd.                          | 06:23 | 3                                          |
| 23     | Longhua                                  | 06:26 | 3                                          |
| 24     | Longcao Rd.                              | 06:29 | 2                                          |
| 25     | Caobao Rd.                               | 06:31 | 2                                          |
| 26     | Guilin Park                              | 06:33 | 2                                          |
| 27     | Hongcao Rd.                              | 06:35 | 3                                          |
| 28     | Hongmei Rd.                              | 06:38 | 2                                          |
| 29     | Donglan Rd.                              | 06:40 | 3                                          |
| 30     | Gudai Rd.                                | 06:43 | 2                                          |
| 31     | Guxin Rd.                                | 06:45 | 2                                          |
| 32     | Qixin Rd.                                | 06:47 | 2                                          |

Shanghai metro line 13:

| Number | Station             | Time  | Using time between adjacent stations (min) |
|--------|---------------------|-------|--------------------------------------------|
| 1      | Jinyun Rd.          | 05:50 |                                            |
| 2      | Jinshajiang Rd.     | 05:52 | 2                                          |
| 3      | Fengzhuang          | 05:56 | 4                                          |
| 4      | South Qilanshan Rd. | 05:59 | 3                                          |
| 5      | Zhenbei Rd.         | 06:01 | 2                                          |
| 6      | Daduhe Rd.          | 06:04 | 3                                          |
| 7      | Jinshajiang Rd.     | 06:08 | 4                                          |
| 8      | Longde Rd.          | 06:10 | 2                                          |
| 9      | Wuning Rd.          | 06:13 | 3                                          |
| 10     | Changshou Rd.       | 06:15 | 2                                          |
| 11     | Jiangning Rd.       | 06:17 | 2                                          |
| 12     | Hanzhong Rd.        | 06:20 | 3                                          |
| 13     | Natural Museum      | 06:21 | 1                                          |
| 14     | West Nanjing Rd.    | 06:24 | 3                                          |
| 15     | Middle Huaihai Rd.  | 06:26 | 2                                          |
| 16     | Xintiandi           | 06:29 | 3                                          |
| 17     | Madang Rd.          | 06:31 | 2                                          |
| 18     | Expo Museum         | 06:33 | 2                                          |
| 19     | Expo Avenue         | 06:36 | 3                                          |

Shanghai metro line 16:

| Number | Station           | Time  | Using time between adjacent stations (min) |
|--------|-------------------|-------|--------------------------------------------|
| 1      | Dishui Lake       | 22:00 |                                            |
| 2      | Harbor Boulevard  | 22:03 | 3                                          |
| 3      | College           | 22:09 | 6                                          |
| 4      | East Huinan       | 22:19 | 10                                         |
| 5      | Huinan            | 22:24 | 5                                          |
| 6      | Wildlife Park     | 22:32 | 8                                          |
| 7      | Xinchang          | 22:39 | 7                                          |
| 8      | East Hangtou      | 22:45 | 6                                          |
| 9      | Hangcheng Hesha   | 22:49 | 4                                          |
| 10     | East Zhoupu       | 22:55 | 6                                          |
| 11     | Luoshan Rd.       | 23:02 | 7                                          |
| 12     | Middle Huaxia Rd. | 23:06 | 4                                          |
| 13     | Longyang Rd.      | 23:11 | 5                                          |

The number of shanghai metro station:

| Serial Number | Station                        | Serial Number | Station                          | Serial Number | Station           |
|---------------|--------------------------------|---------------|----------------------------------|---------------|-------------------|
| 1             | Xinzhuang                      | 122           | Huaning Rd.                      | 243           | Sanmen Rd.        |
| 2             | Waihuan Rd.                    | 123           | Wenjing Rd.                      | 244           | Jiangwan Stadium  |
| 3             | Lianhua Rd.                    | 124           | Minhang Development Zone         | 245           | Wujiaochang       |
| 4             | Jinjiang Park                  | 125           | Gangcheng Rd.                    | 246           | Guoquan Rd.       |
| 5             | Shanghai South Railway Station | 126           | North Waigaoqiao Free Trade Zone | 247           | Tongji University |
| 6             | Caobao Rd.                     | 127           | Hangjin Rd.                      | 248           | Siping Rd.        |
| 7             | Shanghai indoor Stadium        | 128           | South Waigaoqiao Free Trade Zone | 249           | Youdian Xincun    |
| 8             | Xujiahui                       | 129           | Zhouhai Rd.                      | 250           | Hailun Rd.        |
| 9             | Hengshan Rd.                   | 130           | Wuzhou Avenue                    | 251           | North Sichuan Rd. |
| 10            | Changshu Rd.                   | 131           | Dongjing Rd.                     | 252           | Tiantong Rd.      |
| 11            | South Shanxi Rd.               | 132           | Jufeng Rd.                       | 253           | East Nanjing Rd.  |
| 12            | South Huangpi Rd.              | 133           | Wulian Rd.                       | 254           | Yuyuan Garden     |
| 13            | People' s Square               | 134           | Boxing Rd.                       | 255           | Laixinmen         |

|    |                             |     |                                           |     |                             |
|----|-----------------------------|-----|-------------------------------------------|-----|-----------------------------|
| 14 | Xinzha Rd.                  | 135 | Jinqiao Rd.                               | 256 | Xintiandi                   |
| 15 | Hanzhong Rd.                | 136 | Yunshan Rd.                               | 257 | Souh Shanxi Rd.             |
| 16 | Shanghai Railwat<br>Station | 137 | Deping Rd.                                | 258 | Shanghai Library            |
| 17 | North Zhongshan<br>Rd.      | 138 | Beiyangjing Rd.                           | 259 | Jiaotong<br>University      |
| 18 | Yanchang Rd.                | 139 | Minsheng Rd.                              | 260 | Hongqiao Rd.                |
| 19 | Shanghai Circus<br>World    | 140 | Yuanshen Stadium                          | 261 | Songyuan Rd.                |
| 20 | Wenshui Rd.                 | 141 | Century Avenue                            | 262 | Yili Rd.                    |
| 21 | Penpu Xincun                | 142 | Pudian Rd.                                | 263 | Shuicheng Rd.               |
| 22 | Gongkong Rd.                | 143 | Lancun Rd.                                | 264 | Longxi Rd.                  |
| 23 | Tonghe Xincun               | 144 | Shanghai<br>Children' s<br>Medical Center | 265 | Shanghai Zoo                |
| 24 | Hulan Rd.                   | 145 | Linyi Xincun                              | 266 | Hongqiao Airport<br>1       |
| 25 | Gonfu Xincun                | 146 | West Gaoke Rd.                            | 267 | Hongqiao Airport2           |
| 26 | Bao' an Highway             | 147 | Dongming Rd.                              | 268 | Hongqiao Railway<br>Station |
| 27 | West Youyi Rd.              | 148 | Gaoqing Rd.                               | 269 | Disneyland                  |

|    |                                    |     |                           |     |                                 |
|----|------------------------------------|-----|---------------------------|-----|---------------------------------|
| 28 | Fujin Rd.                          | 149 | West Huaxia Rd.           | 270 | Kangxin Highwat                 |
| 29 | Pudong<br>International<br>Airport | 150 | Shangnan Rd.              | 271 | Xiuyan Rd.                      |
| 30 | Haitian Three Way                  | 151 | South Lingyan Rd.         | 272 | Luoshan Rd.                     |
| 31 | The Far East<br>Avenue             | 152 | Oriental Sports<br>Center | 273 | Yuqiao                          |
| 32 | Volley Rd.                         | 153 | Huamu Rd.                 | 274 | Pusan Rd.                       |
| 33 | Chuansha                           | 154 | Longyang Rd.              | 275 | East Sanlin                     |
| 34 | East Huaxia Rd.                    | 155 | Fanghua Rd.               | 276 | Sanlin                          |
| 35 | Mid Chuangxin<br>Rd.               | 156 | Jinxiu Rd.                | 277 | Oriental Sports<br>Center       |
| 36 | Tang Towm                          | 157 | South Yanggao<br>Rd.      | 278 | Longyao Rd.                     |
| 37 | Guanglan Rd.                       | 158 | West Gaoke Rd.            | 279 | Yunjin Rd.                      |
| 38 | Jinke Rd.                          | 159 | Yuntai Rd.                | 280 | Longhua                         |
| 39 | Zhangjiang<br>Hi-Tech Park         | 160 | Yaohua Rd.                | 281 | Shanghai<br>Swimming<br>Stadium |
| 40 | Longyang Rd.                       | 161 | Changqing Rd.             | 282 | Xujiahui                        |
| 41 | Century Park                       | 162 | Houtan                    | 283 | Jiaotong<br>University          |



|    |                                      |     |                     |     |                               |
|----|--------------------------------------|-----|---------------------|-----|-------------------------------|
| 42 | Shanghai Science & Technology Museum | 163 | Mid Longhua Rd.     | 284 | Jiangsu Rd.                   |
| 43 | Century Avenvue                      | 164 | Dong' an Rd.        | 285 | Longde Rd.                    |
| 44 | Dongchang Rd.                        | 165 | Zhaojiabang Rd.     | 286 | Caoyang Rd.                   |
| 45 | Lujiazui                             | 166 | Changshu Rd.        | 287 | Fengqiao Rd.                  |
| 46 | East Nanjing Rd.                     | 167 | Jing' an Temple     | 288 | Zhenru                        |
| 47 | People Square                        | 168 | Changping Rd.       | 289 | Shanghai West Railway Station |
| 48 | West Nanjing Rd.                     | 169 | Changshou Rd.       | 290 | Liziyan                       |
| 49 | Jing' an Temple                      | 170 | Zhenping Rd.        | 291 | Qilianshan Rd.                |
| 50 | Jiangsu Rd.                          | 171 | Lan' gao Rd.        | 292 | Wuwei Rd.                     |
| 51 | Zhongshan Park                       | 172 | Xincun Rd.          | 293 | Taopu Xincun                  |
| 52 | Loushanguan Rd.                      | 173 | No.3 Dahua Rd.      | 294 | Nanxiang                      |
| 53 | Weining Rd.                          | 174 | Xingzhi Rd.         | 295 | Malu                          |
| 54 | Beixinjing                           | 175 | Dachang Town        | 296 | Jiading TXincheng             |
| 55 | Songhong Rd.                         | 176 | Changzhong Rd.      | 297 | Baiyin Rd.                    |
| 56 | Hongqiao Terminal 2                  | 177 | Shangda Rd.         | 298 | Jiading West                  |
| 57 | Hongqiao Railway Station             | 178 | Nanchen Rd.         | 299 | Jiading North                 |
| 58 | East Xujing                          | 179 | Shanghai University | 300 | Jinhai Rd.                    |

|    |                                |     |                        |     |                                          |
|----|--------------------------------|-----|------------------------|-----|------------------------------------------|
| 59 | Shanghai South Railway Station | 180 | Gucun Park             | 301 | Shenjiang Rd.                            |
| 60 | Shilong Rd.                    | 181 | Liuhang                | 302 | Jinjing Rd.                              |
| 61 | Longcao Rd.                    | 182 | Panguang Rd.           | 303 | North Yanggao Rd.                        |
| 62 | Caoxi Rd.                      | 183 | Luonan Rd.             | 304 | Jufeng Rd.                               |
| 63 | Yishan Rd.                     | 184 | Meilanhua              | 305 | Donglu Rd.                               |
| 64 | Hongqiao Rd.                   | 185 | Aerospace Museum       | 306 | Fuxing Island                            |
| 65 | West Yan' an Rd.               | 186 | Lianhang Rd.           | 307 | Aiguo Rd.                                |
| 66 | Zhongshan Park                 | 187 | Jiangyue Rd.           | 308 | Longchang Rd.                            |
| 67 | Jinshajiang Rd.                | 188 | Pujiang Town           | 309 | Ningguo Rd.                              |
| 68 | Caoyang Rd.                    | 189 | Luheng Rd.             | 310 | Jiangpu Park                             |
| 69 | Zhenping Rd.                   | 190 | Lingzhao Xincun        | 311 | Dalian Rd.                               |
| 70 | Zhongtan Rd.                   | 191 | Oriental Sports Center | 312 | Tilan Qiao                               |
| 71 | Shanghai Railway Station       | 192 | Yangsi                 | 313 | International Passenger Transport Center |
| 72 | Baoshan Rd.                    | 193 | Chengshan Rd.          | 314 | Tiantong Rd.                             |
| 73 | Dongbaoxing Rd.                | 194 | Yaohua Rd.             | 315 | Qufu Rd.                                 |
| 74 | Hongkou Football Stadium       | 195 | Zhoujiadu              | 316 | Hanzhong Rd.                             |

|    |                         |     |                          |     |                  |
|----|-------------------------|-----|--------------------------|-----|------------------|
| 75 | Chifeng Rd.             | 196 | South Xizang Rd.         | 317 | West Nanjing Rd. |
| 76 | Dabaishu                | 197 | Lujiabang Rd.            | 318 | South Shanxi Rd. |
| 77 | Jiangwan Towm           | 198 | Laoximen                 | 319 | Jiashan Rd.      |
| 78 | West Yingao Rd.         | 199 | Dashijie                 | 320 | Damuqiao Rd.     |
| 79 | South Changjiang Rd.    | 200 | People' s Square         | 321 | Mid Longhua Rd.  |
| 80 | Songfa Rd.              | 201 | Qupu Rd.                 | 322 | Longhua          |
| 81 | Zhuanghuabang           | 202 | Zhongxing Rd.            | 323 | Longcao Rd.      |
| 82 | Songbin Rd.             | 203 | Northe Xizang Rd.        | 324 | Caobao Rd.       |
| 83 | Shuichan Rd.            | 204 | Hongkou Football Stadium | 325 | Guilin Park      |
| 84 | Baoyang Rd.             | 205 | Quyong Rd.               | 326 | Hongcao Rd.      |
| 85 | Youyi Rd.               | 206 | Siping Rd.               | 327 | Hongmei Rd.      |
| 86 | Tieli Rd.               | 207 | Anshan Xincun            | 328 | Donglan Rd.      |
| 87 | North Jiangyang Rd.     | 208 | Jiangpu Rd.              | 329 | Gudai Rd.        |
| 88 | Yishan Road             | 209 | Huangxing Rd.            | 330 | Guxin Rd.        |
| 89 | Shanghai Indoor Stadium | 210 | Middle Yanji Rd.         | 331 | Qixin Rd.        |
| 90 | Shanghai Stadium        | 211 | Huangxing Park           | 332 | Jinyun Rd.       |
| 91 | Dong' an Rd.            | 212 | Xiangyin Rd.             | 333 | Jinshajiang Rd.  |

|     |                          |     |                            |     |                     |
|-----|--------------------------|-----|----------------------------|-----|---------------------|
| 92  | Damuqiao Rd.             | 213 | Nenjiang Rd.               | 334 | Fengzhuang          |
| 93  | Luban Rd.                | 214 | Shiguang Rd.               | 335 | South Qilanshan Rd. |
| 94  | South Xizang Rd.         | 215 | Middle Yanggao Road        | 336 | Zhenbei Rd.         |
| 95  | Nanpu Bridge             | 216 | Century Avenue             | 337 | Daduhe Rd.          |
| 96  | Tangqiao                 | 217 | Shangcheng Rd.             | 338 | Jinshajiang Rd.     |
| 97  | Lancun Rd.               | 218 | Xiaonanmen                 | 339 | Longde Rd.          |
| 98  | Pudian Rd.               | 219 | Lujiabang Rd.              | 340 | Wuning Rd.          |
| 99  | Century Avenue           | 220 | Madang Rd.                 | 341 | Changshou Rd.       |
| 100 | Pudong Avenue            | 221 | Dapuqiao                   | 342 | Jiangning Rd.       |
| 101 | Yangshupu Rd.            | 222 | Jiangshan Rd.              | 343 | Hanzhong Rd.        |
| 102 | Dalian Rd.               | 223 | Zhaojiabang Rd.            | 344 | Natural Museum      |
| 103 | Linping Rd.              | 224 | Xujiahui                   | 345 | West Nanjing Rd.    |
| 104 | Hailun Rd.               | 225 | Yishan Rd.                 | 346 | Middle Huaihai Rd.  |
| 105 | Baoshan Rd.              | 226 | Guilin Rd.                 | 347 | Xintiandi           |
| 106 | Shanghai Railway Station | 227 | Caohejing Development Zone | 348 | Madang Rd.          |
| 107 | Zhongtan Rd.             | 228 | Hechuan Rd.                | 349 | Expo Museum         |

|     |                  |     |                              |     |                      |
|-----|------------------|-----|------------------------------|-----|----------------------|
| 108 | Zhenping Rd.     | 229 | Xingzhong Rd.                | 350 | Expo Avenue          |
| 109 | Caoyang Rd.      | 230 | Qibao                        | 351 | Dishui Lake          |
| 110 | Jinshajiang Rd.  | 231 | Zhongchun Rd.                | 352 | Harbor Boulevard     |
| 111 | Zhongshan Park   | 232 | Jiuting                      | 353 | College              |
| 112 | West Yan' an Rd. | 233 | Sijing                       | 354 | East Huinan          |
| 113 | Hongqiao Rd.     | 234 | Sheshan                      | 355 | Huinan               |
| 114 | Xinzhuang        | 235 | Dongjing                     | 356 | Wildlife Park        |
| 115 | Chunshen Rd.     | 236 | Songjiang<br>University Town | 357 | Xinchang             |
| 116 | Yindu Rd.        | 237 | Songjiang<br>Xincheng        | 358 | East Hangtou         |
| 117 | Zhuanqiao        | 238 | Songjiang Sports<br>Center   | 359 | Hangcheng Hesha      |
| 118 | Beiqiao          | 239 | Drunken Bai Pond             | 360 | East Zhoupu          |
| 119 | Jianchuan Rd.    | 240 | South Songjiang<br>Station   | 361 | Luoshan Rd.          |
| 120 | Dongchuan Rd.    | 241 | New Jiangwan<br>Town         | 362 | Middle Huaxia<br>Rd. |
| 121 | Jinping Rd.      | 242 | East Yin' gao Rd.            | 363 | Longyang Rd.         |

The transfer station and the corresponding transfer time:

|                                |                        |      |      |      |      |       |       |
|--------------------------------|------------------------|------|------|------|------|-------|-------|
| Shanghai Railway Station       | Transferring direction | 1→3  | 3→1  | 1→4  | 4→1  | 3→4   | 4→3   |
|                                | Time needed            | 11   | 11   | 11   | 11   | 3     | 3     |
| Hanzhong Rd.                   | Transferring direction | 1→12 | 12→1 | 1→13 | 13→1 | 12→13 | 13→12 |
|                                | Time needed            | 5    | 5    | 5    | 5    | 4     | 4     |
| People' s Square               | Transferring direction | 1→2  | 2→1  | 1→8  | 8→1  | 2→8   | 8→2   |
|                                | Time needed            | 9    | 9    | 9    | 9    | 5     | 5     |
| South Shanxi Rd.               | Transferring direction | 1→10 | 10→1 | 1→12 | 12→1 | 10→12 | 12→10 |
|                                | Time needed            | 4    | 3    | 3    | 3    | 3     | 4     |
| Changshu Rd.                   | Transferring direction | 1→7  | 7→1  |      |      |       |       |
|                                | Time needed            | 5    | 4    |      |      |       |       |
| Xujiahui                       | Transferring direction | 1→9  | 9→1  | 1→11 | 11→1 | 9→11  | 11→9  |
|                                | Time needed            | 3    | 3    | 5    | 3    | 7     | 3     |
| Shanghai Indoor Stadium        | Transferring direction | 1→4  | 4→1  |      |      |       |       |
|                                | Time needed            | 5    | 3    |      |      |       |       |
| Caobao Rd.                     | Transferring direction | 1→12 | 12→1 |      |      |       |       |
|                                | Time needed            | 5    | 4    |      |      |       |       |
| Shanghai South Railway Station | Transferring direction | 1→3  | 3→1  |      |      |       |       |
|                                | Time needed            | 3    | 3    |      |      |       |       |
| Xinzhuang                      | Transferring direction | 1→5  | 5→1  |      |      |       |       |
|                                | Time needed            | 3    | 3    |      |      |       |       |
| Longyang Rd.                   | Transferring direction | 2→7  | 7→2  | 2→16 | 16→2 | 7→16  | 16→7  |
|                                | Time needed            | 14   | 10   | 5    | 5    | 6     | 6     |
| Century Avenue                 | Transferring direction | 2→4  | 4→2  | 2→6  | 6→2  | 2→9   | 9→2   |
|                                | Time needed            | 6    | 3    | 3    | 3    | 2     | 3     |
|                                | Transferring direction | 4→6  | 6→4  | 4→9  | 9→4  | 6→9   | 9→6   |
|                                | Time needed            | 3    | 5    | 3    | 5    | 2     | 3     |
| East Nanjing Rd.               | Transferring direction | 2→10 | 10→2 |      |      |       |       |
|                                | Time needed            | 5    | 3    |      |      |       |       |

|                          |                        |      |      |      |      |       |       |
|--------------------------|------------------------|------|------|------|------|-------|-------|
| West Nanjing Rd.         | Transferring direction | 2→12 | 12→2 | 2→13 | 13→2 | 12→13 | 13→12 |
|                          | Time needed            | 8    | 8    | 6    | 4    | 11    | 11    |
| Jiang' an Temple         | Transferring direction | 2→7  | 7→2  |      |      |       |       |
|                          | Time needed            | 3    | 3    |      |      |       |       |
| Jiangsu Rd.              | Transferring direction | 2→11 | 11→2 |      |      |       |       |
|                          | Time needed            | 2    | 2    |      |      |       |       |
| Zhongshan Park           | Transferring direction | 2→3  | 3→2  | 2→4  | 4→2  | 3→4   | 4→3   |
|                          | Time needed            | 3    | 3    | 4    | 3    | 2     | 2     |
| Hongqiao Terminal 2      | Transferring direction | 2→10 | 10→2 |      |      |       |       |
|                          | Time needed            | 5    | 5    |      |      |       |       |
| Hongqiao Railway Station | Transferring direction | 2→10 | 10→2 |      |      |       |       |
|                          | Time needed            | 5    | 5    |      |      |       |       |
| Hongkou Football Stadium | Transferring direction | 3→8  | 8→3  |      |      |       |       |
|                          | Time needed            | 5    | 5    |      |      |       |       |
| Baoshan Rd.              | Transferring direction | 3→4  | 4→3  |      |      |       |       |
|                          | Time needed            | 3    | 3    |      |      |       |       |
| Zhongtan Rd.             | Transferring direction | 3→4  | 4→3  |      |      |       |       |
|                          | Time needed            | 3    | 4    |      |      |       |       |
| Zhenping Rd.             | Transferring direction | 3→4  | 4→3  | 3→7  | 7→3  | 4→7   | 7→4   |
|                          | Time needed            | 6    | 4    | 4    | 4    | 4     | 6     |
| Caoyang Rd.              | Transferring direction | 3→4  | 4→3  | 3→11 | 11→3 | 4→11  | 11→4  |
|                          | Time needed            | 6    | 4    | 4    | 4    | 4     | 6     |
| Jinshajiang Rd.          | Transferring direction | 3→4  | 4→3  | 3→13 | 13→3 | 4→13  | 13→4  |
|                          | Time needed            | 6    | 4    | 4    | 4    | 4     | 6     |
| West Yan' an Rd.         | Transferring direction | 3→4  | 4→3  |      |      |       |       |
|                          | Time needed            | 6    | 6    |      |      |       |       |
| Hongqiao Rd.             | Transferring direction | 3→4  | 4→3  | 3→10 | 10→3 | 4→10  | 10→4  |
|                          | Time needed            | 7    | 5    | 5    | 5    | 5     | 7     |
| Yishan Road              | Transferring           | 3→4  | 4→3  | 3→9  | 9→3  | 4→9   | 9→4   |

|                        |                        |      |      |      |      |      |      |
|------------------------|------------------------|------|------|------|------|------|------|
|                        | direction              |      |      |      |      |      |      |
|                        | Time needed            | 6    | 5    | 5    | 5    | 5    | 6    |
| Longcao Road           | Transferring direction | 3→12 | 12→3 |      |      |      |      |
|                        | Time needed            | 5    | 5    |      |      |      |      |
| Dong' an Rd.           | Transferring direction | 4→7  | 7→4  |      |      |      |      |
|                        | Time needed            | 5    | 6    |      |      |      |      |
| Damuqiao Rd.           | Transferring direction | 4→12 | 12→4 |      |      |      |      |
|                        | Time needed            | 4    | 4    |      |      |      |      |
| South Xizang Rd.       | Transferring direction | 4→8  | 8→4  |      |      |      |      |
|                        | Time needed            | 3    | 6    |      |      |      |      |
| Lancun Rd.             | Transferring direction | 4→6  | 6→4  |      |      |      |      |
|                        | Time needed            | 5    | 8    |      |      |      |      |
| Pudian Rd.             | Transferring direction | 4→6  | 6→4  |      |      |      |      |
|                        | Time needed            | 4    | 7    |      |      |      |      |
| Dalian Rd.             | Transferring direction | 4→12 | 12→4 |      |      |      |      |
|                        | Time needed            | 4    | 6    |      |      |      |      |
| Hailun Rd.             | Transferring direction | 4→10 | 10→4 |      |      |      |      |
|                        | Time needed            | 4    | 6    |      |      |      |      |
| Oriental Sports Center | Transferring direction | 6→8  | 8→6  | 6→11 | 11→6 | 8→11 | 11→8 |
|                        | Time needed            | 4    | 4    | 4    | 4    | 4    | 4    |
| West Gaoke Rd.         | Transferring direction | 6→7  | 7→6  |      |      |      |      |
|                        | Time needed            | 4    | 4    |      |      |      |      |
| Jufeng Rd.             | Transferring direction | 6→12 | 12→6 |      |      |      |      |
|                        | Time needed            | 9    | 5    |      |      |      |      |
| Yaohua Rd.             | Transferring direction | 7→8  | 8→7  |      |      |      |      |
|                        | Time needed            | 4    | 4    |      |      |      |      |
| Mid Longhua Rd.        | Transferring direction | 7→12 | 12→7 |      |      |      |      |
|                        | Time needed            | 4    | 4    |      |      |      |      |
| Zhaojiabang Rd.        | Transferring direction | 7→9  | 9→7  |      |      |      |      |



|                     |                        |       |       |  |  |  |  |
|---------------------|------------------------|-------|-------|--|--|--|--|
|                     | Time needed            | 4     | 5     |  |  |  |  |
| Changshou Rd.       | Transferring direction | 7→13  | 13→7  |  |  |  |  |
|                     | Time needed            | 4     | 4     |  |  |  |  |
| Lujiabang Rd.       | Transferring direction | 8→9   | 9→8   |  |  |  |  |
|                     | Time needed            | 3     | 3     |  |  |  |  |
| Laoximen            | Transferring direction | 8→10  | 10→8  |  |  |  |  |
|                     | Time needed            | 5     | 5     |  |  |  |  |
| Qufu Rd.            | Transferring direction | 8→12  | 12→8  |  |  |  |  |
|                     | Time needed            | 4     | 4     |  |  |  |  |
| Sipin Rd.           | Transferring direction | 8→10  | 10→8  |  |  |  |  |
|                     | Time needed            | 4     | 4     |  |  |  |  |
| Jiashan Rd.         | Transferring direction | 9→12  | 12→9  |  |  |  |  |
|                     | Time needed            | 5     | 3     |  |  |  |  |
| Madang Rd.          | Transferring direction | 9→13  | 13→9  |  |  |  |  |
|                     | Time needed            | 8     | 5     |  |  |  |  |
| Tiantong Rd.        | Transferring direction | 10→12 | 12→10 |  |  |  |  |
|                     | Time needed            | 5     | 6     |  |  |  |  |
| Xintiandi           | Transferring direction | 10→13 | 13→10 |  |  |  |  |
|                     | Time needed            | 6     | 4     |  |  |  |  |
| Jiaotong University | Transferring direction | 10→11 | 11→10 |  |  |  |  |
|                     | Time needed            | 6     | 5     |  |  |  |  |
| Longde Rd.          | Transferring direction | 11→13 | 13→11 |  |  |  |  |
|                     | Time needed            | 3     | 5     |  |  |  |  |
| Longhua             | Transferring direction | 11→12 | 12→11 |  |  |  |  |
|                     | Time needed            | 3     | 5     |  |  |  |  |
| Luoshan Rd.         | Transferring direction | 11→16 | 16→11 |  |  |  |  |
|                     | Time needed            | 3     | 5     |  |  |  |  |

The station inside the inner ring and the probability of being chosen as the terminal station.

|                          | The probability of the passengers outside the inner ring getting off in the inner ring |
|--------------------------|----------------------------------------------------------------------------------------|
| Dong'an Rd.              | 2.0819%                                                                                |
| Zhenping Rd.             | 2.4539%                                                                                |
| Changshou Rd.            | 2.0545%                                                                                |
| Changpin Rd.             | 4.2426%                                                                                |
| Jing'an Temple           | 2.6025%                                                                                |
| Changshu Road            | 2.2772%                                                                                |
| Zhaojiabang Rd.          | 2.0819%                                                                                |
| West Nanjing Rd.         | 3.4387%                                                                                |
| South Shanxi Rd.         | 3.2382%                                                                                |
| Dapuqiao                 | 2.4539%                                                                                |
| Shanghai Railway Station | 3.8406%                                                                                |
| Hanzhong Rd.             | 2.1228%                                                                                |
| People's Square          | 6.3153%                                                                                |
| South Huangpi Rd.        | 2.4539%                                                                                |
| Lujiabang Rd.            | 2.2788%                                                                                |
| South Xizang Rd.         | 2.0954%                                                                                |
| Qufu Rd.                 | 2.0819%                                                                                |
| East Nanjing Rd.         | 4.2426%                                                                                |
| Yuyuan                   | 2.0954%                                                                                |
| Lujiazui                 | 6.3153%                                                                                |
| Dongchang Rd.            | 2.7260%                                                                                |
| Shangcheng Rd.           | 2.4790%                                                                                |
| Pudong Avenue            | 2.2554%                                                                                |
| Century Avenue           | 3.2424%                                                                                |
| Pudian Rd                | 3.0462%                                                                                |

The station outside the inner ring and its corresponding passenger flow volume.

|                                | Passenger flows of each station outside the inner ring (person-time/day) |                          | Passenger flows of each station outside the inner ring (person-time/day) |
|--------------------------------|--------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------|
| East Xujingdong                | 5885.75                                                                  | No.3 Dahua Rd.           | 9645.5                                                                   |
| Hongqiao Railway Station       | 10906.25                                                                 | Xingzhi Rd.              | 5549                                                                     |
| Shonghong Rd.                  | 12895                                                                    | Zhongtan Rd.             | 7732                                                                     |
| Beixinjing                     | 8213.75                                                                  | Shanghai Railway Station | 13589.25                                                                 |
| Loushanguan Rd.                | 9208.5                                                                   | Yanchang Rd.             | 7414.25                                                                  |
| Zhongshan Park                 | 17256.25                                                                 | Pengpu Town              | 12895                                                                    |
| Jinshajiang Rd.                | 9208.5                                                                   | Gongkang Rd.             | 8871.75                                                                  |
| Jinyun Rd.                     | 5183                                                                     | Tonghe Towm              | 12328.5                                                                  |
| Yishan Rd.                     | 5549                                                                     | Hongkou Football Stadium | 9072.25                                                                  |
| Qibao                          | 11472.75                                                                 | Jiangwan Town            | 8434.75                                                                  |
| Jiuting                        | 17256.25                                                                 | West Yingao Rd.          | 5183                                                                     |
| Sijin                          | 5549                                                                     | South Changjiang Rd.     | 5086.25                                                                  |
| Sheshan                        | 5549                                                                     | Century Avenue           | 10212                                                                    |
| Songjiang University Town      | 5183                                                                     | Century Park             | 7414.25                                                                  |
| Shanghai indoor Stadium        | 5183                                                                     | Dalian Rd.               | 5086.25                                                                  |
| Caobao Rd.                     | 7414.25                                                                  | Fanghua Rd.              | 7635.25                                                                  |
| Shanghai South Railway Station | 18678.5                                                                  | Jinxiu Rd.               | 5549                                                                     |
| Jingjiang Park                 | 9208.5                                                                   | South Yanggao Rd.        | 5183                                                                     |
| Lianhua Rd.                    | 18678.5                                                                  | Yaohua Rd.               | 5183                                                                     |
| Waihuan Rd.                    | 5328                                                                     | Chengshan Rd.            | 8505.75                                                                  |
| Xinzhuang                      | 22345.5                                                                  | Yangsi                   | 7635.25                                                                  |
| Caoyang Rd.                    | 10778.5                                                                  | Aerospace Museum         | 8409                                                                     |
| Wuwei Rd.                      | 5183                                                                     | Longyang Rd.             | 13589.25                                                                 |
| Taopu Town                     | 7414.25                                                                  | Gangcheng Rd.            | 5549                                                                     |
| Zhenping Rd.                   | 10212                                                                    | Guanglan Rd.             | 11891.5                                                                  |
| Xincun Rd.                     | 7877                                                                     | Tang Town                | 5086.25                                                                  |

## Appendix X:

The source code 1 of MATLAB:

```
clc, clear
a=zeros(363);
a(1,2)=2;a(2,3)=2;a(3,4)=3;a(4,5)=3;a(5,6)=3;a(6,7)=3;a(7,8)=2;a(8,9)=2;
a(9,10)=2;a(10,11)=2;a(11,12)=2;a(12,13)=3;a(13,14)=2;a(14,15)=2;
a(15,16)=3;a(16,17)=2;a(17,18)=3;a(18,19)=2;a(19,20)=2;a(20,21)=3;
a(21,22)=3;a(22,23)=2;a(23,24)=2;a(24,25)=3;a(25,26)=3;a(26,27)=2;
a(27,28)=2;
%一号线各临界站所需时间
a(29,30)=3;a(30,31)=7;a(31,32)=5;a(32,33)=3;a(33,34)=5;a(34,35)=3;
a(35,36)=3;a(36,37)=16;a(37,38)=3;a(38,39)=2;a(39,40)=4;a(40,41)=2;
a(41,42)=2;a(42,43)=3;a(43,44)=2;a(44,45)=3;a(45,46)=2;a(46,47)=3;
a(47,48)=2;a(48,49)=3;a(49,50)=2;a(50,51)=3;a(51,52)=3;a(52,53)=2;
a(53,54)=3;a(54,55)=3;a(55,56)=2;a(56,57)=3;a(57,58)=3;
%二号线各临界站所需时间
a(59,60)=2;a(60,61)=3;a(61,62)=2;a(62,63)=4;a(63,64)=2;a(64,65)=3;
a(65,66)=2;a(66,67)=2;a(67,68)=2;a(68,69)=3;a(69,70)=2;a(70,71)=3;
a(71,72)=3;a(72,73)=2;a(73,74)=3;a(74,75)=2;a(75,76)=2;a(76,77)=3;
a(77,78)=2;a(78,79)=1;a(79,80)=2;a(80,81)=2;a(81,82)=3;a(82,83)=2;
a(83,84)=5;a(84,85)=2;a(85,86)=2;a(86,87)=2;
%三号线各临界站所需时间
a(88,89)=2;a(89,90)=2;a(90,91)=2;a(91,92)=1;a(92,93)=2;a(93,94)=3;
a(94,95)=2;a(95,96)=3;a(96,97)=2;a(97,98)=2;a(98,99)=2;a(99,100)=3;
a(100,101)=2;a(101,102)=2;a(102,103)=2;a(103,104)=2;a(104,105)=3;
a(105,106)=3;a(106,107)=3;a(107,108)=2;a(108,109)=3;a(109,110)=1;
a(110,111)=3;a(111,112)=2;a(112,113)=2;
%四号线各临界站所需时间
a(114,115)=2;a(115,116)=4;a(116,117)=3;a(117,118)=3;a(118,119)=2;
a(119,120)=3;a(120,121)=2;a(121,122)=3;a(122,123)=3;a(123,124)=2;
%五号线各临界站所需时间
a(125,126)=3;a(126,127)=3;a(127,128)=3;a(128,129)=2;a(129,130)=2;
a(130,131)=2;a(131,132)=2;a(132,133)=3;a(133,134)=2;a(134,135)=2;
a(135,136)=3;a(136,137)=2;a(137,138)=2;a(138,139)=3;a(139,140)=2;
a(140,141)=3;a(141,142)=2;a(142,143)=3;a(143,144)=2;a(144,145)=3;
a(145,146)=2;a(146,147)=3;a(147,148)=2;a(148,149)=2;a(149,150)=2;
a(150,151)=2;a(151,152)=2;
%六号线各临界站所需时间
a(153,154)=3;a(154,155)=2;a(155,156)=3;a(156,157)=2;a(157,158)=3;
a(158,159)=2;a(159,160)=2;a(160,161)=2;a(161,162)=2;a(162,163)=3;
a(163,164)=2;a(164,165)=2;a(165,166)=3;a(166,167)=2;a(167,168)=3;
a(168,169)=2;a(169,170)=2;a(170,171)=3;a(171,172)=2;a(172,173)=2;
a(173,174)=2;a(174,175)=2;a(175,176)=3;a(176,177)=2;a(177,178)=2;
a(178,179)=3;a(179,180)=6;a(180,181)=3;a(181,182)=2;a(182,183)=4;
a(183,184)=2;
```

%七号线各临界站所需时间

a(185,186)=3;a(186,187)=2;a(187,188)=3;a(188,189)=2;a(189,190)=3;  
a(190,191)=2;a(191,192)=2;a(192,193)=2;a(193,194)=2;a(194,195)=3;  
a(195,196)=2;a(196,197)=2;a(197,198)=3;a(198,199)=2;a(199,200)=3;  
a(200,201)=2;a(201,202)=2;a(202,203)=3;a(203,204)=2;a(204,205)=2;  
a(205,206)=2;a(206,207)=2;a(207,208)=3;a(208,209)=2;a(209,210)=2;  
a(210,211)=3;a(211,212)=6;a(212,213)=3;a(213,214)=2;

%八号线各临界站所需时间

a(215,216)=4;a(216,217)=2;a(217,218)=4;a(218,219)=3;a(219,220)=2;  
a(220,221)=2;a(221,222)=2;a(222,223)=2;a(223,224)=3;a(224,225)=3;  
a(225,226)=3;a(226,227)=3;a(227,228)=2;a(228,229)=3;a(229,230)=3;  
a(230,231)=3;a(231,232)=4;a(232,233)=6;a(233,234)=5;a(234,235)=3;  
a(235,236)=4;a(236,237)=4;a(237,238)=3;a(238,239)=3;a(239,240)=4;

%九号线各临界站所需时间

a(241,242)=1;a(242,243)=2;a(243,244)=3;a(244,245)=2;a(245,246)=2;  
a(246,247)=2;a(247,248)=2;a(248,249)=2;a(249,250)=2;a(250,251)=3;  
a(251,252)=2;a(252,253)=2;a(253,254)=3;a(254,255)=2;a(255,256)=3;  
a(256,257)=3;a(257,258)=2;a(258,259)=3;a(259,260)=2;a(260,261)=2;  
a(261,262)=3;a(262,263)=2;a(263,264)=2;a(264,265)=2;a(265,266)=3;  
a(266,267)=4;a(267,268)=1;

%十号线各临界站所需时间

a(269,270)=4;a(270,271)=3;a(271,272)=3;a(272,273)=3;a(273,274)=4;  
a(274,275)=3;a(275,276)=2;a(276,277)=5;a(277,278)=3;a(278,279)=2;  
a(279,280)=2;a(280,281)=2;a(281,282)=4;a(282,283)=2;a(283,284)=4;  
a(284,285)=2;a(285,286)=3;a(286,287)=1;a(287,288)=3;a(288,289)=2;  
a(289,290)=3;a(290,291)=2;a(291,292)=2;a(292,293)=3;a(293,294)=4;  
a(294,295)=5;a(295,296)=4;a(296,297)=2;a(297,298)=4;a(298,299)=3;

%十一号线各临界站所需时间

a(300,301)=4;a(301,302)=2;a(302,303)=3;a(303,304)=2;a(304,305)=3;  
a(305,306)=2;a(306,307)=2;a(307,308)=3;a(308,309)=2;a(309,310)=2;  
a(310,311)=2;a(311,312)=2;a(312,313)=3;a(313,314)=3;a(314,315)=3;  
a(315,316)=3;a(316,317)=3;a(317,318)=2;a(318,319)=3;a(319,320)=2;  
a(320,321)=3;a(321,322)=3;a(322,323)=2;a(323,324)=2;a(324,325)=2;  
a(325,326)=3;a(326,327)=2;a(327,328)=3;a(328,329)=2;a(329,330)=2;  
a(330,331)=2;

%十二号线各临界站所需时间

a(332,333)=2;a(333,334)=4;a(334,335)=3;a(335,336)=2;a(336,337)=3;  
a(337,338)=4;a(338,339)=2;a(339,340)=3;a(340,341)=2;a(341,342)=2;  
a(342,343)=3;a(343,344)=1;a(344,345)=3;a(345,346)=2;a(346,347)=3;  
a(347,348)=2;a(348,349)=2;a(349,350)=3;

%十三号线各临界站所需时间

a(351,352)=3;a(352,353)=6;a(353,354)=10;a(354,355)=5;a(355,356)=8;  
a(356,357)=7;a(357,358)=6;a(358,359)=4;a(359,360)=6;a(360,361)=7;  
a(361,362)=4;a(362,363)=5;

%十六号线各临界站所需时间

$a(16,71)=0; a(16,106)=0; a(71,106)=0;$   
 $a(71,16)=0; a(106,16)=0; a(106,71)=0; \%1. \text{上海火车站} (1,3,4)$   
 $a(15,316)=0; a(15,343)=0; a(316,343)=0;$   
 $a(316,15)=0; a(343,15)=0; a(343,316)=0; \%2. \text{汉中路} (1,12,13)$   
 $a(13,47)=0; a(13,200)=0; a(47,200)=0;$   
 $a(47,13)=0; a(200,13)=0; a(200,47)=0; \%3. \text{人民广场} (1,2,8)$   
 $a(11,257)=0; a(11,318)=0; a(257,318)=0;$   
 $a(257,11)=0; a(318,11)=0; a(318,257)=0; \%4. \text{陕西南路} (1,10,12)$   
 $a(10,166)=0; a(166,10)=0; \%5. \text{常熟路} (1,7)$   
 $a(8,224)=0; a(8,282)=0; a(224,282)=0;$   
 $a(224,8)=0; a(282,8)=0; a(282,224)=0; \%6. \text{徐家汇} (1,9,11)$   
 $a(7,89)=0; a(89,7)=0; \%7. \text{上海体育馆} (1,4)$   
 $a(6,324)=0; a(324,6)=0; \%8. \text{漕宝路} (1,12)$   
 $a(5,59)=0; a(59,5)=0; \%9. \text{上海南站} (1,3)$   
 $a(1,114)=0; a(114,1)=0; \%10. \text{莘庄} (1,5)$   
 $a(40,154)=0; a(40,363)=0; a(154,363)=0;$   
 $a(154,40)=0; a(363,40)=0; a(363,154)=0; \%11. \text{龙阳路} (2,7,16)$   
 $a(43,99)=0; a(43,141)=0; a(43,216)=0;$   
 $a(99,141)=0; a(99,216)=0; a(141,216)=0;$   
 $a(99,43)=0; a(141,43)=0; a(216,43)=0;$   
 $a(141,99)=0; a(216,99)=0; a(216,141)=0; \%12. \text{世纪大道} (2,4,6,9)$   
 $a(46,253)=0; a(253,46)=0; \%13. \text{南京东路} (2,10)$   
 $a(48,317)=0; a(48,345)=0; a(317,345)=0;$   
 $a(317,48)=0; a(345,48)=0; a(345,317)=0; \%14. \text{南京西路} (2,12,13)$   
 $a(49,167)=0; a(167,49)=0; \%15. \text{静安寺} (2,7)$   
 $a(50,284)=0; a(284,50)=0; \%16. \text{江苏路} (2,11)$   
 $a(51,66)=0; a(51,111)=0; a(66,111)=0;$   
 $a(66,51)=0; a(111,51)=0; a(111,66)=0; \%17. \text{中山公园} (2,3,4)$   
 $a(56,267)=0; a(267,56)=0; \%18. \text{虹桥2号航站楼} (2,10)$   
 $a(57,268)=0; a(268,57)=0; \%19. \text{虹桥火车站} (2,10)$   
 $a(74,204)=0; a(204,74)=0; \%20. \text{虹口足球场} (3,8)$   
 $a(72,105)=0; a(105,72)=0; \%21. \text{宝山路} (3,4)$   
 $a(70,107)=0; a(107,70)=0; \%22. \text{中潭路} (3,4)$   
 $a(69,108)=0; a(69,170)=0; a(108,170)=0;$   
 $a(108,69)=0; a(170,69)=0; a(170,108)=0; \%23. \text{镇坪路} (3,4,7)$   
 $a(68,109)=0; a(68,286)=0; a(109,286)=0;$   
 $a(109,68)=0; a(286,68)=0; a(286,109)=0; \%24. \text{曹杨路} (3,4,11)$   
 $a(67,110)=0; a(67,333)=0; a(110,333)=0;$   
 $a(110,67)=0; a(333,67)=0; a(333,110)=0; \%25. \text{金沙江路} (3,4,13)$   
 $a(65,112)=0; a(112,65)=0; \%26. \text{延安西路} (3,4)$   
 $a(64,113)=0; a(64,260)=0; a(113,260)=0;$   
 $a(113,64)=0; a(260,64)=0; a(260,113)=0; \%27. \text{虹桥路} (3,4,10)$   
 $a(63,88)=0; a(63,225)=0; a(88,225)=0;$   
 $a(88,63)=0; a(225,63)=0; a(225,88)=0; \%28. \text{宜山路} (3,4,9)$   
 $a(61,323)=0; a(323,61)=0; \%29. \text{龙漕路} (3,12)$

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a(91,164)=0;a(164,91)=0;%30.东安路(4,7)
a(92,320)=0;a(320,92)=0;%31.大木桥路(4,12)
a(94,196)=0;a(196,94)=0;%32.西藏南路(4,8)
a(97,143)=0;a(143,97)=0;%33.蓝村路(4,6)
a(98,142)=0;a(142,98)=0;%34.浦电路(4,6)
a(102,311)=0;a(311,102)=0;%35.大连路(4,12)
a(104,250)=0;a(250,104)=0;%36.海伦路(4,10)
a(152,191)=0;a(152,277)=0;a(191,277)=0;
a(191,152)=0;a(277,152)=0;a(277,191)=0;%37.东方体育中心(6,8,11)
a(146,158)=0;a(158,146)=0;%38.高科西路 (6,7)
a(132,304)=0;a(304,132)=0;%39.巨峰路 (6,12)
a(160,194)=0;a(194,160)=0;%40.耀华路 (7,8)
a(163,321)=0;a(321,163)=0;%41.龙华中路 (7,12)
a(165,223)=0;a(223,165)=0;%42.肇嘉浜路 (7,9)
a(169,341)=0;a(341,169)=0;%43.长寿路 (7,13)
a(197,219)=0;a(219,197)=0;%44.陆家浜路 (8,9)
a(198,255)=0;a(255,198)=0;%45.老西门 (8,10)
a(201,315)=0;a(315,201)=0;%46.曲阜路 (8,12)
a(206,248)=0;a(248,206)=0;%47.四平路 (8,10)
a(222,319)=0;a(319,222)=0;%48.嘉善路 (9,12)
a(220,348)=0;a(348,220)=0;%49.马当路 (9,13)
a(252,314)=0;a(314,252)=0;%50.天潼路 (10,12)
a(256,347)=0;a(347,256)=0;%51.新天地 (10,13)
a(259,283)=0;a(283,259)=0;%52.交通大学 (10,11)
a(285,339)=0;a(339,285)=0;%53.隆德路 (11,13)
a(280,322)=0;a(322,280)=0;%54.龙华 (11,12)
a(272,361)=0;a(361,272)=0;%55.罗山路 (11,16)
%补充相同站不同线
c=a+a';%实现双向通行
%换乘点如下: (换乘时间不为定值)
c(16,71)=11;c(16,106)=11;c(71,106)=3;
c(71,16)=11;c(106,16)=11;c(106,71)=3;%1.上海火车站 (1,3,4)
c(15,316)=5;c(15,343)=5;c(316,343)=4;
c(316,15)=5;c(343,15)=5;c(343,316)=4;%2.汉中路 (1,12,13)
c(13,47)=9;c(13,200)=9;c(47,200)=5;
c(47,13)=9;c(200,13)=9;c(200,47)=5;%3.人民广场 (1,2,8)
c(11,257)=4;c(11,318)=3;c(257,318)=3;
c(257,11)=3;c(318,11)=3;c(318,257)=4;%4.陕西南路 (1,10,12)
c(10,166)=5;c(166,10)=4;%5.常熟路 (1,7)
c(8,224)=3;c(8,282)=5;c(224,282)=7;
c(224,8)=3;c(282,8)=3;c(282,224)=3;%6.徐家汇 (1,9,11)
c(7,89)=5;c(89,7)=3;%7.上海体育馆 (1,4)
c(6,324)=5;c(324,6)=4;%8.漕宝路 (1,12)
c(5,59)=3;c(59,5)=3;%9.上海南站 (1,3)
c(1,114)=3;c(114,1)=3;%10.莘庄 (1,5)

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c(40,154)=14;c(40,363)=5;c(154,363)=6;  
 c(154,40)=10;c(363,40)=5;c(363,154)=6;%11.龙阳路 (2,7,16)  
 c(43,99)=6;c(43,141)=3;c(43,216)=2;  
 c(99,141)=3;c(99,216)=3;c(141,216)=2;  
 c(99,43)=3;c(141,43)=3;c(216,43)=3;  
 c(141,99)=5;c(216,99)=5;c(216,141)=3;%12.世纪大道 (2,4,6,9)  
 c(46,253)=5;c(253,46)=3;%13.南京东路 (2,10)  
 c(48,317)=8;c(48,345)=6;c(317,345)=11;  
 c(317,48)=8;c(345,48)=4;c(345,317)=11;%14.南京西路 (2,12,13)  
 c(49,167)=3;c(167,49)=3;%15.静安寺(2,7)  
 c(50,284)=2;c(284,50)=2;%16.江苏路(2,11)  
 c(51,66)=3;c(51,111)=4;c(66,111)=2;  
 c(66,51)=3;c(111,51)=3;c(111,66)=2;%17.中山公园(2,3,4)  
 c(56,267)=5;c(267,56)=5;%18.虹桥 2 号航站楼(2,10)  
 c(57,268)=5;c(268,57)=5;%19.虹桥火车站(2,10)  
 c(74,204)=5;c(204,74)=5;%20.虹口足球场(3,8)  
 c(72,105)=3;c(105,72)=3;%21.宝山路(3,4)  
 c(70,107)=3;c(107,70)=4;%22.中潭路(3,4)  
 c(69,108)=6;c(69,170)=4;c(108,170)=4;  
 c(108,69)=4;c(170,69)=4;c(170,108)=6;%23.镇坪路(3,4,7)  
 c(68,109)=6;c(68,286)=4;c(109,286)=4;  
 c(109,68)=4;c(286,68)=4;c(286,109)=6;%24.曹杨路(3,4,11)  
 c(67,110)=6;c(67,333)=4;c(110,333)=4;  
 c(110,67)=4;c(333,67)=4;c(333,110)=6;%25.金沙江路(3,4,13)  
 c(65,112)=6;c(112,65)=6;%26.延安西路(3,4)  
 c(64,113)=7;c(64,260)=5;c(113,260)=5;  
 c(113,64)=5;c(260,64)=5;c(260,113)=7;%27.虹桥路(3,4,10)  
 c(63,88)=6;c(63,225)=5;c(88,225)=5;  
 c(88,63)=5;c(225,63)=5;c(225,88)=6;%28.宜山路(3,4,9)  
 c(61,323)=5;c(323,61)=5;%29.龙漕路(3,12)  
 c(91,164)=5;c(164,91)=6;%30.东安路(4,7)  
 c(92,320)=4;c(320,92)=4;%31.大木桥路(4,12)  
 c(94,196)=3;c(196,94)=6;%32.西藏南路(4,8)  
 c(97,143)=5;c(143,97)=8;%33.蓝村路(4,6)  
 c(98,142)=4;c(142,98)=7;%34.浦电路(4,6)  
 c(102,311)=4;c(311,102)=6;%35.大连路(4,12)  
 c(104,250)=4;c(250,104)=6;%36.海伦路(4,10)  
 c(152,191)=4;c(152,277)=4;c(191,277)=4;  
 c(191,152)=4;c(277,152)=4;c(277,191)=4;%37.东方体育中心(6,8,11)  
 c(146,158)=4;c(158,146)=4;%38.高科西路 (6,7)  
 c(132,304)=9;c(304,132)=5;%39.巨峰路 (6,12)  
 c(160,194)=4;c(194,160)=4;%40.耀华路 (7,8)  
 c(163,321)=4;c(321,163)=4;%41.龙华中路 (7,12)  
 c(165,223)=4;c(223,165)=5;%42.肇嘉浜路 (7,9)  
 c(169,341)=4;c(341,169)=4;%43.长寿路 (7,13)



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c(197,219)=3;c(219,197)=3;%44.陆家浜路 (8,9)
c(198,255)=5;c(255,198)=5;%45.老西门 (8,10)
c(201,315)=4;c(315,201)=4;%46.曲阜路 (8,12)
c(206,248)=4;c(248,206)=4;%47.四平路 (8,10)
c(222,319)=5;c(319,222)=3;%48.嘉善路 (9,12)
c(220,348)=8;c(348,220)=5;%49.马当路 (9,13)
c(252,314)=5;c(314,252)=6;%50.天潼路 (10,12)
c(256,347)=6;c(347,256)=4;%51.新天地 (10,13)
c(259,283)=6;c(283,259)=5;%52.交通大学 (10,11)
c(285,339)=3;c(339,285)=5;%53.隆德路 (11,13)
c(280,322)=3;c(322,280)=5;%54.龙华 (11,12)
c(272,361)=3;c(361,272)=5;%55.罗山路 (11,16)
b=sparse(c); %构造稀疏矩阵 b
n=363;
for i=1:n
    for j=1:n
[x,y,z]=graphshortestpath(b,i,j,'Directed',true,'Method','Dijkstra'); % Directed 是有向的,此处走
的是点 i 到 j 的最短时间
%换乘引起的问题值只会发生在起点和终点上
timeall(i,j)=x;%把最短时间 x 存储到 timeall 数组中
number(i,j)=length(y)-1;%计算共通过多少站路 (包括换乘, 换乘多算一站,去掉起点)
if length(y)>1
p=y(2);m=length(y)-1;q=y(m);
if a(i,p)==0
    number(i,j)=number(i,j)-1;
    timeall(i,j)=timeall(i,j)-a(i,p);
end
if a(q,j)==0
    number(i,j)=number(i,j)-1;
    timeall(i,j)=timeall(i,j)-a(q,j);
end
end
%计算节约的时间即为换乘时间和停靠时间,设每站停靠时间为 1 分钟,停站数为 number(i,j)
tsave(i,j)=number(i,j);
trail=length(y)-1;
for k=1:trail
    s1=y(k);s2=y(k+1);
if abs(s1-s2)>1
    tsave(i,j)=tsave(i,j)+c(s1,s2);
end
end
end
end
%h=view(biograph(b,[],'ShowArrows','on','ShowWeights','on'))
%目标通过循环跑出任意两点间的最短时间,并进行存储, 此处可以根据原线路跑也可以根

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据直达线路跑
%对 tsave 矩阵中所有元素比大小
for i=1:363
    for j=1:363
        z=i+363*(j-1);
        daxiao(z)=tsave(i,j);
    end
end
[paixu,weizhi]=sort(daxiao);
paixu=fliplr(paixu);
weizhi=fliplr(weizhi);
px=paixu(1:10)%选出最大的节省时间
wz=weizhi(1:10);
v=floor(wz/363)+1%终点
w=wz-(floor(wz/363)*363)%起点
%最大时间所对应的起点与终点

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The source code 2 of MATLAB:

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clc, clear
a=zeros(363);
a(1,2)=2;a(2,3)=2;a(3,4)=3;a(4,5)=3;a(5,6)=3;a(6,7)=3;a(7,8)=2;a(8,9)=2;
a(9,10)=2;a(10,11)=2;a(11,12)=2;a(12,13)=3;a(13,14)=2;a(14,15)=2;
a(15,16)=3;a(16,17)=2;a(17,18)=3;a(18,19)=2;a(19,20)=2;a(20,21)=3;
a(21,22)=3;a(22,23)=2;a(23,24)=2;a(24,25)=3;a(25,26)=3;a(26,27)=2;
a(27,28)=2;
%一号线各临界站所需时间
a(29,30)=3;a(30,31)=7;a(31,32)=5;a(32,33)=3;a(33,34)=5;a(34,35)=3;
a(35,36)=3;a(36,37)=16;a(37,38)=3;a(38,39)=2;a(39,40)=4;a(40,41)=2;
a(41,42)=2;a(42,43)=3;a(43,44)=2;a(44,45)=3;a(45,46)=2;a(46,47)=3;
a(47,48)=2;a(48,49)=3;a(49,50)=2;a(50,51)=3;a(51,52)=3;a(52,53)=2;
a(53,54)=3;a(54,55)=3;a(55,56)=2;a(56,57)=3;a(57,58)=3;
%二号线各临界站所需时间
a(59,60)=2;a(60,61)=3;a(61,62)=2;a(62,63)=4;a(63,64)=2;a(64,65)=3;
a(65,66)=2;a(66,67)=2;a(67,68)=2;a(68,69)=3;a(69,70)=2;a(70,71)=3;
a(71,72)=3;a(72,73)=2;a(73,74)=3;a(74,75)=2;a(75,76)=2;a(76,77)=3;
a(77,78)=2;a(78,79)=1;a(79,80)=2;a(80,81)=2;a(81,82)=3;a(82,83)=2;
a(83,84)=5;a(84,85)=2;a(85,86)=2;a(86,87)=2;
%三号线各临界站所需时间
a(88,89)=2;a(89,90)=2;a(90,91)=2;a(91,92)=1;a(92,93)=2;a(93,94)=3;
a(94,95)=2;a(95,96)=3;a(96,97)=2;a(97,98)=2;a(98,99)=2;a(99,100)=3;
a(100,101)=2;a(101,102)=2;a(102,103)=2;a(103,104)=2;a(104,105)=3;
a(105,106)=3;a(106,107)=3;a(107,108)=2;a(108,109)=3;a(109,110)=1;
a(110,111)=3;a(111,112)=2;a(112,113)=2;
%四号线各临界站所需时间
a(114,115)=2;a(115,116)=4;a(116,117)=3;a(117,118)=3;a(118,119)=2;

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$a(119,120)=3;a(120,121)=2;a(121,122)=3;a(122,123)=3;a(123,124)=2;$   
 %五号线各临界站所需时间  
 $a(125,126)=3;a(126,127)=3;a(127,128)=3;a(128,129)=2;a(129,130)=2;$   
 $a(130,131)=2;a(131,132)=2;a(132,133)=3;a(133,134)=2;a(134,135)=2;$   
 $a(135,136)=3;a(136,137)=2;a(137,138)=2;a(138,139)=3;a(139,140)=2;$   
 $a(140,141)=3;a(141,142)=2;a(142,143)=3;a(143,144)=2;a(144,145)=3;$   
 $a(145,146)=2;a(146,147)=3;a(147,148)=2;a(148,149)=2;a(149,150)=2;$   
 $a(150,151)=2;a(151,152)=2;$   
 %六号线各临界站所需时间  
 $a(153,154)=3;a(154,155)=2;a(155,156)=3;a(156,157)=2;a(157,158)=3;$   
 $a(158,159)=2;a(159,160)=2;a(160,161)=2;a(161,162)=2;a(162,163)=3;$   
 $a(163,164)=2;a(164,165)=2;a(165,166)=3;a(166,167)=2;a(167,168)=3;$   
 $a(168,169)=2;a(169,170)=2;a(170,171)=3;a(171,172)=2;a(172,173)=2;$   
 $a(173,174)=2;a(174,175)=2;a(175,176)=3;a(176,177)=2;a(177,178)=2;$   
 $a(178,179)=3;a(179,180)=6;a(180,181)=3;a(181,182)=2;a(182,183)=4;$   
 $a(183,184)=2;$   
 %七号线各临界站所需时间  
 $a(185,186)=3;a(186,187)=2;a(187,188)=3;a(188,189)=2;a(189,190)=3;$   
 $a(190,191)=2;a(191,192)=2;a(192,193)=2;a(193,194)=2;a(194,195)=3;$   
 $a(195,196)=2;a(196,197)=2;a(197,198)=3;a(198,199)=2;a(199,200)=3;$   
 $a(200,201)=2;a(201,202)=2;a(202,203)=3;a(203,204)=2;a(204,205)=2;$   
 $a(205,206)=2;a(206,207)=2;a(207,208)=3;a(208,209)=2;a(209,210)=2;$   
 $a(210,211)=3;a(211,212)=6;a(212,213)=3;a(213,214)=2;$   
 %八号线各临界站所需时间  
 $a(215,216)=4;a(216,217)=2;a(217,218)=4;a(218,219)=3;a(219,220)=2;$   
 $a(220,221)=2;a(221,222)=2;a(222,223)=2;a(223,224)=3;a(224,225)=3;$   
 $a(225,226)=3;a(226,227)=3;a(227,228)=2;a(228,229)=3;a(229,230)=3;$   
 $a(230,231)=3;a(231,232)=4;a(232,233)=6;a(233,234)=5;a(234,235)=3;$   
 $a(235,236)=4;a(236,237)=4;a(237,238)=3;a(238,239)=3;a(239,240)=4;$   
 %九号线各临界站所需时间  
 $a(241,242)=1;a(242,243)=2;a(243,244)=3;a(244,245)=2;a(245,246)=2;$   
 $a(246,247)=2;a(247,248)=2;a(248,249)=2;a(249,250)=2;a(250,251)=3;$   
 $a(251,252)=2;a(252,253)=2;a(253,254)=3;a(254,255)=2;a(255,256)=3;$   
 $a(256,257)=3;a(257,258)=2;a(258,259)=3;a(259,260)=2;a(260,261)=2;$   
 $a(261,262)=3;a(262,263)=2;a(263,264)=2;a(264,265)=2;a(265,266)=3;$   
 $a(266,267)=4;a(267,268)=1;$   
 %十号线各临界站所需时间  
 $a(269,270)=4;a(270,271)=3;a(271,272)=3;a(272,273)=3;a(273,274)=4;$   
 $a(274,275)=3;a(275,276)=2;a(276,277)=5;a(277,278)=3;a(278,279)=2;$   
 $a(279,280)=2;a(280,281)=2;a(281,282)=4;a(282,283)=2;a(283,284)=4;$   
 $a(284,285)=2;a(285,286)=3;a(286,287)=1;a(287,288)=3;a(288,289)=2;$   
 $a(289,290)=3;a(290,291)=2;a(291,292)=2;a(292,293)=3;a(293,294)=4;$   
 $a(294,295)=5;a(295,296)=4;a(296,297)=2;a(297,298)=4;a(298,299)=3;$   
 %十一号线各临界站所需时间  
 $a(300,301)=4;a(301,302)=2;a(302,303)=3;a(303,304)=2;a(304,305)=3;$

$a(305,306)=2;a(306,307)=2;a(307,308)=3;a(308,309)=2;a(309,310)=2;$   
 $a(310,311)=2;a(311,312)=2;a(312,313)=3;a(313,314)=3;a(314,315)=3;$   
 $a(315,316)=3;a(316,317)=3;a(317,318)=2;a(318,319)=3;a(319,320)=2;$   
 $a(320,321)=3;a(321,322)=3;a(322,323)=2;a(323,324)=2;a(324,325)=2;$   
 $a(325,326)=3;a(326,327)=2;a(327,328)=3;a(328,329)=2;a(329,330)=2;$   
 $a(330,331)=2;$   
 %十二号线各临界站所需时间  
 $a(332,333)=2;a(333,334)=4;a(334,335)=3;a(335,336)=2;a(336,337)=3;$   
 $a(337,338)=4;a(338,339)=2;a(339,340)=3;a(340,341)=2;a(341,342)=2;$   
 $a(342,343)=3;a(343,344)=1;a(344,345)=3;a(345,346)=2;a(346,347)=3;$   
 $a(347,348)=2;a(348,349)=2;a(349,350)=3;$   
 %十三号线各临界站所需时间  
 $a(351,352)=3;a(352,353)=6;a(353,354)=10;a(354,355)=5;a(355,356)=8;$   
 $a(356,357)=7;a(357,358)=6;a(358,359)=4;a(359,360)=6;a(360,361)=7;$   
 $a(361,362)=4;a(362,363)=5;$   
 %十六号线各临界站所需时间  
 $a(16,71)=0;a(16,106)=0;a(71,106)=0;$   
 $a(71,16)=0;a(106,16)=0;a(106,71)=0;$ %1.上海火车站 (1,3,4)  
 $a(15,316)=0;a(15,343)=0;a(316,343)=0;$   
 $a(316,15)=0;a(343,15)=0;a(343,316)=0;$ %2.汉中路 (1,12,13)  
 $a(13,47)=0;a(13,200)=0;a(47,200)=0;$   
 $a(47,13)=0;a(200,13)=0;a(200,47)=0;$ %3.人民广场 (1,2,8)  
 $a(11,257)=0;a(11,318)=0;a(257,318)=0;$   
 $a(257,11)=0;a(318,11)=0;a(318,257)=0;$ %4.陕西南路 (1,10,12)  
 $a(10,166)=0;a(166,10)=0;$ %5.常熟路 (1,7)  
 $a(8,224)=0;a(8,282)=0;a(224,282)=0;$   
 $a(224,8)=0;a(282,8)=0;a(282,224)=0;$ %6.徐家汇 (1,9,11)  
 $a(7,89)=0;a(89,7)=0;$ %7.上海体育馆 (1,4)  
 $a(6,324)=0;a(324,6)=0;$ %8.漕宝路 (1,12)  
 $a(5,59)=0;a(59,5)=0;$ %9.上海南站 (1,3)  
 $a(1,114)=0;a(114,1)=0;$ %10.莘庄 (1,5)  
 $a(40,154)=0;a(40,363)=0;a(154,363)=0;$   
 $a(154,40)=0;a(363,40)=0;a(363,154)=0;$ %11.龙阳路 (2,7,16)  
 $a(43,99)=0;a(43,141)=0;a(43,216)=0;$   
 $a(99,141)=0;a(99,216)=0;a(141,216)=0;$   
 $a(99,43)=0;a(141,43)=0;a(216,43)=0;$   
 $a(141,99)=0;a(216,99)=0;a(216,141)=0;$ %12.世纪大道 (2,4,6,9)  
 $a(46,253)=0;a(253,46)=0;$ %13.南京东路 (2,10)  
 $a(48,317)=0;a(48,345)=0;a(317,345)=0;$   
 $a(317,48)=0;a(345,48)=0;a(345,317)=0;$ %14.南京西路 (2,12,13)  
 $a(49,167)=0;a(167,49)=0;$ %15.静安寺 (2,7)  
 $a(50,284)=0;a(284,50)=0;$ %16.江苏路 (2,11)  
 $a(51,66)=0;a(51,111)=0;a(66,111)=0;$   
 $a(66,51)=0;a(111,51)=0;a(111,66)=0;$ %17.中山公园 (2,3,4)  
 $a(56,267)=0;a(267,56)=0;$ %18.虹桥 2 号航站楼 (2,10)

a(57,268)=0;a(268,57)=0;%19.虹桥火车站(2,10)  
a(74,204)=0;a(204,74)=0;%20.虹口足球场(3,8)  
a(72,105)=0;a(105,72)=0;%21.宝山路(3,4)  
a(70,107)=0;a(107,70)=0;%22.中潭路(3,4)  
a(69,108)=0;a(69,170)=0;a(108,170)=0;  
a(108,69)=0;a(170,69)=0;a(170,108)=0;%23.镇坪路(3,4,7)  
a(68,109)=0;a(68,286)=0;a(109,286)=0;  
a(109,68)=0;a(286,68)=0;a(286,109)=0;%24.曹杨路(3,4,11)  
a(67,110)=0;a(67,333)=0;a(110,333)=0;  
a(110,67)=0;a(333,67)=0;a(333,110)=0;%25.金沙江路(3,4,13)  
a(65,112)=0;a(112,65)=0;%26.延安西路(3,4)  
a(64,113)=0;a(64,260)=0;a(113,260)=0;  
a(113,64)=0;a(260,64)=0;a(260,113)=0;%27.虹桥路(3,4,10)  
a(63,88)=0;a(63,225)=0;a(88,225)=0;  
a(88,63)=0;a(225,63)=0;a(225,88)=0;%28.宜山路(3,4,9)  
a(61,323)=0;a(323,61)=0;%29.龙漕路(3,12)  
a(91,164)=0;a(164,91)=0;%30.东安路(4,7)  
a(92,320)=0;a(320,92)=0;%31.大木桥路(4,12)  
a(94,196)=0;a(196,94)=0;%32.西藏南路(4,8)  
a(97,143)=0;a(143,97)=0;%33.蓝村路(4,6)  
a(98,142)=0;a(142,98)=0;%34.浦电路(4,6)  
a(102,311)=0;a(311,102)=0;%35.大连路(4,12)  
a(104,250)=0;a(250,104)=0;%36.海伦路(4,10)  
a(152,191)=0;a(152,277)=0;a(191,277)=0;  
a(191,152)=0;a(277,152)=0;a(277,191)=0;%37.东方体育中心(6,8,11)  
a(146,158)=0;a(158,146)=0;%38.高科西路 (6,7)  
a(132,304)=0;a(304,132)=0;%39.巨峰路 (6,12)  
a(160,194)=0;a(194,160)=0;%40.耀华路 (7,8)  
a(163,321)=0;a(321,163)=0;%41.龙华中路 (7,12)  
a(165,223)=0;a(223,165)=0;%42.肇嘉浜路 (7,9)  
a(169,341)=0;a(341,169)=0;%43.长寿路 (7,13)  
a(197,219)=0;a(219,197)=0;%44.陆家浜路 (8,9)  
a(198,255)=0;a(255,198)=0;%45.老西门 (8,10)  
a(201,315)=0;a(315,201)=0;%46.曲阜路 (8,12)  
a(206,248)=0;a(248,206)=0;%47.四平路 (8,10)  
a(222,319)=0;a(319,222)=0;%48.嘉善路 (9,12)  
a(220,348)=0;a(348,220)=0;%49.马当路 (9,13)  
a(252,314)=0;a(314,252)=0;%50.天潼路 (10,12)  
a(256,347)=0;a(347,256)=0;%51.新天地 (10,13)  
a(259,283)=0;a(283,259)=0;%52.交通大学 (10,11)  
a(285,339)=0;a(339,285)=0;%53.隆德路 (11,13)  
a(280,322)=0;a(322,280)=0;%54.龙华 (11,12)  
a(272,361)=0;a(361,272)=0;%55.罗山路 (11,16)  
%补充相同站不同线  
c=a+a';%实现双向通行

%换乘点如下：(换乘时间不为定值)

c(16,71)=11;c(16,106)=11;c(71,106)=3;  
c(71,16)=11;c(106,16)=11;c(106,71)=3;%1.上海火车站 (1,3,4)  
c(15,316)=5;c(15,343)=5;c(316,343)=4;  
c(316,15)=5;c(343,15)=5;c(343,316)=4;%2.汉中路 (1,12,13)  
c(13,47)=9;c(13,200)=9;c(47,200)=5;  
c(47,13)=9;c(200,13)=9;c(200,47)=5;%3.人民广场 (1,2,8)  
c(11,257)=4;c(11,318)=3;c(257,318)=3;  
c(257,11)=3;c(318,11)=3;c(318,257)=4;%4.陕西南路 (1,10,12)  
c(10,166)=5;c(166,10)=4;%5.常熟路 (1,7)  
c(8,224)=3;c(8,282)=5;c(224,282)=7;  
c(224,8)=3;c(282,8)=3;c(282,224)=3;%6.徐家汇 (1,9,11)  
c(7,89)=5;c(89,7)=3;%7.上海体育馆 (1,4)  
c(6,324)=5;c(324,6)=4;%8.漕宝路 (1,12)  
c(5,59)=3;c(59,5)=3;%9.上海南站 (1,3)  
c(1,114)=3;c(114,1)=3;%10.莘庄 (1,5)  
c(40,154)=14;c(40,363)=5;c(154,363)=6;  
c(154,40)=10;c(363,40)=5;c(363,154)=6;%11.龙阳路 (2,7,16)  
c(43,99)=6;c(43,141)=3;c(43,216)=2;  
c(99,141)=3;c(99,216)=3;c(141,216)=2;  
c(99,43)=3;c(141,43)=3;c(216,43)=3;  
c(141,99)=5;c(216,99)=5;c(216,141)=3;%12.世纪大道 (2,4,6,9)  
c(46,253)=5;c(253,46)=3;%13.南京东路 (2,10)  
c(48,317)=8;c(48,345)=6;c(317,345)=11;  
c(317,48)=8;c(345,48)=4;c(345,317)=11;%14.南京西路 (2,12,13)  
c(49,167)=3;c(167,49)=3;%15.静安寺(2,7)  
c(50,284)=2;c(284,50)=2;%16.江苏路(2,11)  
c(51,66)=3;c(51,111)=4;c(66,111)=2;  
c(66,51)=3;c(111,51)=3;c(111,66)=2;%17.中山公园(2,3,4)  
c(56,267)=5;c(267,56)=5;%18.虹桥 2 号航站楼(2,10)  
c(57,268)=5;c(268,57)=5;%19.虹桥火车站(2,10)  
c(74,204)=5;c(204,74)=5;%20.虹口足球场(3,8)  
c(72,105)=3;c(105,72)=3;%21.宝山路(3,4)  
c(70,107)=3;c(107,70)=4;%22.中潭路(3,4)  
c(69,108)=6;c(69,170)=4;c(108,170)=4;  
c(108,69)=4;c(170,69)=4;c(170,108)=6;%23.镇坪路(3,4,7)  
c(68,109)=6;c(68,286)=4;c(109,286)=4;  
c(109,68)=4;c(286,68)=4;c(286,109)=6;%24.曹杨路(3,4,11)  
c(67,110)=6;c(67,333)=4;c(110,333)=4;  
c(110,67)=4;c(333,67)=4;c(333,110)=6;%25.金沙江路(3,4,13)  
c(65,112)=6;c(112,65)=6;%26.延安西路(3,4)  
c(64,113)=7;c(64,260)=5;c(113,260)=5;  
c(113,64)=5;c(260,64)=5;c(260,113)=7;%27.虹桥路(3,4,10)  
c(63,88)=6;c(63,225)=5;c(88,225)=5;  
c(88,63)=5;c(225,63)=5;c(225,88)=6;%28.宜山路(3,4,9)

```

c(61,323)=5;c(323,61)=5;%29.龙漕路(3,12)
c(91,164)=5;c(164,91)=6;%30.东安路(4,7)
c(92,320)=4;c(320,92)=4;%31.大木桥路(4,12)
c(94,196)=3;c(196,94)=6;%32.西藏南路(4,8)
c(97,143)=5;c(143,97)=8;%33.蓝村路(4,6)
c(98,142)=4;c(142,98)=7;%34.浦电路(4,6)
c(102,311)=4;c(311,102)=6;%35.大连路(4,12)
c(104,250)=4;c(250,104)=6;%36.海伦路(4,10)
c(152,191)=4;c(152,277)=4;c(191,277)=4;
c(191,152)=4;c(277,152)=4;c(277,191)=4;%37.东方体育中心(6,8,11)
c(146,158)=4;c(158,146)=4;%38.高科西路 (6,7)
c(132,304)=9;c(304,132)=5;%39.巨峰路 (6,12)
c(160,194)=4;c(194,160)=4;%40.耀华路 (7,8)
c(163,321)=4;c(321,163)=4;%41.龙华中路 (7,12)
c(165,223)=4;c(223,165)=5;%42.肇嘉浜路 (7,9)
c(169,341)=4;c(341,169)=4;%43.长寿路 (7,13)
c(197,219)=3;c(219,197)=3;%44.陆家浜路 (8,9)
c(198,255)=5;c(255,198)=5;%45.老西门 (8,10)
c(201,315)=4;c(315,201)=4;%46.曲阜路 (8,12)
c(206,248)=4;c(248,206)=4;%47.四平路 (8,10)
c(222,319)=5;c(319,222)=3;%48.嘉善路 (9,12)
c(220,348)=8;c(348,220)=5;%49.马当路 (9,13)
c(252,314)=5;c(314,252)=6;%50.天潼路 (10,12)
c(256,347)=6;c(347,256)=4;%51.新天地 (10,13)
c(259,283)=6;c(283,259)=5;%52.交通大学 (10,11)
c(285,339)=3;c(339,285)=5;%53.隆德路 (11,13)
c(280,322)=3;c(322,280)=5;%54.龙华 (11,12)
c(272,361)=3;c(361,272)=5;%55.罗山路 (11,16)
b=sparse(c); %构造稀疏矩阵 b
n=363;
for i=1:n
    for j=1:n
        [x,y,z]=graphshortestpath(b,i,j,'Directed',true,'Method','Dijkstra'); % Directed 是有向的,此处走
        的是点 i 到 j 的最短时间
        %换乘引起的问题值只会发生在起点和终点上
        timeall(i,j)=x;%把最短时间 x 存储到 timeall 数组中
        number(i,j)=length(y)-1;%计算共通过多少站路 (包括换乘, 换乘多算一站,去掉起点)
        if length(y)>1
            p=y(2);m=length(y)-1;q=y(m);
            if a(i,p)==0
                number(i,j)=number(i,j)-1;
                timeall(i,j)=timeall(i,j)-a(i,p);
            end
            if a(q,j)==0
                number(i,j)=number(i,j)-1;
            end
        end
    end
end

```

```

        timeall(i,j)=timeall(i,j)-a(q,j);
    end
end
%计算节约的时间即为换乘时间和停靠时间,设每站停靠时间为 1 分钟,停站数为 number(i,j)
tsave(i,j)=number(i,j);
trail=length(y)-1;
for k=1:trail
    s1=y(k);s2=y(k+1);
    if abs(s1-s2)>1
        tsave(i,j)=tsave(i,j)+c(s1,s2);
    end
end
end
end
%先输入 QA 数据 (进客量)
A=zeros(363,1);
A(55)=12895;A(233)=5549;A(3)=18678.5;
A(54)=8213.75;A(232)=17256.25;A(4)=9208.5;
A(332)=5183;A(52)=9208.5;A(230)=11472.75;A(5)=18678.5;
A(338)=9208.5;A(51)=17256.25;A(88)=5549;A(89)=5183;A(6)=7414.25;
A(293)=7414.25;A(292)=5183;A(68)=10778.5;
A(174)=5549;A(173)=9645.5;A(172)=7877;A(170)=10212;
A(107)=7732;A(194)=5183;A(193)=8505.75;A(192)=7635.25;A(185)=8409;
A(25)=9438.25;A(23)=12328.5;A(22)=8871.75;A(21)=12895;A(18)=7414.25;A(16)=13589.25;A
(157)=5183;
A(74)=9072.25;A(156)=5549;
A(79)=5086.25;A(78)=5183;A(77)=8434.75;A(102)=5086.25;A(43)=10212;A(41)=7414.25;A(15
5)=7635.25;
A(154)=13589.25;
A(125)=5549;A(37)=11891.5;
A(36)=5086.25;
%输入 QB/Q0 数据 (概率)
B=zeros(1,363);
B(111)=0.034387;B(112)=0.02479;B(113)=0.02479;B(225)=0.034387;
B(89)=0.024289;B(286)=0.02726;B(284)=0.032424;B(282)=0.042426;
B(91)=0.020819;B(69)=0.024539;B(169)=0.020545;B(168)=0.042426;
B(167)=0.026025;B(166)=0.022772;B(165)=0.020819;B(48)=0.034387;
B(11)=0.032382;B(221)=0.024539;B(71)=0.038406;B(316)=0.021228;
B(47)=0.063153;B(12)=0.024539;B(197)=0.02788;B(196)=0.020954;
B(201)=0.020819;B(46)=0.042426;B(254)=0.020954;B(45)=0.063153;
B(44)=0.02726;B(217)=0.02479;B(100)=0.22554;B(216)=0.032424;
B(43)=0.030462;
%换乘
B(106)=0.038406;B(16)=0.038406;%上海火车站
B(109)=0.02726;B(68)=0.02726;%曹杨路

```



```

B(170)=0.024539;B(108)=0.024539;%镇坪路
B(15)=0.021228;B(343)=0.021228;%汉中路
B(50)=0.032424;%江苏路
B(66)=0.034387;B(51)=0.034387;%中山公园
B(65)=0.02479;%延安西路
B(64)=0.02479;B(260)=0.02479;%虹桥路
B(88)=0.034387;B(63)=0.034387;%宜山路
B(7)=0.024289;%上海体育馆
B(10)=0.022772;%常熟路
B(224)=0.042426;B(8)=0.042426;%徐家汇
B(317)=0.034387;B(345)=0.034387;%南京西路
B(164)=0.020819;%东安路
B(94)=0.020954;%西藏南路
B(223)=0.020819;%肇嘉浜路
B(219)=0.02788;%陆家浜路
B(141)=0.032424;B(99)=0.032424;B(43)=0.032424;%世纪大道
B(253)=0.042426;%南京东路
B(13)=0.063153;B(200)=0.063153;%人民广场
B(315)=0.020819;%曲阜路
B(257)=0.032382;B(318)=0.032382;%陕西南路
B(341)=0.020545;%长寿路
B(49)=0.026025;%静安寺
C=A*B;%C 代表任意两点的客运量
D=C.*tsave;
for i=1:363
    for j=1:363
        z=i+363*(j-1);
        daxiao(z)=D(i,j);
    end
end
[paixu,weizhi]=sort(daxiao);
paixu=fliplr(paixu);
weizhi=fliplr(weizhi);
px=paixu(1:10)%选出最大的节省时间
wz=weizhi(1:10);
v=floor(wz/363)+1%终点
w=wz-(floor(wz/363)*363)%起点
%最大时间所对应的起点与终点.

```

The source code 3 of MATLAB:

```

%拟合直线
x=[27 10 47 11 74 18 40 101 45 32 20 37];
y=[29 11 49 12 77 17 42 106 47 36 21 37];
d=polyfit(x,y,1)

```

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# Establishment of Direct Shuttle Train Lines and Optimization of Time Interval

**Abstract:** Nowadays, metro gradually covered each city, and followed by a series of related questions like line optimization. The main purpose of this paper is to solve the problem of building direct shuttle train between two distant stations. Our works contain obtain the reasonable station selection schema, optimize the running interval, and analyze both economic and social benefits. In the end, we get a perfect and appropriate schema.

For problem 1, our goal is to filter out end stations to build direct shuttle train. We use the evaluation system to filter out the main factors that affect the average travel time of passengers. The target city in this paper is Tianjin, therefore we search some relevant data of Tianjin Metro, conduct Entropy Weight Method to determine the weight of each factor, and obtain a scientific expression of evaluation factor function. Then we use relevant software to simulate, compare the results of before and after we build the virtual direct shuttle train. And we reach the conclusion that building a direct shuttle train can significantly save the average travel time of passengers. When the average travel time multiplies the passenger flow, it can show a significant reduction in the overall travel time of the residents.

For problem 2, after we determined the selection of stations and lines, we need to adjust and optimize the running intervals. We quantify the passenger satisfaction and enterprise satisfaction, use target model to maximize the satisfaction of both of them. We form a formula on the basis of other factors and parameters, by forming specific equation of each factor and superimposed simplification, obtain the final model. Using the interval of each period we calculate on the basis of Tianjin Metro Line3, and take the changes of schedule after we build direct shuttle trains into account, then make further adjustment. The resulting increase in economic and social benefits, we analyze these on the aspects of enterprise benefits, traffic benefits, personal benefits, national economic benefits and environmental protection benefits. We put forward the theory, quantify economic benefits and analyze some improvements on the original foundation.

After that, we summarize and analyze the first two problems, then write a letter containing our analysis to our mayor.

**Key Words:** Goal programming model, membership function, factor and benefit analysis, entropy weight method.

# I .Introduction

## 1.1 Background

In the metro networks of today's metropolitan cities, it usually takes a long time if the distance between the origin station and destination of the passenger is too long. If we set up some direct subway lines between these two long-distance stations, that is, no longer stop between the origin and destination, the passengers' time can be saved significantly.

## 1.2 Problems Restatement

To solve the problem above, we will complete the following tasks one by one:

(1)Develop a mathematical model to optimize the choice of the end stations,and evaluate the expected results.

(2)Find out the advantages we may achieve if you adjust the interval between subway trains or build a special direct line, From the economic and social aspects.

(3)Write our mayor a two-page letter, the letter should include our analysis and a train schedule on the second page.

# II .Problem hypothesis

(1)Assuming that the influences of holiday and weather change are not taken into consideration;

(2)Assuming that the data of target city are accurate, and it can reflect the real-time regular pattern of target city.

(3) In each line the demand of passenger is independent and does not interfere with each other.

(4) Passenger demand is independent of the frequency of departure.

(5) The speed of train is constant within the station interval , and there is no special incident occurring.

(6) During the same period , the train's departure interval is unchanged.

(7) The stopping time of each station along the route is fixed.

(8) Within each period of time the waiting time passengers can accept is a certain value.

### III.Symbol Description

|                |                                                                            |                 |                                                                                                 |
|----------------|----------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------|
| $T_1, T_2$     | the start time and end time of the period                                  | $V_{ij}$        | the number of passengers on subway car $j$ between station $i$ and $i+1$                        |
| $n$            | total number of departures during the period                               | $q_{ij}$        | the number of seats subway car $j$ can provide on station $i$                                   |
| $m$            | number of line stations                                                    | $h$             | driving time interval                                                                           |
| $\lambda_i(t)$ | passenger arrival rate in station $i$ at time $t$                          | $C_{\max}$      | Maximum train carrying capacity                                                                 |
| $t_{ij}$       | the time that subway car $j$ arrive at station $i$                         | $L$             | the waiting time passengers can accept                                                          |
| $t'_{ij}$      | the time that subway car $j$ leave station $i$                             | $S$             | number of seats in the carriage                                                                 |
| $\Delta t_i$   | The time trains stay in the $i$ station                                    | $r_0$           | the critical value of the number of people standing in the car without feeling crowded          |
| $\beta$        | The average time passengers need to get on the subway train                | $r_{op}$        | Total operating income during the period                                                        |
| $B_{ij}$       | the number of passengers get on subway car $j$ at station $i$              | $w_1, w_2, w_3$ | Weighting factors of Passenger Waiting Satisfaction, Passenger Comfort, Enterprise Satisfaction |
| $a_i(t)$       | the proportional function of passengers get off subway car at station $i$  |                 |                                                                                                 |
| $A_{ij}$       | the number of passengers get off subway car $j$ at station $i$             | $x_w, x_c, x_e$ | Function of Passenger Waiting Satisfaction, Passenger Comfort, Enterprise Satisfaction          |
| $P_{ij}$       | The number of passengers waiting when subway car $j$ arrive at station $i$ |                 |                                                                                                 |
|                |                                                                            | $C_{\min}$      | the minimum operation income that enterprise can accept                                         |

## **IV.Problem Analysis**

### **4.1 Analysis on the Selection of Target City**

In this paper, Tianjin Metro System is taken as an example to analyze the establishment and verification of the model we build. The problem ask us to analyze metro network of a megapolis, so the first thought is to pick Beijing,Shanghai or Guangzhou, but considering the population of these three cities are too large, the flow of people at all sites are relatively large. And their systems have been running for many years, so they are relatively better. So the effect brought by setting some direct shuttle train may not very significant, therefore we rule out these three cities. Then we start to select cities from the new first-tier cities. Taking the completion of subway construction and flow of people into account, we choose Tianjin as our target city.

### **4.2 Analysis of Problem One**

Problem one ask us to establish a mathematical model to optimize the selection of the origination station and terminal station, and evaluate the expected results.

First of all, we start from station selection, analyze the main reasons that affect our selection, that is, the greater the load is , the more time passengers will wait,etc. And the main factors that affect the time passengers wait include the largest flow of people, number of times passengers have to transfer station, connected OD numbers. In the end, we form the function of evaluation index, the bigger the value is, indicating that this station is more needed to be selected to establish a direct shuttle train.

Second of all, we take the Tianjin Metro System as an example. Taking the data of different stations into the evaluation index function, and pick out the stations that need to be selected to build a direct shuttle train. By comparing the changes of the evaluation index value before and after we build a direct shuttle train, and validate the validity of the model, the expected result is obtained.

### **4.3Analysis of Problem Two**

On the basis of problem one,we start to do some further research on problem two.If we can reasonably choose the subway station,our job is to find out how to optimize the time interval reasonably, as well as after the subway line is adjusted,what kind of the economic and social benefits we can get.First,we use the target model.In the case of direct shuttle train's route has been selected, we maximize satisfaction of passenger and running enterprise.Then we establish the equations, through some analysis,and establish the final model. Then,we analyze the benefits we will obtain,by consulting a large number of data and related formulas,quantifying the results.

## V. Solution of Problem One

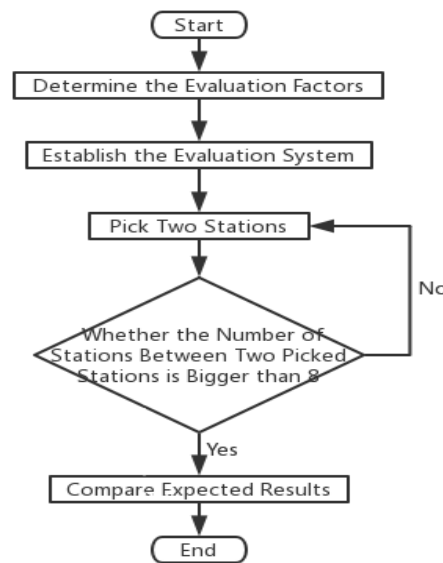
Problem one ask us to establish a mathematical model to optimize the selection of the end stations, and evaluate the expected results. End stations are places where they have to meet the following requirements: When the passengers being transported use a certain transportation and follow a specific route to get to their terminal stations, these terminal stations can suit the passengers' need to switch their transportation or follow other routes. Therefore, we believe that we can accomplish this by the following three steps:

Step 1. Because we have to optimize the selection of stations, we need to consider how to choose from various aspects. Therefore we establish an evaluation index system, screening out the main influence factors.

Step 2. Influence factors we obtained from the first step are taken as independent variables, and then we conduct dimensionless process on these variables. The weight of each variable is determined by Entropy Method, and therefore the evaluation factors function expression can be established. The station which has the largest value is taken as the station to optimize. Meanwhile, because the problem ask us to build direct shuttle train between two long-distance stations, so we refer to some relevant documents, and determine that the number of stations between two picked stations should be more than eight.

Step 3. Take the data of Tianjin Metro System into our function, and compare the changes before and after we build a direct shuttle train through a simulation model, conduct Empirical analysis, and show the expected results.

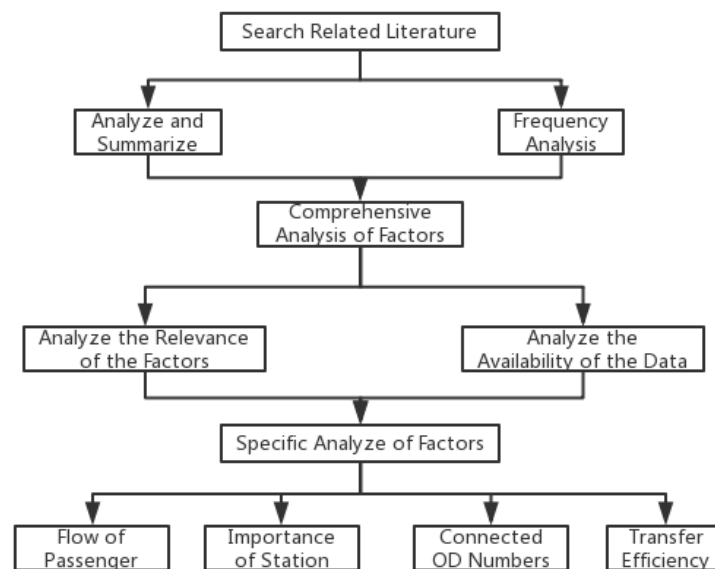
The three steps can be summarized as *Fig 1*:



*Fig 1*

## 5.1 Selection of Evaluation Factors

Through a comprehensive document reference, according to the definition of end stations, compare their conversion, connection, guiding function, we start to filter out some evaluation factor. We discover that the maximum passenger flow, maximum number of OD connections have direct influence with the selection of end stations. At the same time, the importance of stations is also an important factor. The flow chart of the solution to this problem is shown as *Fig 2*.



*Fig 2*

### (1) Maximum Passenger Flow $A_1$

The maximum passenger flow is the decisive factor affecting the selection of end stations. When the maximum passenger flow reaches or exceeds the load level of the station, it means that passengers have to wait a little longer to get on a subway train, it could easily lead to crowded situation. And it also prove that this station need to be optimized from a side, we should build direct shuttle train to alleviate the station.

### (2) The Number of Connected OD $A_2$

OD traffic volume indicates the traffic trips between origin and destination. The contents of survey include distribution of origin and destination, travel mode, travel distance and travel frequency.

### (3) The Importance of Station $A_3$

The stations in the subway network are regarded as the node in the network. More



number of trains go through one particular station, indicating that this station is more important, and this station is more likely to be in between any two stations shortest path. Betweenness can characterize the importance of station and interregional in the networks. The ranges of betweenness is [0,1], when one station is not an intermediate station of any OD pair, the betweenness of this station is 0. When the shortest path of all OD pairs in the network passes through this station, the betweenness of this station is 1. And the definition of the importance of station is as below:

$$A_3 = \sum_{s \neq t \neq v} \frac{\partial_{st}(v)}{\partial_{st}}$$

In the equation above,  $\partial_{st}$  means the number of shortest paths between station  $s$  and station  $t$ ;  $\partial_{st}(v)$  means the number of shortest paths that go through station  $v$  between station  $s$  and station  $t$ .

#### (4) Transfer Efficiency $A_4$

Transport capacity can be used to measure transfer efficiency. Transport capacity can measure the convergence ability and coordination degree of different kinds of transportation in urban hubs, and can reflect the efficiency of passengers transfer between different transportation facilities effectively. If the value of transport capacity is less than 1, then it indicate that the coordination cohesiveness between these two transportation facilities is perfect. Otherwise the coordination between the two transportation facilities needs to be further improved. [1] The formula of the transport capacity in an ideal state is as below:

$$A_4 = \frac{D}{\sum_{k=1}^n C_k \times P_k}$$

In the equation above,  $D$  represents the number of passenger this station needs to gather and evacuate during peak hours;  $C_k$  is the passenger capacity of the  $k$ th transportation mode connected to this station;  $P_k$  is the percentage of the  $k$ th transportation mode.

## 5.2 The Establishment of Evaluation System

### (1) The Pre-treatment of Evaluation Factors

There are many factors mentioned above, and the dimensions of these factors are not the same. Some of them are effect factors, indicating that the bigger these factors are, the better. However, there are some factors, the smaller they are, the better. These factors are called inverse factors. Therefore, they must be pretreated to be

standardized before processing. There are a variety of data preprocessing methods, in this paper we use Min-Max standardize method:

We suppose that  $r_i (i=1,2,\dots,n)$  are effect factors, and their normalization formula is :

$$r'_i = \frac{r_i - \min_i r_i}{\max_i r_i - \min_i r_i}, (i=1,2,\dots,n)$$

Then we suppose that  $e_j (j=1,2,\dots,m)$  are inverse factors, and their normalization formula is :

$$e'_j = \frac{\max_j e_j - e_j}{\max_j e_j - \min_j e_j}, (j=1,2,\dots,m)$$

After the above standardization, the original data are converted to measured value of dimensionless indicators. That is, the values of factors are in the same magnitude, they can be evaluated and analyzed comprehensively.

## (2) Establishment of Evaluation Index System

The entropy weight method is used to determine the weight coefficient in the evaluation system of the site after the standardization of the data.

a. Assuming  $n$  factors,  $m$  observation in different time period, factor values are  $r_{ij} (1 \leq i \leq m, 1 \leq j \leq n)$ , the original data matrix is:

$$R = (r_{ij})_{m \times n} = \begin{pmatrix} r_{11} & \cdots & r_{1n} \\ \vdots & & \vdots \\ r_{m1} & \cdots & r_{mn} \end{pmatrix}$$

b. Using the formula in the preprocessing to normalize the data  $R$ , get the standardized factor matrix:

$$R' = (r'_{ij})_{m \times n} = \begin{pmatrix} r'_{11} & \cdots & r'_{1n} \\ \vdots & & \vdots \\ r'_{m1} & \cdots & r'_{mn} \end{pmatrix}$$

c. Calculating the entropy value of the factors in the site optimized evaluation system, get the entropy value  $H = (h_1, h_2, \dots, h_n)$

$$h_i = -\frac{1}{\ln(m)} \sum_{j=1}^m r'_{ij} \ln(r'_{ij}), 1 \leq i \leq m, 1 \leq j \leq n.$$

d. Calculating the weight value of the factors in the site optimized evaluation system, get the weight value  $W = (w_1, w_2, \dots, w_n)$

$$W_i = \frac{1 - x_i}{n - \sum_{i=1}^n h_i}, 1 \leq i \leq n.$$

e.The final comprehensive evaluation formula is obtained  $G = \sum_{i=1}^n W_i \cdot r_{ij}$

## 5.3Take the Tianjin Subway of example

### 5.3.1Analysis on the Data of Related Sites of Tianjin Metro Line

#### (1) The data characteristics

##### a.Time distribution of maximum passenger flow

The characteristics of time distribution for passenger flow show a certain regularity in a certain time and range.According to the distribution of travel time and travel demand, the analysis of passenger flow are five trends in the time of dynamic changes. The Fig3 of trends are shown:

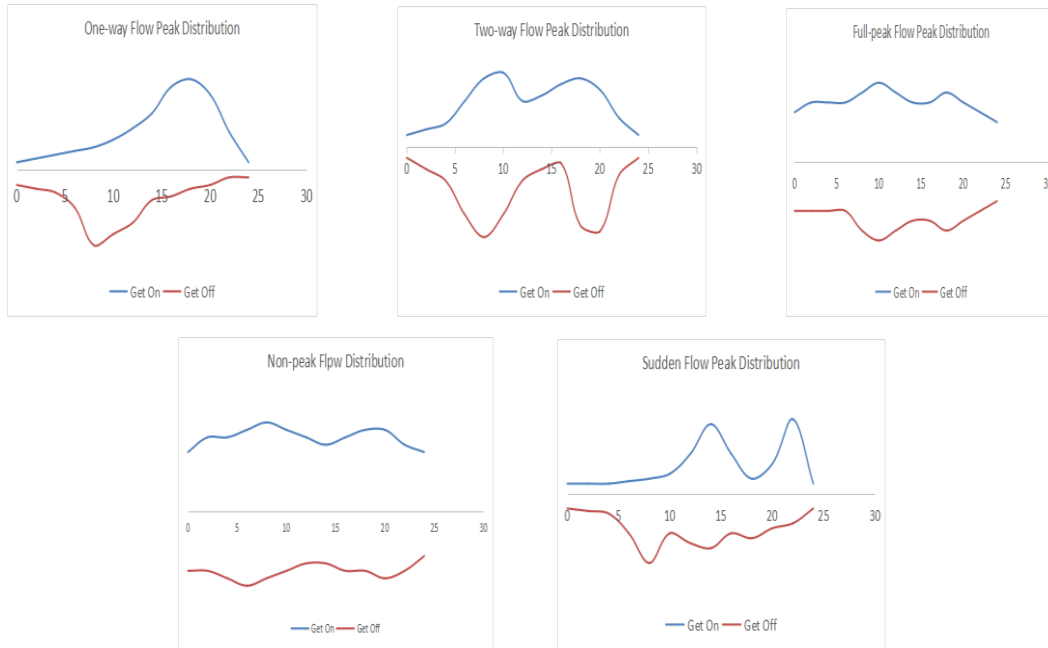


Fig 3 Passenger flow in the time of dynamic changes.

As can be seen from Fig x, the time distribution of passenger flow is very uniform. Rail transit should adopt a combination of direct train and ordinary train.And regardless of the type of passenger flow dynamic trends in the station , the peak is always at 6 A.M. Or 18 P.M. In get off line.So,the maximum person flow in this station is  $A_1 = \max\{a_6, a_{18}\}$ ,  $a_i$  Representing the maximum person flow at a time.According to this, the maximum daily person flow at each site as shown in the Fig X, in which 1-26 correspond to the South Station - Xiao Dian each site.

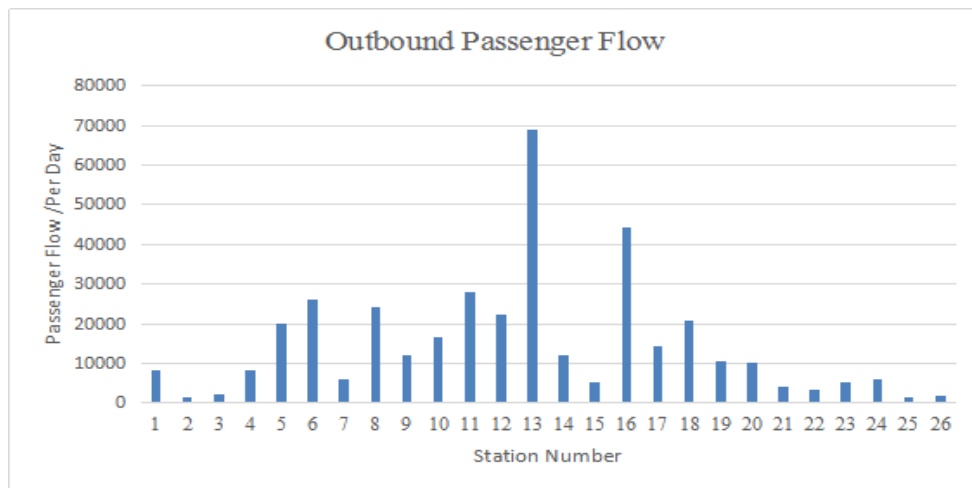


Fig 4

#### b. The Importance of Station $A_3$

Because the formula  $A_3$  is supposed to calculate the number of subway lines go through station, and the proportion of the number of shortest paths go through this station to all shortest paths, therefore from the point of all subway lines in Tianjin, we can obtain the median values of the importance of station of all the station on metro line3. On the basis of definition we can calculate that, only the value of station that it places at the intersection of many subway lines is not zero. Therefore, the value of Yingdaokou station is 0.41, and the value of Tianjin station is 0.88, others are all zero.

#### c. Transfer Efficiency $A_4$

Through the statistics of the proportion of other transforms in this site, calculate the transfer efficiency. The results are as follows:

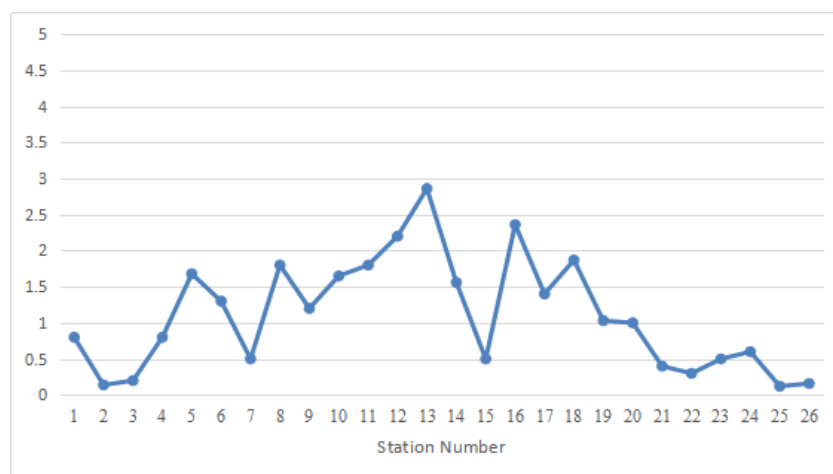


Fig 5

#### (2) Bring to entropy weight model ,determine the objective function expression

Through the pretreatment of the collected original data we determine the weight by using the software MATLAB , the specific code see Appendix 10.1. The results of the

weights are:

|       | The maximum<br>passenger flow $A_1$ | The number of<br>connected OD $A_2$ | The Importance<br>of Station $A_3$ | Transfer<br>Efficiency $A_4$ |
|-------|-------------------------------------|-------------------------------------|------------------------------------|------------------------------|
| $W_i$ | 0.567744233                         | 0.150148639                         | 0.004870176                        | 0.27733789                   |

the objective function expression:

$$G = 0.5677 A_1 + 0.1501 A_2 + 0.0487 A_3 + 0.2773 A_4$$

(3)Select the best end stations

Because the subject requires establish a direct train between two long-distance stations, there must be eight sites between end stations according to the literature.Bring the values to the objective function expression.The larger the values are, the greater it need to establish a direct train.The top ten sites are:

| Station NO. | Station Name               |
|-------------|----------------------------|
| 13          | Yingdaokou station         |
| 16          | Tianjin station            |
| 5           | College Town station       |
| 11          | Wujiayao station           |
| 12          | Xikang Road station        |
| 8           | Hongqi'nanlu station       |
| 6           | Huayuan station            |
| 10          | Tianta station             |
| 18          | Zhongshan Road station     |
| 17          | Golden Lion Bridge station |

So the choices of the end stations to build the direct train have three feasible options: ①College Town station →Yingdaokou station

②College Town station →Tianjin station

③Hongqi'nanlu station →Tianjin station.

And the best option is the first College Town station →Yingdaokou station.

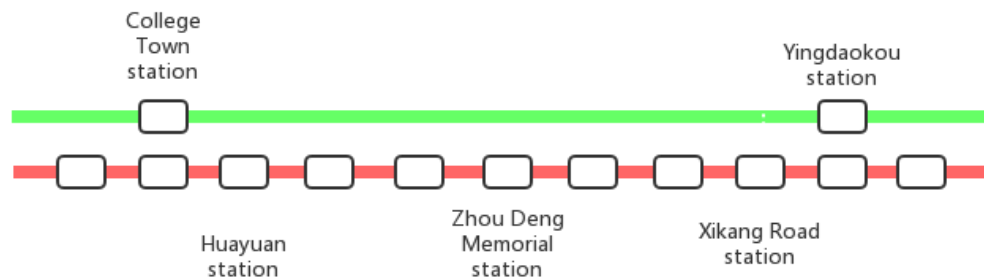


Fig 6

### 5.3.2 Analyze the end stations have been chosen after establishment

Using Vissim software<sup>[7]</sup> to simulate the traffic condition with direct train. And Fig X shows the comparison of the maximum flow in outbound.

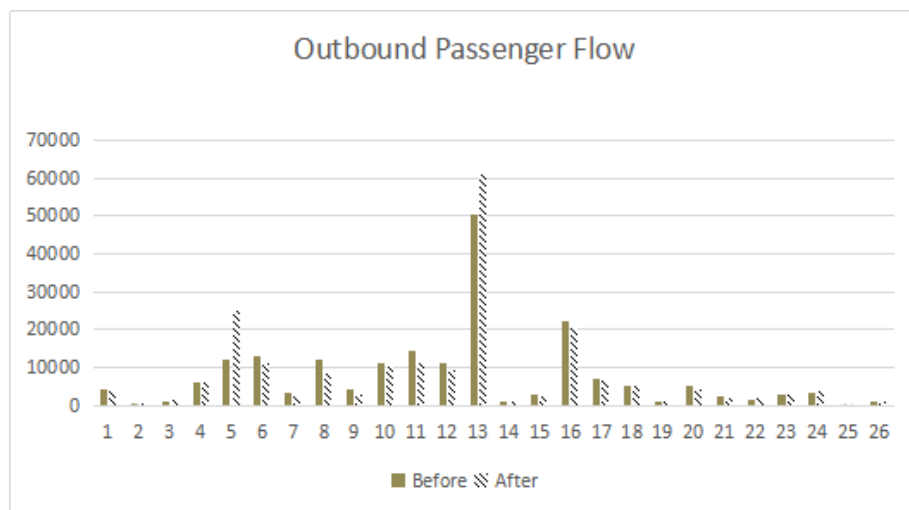


Fig 7

As can be clearly seen from the fig 7, after building the direct train from College Town station to ingdaokou station, the maximum passenger flow at end stations increase suddenly, the maximum passenger flow at stations between end stations and near the stations decline clearly, the maximum person flow at other stations don't have any changes.

There is a Comparison figure of Average Travel Times of Passengers. Before means no direct train, After means with direct train.

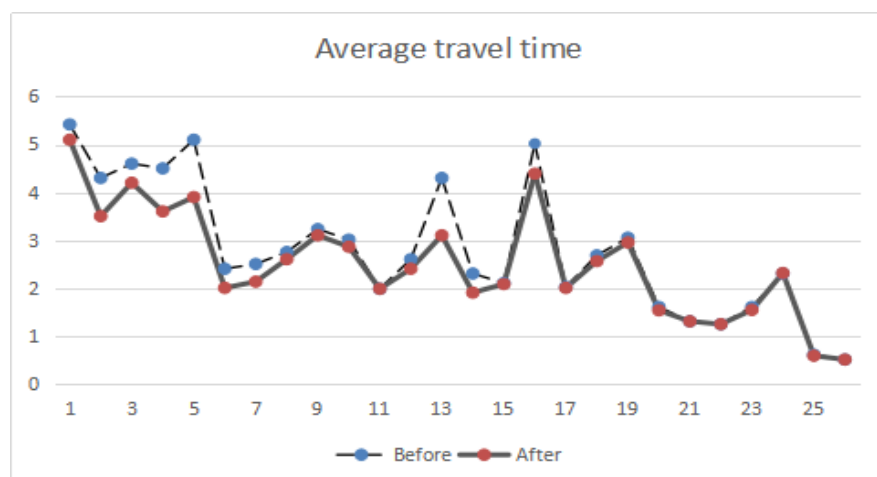


Fig 8

As can be seen from the fig 8, after building the end stations, the average travel time at the end stations declined clearly, and the average travel time at the stations between end stations and near the end stations declined also. From the product of average travel time and the maximum passenger flow in every station, we can see this measure saving a lot of public travel time. There is no doubt that can improve the happiness sense of the city. So it is necessary for urban construction to build the direct train.

## VI. Solution of Problem Two

Determine the intervals of subway lines on the basis of passenger flow can improve the pertinence and effectiveness of rail transit services, thereby improve the quality of rail transit service, and enhance the attractiveness of rail transit to urban residents. However, the formulation of the operating organization program must also take the cost and income of operating enterprise into account. So, the optimization model of the running interval is established to optimize the service level and operating cost.

On the basis of the first problem, we start to conduct some further research on the second problem. If we can reasonably choose the subway station, our job is to find out how to optimize the time interval reasonably, as well as after the subway line is adjusted, what kind of the economic and social benefits we can get. First of all, we use the target model. In the case of direct shuttle train's route has been selected, we make the satisfaction of passenger and running enterprise maximize, which is our goal. Then we establish the equations, through some analysis, and establish the final model. Then, we analyze the benefits we will obtained, by consulting a large number of data and related formulas, quantifying the results.

### 6.1 Divide the Passenger Flow

In the process of analyzing passenger flow, we can divide one day into peak period, normal period, trough period and transition period on the basis of passenger flow, and set a same sending frequency in the same period. Time division can be based on section passenger flow curve, using mathematical clustering method to solve. Or use the method below.

### 6.2 Model Establishment

#### 6.2.1 The Establishment and Analysis of the Model

As described above, the establishment of model should consider the benefits of both passengers and operating enterprise. Therefore, the goal of planning should be getting the maximized weighted average value of passenger waiting satisfaction, passenger comfort and enterprise satisfaction. Based on this, we could obtain the optimization model of the running interval.

$$\max Z = w_1 x_w + w_2 x_c + w_3 x_e$$

$$s.t. \begin{cases} x_w \geq 0.8 \\ h \leq L \\ r_{op} \geq D_{\min} \end{cases} \quad \textcircled{1}$$

### 6.2.2 Calculate the Passenger Waiting Satisfaction

Generally speaking, passengers have a level of tolerance while waiting, if the waiting time is longer than the range of tolerance time, passengers will have negative feelings. Of course, the tolerance time varies depending on the different periods.

We take the passenger waiting time as the measure of passenger waiting satisfaction, the longer the waiting time is, the lower the waiting satisfaction will be. In order to make the algorithm simple, reasonable and persuasive, we select the passenger satisfaction rate as one of the main factors of whether the subway time interval is reasonable. Therefore we can calculate the number of passengers who is unsatisfied with the waiting time, and finish the calculation of the passenger waiting satisfaction. Considering that not all passengers can get on the first train they expected, so we make the following discussion:

When train  $j$  arrives at station  $i$ , the number of unsatisfied passengers  $u_{ij}$  can be divided into three parts:

(1) When train  $j$  arrives at station  $i$ , there are no unsatisfied passengers among waiting people.

$$\text{That is when } p_{ij} - \int_{t_{ij-L}}^{t_{ij}} \lambda_i(t) dt \leq 0, \quad u_{ij} = 0. \quad (2)$$

(2) When train  $j$  arrives at station  $i$ , there are unsatisfied passenger among waiting people, but when train  $j-1$  leaves station  $i$ , there are no unsatisfied passengers in the station.

$$\begin{aligned} \text{That is when } p_{ij} - \int_{t_{ij-L}}^{t_{ij}} \lambda_i(t) dt \leq 0 \quad \text{and} \quad p_{ij-1} - \int_{t_{ij-L}}^{t_{ij-1}} \lambda_i(t) dt - B_{ij-1} \leq 0, \\ u_{ij} = p_{ij} - \int_{t_{ij-L}}^{t_{ij-1}} \lambda_i(t) dt \end{aligned} \quad (3)$$

(3) When train  $j$  arrives at station  $i$ , there are unsatisfied passengers among waiting people, and these passengers have felt unsatisfied when train  $j-1$  get the station.

$$\begin{aligned} \text{That is when } p_{ij} - \int_{t_{ij-L}}^{t_{ij}} \lambda_i(t) dt > 0 \quad \text{and} \quad p_{ij-1} - \int_{t_{ij-L}}^{t_{ij-1}} \lambda_i(t) dt - B_{ij-1} > 0, \\ u_{ij} = p_{ij} - \int_{t_{ij-L}}^{t_{ij-1}} \lambda_i(t) dt - \left( p_{ij-1} - \int_{t_{ij-L}}^{t_{ij-1}} \lambda_i(t) dt - B_{ij-1} \right) \end{aligned} \quad (4)$$

In the equations above:

$$p_{ij} = p_{ij-1} + \int_{t_{ij-1}}^{t_{ij}} \lambda_i(t) dt - B_{ij-1} \quad (5)$$



$$B_{ij-1} = \begin{cases} p'_{ij-1} & q_{ij-1} \geq p'_{ij-1} \\ C_{\max} - q_{ij-1} & q_{ij-1} < p'_{ij-1} \end{cases} \quad (6)$$

$$q_{ij-1} = C_{\max} - (V_{ij-1} - V_{ij-1} a_i t(t_{ij-1})) \quad (7)$$

$$p'_{ij-1} = p_{ij-1} + \int_{t_{ij-1}}^{t_{ij}} \lambda_i(t) dt \quad (8)$$

$$t'_{ij-1} = t_{ij-1} + \beta \cdot B_{ij-1} \quad (9)$$

The above calculation is transitive, it can count the number of unsatisfied passengers using the given initial state. In the end, we can get the passenger waiting satisfaction:

$$x_w = \frac{\sum_{i=1}^m \int_{T_1}^{T_2} \lambda_i(t) dt - \sum_{i=1}^m \sum_{j=1}^n u_{ij}}{\sum_{i=1}^m \int_{T_1}^{T_2} \lambda_i(t) dt} \quad (10)$$

### 6.2.3 Calculate the Passenger Comfort

Passenger comfort can be reflected by passenger density in the train, we can use a membership function to represent it. We might as well set that, the value of satisfaction membership is 1 when this passenger has a seat. When this passenger have to stand in the train, the satisfaction membership decrease as the density increase. Therefore, the standing passengers' satisfaction membership can be represented by function  $\mu(x)$ , and its domain is  $(0, C_{\max} - S]$ . The curve will be like Fig 9

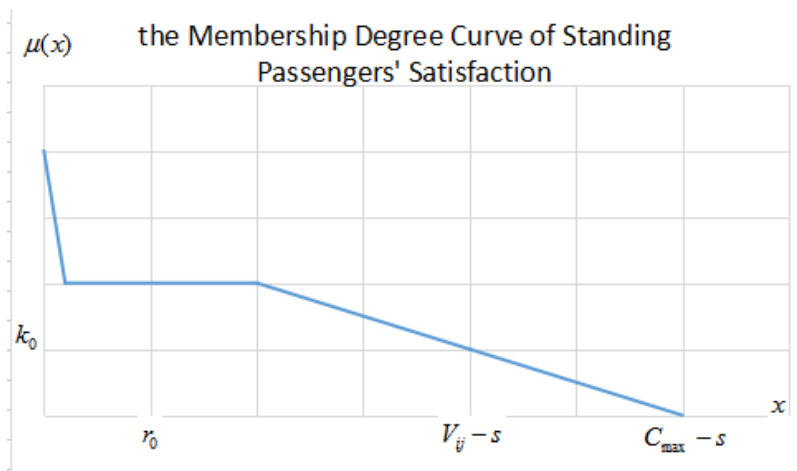


Fig 9

The formula is: 
$$\mu(x) = \begin{cases} k_0 & 0 < v_{ij} - S < r_0 \\ \frac{C_{\max} - S - x}{C_{\max} - S - r_0} k_0 & v_{ij} - S < 0, v_{ij} - S > r_0 \end{cases} \quad (11)$$

In the equations above,  $x$  represents the number of standing people;  $k_0$  represents the satisfaction membership value of standing passengers.

Therefore, when train  $j$  leave station  $i$  to station  $i+1$ , the total value of passenger comfort can be represented as :

$$v_{ij} = \begin{cases} V_{ij} & V_{ij} - S \leq 0 \\ S + (V_{ij} - S)\mu(V_{ij} - S) & V_{ij} - S > 0 \end{cases} \quad (12)$$

From these, we can obtain the calculation formula of passenger comfort:

$$X_c = \frac{\sum_{j=1}^n \sum_{i=1}^{m-1} v_{ij}}{\sum_{j=1}^n \sum_{i=1}^{m-1} V_{ij}} \quad (13)$$

#### 6.2.4 Calculate the Enterprise Satisfaction

Let's assume that the operating enterprise have a minimum expected income of each train. When the actual income is lower than the expected income, the enterprise will feel unsatisfied; when the actual income is higher than the expected income, the enterprise will feel satisfied. Therefore, the enterprise satisfaction can be determined by a 'Semi-Trapezoidal' distributed membership function. The curve will be like

Fig 10.

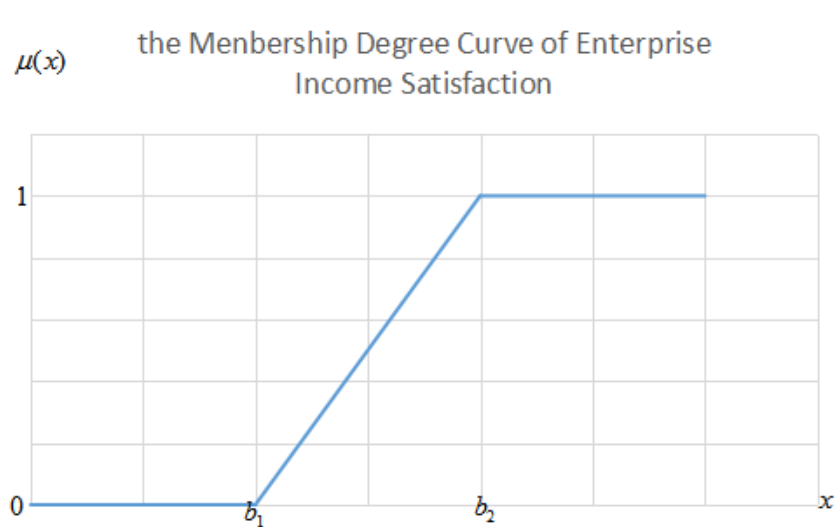


Fig 10

$$\text{The formula is : } \mu(x) = \begin{cases} 0 & 0 < x < b_1 \\ \frac{x - b_1}{b_2 - b_1} & b_1 \leq x \leq b_2 \\ 1 & x > b_2 \end{cases} \quad (14)$$

In the equations above,  $x$  represents ticket revenue,  $b_1$  represents the enterprise's minimum expected ticket revenue,  $b_2$  represents the enterprise's expected ticket revenue.

The calculation formulas of enterprise satisfaction are:

$$x_e = \frac{\sum_{j=1}^n \mu \left( \sum_{i=1}^{m-1} (B_{ij} \cdot \rho) \right)}{n} \quad (15)$$

$$r_{op} = \sum_{i=1}^{m-1} \sum_{j=1}^n B_{ij} \cdot \rho \quad (16)$$

## 6.3 The Application of Model

### 6.3.1 The Application of Model--Take Tianjin Metro Line 3 as an example

The approximate length of Tianjin Metro Line 3 is 33.7km, and the number of stations is 26, the minimum interval can be set as 80s. The annual forecast data of passenger flow are as follow:

| The Annual Passenger Flow of Tianjin Metro Line 3 |                                         |                                       |         |                               | Number of People                      |                                         |                              |
|---------------------------------------------------|-----------------------------------------|---------------------------------------|---------|-------------------------------|---------------------------------------|-----------------------------------------|------------------------------|
| Section<br>Passenger<br>Flow                      | Number of<br>passengers on<br>the train | number of<br>passengers to<br>get off | station |                               | number of<br>passengers to<br>get off | number of<br>passengers on<br>the train | Section<br>Passenger<br>Flow |
| No.1 → No.26                                      |                                         |                                       | Number  | station                       | No.26 → No.1                          |                                         |                              |
|                                                   | 0                                       | 23300                                 | 1       | Tianjin south<br>metro        | 0                                     | 23500                                   | 23500                        |
| 23300                                             | 100                                     | 22900                                 | 2       | Yangwu<br>Village             | 100                                   | 23700                                   | 47100                        |
| 46100                                             | 1600                                    | 19400                                 | 3       | University<br>Industrial Area | 7000                                  | 32200                                   | 79000                        |
| 63900                                             | 3000                                    | 24400                                 | 4       | High-Tech<br>Zone             | 6800                                  | 28700                                   | 100700                       |

|        |         |         |    |                              |         |         |        |
|--------|---------|---------|----|------------------------------|---------|---------|--------|
| 85300  | 3300    | 27600   | 5  | College Town                 | 4800    | 29800   | 123700 |
| 109600 | 3700    | 18200   | 6  | Huayuan                      | 8100    | 26800   | 123700 |
| 124100 | 6800    | 21200   | 7  | Wangdingdi                   | 7900    | 32100   | 145700 |
| 138500 | 8200    | 42900   | 8  | Hongqi'nanlu                 | 6000    | 29800   | 169700 |
| 173200 | 5400    | 76600   | 9  | Zhou Deng<br>Memorial        | 13300   | 64300   | 191600 |
| 244400 | 5100    | 92800   | 10 | Tianta                       | 8900    | 95800   | 249900 |
| 332100 | 5800    | 53200   | 11 | Wujiayao                     | 34000   | 66900   | 332400 |
| 379500 | 44500   | 110800  | 12 | Xikang Road                  | 111300  | 80200   | 390400 |
| 551200 | 258700  | 120800  | 13 | Yingdaokou                   | 57700   | 48100   | 613200 |
| 471700 | 44900   | 31500   | 14 | Heping Road                  | 33700   | 55100   | 419200 |
| 404300 | 30600   | 130100  | 15 | Jinwan Square                | 38500   | 135300  | 435400 |
| 445800 | 116200  | 88100   | 16 | Tianjin station              | 38900   | 93900   | 436600 |
| 503800 | 37300   | 84700   | 17 | Golden Lion<br>Bridge        | 220100  | 114700  | 537000 |
| 413300 | 46200   | 23800   | 18 | Zhongshan<br>Road            | 66700   | 28700   | 441200 |
| 390900 | 56100   | 17500   | 19 | Tianjin north<br>metro       | 54100   | 17700   | 412200 |
| 352300 | 90800   | 20400   | 20 | Tiedong Road                 | 85600   | 21300   | 363200 |
| 281900 | 79900   | 23500   | 21 | Zhangxing<br>Village         | 75300   | 26300   | 249900 |
| 225500 | 45900   | 9200    | 22 | Yinxing Port                 | 54300   | 10500   | 206100 |
| 188800 | 58600   | 2800    | 23 | Tianshili                    | 51800   | 3200    | 157500 |
| 133000 | 64700   | 4900    | 24 | North China<br>Group station | 55600   | 3900    | 105800 |
| 73200  | 26500   | 100     | 25 | Fengchan<br>River            | 61300   | 1800    | 46300  |
| 46800  | 46800   |         | 26 | Xiaodian                     | 46300   | 0       |        |
|        | 1090700 | 1090700 |    |                              | 1094300 | 1094300 |        |

### 6.3.2 The Selection of Parameter

(1)The first train outbound at 6:00 am, and the last train inbound at 22:10 pm.

(2)The total operating time can be divided into five stages based on the flow of passengers. Acceptable waiting time L of each stage are shown below:

|                       |              |              |         |
|-----------------------|--------------|--------------|---------|
| Peak period           | 7:00--9:00   | 17:00--19:00 | L=3 min |
| Sub-peak period       | 9:00--10:00  | 16:00--17:00 | L=6 min |
| High flat peak period | 10:00--12:00 | 14:00--16:00 | L=6 min |
| Low flat peak period  | 6:00--7:00   | 12:00--14:00 | L=6 min |
| Trough period         | 21:00--22:10 |              | L=10min |

(3)Six sections of train are organized for a group, the maximum carrying capacity

is  $C_{\max} = 2500$

(4)The critical number of people standing in the train without feeling crowded is  $r_0 = 1120$  , and now the membership value of passenger satisfaction is  $k_0 = 0.8$  .

(5)Set the estimated average fare income as 5 RMB/person, and the critical value of enterprise's estimate income for each train is 5580 RMB.

(6)Weight values are  $w_1 = 0.4$  ,  $w_2 = 0.3$  ,  $w_3 = 0.3$

### 6.3.3 The Calculation Results of Optimization Model

From the above-mentioned analysis of passenger flow, we can conclude that the intensive sections of passenger flow mainly concentrate between No.5-No.17. Therefore, when we calculate the intervals, we can divide the calculation into three parts.

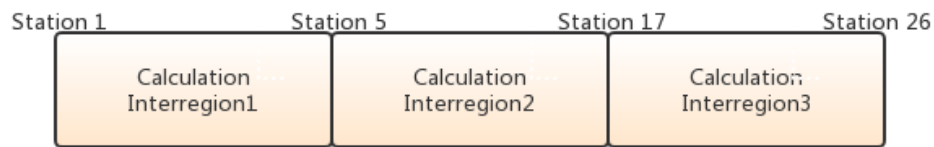


Fig 11

Using MATLAB software to process calculation, and we can get the three intervals time:  $h_1 = 5.5$  ,  $h_2 = 1.5$  ,  $h_3 = 7$  . Combining the results of two models, we can obtain the following results:(unit:min)

## 6.4 Benefits Analysis

Optimize the interval or build direct shuttle train can bring both economic and social benefits.

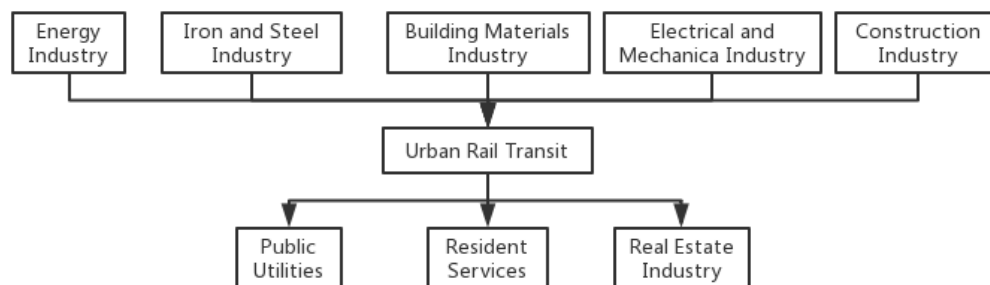


Fig 12

### 6.4.1 Analysis of Economic Benefits

When analyzing the economic benefits of subway optimization, we only analyze and evaluate the economic benefits of the operating enterprises from the enterprises' view, and the impact on the society will be attributed to the social benefit analysis. The following are some main evaluate factor:

#### 6.4.1.1 Total Cost $C$ of Urban Subway Traffic

$$C = C_1 + C_2 + C_3 + C_4$$

##### (1) Cost $C_1$ of Urban Subway Construction

$$C_1 = \sum_{t=1}^m \sum_{k=1}^n (L_k \cdot A_k)$$

In the equation above,  $A_k$  represent unit kilometers cost of the  $kth$  section of urban subway, including tool, materials and equipment acquisition costs, the project all the costs of the cycle,etc[1].  $L_k$  represents the construction mileage of the  $kth$  urban subway,  $n$  represents the number of total road section.

##### (2) Usage Fee $C_2$ of Urban Subway

$$C_2 = \sum_{t=1}^m \sum_{k=1}^n (L_k \cdot M_{kt})$$

In the equation above,  $M_{kt}$  represents the usage fee of the  $kth$  urban subway in the  $tth$  year, including rail train maintenance, traffic management fees, energy costs,etc.  $m$  represents the year limit of urban subway.

##### (3) Overhaul Costs $C_3$ of Urban Subway

Overhaul costs of urban subway refer to the cost of large-scale repair after using the subway for a period of time.

$$C_3 = \sum_{k=1}^l (L_k \cdot R_k)$$

In the equation above,  $l$  represents the number of roads that need a large-scale repair,  $R_k$  represents the overhaul cost of the  $kth$  section.

##### (4) Other Fee $C_4$

Urban subway investment projects should pay taxes, etc, these are also the pre-business investment.

#### 6.4.1.2 Total Operating Income $R$ of Urban Subway

$$R = R_1 + R_2$$

##### (1) Operating Fare Income $R_1$

$$R_1 = \sum_{i=1}^n (T_i \cdot TR_i \cdot L_i)$$

In the equation,  $T_i$  represent the passenger flow in year  $i$  of urban subway.

$TR_i$  represents the charges standard of this city,  $L_i$  represents charge mileage of this city.

##### (2) Other Income $R_2$

In the urban subway, operating enterprises have other corresponding income, such as, advertising revenue inside and outside the station.

#### 6.4.1.3 Economic Efficiency Factors

##### (1) Cost - benefit ratio $FBCR$

It is the ratio of the present value of the annual income to the present value of the annual cost. From the point of economic rationality analysis, the greater the value is, the better it will be.

$$FBCR = \frac{\sum_{i=1}^n B_i (1+t)^i}{\sum_{i=1}^n C_i (1+t)^i}$$

In the equation above,  $B_i$  represents the income amount of the  $i$ th year,  $C_i$  represents the cost amount of the  $i$ th year,  $t$  represents Social discount rate,  $n$  represents the calculation period of this project.

##### (2) Financial Investment Payback Period

Generally speaking, the shorter the payback period is, the better the profitability is, and the risk is smaller.

$$FN = \text{The year in which the cumulative net financial income was incurred} - 1 + \frac{\text{Absolute value of accumulated net income for the previous year}}{\text{Net income for the year}}$$

##### (3) Investment Profit Margin

Investment profit margin refers to a ratio, the ratio indicates the specific value of total annual profit to total investment of this project in a normal operating year. It is the static factor of the project's investment profitability.

$$\text{Investment profit margin} = \frac{\text{Total annual profit}}{\text{The total amount of the project}} * 100\%$$

#### 6.4.2 Analysis of Social Benefits

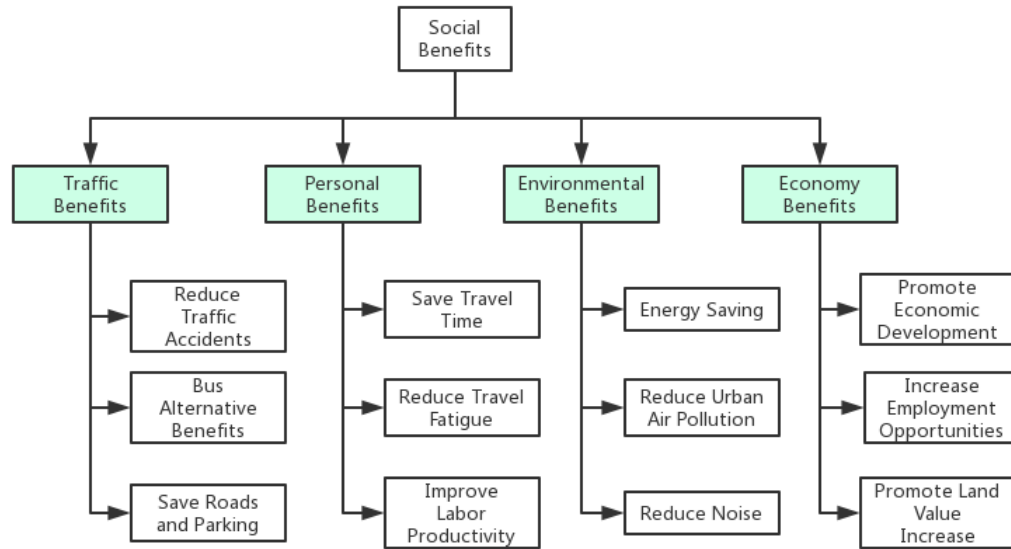


Fig 13

##### 6.4.2.1 Traffic Benefits

###### (1) Reduce Traffic Accidents

As a rail transport, the probability of traffic accidents on subways are much smaller cars, the relevant statistical results as shown below:

|                     | mortality rate | ratio | injury rate | ratio  |
|---------------------|----------------|-------|-------------|--------|
| rail transportation | 0.053          | 1:221 | 0.106       | 1:8080 |
| bus transportation  | 11968          |       | 823.21      |        |

The calculation model is as below:

$$B_3(t) = Q_g \cdot \rho \cdot l$$

In this formula above,  $B_3(t)$  represents the economic benefits of reduction of traffic accidents in year  $t$ ,  $Q_g$  represents passenger volume of subway instead of bus in year  $t$ ,  $\rho$  represents traffic loss per unit of passenger volume,  $l$  represents average transport distance.

###### (2) Alternative bus

Consider the fact that if we do not build the subway, the alternative effect will be brought, like the bus and taxi purchase, road, parking, land operations and other



aspects. Model of each aspect are as follow.

|                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>A. Alternative Bus Purchase Efficiency <math>B_{11}</math></p> $B_{11}(t) = \frac{p_1 \cdot N_1}{n_1 \cdot Y_1} + \frac{p_2 \cdot N_2}{n_2 \cdot Y_2}$                                                                                                                                              | <p>B. The Costs of Saving Road and Parking <math>B_{12}</math></p> $B_{12}(t) = N \cdot S_1 \cdot C + n \cdot S_2 \cdot P$                                                                                                                                                                                                              |
| <p><math>p_1, p_2</math> ——— Purchase Price of Bus and Taxi;</p> <p><math>N_1, N_2</math> ——— the Number of People Taking Bus and Taxi;</p> <p><math>n_1, n_2</math> ——— The Average Number of Passengers Taking Bus or Taxi;</p> <p><math>Y_1, Y_2</math> ——— Depreciation Period of Bus and Taxi</p> | <p><math>N</math> ——— Increased Number of Vehicles</p> <p><math>S_1</math> ——— The Area of Each Car</p> <p><math>C</math> ——— Road Investment on Per Unit</p> <p><math>n</math> ——— Added parking spaces</p> <p><math>S_2</math> ——— The Area of a Single Parking Space</p> <p><math>P</math> ——— The construction cost of per unit</p> |

#### 6.4.2.2 Personal Benefits

##### (1) Saving Passenger Travel Time

The rapidity and punctuality of city subways can help people save time.

##### (2) Reduce Passenger Travel Fatigue, Improve Labor Efficiency

Relatively speaking, urban subway have a better service quality, not easy to produce fatigue, help to improve labor efficiency.

|                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The time passenger save from year <math>t</math>: <math>B_1(t)</math></p> $B_1(t) = 0.5 \times Q_g \left( \frac{1}{v_a} - \frac{1}{v_b} \right) \times l$                                                                                                        | <p>The fatigue reduced from year <math>t</math>, increased labor efficiency <math>B_2(t)</math></p> $B_2(t) = Q_g \times \theta \times \rho$                                                                                                                                                                                 |
| <p><math>Q_g</math> ——— The urban traffic allocated to the bus traffic</p> <p><math>v_a</math> ——— The average speed of the bus;</p> <p><math>v_b</math> ——— The average speed of subway traffic</p> <p><math>l</math> — Average Distance of Urban Mass Transit</p> | <p><math>Q_g</math> ——— The urban traffic allocated to the bus traffic</p> <p><math>\theta</math> ——— Working passenger coefficient;</p> <p><math>\rho</math> ——— The increase coefficient of increased productivity of urban subway traffic passengers higher than that of public transportation passenger productivity</p> |

#### 6.4.2.3 Promote National Economic Development

##### (1) Increase Land Value

The construction of urban subway has changed the land use pattern along the line and brought development to the surrounding areas, bringing significant value - added benefits to the along land. Here, we establish a simple idealized model, assuming that

the distribution of appreciation space around the station is circle, like Fig 14, the center of the each circle is a station, the area of each circle indicates the scope of appreciation space.

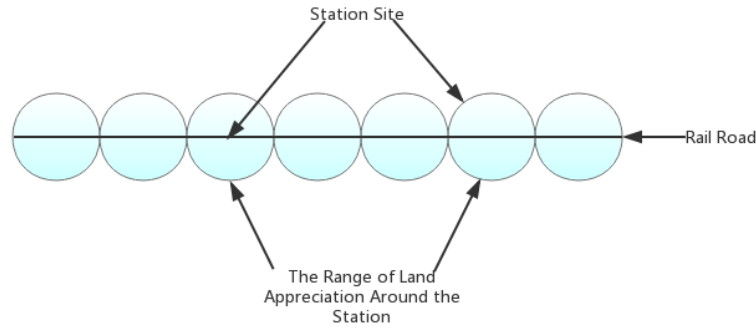


Fig 14

$$B_3(t) = A_3 \times P_t$$

$$A_3 = \pi \times \left[ \frac{L}{(N-1) \times 2} \right]^2 \times N$$

In this formula,  $B_3(t)$  represents the value - added benefits in year  $t$ ,  $A_3$  represents land appreciation area along the subway line,  $P_t$  represents the unit price of land appreciation in year  $t$ ,  $L$  represents the length of rail transit line,  $N$  represents the number of stations.

#### (2)Promote the Growth of Economic

Obviously, the subway construction can lead to a large number of construction of life, production, transportation, education and entertainment facilities. Meanwhile, the early construction will also drive the steel industry and other industries. In addition, the increase in employment is also one of the important benefits.

#### 6.4.2.4 Environmental Benefits

##### (1)The Benefit of Energy Saving

The energy urban metro transportation system uses is polluteless and cheap electricity. Compared to traditional means of transport, subway is polluteless , energy saving and large volume. Based on some material, the unit energy consumption of some main transportation are given in the following table.

|             | energy consumption(KJ per person) | proportion  |
|-------------|-----------------------------------|-------------|
| subway      | 209                               | 1:3.2:11.84 |
| bus         | 670                               |             |
| private car | 2479                              |             |

To do a quantitative calculation of energy-saving benefits, we can use the following model:

$$B_3(t) = l_t \times e_i \times p_{\text{汽}} - l_t \times C_{\text{轨}} \times P_{\text{电}}$$

In this formula:  $e_i$  represents the energy consumed per kilometer of bus and taxi,  $C_r$  represent the power consumption per unit of rail transit,  $P_g, P_e$  represent the shadow price of gasoline and electricity,  $l_t$  represents The mileage of subway transport alternative to public transport system.

## (2)The Benefits of Reducing Urban Pollution

As an excellent green transportation, urban subway is safe, comfortable and less-polluted. It can reduce toxic and harmful gas emissions. See the following table to find out the contribution of each kind of transportation to reduce environmental pollution.

### Comparison of Hazardous Gas Produced by Bus and

Subway(  $g / 100 \text{ person} \cdot \text{km}$ ) [6]

|                         | $CO_x$ | $NO_x$ | $SO_x$ |
|-------------------------|--------|--------|--------|
| bus                     | 12300  | 257    | 21     |
| subway                  | 4600   | 179    | 11     |
| proportion(bus: subway) | 26.7   | 1.4    | 1.9    |

A Quantitative Calculation Model for Environmental Protection:

$$B_4(t) = \sum_{j=1}^n (C_{bi} - C_{si}) \times P_i$$

In this formula,  $n$  represents the number of traffic pollutants,  $C_{bi}$  represents the amount of pollutants  $i$  produced by bus,  $C_{si}$  represents the amount of pollutants  $i$  produced by subway,  $P_i$  represents the standard unit cost of processing this pollutants. In addition, metro traffic has also play a role in reducing noise.

## VII.Solution of Problem Three

### The Letter To Mayor

Dear Mayor,

Thank you for your time to read our letter.

The construction and operation of urban subway rationalization is closely related to our daily life. We have a reasonable demonstration and explanation of the two ideas to set a direct shuttle train between two long-distance stations and optimize the running interval. According to the operation of Tianjin Metro Line 3, we put forward the corresponding program.

The main purpose of the adjustment and optimization is to facilitate the public, but after many aspects of comprehensive consideration, we will also gain economic and social benefits on the basis of the original development, including the economic benefits of various aspects of society, like road traffic mitigation benefits and environmental protection benefits. Besides, there will also be some personal benefits, like travel convenience, reducing fatigue and improving work efficiency, etc. In the long run, this is a worthwhile implementation.

The total length of Tianjin Metro Line 3 is about 33 km, with 26 stations. We come up with a schema on the basis of economy, station conditions, stations' distance and the needs of the public. The following are the three suitable routes for direct shuttle trains: College Town station to Yingdaokou station, College Town station to Tianjin station and Hongqinlan station to Tianjin station. The above three routes, in the case of the difficulty of operation is flat, the demand of building a direct shuttle train between College Road station and Yingdaokou station is the largest, so this line is our key recommendations.

At the same time, set the passenger satisfaction and running enterprise satisfaction as the goal, we optimize interval between trains, putting the schedule of trains between College Town station and Yingdaokou station into the time schedule. Combining common trains with direct shuttle trains ensure that the public can get the most convenience in the routes.

To sum up, we attach the train schedule on the next page for reference, to illustrate the feasibility of the schema.

Best wishes.

Yours Sincerely

**Train Schedule of Tianjin Metro Line 3 (without direct shuttle trains)**

| Station |                             | First train |            | Last train |            |
|---------|-----------------------------|-------------|------------|------------|------------|
|         |                             | No.1→No.26  | No.26→No.1 | No.1→No.26 | No.26→No.1 |
| 1       | Tianjin south metro station | 6:00        |            | 22:10      |            |
| 2       | Yangwu Village station      | 6:02        | 6:57       | 22:12      | 23:02      |
| 3       | University Industrial Area  | 6:04        | 6:55       | 22:14      | 23:00      |
| 4       | High-Tech Zone station      | 6:06        | 6:52       | 22:17      | 22:57      |
| 5       | College Town station        | 6:08        | 6:49       | 22:19      | 22:54      |
| 6       | Huayuan station             | 6:11        | 6:46       | 22:22      | 22:51      |
| 7       | Wangdingdi station          | 6:14        | 6:43       | 22:24      | 22:48      |
| 8       | Hongqi'nanlu station        | 6:16        | 6:41       | 22:26      | 22:46      |
| 9       | Zhou Deng Memorial          | 6:18        | 6:39       | 22:28      | 22:44      |
| 10      | Tianta station              | 6:20        | 6:37       | 22:30      | 22:42      |
| 11      | Wujiayao station            | 6:23        | 6:35       | 22:33      | 22:40      |
| 12      | Xikang Road station         | 6:25        | 6:32       | 22:35      | 22:37      |
| 13      | Yingdaokou station          | 6:27        | 6:30       | 22:38      | 22:35      |
| 14      | Heping Road station         | 6:29        | 6:28       | 22:40      | 22:33      |
| 15      | Jinwan Square station       | 6:30        | 6:26       | 22:41      | 22:31      |
| 16      | Tianjin station             | 6:31        | 6:24       | 22:43      | 22:29      |
| 17      | Golden Lion Bridge station  | 6:33        | 6:22       | 22:46      | 22:27      |
| 18      | Zhongshan Road station      | 6:35        | 6:19       | 22:48      | 22:24      |
| 19      | Tianjin north metro station | 6:37        | 6:18       | 22:50      | 22:23      |
| 20      | Tiedong Road station        | 6:38        | 6:15       | 22:53      | 22:20      |
| 21      | Zhangxing Village station   | 6:40        | 6:12       | 22:55      | 22:17      |
| 22      | Yinxing Port station        | 6:42        | 6:11       | 22:57      | 22:16      |
| 23      | Tianshili station           | 6:44        | 6:09       | 22:59      | 22:14      |
| 24      | North China Group station   | 6:48        | 6:05       | 23:03      | 22:10      |
| 25      | Fengchan River station      | 6:50        | 6:02       | 23:05      | 22:07      |
| 26      | Xiaodian station            |             | 6:00       |            | 22:05      |

**Departure Time of Starting Station(including direct shuttle trains)**

| Period      | Departure Interval(min) | Remarks                                  |
|-------------|-------------------------|------------------------------------------|
| 6:00-9:00   | 1.5                     |                                          |
| 9:00-10:00  | 2                       |                                          |
| 10:00-12:00 | 5.5                     | (1) Number of direct shuttle trains      |
| 12:00-14:00 | 7                       | :                                        |
| 14:00-16:00 | 5.5                     | Number of common trains                  |
| 16:00-17:00 | 2                       | =1:2                                     |
| 17:00-19:00 | 1.5                     | (2) 2 common trains and a direct shuttle |
| 19:00-21:00 | 7                       | trains departure in turn                 |
| 21:00-22:10 | 10                      |                                          |

# VIII. Model Evaluation and Improvement

## 8.1 Model Evaluation and Improvement

### 8.1.1 The evaluation of the Model of Problem One

#### (1) The Advantages of Our Model

Based on the evaluation factor system and the evaluation process, we calculate and analyze the optimal selection scheme of the end stations of Tianjin Metro Line 3, obtain the optimal scheme. Then we conduct simulation of the results we will get after we build direct shuttle trains. The results are globalized, practical, common, and reliable.

#### (2) The Disadvantages of Our Model

① In this model, some data are difficult to obtain, we use Mean Supplement Method to get missing data, it may result in the deviation of the final evaluation result.

② We may ignore some factors in the simulation process, therefore there may be some errors.

### 8.1.2 Improvement on the Model of Problem One

(1) There are many influential factors influencing the station selection of urban subway lines. Although this paper has done some sorting and screening, but the correlation between the factors needs to be classified more rationally.

(2) The evaluation factor system of station selection in urban metro line is limited, and it is affected by other social factors. Therefore, the evaluation system needs to be further improved.

## 8.2 Evaluation of the Model

In the model of optimizing train time interval, the target model is used to maximize the satisfaction of the passengers and the running enterprises. The corresponding mathematical equations are established. After the adjustment, the objective function is obtained. Take the Tianjin Metro Line 3 as example and prove the model. The model has a strong practicality, in which the parameters of the calculation and selection is reliable and accurate. It can be used as a reference basis. For the analysis of its economic efficiency, the quantification of each index can make benefits more visual.

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## X. Appendix

### 10.1 The steps to solve the problem using entropy weight method by MATLAB

#### Step1.Standardize

```
clear
clc
x=[];
a=min(x);
b=max(x);
[n,m]=size(x); k=1/log(n);
for i=1:n
    for j=1:m
        x(i,j)=(x(i,j)-a(j))/(b(j)-a(j))*100;
    end
end
for i=1:n
    for j=1:m
        x(i,j)=(b(j)-x(i,j))/(b(j)-a(j))*100;
    end
end
he=sum(x);
for i=1:n
    for j=1:m
        p(i,j)=x(i,j)/he(j);
    end
```

end

Step2. Determine the weights

function weights = EntropyWeight(R)

    [rows,cols]=size(R);

    k=1/log(rows);

    f=zeros(rows,cols);

    sumBycols=sum(R,1);

    for i=1:rows

        for j=1:cols

            f(i,j)=R(i,j)./sumBycols(1,j);

        end

End

    lnfij=zeros(rows,cols);

    for i=1:rows

        for j=1:cols

            if f(i,j)==0

                lnfij(i,j)=0;

            else

                lnfij(i,j)=log(f(i,j));

            end

        end

end

    Hj=-k\*(sum(f.\*lnfij,1));

    weights=(1-Hj)/(cols-sum(Hj));

end



# The Direct Metro Lines

## Abstract

Nowadays, the construction of urban subway system has entered a stage of great-leap-forward development. It has become the largest scale of traffic system in all urban transportations, with the largest scale of construction and unparalleled speed of development. So it is essential for us to study on urban metro transit system's effectiveness and its assessment methods.

In the first part of the problem, we establish a multi-objective programming model to analyze the impact which the selection of the terminal station may bring to the expected result. Firstly, according to the concept of environmental carrying capacity, a multi-objective function to measure the bearing capacity of line running is introduced. The objective functions consist of a single subway line operation capacity function and function of transfer station operation capacity. Then, we determine the main constraints of a single metro line operation capacity and transfer station operation capacity service factor, and fit the constraints of the relative curves to analyze its rationality. Finally, basing on the multi-objective programming model, we evaluate the expected results in four aspects: subway network planning, passenger flow, passenger flow attraction, and vehicle integrated base.

In the second part of the problem, we divide the problem into two cases. First, in the first case, we only adjust the departure time interval without constructing a direct route. We adjust the departure time interval without affecting the train return route and the punctuality of subway departure time, and use the departure time adjustment algorithm to minimize the sum of subway departure time. And then, in the second case, we construct the direct route without regard to the adjustment of the departure time interval. We choose Nanchang subway line one as our analytic templet, and collect information about the average daily visitors flowrate and the number of existing business circles in several Nanchang metro stations, and then select the stations in which person flow is large and large number of business circles exist. And then we educed the direct line: **Shuanggang Station → Metro Building Station → Bayi Hall Station → Xiejia Village → Olympic Sdadium Station**. Finally, in the two aspects of saving time efficiency and improving labor efficiency, we analyze the economic benefits and social benefits which the adjusted urban metro transit construction may bring to Nanchang city.

In the third part of the problem, we wrote a letter to the mayor. First of all, we summarize the analysis of the departure time. According to the departure time adjustment algorithm, we calculate the departure time of each metro, and then compare the original schedule to adjust the departure time interval. Then, we summarize and analyze the direct line problem of metro, and use the collected different levels of population distribution of each station and the surrounding business circle scopes of each station to determine the nonstop station and build a new direct metro route. Finally, we list the new metro schedule we have designed.

**Keywords:** Multi-objective programming; System efficiency; Run carrying capacity; Metro Lines.

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# **I. Introduction**

## **1.1 Problem background**

In current metro network of a megapolis, it usually takes a long time to transport passengers between two far apart stations. We may suggest that set a direct shuttle train between two long-distance stations to save time of the longhaul passengers.

## **1.2 Statement of problem**

We face mainly four problems:

- Please develop a mathematical framework to optimize the choice of the end stations, and evaluate the expected result of the suggest.
- Explore any advantages you may achieve in terms of economic and social benefits if you adjust the interval between trains or build a special direct line.
- Write a two-page letter to the Mayor of your city summarizing your analysis. Include your train schedule as the second page of your letter.

## **1.3 Our work**

### **1.3.1 Analysis of problem one**

The problem of selecting a mathematical model of the terminal station and evaluating the expected results of the proposal was developed in question one. We main through the establishment of multi-objective planning model to analyze how to choose the terminal. In order to optimize the selection of terminals more scientifically and intuitively, we introduce the multi-objective function, which is the subway carrying capacity, according to the concept of environmental carrying capacity. The objective function mainly includes a single subway line running capacity function, transfer station running capacity function. We want to achieve is the subway running capacity to maximize. The main constraints are the running capacity of a single subway line theory and the transit station operating coefficient. And then fitting the relevant curve to analyze its rationality. Finally, based on the multi-objective planning model of the selected destination, we evaluate the expected results from four aspects: subway network planning, passenger flow, passenger attraction, and vehicle integrated base.

### **1.3.2 Analysis of two**

The problem of adjusting the departure time interval or constructing a direct route to raise the city's economic and social benefits. We will be divided into two cases to consider the situation, the first case only adjust the departure time interval, do not build a direct route. The second case only to build a direct route, do not consider adjusting the departure time interval. For the first case, we do not affect the train return route and the departure time of the premise of adjusting the departure time interval, the use of outbound time adjustment algorithm to train the sum of the outbound time to minimize. For the second case, we do not consider adjusting the departure time interval under the premise of the construction of direct routes. We chose the Nanchang Metro Line 1 as a template, we investigated the various subway stations in Nanchang, the average daily flow of people and the subway station in the vicinity of the existing business district, and then combine the two, select the daily traffic flow and business concentration Associated with more subway stations, and then analyze the corresponding direct routes. Finally, we mainly analyze the adjustment of Nanchang economic benefits and social benefits from the two aspects of saving time efficiency and improving labor efficiency.

### 1.3.3 Analysis of problem three

For questions asked to write a letter to the mayor, summing up the above intervals and the analysis of the direct line, and lists the new metro schedule. In the letter, we first summarize the problem of the interval between the start of the grid, we adjust the departure time interval is mainly based on outbound time adjustment algorithm to calculate the departure time of each train, and then compare the original schedule to adjust the time interval. Then, we summarize the direct line of the subway analysis, we mainly use the site to collect the distribution of population and the various stations around the scope of the district have to select direct access to the site, build a new direct route. Finally, list the new metro schedule.

## II. Symbols, Definitions and Assumptions

### 2.1 Symbols and Definitions

| Symbol              | Definition                                   |
|---------------------|----------------------------------------------|
| $C_{single}$        | a single metro line running capacity         |
| $i$                 | the line number                              |
| $MOBC$              | the metro running capacity                   |
| $C_{transfer}$      | the transit site capacity                    |
| $\alpha_{single}$   | the passage coefficient of single metro line |
| $\alpha_{transfer}$ | the transit station operating coefficient    |
| $k$                 | the total number of transfer points          |
| $t_i$               | the Actual departure time                    |
| $T_i$               | the timetable departure time                 |
| $Q$                 | the saving time benefit                      |
| $B$                 | the labor efficiency benefit                 |

### 2.2 General assumptions

In order to make the model easy to set up and solve, we put forward the following assumptions:

- Assuming that we have access to information from the Internet and data is true and reliable;
- Assume that during the study, the subway speed is relatively stable, no mutation;
- Assuming the construction of the metro line is not affected by the departure time of the impact;
- Assuming that the factors affecting the selection of the terminal are mentioned in this paper, the remaining factors have little or no influence;

### III. Model Establishment and Solution

#### 3.1 Issue I: Develop a mathematical model of the selection of the terminal and evaluate the expected results of the proposal

##### 3.1.1 The multi - objective programming model of selecting the terminal is established

In order to optimize the selection of terminals more scientifically and intuitively .According to the concept of environmental carrying capacity, we introduce a multi-objective function, which is the subway running capacity function.

Metro running capacity function is the function of the subway running system, it not only the structure of the subway running system itself, but also with the outside of the input and output. Considering the subway carrying capacity as a function, it mainly contains two objective functions: Single subway line running capacity and the operational capacity of the transfer stations, the subway running capacity function can be expressed as:

$$MOBC = f(C_{\text{single}}, C_{\text{transfer}}) \quad (3.1)$$

##### ● Objective Function of Running Capacity of Single Metro Line $C_{\text{single}}$

In order to better study the running capacity of a single subway line, first of all, we find the relevant information, obtained a single subway line running capacity of the formula:

$$C_{\text{single}} = \sum_{i=1}^n C_i \quad (3.2)$$

For a single subway line running capacity is  $C_{\text{single}}$  and article  $i$  of the subway line running capacity is  $C_i$ . The line number is  $i$  and number of individual subway lines is  $n$ .

$$C_i = C_0 \times \alpha_{\text{single}} \times \alpha_{\text{transfer}} \times \alpha_{\text{station}} \quad (3.3)$$

The theoretical running capability of a single metro line is  $C_0$ , its size with the subway line design speed; Passage coefficient of single metro line is  $\alpha_{\text{single}}$ , its size with the subway line level; Transit station operating coefficient is  $\alpha_{\text{transfer}}$ , by the two metro lines at the transfer station to determine the operating capacity of the common; Article  $i$  Number of subway lines is  $\alpha_{\text{station}}$ .

##### ● Objective function of transit capacity $C_{\text{transfer}}$

Metro put into use, a single subway lines and other subway lines there are cross-site, known as the transfer station, and the emergence of a single subway line running capacity is to some extent, in order to study the extent of its impact, we Need to calculate the transit capacity of the transit, by looking for information, we have the following formula:

$$C_{\text{transfer}} = \sum_{i=1}^n C_i \quad (3.4)$$

The transit site capacity is  $C_{\text{transfer}}$  and transit factor is  $C_i$ ; Article  $n$  the number of the subway line is  $i$  and number of transit points is  $n$ .

$$C_i = \sum_{j=1}^K C_j \quad (3.5)$$

The total running capacity of different metro lines is  $C_j$ ; Total number of transfer points  $K$  and metro line number is  $j$ .

Summary available:

$$C_{\text{transfer}} = \sum_{i=1}^n \sum_{j=1}^k C_j \quad (3.6)$$

### 3.1.2 Objective function analysis

#### ● *Subway running capacity of the largest*

When the metro is put into operation, if the metro running capacity is small, the metro is prone to congestion, affecting the efficiency of passenger flow. In order to solve this major problem, which is equivalent to improve the metro carrying capacity, that is:

$$\max MOBC = f(C_{\text{single}}, C_{\text{transfer}}) \quad (3.7)$$

#### ● *The minimum number of transit sites*

Metro put into use, the convergence of the metro lines to produce a transit site, and this often leads to pedestrians, vehicle congestion, there will be security risks. Therefore, we take into account the increase in the carrying capacity of the metro, on the other hand to consider the problems caused by the transit site. So our other objective function should make the number of relay stations  $n$  minimum.

### 3.1.3 Establishment of Constraints

#### ● *Constraints on Theoretical Running Capacity of Single metro*

We find the information through the network to understand the theoretical capacity of a single subway line by the subway line design speed, the specific form as follows:

Table 1 The theoretical running capability of a single metro line

| $V(\text{km}/h)$    | 20   | 30   | 40   | 50   | 60   |
|---------------------|------|------|------|------|------|
| $C_0(\text{pcu}/h)$ | 1380 | 1550 | 1640 | 1690 | 1730 |

According to the table, we use matlab programming (see Appendix 1 for the program), the subway line design speed and the theory of a single subway running capacity of the quadratic curve fitting, the following diagram:

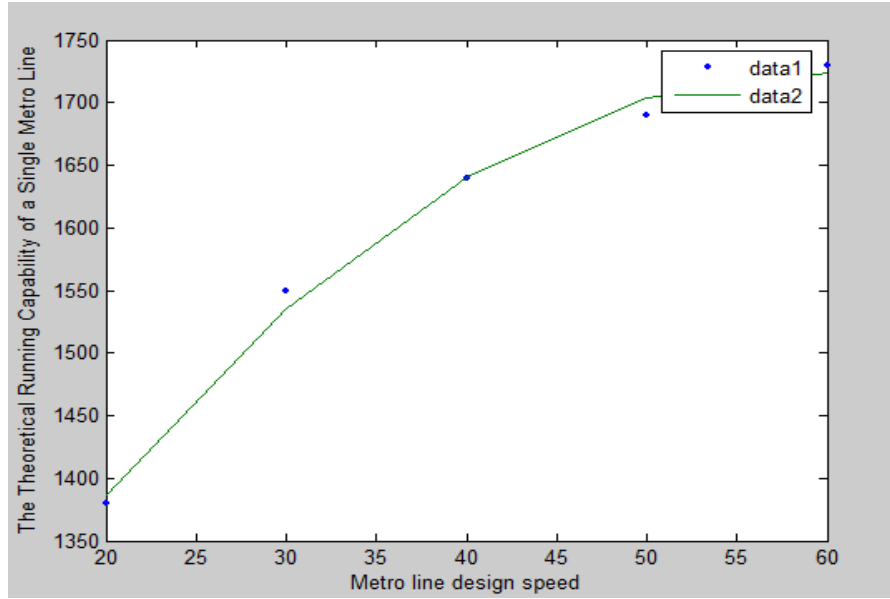


Figure 1 Design speed curve and single line theory running capacity curve fitting

We obtain the secondary relationship between the road design speed and the running capacity of a single metro line:

$$C_0(v) = -0.21v^2 + 26v + 9.6e + 2 \quad (3.8)$$

● **Constraints on operating factors of transfer stations**

We run the formula according to the transit station coefficient of the relevant information to obtain the meaning of the coefficient, and access to its impact factors, as shown in the table:

Table 2 The value of the transit station operating coefficient

| Speed    | The distance between stations ( m ) |      |      |      |      |      |      |      |      |      |
|----------|-------------------------------------|------|------|------|------|------|------|------|------|------|
| $v(m/s)$ | 100                                 | 200  | 300  | 400  | 500  | 600  | 700  | 800  | 900  | 1000 |
| 5.56     | 0.45                                | 0.62 | 0.71 | 0.76 | 0.8  | 0.83 | 0.85 | 0.87 | 0.88 | 0.89 |
| 6.94     | 0.37                                | 0.54 | 0.64 | 0.7  | 0.75 | 0.75 | 0.81 | 0.83 | 0.84 | 0.86 |
| 8.33     | 0.31                                | 0.48 | 0.58 | 0.65 | 0.7  | 0.7  | 0.76 | 0.79 | 0.81 | 0.82 |
| 9.72     | 0.27                                | 0.42 | 0.52 | 0.6  | 0.65 | 0.65 | 0.72 | 0.75 | 0.77 | 0.79 |
| 11.11    | 0.23                                | 0.38 | 0.48 | 0.55 | 0.6  | 0.6  | 0.68 | 0.71 | 0.73 | 0.75 |
| 12.5     | 0.2                                 | 0.34 | 0.43 | 0.5  | 0.56 | 0.56 | 0.64 | 0.67 | 0.69 | 0.72 |
| 13.88    | 0.18                                | 0.3  | 0.39 | 0.46 | 0.52 | 0.52 | 0.6  | 0.63 | 0.66 | 0.68 |

According to the above table we can get: transit station operating coefficient with the subway line design speed is negatively correlated with the distance between the two transit stations are positively correlated. In order to get a more concrete image of the functional relationship between them, we also use Matlab programming (see Appendix 2 procedures), the data in the table fitting, the fitting diagram is as follows:

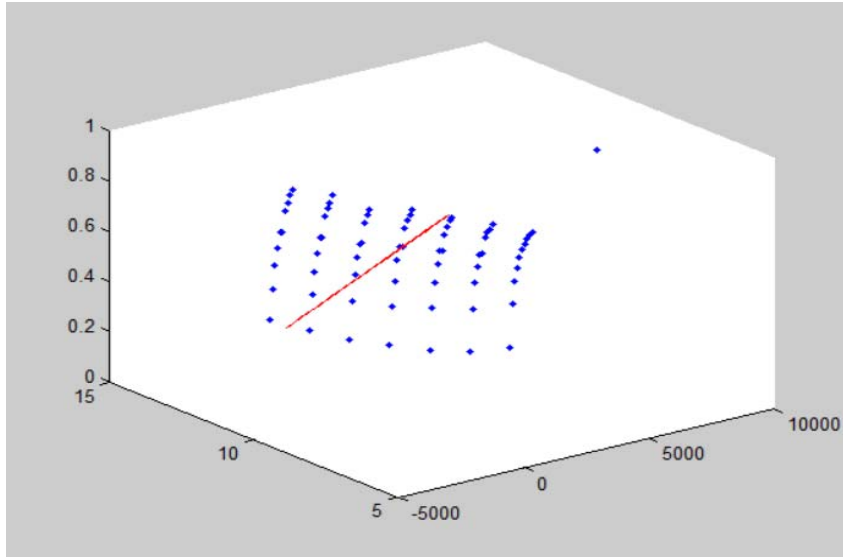


Figure 2 Transfer station running coefficient and speed, the distance fitting diagram

Therefore, we get the transit station operating coefficient and subway line design speed, the distance between the two transit points have the following function:

$$\begin{cases} \Delta x = 1.84\alpha_{\text{transfer}} - 0.46 \\ v = -7.96\alpha_{\text{transfer}} + 14.66 \end{cases} \quad (3.9)$$

#### 3.1.4 A multi-objective programming model is established

Based on the above objective function and constraints, the model is as follows:

$$\begin{aligned} & \max MOBC = f(C_{\text{single}}, C_{\text{transfer}}) \\ & \min n \\ & \text{s.t.} \begin{cases} C_0(v) = -0.21v^2 + 26v + 9.6e + 2 \\ \Delta x = 1.84\alpha_{\text{transfer}} - 0.46 \\ v = -7.96\alpha_{\text{transfer}} + 14.66 \end{cases} \end{aligned} \quad (3-10)$$

#### 3.1.5 The proposed expected results are evaluated based on the multi-objective planning model of the selected destination

##### ● Metro network planning

In the metro construction, the city will be carried out according to their own characteristics of the subway network planning, after the approval by the government department to implement, so in the construction phase, the metro line selection from the terminal point to be based on network planning. As the metro network planning stage, considering the line can be implemented less factors, relatively rough, so in the construction phase to be further combined with the status, planning and other factors, from the implementation point of view the implementation and optimization of line terminal station location.

##### ● Passenger flow

The inter-section passenger flow is the passenger flow between two adjacent stations, and the inter-section passenger flow often has the characteristics of "large middle and small ends". This feature provides the basis for the subway terminal



selection. , The first phase of the project starting from the terminal, and then the rest for the second phase.

Station passenger flow rate determines the size of the station, the terminal from the line selection to bring some impact, because some stations up and down the passenger flow is relatively large, resulting in a larger station, if the station reentry, the station's station size Larger, in order to reduce the station scale, metro line terminal extension.

#### ● ***Passenger attraction***

Transportation hub. There are railway stations, bus stations, airports, ports and so on which have been built, under construction and planned. If the subway terminal is near the transportation hub, the connection between the subway and other modes of transportation can be realized. As the point there are two or more modes of transport transfer, passenger flow scale, high economic efficiency, is the subway line from the end of the preferred location.

The main city center area passenger attraction. According to the subway line passenger flow analysis, the passenger flow has the characteristics of "heart", the main city center area is the passenger density area, so the subway line from the terminal is rarely selected in the main city center area.

Deputy city or town center area passenger attraction. The main city around the city has a number of sub-cities and towns, the downtown area of passenger traffic is relatively large, is the subway from the end point is the ideal choice.

Subway station site. To have built, under construction in the subway station as a subway line from the terminal, to achieve two or more lines of the transfer, the passenger flow scale is larger, can be used as a subway terminal from the choice of location.

Scenic spots. Many tourist cities in China, such as Hangzhou and Wuxi, have successively carried out metro planning and construction. In order to better develop tourism resources and improve the traffic environment of the scenic spots, connecting the scenic spots and the central area, scenic spots as a subway from the end is a good select.

Bus station. There are a number of bus lines in the bus station, and the passenger flow is also large. If the subway terminal is near the bus station, it can achieve the convergence of the two, increase the service radius of the subway and improve the service level of the subway.

In accordance with the passenger flow to attract the point of the passenger flow, the subway line starting from the end of the choice of priorities for the transport hub points, the main city center area passenger attraction, vice city or town center passenger attraction, subway station site, scenic spots and bus station .

#### ● ***Vehicle integrated base***

Vehicle integrated base for the metro vehicles to provide parking and maintenance functions, divided into parking lots and depots. Vehicle integrated base for the metro's largest subsidiary buildings, covers an area of relatively large, of which the vehicle segment covers an area of 16 square kilometers in the above. The comprehensive site of the vehicle is generally located in the suburbs, and distributed in the metro lines at both ends. The integrated base of the vehicles at the stage of the line network will be affected by the relocation of the metro line due to the relocation of houses or other reasons.

## 3.2 Issue II: Adjust the departure interval or establish a direct route, analysis of the economic and social benefits

### 3.2.1 Analysis of issue II

Requiring the adjustment of departure interval or the establishment of direct routes to explore the economic and social benefits can be achieved advantages.

We divide the problem into two cases to solve:

Case 1: only adjust the grid intervals not build straight line.

Case 2: only through the route does not adjust the start interval.

### 3.2.2 Preconditioning of Case I

We have to not affect the train return routes and the premise of the premise of the train departure time to adjust the interval time. We set the actual interval between the two metros to  $h_i (i = 1, 2, 3, \dots, n)$ , the time interval between two columns in the schedule is set to  $D_i$ , minimize metro outbound time sum can be expressed that:

$$\min \sum_{i=1}^n |h_i - D_i| \quad (3.11)$$

$$h_i = t_i - t_{i-1} \quad D_i = T_i - T_{i-1} \quad (3.12)$$

Actual departure time is  $t_i$ , Timetable departure time is  $T_i$

### 3.2.3 Related definitions of case I

The definition of the size of the time interval: the interval is short when  $h_i < D_i$  and the spacing is tight when  $h_i = D_i$  and the interval is a long interval when  $h_i > D_i$ .

### 3.2.4 Build a model for case I

When the metro passes through a group of  $k$  short or tight intervals, followed by  $k+1$  long intervals to extend the short interval, shorten the interval, from the first car start iterating through all the metro:

The first interval for the tight interval or long interval, then it can not extend the first metro outbound time when  $h_1 \geq D_1$ ,  $h_1$  remains unchanged.

The first interval for the short interval, then we must extend the departure time of the first metro when  $h_1 \leq D_1$ , the need to comply with  $h_1 + \sigma_1 \leq D_1$  and  $h_{k+1} \geq D_{k+1}$ .

The first interval is the tight interval, starting from the second metro when  $\sigma_1 = D_1 - h_1$ , this cycle to  $\sigma_1 = D_{k+1} - h_{k+1}$ . The interval of  $k+1$  is a tight interval, the first metro to the group expanded to  $k+1$ , the original group of metro have delays. The process is repeated in a larger group and the frame algorithm is as follows:

**Step 1:** Set up between the two stations running time  $\alpha$ , input parameters  $\alpha_i (i = 1, 2, 3, \dots, n)$

**Step 2:** make  $t_i = \alpha_i (i = 1, 2, 3, \dots, n)$

**Step 3:** make  $x$  for this group of metros the number.  $I$  is the initial value,  $k$  is the number of the last metro in the current group.

**Step 4:** Follow the steps below when  $x \neq n$ .

- If  $h_x \geq D_x$ ,  $x = x + 1$ , then turn to step 4.
- If  $h_x < D_x$ , Obtain the satisfaction of  $\sigma_1$ .
- If  $\sigma_1 = D_1 - h_1$  and  $x = x + 1$ , then turn to step 4.

- If  $t_j + \sigma_1 = \alpha_j^{\max}$  ( $j = 1, 2, 3, \dots, k$ ) and  $x = j + 1$ , then turn to step 4.
- If  $\sigma_1 = h_{k+1} - D_{k+1}$ ,  $h_i = h_i + \sigma_1$  ( $i = 1, 2, 3, \dots, k$ ),  $k = k + 1$ , then turn to step 4.

**Step 5:** Output  $t_i$ , Get the interval time  $h_i$ .

**Step 6:** The time between each site into the different train to get the time, with the original schedule to adjust the time interval.

### 3.2.5 Preconditioning of Case II

Through the construction of direct routes to improve economic and social benefits, the need for each site of the flow of people and business district division, to build a suitable direct route. We select the Nanchang subway as the research object, find the data of Nanchang subway station statistics:

Table 3 Nanchang metro traffic level

| Metro station       | Human traffic level | Metro station        | Human traffic level |
|---------------------|---------------------|----------------------|---------------------|
| Shuanggang          | A                   | Bayi Square          | S                   |
| Jiaoqiao            | A                   | North Dinggong Road  | A+                  |
| Changjiang Road     | A+                  | South Shida Road     | A+                  |
| Zhujiang Road       | A+                  | Pengjiaqiao          | A+                  |
| West Bayi Bridge    | A++                 | Xiejia Village       | A++                 |
| Lvyin Road          | A+                  | Qingshan Lake Avenue | A+                  |
| Huizhan Road        | A++                 | Gaoxin Avenue        | A++                 |
| Metro Building      | S                   | West Aixihu Lake     | A                   |
| Qiushui Square      | A++                 | East Aixihu Lake     | A+                  |
| West Zhongshan Road | A+                  | Taizidian            | A                   |
| Zigu road           | A+                  | Tianxiang Avenue     | A                   |
| Bayi Hall           | S                   | Olympic Stadium      | A+                  |

### 3.2.6 Economic Effects and Population Effects of the Surroundings of Nanchang metro Stations

The empirical results show that the rapid transportation of the subway can produce value-added to the area near the station, and conclude that there are 7 commercial circles: Changbei University, Phoenix, Hongguantang, Chengdong district, high-tech business district.

The following is an analysis of the economic and demographic effects of the seven major business circles.

- Changbei College Circle: high-quality residential land, land-based education research design, students, staff and nearby residents mainly student traffic is relatively large.
- Phoenix Island business district: for large residential areas and supporting business, schools, municipal facilities, health and other land.
- Honggu Tan CBD Business District: Located at the intersection of Honggao Avenue and the west side of Fenggao Avenue, it is located in the heart of Hongguantan CBD. It is surrounded by modern residential area

and supporting commercial area. It has a large number of business, office, residential and passenger flow. A large amount.

- Zhongshan Road shopping district: mainly residential area, the surrounding traffic facilities are more convenient, there are Tengwangge, the Provincial Museum, the Provincial Science and Technology Museum, the city library and residential areas and hotels, cultural relics, historical and cultural tourism commemorative points, , Tourism, office-based.
- Square shopping district: Nanchang is located in the core area, northwest of Nanchang department store, the Northeast is Wal-Mart shopping plaza; southwest of the provincial exhibition center; Bayi Square is the window and the activities of Nanchang center; will attract a large number of commercial passenger , Square passenger flow, office passenger flow and transfer passenger flow, traffic is very large.
- Chengdong shopping district: around the building has a large complex project Hengmao dream Times Square, Chengdong Department Store and Nanchang University campus, to the business, office, residence, colleges and universities. Has a wealth of educational resources for the east of the commercial gathering area.
- High-tech shopping district: Most of the city to be built area, schools, warehouses, planning the region to commercial and residential, administrative office development.



Figure 3 Metro stations around the site distribution



Figure 4 Human traffic and business district combined

We can combine the above chart to get the traffic circle and the surrounding subway station corresponding to the relationship between traffic levels.

Table 4 Metro station traffic level and the corresponding business district

| Metro station traffic level | Business district | Metro station traffic level | Business district |
|-----------------------------|-------------------|-----------------------------|-------------------|
| Shuanggang A                | none              | Bayi Square S               | Square CBD        |
| Jiaoqiao A                  | Changbei CBD      | North Dingon Road A+        | Square CBD        |
| Changjiang Road A+          | Phoenix CBD       | South Shida Road A+         | none              |
| Zhujiang Road A+            | Phoenix CBD       | Pengjiaqiao A+              | ChendonCBD        |
| West Bayi Bridge A++        | none              | Xiejia Village A++          | ChendonCBD        |
| Lvyin Road A+               | none              | Qinshan Lake A+             | Chendon CBD       |
| Huizhan Road A++            | RedBeach CBD      | Gaoxin Avenue A++           | none              |
| Metro Building S            | RedBeach CBD      | West Aixihu Lake A          | none              |
| Qiushui Square A++          | RedBeach CBD      | East Aixihu Lake A+         | none              |
| Zhongshan Road A+           | Zhonshan CBD      | Taizidian A                 | High-tech CBD     |
| Zigu road A+                | Zhonshan CBD      | Tianxiang Avenue A          | High-tech CBD     |
| Bayi Hall S                 | Zhonshan CBD      | Olympic Sdadium A+          | none              |

Combined with the above table and the subway travel speed. Nanchang subway direct line as follows: Shuanggang Station → Metro Building Station →Bayi Hall Station → Xiejia Village →Olympic Sdadium Station. This direct route can bring higher economic benefits and social benefits.

### 3.2.7 Adjusted to bring the economic and social benefits

#### ● To bring people time efficiency

In order to improve the process of regional urbanization, the adjusted metro route will greatly improve the speed of passenger travel, so that it is no longer with the sub-center around the problem due to slow travel and space distance, the high speed of the metro passengers to save the surplus Time to improve overall quality of life.

Saving time benefit:

$$Q = a \times q \times s \times p \times t. \quad (3.13)$$

Percentage of passengers traveling per hour is  $a$  ,Passenger Volume is  $q$  ,Passenger per hour per hour value is  $p$  ,Line distance is  $s$  .

Nanchang to calculate the relevant data:

$$Q = 60\% * 764.2 * 10000 * 60\% * 16 * 20 = 880358400 \text{ (yuan)}$$

- **Improve the efficiency of labor productivity**

Compared with the bus, the subway has a good ride environment, travel fast, Short ride times not only reduce passenger fatigue, but also improve. The efficiency of the work after the passengers have information that passengers uncomfortable in the article. Pieces, every 10 minutes will drive labor productivity by 3% -4%. According to the relevant data, transport fatigue to reduce the value of labor productivity: orbitTraffic is 1.4%, 7% of the bus, then take the metro accordingly High labor productivity 6%.

Labor efficiency is calculated as :

$$B = \beta \times a \times q \times p \times t . \quad (3.14)$$

Take the metro than the bus to reduce the time to improve labor efficiency is  $B$ , Passenger travel trip factor is  $\beta$ , Percentage of passengers traveling per hour is  $a$ . It is concluded that the adjusted subway line can improve the labor efficiency:

$$B = 30\% * 60\% * 764.2 * 10000 * 0.06 * 8 = 660268.8 \text{ (yuan)}$$

### **3.3 Issue III: Write a letter to the Mayor as a summary analysis and a schedule**

#### **3.3.1 Write a letter to the Mayor**

Dear Mayor:

I have the pleasure to tell you that we can change the economic benefits and social benefits of the city by changing the time interval of the metro, so we hope to establish a new metro timetable through the analysis of mathematical model.

We need to maximize the operational efficiency of the route, which needs to be considered based on several factors:

- Average daily traffic per subway station.
- The business situation surrounding each subway station.
- The population at each moment of travel.

At the same time line data for the construction of the most basic basis for the timetable, which includes a number of operations on the number of lines issued, the starting interval, the number of train passengers, train load factor, departure time, arrival time, number of stations, the number of stations.

We select the Nanchang Metro Line as a template, we look for the Nanchang Metro Line 1 of the corresponding site data, average daily traffic and business district situation, and made into a table. Then according to the outbound time adjustment algorithm, calculate the departure time of each train, and then compared with the original schedule to adjust the time interval obtained by using the time interval to obtain the departure time for each site.

We can also build a direct line to improve the economic and social benefits of Nanchang. We use the site we collected the distribution of population levels and the various sites around the scope of the district have to select the choice of direct access to the site.

We can see that Kau Bridge Station and the Yangtze River Road Station in the business district, but the general population cut away from the starting point closer, so do not consider the construction of direct station. Subway station is also located in the center of business district cut population is more intensive, far from the starting point station, so consider the construction of direct station. Bayi Museum Station is located

between the two shopping district and densely populated, the distance from a direct station far, so the station can be considered to build a direct station, thank the village station is far from the commercial and more developed, so the reliability of direct Station, Prince Edward Temple Station and Tianxiang Avenue Station, although in the business district, but closer to the terminal and the population less, so do not be considered.

Your friend.

### 3.3.2 Metro schedule

|                      | The first metro departure |      |                  |      | Last metro departure |       |                  |       |
|----------------------|---------------------------|------|------------------|------|----------------------|-------|------------------|-------|
| Metro station        | starting station          |      | Terminal station |      | starting station     |       | Terminal station |       |
| Shuanggang           | 6:08                      | 6:35 | 7:13             | 7:40 | 21:00                | 20:28 | 22:05            | 21:40 |
| Jiaoqiao             | 6:10                      |      | 7:16             |      | 21:02                |       | 22:08            |       |
| Changjiang Road      | 6:13                      |      | 7:18             |      | 21:05                |       | 22:10            |       |
| Zhujiang Road        | 6:16                      |      | 7:20             |      | 21:08                |       | 22:12            |       |
| West Bayi Bridge     | 6:18                      |      | 7:23             |      | 21:10                |       | 22:15            |       |
| Lvyin Road           | 6:20                      |      | 7:25             |      | 21:12                |       | 22:17            |       |
| Huizhan Road         | 6:22                      |      | 7:27             |      | 21:14                |       | 22:19            |       |
| Metro Building       | 6:24                      | 6:45 | 7:29             | 7:50 | 21:16                | 20:38 | 22:21            | 21:50 |
| Qiushui Square       | 6:28                      |      | 7:31             |      | 21:20                |       | 22:23            |       |
| West Zhongshan Road  | 6:35                      |      | 7:33             |      | 21:27                |       | 22:25            |       |
| Zigu road            | 6:37                      |      | 7:35             |      | 21:29                |       | 22:27            |       |
| Bayi Hall            | 6:39                      | 6:56 | 7:37             | 7:55 | 21:31                | 20:49 | 22:29            | 21:55 |
| Bayi Square          | 6:41                      |      | 7:39             |      | 21:33                |       | 22:31            |       |
| North Dinggong Road  | 6:43                      |      | 7:41             |      | 21:35                |       | 22:33            |       |
| South Shida Road     | 6:45                      |      | 7:43             |      | 21:37                |       | 22:35            |       |
| Pengjiaqiao          | 6:47                      |      | 7:45             |      | 21:39                |       | 22:37            |       |
| Xiejia Village       | 6:49                      | 7:01 | 7:47             | 8:06 | 21:41                | 20:54 | 22:39            | 22:06 |
| Qingshan Lake Avenue | 6:53                      |      | 7:58             |      | 21:45                |       | 22:50            |       |
| Gaoxin Avenue        | 6:55                      |      | 8:00             |      | 21:47                |       | 22:52            |       |
| West Aixihu Lake     | 6:57                      |      | 8:02             |      | 21:49                |       | 22:54            |       |
| East Aixihu Lake     | 7:00                      |      | 8:04             |      | 21:52                |       | 22:56            |       |
| Taizidian            | 7:02                      |      | 8:07             |      | 21:54                |       | 22:59            |       |
| Tianxiang Avenue     | 7:05                      | 7:11 | 8:09             | 8:16 | 21:57                | 21:04 | 23:01            | 22:16 |

**Notes:** the second row of time for the direct metro schedule

## IV. Strength and Weakness

### 4.1 *Strength:*

- The establishment of the model is based on the analysis of the problem when the idea of thinking, first of all to establish the overall objective function, and then the establishment of hierarchical objective function, followed by evaluation of the standard measurement model, the level of progressive easy to understand;
- In the process of solving the model, the method is simple and easy to understand. It has some practicability and extension. In the process of result processing, some tables and graphs are used to make the result clearly reflected;
- We use a scientific method to select from a number of indicators, while some of the quantification of the indicators based on the definition of analogy related to quantitative indicators, and then by the establishment of these indicators to the evaluation system;
- In the establishment of the first question model, we firmly grasp the origin of the same, the principle of moving the end of the establishment of relevant models to prove the end point of the different changes can bring;
- The multi-objective programming model we established has wide applicability, high feasibility, and comprehensively considered the relationship among the metro stations;
- Our model is not affected by the number of subway lines and the different cities, for a variety of urban subway lines of traffic analysis.

### 4.2 *Weakness:*

- In the first issue, we select several aspects of the subway terminal, which have certain subjectivity and discard some aspects with less influence, which reduces the perfection of the evaluation system.
- In the second problem, we only consider the daily traffic volume of each site and the distribution of the surrounding business district, and do not consider the role of schools and tourist attractions in promoting economic benefits and social benefits, so the analysis is not comprehensive;
- Problem three in the construction of direct routes, we only choose a subway line to consider, think more simple.



## V. Future Work

The multi-objective programming model has some applications in resource allocation, planning, production scheduling and so on. It can solve the problem of supplier selection and optimize the performance of supply chain by establishing multi-objective planning model. The development of supply chain channel and the expansion of market demand can solve the problem of optimizing the supply chain in several aspects. , Such as in the production process, manufacturers need to achieve both the highest amount of factory, but also the lowest cost, while the largest profit, environmental standards, transportation and other objectives to meet. As well as in the bus scheduling problems in the application: it is necessary to meet the bus to cover the entire region, but also to meet the number of buses less, while the minimum cost. At the same time in the game arrangement can also be applied.

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## VII. Appendix

### ● Programs and codes I

%Figure 1 The fitting curves of the theoretical running capacity curve of metro line design speed and single metro line

```
x=[20 30 40 50 60];
```

```
y=[1380 1550 1640 1690 1730];
```

```
plot(x,y,'.')
```

```
xlabel('metro line design speed')
```

```
ylabel('The theoretical running capability of a single metro line')
```

### ● Programs and codes II

% Figure 2 Transfer station running coefficient and speed, the distance fitting diagram

```
x=[100 100 100 100 100 100 100 200 200 200 200 200 200 200 300 300 300 300
300 300 300 400 400 400 400 400 400 400 500 500 500 500 500 500 500 600 600
600 600 600 600 600 700 700 700 700 700 700 700 800 800 800 800 800 800 800
900 900 900 900 900 900 900 1000 1000 1000 1000 10000 1000 1000];
```

```
y=[20 25 30 35 40 45 50 20 25 30 35 40 45 50 20 25 30 35 40 45 50 20 25 30 35 40
45 50 20 25 30 35 40 45 50 20 25 30 35 40 45 50 20 25 30 35 40 45 50 20 25 30 35
40 45 50 20 25 30 35 40 45 50 20 25 30 35 40 45 50];
```

```
z=[0.45 0.37 0.31 0.27 0.23 0.2 0.18 0.62 0.54 0.48 0.42 0.38 0.34 0.3 0.71 0.64
0.58 0.52 0.48 0.43 0.39 0.76 0.7 0.65 0.6 0.55 0.5 0.46 0.8 0.75 0.7 0.65
0.6 0.56 0.52 0.83 0.75 0.7 0.65 0.6 0.56 0.52 0.85 0.81 0.76 0.72
0.68 0.64 0.6 0.87 0.83 0.79 0.75 0.71 0.67 0.63 0.88 0.84 0.81 0.77 0.73
0.69 0.66 0.89 0.86 0.82 0.79 0.75 0.72 0.68];
```

```
F=[z;1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1];
```

```
M=F*F'
```

```
N=F*x';
```

```
O=F*y';
```

```
A=(M\N)'
```

```
B=(M\O)'
```

% Linear MATLAB fitting image program

```
x1=1.84*z-0.46;
```

```
y1=-28.67*z+52.79;
```

```
z1=z;
```

```
plot3(x1,y1,z1,'r',x,y,z,'.')
```

# The Direct Metro Lines

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### Abstract

We develop a Q-analysis model and consider different stations as clusters to obtain any two optimal end stations. Those end stations should satisfy three conditions. They are respectively long distance in between, large amount of passenger flow and many middle stations. We get all those two end stations and make them separately starting and terminal point. We categorize new train lines into two sorts, which are both ends are on the same old line and on different lines. For the former one, we decide to utilize the original line and only allocate train schedual to realize. And for the latter one, we consider the situation where passengers will transfer. When the original line does not take a roundabout way, we still make use of the old line and also try to allocate a different subway schedual. When the original route obviously go the long way round, we consider laying new railway to directly connect two stations. Consequenently, we select the following groups of stations. The first group includes five pairs of stations on the same line, which are Maogao bridge-Ande gate, Maogao bridge-Nanjing South, Nanjing-Xinjiekou, Jingtian road-Xinjiekou and Tianrun city-Nannjing South. The second group includes stations that are not on the same road, but the straight distance between does not have much difference with the old line. So we choose not to set up new lines. The metro line in this group is Nanjing-Yuantong. The third group includes stations on different lines and the straight line is clearly shorter. In this group, we choose the line of Ande gate-Kazi gate. Then, we develop the AHP model to evaluate our model. We consider three aspects

of consumers, entrepreneurs and society. For consumers, the key benefit is the hours of passenger's travel, the total distance and transfer index. For entrepreneurs, the key benefit is the bearable intensity and efficiency of the line. For society, the key benefit is city transportation crowdedness, the level of environment pollution, the level of energy consumption and passengers' satisfaction. We endow different indicators with different weight in accord with their importance. After that, we can compare the lines before and after our optimization. We find the new line is evidently better than the old one except for the cost of building the railway. And we can also recover this cost as the passenger flow increases after some time. Finally, we complete the sensitivity analysis, we change the percentage of the increase in the passenger flow after optimization and we find that the change in does not alter the score of evaluation very much. But as far as the economic benefit, the value of will impact the time of retrieving the cost, thus influencing the feasibility of the plan.

Key Words: Q-analysis, AHP

# **1 Introduction**

## **1.1 Restatement of the Problem**

In current metro network of a megapolis ,it usually takes a long time to transport passengers between two far apart stations. We may suggest that set a direct shuttle train between two long-distance stations to save time of the longhaul passengers.

Develop a mathematical framework to optimize the choice of the end stations, and evaluate the expected result of the suggest.

Evaluate the economic and social benefits of the expected result.

Write a two-page letter to the Mayor summarizing the analysis. Include the train schedule as the second page of the letter.

## **1.2 Our Work**

We are required to establish a real, reasonable, useful model to optimize the metro network so as to obtain social and economic efficiency. Our model not only considers choosing the end stations of direct shuttle train but also how to organize the train schedule.

First, We develop a model to select end stations. We take three factors into consideration, including the amount of passenger flow, time taken between two random stations and the number of stations between two. We use Q-analysis to divide train lines connecting two arbitrary stations into several categories, picking out end stations between which we suggest that a direct shuttle train is needed to be set.

And then, based on the model, we get how the amount of passenger flow in direct lines varies with time. For the next step, we arrange the best way the metro runs to minimize the economic cost.

## 2 Assumptions

**The amount of passenger flow in every subway line only changes very slightly.** We found that the data of daily passenger flow on weekdays does not have any obvious change.

**After the direct train line is built, people who shuttle between these two stations will always choose the direct transportation.** Because the direct line makes travel more convenient and save a lot of time, people will always take the direct way.

All the data we have found are reliable.

We only take into consideration two stops on the same subway line.

The time when first subway starts is fixed and no emergency happens.



### 3 Model Design

#### 3.1 Model 1: Q-analysis

| Symbol   | Description                                                                            |
|----------|----------------------------------------------------------------------------------------|
| $Q_{ij}$ | the amount of passenger flow between station i and j before optimization               |
| $t_{ij}$ | the time taken by train from station i to j before optimizaion                         |
| $N_{ij}$ | the number of stations between i and j                                                 |
| $P_{ij}$ | the amount of passenger flow between station i and j after optimizaion(only consid     |
| $U_{ij}$ | the amount of passenger flow between station i and j after optimizaion(consider es     |
| $E_1$    | the efficiency of metro line before optimization                                       |
| $E_2$    | the efficiency of metro line after optimizaion(only consider establishing direct lines |
| $E_3$    | the efficiency of metro line after optimizaion(consider establishing direct lines on d |
| $T_1$    | the time of passenger's outing before optimization                                     |
| $T_2$    | the time of passenger's outing after optimization(only consider establishing direct    |
| $T_3$    | the time of passenger's outing after optimization(consider establishing direct lines   |
| r        | the bearable intensity of the metro line                                               |

Q-analysis is a mathematical framework to describe and analyze structures. These structures are based on sets. This idea was first introduced by Ronald Atkin in the early 1970s. Atkin was a British mathematician teaching at the University of Essex. Crediting the inspiration of his idea to Clifford Dowkers paper (Homology Groups of Relations, Annals of Math, 1952), he

became interested in the algebra of relations in social structures. He tried to explain his idea in both mathematical and also accessible forms to both technical and general audience. His main ideas are reflected in The Mathematical Structure of Human Affairs (1974). That book covers the key ideas in q-analysis and its application to a wide range of examples, like analyzing game of chess, urban structures, politics at university, people and complexes, works of abstract art, and to physics. He contended that q-analysis can be considered as a powerful generalized method wherever we are dealing with relationships among sets.

Denote the evaluation vector of train line  $L_{ij}$  as  $(Q_{ij}, t_{ij}, N_{ij})$ . And all train lines are illustrated in the graph below (Figure 1). We use absolute length to measure the distance between any two points. Also, we use the nearest distance method to calculate the difference between any two categories. The evaluation function can be written as

$$d(L_{ij}-L_{mn})=\|Q_{ij}-Q_{mn}\|+\|t_{ij}-t_{mn}\|+\|N_{ij}-N_{mn}\|$$

$$D(G_p, G_q)=\min\{d(L_{ij}-L_{mn})\}$$

$$L_{mn} \in G_q$$

$$L_{ij} \in G_p$$

Figure 1: the unit length of x axis shows the amount of passenger flow of 200 million people

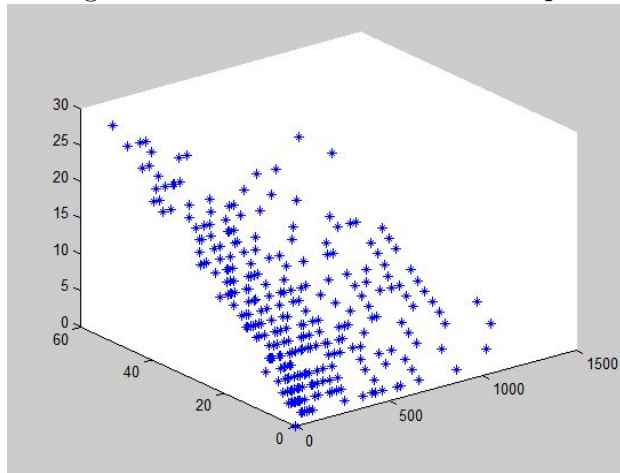
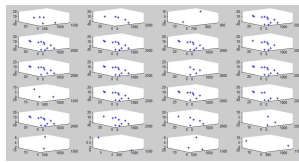


Figure 2: the unit length of x axis shows the amount of passenger flow of 200 million people



We can get the matrix  $A$  from the distance formula and do the cluster analysis with the matrix  $A$ . Then, we can draw the cluster graph. We divide all the routes into 24 kinds. The graph is as below (Figure 2). According to the classification, we pick some of them which satisfy the problem requirement. All the chosen stations are illustrated below in graph.

From these graphs, we can notice that 18 categories from 24 satisfy the requirement, so we pick those points out. At last, we select five groups of stations, which are Maogao bridge-Ande gate, Maogao bridge-Nanjing, Nanjing-Xinjiekou, Jingtian road-Xinjiekou, Tianrun city-Nanjing.

We derive the following results by calculation.

#### **the total time of passenger's outing**

The total time of passenger's outing on the same line before the optimization

$$T1=43\text{min}$$

The total time of passenger's outing on the same line after the optimization

$$T2=34\text{min}$$

We process the total time by normalization and we have

$$\frac{T1}{43}=1$$

$$\frac{T2}{43}=0.79$$

#### **the bearable intensity of the line**

The bearable intensity of the line before the optimization

$$I_1=\frac{Q1}{\sum I}=1.1$$

By searching materials, we estimate that

$$=1.13$$

The bearable intensity after optimization

$$_2=\frac{Q2}{\sum l}=\frac{\times Q1}{\sum l}=1.243$$

After the normalization

$$\frac{1}{1.243}=0.88$$

$$\frac{2}{1.243}=1$$

### **the efficiency of the line**

The efficiency of the line before optimization

$$E_1=\frac{Q1}{\sum q1ij \times lij}=0.7$$

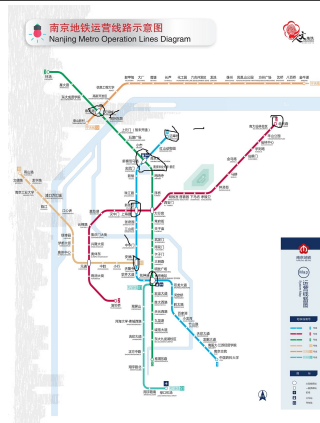
The efficiency of the line after optimization

$$E_2=\frac{Q2}{\sum q2ij \times lij}=0.87$$

After the non-dimensionalization of the line efficiency

$$\frac{E1}{0.87}=0.8$$

$$\frac{E2}{0.87}=1$$



### 3.2 Model 2: the Evaluation of Model 1 by AHP

### 3.2 Model 2: the Evaluation of Model 1 by AHP

It has particular application in group decision making, and is used around the world in a wide variety of decision situations, in fields such as government, business, industry, healthcare, shipbuilding and education.

Rather than prescribing a "correct" decision, the AHP helps decision makers find one that best suits their goal and their understanding of the problem. It provides a comprehensive and rational framework for structuring a decision problem, for representing and quantifying its elements, for relating those elements to overall goals, and for evaluating alternative solu-

tions.

Users of the AHP first decompose their decision problem into a hierarchy of more easily comprehended sub-problems, each of which can be analyzed independently. The elements of the hierarchy can relate to any aspect of the decision problem tangible or intangible, carefully measured or roughly estimated, well or poorly understood anything at all that applies to the decision at hand.

Once the hierarchy is built, the decision makers systematically evaluate its various elements by comparing them to each other two at a time, with respect to their impact on an element above them in the hierarchy. In making the comparisons, the decision makers can use concrete data about the elements, but they typically use their judgments about the elements' relative meaning and importance. It is the essence of the AHP that human judgments, and not just the underlying information, can be used in performing the evaluations.

The AHP converts these evaluations to numerical values that can be processed and compared over the entire range of the problem. A numerical weight or priority is derived for each element of the hierarchy, allowing diverse and often incommensurable elements to be compared to one another in a rational and consistent way. This capability distinguishes the AHP from

other decision making techniques.

In the final step of the process, numerical priorities are calculated for each of the decision alternatives. These numbers represent the alternatives' relative ability to achieve the decision goal, so they allow a straightforward consideration of the various courses of action.

Several firms supply computer software to assist in using the process. We need to assess the optimized train lines, so we use AHP model to appraise the result of model 1. And we consider three factors of passengers, the components that impact enterprise effectiveness( the bearable intensity and the efficiency of the line).

### **the travel time of passengers**

As the travel distance rises, the total time of outings denoted by  $T$  is the most valued factor. The time is composed of two parts, the time on the train denoted by  $t_1$  and the time waiting for the train denoted by  $t_2$ . The formula of this indicator is as below

$$T=t_1+t_2$$

### **the bearable intensity of the line**

The bearable intensity refers to the ratio of daily passenger flow of subway line and the whole length of the line. This indicator reflects how much



passenger flow every unit line is able to carry. The shorter the length of the line, the greater the daily passenger flow, and the higher the bearable intensity is. It is used to assess the efficiency and the economic of the line. By the calculating formual, we know the intensity and the passenger flow is related to the total length.

$$= \frac{Q}{\sum l}$$

In the formula above,  $l$  denotes the bearable intensity of the line,  $Q$  denotes the daily passenger flow of the line, and the total length of the line is denoted as  $\sum l$ .

### **the efficiency of the line**

The calculating formula is demonstrated by (3.8). This indicator is concerned with total amount of passenger flow and the length of the passengers' outing. And the efficiency of subway company is better when the total amount of passenger flow is larger and the time of the outings is shorter. And the elements affecting these two variables are already analyzed before.

$$E = \frac{Q}{\sum q_{ij} \times l_{ij}}$$

In the formula above,  $E$  denotes the efficiency of the line,  $Q$  denotes the amount of passenger flow in subway line, and the total length of the passengers' outing is denoted by  $\sum q_{ij} \times l_{ij}$ . And  $q_{ij}$  is the section flow of adjacent stations  $i$  and  $j$ .  $l_{ij}$  is the distance between stations  $i$  and  $j$ .

Let the weighted vector (the total time of passenger's outing, the bearable

intensity of the line, the efficiency of the line) $=(-0.3, 0.3, 0.4)$

By calculation, we have

The estimated score of the line before optimization is 0.284.

The estimated score of the line after optimization is 0.316.

To conclude, we can see it from the three value of indicators above that the line of which we build direct train line is strictly better than the former one. Although we can not have the cost of establishing the new railway, we can reckon that the economic benefit brought by the direct lines will exceed the cost after a period of time as passengers taking new lines grow.

### 3.3 Model 3: the Improved Q-analysis

In this part, we almost remain all the process of the Q-analysis except that we make transferring lines possible in our new model. By cluster analysis, we choose the train lines to transform as shown in the graph. All the stations that we choose can be divided into these three sorts

#### The first group

The first group includes five pairs of stations on the same line, which are Maogao bridge-Ande gate, Maogao bridge-Nanjing South, Nanjing-Xinjiekou, Jingtian road-Xinjiekou and Tianrun city-Nannjing South.



Figure 4: the distribution of every kinds of points after clustering

### The second group

The second group includes stations that are not on the same road, but the straight distance between does not have much difference with the old line. So we choose not to set up new lines. The metro line in this group is Nanjing-Yuantong.

### The third group

The third group includes stations on different lines and the straight line is clearly shorter. In this group, we choose the line of Ande gate-Kazi gate.

## 3.4 Model 4: the Evaluation of the Model 3 by AHP

We use AHP model to evaluate our result of model 3. Besides the three factors we consider in the model 2, we add another two factors of the total distance of passenger's outing denoted by  $S$  and transfer index. And as we

know the total distance include the distance covered by train and by walk.  
However, we only consider the distance taken on the train.

### **the total time of passenger's outing**

The total time of passenger's outing before optimization

$$T_1=43\text{min}$$

The total time of passenger's outing after optimization

$$T_3=28\text{min}$$

After the normalization, we have

$$\frac{T_1}{43}=1 \text{ and } \frac{T_3}{43}=0.65$$

### **the total distance of passenger's outing**

The total distance before optimization

$$S_1=21 \text{ kilometer}$$

The total distance after optimization

$$S_2=16 \text{ kilometer}$$

After the normalization, we have

$$\frac{S_1}{21}=1 \text{ and } \frac{S_2}{21}=0.76$$

**the bearable intensity of the line**

The bearable intensity before optimization

$$I_1 = \frac{Q_1}{\sum l} = 1.1$$

By searching materials, we estimate that

$$I_1 = 1.27$$

The bearable intensity after optimization

$$I_3 = \frac{Q_3}{\sum l'} = \frac{\sum Q_1}{\sum l'} = 1.33$$

After the normalization, we have

$$\frac{I_1}{I_3} = 0.827$$

$$\frac{I_3}{I_3} = 1$$

**the efficiency of the line**

The efficiency before optimization

$$E_1 = \frac{Q_1}{\sum q_{1ij} \times l_{ij}} = 0.7$$

The efficiency after optimization

$$E_3 = \frac{Q_3}{\sum q_{3ij} \times l_{ij}} = 0.88$$

After the non-dimensionalization, we have

$$\frac{E1}{0.88}=0.79$$

$$\frac{E3}{0.88}=1$$

### **transfer index**

The transfer index before optimization

$$_1=\frac{Q1}{\sum OD1}=1.3$$

The transfer index after optimization

$$_2=\frac{Q2}{\sum OD2}=1.1$$

After the normalization, we have

$$\frac{1}{1.3}=1$$

$$\frac{2}{1.3}=0.846$$

The weighted vector of indicators is

(total time of passenger's outing, total distance of passenger's outing, the  
bearable intensity of the line, the efficiency of the line, transfer  
index)=(-0.15, -0.15, 0.3, 0.24, -0.16)

The estimated score before optimization is -0.0223.

The estimated score after optimization is 0.193.

From the value of the five indicators above, we can see it that the line of which we build direct train line is strictly better than the former one in all five aspects. Although we can not have the cost of establishing the new railway, we can reckon that the economic benefit brought by the direct lines will exceed the cost after a period of time as passengers taking new lines grow.

## 4 Sensitivity Analysis

We try to find the impact of the percentage of the increase in the daily passenger's flow after optimization. So we alter the value of  $\alpha$  we set before.

**When we only consider connecting two end stations on the same line** We change the original  $\alpha$  to 1.20. And we derive that the estimated score before optimization is 0.269 and the estimated score after optimization is 0.316.

**When we consider connecting two end stations on different lines** We change the original  $\alpha$  to 1.35. And we derive that the estimated score before optimization is -0.0484 and the estimated score after optimization is 0.193.

In conclusion, the value of  $\alpha$  does not change the estimated score of the

transportation line very violently. But as far as the economic benefit, the transformation of will influence the time we retrieve the cost of building railways, thus impacting the effectiveness of the plan.

## 5 Conclusion

We develop a Q-analysis model and consider different stations as clusters to obtain any two optimal end stations. Those end stations should satisfy three conditions. They are respectively long distance in between, large amount of passenger flow and many middle stations. We get all those two end stations and make them separately starting and terminal point. We categorize new train lines into two sorts, which are both ends are on the same old line and on different lines. Consequently, we select the following groups of stations. The first group includes five pairs of stations on the same line, which are Maogao bridge-Ande gate, Maogao bridge-Nanjing South, Nanjing-Xinjiekou, Jingtian road-Xinjiekou and Tianrun city-Nanjing South. The second group includes stations that are not on the same road, but the straight distance between does not have much difference with the old line. So we choose not to set up new lines. The metro line in this group is Nanjing-Yuantong. The third group includes stations on different lines and the straight line distance is clearly shorter. And we establish direct lines in between. In this group, we choose the line of Ande gate-Kazi gate. Then, we develop the AHP model to



evaluate our model. We consider three aspects of consumers, entrepreneurs and society. For consumers, the key benefit is the hours of passenger's travel, the total distance and transfer index. For entrepreneurs, the key benefit is the bearable intensity and efficiency of the line. For society, the key benefit is city transportation crowding, the level of environment pollution, the level of energy consumption and passengers' satisfaction. After that, we can compare the lines before and after our optimization. We find the new line is evidently better than the old one except for the cost of building the railway. And we can also recover this cost as the passenger flow increases after some time. Finally, we complete the sensitivity analysis, we change the percentage of the increase in the passenger flow after optimization and we find that the change in  $\alpha$  does not alter the score of evaluation very much. But as far as the economic benefit, the value of  $\alpha$  will impact the time of retrieving the cost, thus influencing the feasibility of the plan.

## 6 Strengths and Weaknesses

### 6.1 Strengths

We successfully use the Cluster-analysis to get the optimal choice of end stations.

We also use the AHP model to evaluate the economic and social benefits of adding the direct train line

## 6.2 Weaknesses

This model does not involve how to connect two end points on two separate subway lines.

We did not consider the situation that the amount of passenger flow will be different on weekdays, weekends and holidays.

We did not think of the condition that railway will be rebuilt.

We can not give a reasonable cost of building new railway lines.

## A Letter to the Mayor of Nanjing

November 27, 2016

Dear Sir or Madam:

We are three college students who concerned about the development and the construction of the city. We've found that with the increase of the urban population and the travel volume of the citizens, the increasingly serious city's congestion arises much more concern. To ease this situation, we propose some suggestions here.

We've found that in current metro network of a megapolis ,it usually takes a long time to transport passengers between two far apart stations. So, to solve this problem, we suggest that we can set a direct shuttle train between two long-distance stations with large passenger flow to save time of the longhaul passengers. By rigorous and scientific argument, we've found that after setting the direct shuttle train, the score of the line network increases significantly, by about 10%.

In spite of our inaccurate estimate of the cost of the construction of the new lines, the passenger flow increases by 13% and 27% after postoptimality, and the line network operation efficiency can be promoted dramatically. And because the passengers travel more convenient, the transport satisfaction also improves. Further more, the direct shuttle train can decrease the energy consumption due to the stops in between end stations, and can protect the environment.

To sum up, we think that the network after postoptimality has more

social and economic benefit. We hope that you can consider our proposal to improve the metro network further.

Yours sincerely,

Team 2759

| Line                        | First Train | Last Train | Time Interval(min) |    |
|-----------------------------|-------------|------------|--------------------|----|
| Maogao bridge-Ande gate     | 6:00        | 23:27      | 12                 | 20 |
| Maogao bridge-Nanjing South | 6:00        | 23:18      | 12                 | 20 |
| Nanjing-Xinjiekou           | 6:00        | 23:25      | 12                 | 20 |
| Jingtian road-Xinjiekou     | 6:00        | 23:27      | 12                 | 20 |
| Tianrun city-Nanjing South  | 6:00        | 23:22      | 12                 | 20 |
| Nanjing-Yuantong            | 6:00        | 23:48      | 5                  | 10 |
| Ande gate-Kazi gate         | 6:00        | 23:58      | 4                  | 8  |

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# Antibiotic Resistance Crisis ? A Simulation Method

## Abstract

To figure out the relationships in this intricate system, we build a dynamic simulation model. In this comprehensive model, we divide the system into three main parts and model them respectively: patient model, hospital model and pharmaceutical company model. We abstract the most concerned interests of the three parts inside of the system: the sales volume of antibiotic, the average treatment time and the resistance. Then we describe the two outside influential factors : the regulated circulation of antibiotics and treatment plans. In this model, we simulate the interactions of each parts and their internal behaviors simultaneously, which is based on the external factors.

To investigate the development without any government intervention, we adapt the Logistic model for prediction. Specifically, we collect the data on average resistance in New York area from 1999 to 2012. We predict the future resistance in New York area. Using the series of data as the input of the above comprehensive model, and we analyse the outputs after we simulate the dynamic process, which are : the resistance rate, the sales volume and treatment time. They all grow rapidly. The resistance even raise up to 70%. Hence making the policy is imperative.

We extract the two major concerns from three parts: average treatment time and sales volume of antibiotics. We combine them as comprehensive evaluation index of system. We formulate a range of policies to regulate the external factors. We observe that comprehensive evaluation value of system varies with different policies. We conclude that our policy is : to control the circulation of antibiotics in New York area from 100 to 120 standard units per 1000/pop.

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## Introduction

These days the antibiotics misuse has drawn people's attention. Today, Bacteria that have evolved to withstand antibiotics kill 700,000 people each year. Significant concerns have been expressed following the spread of resistance. The World Health Organization(WHO) [1] organized meetings, but experts got bogged down. Since the experts can hardly regulate antibiotic smoothly and have no idea how to measure resistance, we analyse the system factors including hospital administrators, patients, pharmaceutical companies, etc. Measurement of different relationships requires an precise model and a suitable method.

So our goal is pretty clear:

- Model the system to describe the conflict of interest.
- Devise metrics to evaluate the influences on each side's interest.
- Develop a specific method to measure the antibiotic resistance.
- Predict the trend toward the crisis without government intervention.
- Write a brief report and propose a set of policies based on model.

Our approach is:

- Simplify these factors that can affect both supply and demand.
- Search the literature related to antibiotic resistance.
- Collect experimental data from WHO reports and other sources.
- Build a model to simulate the real situation and predict the possible resistance trend.
- Interpret the practical circumstances with predicted results.
- Do further discussion based on our work.

## The Comprehensive Model Of System

The comprehensive model is abstracted from the practical process (See Figure 1 below) , i.e., a dynamic simulation of conflict of interests from both sides. We divide the whole system into three parts: the patients, the hospitals and pharmaceutical companies. Each part has its own interest. The whole system consists of three kinds of models, and we will thoroughly introduce them afterwards.

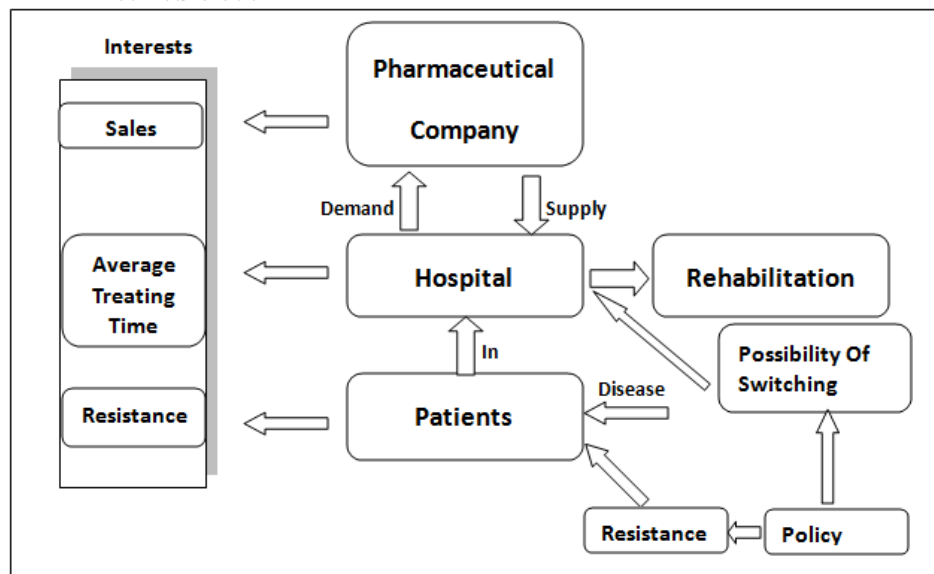


Fig 1: The Comprehensive Model

## The Patient Model

### General Assumption

There is a wide range of antibiotics and bacteria in the medical world, and their intricate interactions make it extremely tough to collect complete data set. To save effort, we assume that

- The colonized patients in our research only carry *Klebsiella pneumoniae*.
- For those who carry other kinds of germs, we only need to modify few parameters.
- Resistance are uneven distributed in different areas, so our research is conducted only in one area.

### Poisson Stream On Patients' Arrival

Suppose  $N(t)$  is the numbers of patients reaching the hospital during the interval  $[0, t)$  ( $t > 0$ ), and  $P_n(t_1, t_2)$  stands for the possibility of  $n (\geq 0)$  patients, i.e.,

$$P_n(t_1, t_2) = P\{N(t_2) - N(t_1) = n\} \quad (t_2 > t_1, n \geq 0)$$

when  $P_n(t_1, t_2)$  is qualified for the following three conditions, we say that the arrival of patients forms Poisson stream. The three conditions are:

- 1 The patients arrivals are independent in non-overlapping intervals, this property is called non-after-effect property.



2 For sufficiently small  $\Delta t$ , the possibility of one patient's arrival and  $t$  are irrelevant in time interval  $[t, t + \Delta t)$ , and it is about proportional to the length of interval  $\Delta t$ , i.e.,

$$P_1(t, t + \Delta t) = \lambda \Delta t + o(\Delta t)$$

where  $o(\Delta t)$ , while  $\Delta t \rightarrow 0$ , is a higher order infinitesimal related to  $\Delta t$ .  $\lambda (> 0)$  is a constant, which represents the possibility of one patient's arrival per unit time, called probabilistic strength.

3 For sufficiently small  $\Delta t$ , the possibility of two or more patients' arrivals is minimal, so that it can be neglected, i.e.,

$$\sum_{n=2}^{\infty} P_n(t, t + \Delta t) = o(\Delta t)$$

Under these conditions, we study on probability distribution of patients' arriving number  $n$ .

Under condition 2, we can count the time from 0, denoted as  $P_n(0, t) = P_n(t)$ .

Under condition 1 and 2, then

$$P_0(t, t + \Delta t) = P_0(t) P_0(\Delta t)$$

$$P_n(t, t + \Delta t) = \sum_{k=0}^n P_{n-k}(t) P_k(\Delta t), \quad n = 1, 2, \dots$$

Under condition 2 and 3

$$P_0(\Delta t) = 1 - \lambda \Delta t + o(\Delta t)$$

therefore

$$\begin{aligned} \frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} &= -\lambda P_0(t) + \frac{o(\Delta t)}{\Delta t} \\ \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} &= -\lambda P_n(t) + \lambda P_{n-1}(t) + \frac{o(\Delta t)}{\Delta t} \end{aligned}$$

According to the above two formula, we let the limit of  $\Delta t$  tends to 0. When the function involved in the assumption is derivable, we get differential equations that follows:

$$\begin{aligned} \frac{dP_0(t)}{dt} &= -\lambda P_0(t), \\ \frac{dP_n(t)}{dt} &= -\lambda P_n(t) + \lambda P_{n-1}(t), \quad n = 1, 2, \dots \end{aligned}$$

Choose the initial value  $P_0(0) = 1, P_n(0) = 0 (n = 1, 2, \dots)$ , easily we solve  $P_0(t) = e^{-\lambda t}$ . Then we let  $P_n(t) = U_n(t) e^{-\lambda t}$ , and we get differential equations which  $U_0(t)$  and other  $U_n(t)$  satisfy, i.e.

$$\frac{dU_n(t)}{dt} = \lambda U_{n-1}(t), \quad n = 1, 2, \dots$$

$$U_0(t) = 1, \quad U_n(t) = 0$$

Easy to work out

$$P_n(t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 1, 2, \dots$$

Just as what we learned in the theory of probability, random variable  $\{N(t) = N(s+t) - N(s)\}$  obey the Poisson distribution. Its expectation and variance respectively are:

$$E[N(t)] = \lambda t, \quad \text{Var}[N(t)] = \lambda t$$

When the process of input is Poisson stream, thus the time interval  $T$  of patients' successive arrival surely obey the Poisson distribution. Because

$$P\{T > t\} = P\{\text{number of calls} = 0, \text{in } [0, t]\} = P_0(t) = e^{-\lambda t}$$

Then we take  $F(t)$  to express the distribution function of  $T$ , thus

$$P\{T > t\} = F(t) = \begin{cases} 1 - e^{-\lambda t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Yet the distribution function is

$$f(t) = \lambda e^{-\lambda t}, t > 0$$

For Poisson stream,  $\lambda$  represents the average quantity of arriving patient in unit time, and thereby  $\frac{1}{\lambda}$  express the average time intervals while patients successive arrive, which equates with the meaning of  $ET$ .

### Antibiotic Resistance

Every patient with germ exhibits resistance at different degree. From clinical data[7] we know that the average resistance *Klebsiella pneumoniae* in the New York area. We hold that the patients' resistances are randomly distributed at  $(0, 1)$ , and the mean value is used as average resistance. The figure below shows the variation of resistance according to data.

### Recovery Rate

We assume that patients can be treated with cephalosporins or second-line drugs[3]. The recovery rate of cephalosporins treatment equals  $f(c)$ , related to categories of resistance, i.e.,

$$R_c = \frac{R}{1 - Res}$$

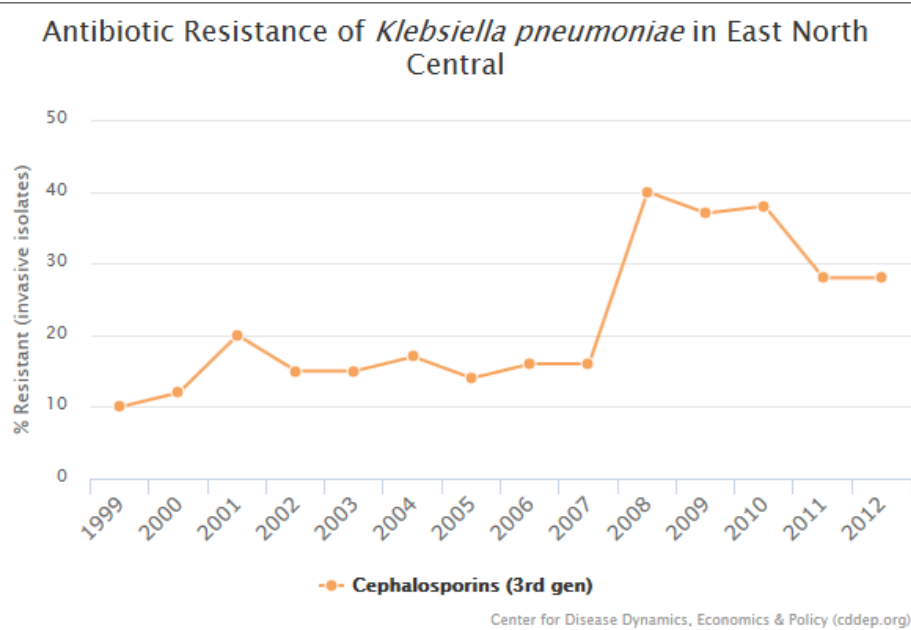


Fig 2: The Average Resistance

where  $Res$  is the resistance of patients carrying *Klebsiella pneumoniae* (from 0 % to 100%) ,  $R_c$  is recovery rate of cephalosporins treatment and  $R$  is rate of cephalosporins treatment for patients with germ of no resistance.

While treating with Second-line drugs, where there is no resistance. The recovery rate of second-line drugs treatment is a constant  $R_s$ .

The rehabilitation condition can be presented as

$$Reh = t_c * R_c + t_s * R_s$$

where  $Reh$  is for rehabilitation condition,  $t_c$  is for cephalosporins treatment time,  $t_s$  is for second-line drugs treatment time,  $R_c$  and  $R_s$  is same as above.

## The Hospital Model

We assume the hospital is a closed environment into which patients enter and leave. Within the hospital, patients are with respect to colonization and treatment with one of two drugs for the target bacteria (*Klebsiella pneumoniae*).

Using stochastic simulation model, we estimate average treatment time of patients in hospital. We study how the average treatment time vary with different factors involved in the system. Treatment is initiated with the cephalosporins. Patients may be switched to the second-line drug in one of two ways:

- (i) At random, with a probability of switching each day. The probability

depends on treatment effect, which changes with time. It is calculated via

$$P_{sw} = 0.5 * \left[ \left( 1 - \frac{Reh}{t_c * R_c} \right) + t_c * R_c \right]$$

(ii) Directly, when the patient remains on the Cephalosporins until she/he has been treated  $2T$  (two times longer than those without resistance), doctor will think she/he is colonized with bacteria resistant to that drug.

$U$  represents patients without being colonized, and  $R$  is patients colonized with bacteria. On any given day, colonized patients carrying the bacteria are of three states with respect to treatment:  $R_0$  expresses "untreated",  $R_1$  expresses "treated with *Klebsiella pneumoniae*",  $R_2$  is "treated with second-line drugs". The letters,  $U$ ,  $R_0$ ,  $R_1$ , etc, are both the designations of these patient states and their numbers (proportions) in the hospital. See Figure for a diagram of this model. The parameters and variables in this basic model are separately defined in Table 1 that follows.

Tab 1: Parameters and their default values

| Symbol   | Description                                   | Value In Simulation                                                       |
|----------|-----------------------------------------------|---------------------------------------------------------------------------|
| $e_R$    | Possibility of patient's admission            | $P_n(t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 1, 2, \dots$ |
| $P_{sw}$ | Possibility of switch to different medication | $Reh = t_c * R_c + t_s * R_s$                                             |
| $R_c$    | Recovery rate of cephalosporins treatment     | $R_c = \frac{R}{1 - Res}$                                                 |
| $R_s$    | Recovery rate of second-line drugs treatment  | 0.1                                                                       |
| $x$      | Possibility of self-healing                   | 0.001                                                                     |

## The Pharmaceutical Company Model

The cephalosporins and second-line drugs provider is the pharmaceutical company. According to the related reports, the main profit source[6] of pharmaceutical company is antibiotics (where it is cephalosporins), so the main interest they care about is sales volume of antibiotics. Therefore, the production of antibiotics directly depends on the demand of local hospitals.

## The Prediction Model

We collect related data on *Klebsiella pneumoniae*'s average resistance to cephalosporins in New York area from 1999 to 2012[7]. The existing data of years is approached by fitting analysis using the Logistic model, and we will predict the development trend of resistance without any governmental intervention, while testing the fitting result with  $\chi^2$  test. Finally we get the prediction formula

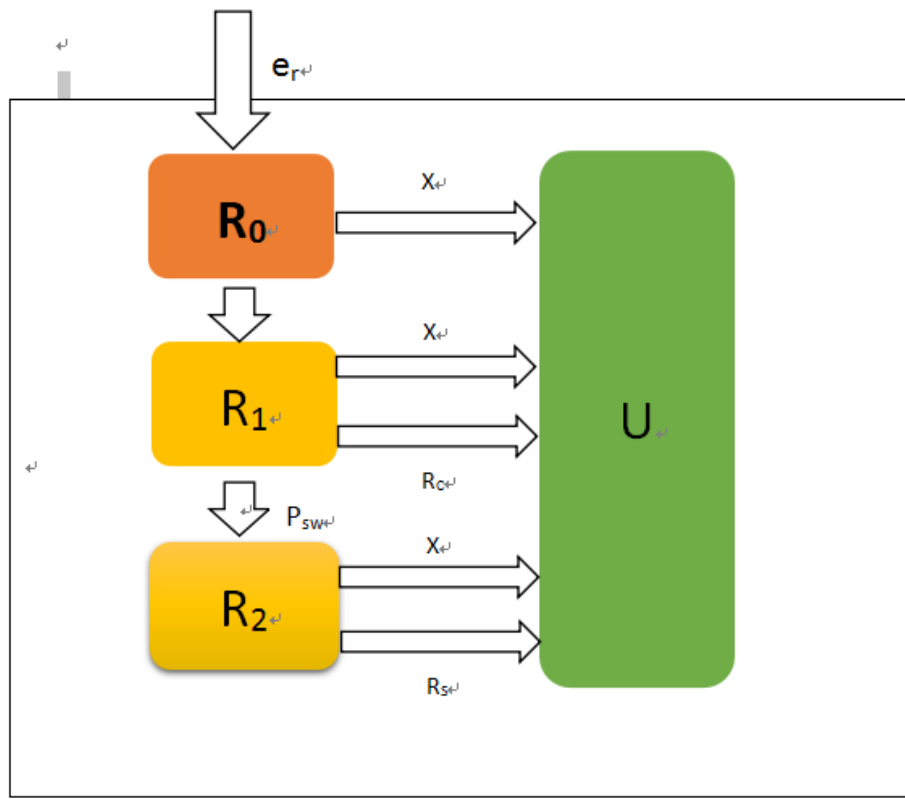


Fig 3: The Switching Process

of average resistance to *Klebsiella pneumoniae* without government intervention. We adapt the Logistic model[4] to approach the fitting analysis of data on cephalosporins resistance of *Klebsiella pneumoniae*. Here is the Logistic model:

$$\frac{dx}{dt} = r * x(t) \left[ 1 - \frac{x(t)}{N} \right] \quad (1)$$

Where  $x(t)$  represents the quantity of population at time  $t$ ,  $r$  is the intrinsic rate of increase (i.e., growth rate minus mortality rate),  $N$  is a constant, being the environmental carrying capacity. Using the theory for differential equation stability we know that[5],  $x = 0$  is unstable equilibrium point, and  $x' = N$  is stable equilibrium point. Seeking the analytical solution of the Logistic model, we use the method of separation of variables, and we employ separation of variables of differential equation, to seek the analytical solution of the above equation:

$$x(t) = \frac{N}{1 + \left[ \frac{N}{x_0} - 1 \right] e^{-rt}} \quad (2)$$

Where  $x_0$  is primary resistance rate,  $t$  is year.  $r$  is a constant called natural

increase rates of resistance rates,  $N$  is a constant as well, called environmental carrying capacity, where is maximal resistance rate.

We choose  $t_0, t_1, t_2$  for resistance rates in three years  $x_0, x_1, x_2$ , and let  $t_1 - t_0 = t_2 - t_1 = \tau$  to solve the parameter  $r, N$  of the Logistic model:

$$r = \frac{1}{\tau} \ln \frac{\frac{1}{x_0} - \frac{1}{x_1}}{\frac{1}{x_1} - \frac{1}{x_2}}, \quad N = \frac{1 - e^{-r\tau}}{\frac{1}{x_1} - \frac{1}{x_0} e^{-r\tau}} \quad (3)$$

Then we plug formula (2) into formula (3), thereby we get the Logistic model of bacterial resistance.

We substitute this model into formula (2) to get predicted values, and we use fitness  $\chi^2$  test to employ goodness-of-fit test on fitting results. Here is the equation of goodness-of-fit test:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E} \quad (4)$$

Where  $O_i$  is the actual value in time  $i$ ,  $E_i$  is the actual predicted value in time  $i$ . If  $\chi^2 \geq \chi_{0.05}^2$ , i.e.  $P \leq 0.05$ , which indicates that the equation has a bad fitting result, means the actual value and predicted value does not fit, so it cannot be used for predicting the future, and vice versa. If  $\chi^2 \leq \chi_{0.05}^2$ , i.e.  $P \geq 0.05$ , which suggests that the equation has a good fitting result, means the actual value and predicted value fit well, we can use it for predicting the future. The value of  $\chi_{0.05}^2$  is looked up on  $\chi^2$  table as degree of freedom  $f = n - 1$ ,  $n$  is logarithm of actual value and predicted value.

## Result

The first three points of the drug resistance data in New York from 1999 to 2012 were selected to calculate the parameters  $r, N$  for the prediction formula. And the result formula is:

$$x(t) = \frac{0.27}{1 + \left(\frac{0.27}{0.12} - 1\right) e^{-0.38t}}$$

Then, the two parameters were used to predict the resistance data from 2002 to 2012, and then compared the resistance data which were predicted using the formula just now with the actual data. And the results shown in Figure:

From the figure, except the years from 2007 to 2008, where the original data suddenly increased, the gap between the forecast data and the actual data is relatively small. It is noteworthy that the data obtained from Internet is not a natural developing trend of resistance, the United States has carried out the policy of antibiotic control many years ago. However, the chi-square test showed that

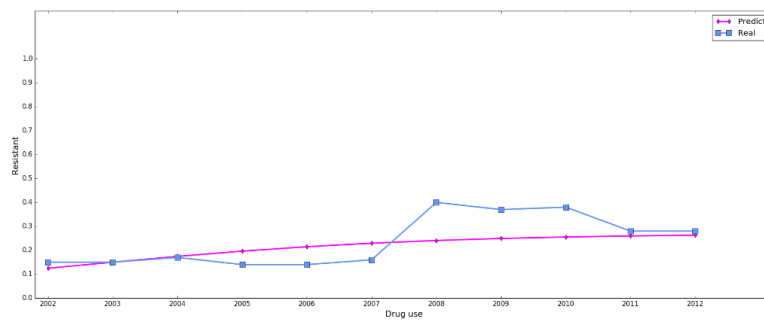


Fig 4: Prediction Of Resistance

$\chi^2$  between the prediction values and actual values was 0.98, which was much higher than that of  $\chi^2_{0.05}$ . Therefore, the predictive result was acceptable, indicating that the formula predicted the drug resistance trend under a controlled state successfully, so it could be used to predict the development of drug resistance respectively. In order to predict the drug resistance development with a free market, we need to select those points which were close to the natural development trend from the drug resistance data in New York from 1999 to 2012. By looking at biology-related papers, we can obtain the natural development of the resistance from experimental data. For this point, by comparison, we select the data of 2007, 2008, 2011 three years which were close to a natural state to predict the trend of resistance development for the next years. Through these three points, we again calculated the parameters  $r$ ,  $N$ , with which to calculate the drug resistance and simulate treatment time in the future. The forecast results are shown as follows:

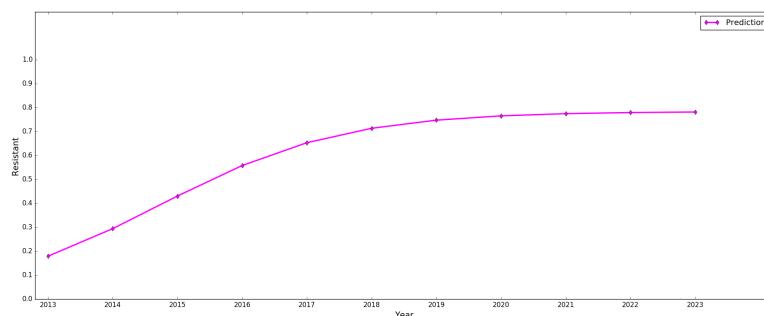


Fig 5: Predict Resistance In Next 10 Years

It can be seen that the results were in line with our expectations. Drug resistance grow faster in the next few years, and then tends to be gentle, which corresponds with the trend of Treatment time. At the same time, the results al-

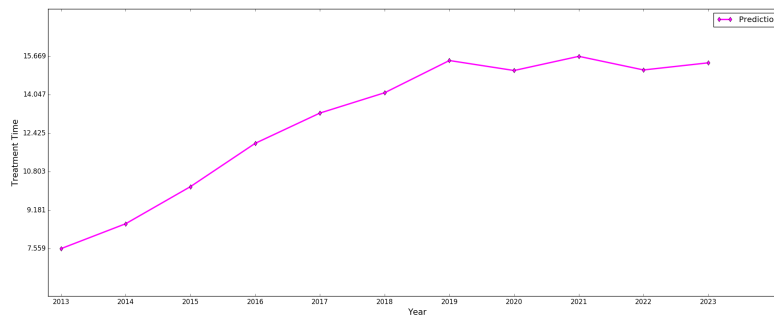


Fig 6: Prediction of Treatment Time

so show that without government regulation, the drug resistance of New York area will grow rapidly. When the drug resistance ratio increases to 60%, or even 70% or more, the existence of this antibiotic itself will lose its meaning. And at that time, the government will have to spend a lot of manpower and financial resources to develop another new and effective drug.

## Policy-Making Model

First we define a index *StatusIndex* to evaluate the conditions of model. The two major concerns of this comprehensive model are :

- The sales volume of antibiotics in pharmaceutical company.
- The mean treatment time of patients.

Both of two data sets are not in the same order of magnitude, therefore we normalize the index (1) and (2) according to the following equations:

$$sales_{(n)} = \frac{sales - \min(sales)}{\max(sales) - \min(sales)}$$

$$mTime_{(n)} = \frac{mTime - \min(mTime)}{\max(mTime) - \min(mTime)}$$

Here we define *StatusIndex*:

$$StatusIndex = mTime_{(n)} + \frac{1}{sales_{(n)}}$$

We collect usage amount of cephalosporins in New York area from 1999 to 2012[7] as the following figure 7 shows.



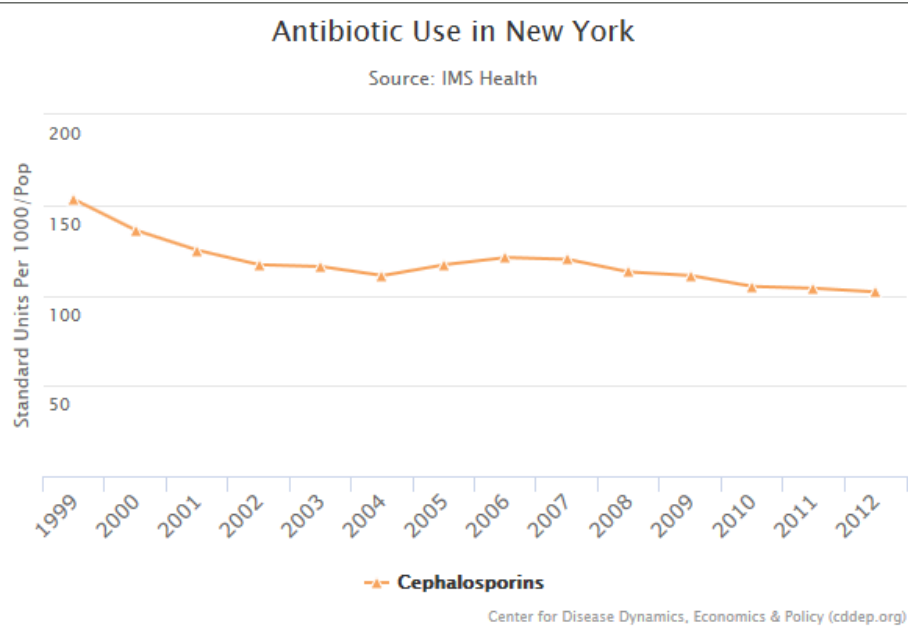


Fig 7: The Usage Amount Of Cephalosporins In New York Area

We use the correlation test to examine usage amount of antibiotics and average resistance in this area, the result suggests that both have relatively strong correlation. So we fit the usage amount and resistance, thus gain the function of them. The first policy we would make is to control the use of antibiotic in the local area, the suggested value is proposed by our comprehensive model specifically.

The methodology is:

- We adjust continuously to restrict the quantity of antibiotics, and get the change value of resistance by the function of usage amount and resistance.
- Then we input the result to comprehensive simulate model, let it outputs the two major concerns of model, and calculate the *StatusIndex* from above equation.
- We plot the graph the function of *UsageAmount* – *StatusIndex*, and seek the optimal solution.
- The limit value government suggested should be given afterwards.

In order to determine the optimal policy, it is necessary to determine the relationship between the quantity of antibiotics flux allowed to circulate in the market and the duration of treatment, drug resistance and other factors Firstly, the relationship between the quantity of antibiotic flux and drug resistance in the New York area from 1999 to 2012 was fitted. This relationship will be used

to simulate the involution index under different controlling levels of policies. By observing the relationship between drug resistance and dosage, the type function we use to fit it is:

$$y = a \ln(b * x)$$

And the fitting result is shown as follow:

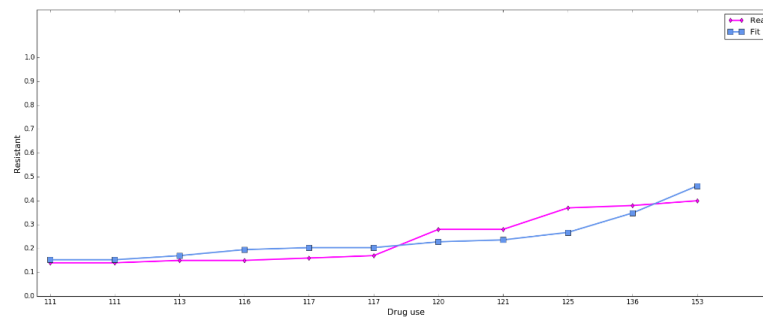


Fig 8: Resistance & Circulation Of Antibiotics

The result shows that chi-square test value of the prediction was 0.99, which was much larger than  $\chi^2_{0.05}$ , so the result was acceptable. The fitting parameters of the type function were that  $a = 0.96$  and  $b = 0.01$ . Next, to determine the optimal policy, we vary the quantity of drug flux from 80 to 200, and use the fitting function to calculate resistance with this flux. Then with the resistance, we simulate the mean treatment time of the patients and the sale of antibiotics in the hospital which were in order to measure the involution index. The simulation results are shown in Fig 9.

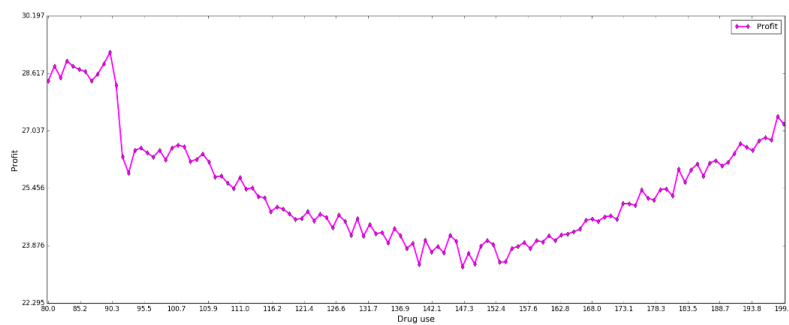


Fig 9: The Evaluation Index Of Hospital

## Discussion

### Strengths

- Focus on these three main factors of the system, and neglect other noises , thus emphasize the essential of problem.
- The dynamic simulation well presents the process of each parts and their interactions, so that the policy-maker of government thoroughly understand these strategies on both sides.
- Our initial fitness result appears not good enough, so that we change a few parameters. During this trial-and-error process, our model can be more valid.

### Weakness

- The function to predict the development of drug resistance is sensitive to the original data, and further optimization can be designed by providing more details. In this paper, the data were obtained from a regulated market by the government.
- During the simulation of the hospital, the probability with which a doctor try to find an alternative drug here is a rough estimate. Specifically, we need to consider the level of a doctor and the actual disease situation. Here is a bit simplified, in order to obtain a better generalization performance.

## Conclusion

To this end, in order to coordinate the interests of the hospital and market development, and effectively control the rise of drug resistance simultaneously, the government can control quantity of the antibiotics flux in the market to an interval between 100 to 120, then drug resistance ratio of this area should be below 23%. At the same time, the health workers should raise their prevention awareness, try to detect drug-resistant patients as early as possible and seek to alternative drugs timely. At the same time, in order to prevent a sudden outbreak of drug resistance, the research institutions should step up new antibiotics or alternative drug research. Equally important, hospitals and health institutions should strengthen linkages and facilitate information sharing.

To this end, we recommend that the government develop the following policies (in the case of the New York area):

- Restrictions on the area of drug resistance in the circulation between 100 to 120 standard units per 1000/pop, and restrict the usage of non-medical antibiotics (which is set for farmers) .

- Advance development and use of rapid and innovative diagnostic tests for identification of resistant bacteria.
- Accelerate basic and applied research for new antibiotics and other therapeutics, and vaccines.
- Promote collaboration among different research institutions for antibiotic resistance prevention, surveillance and control.

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# Appendix

```
# -*- coding: utf-8 -*-
from scipy import stats
import numpy as np
from numpy.random import rand
import math
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import chisquare
from scipy.optimize import curve_fit

pre_rst = [0.1, 0.12, 0.2, 0.15, 0.15, 0.17, 0.14, 0.14, 0.16,
           0.4, 0.37, 0.38, 0.28, 0.28]
pre_use = [153, 136, 125, 117, 116, 111, 117, 121, 120, 113, 111, 105,
           104, 102]

def cal_r(t, x0, x1, x2):
    r = (1/t)*math.log((1/x0-1/x1)/(1/x1-1/x2))
    return r

def cal_n(r, t, x0, x1):
    n = (1-math.e**(-r*t))/(1/x1-math.e**(-r*t)/x0)
    return n

def cal_rst(t, r, n):
    return n/(1+(n/0.1-1)*math.e**(-r*t))

r=cal_r(2.0, 0.14, 0.28, 0.37)
n=cal_n(r, 2.0, 0.14, 0.37)

pdt_rst = []
years = len(pre_rst)-2
for i in range(1, years, 1):
    pdt_rst.append(cal_rst(i, r, n))

# pre_use = pre_use[:len(pre_use)-3]
```

```

pre_use = sorted(pre_use)
pdt_rst = sorted(pdt_rst)
# z1 = np.polyfit(pre_use, pdt_rst, 3)#用 3 次多项式拟合
# p1 = np.poly1d(z1)
# 模拟医院，医生，病人的动作
def func(x, a, b):
    return a*np.log(b*x)
popt, pcov = curve_fit(func, pre_use, pdt_rst)
a=popt[0]#popt 里面是拟合系数，读者可以自己 help 其用法
b=popt[1]
print a,b
print 'func:', func(147, a, b)
print 'optimal:', func(120, a, b)
class Patient:
    # p_id 为病人的病号，drug 为当前用药, resistant 为抗药性，recover 为恢复状况，cure_days 为已经治疗的天数
    def __init__(self, pid, drug, resistant, recover, cure_days):
        self.pid = pid
        self.drug = drug
        self.resistant = resistant
        self.recover = recover
        self.cure_days = cure_days

RESISTANT = 0.5
T1 = 5
T2 = 10
DAY_PATIENTS = 30 # 每天 30 个病人

p_id = 0

sim_cure_days = []
sim_d1_sale = []

pre_use = [153, 136, 125, 117, 116, 111, 117, 121, 120, 113, 111, 105, 104, 102]
mean_use = np.mean(pre_use)

# 描述进入医院的病人
def ent_hpt(patients):
    # day_rst = stats.poisson.pmf(np.arange(DAY_PATIENTS), RESISTANT) # 每天病人的抗药性分布

```

```

day_rst = rand(DAY_PATIENTS)
tmp = np.mean(day_rst)
for i in range(len(day_rst)):
    day_rst[i] = day_rst[i]*RESISTANT/0.5
day_ptn = map(lambda x: Patient(p_id, 1, x, 0.0, 0.0), day_rst)
patients += day_ptn
return patients

def cure(patients, rec_ptn, drug1_sale, drug2_sale, use, total_cure_days):
    tmp = []
    for p in patients:
        if p.drug == 1 and p.cure_days > 0:
            # cdp = use*((0.5*(1-p.recover/p.cure_days) +
0.5*p.cure_days/T1)**2)
            # cdp = (use*(1-p.recover/p.cure_days))**2
            cdp = 0.3*((1-use)*(1 - p.recover / p.cure_days))
            if rand() < cdp:
                p.drug = 2
        rec_ratio = 0
        if p.drug == 1:
            rec_ratio = (float)((1-p.resistant))/T1
            drug1_sale += 1
        if p.drug == 2:
            rec_ratio = 1.0/T2
            drug2_sale += 1
        p.recover += rec_ratio
        p.cure_days += 1
        if p.recover >= 1:
            tmp.append(p)
            rec_ptn.append(p)
            total_cure_days.append(p.cure_days)
    for p in tmp:
        patients.remove(p)
    return patients, rec_ptn, drug1_sale, drug2_sale, total_cure_days

def sim_hosp(use):
    all_patients = []
    rec_patients = []
    cure_days = []
    d1_sale = 0
    d2_sale = 0
    for i in range(30):

```

```

    patients = ent_hpt(all_patients)

    while all_patients:
        patients, recover_patients, dls, d2s, cure_days = cure(patients,
rec_patients, dl_sale, d2_sale, use, cure_days)
        dl_sale = dls
        dl_count = 0
        for p in rec_patients:
            if p.drug == 1:
                dl_count += 1
        sim_cure_days.append(np.mean(cure_days))
        sim_dl_sale.append(dls)
        # print '平均治疗时间: ', np.mean(total_cure_days)
        # print '抗生素销售: ', dl_sale
        # print '二线药物销售: ', d2_sale
        # print '坚持使用抗生素病人数量: ', dl_count

# 市场自由发展
next_cure_days = []
next_dl_sale = []
max_use = np.mean(pre_use)
RESISTANT = func(max_use, a, b)

curr_use = max_use
pdt_rst = [0.1792628046926503, 0.2943181818181818, 0.43063362673704436,
0.558124635993011, 0.6535518026442507, 0.713738475784049, 0.7476984370881284,
0.7656659431653637, 0.7748492466183726, 0.7794600206940622,
0.7817543053571258]

for rst in pdt_rst:
    print rst
    RESISTANT = rst
    sim_hosp(curr_use/200)
print sim_cure_days
print sim_dl_sale

sim_uses = np.arange(80, 200, 1)
min_use = np.min(sim_uses)
max_use = np.max(sim_uses)
for use in sim_uses:
    RESISTANT = func(use, a, b)
    sim_hosp(use)
    print use, '==>', RESISTANT, '==>', sim_cure_days[len(sim_cure_days)-1]

```



```

print sim_cure_days
print sim_dl_sale

dls_min = np.min(sim_dl_sale)
dls_max = np.max(sim_dl_sale)
sim_dl_sale = map(lambda x: x*1.0/dls_max, sim_dl_sale)
cd_min = np.min(sim_cure_days)
cd_max = np.max(sim_cure_days)
# sim_cure_days = map(lambda x: x*1.0/cd_max, sim_cure_days)

profits = []

for i in range(len(sim_cure_days)):
    profits.append(sim_cure_days[i]+10.0/sim_dl_sale[i])

print profits

uses = sim_uses
min_pft = np.min(profits)
max_pft = np.max(profits)
print max_pft, min_pft
plt.figure(figsize=(30, 9), dpi=80)
plt.yticks(np.linspace(min_pft-1, max_pft+1, num=(int) (max_pft-
min_pft)+1, endpoint=True))
plt.xticks(np.linspace(min_use, max_use, num=(max_use-
min_use)/5+1, endpoint=True))
plt.ylim(min_pft-1, max_pft+1)
plt.xlim(min_use-0.2, max_use+0.2)
plt.ylabel('Profit', fontsize=15)
plt.xlabel('Drug use', fontsize=15)
plt.plot(uses, profits, label='Profit', marker='d', linewidth=2.5, markersize=6,
color='fuchsia')
#
plt.plot(X, fit_vals, label='Fit', marker='s', linewidth=2.5, markersize=10, colo
r='cornflowerblue')
plt.legend(loc='upper right')
plt.show()
plt.figure(figsize=(30, 9), dpi=80)
plt.yticks(np.linspace(min_pft-1, max_pft+1, num=(int) (max_pft-
min_pft)+1, endpoint=True))
plt.xticks(np.linspace(min_use, max_use, num=(max_use-

```

```

min_use)/5+1, endpoint=True))
plt.ylim(min_pft-1, max_pft+1)
plt.xlim(min_use-0.2, max_use+0.2)
plt.ylabel(' Profit', fontsize=15)
plt.xlabel(' Drug use', fontsize=15)
plt.plot(uses, profits, label=' Profit', marker=' d', linewidth=2.5, markersize=6,
color=' fuchsia')
#
plt.plot(X, fit_vals, label=' Fit', marker=' s', linewidth=2.5, markersize=10, colo
r=' cornflowerblue')
plt.legend(loc=' upper right')
plt.show()

# -*- coding: utf-8 -*-
import math
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import chisquare
from scipy.optimize import curve_fit

pre_rst = [0.1, 0.12, 0.2, 0.15, 0.15, 0.17, 0.14, 0.14, 0.16,
0.4, 0.37, 0.38, 0.28, 0.28]
pre_use = [153, 136, 125, 117, 116, 111, 117, 121, 120, 113, 111,
105, 104, 102]

def cal_r(t, x0, x1, x2):
    r = (1/t)*math.log((1/x0-1/x1)/(1/x1-1/x2))
    return r

def cal_n(r, t, x0, x1):
    n = (1-math.e**(-r*t))/(1/x1-math.e**(-r*t)/x0)
    return n

def cal_rst(t, r, n):
    return n/(1+(n/0.1-1)*math.e**(-r*t))

# pre_rs =
[cal_r(1, 0.1, 0.12, 0.2), cal_r(1, 0.14, 0.16, 0.4), cal_r(2, 0.14, 0.16, 0.37)]
# pre_ns =

```

```

[cal_n(pre_rs[0], 1, 0.1, 0.2), cal_n(pre_rs[1], 1, 0.14, 0.4), cal_n(pre_rs[
2], 2, 0.14, 0.37)]
#
# print pre_ns
r=cal_r(2.0, 0.12, 0.15, 0.17)
n=cal_n(r, 2.0, 0.12, 0.17)
print 'r=',r
print 'n=',n
pdt_rst = []
years = len(pre_rst)-2
for i in range(1, years, 1):
    pdt_rst.append(cal_rst(i, r, n))
next_rsts = []
for i in range(12, 22, 1):
    next_rsts.append(cal_rst(i, r, n))
print 'next_years:', next_rsts
data_names = range(2002, 2013, 1)
X = range(len(data_names))
print 'pdt_rst', pdt_rst
plt.figure(figsize=(30, 9), dpi=80)
plt.yticks(np.linspace(0, 1, num=11, endpoint=True))
plt.xticks(range(len(data_names)), data_names)
X = range(len(data_names))
plt.ylim(0, 1.2)
plt.xlim(-0.2, len(data_names)+0.2)
plt.xlabel('Drug use', fontsize = 15)
plt.ylabel('Resistant', fontsize=15)
plt.plot(X, pdt_rst, label='Predict', marker='d', linewidth=2.5, markersiz
e=6, color='fuchsia')
plt.plot(X, pre_rst[3:], label='Real', marker='s', linewidth=2.5, markersi
ze=10, color='cornflowerblue')
plt.legend(loc='upper right')
plt.show()

# pre_use = pre_use[:len(pre_use)-3]
# pre_use = sorted(pre_use)
# # print pre_use
# pdt_rst = sorted(pre_rst[3:])
# def func(x, a, b):
#     return a*np.log(b*x)
# popt, pcov = curve_fit(func, pre_use, pdt_rst)
# a=popt[0]
# b=popt[1]
# print a, b

```

```

# fit_vals = []
# for i in range(len(pre_use)):
#     fit_vals.append(func(pre_use[i], a, b))
# print len(pre_use), len(fit_vals)
# # z1 = np.polyfit(pre_use, pdt_rst, 3)# 用3次多项式拟合
# # p1 = np.poly1d(z1)
# # print(p1) # 打印拟合多项式
# # fit_vals = p1(pre_use)
#
# pre_rst_plt = pre_rst[3:]
#
# print chisquare(f_obs=pre_rst_plt, f_exp=pdt_rst)
#
#
#
# data_names = pre_use
#
#
# plt.figure(figsize=(30, 9), dpi=80)
# plt.yticks(np.linspace(0, 1, num=11, endpoint=True))
# plt.xticks(range(len(data_names)), data_names)
# X = range(len(data_names))
# plt.ylim(0, 1.2)
# plt.xlim(-0.2, len(data_names)+0.2)
# plt.xlabel('Drug use', fontsize = 15)
# plt.ylabel('Resistant', fontsize=15)
#
# plt.plot(X, pdt_rst, label='Real', marker='d', linewidth=2.5, markersize=6,
# , color='fuchsia')
#
# plt.plot(X, fit_vals, label='Fit', marker='s', linewidth=2.5, markersize=10, color='cornflowerblue')
# plt.legend(loc='upper right')
# plt.show()
pdt_rst = [0.1792628046926503, 0.2943181818181818, 0.43063362673704436,
0.558124635993011, 0.6535518026442507, 0.713738475784049,
0.7476984370881284, 0.7656659431653637, 0.7748492466183726,
0.7794600206940622, 0.7817543053571258]
data_names = range(2013, 2024, 1)
plt.figure(figsize=(30, 9), dpi=80)
plt.yticks(np.linspace(0, 1, num=11, endpoint=True))
plt.xticks(range(len(data_names)), data_names)
X = range(len(data_names))
plt.ylim(0, 1.2)

```

```
plt.xlim(-0.2, len(data_names)+0.2)
plt.xlabel('Year', fontsize = 15)
plt.ylabel('Resistant', fontsize=15)
plt.plot(X, pdt_rst, label='Prediction', marker='d', linewidth=2.5, marker
size=6, color='fuchsia')
#
plt.plot(X, fit_vals, label='Fit', marker='s', linewidth=2.5, markersize=1
0, color='cornflowerblue')
plt.legend(loc='upper right')
plt.show()
```

---

# Resolve Antibiotic Crisis With Science Strategies and Policies

## Abstract:

This article mainly studies how to deal with antibiotics crisis. We build two economic models to evaluate antibiotics' interest chain and a SD model to predict antibiotics crisis. Finally, we come up with a portable management systems and put forward reasonable reward policy based on our models.

In order to solve **Task one**, we establish two antibiotics interest chain models. One is at the point of medical industries while the other is of stock-breeding industries. Then, we construct two models. The price model basing on goal programming describes the supply and demand of antibiotics. The governments' policies model demonstrates the impacts of four economic policies on the antibiotics interest chain, including *price*, *tax*, *loan interest rate* and *production restrict policy*.

In order to solve **Task two**, we combine the theoretical hypothesis with existing data, using *Vensim* software to create the relationship graph. According to variables, we build the function equations to connect different variables. Regarding time axis from 2010 to 2050 as independent variables. Through simulation, we test the results' consistency with actual data. Also, we classify three most vertical variables: the effectiveness of antibiotics, patients' number and the price of antibiotics. Finally, we come up with the summary that antibiotics crisis is not noticeable in a short time. However, the crisis will become devastating in an exponential tendency over time.

In order to solve **Task three**, we design a portable management system. We come up with a report, which consists of the antibiotic circle, two models on task one, and the expected effort of models. After utilizing the data from existing papers, we simulate the given policies' affects on the models. At the end of the report, we draw four conclusions: Government should set the ceiling and floor price of antibiotic; The tax should be increased by approximately 10%; The loan interest rate should be increased to 4%; Governments should set production restriction to 8%.

In order to solve **Task four**, we reveal the essential causes of farmers' abusing antibiotics drugs while breeding livestock. Two spiritual reward policies and four material reward policies were given.

## Key words:

**Antibiotics crisis, System Dynamics, Interest Chain, Goal Programming, Vensim, Government Policies**

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# | Introduction

## 1.1 Backgrounds

Antibiotic is a type of antimicrobial drug produced by microbes (mostly fungi or bacteria), which can be used in the treatment and prevention of bacterial infections<sup>①</sup>. In 1928, Alexander Fleming discovered penicillin and Florey and Chain purified it in 1942. For the first few years, antibiotics were widely used for the purpose of curing recurring illness like tuberculosis and syphilis.

However, their effectiveness and easy access caused overuse, especially in the field of livestock raising. Abusing antibiotics has led to many problems as side-effects, super infection, and the one we most worried about——antibiotic resistance. Nowadays antibiotic resistance is suspected to be the arch-criminal of superbugs. According to a recent investigation, the use of antibiotics has increased by thirty-six percent in the past ten years and is still growing rapidly. Abusing antibiotics led to a complete catastrophe. 700,000 people die of the infection of superbugs each year. Some of them lose their lives because of taking antibiotic drugs without restraint. Others are newborn babies who digest the proteins eaten by their mothers.

In recent years, Drug Regulatory Authorities (DRA) makes an effort to supervise the use and abuse of antibiotics but cannot solve the problem thoroughly, for the interest chain behind the market is hard to be cut off. Given this circumstance, some policies executed by relative department are urgent.

## 1.2 What do we need to do?

- For task one, we need to use some view of economic to consider the antibiotic's supply and demand in the whole market, then we should build interest chain and some modules to express the whole chain.
- For task two, we need to build a model under the perfect competitive markets, and use the existing data to predict the develop trend towards the crisis. So that we could see the mechanism of the antibiotic crisis more clearly.
- For task three, we need to build an antibiotic management system and write a report. Using the existing models in task 1 and task 2 to simulate what impact policies will make and propose some effective policies.
- For task four, we need to give some reward policies for farmers, so that they will reduce the using of antibiotics which is often used to feed cattle.

① Wikipedia

## **II Analysis of the Problems**

### **2.1 Task one**

For task one, we consider the demand and supply relationships in market, and build two interest chains of antibiotics. In order to solve the problem more efficient, we simplified them into a single antibiotics chain. We then formulate two models to account for the influence of the price and government policies toward the antibiotics' interest chain, as well as how this change would affect the overall profit and economic welfare of the interest chain.

### **2.2 Task two**

On the basis of perfect competitive markets' characters, we simplify antibiotics' markets into a free market without governments' interference. As a special commodity, there's an extensive and intricate networking. After integrating and summarizing repetitively, we finally ascertain thirteen relevant variables. By using Vensim software, we construct a reasonable causality relationship graph. Finally, we select those can represent prediction process to be the support of our model.

### **2.3 Task three**

For task three, we regard ourselves as members of Food and Drug Administration, to help resist antibiotic crisis and superbugs, we build an antibiotics management system. According to the research and analysis in task 1 and task 2, we test our model and give four policies that can help the government win the war with superbugs.

### **2.4 Task four**

The abuse of antibiotics in stock raising is the same as it in the clinical medicine. Though governments used to adopt punishment policies, the result wasn't satisfying. So we need to analyze the reasons why farmers keep using antibiotics and come up with a reward policy accordingly.

### III Assumptions and Symbol Definitions

#### 3.1 Modeling Assumptions

- We regard pharmaceutical companies and farmers as manufacturers in the interest chain while doctors and meat-producing factories as suppliers.
- The number of ingredients described from suppliers by manufacturers is decided by their motivation to maximize the interest.
- The number of ingredients provided by suppliers is also decided by their motivation to maximize the interest.
- The amounts provided by suppliers and by manufacturers are equal.
- We give tacit consent to all the antibiotics are of one kind.
- We assume that the supply and demand will finally get balanced.

#### 3.2 Symbol Definitions

| Variables  | Variable Meaning                                 |
|------------|--------------------------------------------------|
| $v_{jt}$   | Sale of antibiotic j in phase t                  |
| $z_{jt}^L$ | Antibiotic inventory j in phase t                |
| $b_{it}$   | Ingredients demand of antibiotic i in phase t    |
| $x_{it}$   | Suppliers' production in phase t                 |
| $z_{jt}$   | Production of antibiotic j in phase t            |
| $y_{it}^L$ | Ingredients inventory of antibiotic i in phase t |
| $l_{it}$   | Supply of antibiotic products i in phase t       |
| $x_{it}^L$ | Suppliers' products in phase t                   |

*Table 1: Symbol Definitions*

## IV Antibiotics' Interest Chain

### 4.1 The Description of Antibiotics' Interest Chain

Interest chain is a kind of business relationships, which can be used to describe the order of operations and functions of each participant. Through economic connection, the interest is flowing. In the mean time, market games help them to maximize their benefits. As participants in the medicine market, pharmaceutical companies, doctors, patients, governments, farmers who feed livestock with antibiotics and meat-producing factories, have a huge influence on every link of the interest chain.

The interest chain is the origin of abusing antibiotics. Thus, we need to make some intensive analysis of the interest chain behind antibiotics.

### 4.2 The Structure and Characters of Antibiotics' Interest Chain

The participants of antibiotics' interest chain are connected by antibiotics supply and antibiotics demand. We classify them into six parts:

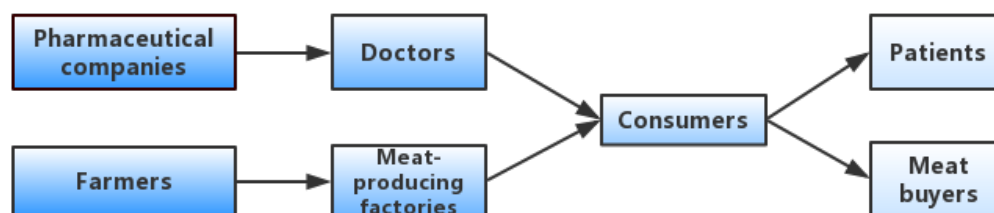


Figure 1 Specific Interest Chain

- **Governments**

Governments have the capability of supervising policies and the regulations, as well as adjusting financial environment and economic strategies to maintain both economic interests and social interests.

- **Pharmaceutical companies**

Pharmaceutical companies develop and sell antibiotics to gain interests. Their products revolutionize the whole interest chain, thus pharmaceutical companies are a significant node.

- **Doctors**

Doctors' prescriptions decide whether the patients will take antibiotic drugs or not, and the amounts or types they use. Their behaviors have a lot to do with the consumption of antibiotics.

- **Farmers**

Some farmers feed antibiotics to livestock for the purpose of curing the disease quickly. Though we didn't take in antibiotics directly, on the basis of the food chain theory, after eating the livestock fed by them, antibiotics still can accumulate in our body and aggravate antibiotic resistance crisis.

- **Meat-producing factories**

Meat-producing factories are those who deal with meat and sell them to the market, are similar to pharmaceutical in the interest chain.

- **Consumers**

We divide all the consumers into two parts: some of them are patients, they are aware of that they are using antibiotics prescribed by their doctors; Others are consumers who buy the proteins raised by antibiotics accidentally, which means they totally have no idea what they're eating.

### 4.3 The Interest Expectations of Each Participant

| Participants                    | Expectations                                                                              |
|---------------------------------|-------------------------------------------------------------------------------------------|
| <b>Governments</b>              | Manage the market and ensure the safety of drugs                                          |
|                                 | Control the danger of abusing drugs and avoid abusing antibiotics                         |
|                                 | Promote the progress of pharmaceutical technology to decrease the reliance on antibiotics |
| <b>Pharmaceutical Companies</b> | Increase the productivity of drugs                                                        |
|                                 | Gain more benefits                                                                        |
|                                 | Explore new antibiotics                                                                   |
| <b>Doctors</b>                  | Cure the patients quickly to relieve their pain                                           |
|                                 | Maximize the benefits                                                                     |
| <b>Farmers</b>                  | Preventing the livestock from be infected by bacteria                                     |
|                                 | Provide more proteins                                                                     |
| <b>Meat-producing Factories</b> | Increase the productivity of proteins                                                     |
|                                 | Reduce the cost of purchasing ingredients                                                 |
|                                 | Gain more benefits                                                                        |
| <b>Consumers</b>                | Direct: Recover soon and reduce the expenses                                              |
|                                 | Indirect: Buy tasty meat in a low price                                                   |

Table 2: The Interest Expectations of Each Participant

#### 4.4 An Interest Chain Model Based on Supply and Demand

We consider the interest chain an interrelated whole, all the participants form a supply-demand relationship. According to the analysis above, antibiotics suppliers and antibiotics demanders have some conflicts between each other, but they need to cooperate to maximize the interest.

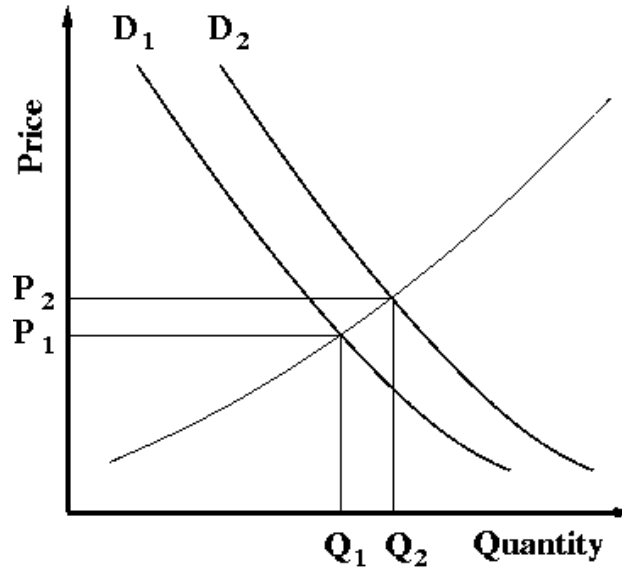


Figure 2 Supply and Demand Curve

We suppose both the price of antibiotics ingredients  $r_{ht}$  ( $h = 1 \dots H$ ) and the price of antibiotics  $p_{jt}$  ( $j=1, \dots, J$ ) are uncertain.

We set up a goal-programming model [1] to describe the purpose of each participant on the antibiotics' interest chain.

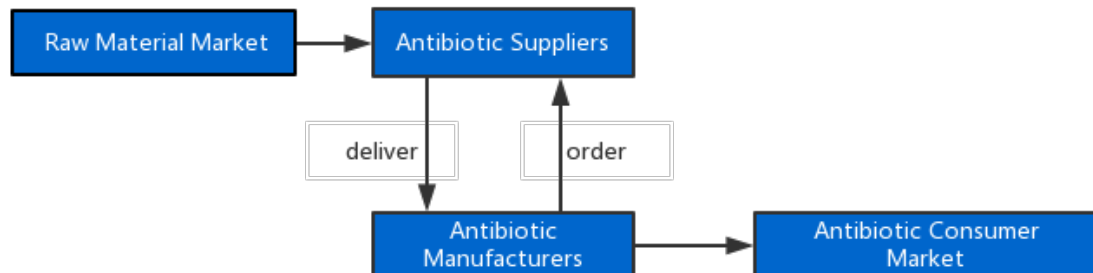


Figure 3 Simplified Interest Chain

##### **Objective Function One: The operational coordination of participants**

The delivery offered by antibiotics suppliers should satisfy the need of antibiotics manufacturers.

We can express the antibiotics' value chain in a goal-programming model:

$$\text{Min } P_T \times \sum_{t=1}^T \sum_{i=1}^I (d_{it}^- + d_{it}^+) \quad (4.1)$$

$$\text{s.t. } -b_{it} + l_{it} + d_{it}^- - d_{it}^+ = 0 \quad i, t \quad (4.2)$$

$P_T$  means priority factor, while  $d_{it}^-$  and  $d_{it}^+$  means the shortage and surplus of antibiotics demand.

##### **Objective Function Two: The target profit for manufacturers**

For antibiotics manufacturers, their goal is to gain their best interest

We can formulate a model as follow

$$C^P = \sum_{t=1}^T (\sum_{j=1}^J (p_{jt} v_{jt} - c_j^z z_{jt} - h_j^z z_{jt}^L) - \sum_{i=1}^I (h_i^y y_{it}^L + q_{it} b_{it})) \rightarrow \max \quad (4.3)$$

We make this model expressed in goal programming:

$$\text{Min } P_P \times d_P^- \quad (4.4)$$

$$\text{s.t. } C^P + d_P^- - d_P^+ = M_P \quad (4.5)$$

In this model,  $P_T$  is priority factor,  $M_P$  is the best interest that antibiotics manufacturers long for, it's also a constant.  $d_P^-$  and  $d_P^+$  are the shortage and surplus of the antibiotics target profit.

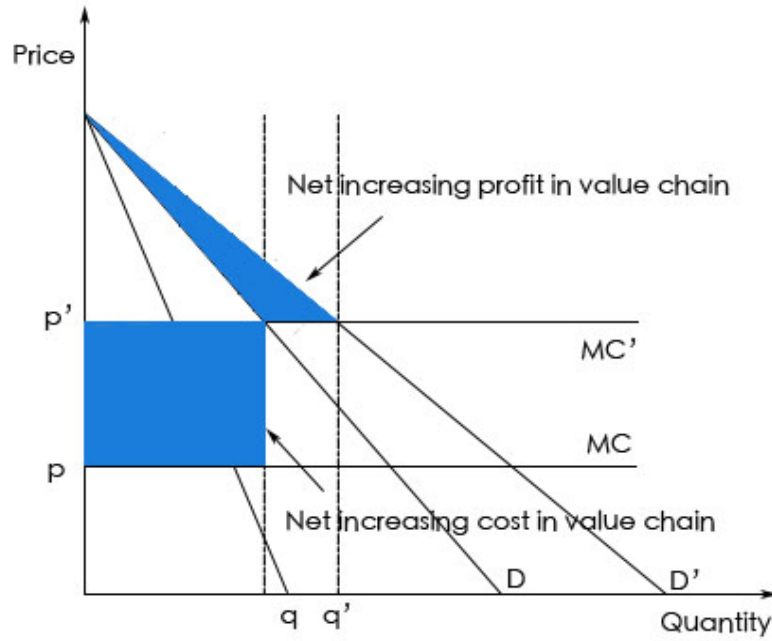


Figure 4 Marginal Rate of Return Curve

### Objective Function Three: The target profit for suppliers

The model for purpose 3 is as follow:

$$C^S = \sum_{t=1}^T \sum_{i=1}^I (q_{it} l_{it} - (c_j^x + \sum_{h=1}^H r_{ht} s_{hi}^r) x_{it} - h_i^x x_{it}^L) \rightarrow \max \quad (4.6)$$

In a goal programming way:

$$\text{Min } P_S \times d_S^- \quad (4.7)$$

$$\text{s.t. } C^S + d_S^- - d_S^+ = M_S \quad (4.8)$$

$P_T$  means priority factor.  $M_P$  is the best interest that antibiotics suppliers long for, it's also a constant.  $d_P^-$  and  $d_P^+$  are the shortage and surplus of the antibiotics target profit.

During the process of the operation of antibiotics interest chain, the priority order is:  $P_T \gg P_P = P_S$ .

Thus, we can summarize a model as follow:

### Objective function:

$$\text{Min } [P_T \times \sum_{t=1}^T \sum_{i=1}^I (d_{it}^- + d_{it}^+) + P_P \times d_P^- + P_S \times d_S^-] \quad (4.9)$$

**Restraints are as follow**

The production capacity constraint of antibiotics manufacturers in every phase:

$$\sum_{j=1}^J a_j^k z_{jt} \leq K_{max} \forall t \quad (4.10)$$

The stock constraint of antibiotics manufacturers in every phase:

$$z_{jt}^L = z_{j, t-1}^L + z_{jt} - v_{jt} \forall j, t \quad (4.11)$$

$$z_{j0}^L = z_{j0}^{L'} \forall j \quad (4.12)$$

$$\sum_{j=1}^J o_j^z z_{jt}^L \leq z^{Lmax} \forall t \quad (4.13)$$

The production capacity constraint of antibiotics manufacturers:

$$\sum_{i=1}^I a_i^g x_{it} \leq G_{max} \forall t \quad (4.14)$$

The stock constraint of antibiotics suppliers:

$$x_{it}^L = x_{i, t-1}^L + x_{it} - l_{it} \forall i, t \quad (4.15)$$

$$x_{i0}^L = x_{i0}^{L'} \forall i \quad (4.16)$$

$$\sum_{i=1}^I o_i^z x_{it}^L \leq x^{Lmax} \forall i, t \quad (4.17)$$

No negativity constraint:

$$z_{jt}, z_{jt}^L, y_{it}^L, b_{it}, v_{jt}, l_{it}, x_{it}, x_{it}^L \geq 0 \forall i, j, t \quad (4.18)$$

$$v_{jt}', x_{it}', z, y, \xi_{jt}, \zeta_{it} \geq 0 \forall i, j, t \quad (4.19)$$

$$d_{it}^-, d_{it}^+, d_p^-, d_p^+, d_s^-, d_s^+ \geq 0 \forall i, j, t \quad (4.20)$$

From this model, we can clearly see that the whole antibiotic interest chain's profit system is related to many parts and every parts, for their own sake, want to make their profit maximum.

So the whole chain is under the constant adjustment. And it's the markets' supply and demand power provide motivation for the stimulation. The market will finally be balanced when supply equals demand.

## 4.5 The Interest Chain Model With Government's Policies

Governments' policies have a huge effect on the interest chain. Therefore, we take governments' price policy, tax policy, interest policy, and production restraint into consideration in order to evaluate policies' effect on the interest chain.

Let's suppose the economic welfare<sup>[2]</sup> of producers is  $EW_1$  and it of suppliers is  $EW_2$ . The investment economic welfare on developing new antibiotics due to the antibiotics resistance crisis is  $EW_3$ . So, the total effects of policies are:



$$\begin{aligned} &\text{Max EW} \\ &= (EW_1(Z_1) + EW_2(Z_1) + EW_3(Z_1) + (EW_1(Z_2) + EW_2(Z_2) + EW_3(Z_2) + (EW_1(Z_3) + \\ &EW_2(Z_3) + EW_3(Z_3) \end{aligned}$$

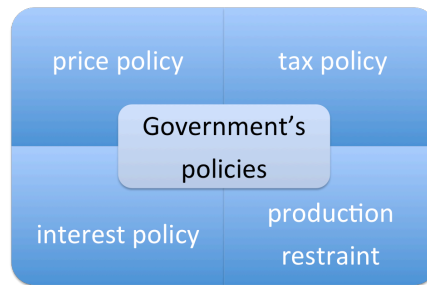


Figure 5 Government's policies

### ① Government's Policies & Antibiotic Producers

We can use the producer surplus formula to get this result:

$$EW_1 = PQ_C(1 + z_1)(1 + z_2)(1 + z_3)(1 + z_4) - \int_0^{Q_C(1+z_4)} (e + fQ)(1 + z_3)dQ \quad (4.21)$$

P means target price, QC means the production quota of antibiotics. The supply function is:

$$MC(P) = e + fQ \quad (4.22)$$

The price of antibiotics is decided by the open market, the formula is:

$$Q' = (1/\varepsilon_S + 1/\varepsilon_D) / (1/(\varepsilon_S(1 + \alpha)) + 1/(\varepsilon_D(1 + \beta))) Q_0 \quad (4.23)$$

$$P = P_0(1 - 1/\varepsilon_S) + P_0/(\varepsilon_S(1 + \alpha) \times Q_0) Q' \quad (4.24)$$

$\varepsilon_S$ ,  $\varepsilon_D$  mean the elasticity of supply and demand price,  $P_0$ ,  $Q_0$  mean balanced volume of transactions and balanced price,  $\alpha$ ,  $\beta$  mean growth rates of supply chain and demand chain.

### ② Government's Policies & Antibiotic Suppliers

The formula to calculate producers' surplus is:

$$EW_2 = (P_i S_i - \int_0^{S_i} (a_i(1 + \rho z_1) S^{b_i}) dS) \quad (4.25)$$

$P_i$  is the average price of producing antibiotics,  $S_i$  is the average volume of transaction.

The supply function is:

$$P_i = a_i(1 + \rho z_1) S^{b_i} \quad (4.26)$$

$\rho$  is the portion that the cost of ingredients takes in.

### ③ Government's Policies & Developing New Antibiotics

The new-developed economic welfare is:

$$EW_3 = EW_1/Q_C \times Q_A \times dA \quad (4.27)$$

The recoverable rate of investment:

$$r = \text{net profit/investment} \quad (4.28)$$

$$= \left( \frac{PQ_C(1 + z_1)(1 + z_2)(1 + z_4) - \int_0^{Q_C(1+z_4)} (e + fQ)(1 + z_3)dQ}{Q_R \div Q_A} \right) \div (I_A n)$$

$Q_C$  is the limitation of production,  $Q_R$  is the recoverable rate of investment,  $Q_A$  is the individual plant of antibiotics,  $I_A$  is individual investment.

**The solution of this model will be given in Task 3**

## V Development trend of Antibiotic Crisis

### 5.1 Theories of SD Models

System dynamics is a method of predicting long-running trends by using computer simulation. This method asks us to build up SD models, flow diagram and functional equations.

SD models have their own preponderances:

A large number of variables can be contained in SD models

- SD models enable us to fully understand the inner structure of the system
- SD models have the capacity to deal with non-linear and multi-feedback problems

### 5.2 Causality Relationship Graph

Under the premise of perfect competitive market, the trend of antibiotics crisis mainly covers the effective rate of antibiotics, the price changes of antibiotics and the number of illness people caused by using antibiotics. By analyzing statistics of each year and the theory of system dynamics, we choose thirteen variables. And take the balance of supply and demand, build up the causality of each factor(Pic 1). ***Here are the feedback loops:***

- The price of antibiotics ↓ → Antibiotic usage amount in hospital ↑ → Antibiotic receiver number ↑ → Effective rate ↓ → Antibiotic effective people ↓ → Patients number ↑ → The price of antibiotics ↑
- The price of antibiotics ↑ → Patients number ↑ → Antibiotic supply amount ↑ → The price of antibiotics ↓
- Patients number ↑ → The price of antibiotics ↑ → Antibiotic development rate ↑ → Effective rate ↑ → Bacteria increasing rate ↓ → Bacteria scale ↓ → Patients number ↓
- Patients number ↓ → The price of antibiotics ↓ → Antibiotic usage amount with farmers and doctors ↑ → Antibiotic receiver number ↑ → Patients number ↓
- Bacteria scale ↑ → The price of antibiotics ↑ → Antibiotic development rate ↑ → Effective rate ↑ → Bacteria increasing rate ↓ → Bacteria scale ↓

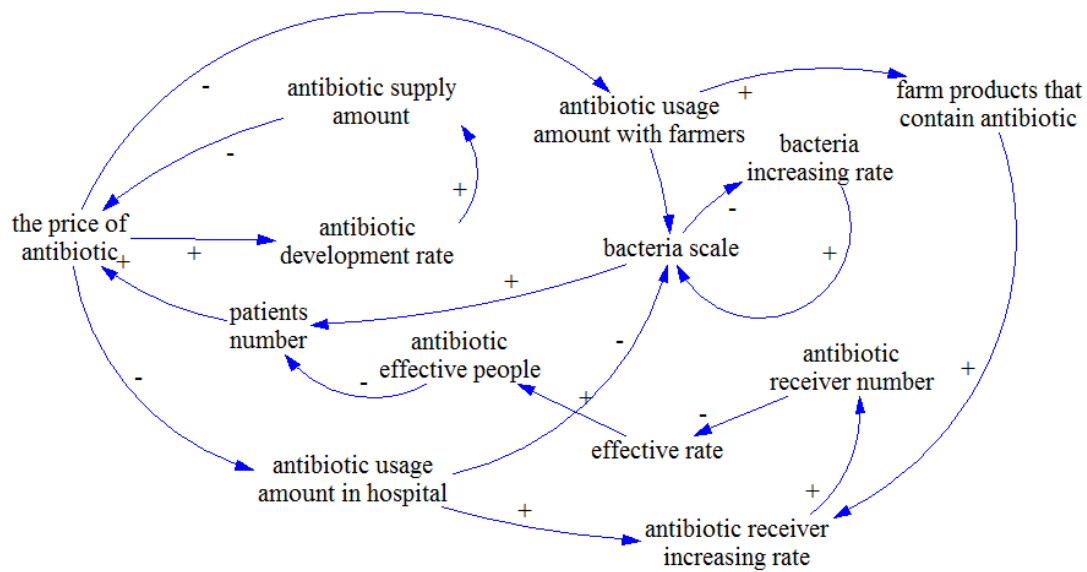


Figure 6 Causality Relationship of Perfect Competitive Market Antibiotic Crisis Prediction SD Simulation Model

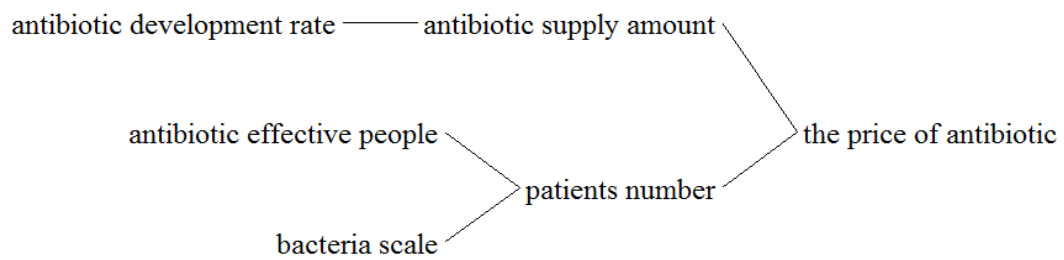


Figure 7 The Price of Antibiotic: Causes Tree

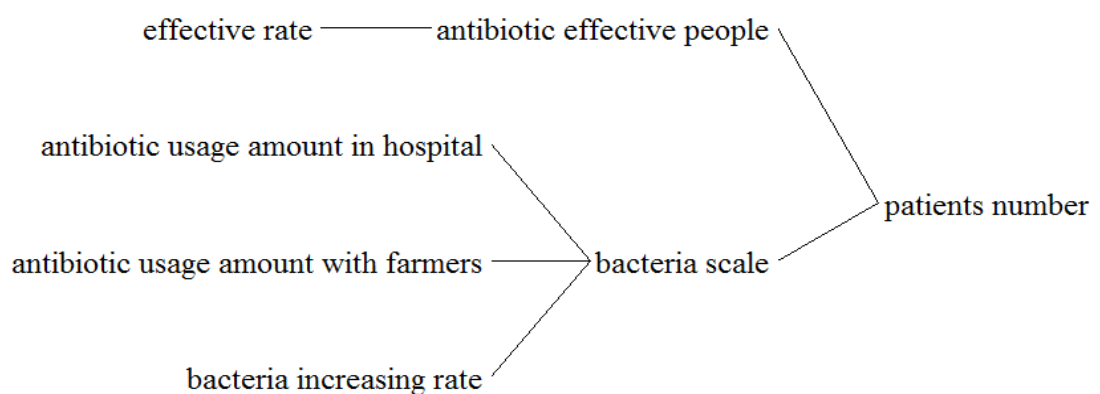


Figure 8 Patients Number: Causes Tree

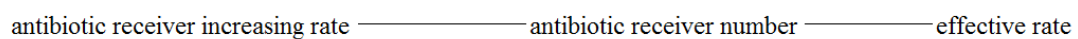


Figure 9 Effective Rate: Causes Tree

### 5.3 Flow Diagrams and Equations

According to the relationship between factors of perfect competitive market crisis prediction system, we design a flow diagram:

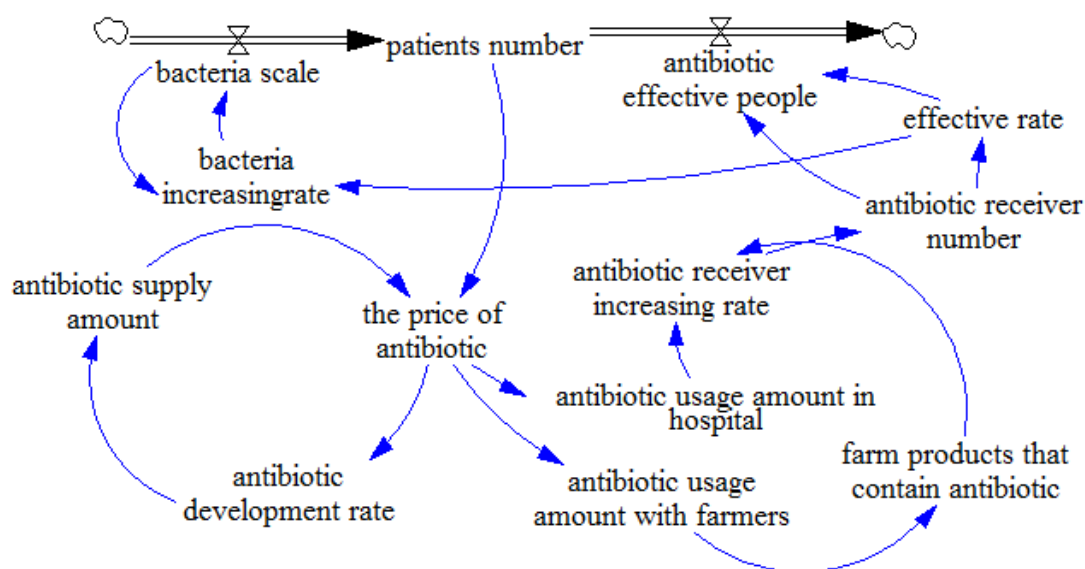


Figure 10 Antibiotics Crisis Prediction System SD Simulation Model

| Dependent variable(y)                              | Equations                                                                                                                                                   |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Patients number</b>                             | $Y = \text{INTEG}(\text{Bacteria scale} * \text{Coefficient} - \text{Antibiotic effective people, the initial number of patients})$                         |
| <b>Effective rate</b>                              | $Y = \text{Antibiotic receiver number} / \text{Antibiotic effective people}$                                                                                |
| <b>The price of antibiotics</b>                    | $Y = \text{LOOKUP}(\text{Patients number} * \text{Demanding coefficient} / \text{Antibiotic supply} * \text{Supply coefficient, The price of antibiotics})$ |
| <b>Bacteria scale</b>                              | $Y = \text{INTEG}(\text{Bacteria scale} * \text{Bacteria increasing rate, the initial number of bacteria scale})$                                           |
| <b>Bacteria increasing rate</b>                    | $Y = \text{Increasing coefficient} * (1 - \text{effective rate})$                                                                                           |
| <b>Antibiotic develop rate</b>                     | $Y = \text{Rate index} * \log(1 + 1 / \text{The price of antibiotics}) * 100\%$                                                                             |
| <b>Antibiotic supply</b>                           | $Y = \text{INTEG}(\text{Antibiotic supply} * \text{Antibiotic development rate, Antibiotic supply})$                                                        |
| <b>Medical antibiotics usage</b>                   | $Y = \text{Medical demand} / \text{The price of antibiotics}$                                                                                               |
| <b>Agricultural antibiotics usage</b>              | $Y = \text{Maximum cost of agriculture} / \text{The price of antibiotics}$                                                                                  |
| <b>Increase rate of antibiotic Receiver number</b> | $Y = (1 - 1 / \text{Agricultural antibiotics usage}) * 1 + (1 - 1 / \text{Medical antibiotics usage}) * \text{Correlation coefficient}$                     |

|                                                  |                                                                                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Antibiotic receiver number</b>                | $Y = \text{INTEG}(\text{Antibiotic receiver number} * \text{Increasing rate, the initial number of Antibiotic receiver number})$ |
| <b>Antibiotic effective people</b>               | $Y = \text{Antibiotic receiver number} * \text{effective rate}$                                                                  |
| <b>Agricultural products contain antibiotics</b> | $Y = \text{Agricultural antibiotics usage} * \text{Correlation coefficient}$                                                     |

Table 3: Variables at Original Status

| Bacteria scale                          | Morbidity unit*100      |
|-----------------------------------------|-------------------------|
| Antibiotic receiver number              | Group organization*1000 |
| Total demand of antibiotics in hospital | Usage unit*10000        |
| Maximum cost of agriculture             | Cash unit*10000         |
| Price                                   | Cash unit*100           |
| Patients number                         | Group organization*100  |

Table 4: Variables at Basic Status

## 5.4 Simulation And The Results

### 5.4.1 Test of the model

In order to know whether the established model is qualified to reflect the essential characters and its accordance to reality, we must test the validity of the model.

Firstly, we take year 2010 as a start, and then gain the analog value from 2010 to 2016. Secondly, we adjust the parameters repetitively to make it closer to the reality. Finally, the probability of errors being no more than 5% is 80.45%, which verifies the validity of this model. Henceforth, this model is apt to predict the trend of the system.

### 5.4.2 Simulation

We select antibiotic effective rate, patients' number (relevant to antibiotics), and the price of antibiotics as representative index. Also, we set the time border from 2010 to 2015, time step is one year, and get the picture as following:

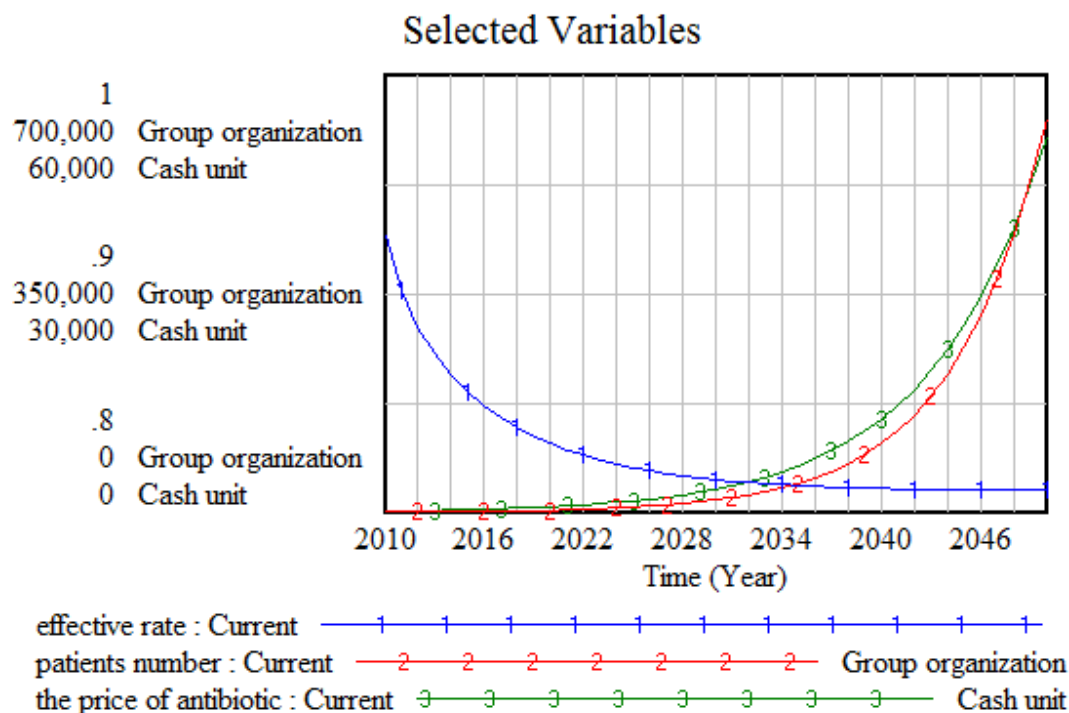


Figure 11 Tendency prediction of Antibiotics crisis under perfect competitive market

From figure 11, we discover that as time goes by, the patients number(relevant to antibiotics) and antibiotics price under perfect competitive market is growing slowly before 2030. After 2030, they begin to soar. On the contrary, antibiotics effectiveness has a trend of slumping before 2030 and is still decreasing after 2030. The concrete statistics are in the chart.

The potential risk under perfect competitive market is hard to figure out in a short time. But once begin to aggravate, the crisis becomes enormous.

Furthermore, if the effective power goes down, will lead to more bacteria residue, which could encourage the bacteria turns into the superbugs and make the disease more severe. At the end of this control, maybe it will be extremely hard for human beings find effective medicine to fight against superbugs.

At the same time, the increase of patient will increase the whole societies' medical load, and decrease the people's trust toward the government.

The most severe problem is the increase price of antibiotics, when it comes to a price that people can't afford, the medical conflict will turn into social conflict, which, at last become the beginning of social instability.

## VI Report of Antibiotic Management System

### Government Policy Report of Antibiotic Resistance

#### 1 Backgrounds

For many years, people get benefit from using antibiotics, for it's high efficiency and good effect. But with the development of science and technology, more and more people get concerned about the big damage of antibiotic. Now we have come to a stage where we must make some efforts to prevent the development of superbugs. To solve this problem, we build *price model* and *government policy model* to describe antibiotics' value chain, as well as give some effective policies that may works.

#### 2 Vicious Antibiotic Circle and Simulate

There is a vicious circle in antibiotics' interest chain: pharmaceutical companies in order to resist new superbugs may develop new antibiotics constantly; hospitals for the sake of profit often offer new-developed drugs; patients want to recover faster so they also want to buy new-developed antibiotics, so their antibiotics resistance get more and more serious, which in turns encourage pharmaceutical companies to development more antibiotics.

To forecast this circle, we use system dynamic method to describe this process. Using *Vensim Software* to simulate the development trend towards the crisis under perfect competitive markets. The simulate results shows patients relevant to antibiotics will booming in the future and in **2030** will comes to a great catastrophe.

#### 3 Antibiotics' Interest Chain Models

##### Model I Price model

To make the interest chain more explicit, we build a supply and demand model that based on price. This model mainly illustrates the balance result the market will finally achieve. In price models, we consider three objective functions:

- *The operational coordination of participants*
- *The target profit for manufacturers*
- *The target profit for suppliers*

Also we make some restraints include *production capacity constraint*, *inventory constraint* and *no negativity constraint*. Through this model, we could approximately evaluate when the whole interest chain get balance based on pursuit of profit.

##### Model II Government Policy Model

However, as antibiotics belong to special medicine, government must take

some effort to manage the produce and the use of antibiotics. Using the theory of welfare economics, we build a model considering the government policy.

#### 4 The Expected Effect of Model

For model II, we consider four government policy factors that may influence the whole interest chain: *price, tax, interest, production restriction*.

After consulting existing research results, we determine the relevant parameters' value (See appendix), we use Matlab to calculate the exact value and draw drama, shown as follows:

(EW<sub>1</sub> represents economic welfare of antibiotics producers; EW<sub>2</sub> represents economic welfare of suppliers; EW<sub>3</sub> represents the economic welfare on developing new antibiotics)

##### ① Price Policy

According to our model, under the tax rate of 65%, the price change impact on EW<sub>i</sub> are shown as follows:

| Price Change(%) | EW <sub>1</sub> | EW <sub>2</sub> | EW <sub>3</sub> | EW    |
|-----------------|-----------------|-----------------|-----------------|-------|
| -60.0           | -9.71           | 17.53           | 0.00            | 7.82  |
| -40.0           | -7.32           | 14.01           | 0.00            | 6.70  |
| -20.0           | -4.93           | 10.49           | 0.00            | 5.57  |
| 0.0             | -2.54           | 6.98            | 0.00            | 4.44  |
| 20.0            | -0.15           | 3.46            | -0.13           | 3.18  |
| 40.0            | 2.24            | -0.06           | 4.07            | 6.25  |
| 60.0            | 4.63            | -3.58           | 12.62           | 13.67 |

Table 5: The Impact of Price Policy on Antibiotics Management System

From this table, we can easily see both the reduce and increase of antibiotic price will increase the whole interest chain's economic welfare. Decrease price could make the economic welfare increase as well as control the development of new medicine. Thus, government set ceiling price is an effective way for antibiotic resistance crisis.

##### ② Tax Policy

Tax is an important measure for government to control the whole interest chain. According to our model, the tax change impact on EW<sub>i</sub> are shown as follows:

| Tax Change(%) | EW <sub>1</sub> | EW <sub>2</sub> | EW <sub>3</sub> | EW    |
|---------------|-----------------|-----------------|-----------------|-------|
| 0.0           | 12.67           | 6.98            | 0.00            | 19.65 |
| 10.0          | 9.96            | 6.98            | 0.00            | 16.93 |
| 20.0          | 7.24            | 6.98            | 1.99            | 16.20 |
| 30.0          | 4.52            | 6.98            | 5.00            | 16.50 |
| 40.0          | 1.81            | 6.98            | 1.73            | 10.51 |
| 50.0          | -0.91           | 6.98            | -0.38           | 5.69  |
| 56.0          | -2.54           | 6.98            | 0.00            | 4.44  |
| 60.0          | -3.62           | 6.98            | 0.00            | 3.35  |

Table 6: The Impact of Tax Policy on Antibiotics Management System

From the results, we can see when the tax rate above 50%, the EW<sub>1</sub> turns to



negative, which means the tax rate shouldn't exceed 50%, also, we can discover that when the tax rate is at approximate 10%, the economic welfare is at the optimum point, the exact optimum point need further calculation.

### ③ **Loan Interest Policy**

Government could stimulate the loan interest rate to control the produce and development of antibiotics. According to our model, the loan interest rate change impact on  $EW_i$  are shown as follows:

| <b>Interest<br/>change(%)</b> | <b>EW1</b> | <b>EW2</b> | <b>EW3</b> | <b>EW</b> | <b>EW(tax 20%)</b> |
|-------------------------------|------------|------------|------------|-----------|--------------------|
| 0                             | -1.22      | 6.98       | -0.61      | 5.14      | 19.82              |
| 4                             | -1.75      | 6.98       | -0.52      | 4.7       | 17.42              |
| 6                             | -2.01      | 6.98       | -0.4       | 4.56      | 16.3               |
| 10                            | -2.54      | 6.98       | 0          | 4.44      | 14.22              |
| 12                            | -2.8       | 6.98       | 0          | 4.17      | 13.95              |
| 16                            | -3.33      | 6.98       | 0          | 3.65      | 13.43              |
| 20                            | -3.86      | 6.98       | 0          | 3.12      | 12.9               |

*Table 7: The Impact of Loan Interest Policy on Antibiotics Management System*

From the results, we can clearly see that when the tax rate is 65%, the loan interest rate's change doesn't matter that much. So we change the tax rate into 20%, under this level, the loan interest can influence the whole antibiotics interest chain.

### ④ **Production Restriction**

Government has the power to make some production restriction, so that it can reduce the antibiotic production from the source. According to our model, the production restriction change impact on  $EW_i$  are shown as follows:

| <b>Production<br/>Restriction (%)</b> | <b>EW<sub>1</sub></b> | <b>EW<sub>2</sub></b> | <b>EW<sub>3</sub></b> | <b>EW</b> | <b>EW<br/>(tax 20%)</b> |
|---------------------------------------|-----------------------|-----------------------|-----------------------|-----------|-------------------------|
| -16.0                                 | -2.13                 | 6.98                  | -0.39                 | 4.45      | 13.06                   |
| -12.0                                 | -2.23                 | 6.98                  | -0.29                 | 4.45      | 13.35                   |
| -8.0                                  | -2.33                 | 6.98                  | -0.20                 | 4.45      | 13.64                   |
| -4.0                                  | -2.44                 | 6.98                  | -0.10                 | 4.44      | 13.93                   |
| 0.0                                   | -2.54                 | 6.98                  | 0.00                  | 4.44      | 14.22                   |
| 4.0                                   | -2.64                 | 6.98                  | 0.00                  | 4.34      | 15.30                   |
| 8.0                                   | -2.74                 | 6.98                  | 0.00                  | 4.24      | 16.39                   |
| 12.0                                  | -2.84                 | 6.98                  | 0.00                  | 4.13      | 17.47                   |
| 16.0                                  | -2.94                 | 6.98                  | 0.00                  | 4.03      | 18.56                   |

*Table 8: The Impact of Production Restriction on Antibiotics Management System*

From the table, we read that when the tax rate is 56%, along with the production restriction increase, the whole antibiotic economic welfare will decrease. But when the tax rate turns into 20%, the production restrict will increase the whole systems' economic welfare.

## 5 *Policy suggestions according to model*

With the analysis above, we have come into some policy suggestions:

- ***Government Pricing***

Government should set the ceiling price and the floor price of antibiotic. By given the range of factory price and the market price, government could control the whole interest chain. This method is also described as Keynesianism.

- ***Increasing Tax***

Increasing the tax of antibiotics is an effective way towards the antibiotics' producers and suppliers, such as pharmaceutical companies, doctors and farmers, according to our model, the tax should be increased by approximately 10%.

- ***Increasing Loan Interest Rate Related to Antibiotics***

For the companies that produce or sell antibiotics, government should increase their loan interest rate, the exact interest rate change should consider tax rate. For example under the tax rate of 20%, we suggest the loan interest rate should be increased to 4%.

- ***Making Production Restriction***

Government should set the production restriction in order to prevent the company produce antibiotics more than necessary, the amount of production restriction depend on the tax rate. For example, under the tax rate of 20%, government should set production restriction to 8%.

## VII Encouragement Policies for Framers

### 7.1 Reason Analysis

Antibiotics are welcomed by farmers when raising cattle. But they place no limits on using them. The reasons why farmers insist on using antibiotics to raise the cattle are as follow:

- ***Lack of professional knowledge***

Most farmers didn't get enough education so they have no idea what great catastrophe will happen because of their abusing antibiotics.

- ***Desire for higher profits***

Antibiotics are inexpensive and easy accessible. Cattle fed with antibiotics will not get infected easily and grow in a fast speed, which means more profits for farmers.

- ***Insufficient and unprofessional disease prevention infrastructure***

Sometimes the cattle get sick and it's hard to get a professional vet to cure them, the farmers have to use antibiotics to cure them.

### 7.2 Specific Encourage policies

*Henceforth, we come up with some reward policies:*

- ***Spiritual Incentives toward farmers***

- ◆ Issue certificates in not using antibiotics to honor them.
- ◆ Ask those farmers who don't use antibiotics to share their experience with others.

- ***Material Incentives toward farmers***

- ◆ Offer free courses on knowledge of stock raising to inform farmers of the terrible outcomes of abusing antibiotics and the suitable condition to use them properly.
- ◆ Lower the taxes of those who promise not to use antibiotic drugs
- ◆ Give out free books to farmers about how to raise stock without antibiotics, like how to prevent animal diseases.
- ◆ Give out fine varieties of stock to those promise not to use antibiotic drugs to help them get higher quality of proteins without using antibiotics.

## VIII Analysis of the Result

### 8.1 Modles' Strengths

- Simplify the complicated relationship, use a few variables to reflect the real condition well.
- The two models make full use of existing data to solving the problems.

### 8.2 Modules' Weakness

Although the model is very reasonable, there still have some disadvantages.

- Insufficient real statistic, a little of them was under the process of simplification.
- We have ignored a few factors that exert a little influence on models

### 8.3 Modules' Improvements

Due to the limit of time, there are still some weakness. We can make the following improvements:

- Search some big data base to attain more statistics.
- Take more factors to build a more thorough model to make the result more reliable.

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## X Appendix

### / Goal-programming Objection and Restricts

| Objective fuction                                                                  | Restricts                                                                                                |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Purpose 1: The opertational coordination of participants in antibiotic value chain |                                                                                                          |
| Purpose 2: The target profit for antibiotic manufactuers                           | 1. The production capacity constraint for manufacturers<br>2. The inventory constraint for manufacturers |
| Purpose 3: The target profit for antibiotic suppliers                              | 3. The production capacity constraint for suppliers<br>4. The inventory constraint for suppliers         |

### // Relative Parameter at Original Status of Policy model

| Parameter | Value | Parameter  | Value   |
|-----------|-------|------------|---------|
| $z_1$     | 0     | $Q_A$      | 10      |
| $z_2$     | -56%  | $I_A$      | 211     |
| $z_3$     | 10%   | $r_0$      | 0       |
| $z_4$     | 0     | $n$        | 20      |
| $P$       | 485   | $\rho$     | 0.5     |
| $Q_c$     | 560   | $a_1$      | 551.3   |
| $e$       | 70    | $a_2$      | 493.67  |
| $f$       | 0.59  | $a_3$      | 2367.81 |
| $P_{s1}$  | 1500  | $b_1$      | 0.2122  |
| $S_1$     | 111.7 | $b_2$      | 0.1345  |
| $P_{s2}$  | 800   | $b_3$      | 0.1963  |
| $S_2$     | 36.12 | $\alpha$   | 6.29    |
| $P_{s3}$  | 5000  | $\mu$      | 33.43   |
| $S_3$     | 45.04 | $\delta$   | 44%     |
| $Q_R$     | 700   | $\delta_0$ | 10%     |

### ***III The Simulation of Effective Rate, Patients Number and Antibiotic price in 2010——2048***

| Time<br>(Year) | Effective<br>rate | Patients number<br>(Group organization) | Antibiotic price<br>(Cash unit) |
|----------------|-------------------|-----------------------------------------|---------------------------------|
| 2010           | 0.926250          | 273.8                                   | 100.00                          |
| 2011           | 0.901685          | 366.7                                   | 142.38                          |
| 2012           | 0.884799          | 491.6                                   | 189.97                          |
| 2013           | 0.872343          | 648.2                                   | 242.01                          |
| 2014           | 0.862681          | 839.1                                   | 298.14                          |
| 2015           | 0.854916          | 1069.0                                  | 358.70                          |
| 2016           | 0.848512          | 1343.8                                  | 424.67                          |
| 2017           | 0.843140          | 1671.4                                  | 497.44                          |
| 2018           | 0.838581          | 2061.2                                  | 578.66                          |
| 2019           | 0.834680          | 2524.6                                  | 670.21                          |
| 2020           | 0.831326          | 3075.1                                  | 774.11                          |
| 2021           | 0.828483          | 3729.2                                  | 892.59                          |
| 2022           | 0.825933          | 4506.2                                  | 1028.14                         |
| 2023           | 0.823768          | 5429.5                                  | 1183.52                         |
| 2024           | 0.821891          | 6526.6                                  | 1361.85                         |
| 2025           | 0.820264          | 7830.5                                  | 1566.68                         |
| 2026           | 0.818852          | 9380.4                                  | 1802.06                         |
| 2027           | 0.817626          | 11223.0                                 | 2072.64                         |
| 2028           | 0.816562          | 13414.0                                 | 2383.71                         |
| 2029           | 0.815637          | 16019.5                                 | 2741.40                         |
| 2030           | 0.814835          | 19118.3                                 | 3152.69                         |
| 2031           | 0.814137          | 22804.3                                 | 3625.65                         |
| 2032           | 0.813531          | 27189.2                                 | 4169.54                         |
| 2033           | 0.813005          | 32405.6                                 | 4795.00                         |
| 2034           | 0.812547          | 38612.8                                 | 5514.27                         |
| 2035           | 0.812149          | 45998.3                                 | 6341.42                         |
| 2036           | 0.811804          | 54786.7                                 | 7292.65                         |
| 2037           | 0.811503          | 65245.2                                 | 8386.55                         |
| 2038           | 0.811242          | 77691.8                                 | 9644.53                         |
| 2039           | 0.811015          | 92505.0                                 | 11091.20                        |
| 2040           | 0.810817          | 110135.0                                | 12754.90                        |
| 2041           | 0.810646          | 131120.0                                | 14668.10                        |
| 2042           | 0.810497          | 156096.0                                | 16868.40                        |
| 2043           | 0.810367          | 185826.0                                | 19398.60                        |
| 2044           | 0.810254          | 221213.0                                | 22308.40                        |
| 2045           | 0.810156          | 263336.0                                | 25654.70                        |

---

|      |          |          |          |
|------|----------|----------|----------|
| 2046 | 0.810071 | 313748.0 | 29502.90 |
| 2047 | 0.809997 | 373165.0 | 33928.30 |
| 2048 | 0.809932 | 444217.0 | 39017.50 |
| 2049 | 0.809876 | 528797.0 | 44870.20 |

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# The antibiotic resistance analysis based on interest chain

## Abstract

In this paper, we construct an antibiotic interest chain, and then use the differential equation model and multiple regression model to analyze the antibiotic resistance crisis.

Firstly, based on the supply and demand of interest chain, an economic model describing the relationship between supply and demand in the market is established using differential equation. The simple supply and demand model and the inventory model of supply and demand were analyzed qualitatively and stability. Analyze the sensitivity of the various nodes in the interest chain by changing the supply and demand of antibiotics.

Secondly, in the perfect competitive market, using the least squares estimate, the multiple regression models are established to describe the development of the antibiotic market, and further analyze the changes of the antibiotic resistance.

Thirdly, based on the established model and related analysis, a feasible management system was established to rectify the abuse of antibiotics.

Finally, we analyze farmers' willingness to be compensated qualitatively. Using Douglas production function to analyze the related factors, and put forward a set of incentive policies to reduce the use of antibiotics in aquaculture industry.

In summary, this paper described the dynamic interest change of each supply group under the supply and demand of antibiotics, achieved the goal of defeating the antibiotic resistance crisis with changing supply and demand, and using the model established in this paper to verify the regulation of the antibiotics use in livestock finally.

**Key words:** interest chain, differential equation model, multiple regression model, sensitivity analysis

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# 1. Introduction

"Antibiotic resistance" is a hot topic in the moment. Every year, the number of drug resistant bacteria can kill up to 700 thousand, more terrible bacterial strains still continue to spread in the world. Researchers worry that we're going to be in the stage of a lack of antibiotics, that is, one day, the bacteria that infect us may have a drug resistance to every antibiotic.

In order to analyze the causes of the antibiotic resistance crisis better, we are asked to complete the following tasks:

- Develop a model that provides a interest chain for use of antibiotics and the sensitivity of parties. You may need to consider the dynamic nature of the factors that affect both supply and demand in your modeling process.
- Predict the development trend towards the crisis under perfect competitive markets.
- Design a portable management system for the use of antibiotics is the most urgent affair for superbug. ICM-FDA has been asked to attend a policy strategy meeting where your team is asked to write a report on your model and propose a set of policies.
- Develop a reward policy for formers to encourage them to raise cattle without antibiotic drugs.

# 2. Assumptions

- The external natural factors do not influence price changes
- All enterprises are in the normal competition condition
- The supply chain of antibiotic is stable
- Market changes in the ideal situation

# 3. Supply-demand Models

## 3.1 The prices of antibiotic products based on the dynamic nature

Pharmaceutical produce antibiotics, and sold them to hospitals, wholesale market and retail market to profit. These institutions gain profit by buying and selling and they are the supply side. Farmers and patients are the demand side. An antibiotic interest chain was constructed, as follows:

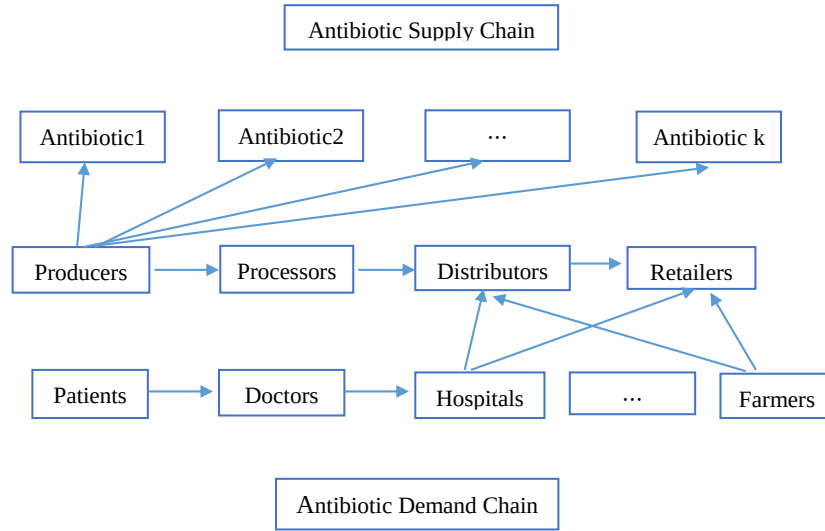


Figure.3.1 Antibiotic supply - demand chain

### 3.1.1 Demand function is the linear function of prices

Assume that  $p(t)$  is the price of some goods at time  $t$ ,  $S(t)$  and  $D(t)$  respectively represent the amount of supply and demand at moment  $t$ .  $D_0$  is the greatest quantity demand.  $S_0$  is the smallest quantity supplied, among them,  $\alpha$  is a rate which stimulates production as supply increase with the price.  $\beta$  is a rate which reduces the power to purchase as demand increase with the price. The speed of price increase  $dp/dt$  is proportional to  $D - S$ .  $\mu$  is the ratio of its rising and falling. Prices' rising and falling are decided directly by the supply of the manufacturers and the demands of the consumers. As follows is the specific model,

$$\begin{cases} \frac{dp}{dt} = \mu(D(t) - S(t)), \quad \mu > 0 \\ S(t) = S_0 + \alpha p(t), \quad \alpha > 0 \\ D(t) = D_0 - \beta p(t), \quad \beta > 0 \end{cases}, \quad (3.1)$$

Its solution is:

$$p(t) = c_0 e^{-kt} + \frac{h}{k} = c_0 e^{-kt} + \frac{D_0 - S_0}{\alpha + \beta}, c_0 \in R, \quad (3.2)$$

$C_0$  is decided by the initial price. This solution is going to be  $\frac{D_0 - S_0}{\alpha + \beta}$ .

### 3.1.2 Demand function is a quadratic function of the price

When purchasing power price is less than the threshold of  $p_0$ , demand stimulates prices, and then the amount of purchasing is increased as price rising. When purchasing power price is more than the threshold

of  $p_0$ , price increases and demand decreases. Establish a demand function is:

$$D(t) = D_0 + (ap^2 + bp + c), a < 0, \quad (3.3)$$

Models are as follows:

$$\begin{cases} \frac{dp}{dt} = \mu(D(t) - S(t)), \mu > 0 \\ S(t) = S_0 + \alpha p(t), \alpha > 0 \\ D(t) = D_0 + (ap^2 + bp + c), a < 0 \end{cases}, \quad (3.4)$$

Among that,  $a, b, c$  are the coefficients of the quadratic function. If the following condition is met:

$$\begin{cases} (1) b \geq a > 0 \\ (2) 2b\alpha - \alpha^2 < -4a(D_0 - S_0) < b^2 \\ (3) \alpha^2 - 2b\alpha - 4a(D_0 - S_0) > 0 \\ (4) \frac{-D_0}{a} > \frac{\alpha^2 + \alpha^2 - 2b\alpha - 4a(D_0 - S_0) + 2\alpha\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}}{4a^2} \\ (5) a < 0, b > 0, c < 0, b^2 = 4ac \end{cases}, \quad (3.5)$$

This model will produce two singular points, which are  $P_1^*$  and  $P_2^*$ . And  $P_1^*$  is a unstable nodal point while  $P_2^*$  is a stable nodal point. Among them:

$$\begin{cases} p_1^* = \frac{-B + \sqrt{B^2 - 4AC}}{2A} = -\frac{-(b - \alpha) + \sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}}{2a} \\ p_2^* = \frac{-B - \sqrt{B^2 - 4AC}}{2A} = -\frac{-(b - \alpha) - \sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}}{2a} \end{cases}, \quad (3.6)$$

If condition (4) becomes:

$$\frac{-D_0}{a} > \frac{\alpha^2 + \alpha^2 - 2b\alpha - 4a(D_0 - S_0) - 2\alpha\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}}{4a^2}, \quad (3.7)$$

then this model will produce only one singular point and it's a stable nodal point. Numerical simulation is shown in fig.1 and fig.2. The following are parameters:

$$\mu = 0.05, \alpha = 2, \beta = 4, D_0 = 100, S_0 = 50, a = -5, b = 40, c = -80$$

### 3.1.3 Supply and demand model with inventory

1. The demand function is a linear function of the price

In fact, in many cases there are dealers in the intermediate link, which will hoard goods, and came the result of market manipulation phenomenon, the specific example is the holding of antibiotics which is going

to make it expensive and unstable. So we can improve the model further by define a stock function  $H(t)$ , which represents the warehouse inventory, and is the excess demand in supply from the amount of time 0 to time t. At this time the price rise and fall is determined directly by the current demand and inventory. Consumers directly affect price move together with the dealer. No longer directly affection from the factory supply, the rise and fall of prices is proportional to  $D - H$ , which can be displayed exactly as follows:

$$\begin{cases} H(t) = \int_0^t (S(t) - D(t))dt \\ \frac{dp}{dt} = \mu(D(t) - H(t)), \mu > 0 \\ S(t) = S_0 + \alpha p(t), \alpha > 0 \\ D(t) = D_0 + (ap^2 + bp + c), a < 0 \end{cases}, \quad (3.8)$$

Analysis the model, system (3.8) exists the only singular point  $(\frac{D_0 - S_0}{\alpha + \beta}, 0)$ , and the stable node when

$\mu^2 \beta^2 - 4\mu(\alpha + \beta) > 0$ , the stable focus when  $\mu^2 \beta^2 - 4\mu(\alpha + \beta) < 0$ . The parameters are those

$$\mu = 0.05, \alpha = 2, \beta = 4, D_0 = 100, S_0 = 50,$$

2. Demand function is a quadratic function of the price

The same method. The inventory should also be consider the demand function for the price, also the quadratic function of the price, and the model can be presented as follows:

$$\begin{cases} H(t) = \int_0^t (S(t) - D(t))dt \\ \frac{dp}{dt} = \mu(D(t) - H(t)), \mu > 0 \\ S(t) = S_0 + \alpha p(t), \alpha > 0 \\ D(t) = D_0 + (ap^2 + bp + c), a < 0 \end{cases}, \quad (3.9)$$

when it meet the condition of

$$\begin{cases} b \geq a > 0 \\ 2b\alpha - \alpha^2 < 4a(D_0 - S_0) < b^2 \\ 2b\alpha < -4(D_0 - S_0) \\ \alpha^2 - 2b\alpha < 4a(D_0 - S_0) > 0 \\ \frac{-D_0}{a} > \frac{\alpha^2 + \alpha^2 - 2b\alpha - 4a(D_0 - S_0) + 2\alpha\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}}{4a^2} \\ a < 0, c < 0, b^2 = 4ac \end{cases}, \quad (3.10)$$

The system (3.9) will produce two singularities  $(P_1^*, 0)$  and  $(P_2^*, 0)$ . The first singular point  $(P_1^*, 0)$  is a saddle point when

$$(\mu\alpha - \mu\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}) - 4\mu\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)} > 0, \quad (3.11)$$

and the second point is a stable pitch point, when

$$(\mu\alpha - \mu\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}) - 4\mu\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)} < 0, \quad (3.12)$$

the second point is a stable focus point.

If the condition five is changed into

$$\frac{-D_0}{a} > \frac{\alpha^2 + \alpha^2 - 2b\alpha - 4a(D_0 - S_0) - 2\alpha\sqrt{\alpha^2 - 2b\alpha - 4a(D_0 - S_0)}}{4a^2}, \quad (3.13)$$

then in the system (3.9), there is only a singularity  $(P_2^*, 0)$ , the stable node or the stable focus point. It's parameters are

$$\mu = 0.05, \alpha = 2, \beta = 4, D_0 = 100, S_0 = 50, a = -5, b = 40, c = -80 \quad (3.14)$$

### 3.2 Enterprise profit distribution of supply chain members

The main text illustrated the basic model of the supply chain management based on the antibiotic products supply chain management theory, studied the main distribution subjects, objects and the mediation means, pinpointed the principles that shall be followed and the importance of the efficient allocation of interests, also using the game theory, value and network analysis and other scientific theory to research the antibiotic products supply chain from the symbiosis and the benefit distribution and the development countermeasure, etc.

The symbols which appear in the process of constructing the model are explained as follows:

$P_1$ : The product prices the antibiotic producers set for the processing manufactory;

$P_2$ : The product prices the antibiotic processors set for the distributors;

$P_3$ : The product prices the distributors set for the consumer;

$C_1$ : The equilibrium production cost antibiotic producer;

$C_2$ : The equilibrium purchasing cost antibiotic processors;

$C_3$ : The equilibrium pin cost antibiotic products distributor;

$W_1$ : The profits of the antibiotic producer;

$W_2$ : The profits of the antibiotic processor;

$W_3$ : The profits of distributors equalization pin cost;

$W$ : The overall interests of antibiotic products supply chain

$k_1$ : The loss rate of antibiotic production in the process of productive process

$k_2$ : The loss rate of antibiotic production in the process of workpiece process

$k_3$ : The loss rate of antibiotic production in the process of distribution process

$Q$ : The sales volume.

Based on the assumption above we can easily arrive at the price and demand function:

$$P_3 = a - bQ, \quad a, b \geq 0, \quad (3.15)$$

among which a and b are constants.

And we can obtain the profits of producers of antibiotic products

$$W_1 = \frac{P_1 Q}{(1+k_3)(1-k_2)} - \frac{C_1 Q}{(1-k_3)(1-k_2)(1-k_1)}, \quad (3.16)$$

the profits of processor is

$$W_2 = \frac{P_2 Q}{1-k_3} - \frac{P_1 Q}{(1-k_3)(1-k_2)} - \frac{C_2 Q}{(1-k_3)(1-k_2)}, \quad (3.17)$$

the profits of the distribution partner is

$$W_3 = P_3 Q - \frac{P_2 Q}{1-k_3} - \frac{C_3 Q}{1-k_3}, \quad (3.18)$$

suppose  $n_1 = \frac{1}{1-k_3}$ ,  $n_2 = \frac{1}{1-k_2}$ ,  $n_3 = \frac{1}{1-k_1}$ , then

$$\begin{cases} W_1 = (P_1 - C_1 n_3) n_1 n_2 Q \\ W_2 = (P_2 - P_1 n_2 - C_2 n_2) n_1 Q, \\ W_3 = (P_3 - P_2 n_1 - C_3 n_1) Q \end{cases} \quad (3.19)$$

so the overall interests of the antibiotic supply chain is

$$W = W_1 + W_2 + W_3 = (P_3 - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1) Q, \quad (3.20)$$

### 3.2.1 The member enterprises don't cooperate

Each independent economic entity in the interest line is pursuing benefit maximization, and each enterprises is not willing to cooperate and decide with its own interests. The alliance formed a 3-level complete information dynamic game, with the help of the theory of Stackelberg, and the member companies relying on the negotiation price to achieve the overall balance, in the supply chain the producer as a core, the producer dominated the whole confederation and the process merchant, distribution partner as a distribution partner. The antibiotic producer first conducted the decision making of price  $P_1$ , the processing merchant and distribution partners as a subordinate position have to follow the dominant setting price to adjust theirs.

We can reverse solution by concluding, so we have



$$\begin{cases} P_1 = \frac{a + C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1}{2 n_1 n_2} \\ P_2 = \frac{3a + C_1 n_1 n_2 n_3 + C_2 n_1 n_2 - 3C_3 n_1}{4 n_1} \\ P_3 = \frac{7a + C_1 n_1 n_2 n_3 + C_2 n_1 n_2 + C_3 n_1}{8} \\ Q = \frac{a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1}{8b} \end{cases}, \quad (3.21)$$

Thus the producers, processors, distributors of antibiotic and the whole interests of the supply chain respectively are

$$\begin{cases} W_1 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{16b} \\ W_2 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{32b} \\ W_3 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{64b} \\ W = \frac{7(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{64b} \end{cases}, \quad (3.22)$$

### 3.2.2 Partial member enterprises cooperate

As the market competition intensifies, the enterprises gradually realize that only by cooperate with other enterprises can they achieve the goal of profit, and the specialty of the antibiotic also determines that cooperators need value the partnership between enterprises, there are mainly two ways which have operability and efficiency between part member enterprises. Which is the cooperation between producers and processors or alliance cooperation between processors and distributors, points out that producers and distributors, without direct contact, their cooperation interests is the sum of interests for both non-cooperative situations, which can be listed as follows:

$$W_1 + W_3 = \frac{5(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{64b}, \quad (3.23)$$

1) Alliance cooperation of antibiotics producers and processors.

Antibiotics producers and constitutes constitute a two stage dynamic game between each other, and the antibiotic producers and processors first to make the optimal price decision to the price of distributors, and the distributors to make their own price decision to sell antibiotics to consumers in order to achieve the optimization of their own interests.

And the cooperation benefits of antibiotics producers and processors is

$$W_1 + W_2 = (P_2 - C_1 n_2 n_3 - C_2 n_2) n_1 Q, \quad (3.24)$$

profits of antibiotic distributors is

$$W_3 = (P_3 - P_2 n_1 - C_3 n_1),$$

reverse conclude solutions with Stackelberg way, we can obtain

$$\begin{cases} P_2 = \frac{a - C_1 n_1 n_2 n_3 + C_2 n_1 n_2 - C_3 n_1}{2n_1} \\ P_3 = \frac{3a + C_1 n_1 n_2 n_3 + C_2 n_1 n_2 + C_3 n_1}{4}, \\ Q = \frac{a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1}{4b} \end{cases} \quad (3.25)$$

Coordination benefits of antibiotics producer and processor, overall interests of the distributors and supply chain of antibiotics respectively are

$$\begin{cases} W_1 + W_2 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{8b} \\ W_3 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{16b} \\ W = \frac{3(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{16b} \end{cases} \quad (3.26)$$

## 2) Alliance cooperation of antibiotics processors and distributors

Cooperation of the antibiotics processors and distributors formed the second stage dynamic game as well. Antibiotics producers supply antibiotics first as a source of supply chain, then antibiotics processors and distributors to make the optimal pricing decision to optimize cooperation benefit.

And finally the profit of antibiotics produce is

$$W_1 = (P_1 - C_1 n_3) n_1 n_2 Q, \quad (3.27)$$

the cooperate benefit of antibiotic processor and distributors is

$$W_2 + W_3 = (P_3 - P_1 n_1 n_2 - C_3 n_1 - C_2 n_1 n_2) Q, \quad (3.28)$$

so we can have the solution in a similar way based on the Steckelberg game model:

$$\begin{cases} P_1 = \frac{a + C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1}{2n_1 n_2} \\ P = \frac{3a + C_1 n_1 n_2 n_3 + C_2 n_1 n_2 + C_3 n_1}{4}, \\ Q = \frac{a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1}{4b} \end{cases} \quad (3.29)$$

So the interests of the antibiotic producers, cooperate interests of antibiotics processors and distributors, the whole interests of the whole antibiotics supply chain respectively are

$$\begin{cases} W_1 = \frac{(a - C_1n_1n_2n_3 - C_2n_1n_2 - C_3n_1)^2}{8b} \\ W_2 + W_3 = \frac{(a - C_1n_1n_2n_3 - C_2n_1n_2 - C_3n_1)^2}{16b}, \\ W = \frac{3(a - C_1n_1n_2n_3 - C_2n_1n_2 - C_3n_1)^2}{16b} \end{cases} \quad (3.30)$$

### 3.2.3 All member enterprises cooperate

All the member enterprises of the antibiotics supply chain will negotiated price together optimizing the whole interests of all alliances when they all participate in the alliance cooperation. And the whole interests of antibiotics supply chain is

$$W = (a - bQ - C_1n_1n_2n_3 - C_2n_1n_2 - C_3n_1)Q, \quad (3.31)$$

in order to achieve the desired max interest, after Derivation of Q

$$\begin{cases} Q = \frac{a - C_1n_1n_2n_3 - C_2n_1n_2 - C_3n_1}{2b} \\ P_3 = \frac{a + C_1n_1n_2n_3 + C_2n_1n_2 + C_3n_1}{2} \end{cases}, \quad (3.32)$$

and the whole interests of the supply chain is

$$W = \frac{(a - C_1n_1n_2n_3 - C_2n_1n_2 - C_3n_1)^2}{4b}, \quad (3.33)$$

Replace the antibiotic producers, processors and distributors with A B C, and let

$$\frac{(a - C_1n_1n_2n_3 - C_2n_1n_2 - C_3n_1)^2}{64b} = K(\text{constant}), \text{ the benefit comparison table of each member enterprise}$$

of supply chain when cooperate and non-cooperate is as follows:

Table3.1 The interest comparison of Cooperation and collaboration

| Cooperate Form                                 | Member enterprises | interests | Whole interests of supply chain |
|------------------------------------------------|--------------------|-----------|---------------------------------|
| Respectively independent                       | A                  | 4K        | 7K                              |
|                                                | B                  | 2K        |                                 |
|                                                | C                  | K         |                                 |
| Two parties corporate<br>One party independent | A, B               | 5K        | 7K                              |
|                                                | C                  | 2K        |                                 |
|                                                | A,C                | 8K        | 12K                             |
|                                                | B                  | 4K        |                                 |
|                                                | B,C                | 4K        |                                 |
|                                                | A                  | 8K        |                                 |
| All members cooperate                          | A,B,C              | 16K       | 16K                             |

### 3.2.4 Conclusions of member enterprises cooperation

The benefit of the whole supply chain can be optimized when all the participating members work together, on the contrary, the least among all the participating members.

The whole interests of supply chain of which producers and processors of antibiotic work together is equal to that processors and distributors of antibiotic cooperate together.

In order to maximize the overall interest, the member enterprises should maintain a good and sustainable partnership which can be improved by incentive mechanism and other measures of all member enterprises to carry out communication. At the same time, the value-added benefits of cooperation should be scientifically and reasonably allocated to maintain the stability of the supply chain.

### 3.2.5 The distribution method of antibiotic supply chain benefit based on Shapley value

Assuming  $N=\{A, B, C\}$ , cooperation between antibiotic enterprise A and B record as A+B, and so on are denoted as B+C, A+C. Member enterprises A, B and C cooperate regard as A+B+C,  $\phi_1(v), \phi_2(v), \phi_3(v)$  respectively represents the benefits of first allocation according to the Shapley value. We can calculate the Cooperative interests of three level supply chain member enterprises based in the Table3.1, which can be listed as follows:

Table3.2 Benefits of the members of the antibiotic supply chain

| Member enterprises participating in cooperation | Benefits |
|-------------------------------------------------|----------|
| Antibiotic producer(A)                          | 4K       |
| Antibiotic processor(B)                         | 2K       |
| Antibiotic distributor(C)                       | K        |
| producer and processor (A+B)                    | 8K       |
| producer and distributor (A+C)                  | 5K       |
| processor and distributor (B+C)                 | 4K       |
| producer and processor + distributor (A+B+C)    | 16K      |

So we can obtain interest of each member enterprises based on the Shapley value method, which respectively are the following:

$$\begin{aligned}\phi_1(v) &= \frac{(3-1)!(1-1)!}{3!} \times 4K + \frac{(3-2)!(2-1)!}{3!} \times [(8K - 2K) + (5K - K)] \\ &+ \frac{(3-3)!(3-1)!}{3!} \times (16K - 4K) = 7K\end{aligned}$$

$$\begin{aligned}\phi_2(v) &= \frac{(3-1)!(1-1)!}{3!} \times 2K + \frac{(3-2)!(2-1)!}{3!} \times [(8K - 4K) + (4K - K)] \\ &+ \frac{(3-3)!(3-1)!}{3!} \times (16K - 5K) = \frac{11K}{2}\end{aligned}$$

$$\phi_3(v) = \frac{(3-1)!(1-1)!}{3!} \times K + \frac{(3-2)!(2-1)!}{3!} \times [(5K-4K) + (4K-2K)] \\ + \frac{(3-3)!(3-1)!}{3!} \times (16K-8K) = \frac{7K}{2}$$

The added benefits of antibiotic producers are  $\Delta W_1 = 7K - 4K = 3K$

The added benefits of antibiotic processors are  $\Delta W_1 = 7K - 4K = 3K$

The added benefits of antibiotic distributors are  $\Delta W_1 = 7K - 4K = 3K$

The added benefits of whole supply chain of antibiotic are  $\Delta W_1 = 7K - 4K = 3K$

### 3.3 Sensitivity analysis of the risk

As the antibiotic resistance risk is mainly caused by human unlimited use of the drugs driven by money interests, so the sensitivity of the risk is how stable the interest structure. Which means the more stable the interest structure is, the less sensitivity the antibiotic resistance will be. So what we are to do to test the sensitivity of the antibiotic resistance can be replace to test the stability of the interest structure.

Based on the thesis above, we can simply divide the whole parties into three parties which can be listed as the producer, processor and distributors to be the supply chain. And the demand chain such as hospitals patients and the other organizations. Assuming that all those supply chain and demand chain have to be monitored by the government, or the Food and Drug Administrations.

So we can define the matrix A which the element  $a_{ij}$  means the total number of volume supply chain of number i, supply the antibiotics to demand party of number j. So based on the data of matrix A, we can easily know the matrix norm in order to assess the stability of the supply chain structure. And here we did classification of interest chain and demand chain.

Table3.3 The parties divided by supplier and demander

| Regulatory Party: Food and drug administration, government |                             |
|------------------------------------------------------------|-----------------------------|
| Provider of the antibiotics                                | Demander of the antibiotics |
| Pharmaceutical companies                                   | Hospital                    |
| Processor of the antibiotics                               | Framer                      |
| Distributors of antibiotics                                | Patients                    |

Table3.4 The parties divided by whether willing to use it

| Dominant use | Initiative restricted party  | Can be on both sides of the party |
|--------------|------------------------------|-----------------------------------|
| Doctors      | Researchers                  | Farmers                           |
|              | Food and Drug Administration | patients                          |

Table3.5 The type of antibiotics

| The types of antibiotics |
|--------------------------|
|--------------------------|

|           |                 |                |
|-----------|-----------------|----------------|
| man-rated | animals-rated t | Insects--rated |
|-----------|-----------------|----------------|

Assuming that each member enterprises the supply chain do not cooperate. So we can get the benefits of their own as formula (3.21) and (3.22), which can be list as follows

$$\left\{ \begin{array}{l} P_1 = \frac{a + C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1}{2 n_1 n_2} \\ P_2 = \frac{3a + C_1 n_1 n_2 n_3 + C_2 n_1 n_2 - 3C_3 n_1}{4 n_1} \\ P_3 = \frac{7a + C_1 n_1 n_2 n_3 + C_2 n_1 n_2 + C_3 n_1}{8} \\ Q = \frac{a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1}{8b} \end{array} \right. \left\{ \begin{array}{l} W_1 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{16b} \\ W_2 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{32b} \\ W_3 = \frac{(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{64b} \\ W = \frac{7(a - C_1 n_1 n_2 n_3 - C_2 n_1 n_2 - C_3 n_1)^2}{64b} \end{array} \right.$$

And in what way the data will change prominently, we can simulate it from the whole data domain of definition, confirming that the risk trend will continue if the hospital continue obtaining high profits, which make the matters worse.

### 3.4 The development trend towards the crisis

#### 3.4.1 Background and Basic information about complete market

In a fully competitive market, there is no open competition between economic entities. When consumers and producers are making decisions, they don't really need to think about the influences what their actions will have on the rest of the market. The reason is that in a perfectly competitive market, individual economic units of production is so small compared with the whole market, so that their behavior have tiny influence on the other buyers and sellers, and hard to be visible.

In the case of making the appropriate decision other companies without considering the response of the other enterprises. In a fully competitive market, participants are competing with each other. The place they made their decision are exactly in what they seem to be a fixed or relatively fixed place, In a perfectly competitive market, there must be a large number of sellers so that no one who is able to exert a greater impact on prices. In fact, this is what we say to sellers in western economics in accordance with the price taker, decided by the market price of which no matter how many products production can be sold. Actually, the seller face the certain demand which in the demand curve a horizontal demand curve. The same situation, the second requirements of the fully competitive market is the number of buyers must be sufficient, each buyer in which he size is very small has no ability to affect the price of the product. That is to say, all the buyers are the price recipient.

##### 1. The analysis of perfect competition market

In the perfect competitive market, it is quite easy to assume that the person is in or out of the industry. If the price of the product is higher than the cost, then it will produce economic profit, the resources will be used to start a new business or to expand the existing production capacity of enterprises. If the producer's profit is lower than the average, it is nature that the resources to be transferred from the industry, for the production of other products that can get more profits. Perfectly competitive market is in fact a product that

is assumed to have no difference. There is no difference between a company's products and other products. In this way, consumers' buying decisions are made entirely on the basis of price. If the price of an enterprise is higher than the price determined by the market, it will not be able to attract more buyers. As the manufacturer is in accordance with the market price to the production of all the products are sold out, naturally it is not necessary to reduce the price.

As a result, there is also a feature of the complete competition, enterprises always produced at the lowest cost from the long-term perspective. As has been illustrated previously, the maximum profit yield is at the lowest point of the average cost curve. This result does not mean that the competitiveness of enterprise efficiency will be higher than other types of market structure of enterprises. But it does mean that if the enterprise can use technology under certain circumstances, the economic factors under full competition requires the producer to produce the lowest cost per unit product.

### 3.4.2 Predict the development trend

As the antibiotic resistance is directly relate to some factors, so we can use the multiple linear regression model get the detail connection between the resistance and other factors like the total amount of antibiotic producers and producers , and also the number of distributers of antibiotic, also the hospitals and patients, farmers and other group relate to the antibiotics.

All those factors are influencing the influence risk directly or indirectly, so set these factors as k independent variables  $x_1, x_2, \dots, x_k$  , and the antibiotic resistance risk value as y. The n groups of observation values  $y_a, x_{1a}, x_{2a}, \dots, x_{ka}, a = 1, 2, \dots, n$  , so the multiple linear regression model for the structure of the form is:

$$y_a = \beta_0 + \beta_1 x_{1a} + \beta_2 x_{2a} + \dots + \beta_k x_{ka} + \varepsilon_a, \quad (3.34)$$

In which  $\beta_0, \beta_1, \dots, \beta_k$  are the undetermined parameters,  $\varepsilon_a$  is a random variable. If  $b_0, b_1, \dots, b_k$  respectively are fitting values of  $\beta_0, \beta_1, \beta_2, \dots, \beta_k$  . Then the regression equation is

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_k x_k, \quad (3.35)$$

In which  $b_0$  is a constant,  $b_1, b_2, \dots, b_k$  are partial regression coefficients. And the meaning of partial regression coefficient  $b_i$  is that while the rest of the independent variables  $x_j (j \neq i)$  are all constants, the independent variable  $x_i$  means the average numerical change of dependent variable Y while the every change unit of independent variable  $x_i$  , based on the principle of least square method. The estimated value  $b_i$  of variable  $\beta_i (i = 0, 1, 2, \dots, k)$  shall let

$$Q = \sum_{a=1}^n \left( y_a - \hat{y}_a \right)^2 = \sum_{a=1}^n \left[ y_a - (b_0 + b_1 x_{1a} + b_2 x_{2a} + \dots + b_k x_{ka}) \right]^2 \rightarrow \min, \quad (3.36)$$

According to the necessary condition of extreme value,

$$\begin{cases} \frac{\partial Q}{\partial b_0} = -2 \sum_{a=1}^n (y_a - \hat{y}_a) = 0 \\ \frac{\partial Q}{\partial b_j} = -2 \sum_{a=1}^n (y_a - \hat{y}_a) x_{ja} = 0 (j = 1, 2, \dots, k) \end{cases}, \quad (3.37)$$

And let it all unfurl, the equations become the following normal equations system,

$$\left\{ \begin{array}{l} nb_0 + \left(\sum_{a=1}^n x_{1a}\right)b_1 + \left(\sum_{a=1}^n x_{2a}\right)b_2 + \dots + \left(\sum_{a=1}^n x_{ka}\right)b_k = \sum_{a=1}^n y_a \\ \left(\sum_{a=1}^n x_{1a}\right)b_0 + \left(\sum_{a=1}^n x_{1a}^2\right)b_1 + \left(\sum_{a=1}^n x_{1a}x_{2a}\right)b_2 + \dots + \left(\sum_{a=1}^n x_{1a}x_{ka}\right)b_k = \sum_{a=1}^n x_{1a}y_a \\ \left(\sum_{a=1}^n x_{2a}\right)b_0 + \left(\sum_{a=1}^n x_{1a}x_{2a}\right)b_1 + \left(\sum_{a=1}^n x_{2a}^2\right)b_2 + \dots + \left(\sum_{a=1}^n x_{2a}x_{ka}\right)b_k = \sum_{a=1}^n x_{2a}y_a \\ \dots \\ \left(\sum_{a=1}^n x_{ka}\right)b_0 + \left(\sum_{a=1}^n x_{1a}x_{ka}\right)b_1 + \left(\sum_{a=1}^n x_{2a}x_{ka}\right)b_2 + \dots + \left(\sum_{a=1}^n x_{ka}^2\right)b_k = \sum_{a=1}^n x_{ka}y_a \end{array} \right. , \quad (3.37)$$

and if the introduction of the vector and matrix  $b = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ \dots \\ b_k \end{pmatrix}$ ,  $Y = \begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{pmatrix}$ ,  $X = \begin{pmatrix} 1 & x_{11} & x_{21} & \dots & x_{k1} \\ 1 & x_{12} & x_{22} & \dots & x_{k2} \\ 1 & x_{13} & x_{23} & \dots & x_{k3} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_{1n} & x_{2n} & \dots & x_{kn} \end{pmatrix}$ , so

$$A = X^T X = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ x_{11} & x_{12} & x_{13} & \dots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \dots & x_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ x_{k1} & x_{k2} & x_{k3} & \dots & x_{kn} \end{pmatrix} \begin{pmatrix} 1 & x_{11} & x_{21} & \dots & x_{k1} \\ 1 & x_{12} & x_{22} & \dots & x_{k2} \\ 1 & x_{13} & x_{23} & \dots & x_{k3} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_{1n} & x_{2n} & \dots & x_{kn} \end{pmatrix}$$

$$= \begin{pmatrix} n & \sum_{a=1}^n x_{1a} & \sum_{a=1}^n x_{2a} & \dots & \sum_{a=1}^n x_{ka} \\ \sum_{a=1}^n x_{1a} & \sum_{a=1}^n x_{1a}^2 & \sum_{a=1}^n x_{1a}x_{2a} & \dots & \sum_{a=1}^n x_{1a}x_{ka} \\ \sum_{a=1}^n x_{2a} & \sum_{a=1}^n x_{1a}x_{2a} & \sum_{a=1}^n x_{2a}^2 & \dots & \sum_{a=1}^n x_{2a}x_{ka} \\ \dots & \dots & \dots & \dots & \dots \\ \sum_{a=1}^n x_{ka} & \sum_{a=1}^n x_{1a}x_{ka} & \sum_{a=1}^n x_{2a}x_{ka} & \dots & \sum_{a=1}^n x_{ka}^2 \end{pmatrix},$$

and  $B = X^T Y = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ x_{11} & x_{12} & x_{13} & \dots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \dots & x_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ x_{k1} & x_{k2} & x_{k3} & \dots & x_{kn} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \dots \\ y_n \end{pmatrix} = \begin{pmatrix} \sum_{a=1}^n y_a \\ \sum_{a=1}^n x_{1a}y_a \\ \sum_{a=1}^n x_{2a}y_a \\ \dots \\ \sum_{a=1}^n x_{ka}y_a \end{pmatrix}$ , so the formula can be re write as



$Ab = B$ . The result of solving the equation A is  $b = A^{-1}B = (X^T X)^{-1} X^T Y$ .

And if the introduction of mark

$$\begin{cases} L_{ij} = L_{ji} = \sum_{a=1}^n (x_{ia} - \bar{x}_i)(x_{ja} - \bar{x}_j)(i, j = 1, 2, \dots, k) \\ L_{iy} = \sum_{a=1}^n (x_{ia} - \bar{x}_i)(y_a - \bar{y})(i = 1, 2, \dots, k) \end{cases}, \quad (3.38)$$

The normal equations can be written as

$$\begin{cases} L_{11}b_1 + L_{12}b_2 + \dots + L_{1k}b_k = L_{1y} \\ L_{21}b_1 + L_{22}b_2 + \dots + L_{2k}b_k = L_{2y} \\ \dots \\ L_{k1}b_1 + L_{k2}b_2 + \dots + L_{kk}b_k = L_{ky} \\ b_0 = \bar{y} - b_1\bar{x}_1 - b_2\bar{x}_2 - \dots - b_k\bar{x}_k \end{cases}, \quad (3.39)$$

So, solve the equation set (3.39) will get regression coefficients.

## 3.5 Portable management system of antibiotics

### 3.5.1 Detail information about the management system

The management mainly includes the manager of the supply chain of antibiotic and the management of producing and distributing and its applications.

Clinical trials drugs used in clinical trials are those what called trial drugs, control medicines or placebo. Clinical trials are an important part of the research and development of new drugs. And the production quality directly affects the safety and effectiveness of clinical trials, which has important significance in clinical trials with drug administration and safety, is an extremely important link of drug quality supervision in the whole chain.

#### 1. Drug regulatory model of the CPS

Due to the less use of CPS for drug supervision, we compared the applications in different other areas through analysis and adopted the design idea based on model structure, the possibility of applying the CPS in drug supervision is guarantee, Combining the advantages deep integration and high relation of CPS information and physical environment and presenting the CPS drug regulatory of architecture model based on the Internet of things system.

The CPS system model includes the following basic rules:

1. CPS model should be established on the basis of system block modeling, on the basis of the requirements of the scale and complexity of CPS;
2. Handshake connection between internal subsystem needs to be corresponded;
3. Information and the connector of part physical is definable and achievable;
4. CPS model is established by hybrid system which can be continuous, discrete even both,

### 3.5.2 Report and set of policies

Providing a certain amount of social security and cash compensation for the economic loss of farmers to reduce using antibiotic.

#### **Policies for enterprises:**

##### 1. Antibiotics insurance

In considering the insurance organization survey, underwriting, claims ability, especially on the basis of the construction of the basic level service system to determine the best insurance underwriting institutions, and comply with the following principles.

##### **Classified underwriting**

The varieties central financial subsidies are determined by the underwriting institution according to the planting, breeding, forest three varieties, are not allowed to cross cover in principle. In which a large class of insurance variety can only be insured by one company, and one insurance company can cover multiple varieties.

##### **Three years without changing**

The Insurance Company underwriting a variety of central financial subsidies, once determined, the share structure should remain unchanged for at least three years in principle within the scope its county.

##### **Exit when illegal**

After confirmed investigated and dealt with by Finance, China Insurance Regulatory Commission, audit, discipline inspection and supervision, judicial and other departments, the underwriting qualifications in the original area will be canceled from the next year.

##### 2. Policies support

Arranging the transformation of pharmaceutical companies to become a new driving force for economic growth, give them logistics support. Sound regional ecological compensation legal system, and take a variety of compensation methods by according to the wishes of the enterprise to compensate for the actual degree of loss as average amount of compensation in order to narrow the gap between the different enterprises.

Also set up the compensation mechanism of public participation, expert argumentation and government decision making consultation on the basis of local environmental laws and regulations, relevant administrative rules and regulations.

##### 3. Technical training

##### **Management knowledge training**

Provide the enterprises the management knowledge to understand the market, use the market and integrate into the market, also improve the information collection, information analysis and judgment and utilization ability, so they can avoid blind production or high yield but less harvest.

##### **Training of technical skill knowledge**

The modern production of antibiotic cannot be separated from the scale, standardization and intensification produce. It's is very important to provide them technology to improve the interests. Which will help enterprises transform and limit the antibiotic production.

#### **Policies for ordinary users:**

Let the public know that it's bad to use plenty of antibiotics at any time, and develop Incentive mechanism to use appropriate drugs while irrational use of antibiotics will pay more before.

## 3.6. Encouraging breeding policy

### 3.6.1 Econometric model

From the qualitative analysis, farmers' willingness to pay is not only affected by the internal factors such as individual characteristics, family characteristics, but also affected by the external factors, such as geographical location, policies and regulations, sales market, climate conditions and so on. Therefore, this paper uses the multi variables of the Douglas production function to analysis the relevant econometric factors. Specific models are as follows:

$$\log(y_i) = \sum_{j=1}^n a_{ji} (Ant \times \log(X))_i + \sum_{j=1}^n \beta_{ji} Z_i + \varepsilon_i$$

In the formula,  $y$  is the willingness of farmers to pay the amount of compensation.  $\alpha$  is the internal factor coefficient of the first  $j$  to affect the willingness of farmers to be repaid.  $X$  is a variety of internal factors.  $\beta_i$  is the first  $j$  kind of external factors that affect the willingness of farmers to be repaid.  $Z$  is a variety of external factors.  $Ant$  is the core variable antibiotic.  $\varepsilon_i$  is a random disturbance term. Access to relevant information, to assess the willingness of farmers to measure table, see Table 3.6:

Table 3.6 Correlations between antibiotics and factors

| Relevance  | Education level | Annual income | Revenue attraction | climatic conditions |
|------------|-----------------|---------------|--------------------|---------------------|
| Antibiotic | +               | -             | -                  | +                   |

The use of antibiotics resulted in a significant negative correlation with farmers' willingness to pay, and the amount of compensation should be increased. The higher education level is, the lower the degree of use of antibiotics is. Farmers' annual income reflects the degree of dependence on antibiotics, the higher proportion of the income of the family income is, the higher the degree of dependence on antibiotics is. The lower the income attraction is, the less the more antibiotics are, and the lower the expected standard of compensation is. Climate conditions allow farmers to reduce the use of antibiotics.

### 3.6.2 Incentive policy

Farmers use antibiotics for two reasons, one is to reduce the loss caused by the death of livestock, the two is to make animals grow faster.

Firstly, agricultural insurance can be established in order to encourage farmers to reduce the use of antibiotics. It provides a certain social security and cash compensation for farmers to reduce the economic losses caused by the use of antibiotics.

Secondly, we can give some policy supports, such as the establishment of regional special funds. Take a variety of compensation methods. According to the table 4.1, the higher the education degree of the farmers is, the less the use of antibiotics is, and the more compensation is. Farming income accounted for the proportion of household income, the more the use of antibiotics is, the less the compensation is. The more the income of non-breeding is, the less the use of antibiotics is, and the more compensation is. The more suitable climatic conditions are, the less the use of antibiotics is, and the more compensation is.

Finally, we can provide farmers with related Breeding training to reduce the loss of breeding.

## 4. Strengths and weaknesses

### 4.1 Strengths

- This model is based on quantitative analysis, so the results are objective and effective.
- This model can precisely calculate the interest of each part of the supply chain.
- Besides of accuracy, our model deals with the problem in a relatively shorter period of time, especially the simple and efficient algorithm can save a huge amount of time.
- The model is used to analysis the antibiotic resistance and predict the trend, which is valuable to the sociality and easy to manipulate. Save a huge amount of time and expense. With shorter decision time and less substantial searches, our model is of comparably higher practical value.
- It is shown that the weights set in prior have little impact on the model. Hence, we do not need worry about the divergence of the results with different parameters.

### 4.2 Weaknesses

- There is the possibility that the model may have to deal with huge amount of data.
- Some of our measures are based on unrealistic foundations, e.g. we do not distinguish drifters, however, in reality, bodies are used more to assist search.
- Some of the parameters are hard to confirm, so we have to set humanity.

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| T1 _____            | <b>2885</b>         | F1 _____            |
| T2 _____            |                     | F2 _____            |
| T3 _____            | Problem Chosen      | F3 _____            |
| T4 _____            | <b>C</b>            | F4 _____            |

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**2016 Ninth "Certification Cup" Mathematical Mathematics Modeling  
Challenge Summary Sheet**

## The Campaign On Antibiotics

### Summary

In this paper, we employ mathematical models to construct interest chains, analyze the sensitivity of the parties, predict the trend of the antibiotics crisis, and design a portable management system as well as propose a series of reward policies for farmers with rational use of antibiotics.

For Task1, we found that there are two interest chains (Doctors-Patients and Farmers-Drugstore) in the model constructed to analyze the mechanism of antibiotics in the market. In order to demonstrate the interest chains, we establish an adjacency matrix on variation of parties profits under policy intervention. At the same time, the Analytic Hierarchy Process (AHP) is used to evaluate the sensitivity of different parties. In this model, a key assumption is that the sensitivity of parties is directed at policy. We then compute the weight of parties to the policy, and get an order: pharmaceutical companies>hospital>patients>doctors>farmers>government.

For Task2, through using differential equation to imitate the Logistic model, we first analyze the interest chain between doctors and patients and conclude that the trend of crisis in a perfectly competitive market will increase with time. However, in order to consider the parties cognition of crisis, we construct two systems as another model. Doctors-Patients is an adjustable chain system, while the Farmers-Drugstore is a positive feedback system. Therefore, under the mechanism analysis, we conclude that with no policy intervention in a perfect competitive market, the antibiotic crisis will intensify.

For Task3, we design a management system of using antibiotics as an  $EM$  evaluation model. It defines an economic index  $E$  through the antibiotic interest chain. According to the average daily consumption of the local usage amount of antibiotics and the criteria of the Safety and Health Administration, we get the medical index  $M$  and medical safety index  $M_0$ . Under three cases  $S1(M < M_0)$ ,  $S2(M > M_0, E < M)$   $S3(M > M_0, M > E)$ , we can make a policy adjustment in different direction. Finally, we give a report on the model and propose a set of policies.

For Task4, in order to design a series of reasonable reward policy for farmers, the Maslow's demand theory was imitated for analyzing the demands of farmers raising livestock to get a Farmer-Demand model. The demand of farmers can be divided into four layers from low to high. Only when it meets the low-level, farmers will try to meet the next level. According to this, we have set a series of appropriate policies for farmers to meet all levels, and thus encourage them to use antibiotics less.

**Keywords:** Antibiotics    Differential equations    Mechanism analysis    Positive Feedback System    Maslow's Demand Theory

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November 28, 2016

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# 1 Introduction

## 1.1 Background

Referring to Antibiotics, it can not only kill bacteria but also plays an important role in inhibiting or killing the mold, mycoplasma, chlamydia and other pathogenic microorganisms. Antibiotics can be produced by certain microorganisms in the process of growth-reproduction. Also, some of them are completely synthetic or partially synthetic. Generally speaking, antibiotics are drugs used to treat a variety of bacterial infections or pathogenic microbial infections.

When antibiotics are honored to be the winner in the fight with tiny bacteria, however, all the modern medical measures and people now are in a spot. For example, more than 100 million people are infected with gonorrhea - a sexually transmitted disease - every year. Gonorrhea was easy to treat, but now gonorrhea bacteria have evolved out of the super-resistant strains, and spread throughout the world; In African hospitals, patients with super-drug-resistant tuberculosis are usually cureless.

From these examples, we seem to catch a glimpse of a "Doomsday scene" in the antibiotic era. The World Health Organization also predicted that, without strong measures, the "doomsday scenario" would become a reality in this century.

Aware of this threat to mankind, government and health officials have begun to pay attention to, and strive to solve this problem. Although it seems to be a scientific problem - bacteria and drugs, but in fact it is a matter of economic and even national security, global issues.

# 2 Nomenclatures

- Interest chain : a chain of people to generate upstream and downstream interests.
- Sensitivity : a change in the a factor resulting in a change in a outcoming.
- Perfectly competitive market : a market structure which in full competitions without any obstruction and interferences.
- Conditions for a perfectly competitive market:  
 Competition is not subject to government policy intervention.  
 The production of atibioticis far greater than market demand.  
 Resources are free to flow in the market.  
 The information is complete.
- Antibiotic resistance : antibiotics have lost the ability to effectively control or kill bacterial growth, that is, the bacteria are "resistant" and continue to multiply in the presence of antibiotic treatment levels.
- Economic Index (E):The relative number of the current total economic value flow-  
 ing into the entire interest chain to the sum of total expenditure of the patient on  
 medical costs and total input of the farmer on the use of antibiotics to feed livestock.

- Medical Index(M): The medical effect brought by the use of antibiotics in the region .

### 3 Basic Assumptions

- Assuming that the sensitivity of the parts which in interest chain to the supply and demand of antibiotics is only affected by policy.
- Assuming that the number of livestock which feed by each farmer is same .
- The price of all the livestock which sold by farmers in the perfectly competitive market is the same.

### 4 Variables and Definition

Table 1: Variables

| Symbol                             | Description                                                                                                            |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <i>Variables in Task1</i>          |                                                                                                                        |
| $P_1$                              | The patient                                                                                                            |
| $P_2$                              | The farmer                                                                                                             |
| $P_3$                              | The doctor                                                                                                             |
| $P_4$                              | The hospital administration                                                                                            |
| $P_5$                              | The pharmaceutical company                                                                                             |
| $P_6$                              | The government                                                                                                         |
| $B = (b_{ij})_{6 \times 6}$        | The relationship of interest among six groups from $P_i$ to $P_j$                                                      |
| $b_{ij} \quad (b_{ij} \in [0, 1])$ | Under the current policy, the transmission of proportion profits by using antibiotics from $P_i$ to $P_j$              |
| $\Delta_{ij}$                      | Under government intervention, the impact of policy on $P_i$ and $P_j$                                                 |
| <i>Variables in Task2</i>          |                                                                                                                        |
| $A$                                | Judging matrix                                                                                                         |
| $\lambda_{max}$                    | Greatest eigenvalue of A                                                                                               |
| $CI$                               | Indicator of consistency check                                                                                         |
| $CR$                               | consistency ratio                                                                                                      |
| $RI$                               | Random consistency index                                                                                               |
| $CW$                               | Weight vector for criteria level                                                                                       |
| $PW$                               | Weight vector for parties level                                                                                        |
| $n$                                | The dimension of the matrix                                                                                            |
| $F_1$                              | A farmer who does not use the antibiotics in the proceeding of feeding livestock.                                      |
| $F_2$                              | A farmer who uses the antibiotics to improve the percentage of animal survival in the proceeding of feeding livestock. |
| $num$                              | The number of livestock which feed by each farmer.                                                                     |
| $m$                                | The total money of $F_1$ spent in the breeding process.                                                                |
| $q$                                | The average cost of antibiotics per animal.                                                                            |
| $m + num \cdot q$                  | The total money of $F_2$ spent in the breeding process .                                                               |

|                           |                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| $r_1$                     | The survival rate of livestock which feeded by $F_1$ .                                                           |
| $r_2$                     | The survival rate of livestock which feeded by $F_2$ .                                                           |
| $v$                       | The selling price of the livestock in the market.                                                                |
| <i>Variables in Task3</i> |                                                                                                                  |
| $N_1$                     | The total population in a certain area.                                                                          |
| $Q(t)$                    | The number of people who used antibiotics at time $t$ .                                                          |
| $Q_1(t)$                  | The number of people with antibiotics resistance in the area at time $t$ .                                       |
| $Q_0$                     | The number of people who used antibiotics in the initial time ( $t = 0$ ).                                       |
| $N_1$                     | The average number of patients treated per day in all hospitals in the area.                                     |
| $\alpha$                  | The average number of patients treated per day in all hospitals in the area.                                     |
| $\mu(t)$                  | The number of patients previously treated with antibiotics among the patients who were hospitalized on day $t$ . |
| $\Delta t$                | The average time interval for a person need to go to the hospital.                                               |
| $t_m$                     | The average time for a person to develop drug resistance in the region.                                          |
| $W_1$                     | The total expenditure of the patient on medical costs.                                                           |
| $W_2$                     | The total input of the farmer on the use of antibiotics to feed livestock.                                       |
| $W_0$                     | The sum of $W_1$ and $W_2$ .                                                                                     |
| $W$                       | The current total economic value flowing into the entire interest chain .                                        |
| $X$                       | The usage amount of antibiotics.                                                                                 |
| $X_0$                     | The usage amount of antibiotics for human beings by the local medical institutions can not exceed .              |
| $\beta$                   | The coefficients of the $X - t$ function                                                                         |
| $g$                       | A coefficient.                                                                                                   |
| $h$                       | A coefficient.                                                                                                   |
| $M$                       | Medical index.                                                                                                   |
| $M_0$                     | risk threshold of medical index locally                                                                          |
| $D$                       | The ideal point of $(1, 1)$ in coordinates .                                                                     |
| $S$                       | The area in coordinate .                                                                                         |

## 5 Methodology

### 5.1 Task 1

#### 5.1.1 One model of interest chain

In order to establish the interests chain of the use of antibiotics, according to the circulation of antibiotics in the market, we reverse the direction of the circulation of antibiotics to form the interest chain. In the market, the government which is in the central

position influences the operation of the market through the policy . On the one hand, the antibiotics developed by pharmaceutical companies flow into the hospital through the purchase of hospital administrators and eventually flow into the patient through the doctor. On the other hand, antibiotics developed by pharmaceutical companies influx into the veterinary station. In order to reduce disease infection, farmers feed livestock with antibiotics or fodder containing antibiotics. By analyzing the circulation of antibiotics in the market, we present interest chain for the use of antibiotics in Figure 1 below.

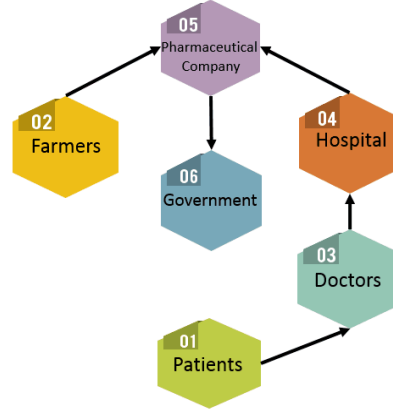


Figure 1: Interest Chain

### 5.1.2 Another model of interest chain

In order to demonstrate the interest chains, we establish an adjacency matrix on variation of parties profits under policy intervention.

$$B_{ij} = \begin{bmatrix} 1 & b_{12} & b_{13} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & b_{25} & 0 \\ 0 & 0 & 1 & b_{34} & 0 & 0 \\ 0 & 0 & 0 & 1 & b_{45} & 0 \\ 0 & 0 & 0 & 0 & 1 & b_{56} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

### 5.1.3 Analysation of sensitivity

When we try to analyze the sensitivity of the parties of the interest chain to the policy through analyzing the weight of each part in the interest chain ,considering the subjectivity is too strong and poorly-conceived. So we decided to combine the weighting coefficients of all the indicators in the evaluation system through the Analytic Hierarchy Process ( AHP) to analyze the sensitivity of parties to the policy .

### The three-layer hierarchy structure

The three-layer hierarchy structure that contains the criteria layer and parties layer is shown in Figure 2.

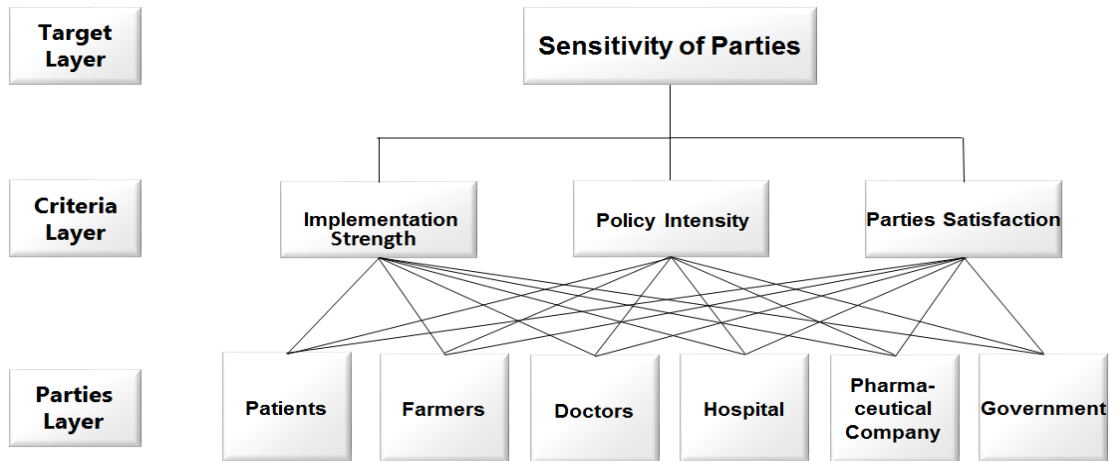


Figure 2: AHP

### Obtaining the index weights

- In order to compare the effect of the criteria layer on the target layer, we take two factors  $C_i$  and  $C_j$  at each time, and use  $a_{ij}$  to denote the influence of  $C_i$  and  $C_j$  on the target layer. All the comparison results can be expressed by pairwise comparison matrix  $A = a_{ij}$  :

$$a_{ij} = a_{ik}a_{kj}$$

where  $a_{ij}$  is set according to 1-9 method.

- Calculate the eigenvalues and eigenvectors for the greatest eigenvalue  $\lambda$ , then we use the size of value  $\lambda_n$  to measure the degree of inconsistency of Matrix A.
- Do a consistency check. The indicator of consistency is:

$$CI = \frac{\lambda_n}{n - 1}$$

Where n is the dimension of the matrix.

The expression of consistency ratio is :

$$CR = \frac{CI}{RI}$$

- Having confirmed the weighting coefficients of all the indicators in the evaluation system, now we quantify the importance of coaches.  $CW$  indicates the weight vector of the criteria layer C to the target layer  $PW_i$  denotes the weight vector of the parties layer P to the criteria layer C. The formula for the total ranking weight can be calculated as follow:

$$PW_{sum} = [ PW_1 \quad PW_2 \quad PW_3 ] \cdot CW$$

## Results and analysis

Through the above analysis, we obtain the following results.

- Judging matrix:

$$A = \begin{bmatrix} 1 & 3 & 5 \\ \frac{1}{3} & 1 & 2 \\ \frac{1}{5} & 2 & 1 \end{bmatrix}$$

- Weight vector of criteria level:

$$CW = [ 0.6571 \quad 0.1466 \quad 0.1963 ]$$

For this level,  $CI = 0.0816$ ,  $CR = 0.0907$ , satisfying the consistency check  $\frac{CI}{CR} < 0.1$

- Weight vector of parties layer:

— Implementation Strength :

$$PW_1 = [ 0.0377 \quad 0.0479 \quad 0.1352 \quad 0.2121 \quad 0.4884 \quad 0.0827 ]$$

— Policy Intensity:

$$PW_2 = [ 0.0272 \quad 0.0375 \quad 0.1415 \quad 0.2664 \quad 0.4096 \quad 0.01178 ]$$

— Parties Satisfaction:

$$PW_3 = [ 0.4713 \quad 0.2735 \quad 0.0350 \quad 0.0641 \quad 0.0994 \quad 0.0567 ]$$

CI and CR in Parties Layer

| Parties to Criteria Layer       | CI     | CR     |
|---------------------------------|--------|--------|
| <i>Parties to C<sub>1</sub></i> | 0.0080 | 0.0064 |
| <i>Parties to C<sub>2</sub></i> | 0.1083 | 0.0874 |
| <i>Parties to C<sub>3</sub></i> | 0.0789 | 0.0637 |

For this layer, we can get that all of these weight vectors satisfy  $\frac{CI}{CR} < 0.1$  through the table 5.

Using MATLAB to run the program, we got the final results. And then, we summarized the rank of sensitivity of parties in table 6 as follows:

Table 2: The rank of sensitivity

| Rank | Parties | Sensitivity of parties |
|------|---------|------------------------|
| 1    | $P_5$   | 0.4005                 |
| 2    | $P_4$   | 0.1910                 |
| 3    | $P_1$   | 0.1187                 |
| 4    | $P_3$   | 0.1164                 |
| 5    | $P_2$   | 0.0907                 |
| 6    | $P_6$   | 0.0820                 |

Through the table we can see the sensitivity of pharmaceutical companies is strongest and the sensitivity of hospital administrators, patients, doctors, farmers weakened in turn and the sensitivity of the government is minimum.

## 5.2 Task2

In order to analyze the trend of crises in a perfectly competitive market, we decided to proceed from the analysis of the interest chain constructed in the first question. In fact, the interests of the chain we constructed in the first question contains two chain of interests:

*farmer*  $\longrightarrow$  *pharmacy*  $\longrightarrow$  *pharmaceutical company*

*patients*  $\longrightarrow$  *doctor*  $\longrightarrow$  *hospital administrator*  $\longrightarrow$  *pharmaceutical company*

Next we build the first model to analyze possible crisis trends from the first interest chain.

### 5.2.1 Model One

For people in a certain area, we assume that the total population  $N$  is a constant. Our system neither consider the death nor migration, so the crowd can be divided into two groups: Group1 have used antibiotics and Group2 have never used antibiotics. Then we can get that  $Q(t)$  which denotes the number of people who used antibiotics at time  $t$  is a continuous, differentiable function.

In a statistical sense:

$$\mu(t) = e^{-\frac{t^2}{2}} \quad (1)$$

Assume the number of people who used antibiotics was  $Q_0$  in the initial time( $t = 0$ ),then:

$$\frac{dQ(t)}{dt} = \alpha \cdot N_1 \cdot \mu(t) \quad (2)$$

$$Q(0) = Q_0 \quad (3)$$

The result of solving the equation(1)(2)(3) is :

$$Q(t) = \alpha \cdot N_1 \cdot \int_0^t e^{-\frac{\sigma^2}{2}} d\sigma + Q_0 \quad (4)$$

The function graph of the result is Figure 3 as follows:

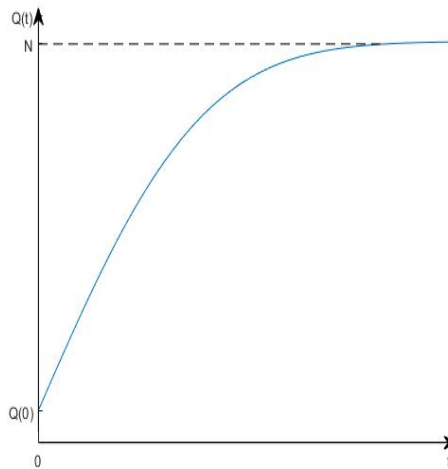


Figure 3: Funtion  $Q(t)$

Obviously, as time goes by, the number of people using antibiotics is growing. And in the process, the event that the same person receiving antibiotic treatment repeatedly continues to occur. We assume that superbug will burst in a normal persons body if he receives  $k$  antibiotic treatment so that antibiotics may exhibit resistance. We suppose that the number of people with antibiotics resistance in the area is  $Q_1(t)$ , and on average that once in a time  $\Delta t$  everyone in the area need to go to the hospital.

The average time for a person to develop drug resistance in the region is  $t_m$  calculated as follows:

$$t_m = \frac{k \cdot \Delta t}{\alpha} \quad (5)$$

When  $t \geq t_m$ ,

$$Q_1(t) = Q(t - t_m) \quad (6)$$

When  $0 < t < t_m$ , the number of people with drug resistance in the region should have increased very slowly, and as time  $t \rightarrow t_m$ , the number of people with resistance appears to be increasing at a faster rate. So,  $Q_1(t)$  should satisfy the following conditions:

$$\begin{cases} \frac{dQ_1(t)}{dt} > 0 \\ \frac{d^2Q_1(t)}{dt^2} > 0 \end{cases} \quad 0 < t < t_m$$

So the function figure of  $Q_1(t)$  can be shown in Figure 4 as follows:

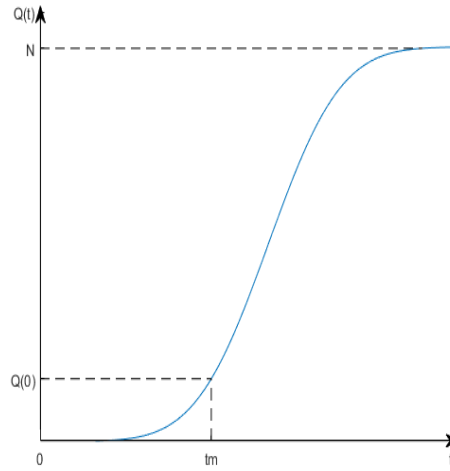


Figure 4: Function  $Q_1(t)$

We can regard the number of people with drug resistance as a degree of crisis, so under the condition of free competition, the trend of the crisis in the region grows slowly at the initial time. But once the time reaches time  $t_m$ , the real crisis came, and people began to alert to that. The measures they take may lead to decreasing superbugs growth rate, but the crisis will still be rising. We think that once the number of drug-resistant population is equal to the total number of people, the end of this region will come.



### 5.2.2 Model Two

Since the model above does not take people cognitive level into account, in the model below, we analyze the flow of antibiotics in the community from another point of view and believe it can be simulated with a system in the absence of government policy intervention. The system operating mechanism is shown in Figure 5 as follow:

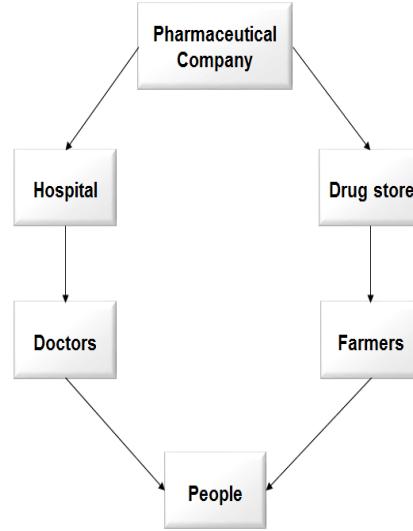


Figure 5: The System Operating Mechanism

We reckoned that the purchase of antibiotics by farmers is a motivating factor in this system. The following explains the behavioral trends of farmers using antibiotics by constructing a basic model:

Suppose there are two farmers  $F_1$  and  $F_2$  and they two keep the same number of livestock. We all know that there is a certain probability of death of livestock in the breeding process, assuming  $F_1$  farmers without using of antibiotics spent a total of  $m$  in the breeding process which survival rate of livestock is  $r_1$ . In order to improve animal survival,  $F_2$  use antibiotics and spent a total of  $m + nq$ . In the proceeding of  $F_2$ , the survival rate of livestocks is  $r_2$ . Finally, the farmer sells the surviving livestock. Thus the gains of  $F_1$  and  $F_2$  are as follows:

$$W_{F_1} = v \cdot r_1 \cdot n - m \quad (7)$$

$$W_{F_2} = v \cdot r_2 \cdot n - m - n \cdot q \quad (8)$$

So, we can get that :

$$W_{F_1} - W_{F_2} = v \cdot (r_2 - r_1) \cdot n - n \cdot q \quad (9)$$

According to statistics, after that the EU implemented a comprehensive "ban" in aquaculture industry in 2006, the survival rate of a hog reduced by 1.5 percent. The total price of antibiotics required for a hog which grows from 23kg to 13 kg is 1.65 dollars, while the sale price of per catty is 1.25 dollars. So the selling price of a hog is  $113 \times 2 \times 2.5 = 282.5$ . From the data above,

$$p = 1.25, v = 282.5, r_2 - r_1 = 1.5\%$$

By calculating ,the following inequation proved to be true :

$$(r_2 - r_1) > \frac{q}{v} \quad (10)$$

So,we can get  $W_{F_2} > W_{F_1}$ .Farmers have found that using antibiotics to raise livestock obtains greater profits, which will encourage them to expand the use of antibiotics.

The variation of the antibiotic social flow system can be expressed as follows;

Firstly, we suppose that move the interest chain(*farmer*  $\rightarrow$  *pharmacy*  $\rightarrow$  *pharmaceutical company*)out of the system in Figure6:

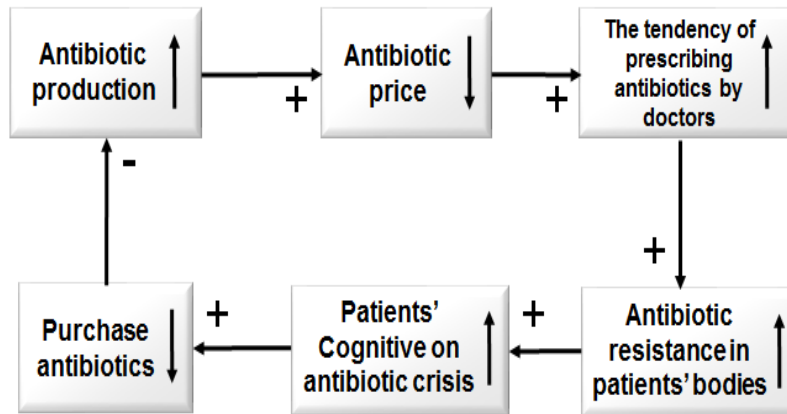


Figure 6: The System Without Farmer

We believe that when the patient is aware that excessive antibiotics can lead to antibiotic resistance, they will try to ask the doctor not to prescribe antibiotics. Then the amount of antibiotics seem to be descend. Thus, according to their own knowledge to limit the use of antibiotics , this is a self-regulating system. Therefore, without farmers in the system, we believe that the antibiotic crisis will not occur.

Add the interest chain(*farmer*  $\rightarrow$  *pharmacy*  $\rightarrow$  *pharmaceutical company*)into the system.We can get Figure7:

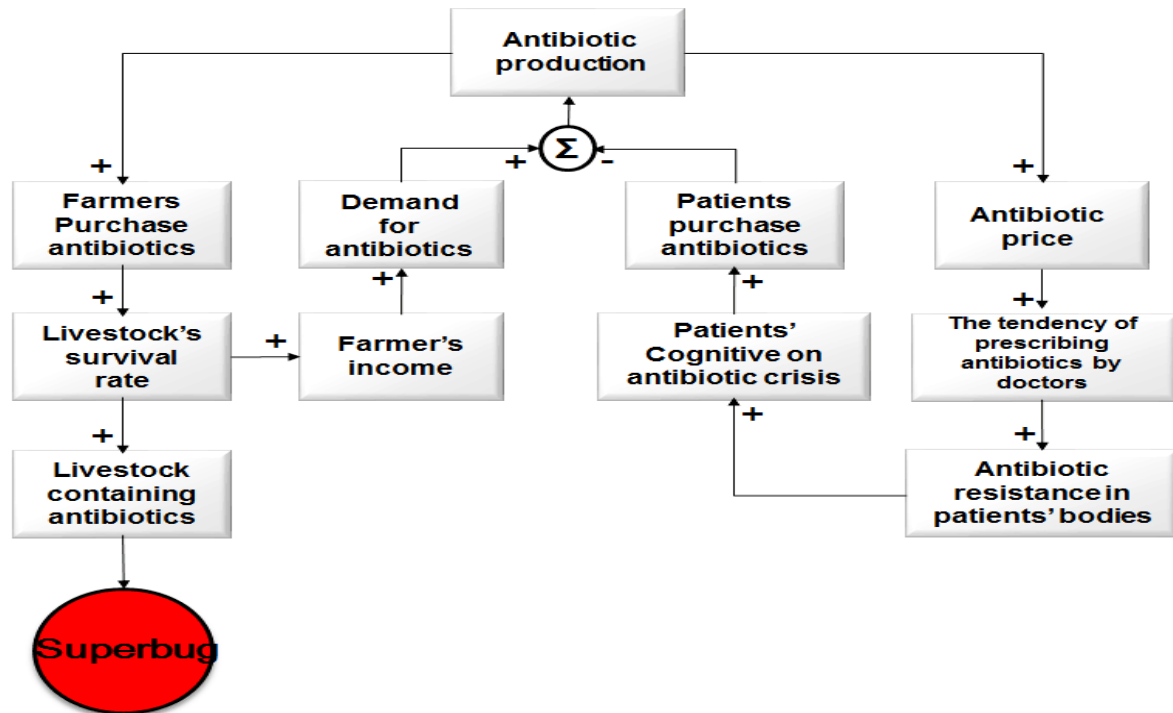


Figure 7: The Entire Interest Chain System

When adding the interest chain: *farmer*  $\longrightarrow$  *pharmacy*  $\longrightarrow$  *pharmaceutical company* into the system, even if the patient will be aware of antibiotic resistance and thus reduce the purchase of antibiotics, under a perfectly competitive market, farmers will still regard their own interests as the most important goal. Thus leading to a positive feedback system. In this positive feedback system, the more antibiotics are used, the more increasing resistance of antibiotics will be. Then, we doom that a crisis of superbug will burst.

Therefore, in the conditions of perfect competition, the government doesn't enact any policy on the amount of using antibiotics for parties. This will lead to a burst of superbug that increases the antibiotic crisis.

### 5.3 Task3

To design a portable management system for use of antibiotics is the most urgent affair for superbug. According to the current social situation of using antibiotics, the system tries to promulgate a series of policies, so that it will enable the status of the use of antibiotics to be in a good direction.

We evaluated the current use of antibiotics in two ways: **Economic index(E)** and **Medical index (M)**.

**Economic index(E)**: The relative number of the current total economic value flowing into the entire interest chain to the sum of total expenditure of the patient on medical costs and total input of the farmer on the use of antibiotics to feed livestock.

From the interest chain we can learn that the initial interests of the entire interest chain is from the antibiotics consumers (patients and farmers). Imitating the food chain, the economic benefits obtained by the use of antibiotics is also transmitting layer and layer

upward. So the current total economic value flowing into the entire interest chain is:

$$W = W_1 \cdot b_{13} + W_2 \cdot b_{25} \quad (11)$$

$W_1$  represents the total expenditure of the patient on medical costs, and  $W_2$  represents the total input of the farmer on the use of antibiotics to feed livestock.  $B_{ij}$  represents the relationship of interest among six groups from  $P_i$  to  $P_j$ . So we have the following two equations:

$$W_0 = W_1 + W_2 \quad (12)$$

$$E = \frac{W}{W_0} \quad (13)$$

When  $E = 0$ , it means that patients and farmers are all do not use antibiotics. The interest chain will be destroyed. When  $E = 1$ , it means patients and farmers are all using antibiotics. The economic efficiency reaches the maximum. When  $0 < E < 1$ , the greater  $E$  is, the better the economic benefits of using antibiotics is.

From the second problem we can see that with the antibiotics unconstrained abuse, the whole community medical index will be greatly damaged, and eventually it will lead to a drug-free situation. However, due to the current certain diseases fail to avoid using antibiotics, to ban on the use of antibiotics completely is unscientific.

Lets first consider the relationship between the current use of antibiotics in the community and parties using antibiotics. We assume the use of antibiotics is fix and it can be described as a symbol  $X$ . According to the effect of using antibiotics, the more the antibiotics are used, the larger amount of germs can be swallowed. It can be demonstrate as  $hX (h > 0)$ . Also, the effect that patients can be cured increases. At the same time, with the increase in the amount of bacteria in the humans body, bacteria begins to variate, which leads to the decreasing number of bacteria that can be killed. It can be demonstrate as  $\frac{g}{x} (g > 0)$ . Also, antibiotic treatment decreased, which will need more antibiotic. This is a vicious circle, and the treatment of antibiotics continue to decline which will finally tend to 0.

Therefore, we use the following function to express the relationship between antibiotic medical index and antibiotic use in a region:

$$M = \frac{1}{2\sqrt{hg}} \cdot \frac{1}{hX + \frac{g}{X}} \quad (14)$$

The fuction figure is as follows:

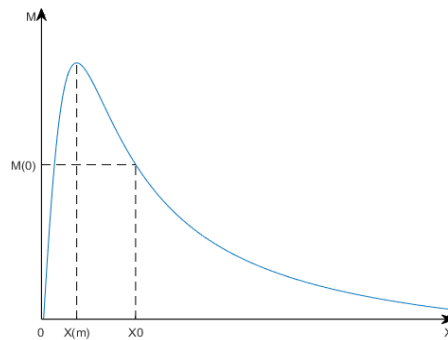


Figure 8: Founction  $M$

When the amount of antibiotics used  $X$  is equal to  $(\sqrt{\frac{g}{h}})$ , medical index reach the best. At the same time, we assume that the usage amount of antibiotics for human beings by th local medical institutions can not exceed the amount  $X_0$ .  
When  $X = N_1 \cdot X_0$ ,

$$M_0 = \frac{1}{2\sqrt{hg}} \cdot \frac{1}{hNX_0 + \frac{g}{N_1X_0}} \quad (15)$$

$M_0$  represents a medical index risk threshold locally. We should ensure the stability of the medical environment in the region, so  $M$  must be larger than  $M_0$ .

Therefore, the overall benefit of antibiotic use in a region reach its ideal state when  $E = 1$  and  $M = 1$ , that is,  $(E, M) = (1, 1)$ . However, the actual use of antibiotics is difficult to achieve this ideal state. Whatever, with respect to the medical index of a region, it always owns a minimum bottom line. Thus  $M > M_0$ , otherwise the region's medical system will face the risk of collapse. Therefore, when we assess the overall situation of an area, we can make the appropriate changes according to the situation. Here we divide it into the following three situations:

**State 1:**  $M < M_0$  : The current state of the medical index is lower than the minimum standard. So it is imperative to quickly improve the medical index. Due to the use of antibiotics,  $M$  and  $E$  always cannot increase at the same time. At this time, we will not consider the economic index temporarily and improve the value of  $M$  as much as possible, that is, try best to control the use of  $X$ , making  $M > M_0$ .

**State 2:**  $M > M_0$  and  $M < E$  : At this point the medical index is less than the economic index. In order to improve the comprehensive efficiency for the use of antibiotics, we should try to develop towards  $D(1, 1)$ . It is better to make a preference to treatment benefits than to economic benefits, and adjust the policy according to it.

**State 3:**  $M > M_0, M > E$  : In contrast to state 2, It is better to make a preference to economic benefits than to treatment benefits, and adjust the policy according to it.

The  $E - M$  Figure is as follows:

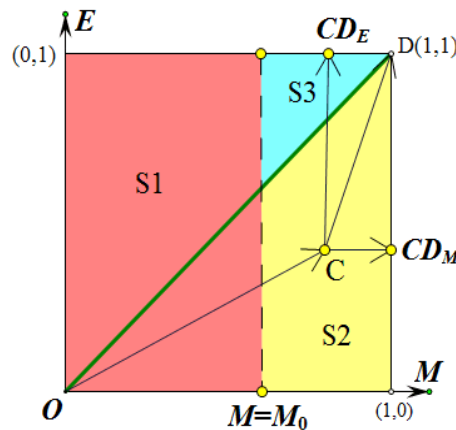


Figure 9:  $EM$  Model

Let us take state 2 as an example. Suppose that a region reaches  $(E, M) = C$  which in state 2 after evaluated. In this case, it should develop towards  $CD$ . After the orthogonal decomposition vector,  $|CD_M| < |CD_E|$ , should be more emphasis on economic development. In order to increase  $E$ , we can enact policies to increase  $\Delta_{13}$   $\Delta_{25}$ , for example, by

reducing the corresponding taxes on antibiotics. Let us take state 2 as an example. Suppose that a region reaches state 2 evaluated. In this case, it should develop towards  $CD$ . After the orthogonal decomposition vector,  $|CD_M| < |CD_E|$ , should be more emphasis on economic development. In order to increase  $E$ , we can enact policies to increase  $\Delta_{13}$   $\Delta_{25}$ , for example, by reducing the corresponding taxes on antibiotics.

With the analysis above, we provide a series of policy principles in the report.

#### 5.4 Task4

Based on the previous analysis, we have concluded that in the fully competitive market, the interest chain with farmers will intensify the emergence of antibiotic-resistant crisis. Therefore, in order to alleviate the emergence of antibiotic resistance crisis, government needs to formulate policies to intervene the market. However, in the AHP analysis of the first question, we conclude that the sensitivity of farmers to policy is relatively low. At the same time, simply policies to restrict the use of antibiotics will have a certain impact on their economy. Therefore, we believe that the government is necessary to introduce more incentive policies to increase the life satisfaction of farmers so that farmers are more willing to reduce the use of antibiotics. In order to learn what kind of incentive policies should be developed, next, imitating the Maslow's theory of demand, we start an analysis of the needs of farmers feeding cattle and get a Farmers-Demand model as follows:

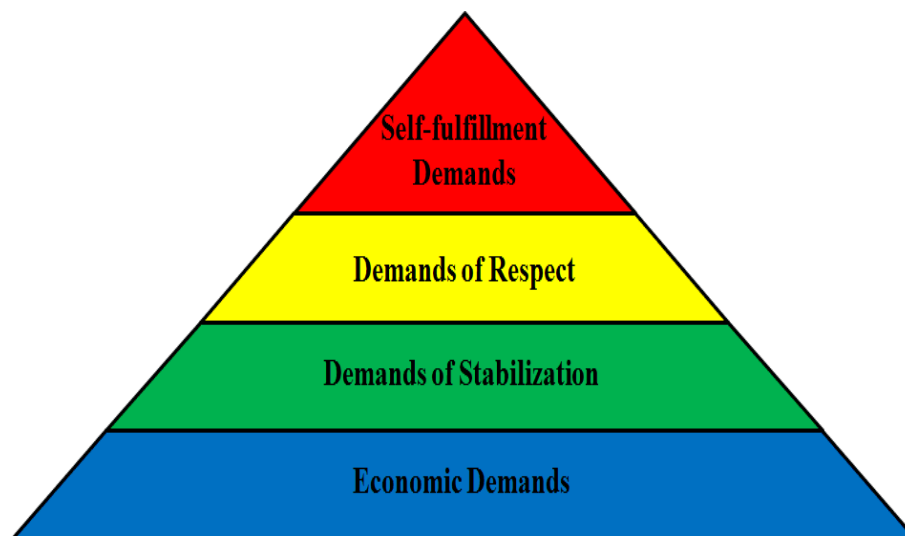


Figure 10: Farmers-Demand Model

The demand of farmers can be divided into four layers from low to high. Only when it meets its low-level, the farmers will seek to meet the needs of the next level. We can set a series of appropriate policies to enable farmers to meet needs of all levels, and thus encourage them not to use antibiotics to raise cattle.

##### **Economic Demands:**

Because the farmer groups make profits by selling livestock, cattle are, in a way, commodities, and they belong to the category of businessmen. Therefore, the most basic need is economic profit.

##### **Economic policies:**

- **Subsidy policy for the introduction of improved livestock:** Encourage farmers to introduce fine varieties of livestock. Production of fine varieties of livestock can receive a certain amount of subsidies.
- **Compulsory immunization subsidy policy for major animal diseases:** Compulsory immunization policy for major animal diseases; Vaccine funds shall be shared proportionately by the central and local governments, and the cost of compulsory immunization vaccine shall not be paid by farms.
- **Subsidy policy for the harmless livestock and poultry breeding in the breeding sector:** The large-scale farm with dead livestock and poultry shall be subsidized with the sound disposal treatment subsidy, whose subsidies shall be jointly borne by the central and local governments.
- **Incentives for the use of antibiotic substitutes:** In accordance with the receipt of tickets, the Government shall provide incentives for the purchase of antibiotic substitutes by the farming personnel. The fund shall be shared in proportion by the central government and local finance.

#### **Demands of Stabilization:**

To meet the basic economic profit, farmers hope that their profit-making way is durable, not short-lived. In other words, farmers want to keep their own methods of cattle can be stable and healthy growth, and gain profits. The use of antibiotics to ensure the health of cattle and reduce the cost of its feeding .

#### **Stability Policy:**

- Agricultural insurance support policy: The seed insurance shall be shared in proportion by the central government and local finance.
- Agricultural insurance support policy: The seed insurance shall be shared in proportion by the central government and local finance.

#### **Demands of Respect :**

For some areas of occupational discrimination, once the farmers have a stable income, they also want to be respected in other sectors of people. Therefore, it is better to promote a new agricultural landscape, making the farmers raising cattle scientifically well-known, and expand their market influence.

#### **Promotion Policy:**

- Increase the propaganda of the risk of antibiotics to raise awareness of the risk of antibiotic resistance crisis.
- Actively guide the breeding retail to do scale farming, and join some high-tech products for livestock environment for intelligent control.
- Introduce some highly educated personnel on grass-roots culture of high-tech pollution-free farming training.

#### **Self-fulfillment Demands:**

The demand for self-realization is the highest level, which refers to the realization of farmer ideals, aspirations, maximum ability of individuals. In this level, they are willing

to development contribution to their own strength and strive to realize their potential to make themselves closer to the pinnacle of their field.

#### Incentive policies:

- For those farmers who are able to explore a more environmentally friendly, healthier and more effective way of feeding without the use of antibiotics, they are worth validating by the relevant authorities and awarding a "Best New Farmer Model".

## 6 Sthrengths and Weakness

### 6.1 Strengths

- The interest chain showed by using of adjacency matrix which takes the profit ratio as elements , but also can clearly response to the impacts on two adjacent individuals in interest chain due to policy changes.
- The model can be used to simulate the relationship between the patient and the hospital through using differential equation to imitate the Logistic model. The time of the disease is  $t_m$ , and the number of the super bacteria population can be calculated as time goes, which can reflect the trend of the crisis more accurately.
- The comprehensive-benefits coordinate graph of (antibiotic management system)Model III is easy to operate ,which enable the users to draw the direction of policy which needs to adjust in the region.
- In Task 4,we analyze the demand of farmer groups according to Maslow's theory,take all-round consideration into the basic needs of farmers, develop the more convincing reward policies.

### 6.2 Weaknesses

- The data of Task 2 are obtained in the mean sense, while the changes of actual data should be volatile.So there is a certain error in the selection of the data.
- The economic index in Task3 needs to obtain the input of farmers and patients, and the corresponding profit ratio is more difficult. The demarcation of medical safety index  $M_0$  is affected by comprehensive impacts including the scale of local hospital which is more difficult to assess accuratly.



## 7 Report

### Report On $EM$ Model And A Series Of Policies

#### Introduction

According to our analysis of the use of antibiotics, we have designed a portable management system using antibiotics-EM model, and put forward to a corresponding series of policies. The report is going to analyze the process of our management system and finally propose a series of policies on parties in the interest chain.

#### System Theory

**Step1:** Evaluate the corresponding economic index  $E$  of antibiotic use. According to the chain of interest caused by the use of antibiotics to a region, we choose the ratio of the input of the antibiotic to the total input of the consumers as the economic index  $E(0 \leq E \leq 1)$ . The greater the  $E$  is, the better the economic benefit of the current use of antibiotics in the region is.

**Step 2:** Evaluate the corresponding medical index  $M$  of the use of antibiotics; According to the characteristics of antibiotics will occur with drug resistance, we defined the medical index  $M(0 \leq M \leq 1)$  for the region daily antibiotics of the amount used. According to the daily dose of antibiotics we can get the  $M$  of the region. The larger the value of  $M$  is, the better medical benefits it will get. At the same time, according to the maximum use determined by the local health security bureau, we can get the safety index  $M_0$ . That is to say,  $M > M_0$  is on behalf of that the current use of antibiotics is safe, and  $M < M_0$  is on behalf of that the current regional use of antibiotics is at risk.

**Step3:** According to the survey of actual situation of a region, we can get its comprehensive benefits coordinates  $(E, M)$ . Three cases we have defined in the model provide in the Figure 9 below:

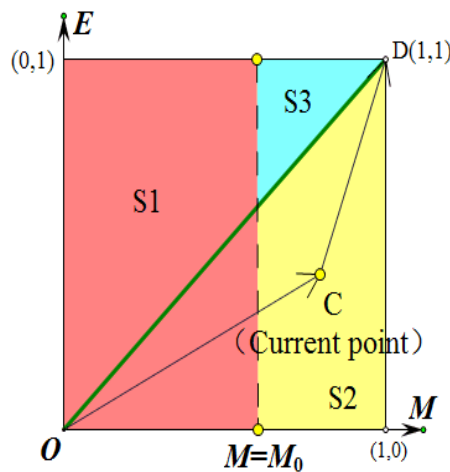


Figure 11: Farmers-Demand Model

we can give the direction of adjustment policy:

- **For pharmaceutical companies:** Encourage the development of new drugs, which will increase the human medical index.

- **For the hospital:** Establish scientific system rational allocation of performance, rewards and penalties. Improve the enthusiasm of the rational use of antimicrobial drugs and medical staff.
- **For doctors:** Regulate the frequency of its use of antibiotics strictly. Do not use antibiotics in case of that they have the same cure effect as other drugs.
- **For the general population:** Do not buy antibiotics privately; Do not take the initiative to request for antibiotics; Do not casually stop the medicine.
- **For farmers:** Ensure that using antibiotics for animals only for the treatment of infectious diseases and should be under the supervision of veterinarians; The amount of antibiotics should meet the standard of the World Health Organization.
- **For the Government:** Develop a strong national action plan to address antibiotic resistance; Improve monitoring antibiotic resistance infection; Strengthen infection prevention and control measures.

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# Control the antibiotic to save more lives

November 28, 2016

## Summary

Also known as antimicrobial drugs, antibiotics have saved countless lives. Misuse and overuse of these drugs, however, have contributed to a phenomenon known as antibiotic resistance. Control of the abuse of antibiotics becomes the imperative to solve the problem of super bacteria.

To solve the problem of the antibiotic abuse, we establish the interest chain to analyze the source of the antibiotic abuse. Then we use the comprehensive competition model to confirm the damage of the antibiotic abuse. Finally, we give several suggestion to the government to better manage the circulation of antibiotics.

To task 1, according to the general drug distribution process, we establish the interest chain for use of antibiotics. Then, we use the correlation coefficient matrix to analyze the dynamic factors to the interest chain. One of the innovation is that we use the benefit of each parties to represent the the sensitivity to the interest chain.

To task 2, we establish the Markov chain model to predict the trend of the number of patients with time. Then, we choose 5 yeras to be a period of the antibiotic's circulation .By using the correlation matrix and taking the hysteresis effect into account ,we get the development trend of antibiotics in the perfectly competitive market.

To task 3, based on the comprehensive comprtiton model in task 2, we give some reward policies on the perspective of the government.

To task 4, after referring to the related articles and study, we give the formers reward policy to encourage them use antibiotics less.

This paper take many dynamic factors into consideration, and analyze the correlation between these factors. In practical application the models obtain better adjustability.

**Keywords:** antibiotic control, dynamic interest chain, comprehensive competition model, linear correlation matrix

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# 1 Introduction

## 1.1 Problem Statement

As is known to all, antibiotics are drugs used for infections caused by bacteria, and antibiotics have saved countless lives since the World War II. However, misuse and overuse of the drugs have contribute to a phenomenon known as antibiotic resistance. It means that bacteria have evolved to withstand antibiotics. To fight with these new type of bacteria which is also known as superbug, scientists have to study new antibiotics. As a consequence, the fight between the human and the bacteria will fall into a vicious circle, and more and more antibiotics will lose their curative effect on bacteria.

The only way to prevent the disastrous consequence is to slow down the spread of antibiotics, and reduce the use of antibiotics unless it is necessary. A big block on our way to limit the use of antibiotics is the unawareness of doctors and formers. In order to seek the best benefit, a majority of them use the antibiotics unlimitedly, regardless of the damage and the potential crisis. So our team will develop a better understanding of the factors involved with antibiotic resistance crisis. First of all, we will develop a dynamic interest chain to illustrate the component of people who motivate the antibiotic spreading. Whats more, we shall predict the tendency of the crisis. So that people will have a better knowledge of the potential danger the crisis would bring about. Of course, some feasible suggestions will be put forward to limit the use of antibiotics.

## 1.2 Related Work

In recent years, there has been more and more concerns on the abuse of the antibiotics and the potential crisis it would cause. The FDA (Food and Drug Administration) has pay a lot of attention on the problem and has released many reports about combating antibiotic resistance. And in china, CFDA set up a series of regulations to manage the antibiotic market in 2004. For example, the retail drugstore can only sell the prescription medicine when it get the specific certification.

Many works have done to evaluate the danger of antibiotics abuse. According to the statistics, in china, the patients who use antibiotic drugs when in the hospital account for about 58%. The data has been far above the world standard 30%. The main curse for the result may be the unreasonable clinical use, the wrong choose of drugs and the lax control of preventive medicine. Despite these results, we have to further the evaluation models, and give some feasible suggestions to the ICM-FDA.

# 2 Assumptions and Notations

## 2.1 Assumptions

- Regardless of changes in the price of antibiotics.
- Patients buy antibiotics only for treatment.
- The price of the cattle that raised by the formers is fixed.

- The raw material prices of the antibiotic is unchanged.
- Evaluations of non-quantifying factors in medical institutions are average

## 2.2 Notations

We use a list of symbols (cf. Table 1) for simplification of expression.

Table 1: Notations

| Symble         | Definition                                                   |
|----------------|--------------------------------------------------------------|
| $x_i$          | the first $i$ dynamic factor before the non-dimensional data |
| $y_i$          | the first $i$ dynamic factor after the non-dimensional data  |
| $\gamma_1$     | highest satisfaction                                         |
| $\gamma_2$     | lowest satisfaction                                          |
| $\Delta\gamma$ | the range of satisfaction changes                            |
| $R$            | linear correlation matrix                                    |
| $Z$            | principal component matrix                                   |
| $\chi_i$       | people's state random variables                              |

## 3 The Dynamic Interest Chain For Antibiotics

The widespread use of the antibiotics which is represented by penicillin caused the sharp decline in infectious diseases in the 20th century. However, as the price, the superbug which also known as the resistant bacteria experienced a widely distribution process. And the resistant mechanisms has become more complicated, the resistant bacteria spectrum also become multiple.

In order to control the crisis, we have to analyze the problem from the source of the widespread use of the antibiotic. whats more, we also need to figure out who push the industrial chain develop so fast.

### 3.1 Establishment of the interest chain

On the basis of reference for other industry interest chain, we divide the interest chain for the antibiotics into four parts: Producer, Distribution process and Consumer[1]. Different parts has different functions and also has different benefits from the antibiotics.

The producer mainly includes the production enterprise which produce the antibiotics and sell the antibiotics to the market. The distribution process includes the distribution enterprise, the hospital and the retail drugstore. Due to the strict control from the government on the circulation of the antibiotics, the distribution process can only be completed by the authorized hospitals or enterprises. And the doctors prescription is essential when you want to buy the antibiotic medicine. So, the doctor is the important terminal conductor for the use of antibiotics. The last part of the chain is the consumer. Patients and Formers are both included into this part. According to the related study, the majority use of the antibiotics is from the formers who raise cattle for sell.

Finally, we establish the interest chain for the use of antibiotics which is vividly shown in the Figure 1.

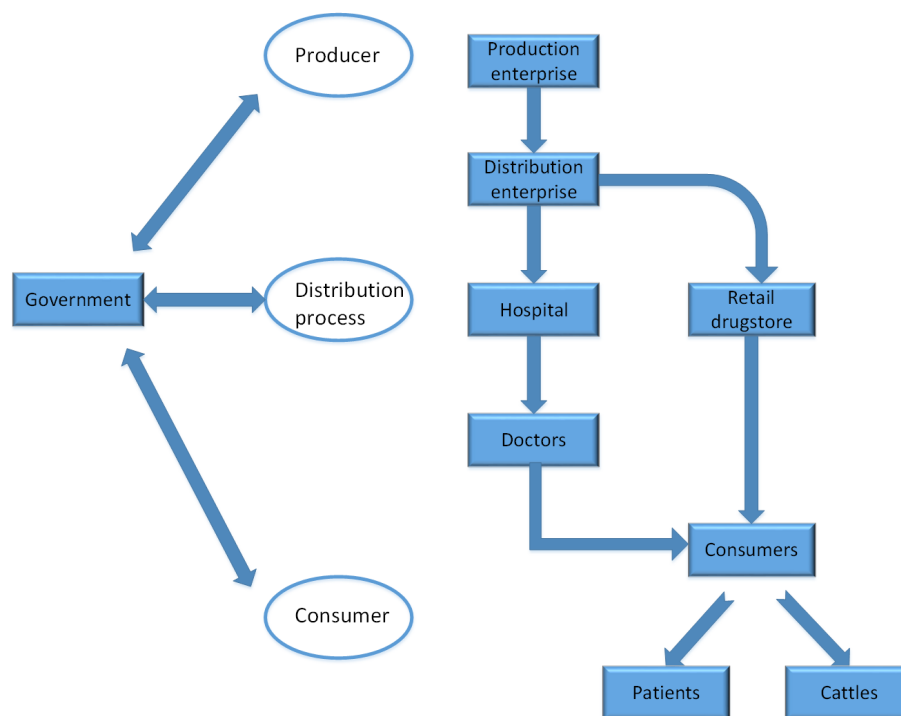


Figure 1: The interest chain for the use of antibiotics

### 3.2 Selection of the dynamic factors

To extend the study of the components in the interest chain, we have to analysis the dynamic factors. First of all, many factors can affect the producers. The demand for medical institutions and the cost of production may affect the output of the antibiotics. The government subsidies, the competition of enterprises and the reputation of the enterprise can affect the benefits of the enterprise indirectly.

To the distribution process, the most important factor may be the control from the government. Because of the lax supervision to the sell of the antibiotics, the demands from consumers can account for a large proportion of the profile of medical institutions.

The doctors has two sides in the sell of the antibiotics. On one hand, they want to get more profile; on another hand, they have their own social responsibility.

When it comes to the patients, the consumption level in their medical cure and the doctors advice are the most important factors. The main purpose for formers to use the antibiotics is to short the culturing period. So, we can see the main dynamic factors that affect the interest chain in the table2.

According to the table and the anlysis of different parts, we can figure out the factor structure of each part.

- The dimensionless processing of data



Table 2: Dynamic factors

| Components who get profile | Dynamic factor                                                                                                                                      |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Production enterprise      | government subsidies<br>the demand for medical institutions<br>cost of production<br>the reputation of the enterprise<br>competition of enterprises |
| Medical institution        | government subsidies<br>the demand for consumers<br>the government certification                                                                    |
| Doctors                    | social responsibility<br>the demand for patients<br>rebates from the drug producers<br>regulations of Antibiotic use                                |
| Patients                   | consumption level<br>government subsidies<br>the doctor's advice                                                                                    |
| Formers                    | feeding cost<br>government subsidies<br>culturing period                                                                                            |

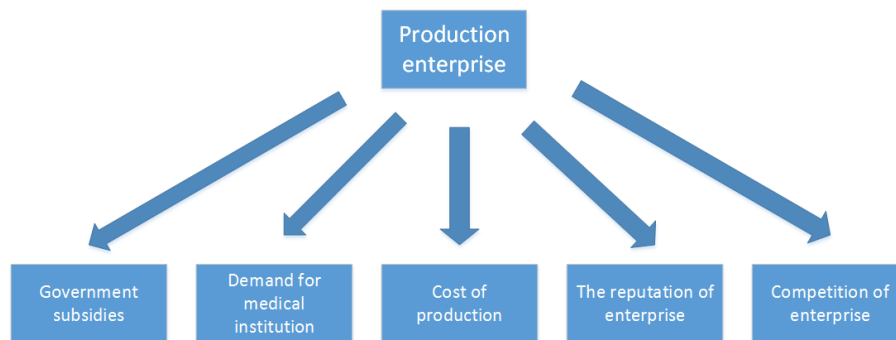


Figure 2: Factors for the production enterprise

To analyze the different factors in one model, we have to make the factors dimensionless. There are many ways to unified the dimension. To simplify the calculation process, we use the extremum method. If we use  $x_i$  to represent the former data, then we get the data after dimension  $y_i$  :

$$y_i = k \frac{x_i - \min x_i}{\max x_i - \min x_i} + q \quad (1)$$

Among the equation,  $k$  and  $q$  are the control parameter.

Except the quantifiable factors, there are many factors cannot be quantified. We choose the reputation of the enterprise as an example to undertake the scoring systems. When it comes to the reputation of the enterprise, we often take the satis-

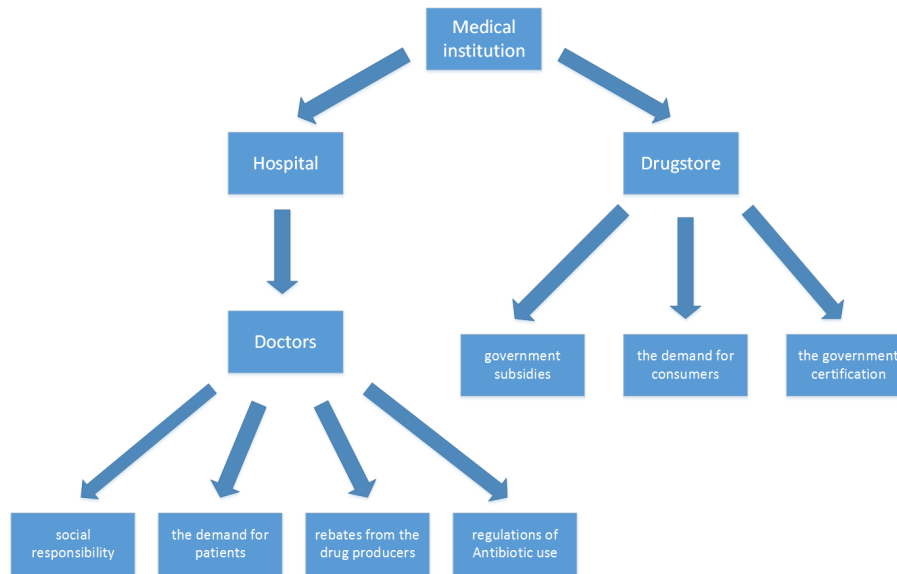


Figure 3: Factors for the medical institution

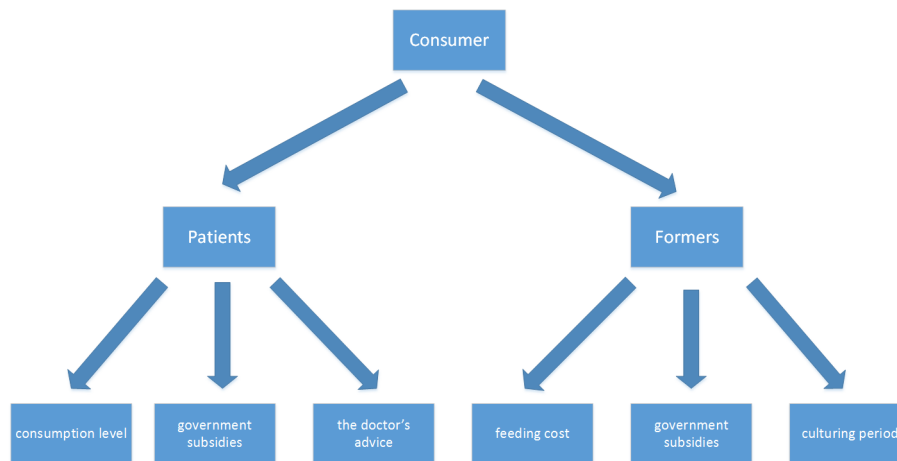


Figure 4: Factors for the consumer

faction of people as an alternative. Then, we can do a survey to let consumers give their about the enterprise. And the total score is ten.

$$\Delta \gamma = \frac{\gamma_2 - \gamma_1}{10} \quad (2)$$

$\Delta \gamma$  means the satisfaction varying range.

And the results can be divided into several levels, which can be seen in the table 3. The quantified datas can also be nondimensionalized by the method used in equation 1.

- **establishment of the correlation coefficient matrix**

To figure out the effect among the factors, we establish the correlation coefficient matrix[2]. Then we can assume the correlation coefficient matrix can be represented as follows.

Table 3: The evaluation of non-quantification factors

| score                           | levels        |
|---------------------------------|---------------|
| $\gamma_1$                      | 1 (very bad)  |
| $\gamma_1 + 5 \triangle \gamma$ | 5 (medium)    |
| $\gamma_2$                      | 10(very good) |

$$\begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{nn} \end{bmatrix}$$

Through the establishment of the correlation matrix model, we can calculate for each factor with corporate interests between two correlation coefficient matrix. But we still found that some of the relevant risk coefficient changing with corporate interests range is small, even can be neglected. The result of multivariate function relation equation is complex, often do not have a final solution. To eliminate the possibility of no solution, we need to modify the factors that has little impact on the fuction. As a consequence, we establish the expert-principal component model. The linear correlation matrix can be calculated by the linear coefficient model.

In order to calculate the characteristic equation, let :

$$|\eta I - R| = 0 \quad (3)$$

The eigenvalue can be represented as  $\eta$ . The corresponding eigenvector is  $|E_i| = 1$ . After getting the eigenvalue, we can calculate the contribution rate and cumulative contribution rate. When the contribution rate and cumulative contribution rate are range from 85% ~ 95%. The corresponding factor can be considered as a proper result.

And the contribution rate can be represented as:

$$\frac{\lambda_i}{\sum_{k=1}^n \lambda_k} (i = 1, 2, \dots, n) \quad (4)$$

The cumulative contribution rate can be represented as:

$$\frac{\sum_{k=1}^i \lambda_k}{\sum_{k=1}^n \lambda_k} (i = 1, 2, \dots, n) \quad (5)$$

By calculating the contribution rate and cumulative contribution rate, we can confirm the principal component. Other factors except the principal components can be ignored. And the principal component can be represented as matrix:

$$\begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1p} \\ z_{21} & z_{22} & \cdots & z_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \cdots & z_{np} \end{bmatrix}$$

## 4 Comprehensive competition model

### 4.1 The analysis

For Problem 2, we build the chain of antibiotic interests in the context of a perfectly competitive market to predict crisis trends. As the market is perfectly competitive, the entire chain of antibiotic interest chains is driven by consumer demand.

In order to prevent errors and data distortion, this paper takes 5 years as a cycle. We use the Markov chain model to predict the trend of the number of patients with time to study and change in response to consumer demand. Farmers demand for antibiotics with a wide range of coverage and fuzzy characteristics. So we use the extreme method. The total demand of patients and farmers is the demand chain.

After solving Problem one, we can see that consumer demand has certain connection with other dynamic factors. Therefore, by establishing the correlation matrix between consumer demand and dynamic factors in 5 years, the correlation coefficient is used to establish the relationship between consumers. The change of demand leads to the change of other dynamic factors.

Finally, it brings the producer interest function and the middleman interest function established in question one and sums up the change trend of interest. In view of the crisis situation, this paper shows the development of the crisis in the perfect competitive market by modifying the profit function because the sales quantity function is difficult to be determined due to the sales quantity of antibiotics, and because there is lagging interest between interest and sale trend.

### 4.2 The assumptions[4]

- A large number of buyers and sellers .
- All consumers and producers know all prices of products and utilities each person would get from owning each product.
- The products are perfect substitutes for each other.
- No barriers to entry or exit.
- Firms sell where the most profit is generated, where marginal costs meet marginal revenue.
- No participant with market power to set prices.

### 4.3 The establishment of the Markov chain model

Markov chain model is used to solve a system known now, in the future has nothing to do with the history of the later, time, state are discrete random transfer process and the

human condition is in line with the above changes. As the actual number of patients on the demand for antibiotics is difficult to achieve statistical data, so this article reflects the patient's side by the prevalence of antibiotic demand.

Since the illness is a function of time, the time is discretized. For each discrete time, the state of a person is represented by a random variable  $\chi_i$ . When  $\chi_i = 1$ , people are in need of the antibiotics. When  $\chi_i = 0$ , people aren't in need of the antibiotics. We can use  $a_i(n) = P(\chi_n = i)$  to represent the state probability.

The probability of transition from state  $\chi_i = 0$  to state  $\chi_i = 1$  is  $P_{12} = (\chi_{n+1} = 1 | \chi_n = 0)$ . To sum up, the Markov chain equation is as follows:

$$\begin{cases} a_i(n+1) = \sum_{j=1}^k a_j(n)P_{ji} \\ \sum_{i=1}^k a_i(n) = 1 \\ P_{ij} \geq 0 \\ \sum_{j=1}^k P_{ij} = 1 \\ a(n) = (a_1(n), a_2(n), a_3(n), \dots) \\ P = p_{ij \ k \times k} \end{cases}$$

Then we get the probability of people getting sick at different time points:

$$\begin{cases} a(n+1) = a(n)P \\ a(n) = a(0)P^n \end{cases}$$

Where  $P$  denotes the transition probability matrix.

Similarly, the probability of bacterial resistance to antibiotics,  $q$ , we can establish the Markov chain:

$$\begin{cases} b_i(n+1) = \sum_{j=1}^k b_j(n)q_{ji} \\ \sum_{i=1}^k b_i(n) = 1 \\ q_{ij} \geq 0 \\ \sum_{j=1}^k q_{ij} = 1 \\ b(n) = (b_1(n), b_2(n), b_3(n), \dots) \\ q = q_{ij \ k \times k} \end{cases}$$

The probability of bacterial resistance to antibiotics at different time points is:

$$\begin{cases} b(n+1) = b(n)q \\ b(n) = b(0)q^n \end{cases}$$

In this paper, we choose the extreme value method, which finds that the farmers at any time demand is equal to the threshold, the farmers demand function is as follows:

$$\beta = b \tag{6}$$

To sum up the consumer demand chain in the interests of the same time as a function of the following:

$$\varepsilon = S \times a(0)P^n + S \times b(0)q^n + b \quad (7)$$

Where  $\varepsilon$  represents the consumer demand. Then, we calculate the time-varying demand which is shown in the figure5.

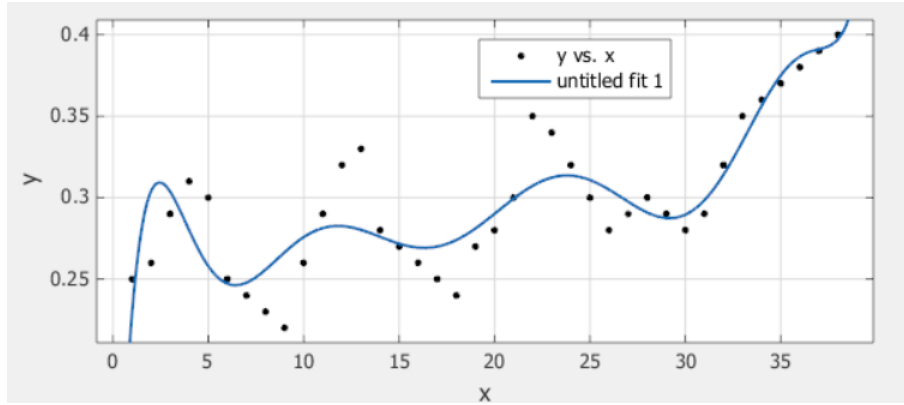


Figure 5: The time-varying demand

Due to market interests and product circulation has a lag, so we need to add a lag coefficient  $\theta$  of the time axis. The function of the crisis development trend in the fully competitive market is:

$$\omega = \alpha_0 + \alpha_1(U_{1+\theta} + \Delta U_{1+\theta}) + \alpha_2(U_{2+\theta} + \Delta U_{2+\theta}) + \dots + \alpha_{s-1}(U_{s-1+\theta} + \Delta U_{s-1+\theta}) + \varepsilon + \Delta \varepsilon \quad (8)$$

The antibiotic circulation was calculated as figure 6.

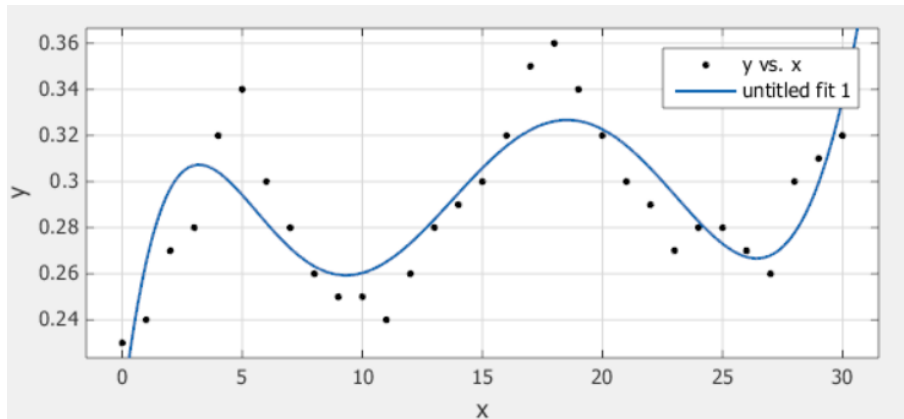


Figure 6: The antibiotic circulation

By comparing the trend of crisis with the demand of consumers, we can see that in a perfectly competitive market, there is no government function to control the circulation of antibiotics. Because of the self-regulation of market, the crisis develops in a period of time. A horizontal line swings up and down. With the prolongation of time and the abuse of antibiotics, the market's self-regulation is rapidly broken, the crisis is rising rapidly, which seriously endangers the survival of human society. Therefore, the abuse of antibiotics should be regulated and contained.

## 5 Reward policy for formers

According to the models we established in the former questions, we give the government some reward policies as follows.

- **The agricultural income subsidies policy**

1. The direct subsidies to agricultural income. In order to stabilize and increase farmers' income and reduce the loss bringing to the farmers' income policy changes, the government of individual farmers and farmers farms in public finance transfer payment directly. The key is to take production cost subsidy.

2. The indirect subsidies to agricultural income. Agricultural income indirect finance subsidy policy is mainly determined by the government planning target support price to encourage production to indirectly protect farmers get higher income.

- The government subsidies the farmer in mountainous areas in order to reduce the production costs in these areas.
- Subsidies for the purchase of aquaculture production. Farmers who do not use antibiotics to raise pigs or raise chickens can get subsidies from the central government for the purchase of agricultural machinery and the construction of agricultural facilities.
- Credit for agricultural production. The government provides low-interest and interest-free loans to farmers through credit institutions such as agricultural credit banks and agricultural credit cooperatives so that producers can obtain the necessary funds for purchasing production materials and agricultural machinery.
- Breeding insurance. The insurance company provides a certain proportion of compensation when the farmer suffers loss from insurance coverage. In the event of major disasters, the insurance companies provide full insurance subsidies.
- Strengthen policy subsidies. The government to use antibiotics in line with the testing standards of units in accordance with the output value of subsidies incentives.

## 6 Strengths and Weaknesses

- **Strengths**

In this paper, we adopt the profile of the various parts of the interest chain for the abuse of antibiotics as their sensitivity to the interest chain. We use the linear correlation matrix to determine the degree of correlation between the various factors, and use quantify the data processing system to make the non-quantitative data quantified.

- **Weaknesses**

In this paper, we emphasize the theory and model and lake large amounts of data support. What's more, in this paper, many ideal assumptions are made, and many perturbations are neglected. In practice, errors may occur often.

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